OSTERWALDER–SCHRADER AXIOMS FOR THREE-DIMENSIONAL SCALAR QUANTUM ELECTRODYNAMICS AT FIXED PHOTON MASS

Abstract. King constructed the continuum, infinite-volume limit of three-dimensional scalar quantum electrodynamics, the $U(1)$ (abelian) Higgs model, along Neumann–Dirichlet cubes at block factor 5 [15, 16]. For his printed state at fixed photon mass we prove the Osterwalder–Schrader axioms: regularity E0″ with absolute constants, invariance under the full Euclidean group, reflection positivity about every plane, symmetry, and the measure with its ergodic decomposition, on whose components the cluster axiom holds in all its forms. The existence of King's cube limits, the existence clause of his Theorem 2.1, is proved from his renormalisation-group analysis in a ratio form in which the vacuum energies cancel; rotation invariance is proved by comparing King's cubes with periodic tori carrying a smooth mass wall on two lattices whose orientations differ by arctan(3/4). With real scalar sources the same holds jointly for the curvature and the renormalised Wick square on a tube of small Wick amplitudes, without a Euclidean clause. The curvature results hold on King's whole coupling range; the Wick results hold under a one-sided sign clause of the Wick leg, which we prove on explicit coupling regions. The limit of vanishing photon mass is left to later work. This is the short form of the paper: the complete proofs of all technical estimates are given in the full version, whose numbering this text keeps.

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1. Introduction and main results

1.1. **King's construction.** King [15, 16] constructed three-dimensional scalar quantum electrodynamics with a massive photon (the $U(1)$ Higgs model in King's terminology) as a continuum and infinite-volume limit of lattice models. The lattice theory lives on $\Lambda_k = \varepsilon_k \mathbb{Z}^3$ with spacings $\varepsilon_k = 5^{-k}$; its action has a photon A in Feynman form with mass $\mu_0 > 0$, an N-component scalar ϕ with quartic coupling $\lambda > 0$ and bare mass $m > 0$, minimally coupled with charge $e > 0$, and one mass counterterm $\delta m^2_{\varepsilon_k}$. The counterterm is King's free-boundary one: it is computed with infinite-lattice propagators, so it is one number per spacing, the same at every site of every box and every torus. The ultraviolet problem is settled in [15] with Bałaban's renormalisation group [1, 2, 3, 4] at block factor 5: on a finite volume the effective actions converge as $k \to \infty$. The infinite-volume problem is treated in [16] with the correlation inequalities of Brydges, Fröhlich and Seiler [7]. King's finite-volume states are Neumann–Dirichlet states on the cubes $C_J = [-MJ, MJ)^3$, M his large-block integer, "a large integer" [15, §2.3]: the photon gradient terms are kept only inside the cube (Neumann), and the scalar field vanishes on a boundary layer (Dirichlet). The limit is iterated, first $k \to \infty$ at fixed J, then $J \to \infty$; the second limit exists because the cube states are monotone in J on a cone of observables, by [7].

This paper proves the Osterwalder–Schrader axioms for King's printed state at fixed photon mass, with full Euclidean invariance, and the corresponding statements for the joint field of the curvature and the renormalised Wick square on a tube of small Wick amplitudes. It is self-contained in the following sense. The existence of King's cube limits, the existence clause of his Theorem 2.1 [16], is proved in the existence part of the paper (Section 5, and Section 6 for the version with scalar sources): King's route, the comparison of two spacings at a common block scale, is run in a ratio form in which the vacuum energies cancel, so that the convergence of the vacuum energies, which his printed argument uses and which is not available on the cube, is not needed. The passage from the cube limits to the axioms in [16] is asserted at three places, and at each we proceed differently. First, King's Neumann–Dirichlet cube form is not reflection symmetric about any plane, and its natural symmetrisation is not reflection positive (an exact counterexample at $e = 0$, Remark 9.7), so his reflection positivity goes through a modified box state; that this box state is reflection positive is his Proposition 3.3, which he derives from [7, Cor. 5.4], whose hypotheses do not cover the finite-box Gaussian of the modified action. Proposition 3.3 is true, and it is proved here (Lemma 9.3) rather than derived from [7, Cor. 5.4]. Second, the inequality by which he passes from the box states back to the cube limit holds in the reverse direction for the class of observables his argument needs: weakly, at every coupling, for the box state with the site-independent counterterm, and strictly at $e = 0$ (Lemma 9.6); it is replaced by a two-sided sandwich between cube states with a margin of order μ_0^{-1} (Lemmas 9.9 and 9.10). Third, the continuum subsequences through which he reaches the limit theory are built here (Remark 14.15). With scalar sources, finally, his Theorem 2.1 needs a vacuum counterterm that his Wick ordering does not contain; we work with a scheme a and prove the existence of the cube limits with sources in that scheme.

The abstract measure theory behind the probabilistic axioms (the realisation of the limit functional as a measure on a space of distributions, its ergodic decomposition, reconstruction in a coordinate direction, the weak clustering of the components, and clustering in every direction under rotation invariance) is proved in full in the full version. Rotation invariance is proved in the rotation part of this paper, whose main theorem (Theorem 15.1) precedes the main theorem of the paper: King's cubes are compared with periodic tori carrying a smooth mass wall, the walled tori on two lattices whose orientations differ by $\arctan(3/4)$ are compared, and King's state is identified as the lower envelope of the walled tori through a heavy-shell decoupling estimate on King's block-spin flow with the shell mass inside the matter covariance.

1.2. **The two cube conventions.** King's cube state $\mathrm{ND}^K_{k,J}$ (Definition 2.2) has its photon variables on $X_J = C_J \cap \Lambda_k$ and its scalar alive on $\{x \in X_J : \ x_\nu < MJ - \varepsilon_k \text{ for all } \nu\}$, that is, killed on the three top face layers T_J of the cube. The comparison cube state $\mathrm{ND}^{(a)}_{k,J}$ (Definition 2.1) has the scalar alive on all of X_J. The Wick-tube lemmas and the lattice comparisons of the rotation part are written for the comparison convention, while King's printed state is on his own. The two are interlaced: for observables of the cone class, with supports at ℓ^∞-distance > 1 from the complement of the cube,

$$\langle\Theta\rangle^K_{k,J} \ \le \ \langle\Theta\rangle^{(a)}_{k,J} \ \le \ \langle\Theta\rangle^K_{k,J+1},$$

and reversely for the opposite class (Lemma 4.2). Hence both families have the same J-limit at fixed k; the comparison family's fixed-cube limits inferior and superior are squeezed to King's printed state Z^K_∞ as $J \to \infty$, and every continuum-subsequence limit of the comparison family equals Z^K_∞ (Corollary 4.4). What does not transfer is the existence of the fixed-J limit $k \to \infty$ itself: on one convention it does not imply it on the other. This is why the existence theorems are proved on King's convention, where the printed state lives and where King's block-spin flow runs; the Wick tube, proved on the comparison convention, is carried to King's convention by a transfer theorem (Theorem 21.3).

1.3. **The main theorem.** Sources are real 2-forms $f_j \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$ with coefficients z_j and real scalar test functions $h_j \in C^\infty_c(\mathbb{R}^3)$ with coefficients w_j; $F(f)$ is the curvature paired with f, $W^a(h)$ the renormalised Wick square in the scheme a, and $g(z) = \sum_j (\mathrm{Re}\, z_j) f_j$, $k(\zeta) = \sum_j (\mathrm{Re}\, w_j) h_j = k_+ - k_-$, with the norms $\|\cdot\|_2$, $|\cdot|_4$, $|\cdot|_w$ and the tube $\mathcal{D}_{\sigma_+,H}$ of Definitions 2.4 and 2.6. The Schwinger functions S_n, $S_{n,p}$ are the Taylor coefficients of the limit functionals at 0. A scheme a is admissible if $a = z$, the reference scheme, or $z_k - a_k$ converges (Definition 3.7). The following summary is restated with every clause in full and proved as Theorem 21.1.

Theorem 1.1 (Main Theorem, summary)**.** *Let $e > 0$ and $\lambda, m, \mu_0 > 0$ be arbitrary, and let King's large-block integer satisfy $M \in 50\mathbb{N}$ and $M \ge M_{\min}$ (Remark 2.3); the objects below do not depend on such M, the limits of (B) on the Wick tube coinciding where both are defined.*

(A) The curvature field. *King's printed state $Z^K_\infty(f, 0) = \lim_J \lim_k \langle e^{F(f)}\rangle^K_{k,J}$ exists for every real 2-form f, extends to an entire function of the coefficients of every finite family, and satisfies:*

(i) (OS0) $|Z^K_\infty(\sum_j z_j f_j, 0)| \ \le \ e^{\|g(z)\|_2^2}$ *for all $z \in \mathbb{C}^N$, with no dependence on μ_0, e, λ, m or M, and $|Z^K_\infty(\sum_j z_j f_j, 0)| \ \le \ \exp\big(\frac{1}{2}\mu_0^{-2}\|\sum_j (\mathrm{Re}\, z_j) d^* f_j\|_2^2\big)$; for real test functions $|S_n(f_1, \dots, f_n)| \ \le \ (\sqrt{2}e)^n (n!)^{1/2} \prod_j \|f_j\|_2 \ \le \ (e\pi/2)^n (n!)^{1/2} \prod_j |f_j|_4$; the S_n are tempered distributions and satisfy the regularity axiom* E0″ *with $s = 4$, the constant $\sigma_n = (e\pi)^n (n!)^{1/2}$ for complex $f_j \in \mathcal{S}$, and every $\beta > \frac{1}{2}$.*

(ii) (OS1) *Invariance under the full Euclidean group $\mathbb{R}^3 \rtimes O(3)$: under all translations of $\mathbb{R}^3$, under the cubic group O_h about every point, and under all rotations, F transforming as a 2-form; the S_n are Euclidean covariant.*

(iii) (OS2) *Reflection positivity about every plane $\{x \cdot n = c\}$, n a unit vector, $c \in \mathbb{R}$, in matrix form for exponentials with arbitrary complex curvature coefficients and in polynomial form.*

(iv) (OS3) *S_n is real and symmetric, and $S_n = 0$ for odd n.*

(v) (The measure) *For every even $r \geq 14$ there is exactly one Borel probability measure μ_F on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ with moments S_n and characteristic functional $Z^K_\infty(\mathrm{i}\,\cdot,0)$ on $\mathcal{S}$; its image on H^{-r}_{loc} is determined by the characteristic functional and by the moments. The laws of the localised lattice fields χF_{Λ_k} under $\mathrm{ND}^K_{k,J}$ converge weakly on $H^{-r}_{\mathrm{per}}(Q')$, Q' a box containing $\operatorname{supp}\chi$, along the full sequence $k\to\infty$ at every fixed $J\geq J_0(\chi)$, and then as $J\to\infty$, to the law of χF under μ_F.*

(vi) (OS4 on the ergodic components) *μ_F is $\mathbb{R}^3\rtimes O(3)$-invariant and $\mu_F=\int \mu_\xi\,d\nu(\xi)$ with translation-ergodic components; ν-almost every component is $\mathbb{R}^3\rtimes O(3)$-invariant, reflection positive about every plane, clusters in the mean and under diagonal separation, has a unique joint vacuum in every direction, satisfies* OS4 *in all its forms for every choice of the time direction, and has real symmetric moments with a factorial growth bound with component constants; the decomposition is O_h-invariant, each component being a mixture of $k\in\{1,2,3,6\}$ ergodic measures. The vanishing of the odd moments is not claimed for a component.*

(B) The joint field on the Wick tube. *Let the couplings satisfy the sign clause* (T3) *of Hypothesis 3.12, $\sigma_1(M_1)>0$ for some $M_1\in(0,m)$; in particular, let them lie in one of the explicit regions of Proposition 22.14. Let a be an admissible scheme. Then for every family of sources supported in $Q_R=[-R,R]^3$ the joint functional of (F,W^a) has a limit G_∞ on the tube $\mathcal{D}_{\sigma_+,H}$, with*

$$\sigma_+=s_1,\qquad H=\tfrac{s_1}{16},\qquad \kappa=\tfrac{C_U}{s_1},$$

and K_- determined by the cell constant B_{cell} (Equation (30)), where $s_1\in(0,\sigma_1(M_1))$ is the amplitude of the extensive bound and C_U its constant (Equation (28) and Proposition 12.6); the constants depend on (e,λ,m,μ_0,M,a) and on the auxiliary choices M_1, χ_0, s_1, and on nothing else, and they are not uniform in μ_0. The joint Schwinger functions $S_{n,p}$ satisfy (i)–(vi) *in the following form:*

(i) *$|G_\infty(\zeta)|\leq\exp\big(\|g\|_2^2+\kappa\|k_+\|_{L^1}+K_-|k_-|_w\big)$ on $\mathcal{D}_{\sigma_+,H}$, and for real test functions $|S_{n,p}|\leq e^{K'\sigma_0}(\sqrt{2}e)^n(n!)^{1/2}(e/\sigma_0)^p\,p!\prod_i\|f_i\|_2\prod_j|h_j|_w$ with $\sigma_0=H/4$ and $K'=\kappa\pi^2+K_-$ (a factor 2^{n+p} for complex test functions); $\mathrm{E0}''$ holds with $s=4$ and every $\beta>1$.*

(ii) *Translations and O_h about every point hold as identities on the tube, W^a transforming as a scalar; there is no Euclidean clause.*

(iii) *Reflection positivity about every coordinate plane for families whose Wick amplitudes satisfy $\mathsf{a}(\zeta_i)<\sigma_W=\sigma_+/8$, in particular for all purely imaginary Wick coefficients, and the polynomial form.*

(iv) *$S_{n,p}=0$ for odd n.*

(v), (vi) *As in* (A) *for the measure on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2\oplus\mathbb{R})$, $r\geq14$ even, with the scheme-shifted lattice field in the convergence clause, and with the weak form of* OS4 *(ergodicity under $\mathbb{R}^3$ and clustering in coordinate directions) on ν-almost every component; there is no Euclidean clause.*

The same statements hold for the tube object G^K_∞ of King's printed state on his own convention (Theorem 21.3 and Proposition 21.4).

The μ_0-free bound in (A)(i) is absolute: it depends on none of the parameters of the model. The μ_0^{-2} bound is the finite-volume Gaussian bound carried to the limit. Part (B) is the headline form of the Wick leg; the extension of (B) to arbitrary Wick amplitudes, under the hypothesis (W2) at every amplitude, with the only Euclidean clause of the Wick leg, is stated separately as Theorem 23.7 among the open problems.

1.4. **The massless limit.** The photon mass μ_0 is fixed throughout: King's construction, and every statement above, is at $\mu_0>0$, and the constants of the Wick leg are not uniform in μ_0. The limit of vanishing photon mass, along a family whose counterterm and scheme are frozen at a reference mass, is not treated in this paper; it is listed among the open problems as later work.

1.5. **The coupling range.** The curvature leg holds on all of King's range: $e > 0$ and $\lambda, m, \mu_0 > 0$ arbitrary. Every smallness condition in [15, 16, 1, 2, 3, 4] is a cap on the running couplings at a stopping scale that is chosen after the point (e, λ, m, μ_0) and the datum, so the quantifier over the couplings is the one over the sources, and no sentence "e, λ sufficiently small" occurs (Lemma 22.1; Remark 22.15). No relation $\mu_0 \le m$ is used or needed. The only coupling condition in the paper is the one-sided sign clause (T3) of the Wick leg, $\sigma_1(M_1) > 0$ for some $M_1 \in (0, m)$; it fails exactly when m minimises King's free-boundary bare coefficient α_∞ on $(0, m]$, and to one-loop order it holds if and only if

$$e^2 < e_c^2(\lambda, m, \mu_0) = \max\Big\{\frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1+\frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1+\frac{\lambda(N+2)}{\pi m}\Big)\Big\}$$

(Equation (60)), so that a half-line of couplings fails it at one loop. Beyond one loop, (T3) is proved on explicit coupling regions, at $\mu_0 = m$ on a region containing $0 \le \lambda, e^2 \le 0.1m$ and at every $\mu_0 \in (0, m]$ on explicit boxes (Proposition 22.14); the Wick leg is stated on those regions, and more generally wherever (T3) holds. The mass clause in Bałaban's upper bound, a cap on the dimensionless scalar mass at the stopping scales where his Proposition 3.1 is used, is part of that proposition's statement and is one more one-sided cap on the stopping scale, not a condition on the couplings (Remark 22.2).

1.6. **What is proved where.** The inputs of the main theorem are collected in Section 3: the existence theorems for King's cube limits without and with scalar sources (Theorems 3.3 and 3.6), proved in the existence part; the rotation theorem (Theorem 3.14), proved in the rotation part; Bałaban's torus stability bounds and the c-number lemma in the form the Wick leg uses (Proposition 3.9 and Lemma 3.10); and the coupling condition (T3). Everything else is proved in the sections between. Open are only the optional extensions: the Wick sector at arbitrary amplitudes ((W2), Form (A), Theorem 23.7), the relation of King's state to the limits of periodic-torus states (Definition Q.1 of the full version) and its uniqueness and ergodicity, the limit of vanishing photon mass, and non-triviality; they are collected in the final section.

1.7. **Structure of the paper.** The model, King's cube, the comparison cube, the sources and the standing lemmas (the correlation inequalities of [7], with the two inequalities that are only sketched there proved in full, Gaussian domination, cutting and the elementary facts) come first, then the inputs with their provenance. The interlacing of the two cube conventions (Lemma 4.2 and Corollary 4.4) and its three geometry margins follow, and then the existence part: King's existence theorem on the cube (Section 5) and its version with scalar sources. The curvature sector is then built up in the order of the axioms: regularity with the absolute constants (Theorems 7.4 and 7.6), translations, the cubic group and the rotation clause (Theorems 8.5 and 8.7), reflection positivity through the modified box and its passage to the limit with OS3 and the polynomial form (Lemma 9.3, Theorem 10.6, and Propositions 10.9 and 10.10), and the measure with its ergodic decomposition (Theorem 11.1). The Wick tube is treated in two sections, the torus inputs and the extensive bound (Propositions 12.2 and 12.6) and the uniform tube bound with its limits (Theorem 13.10 and Lemma 13.11); continuum subsequences and the states they define, with no existence input, form a section of their own (Theorem 14.9). The rotation part follows: the geometry of the two lattices and the walled tori, the admissible schemes and the full object, the torus comparison and its reduction to one-sided bounds at one spacing, the closing estimate and the invariance theorems, and the heavy shell in two sections; the corrections to King's papers that it uses are collected in the full version. The main theorem is assembled in Theorem 21.1, and the coupling range is established in Lemma 22.1 and Proposition 22.14. The open problems close the paper. This is the short form of the paper: the complete proofs of every statement are given in the full version, whose numbering of statements and equations this text keeps, so that a statement quoted from there can be found under the same number; a statement printed only there is cited as such. The unnumbered section before the open problems outlines, appendix by appendix, what the full version proves and how, among other things the measure theory on spaces of distributions, the ergodic decomposition, reconstruction and

clustering, clustering under rotation invariance and the two-loop majorants of the lattice mass counterterm.

2. The model, King's cube and the standing lemmas

2.1. **Couplings, lattices and the action.** Throughout, $e > 0$ and $\lambda, m, \mu_0 > 0$ are the charge, the quartic coupling, the bare scalar mass and the photon mass; no further condition on the couplings is imposed, since every smallness condition of [15, 16, 1, 2, 3, 4] is a cap on the running couplings at a stopping scale chosen after the point (e, λ, m, μ_0) and after the datum (the section on the coupling range). The block factor is 5: the spacings are $\varepsilon_k := 5^{-k}$, the lattices $\Lambda_k := \varepsilon_k \mathbb{Z}^3$ with the standard frame (e_α), and M is King's large-block integer, "a large integer" [15, §2.3], the integer of [4], with $M \in 50\mathbb{N}$ and $M \geq M_{\min}$. Here $M_{\min}$ is the least integer such that every integer $M \geq M_{\min}$ satisfies the lower bounds placed on the large-block integer, or on the side of the big or of the large blocks, in the full version (Lemma K.17, Proposition K.20, Convention K.87, Proposition K.227, Lemma F.163 and Lemma K.335 of the full version, its hypothesis (H1) on the torus bounds, and its sections on the printed inputs on a lattice torus fitted to the blocks, on the order of the constants of the shell bounds and on the walk-sum bounds). Each of these bounds holds for all M above a threshold that depends only on the couplings and on constants fixed before M, not on the spacing, the cube, the torus or the sources; so $M_{\min} < \infty$, and $M_{\min} \geq 50$ by the walk-sum bounds of the full version. The divisibility $50 \mid M$ enters twice: $M/5$ is even in Lemma K.335 of the full version, and $2M \in 25\mathbb{N}$ in the walk-sum bounds there. M is fixed after the couplings. Where an argument follows [15, 16], a *large block* is King's block of M^3 sites of the lattice of its scale [15, §2.3]; where it follows [1, 2, 3, 4], these cubes are called *big blocks*, and a large block is a cube of $5M$ sites per side [1, p. 607] ((H1) of the full version). King's cubes are $C_J := [-MJ, MJ)^3$, $J \in \mathbb{N}$, and $X_J := C_J \cap \Lambda_k$ (the spacing understood from the context). A *King torus* is $\mathbb{R}^3/(\ell\mathbb{Z})^3$ with $\ell \in 2M\mathbb{N}$; Λ_k descends to it, with $\ell\, 5^k$ sites per side.

The scalar field is $\phi(x) = r_x e^{\mathrm{i}\vartheta_x} \in \mathbb{C}$ and the photon field a real $A_\alpha(x)$ for every site x and direction α; the bond $\langle x, x + \varepsilon e_\alpha\rangle$ carries $A_\alpha(x)$ and the reversed bond $-A_\alpha(x)$. King's action [16, (2.5)] at spacing ε is

$$S = \tfrac{1}{2}\sum_{\alpha,\beta}\sum_{x} \varepsilon \big(A_\alpha(x+\varepsilon e_\beta) - A_\alpha(x)\big)^2 + \frac{\mu_0^2}{2}\sum_x \varepsilon^3 |A(x)|^2 + \tfrac{1}{2}\sum_\beta \sum_x \varepsilon \left|e^{\mathrm{i}e\varepsilon A_\beta(x)}\phi(x+\varepsilon e_\beta) - \phi(x)\right|^2$$
$$+ \sum_x \varepsilon^3 \Big[\tfrac{1}{2}\big(m^2 + \delta m_\varepsilon^2\big) r_x^2 + \lambda r_x^4\Big],$$

the photon in Feynman form with mass μ_0, and the hopping term of a bond $\langle x, y\rangle$, $y = x + \varepsilon e_\beta$, equal to $\frac{\varepsilon}{2}(r_x^2 + r_y^2) - \varepsilon r_x r_y \cos(\vartheta_x - e\varepsilon A_\beta(x) - \vartheta_y)$: a *diagonal part* and a *cosine part*. The counterterm δm_ε^2 is King's free-boundary counterterm [16, p. 330], fixed by his two-point graphs of order at most four in (e, λ) at zero momentum with the propagators of the infinite lattice $\varepsilon\mathbb{Z}^3$: it is *one number per spacing*, the same at every site of every box and torus. The Wick constant is the free one, $\mathsf{W}_\varepsilon := (-\Delta_{\varepsilon\mathbb{Z}^3} + m^2)^{-1}(0,0)$. This *free convention* is used on cubes and tori alike unless said otherwise; King's own torus convention appears only inside the torus inputs of the Wick leg.

2.2. **The states.**

Definition 2.1 (Neumann–Dirichlet boxes, killed sites, the comparison cube)**.** For a finite set $X \subset \Lambda_k$ the *Neumann–Dirichlet state* $\langle\cdot\rangle_{k,X}^{\mathrm{ND}}$ is the normalised measure $Z^{-1}e^{-S}\prod_{x\in X} d^2\phi(x) \prod_{x\in X,\alpha} dA_\alpha(x)$ with $\phi = 0$ at every site outside X (Dirichlet condition: a bond leaving X contributes only its diagonal part $\frac{\varepsilon}{2}r_x^2$ at its end $x \in X$), photon gradient terms only for pairs $x, x + \varepsilon e_\beta$ both in X, and the photon mass term on X (Neumann condition). For $U \subset X$ the state *with the matter killed on* U replaces the single-site measure $d^2\phi(x)$ times its weight by the point mass at $\phi(x) = 0$ for $x \in U$; every hopping term with an endpoint in U then reduces to its diagonal part at the other endpoint, and the photon variables are unchanged. The *comparison*

cube is $\mathrm{ND}^{(a)}_{k,J} := \langle\cdot\rangle^{\mathrm{ND}}_{k,X_J}$, written $\langle\cdot\rangle^{(a)}_{k,J}$: the scalar is alive on all of X_J, with Dirichlet planes $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ$.

Definition 2.2 (King's RG cube)**.** Let $\mathrm{alive}^K_J := \{x \in X_J : \ x_\mu < MJ - \varepsilon_k \ \forall\mu\}$ and $T_J := X_J \setminus \mathrm{alive}^K_J$, the three top face layers. *King's RG cube* $\mathrm{ND}^K_{k,J}$, written $\langle\cdot\rangle^K_{k,J}$, is the Neumann–Dirichlet state on X_J with the matter killed on T_J: the photon variables $A_\alpha(x)$, $x \in X_J$, their gradient terms for pairs in X_J and their mass term on X_J are those of the comparison cube; the scalar is alive on alive^K_J and $\phi = 0$ on every other site of Λ_k, in particular on T_J. Its Dirichlet planes are $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ - \varepsilon_k$, at separation $2MJ$; the Neumann image planes of the photon, $x_\mu = -MJ - \varepsilon_k/2$ and $x_\mu = MJ - \varepsilon_k/2$, are the same in both conventions.

The two conventions differ exactly by the matter on T_J; the photon Gaussian is the same. King's RG cube is the box [16, (3.3)–(3.4)] with the Dirichlet planes of [16, (3.9)] and the photon reflection planes of [16, (3.7)], the geometry of his renormalisation group; his Theorem 3.2 and his modified box [16, (3.38)] are written in the comparison convention. The two families are interlaced in Section 4; the existence theorems of this paper are posed on ND^K, while the rotation part and the cell lemmas of the Wick leg are written for the comparison family. The *periodic state* $\langle\cdot\rangle_{k,T}$ on a King torus T has the action summed over the finite graph $\Lambda_k/(\ell\mathbb{Z})^3$, in the free convention.

Remark 2.3 (independence of M). King's cube state depends on (M, J) only through MJ (Definition 2.2); the spacing, the action, the counterterm, the Wick constant and the schemes (Definition 2.4) do not involve M. Let M and $\tilde{M}$ both satisfy the conditions above. Their least common multiple $\check{M}$ satisfies them too, and for $X \in \{M, \tilde{M}\}$ the cubes of $\check{M}$ are cubes of X: $[-\check{M}J, \check{M}J)^3 = [-XJ_X, XJ_X)^3$ with $J_X := (\check{M}/X)J \in \mathbb{N}$. Hence, whenever an iterated limit $\lim_J \lim_k$ of the following sections exists for M and for $\tilde{M}$, the two limits agree with the limit along the cubes of $\check{M}$, so they coincide; for limits on a tube that depends on M, they coincide on the intersection of the two tubes. Thus the limit objects of Theorem 21.1 do not depend on the choice of M among the integers $M \in 50\mathbb{N}$ with $M \geq M_{\min}$: such a choice changes at most the tube on which a limit is asserted, through the constants of the tube, and two such limits agree where both are defined. Moreover, for every integer $M^\sharp \geq 1$ and every $J \in M\mathbb{N}$, $[-M^\sharp J, M^\sharp J)^3 = C_{M^\sharp J/M}$. So along $J \in M\mathbb{N}$ the iterated limits for the cubes $[-M^\sharp J, M^\sharp J)^3$ exist and equal those for M. Along the full sequence $J \in \mathbb{N}$ they are asserted only for $M^\sharp$ satisfying the conditions above.

BFS measures. A *BFS measure* on a finite set V of sites is a probability measure

$$Z^{-1} \exp\Big(\sum_{\{x,y\}\in B} c_{xy}\, r_x r_y \cos\big(\vartheta_x - \vartheta_y - \ell_{xy}(A)\big)\Big) \prod_{x\in V} d\rho_x(r_x)\, d\vartheta_x\, d\gamma(A),$$

with B a set of pairs of sites, $c_{xy} \geq 0$, each ρ_x a finite measure on $[0,\infty)$ with $\int e^{\alpha r^2} d\rho_x < \infty$ for all $\alpha \geq 0$ (point masses at 0, that is killed sites, included), γ a non-degenerate centred Gaussian measure on $\mathbb{R}^m$ (the photon variables), and linear functionals ℓ_{xy}. Every state above is a BFS measure: $c_{xy} = \varepsilon$ for the bonds present, the single-site measures contain $e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]}$ and the diagonal parts $\frac{\varepsilon}{2} r_x^2$ of all bonds at x (those leading out of the box or into a killed site included); the photon form is positive definite because $\mu_0 > 0$, and the exponential moments are finite because $\lambda > 0$. So is every measure obtained from such a state by multiplying some c_{xy} or some photon gradient terms by $t \in [0,1]$, adding $s|\ell(A)|^2$ ($s \geq 0$) to the action, multiplying single-site measures by $e^{\varepsilon^3 q(x) r^2}$ with real q, or killing sites. A BFS measure is *of King type* if $c_{xy} \in [0, \varepsilon]$ and each ρ_x is a point mass at 0 or has the density $r\, e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]-\varepsilon^3 w(x) r^2 - \frac{\varepsilon}{2} n_x r^2}$ with $w(x) \geq 0$, $n_x \in \{0,\dots,6\}$; every state above, and every measure obtained from one by the operations just listed with $q \leq 0$, is of King type. Every state above is invariant under the *charge conjugation* $(A, \vartheta) \mapsto (-A, -\vartheta)$, which changes the sign of every field strength and fixes r.

2.3. **Sources, schemes and the objects.**

Definition 2.4 (sources, schemes, classes, transport)**.** A *curvature source* is a real 2-form $f \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$, a *scalar source* a real $h \in C_c^\infty(\mathbb{R}^3)$. For the plaquette $p=(x;\alpha<\beta)$ with centre $c(p)$ let $F(p) := \varepsilon^{-1}(A_\alpha(x)+A_\beta(x+\varepsilon e_\alpha)-A_\alpha(x+\varepsilon e_\beta)-A_\beta(x))$ and

$$F_{\Lambda_k}(f) := \sum_p \varepsilon^3 f_{c(p)}(e_\alpha,e_\beta)F(p), \qquad \mathsf{S}_{\Lambda_k}(h) := \sum_x \varepsilon^3 h(x), \qquad :\phi^2:(h) := \sum_x \varepsilon^3 h(x)\big(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon\big),$$

the sums over the plaquettes (corners in the box), respectively the sites, of the region. A *scheme* is a real sequence $a=(a_k)$, a_k depending only on ε_k and the couplings; $W^a(h) := \sum_x \varepsilon^3 h(x)(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon - a_k)$. For a finite family $\zeta=(z_j,f_j;w_j,h_j)_j$ with complex coefficients put $\Theta(\zeta) := \exp(\sum_j z_j F(f_j) + \sum_j w_j{:}\phi^2{:}(h_j))$ and, for a Neumann–Dirichlet state on a box X (killed sites allowed),

$$Z^a_{k,X}(\zeta) := \Big\langle \exp\Big(\sum_j z_j F(f_j) + \sum_j w_j W^a(h_j)\Big)\Big\rangle^{\mathrm{ND}}_{k,X} = e^{-\sum_j w_j a_k \mathsf{S}_{\Lambda_k}(h_j)}\big\langle\Theta(\zeta)\big\rangle^{\mathrm{ND}}_{k,X};$$

$Z^a_{k,J} := Z^a_{k,X_J}$ for the comparison cube, $Z^{a,K}_{k,J}$ for King's RG cube, and $N^a_{k,X}(0,q) := e^{-(a_k+2\mathsf{W}_\varepsilon)\mathsf{S}(q)}\big\langle e^{\sum_x \varepsilon^3 q(x) r_x^2}\big\rangle^{\mathrm{N}}_{k,}$ for one real scalar source. At $h=0$ the multiplier is 1 and a is dropped: $Z_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$, $Z^K_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ for $z\in\mathbb{C}^n$. The *cone* is $\mathcal{K} := \{(f,g) : f \text{ a real 2-form},\ g\ge 0\}$ with the observable $\Theta(\mathrm{i}f,g)$ (*class* $\mathcal{I}$); *class* $\mathcal{D}$ consists of the observables $\Theta(f,-g)$, $(f,g)\in\mathcal{K}$. Transport: $(O\cdot h)(x) := h(O^{-1}x)$, $(O\cdot f)_x(u,v) := f_{O^{-1}x}(O^{-1}u,O^{-1}v)$ for $O\in O(3)$; $\tau_b h := h(\cdot - b)$, $(\tau_b f)_x := f_{x-b}$. For $\nu\in\{1,2,3\}$, $c\in\mathbb{R}$ let $\rho=\rho_{\nu,c}$, $x\mapsto x-2(x_\nu-c)e_\nu$; $\theta h := h\circ\rho$, $\theta f := \rho^* f$, $(\rho^* f)_x(u,v) := f_{\rho x}(\rho u,\rho v)$ (the components $f_{\mu\nu}$, $\mu\ne\nu$, change sign); $\theta\zeta^* := (\bar z_j,\rho^* f_j;\bar w_j,h_j\circ\rho)_j$, and $\zeta\sqcup\zeta'$ is the union of two families. The *amplitude* is $\mathsf{a}(\zeta) := \sum_j |\operatorname{Re} w_j|\,\|h_j\|_\infty$. The *reference scheme* is $z=a^\sharp$, $a^\sharp_k := e^2 X_{\varepsilon_k}$, X_ε the coefficient of e^2 in the perturbative expansion of $\langle|\phi(0)|^2\rangle$ on $\varepsilon\mathbb{Z}^3$ at $\lambda=0$ with the order-e^2 part of δm^2_ε inserted.

By charge conjugation $\langle\Theta(\mathrm{i}f,g)\rangle = \langle\cos F(f)\, e^{:\phi^2:(g)}\rangle$ is real for every state above, and $Z^a_{k,X}(-z,w) = Z^a_{k,X}(z,w)$.

Remark 2.5 (schemes). The Wick leg is stated for the reference scheme z, or for an admissible scheme a (Definition 3.7: z_k-a_k converges, hence is bounded). If the fixed-cube limits exist in $(0,\infty)$ for both a and z at a real scalar source, the scheme is admissible: for $0\le h\in C_c^\infty$ with $\int h>0$ supported well inside some C_J, $Z^z_{k,J}(0,h)/Z^a_{k,J}(0,h) = e^{-(z_k-a_k)\mathsf{S}_{\Lambda_k}(h)}$ converges to a number in $(0,\infty)$, and $\mathsf{S}_{\Lambda_k}(h)\to\int h>0$. What the Wick leg uses is only $\sup_k|z_k-a_k|<\infty$.

Definition 2.6 (norms and the tube)**.** For a family with sources $f_1,\dots,f_n$, $h_1,\dots,h_p$ supported in $Q_R := [-R,R]^3$ and $\zeta=(z,w)\in\mathbb{C}^{n+p}$ put $g(\zeta) := \sum_i(\operatorname{Re} z_i)f_i$ and $k(\zeta) := \sum_j(\operatorname{Re} w_j)h_j = k_+ - k_-$ (pointwise positive and negative parts). Lattice norms: $\|g\|^2_{2,k} := \sum_p \varepsilon_k^3 g_{c(p)}(e_\alpha,e_\beta)^2$ over the plaquettes of Λ_k, $\|q\|_{1,k} := \sum_{x\in\Lambda_k}\varepsilon_k^3|q(x)|$. Continuum norms: $\|g\|_2^2 := \sum_{\alpha<\beta}\|g_{\alpha\beta}\|^2_{L^2}$; $|q|_w := \sup_x(1+|x|^2)^2|q(x)|$; $|q|_4 := \sup_{x,|\gamma|\le 4}(1+|x|^2)^2|D^\gamma q(x)|$ and $|g|_4 := (\sum_{\alpha<\beta}|g_{\alpha\beta}|_4^2)^{1/2}$; on scalar functions in $\mathcal{S}(\mathbb{R}^3)$, $|q|_4=|q|_4$, where $|q|_p := \sup_{x,|\gamma|\le p}|(1+|x|^2)^{p/2}D^\gamma q(x)|$ is the norm of [20, (2.1), p. 284]. For $\sigma,H>0$ the *tube* is

$$\mathcal{D}_{\sigma,H} := \big\{\zeta\in\mathbb{C}^{n+p} :\ \|k_+(\zeta)\|_\infty<\sigma/2,\ \ \|k_-(\zeta)\|_\infty<H/2\big\}.$$

Since $s\mapsto\|(\sum_j s_j h_j)_\pm\|_\infty = \sup_x(\pm\sum_j s_j h_j(x))_+$ is a supremum of convex functions of $s=\operatorname{Re} w$, hence convex and continuous, $\mathcal{D}_{\sigma,H}$ is the preimage under $\zeta\mapsto\operatorname{Re} w$ of the intersection of two open convex sublevel sets: open, convex, hence connected; it contains 0 and every ζ with purely imaginary w; $\|k(\zeta)\|_\infty<\min(\sigma,H)/2$ implies $\zeta\in\mathcal{D}_{\sigma,H}$; its real points form an open subset of $\mathbb{R}^{n+p}$. The tube of the Wick leg is $\mathcal{D}_{\sigma_+,H}$, with σ_+ and H fixed there.

The objects. For real 2-forms $f_1, \dots, f_n$ let $J_0(f)$ be the least J such that every support lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$. King's *cube limits* and *printed state* are

$$Z_J^K(z) := \lim_{k\to\infty} Z_{k,J}^K(z) \quad (J \geq J_0(f)), \qquad Z_\infty^K\Big(\sum_j z_j f_j, 0\Big) := \lim_{J\to\infty} Z_J^K(z),$$

whenever the limits exist: their existence at real z is Theorem 3.3, and their existence at every $z \in \mathbb{C}^n$, entire in z, is derived from it in the curvature-sector section. The object is scheme-free. For general sources in a scheme a the *tube object* is $G_J(\zeta) := \lim_k Z_{k,J}^a(\zeta)$, $G_\infty(\zeta) := \lim_J G_J(\zeta)$, $\zeta \in \mathcal{D}_{\sigma_+,H}$, whenever these limits exist (the Wick-leg sections); where the full object $Z_\infty^a := \lim_J \lim_k Z_{k,J}^a$ of the optional extension (Theorem 23.7) exists it agrees with G_∞ on the tube, being the same iterated limit.

2.4. **Elementary facts.**

Lemma 2.7 (elementary facts)**.** (E1) *For $\varepsilon \leq 1$ and any translate Γ of $\varepsilon\mathbb{Z}^3$: $\sum_{x\in\Gamma} \varepsilon^3(1+|x|^2)^{-4} \leq C_0 := (5/2)^4\pi^2/8 < 49$ and $\sum_{x\in\Gamma} \varepsilon^3(1+|x|^2)^{-2} \leq C_0' := (5/2)^2\pi^2 < 62$.* (E2) *$\|q\|_{L^1} \leq \pi^2|q|_w$, $\|q\|_{L^2} \leq \frac{\pi}{2\sqrt{2}}|q|_4$, $\|q\|_\infty \leq |q|_w \leq |q|_4$, $\|g\|_2 \leq \frac{\pi}{2\sqrt{2}}|g|_4$; $|\cdot|_w$ is a norm, so $|k(\zeta)|_w \leq \sum_j |\operatorname{Re} w_j||h_j|_w$ and $|k_\pm|_w \leq |k|_w$.* (E3) *Sampling is evaluation at points: it commutes with pointwise positive and negative parts, and a sampled sup is at most the continuum sup, so $\zeta \in \mathcal{D}_{\sigma,H}$ for the continuum sources implies the same bounds for the sampled ones. For continuous compactly supported g, q: $\|g\|_{2,k} \to \|g\|_2$, $\|q\|_{1,k} \to \|q\|_{L^1}$, and for every k: $\|g\|_{2,k}^2 \leq C_0|g|_4^2$, $\|q\|_{1,k} \leq C_0'|q|_w$.* (E4) *$n^{n/2} \leq e^{n/2}(n!)^{1/2}$.*

Proof. (E1) For y in the cube of side ε centred at $x \in \Gamma$, $|x-y|^2 \leq \frac{3}{4}$ and $|y|^2 \leq 2|x|^2+2|x-y|^2 \leq 2|x|^2+\frac{3}{2}$, so $1+|y|^2 \leq \frac{5}{2}(1+|x|^2)$ and $(1+|x|^2)^{-r} \leq (5/2)^r(1+|y|^2)^{-r}$. Integrate over the cube and sum over the disjoint cubes: $\sum_{x\in\Gamma} \varepsilon^3(1+|x|^2)^{-r} \leq (5/2)^r \int_{\mathbb{R}^3}(1+|y|^2)^{-r}dy$, with $\int(1+|y|^2)^{-4} = 4\pi \cdot \frac{\pi}{32} = \frac{\pi^2}{8}$ and $\int(1+|y|^2)^{-2} = 4\pi \cdot \frac{\pi}{4} = \pi^2$. (E2) $\|q\|_{L^1} \leq |q|_w \int(1+|y|^2)^{-2}$, $\|q\|_{L^2}^2 \leq |q|_4^2 \int(1+|y|^2)^{-4}$; the rest is immediate. (E3) Riemann sums for the convergences; the plaquette centres of one orientation form a translate of $\varepsilon_k\mathbb{Z}^3$, so (E1) bounds $\sum_{p:(\alpha,\beta)} \varepsilon_k^3 g_{c(p)}(e_\alpha, e_\beta)^2$ by $C_0|g_{\alpha\beta}|_4^2$, and the three orientations sum to $C_0|g|_4^2$; the last bound is (E1) with $r = 2$. (E4) is $n^n \leq e^n n!$. □

Lattice-to-continuum. On a box X let $\langle A, B\rangle_k := \sum_{x\in X}\sum_\alpha \varepsilon^3 A_\alpha(x)B_\alpha(x)$, and for a real 2-form g whose non-vanishing plaquettes have their corners in X define d_k^*g by $F_{\Lambda_k}(g) = \langle d_k^*g, A\rangle_k$. Each component $(d_k^*g)_\gamma$ is a backward difference quotient of the sampled components $g_{\gamma\beta}$, $\beta \neq \gamma$; for smooth compactly supported g it converges uniformly to $(d^*g)_\gamma$, up to the orientation sign convention of the components, at the rate $O(\varepsilon_k)$ of Taylor's theorem. Hence

$$\|d_k^*g\|_{2,k} \to \|d^*g\|_{L^2}, \qquad \sup_k \|d_k^*g\|_{2,k} < \infty, \qquad \|g\|_{2,k} \to \|g\|_2 \qquad (k \to \infty). \tag{1}$$

2.5. **Correlation inequalities.** For a BFS measure, a real linear functional F of A and $g \geq 0$ on V put $\Theta_{F,g} := e^{\mathrm{i}F}e^{\sum_x g_x r_x^2}$, $\Theta_{F,g}^- := e^F e^{-\sum_x g_x r_x^2}$, and $\langle X; Y\rangle := \langle XY\rangle - \langle X\rangle\langle Y\rangle$. Since $F(f)$ is a real linear functional of A and $e^{:\phi^2:(g)}$ a positive constant times $e^{\sum_x \varepsilon^3 g(x) r_x^2}$, class $\mathcal{I}$ consists of multiples of observables $\Theta_{F,g}$ and class $\mathcal{D}$ of multiples of observables $\Theta_{F,g}^-$.

Lemma 2.8 (correlation inequalities)**.** *Let $\langle\cdot\rangle$ be a BFS measure, F a real linear functional of A, $g \geq 0$, $\Theta = \Theta_{F,g}$, $\Theta^- = \Theta_{F,g}^-$. Then $\langle\Theta\rangle$ is real, and*

- (a) *$\langle\Theta; |\ell(A)|^2\rangle \leq 0$ for every complex linear functional ℓ of A; hence adding $s|\ell(A)|^2$ to the action, $s \geq 0$, does not decrease $\langle\Theta\rangle$, and deleting a photon gradient term does not increase it;*
- (b) *$\langle\Theta; r_x r_y \cos(\vartheta_x - \vartheta_y - \ell_{xy}(A))\rangle \geq 0$; hence decreasing c_{xy} does not increase $\langle\Theta\rangle$;*
- (c) *$\langle\Theta; r_x^2\rangle \geq 0$; hence multiplying ρ_x by e^{-sr^2}, $s \geq 0$, does not increase $\langle\Theta\rangle$, and killing a set U of sites does not increase it;*
- (d) *for Θ^- the three inequalities and the monotonicity statements hold with the opposite signs (increase and decrease interchanged);*

(e) *if the pairs in B and the Gaussian γ do not couple the fields at the sites of $X \subset V$ (and the photon variables attached to them) to the others, the measure is a product, and for an observable depending only on the fields in X the expectation equals the expectation in the X-factor.*

Proof. (a)–(c) are [7, Thms. 6.1, 6.2(1),(2)], whose hypotheses are those of a BFS measure: an arbitrary finite lattice, $c_{xy} \geq 0$, finite single-site measures on $[0,\infty)$ with $\int e^{\alpha r^2} d\rho_x < \infty$ for all $\alpha \geq 0$ (a point mass at 0 is one), an arbitrary centred Gaussian with covariance $C \geq 0$, and $g \geq 0$ (so $g = 0$ is included). Theorem 6.1 states $\langle P(r)\cos(n\cdot\vartheta + f\cdot A); Q(r)\cos(m\cdot\vartheta + h\cdot A)\rangle \geq 0$ for polynomials P, Q in the r_x with positive coefficients; (a), (b) follow as in the printed proof of 6.2(1),(2): $e^{\sum g_x r_x^2}$ is a limit of such polynomials, $e^{\mathrm{i}F}$ may be replaced by $\cos F$ because the expectation and both observables are invariant under $(A,\vartheta) \mapsto (-A,-\vartheta)$ (which also gives the realness), and $|\ell(A)|^2 = \ell_1(A)^2 + \ell_2(A)^2$ with $\ell_r(A)^2 = \lim_{t\downarrow 0} t^{-2}[1-\cos(t\ell_r(A))]$; (c) is Theorem 6.1 with $Q = r_x^2$ and $m = 0$, $h = 0$. Each monotonicity statement follows by differentiating in the parameter, every interpolating measure being a BFS measure. For the killing statements of (c) and (d), let $x \in V$ and let $\langle\cdot\rangle_x$ be the measure with ρ_x replaced by the point mass at 0, in which every term $c_{xy} r_x r_y \cos(\cdot)$ vanishes. Decreasing c_{xy} to 0 for every pair $\{x,y\} \in B$ containing x does not increase $\langle\Theta\rangle$, by (b), and does not decrease $\langle\Theta^-\rangle$, by (d). In the resulting measure (r_x, ϑ_x) is coupled to nothing, so by (e) it is the product of $\rho_x(dr_x)\, d\vartheta_x$ and of the law of the other variables, which is that of $\langle\cdot\rangle_x$; the expectations of Θ and Θ^- in it are $m_x^+\langle\Theta\rangle_x$ and $m_x^-\langle\Theta^-\rangle_x$ with $m_x^\pm := \int e^{\pm g_x r^2} d\rho_x / \int d\rho_x$, so that $m_x^- \leq 1 \leq m_x^+$. Since $\Theta^- > 0$, $\langle\Theta^-\rangle_x \geq m_x^-\langle\Theta^-\rangle_x \geq \langle\Theta^-\rangle$; and $\langle\Theta\rangle_x \geq 0$ (multiplying the single-site measures by $e^{g_y r_y^2}$ gives a BFS measure, in which the expectation of $e^{\mathrm{i}F}$ is non-negative by identity (b) of Lemma 2.9, whose proof does not use the present lemma), hence $\langle\Theta\rangle_x \leq m_x^+\langle\Theta\rangle_x \leq \langle\Theta\rangle$. Several sites are killed one at a time; no condition on ρ_x beyond the definition of a BFS measure is used.

(d) is stated in [7, Thm. 6.2(3),(4)] for the photon and the hopping inequality with a sketched proof, and its r^2-inequality is [16, (3.28)], stated there without proof; we prove all three. Let μ be the unnormalised measure and $\omega' = (r', \vartheta', A')$ an independent copy of $\omega = (r, \vartheta, A)$; then $(\int d\mu)^2\langle\Theta^-; Y\rangle = \iint \Theta^-(\omega)(Y(\omega) - Y(\omega'))\, d\mu(\omega) d\mu(\omega')$. Write $r = u - v$, $r' = u + v$ (so $u_x \geq |v_x|$), $\vartheta = \vartheta_+ + \vartheta_-$, $\vartheta' = \vartheta_+ - \vartheta_-$, $A = a_+ + a_-$, $A' = a_+ - a_-$. Then $a_\pm = (A \pm A')/2$ are independent with the common centred Gaussian law of covariance $C/2$, and the radial law $\prod_x d\rho_x(u_x - v_x) d\rho_x(u_x + v_x)$ on $\{u \geq |v|\}$ is invariant under $v_x \mapsto -v_x$ at each site. With $\chi_{xy} := \vartheta_{+,x} - \vartheta_{+,y} - \ell_{xy}(a_+)$ and $\psi_{xy} := \vartheta_{-,x} - \vartheta_{-,y} - \ell_{xy}(a_-)$ the unprimed phase is $\chi + \psi$, the primed one $\chi - \psi$, and

$$r_x r_y \cos(\chi+\psi) + r'_x r'_y \cos(\chi-\psi) = 2(u_x u_y + v_x v_y)\cos\chi\cos\psi + 2(u_x v_y + v_x u_y)\sin\chi\sin\psi,$$
$$r_x r_y \cos(\chi+\psi) - r'_x r'_y \cos(\chi-\psi) = -2(u_x v_y + v_x u_y)\cos\chi\cos\psi - 2(u_x u_y + v_x v_y)\sin\chi\sin\psi,$$

$r_x^2 - r_x'^2 = -4u_x v_x$, $\ell(A)^2 - \ell(A')^2 = 4\ell(a_+)\ell(a_-)$ for real ℓ, and $\Theta^-(\omega) = e^{F(a_+)} e^{F(a_-)} e^{-\sum_x g_x(u_x^2+v_x^2)} e^{2\sum_x g_x u_x v_x}$. Expand $\exp(\sum_b c_b[\text{first line}])$ and $e^{2\sum g_x u_x v_x}$ in power series (absolutely convergent at fixed spacing). Every term of the integrand is a sign σ times a monomial in (u, v) with non-negative coefficient, times the even radial factor $e^{-\sum g_x(u_x^2+v_x^2)}$, times $V(\vartheta_+, a_+)V(\vartheta_-, a_-)$ with one real function V (a product of $e^{F(\cdot)}$, of factors $\cos\chi_b$ or $\sin\chi_b$, and of $\ell(\cdot)$ for the photon slot). The photon integral of $V(\vartheta_+, a_+)V(\vartheta_-, a_-)$ factorises because a_+, a_- are independent and identically distributed. The angular integral factorises by the following argument: per site $d\vartheta\, d\vartheta' = 2\, d\vartheta_+ d\vartheta_-$, and $[0, 2\pi)^2$ in the (ϑ, ϑ')-variables is a fundamental domain of the lattice L generated by (π, π) and $(\pi, -\pi)$ in the $(\vartheta_+, \vartheta_-)$-plane, which contains $(2\pi\mathbb{Z})^2$ with index 2; $V(\vartheta_+, \cdot)V(\vartheta_-, \cdot)$ is L-invariant, because the simultaneous shift of $\vartheta_{+,x}$ and $\vartheta_{-,x}$ by π at one site changes every χ_b and ψ_b with $b \ni x$ by π and leaves $\cos\chi_b\cos\psi_b$ and $\sin\chi_b\sin\psi_b$ unchanged; hence $\int_{[0,2\pi)^2} VV\, d\vartheta\, d\vartheta' = 2\int_{L\text{-domain}} VV\, d\vartheta_+ d\vartheta_- = \int_{[0,2\pi)^2} VV\, d\vartheta_+ d\vartheta_- = (\int_0^{2\pi} V)^2$. The radial integral of a monomial vanishes if its degree in some v_x is odd, by the $v \mapsto -v$ symmetry, and is non-negative otherwise, since $u_x \geq |v_x|$. So every term has the sign σ: $\sigma = -1$ for $Y = r_x^2$ (from $-4u_x v_x$) and for the hopping term (second line), $\sigma = +1$ for $Y = \ell(A)^2$ (for complex ℓ, real and

imaginary parts). Hence $\langle\Theta^-; r_x^2\rangle \le 0$, $\langle\Theta^-; r_x r_y \cos(\cdot)\rangle \le 0$, $\langle\Theta^-; |\ell(A)|^2\rangle \ge 0$. Monotonicity and killing follow as in (a)–(c), with (d)'s hopping inequality in place of (b). (e) is clear. □

Lemma 2.9 (current representation and Gaussian domination)**.** *Let $\langle\cdot\rangle$ be a BFS measure whose Gaussian has covariance C. Integrating out the matter at fixed A, the matter integral is $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q), A\rangle$, q running over the integer-valued divergence-free currents on B, $w(q) \ge 0$, $\sum_q w(q) < \infty$, $\langle\kappa(q), A\rangle := \sum_{xy} q_{xy}\ell_{xy}(A)$. Let $\mathbb{E}^\sharp$ be the probability measure on currents with weights proportional to $w(q)e^{-\langle\kappa(q),C\kappa(q)\rangle/2}$. Then for $v \in \mathbb{R}^m$, $s \in \mathbb{R}$ (C is symmetric, so $\langle Cv, \kappa\rangle = \langle v, C\kappa\rangle$):*

$$\text{(a)} \qquad \langle e^{s\langle v,A\rangle}\rangle = e^{s^2\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp\cos\big(s\langle Cv,\kappa\rangle\big) \le e^{s^2\langle v,Cv\rangle/2},$$

$$\text{(b)} \qquad \langle e^{\mathrm{i}\langle v,A\rangle}\rangle = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp\cosh\big(\langle Cv,\kappa\rangle\big) \;\ge\; e^{-\langle v,Cv\rangle/2} > 0.$$

Proof. Each bond factor expands as $e^{c_{xy}r_xr_y\cos t} = \sum_{q\in\mathbb{Z}} I_q(c_{xy}r_xr_y)e^{\mathrm{i}qt}$ with $I_q \ge 0$, $I_{-q} = I_q$, $\sum_q I_q(c) = e^c$; after the radial integrals the multiple series converges absolutely, dominated by $\prod_x e^{\frac12\sum_y c_{xy}r_x^2}$, integrable against the single-site measures. Integrating the angles forces $\sum_y q_{xy} = 0$ at every site (at a killed site $r_x = 0$ and $I_q(0) = \delta_{q0}$, so the currents vanish on its bonds); pairing q with $-q$ gives Ψ with $w(q) \ge 0$. For a centred Gaussian γ with covariance C and $a \in \mathbb{C}^m$, $\int e^{\langle a,A\rangle}d\gamma = e^{\langle a,Ca\rangle/2}$; with $a = sv \pm \mathrm{i}\kappa$ and the average of the two signs, $\int e^{s\langle v,A\rangle}\cos\langle\kappa, A\rangle\,d\gamma = e^{s^2\langle v,Cv\rangle/2-\langle\kappa,C\kappa\rangle/2}\cos(s\langle v, C\kappa\rangle)$. Sum over q with the weights $w(q)$, divide by the same sum at $s = 0$ (positive), and use $|\cos| \le 1$. For the imaginary exponent take $a = \mathrm{i}v \pm \mathrm{i}\kappa$: $\int e^{\mathrm{i}\langle v,A\rangle}\cos\langle\kappa, A\rangle\,d\gamma = \frac12\big[e^{-\langle v+\kappa,C(v+\kappa)\rangle/2} + e^{-\langle v-\kappa,C(v-\kappa)\rangle/2}\big] = e^{-\langle v,Cv\rangle/2-\langle\kappa,C\kappa\rangle/2}\cosh\langle v, C\kappa\rangle$; the same sum over q and the same division give the second identity, and $\cosh \ge 1$ gives the lower bound. □

The lemma applies to every state of this paper at every k, J, $\mu_0 > 0$, with or without a real scalar source absorbed in the single-site measures (the photon covariance is unchanged).

Lemma 2.10 (killing)**.** *Let $\langle\cdot\rangle$ be a BFS measure of King type, U a set of alive sites, $\chi : U \to (0,\infty)$, and $\langle\cdot\rangle_{U,s}$ the measure with ρ_x multiplied by $e^{-s\varepsilon^3\chi(x)r_x^2}$ for $x \in U$. Let Θ be continuous with $|\Theta(A,\phi)| \le e^{\ell(A)}e^{C\sum_x r_x^2}$ for a real linear functional ℓ and $C \ge 0$. Then $\langle\Theta\rangle_{U,s} \to \langle\Theta\rangle_U$ as $s \to \infty$, $\langle\cdot\rangle_U$ the measure with ρ_x replaced by the point mass at 0 for $x \in U$ (the state with the matter killed on U).*

Proof. The unnormalised density is $e^{-Q(A)/2}dA\prod_{x\text{ alive}} w_x(r_x)r_x dr_x d\vartheta_x \prod_b e^{c_b r_x r_y\cos(\cdot)}\prod_{x\in U} e^{-s\varepsilon^3\chi(x)r_x^2}$, Q positive definite, killed sites carrying no variables. Substitute $r_x = (s\varepsilon^3\chi(x))^{-1/2}u_x$ for $x \in U$; the Jacobians cancel between numerator and denominator. *Domination for $s \ge s_0$:* at $x \in U$, $w_x(r_x) \le e^{\varepsilon^3|\delta m_\varepsilon^2|r_x^2/2}$ (the other exponents in w_x are ≤ 0), $|\Theta|$ contributes $e^{Cr_x^2}$, and each of the at most six bond weights at x is at most $e^{\frac{\varepsilon}{2}r_x^2+\frac{\varepsilon}{2}r_y^2}$ ($0 \le c_b \le \varepsilon$); in the variable u_x these factors together are $\le e^{u_x^2/2}$ once $s \ge s_0(\varepsilon, m, \delta m_\varepsilon^2, C, \min\chi)$, so the U-factor is at most $e^{-u_x^2/2}u_x$, integrable on $[0,\infty)\times[0,2\pi)$; at an alive $y \notin U$ the factor is at most $w_y(r_y)e^{(C+3\varepsilon)r_y^2}r_y$, integrable since $\lambda > 0$; the photon factor $e^{\ell(A)}e^{-Q(A)/2}$ is integrable since Q is positive definite. *Pointwise limit:* at fixed $(u, \vartheta, A, r_{V\setminus U})$, $r_x \to 0$ for $x \in U$, so $\Theta \to \Theta|_{\phi_U=0}$ by continuity, every bond weight with an endpoint in U and every $w_x(r_x)$, $x \in U$, tends to 1, and the U-factor tends to $e^{-u_x^2}u_x\,du_x d\vartheta_x$, whose integral cancels between numerator and denominator. By dominated convergence the ratio tends to $\int\Theta|_{\phi_U=0}d\mu_U/\int d\mu_U$, where μ_U has the cosines at U removed, the weights w_y (which already contain all diagonals at y) and the photon unchanged, and the point mass at U: that is $\langle\cdot\rangle_U$, with a positive denominator. □

Lemma 2.11 (cutting)**.** *Let $\langle\cdot\rangle_Y$ be a Neumann–Dirichlet box state on Λ_k, possibly with killed sites, or a periodic state on a King torus, and let X be a set of sites of Y such that X and its nearest neighbours lie in one fundamental cell (a clause in torus language: for a box state it has no content beyond $X \subset Y$). Let $\langle\cdot\rangle_X^{\mathrm{ND}}$ be the Neumann–Dirichlet state on X whose single-site measures (killed sites and the diagonal parts of the bonds leaving X included) are those of $\langle\cdot\rangle_Y$.*

Let $(f,g)\in\mathcal{K}$ be such that every plaquette and every site at which f, respectively g, does not vanish lies in X. Then, at every lattice spacing,

$$\langle\Theta(\mathrm{i}f,g)\rangle_X^{\mathrm{ND}}\le\langle\Theta(\mathrm{i}f,g)\rangle_Y,\qquad\langle\Theta(f,-g)\rangle_X^{\mathrm{ND}}\ge\langle\Theta(f,-g)\rangle_Y,$$

and the same for Z^a. The first inequality uses Lemma 2.8(a),(b),(e) only; the second uses (d),(e). If no site of X is killed, $\langle\cdot\rangle_X^{\mathrm{ND}}$ is the state of Definition 2.1.

Proof. Delete from Y every photon gradient term coupling a site of X to a site outside X, and the cosine parts of all hopping terms between X and its complement, keeping their diagonal parts $\frac{\varepsilon}{2}(r_x^2+r_y^2)$. By Lemma 2.8(a),(b) this does not increase the first expectation, by (d) it does not decrease the second, and every intermediate measure is a BFS measure. The result is a product (Lemma 2.8(e)) whose X-factor is $\langle\cdot\rangle_X^{\mathrm{ND}}$: the diagonal parts of the cut bonds are exactly the Dirichlet terms and the remaining photon terms the Neumann ones of X; the observable depends on the X-variables only. The free Wick constant, δm_ε^2 and the scheme multiplier are the same on both sides. □

With $Y=X_{J+1}$, $X=X_J$ this is King's monotonicity in the volume [16, Thm. 3.2, (3.31)–(3.33)] between comparison cubes, in the form his proof proves: with one Wick constant for all J, the free subtraction King himself switches to at the start of that proof [16, p. 333]. With the Dirichlet Wick constant of [16, (2.13)], which increases with J, the c-number factor moves against the monotone factor and the monotonicity does not follow; that form is not used. The box version with killed sites and the monotonicity of King's RG cubes are in Section 4.

2.6. **Algebra of the scheme, covariance, reflection, continuity.**

Lemma 2.12 (algebra of the scheme)**.** *Let a be a scheme and Z^a the functional of a state above on a box at spacing ε_k.*

(a) *For real coefficients w the multiplier $e^{-\sum_j w_j a_k \mathsf{S}(h_j)}$ is a positive constant depending only on k, w and $\mathsf{S}_{\Lambda_k}(h_j)$. Hence the inequalities of Lemmas 2.8 and 2.11 between two states on the same lattice with the same sources hold for Z^a in the same direction.*

(b) *$|Z^a(\zeta)|\le Z^a(\operatorname{Re}\zeta)$, and $\ln Z^a$ is convex in the real coefficients.*

(c) *$Z^a(f,h)\,Z^a(-f,-h)\ge 1$ for real (f,h).*

(d) *For real f and real $k_1,k_2\ge 0$: $Z^a(f,k_1-k_2)\le Z^a(2f,-2k_2)^{1/2}\,Z^a(0,2k_1)^{1/2}$.*

Proof. (a) is immediate. (b) $|\Theta(\zeta)|=\Theta(\operatorname{Re}\zeta)$ and the multiplier has modulus $e^{-\sum_j(\operatorname{Re}w_j)a_k\mathsf{S}(h_j)}$; convexity is Hölder's inequality. (c) $\langle e^X\rangle\langle e^{-X}\rangle\ge 1$ by Cauchy–Schwarz, and the multipliers cancel. (d) Cauchy–Schwarz gives $\langle e^{F+:\phi^2:(k_1)-:\phi^2:(k_2)}\rangle\le\langle e^{2F-2:\phi^2:(k_2)}\rangle^{1/2}\langle e^{2:\phi^2:(k_1)}\rangle^{1/2}$, and $e^{-a_k\mathsf{S}(k_1-k_2)}=(e^{2a_k\mathsf{S}(k_2)})^{1/2}(e^{-2a_k\mathsf{S}(k_1)})^{1/2}$. □

Lemma 2.13 (passive covariance)**.** *Let $O\in O(3)$, Λ a lattice of spacing ε with frame (f_α), and X a box or a torus for Λ. Equip $O\Lambda$ with the frame (Of_α) and OX with the transported boundary conditions. Then $\langle\Theta(z\,O\cdot f,w\,O\cdot h)\rangle_{O\Lambda,OX}=\langle\Theta(zf,wh)\rangle_{\Lambda,X}$ and $\mathsf{S}_{O\Lambda}(O\cdot h)=\mathsf{S}_\Lambda(h)$, likewise for families and for Z^a. On a torus the periodic model does not change if the frame (f_α) is replaced by $(\pm f_{\pi(\alpha)})$, π a permutation.*

Proof. The map $\phi'(Ox)=\phi(x)$, $A'_\alpha(Ox)=A_\alpha(x)$, components taken in the frames (f_α) and (Of_α), is a bijection of the configuration spaces preserving Lebesgue measure; the action is written in the frame, and δm_ε^2, W_ε, a_k depend on ε only; plaquettes, centres and field strengths are transported, and $(O\cdot f)_{Oc}(Of_\alpha,Of_\beta)=f_c(f_\alpha,f_\beta)$. A permutation of the frame relabels the components; reversing f_α replaces $A_\alpha(x)$ by $-A_\alpha(x-\varepsilon f_\alpha)$, a bijection of the torus variables preserving the action, which changes the sign of both $F(p)$ and $f_{c(p)}(f_\alpha,f_\beta)$ and preserves the set of plaquette centres. □

Lemma 2.14 (the reflection sandwich at fixed spacing, comparison convention)**.** *Let $\sigma_3=\operatorname{diag}(1,1,-1)$ and let $(f,g)\in\mathcal{K}$ be supported at ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_{J-1}$. Then for every $k\ge 0$*

$$Z^a_{k,J-1}(\mathrm{i}f,g)\le Z^a_{k,J}(\mathrm{i}\,\sigma_3\cdot f,\sigma_3\cdot g)\le Z^a_{k,J+1}(\mathrm{i}f,g),$$

and the reversed inequalities hold for $Z^a(f,-g)$.

Proof. σ_3 preserves Λ_k and S_{Λ_k} and maps X_J onto $X_+ := (C_J + \varepsilon_k e_3) \cap \Lambda_k$. By Lemma 2.13, $Z^a_{k,J}(\mathrm{i}\sigma_3 \cdot f, \sigma_3 \cdot g)$ is the functional with sources (f, g) of the Neumann–Dirichlet state on X_+ in the frame $(e_1, e_2, -e_3)$. Written in the frame (e_α), with $A'_3(x) = -A_3(x - \varepsilon e_3)$, this is the BFS measure R in which ϕ, A_1, A_2 live on X_+ and A_3 on $X_+ - \varepsilon e_3 = X_J$, with Dirichlet conditions for ϕ on X_+, photon gradient and mass terms within these sets, and the hopping term in direction 3 the standard one of the bond $\langle y - \varepsilon e_3, y\rangle$; the plaquette terms $(x; \alpha, 3)$ pick up two sign changes and keep their centres, so the observable is $\Theta(\mathrm{i}f, g)$ in the standard frame. R is obtained from the Neumann–Dirichlet state on X_{J+1} by deleting photon gradient terms and hopping cosines and factorising, since $X_+ \cup X_J \subset X_{J+1}$, and the state on X_{J-1} is obtained from R in the same way, since $X_{J-1} \subset X_J \cap X_+$; the diagonal parts of the cut bonds are the Dirichlet terms of the smaller state, and the sources stay inside X_{J-1}. Lemma 2.8(a),(b),(e), respectively (d),(e), give the directions, and Lemma 2.12(a) carries them to Z^a. □

Lemma 2.15 (continuity of convex functions)**.** *Let U be a non-empty open convex subset of a Fréchet space and $\Phi : U \to \mathbb{R}$ a convex Borel function. Then Φ is continuous on U.*

Proof. U is a Baire space and $U = \bigcup_{j\in\mathbb{N}}\{\Phi \le j\}$, so some $S := \{\Phi \le j\}$ is non-meagre in U; being Borel, S has the Baire property [13]: $O \setminus \mathcal{M} \subset S$ for a non-empty open $O \subset U$ and a meagre $\mathcal{M}$. Pick $x_0 \in O$ and a balanced open neighbourhood W of 0 with $x_0 + W + W \subset O$. For $w \in W$ and $y := x_0 + w$, $y + W \subset O$, and the point reflection $\iota(x) := 2y - x$ maps $y + W$ onto itself and preserves meagre sets; so $S \cap (y + W)$ and $\iota(S \cap (y + W))$ are comeagre in $y + W$ and intersect: there is $u \in S$ with $2y - u \in S$, and $\Phi(y) \le \frac{1}{2}(\Phi(u) + \Phi(2y - u)) \le j$. Hence $\Phi \le j$ on $x_0 + W$. A convex function on an open convex set bounded above near one point is bounded above near every point $y \in U$ (choose $z \in U$ with y strictly between x_0 and z, and take the convex combinations of $x_0 + W$ with z), and a convex function locally bounded above is continuous [24]. □

3. The inputs

This section states, in the form in which the later sections consume them, the results on which the theorems of this paper rest, and says where each is proved. Three are theorems of this paper proved in later parts: the existence of King's cube limits, without and with scalar sources (Theorems 3.3 and 3.6; the existence part, Sections 5 and 6), and the rotation theorem (Theorem 3.14; the rotation part, Section 15, whose main theorem is Theorem 15.1). Two are the torus inputs of the Wick leg, Bałaban's stability bounds and the c-number lemma (Proposition 3.9 and Lemma 3.10), established in the full version. One is a condition on the couplings, the sign clause (T3) (Hypothesis 3.12), proved on explicit coupling regions in Proposition 22.14; it is the only condition on the couplings anywhere in the paper. The optional extension of the Wick leg to arbitrary amplitudes, together with the hypotheses it needs ((W2), (R_W), (FA^K)), is stated in the final section (Section 23); no clause of the main theorem uses them. From the literature, the paper uses the correlation inequalities of [7], the chessboard estimate of [9], the reconstruction theorems of [19, 20], Bałaban's torus analysis [1, 2, 3, 4] and King's papers [15, 16], each at the place where it is applied (Remark 3.17); the general results on the measure, its ergodic decomposition, reconstruction and clustering are proved in the full version.

3.1. **Objects and conventions.** The two cube conventions of Definitions 2.2 and 2.1 are King's convention ND^K (the scalar field killed on the three top face layers T_J of the cube C_J, the convention of the state King prints) and the comparison convention $\mathrm{ND}^{(a)}$ (the scalar alive on all of X_J), in which the tube results (Theorem 13.10, Lemma 13.11) and the lattice comparisons of the rotation part are written. The two are interlaced by Lemma 4.2. Which convention a statement is posed on is part of it:

- The existence theorems Theorems 3.3 and 3.6 are posed on King's convention ND^K, on which King's renormalisation-group flow runs (Remark 5.3). By Corollary 4.4 the existence of the fixed-cube limits on one convention does not imply it on the other; the comparison family's fixed-cube limits are reached through that corollary, and the statements proved for

the comparison family under the assumption that they exist (Definition 3.1; on the Wick tube Definition 3.2) serve as templates for King's convention.

- The rotation theorem Theorem 3.14 is proved in the rotation part for King's printed state Z_∞^K on King's convention (Theorem 15.1) and is carried to the limit states of the comparison family through Corollary 4.4; its Wick-sector form is posed on the comparison family (Theorems 18.9 and 18.11).
- Proposition 3.9 and Lemma 3.10 and (T3) are statements about King's tori (Definition 12.1) and about c-numbers.

Throughout, $Z^a_{k,J}(z,w)$ denotes the generating functional of Definition 2.4 with curvature coefficients z and Wick coefficients w in the scheme a, on the convention in question (written $Z^{a,K}_{k,J}$ when the convention ND^K is to be stressed); $z = a^\sharp$ is the reference scheme of Remark 2.5; $\langle\cdot\rangle^K_{k,J}$ is King's cube expectation at spacing ε_k.

Definition 3.1 (convergence of the comparison family at fixed cubes)**.** The comparison family is *convergent at fixed cubes* if for every finite family $f_1,\dots,f_n$ of real 2-forms in $C^\infty_c(\mathbb{R}^3;\Lambda^2)$, every $J \ge J_0(f)$ and every $x \in \mathbb{R}^n$ the limit $\lim_k \langle e^{\sum_j x_j F(f_j)}\rangle^{(a)}_{k,J}$ exists and lies in $(0,\infty)$.

This is the clause of Theorem 3.3 read for the family $\mathrm{ND}^{(a)}$. It is not a result of this paper and is not needed for the main theorem: the statements below that assume it are proved for the comparison family as templates, and their forms for King's printed state are proved under Theorem 3.3; the continuum-subsequence section (Theorem 14.9) removes the dependence on the fixed-cube limits altogether.

Definition 3.2 (convergence of the comparison family at fixed cubes on the tube)**.** The comparison family is *convergent at fixed cubes on the tube* (in a scheme a) if the clause of Theorem 3.6 holds for it: for every finite family of real 2-forms f_j and real functions h_i in C^∞_c, every $J \ge J_0(f,h)$ and every real coefficient vector (z,w) whose positive scalar part $k_+ = (\sum_i w_i h_i)_+$ satisfies $2\|k_+\|_\infty < s_W$, the limit $\lim_k Z^a_{k,J}(z,w)$ on the convention $\mathrm{ND}^{(a)}$ exists and lies in $(0,\infty)$.

This is the Wick-sector counterpart of Definition 3.1, with the same status: it is not a result of this paper and is not needed for the main theorem, and the two conventions' clauses do not imply one another by the inputs of this paper (Corollary 4.4, Remark 5.10). The tube statements that assume it (Theorem 13.10(i), Lemma 13.11, Proposition 10.5, Theorem 10.7, Proposition 14.12(c)) are templates for the comparison family; their forms for King's convention, on which the limits exist by Theorem 3.6, are Theorem 21.3 and Proposition 21.4.

3.2. **Existence of King's cube limits.**

Theorem 3.3 (King's cube limits, curvature)**.** *For every finite family $f_1,\dots,f_n$ of real 2-forms in $C^\infty_c(\mathbb{R}^3;\Lambda^2)$, every $J \ge J_0(f)$, where $J_0(f)$ is the least J such that every* $\operatorname{supp} f_j$ *lies at ℓ^∞-distance greater than* 1 *from $\mathbb{R}^3 \setminus C_J$, and every $z \in \mathbb{R}^n$, the limit*

$$\lim_{k\to\infty} \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$$

exists and lies in $(0,\infty)$.

Proved in Section 5 (Theorem 5.9), from King's renormalisation-group analysis of the cube.

Remark 3.4 (on Theorem 3.3). (i) *Existence clause only.* Theorem 3.3 is used only through the existence of the fixed-cube limit at real coefficients: no constants clause, no uniformity in J and no locally uniform convergence in z enter from it. Every bound used on the limit is proved here, with absolute constants, in Theorem 7.4; the monotonicity in the cube, the J-limit, the entire extension and the translations are proved in Lemma 4.2, Theorem 7.4, Lemma 7.7 and Theorem 8.7. (ii) *Origin.* This is the existence clause of King's Theorem 2.1 [16] at zero scalar source, for the state he prints. The proof in Section 5 follows King's route, the comparison of two spacings at a common block scale, in a ratio form in which the vacuum energies cancel;

what it uses of Bałaban's block-factor-5 analysis [1, 2, 3, 4] is stated there. The mass clause of [2, Prop. 3.1], upstream of it, is part of that proposition's statement and is cited under it (Remark 22.2); it is not a condition of this paper. (iii) *Where it is used.* The whole curvature sector (Theorems 7.4, 8.7, 10.6, 11.1), the identification of the LIM states with King's state (Corollary 14.13) and the main theorem (Theorem 21.1).

Remark 3.5 (the cube). The cube on which the existence theorem is proved is King's RG cube C_J^K of Definition 2.2, with King's killing of the scalar on the top layers T_J; it is the only cube on which King's block-spin flow exists (Remark 5.3). Every placement of a comparison-cube result on King's cube goes through Lemma 4.2 and its corollaries.

Theorem 3.6 (King's cube limits with scalar sources)**.** *Let $z = a^\sharp$ be the reference scheme and let $s_W := s_1$ be the amplitude of Proposition 12.6. For every finite family of real 2-forms $f_j \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ and real functions $h_i \in C_c^\infty(\mathbb{R}^3)$, every $J \geq J_0(f,h)$ and every real coefficient vector (z,w) whose positive scalar part $k_+ := \big(\sum_i w_i h_i\big)_+$ satisfies $2\|k_+\|_\infty < s_W$, the condition of the tube $\mathcal{D}_{\sigma_+,H}$ with $\sigma_+ = s_W$ (the negative scalar part is unrestricted), the limit $\lim_{k\to\infty} Z^z_{k,J}(z,w)$ exists and lies in $(0,\infty)$.*

Proved in Section 6 (Theorem 6.17 and Corollary 6.18), by the route of Section 5 with the scheme c-number carried along.

Definition 3.7 (admissible schemes)**.** A scheme a (Remark 2.5) is *admissible* if $a = z$ or $\lim_k(z_k - a_k)$ exists. For an admissible scheme Theorem 3.6 holds for $Z^a_{k,J}$ in place of $Z^z_{k,J}$ (Lemma 2.12(a)), and $\sup_k |z_k - a_k| < \infty$.

Remark 3.8 (on Theorem 3.6). (i) At zero Wick coefficient Theorem 3.6 is Theorem 3.3. (ii) *Origin.* Theorem 3.6 is *not* King's Theorem 2.1 with scalar sources: as printed, in King's Wick ordering, that statement fails, and the state with scalar sources exists only after a vacuum counterterm, which is what the scheme supplies (Remark 2.5). The threshold on the positive scalar part is the one at which the positive-source bound of the Wick leg is available (Proposition 13.2); what happens above it is the subject of the optional extension (Hypothesis 23.1). (iii) One instance of Theorem 3.6 is named separately because the statements on continuum subsequences need it alone: ($\mathrm{B}_{\mathrm{cell}}$) (Lemma 14.4; one cube, one negative Wick source). (iv) *Where it is used.* The Wick tube (Proposition 12.6, Theorem 13.10), its transfer to King's convention (Theorem 21.3), the Wick sector of Theorem 14.9, and Form (B) of the main theorem (Theorem 21.1, Proposition 21.4).

3.3. The torus inputs of the Wick leg.

Proposition 3.9 (torus stability at two scalar masses)**.** *For the scalar masses $M = m$ and $M = M_1$ (the auxiliary mass fixed in Proposition 12.6) there are constants $E_\pm(M) < \infty$, a tail K_1 and, for each $k \geq K_1$, a $J_*(k)$ such that the source-free partition functions of King's tori T (Definition 12.1) at spacing ε_k and at scalar mass M satisfy, in King's torus convention,*

$$e^{-E_-(M)|T|} \leq Z^{(M)}_{k,T} \leq e^{E_+(M)|T|} \qquad (k \geq K_1,\ J \geq J_*(k)).$$

These are Bałaban's lower and upper stability bounds, [1, (1.14)] and [2, (1.3)], at a fixed scalar mass, read at the two masses m and M_1 and in King's torus normalisation of the counterterm; the form used here is established in the full version (Theorem F.1 there). Proposition 3.1 of [2], behind the upper bound, is stated under a cap on the dimensionless scalar mass at the scales at which it is applied; it is cited under that cap, one more one-sided cap on the stopping scale and no condition on the bare mass (Remark 22.2). Only a tail in k and, for each k, large J are used (Proposition 12.6, Theorem 13.10).

Lemma 3.10 (the c-number input)**.** *Let $\alpha_k^\infty(M) = M^2 + \delta m_k^2(M)$ be King's free-boundary bare coefficient at scalar mass M and spacing ε_k, in the reference scheme z. For the auxiliary mass M_1 of Proposition 12.6, the quantities*

$$\sigma_1(M_1) := \liminf_{k\to\infty} \tfrac{1}{2}\big(\alpha_k^\infty(m) - \alpha_k^\infty(M_1)\big) \qquad \text{and} \qquad \beta_1(M_1)$$

are finite, where $\beta_1(M_1)$ is the second constant entering Proposition 12.6.

The proof is given in the full version (Theorem F.168 there).

Remark 3.11 (on Lemma 3.10). (i) This is a statement about lattice Feynman integrals: the difference of two mass counterterms at the same photon mass. The limit $\alpha_\infty(M) = \lim_k \alpha_k^\infty(M)$ and its description as a function of the couplings are discussed in Lemma 22.1. (ii) *Where it is used.* Proposition 12.6 (the constants $\sigma_+ = s_1$, H, κ of the tube) and the coupling range, Lemma 22.1.

Hypothesis 3.12 ((T3): the coupling condition)**.** $\sigma_1(M_1) > 0$ for some $M_1 \in (0, m)$. The couplings (e, λ, m, μ_0) are said to lie in the *coupling region* of this paper when (T3) holds. By Proposition 22.14 the coupling region contains the explicit regions listed there: at $\mu_0 = m$ a region containing $0 \le \lambda, e^2 \le 0.1m$, and at every $\mu_0 \in (0, m]$ explicit boxes.

This is the condition on the couplings under which the Wick leg of the main theorem is stated; the curvature leg carries no condition on the couplings.

Remark 3.13 (on (T3)). (i) (T3) is the only condition on the couplings anywhere in this paper, and it is one-sided: it fails exactly on the set $E_W^<$ of couplings (e, λ, m, μ_0) at which m minimises $M \mapsto \alpha_\infty(M)$ on $(0, m]$ (Lemma 22.1). At one loop it holds if and only if

$$e^2 < e_c^2(\lambda, m, \mu_0) = \max\Big\{ \frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1 + \frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1 + \frac{\lambda(N+2)}{\pi m}\Big)\Big\},$$

N the number of real components of the scalar field (equation (60)); so at one loop a half-line of values of e^2 fails. Beyond one loop, (T3) is *proved* on explicit coupling regions (Proposition 22.14): at $\mu_0 = m$ on a region containing $0 \le \lambda, e^2 \le 0.1m$, and at every $\mu_0 \in (0, m]$ on explicit boxes; the main theorem is stated on the region where it holds. The two-sided variant and the set $E_W \subseteq E_W^<$ are described in Lemma 22.1. (ii) *Where it is used.* Proposition 12.6 and Lemma 22.1.

3.4. **The rotation theorem.** For a family f of real 2-forms and $J \ge J_0(f)$ let L_J^f be the set of locally uniform limits on $\mathbb{C}^N$ of subsequences of $(a_{k,J}^{(a)})_k$, $a_{k,J}^{(a)}(z) = \langle e^{\sum_j z_j F(f_j)} \rangle_{k,J}^{(a)}$ (the comparison family; by Lemma 7.3 and Montel's theorem the set is non-empty).

Theorem 3.14 (rotation invariance of the comparison family's limit states)**.** *For every finite family f of real 2-forms in $C_c^\infty(\mathbb{R}^3; \Lambda^2)$ and every choice $\zeta_J \in L_J^f$, $J \ge J_0(f)$, the limit $Z_\infty^{\mathrm{LIM}}(\sum_j z_j f_j, 0) := \lim_{J\to\infty} \zeta_J(z)$ exists locally uniformly on $\mathbb{C}^N$, is entire and does not depend on the choices. The functional $Z_\infty^{\mathrm{LIM}}(\cdot, 0)$ is invariant under the Euclidean group $\mathbb{R}^3 \rtimes O(3)$: $Z_\infty^{\mathrm{LIM}}(\gamma \cdot f, 0) = Z_\infty^{\mathrm{LIM}}(f, 0)$ for every Euclidean motion γ and every complex coefficient vector, F transforming as a 2-form.*

Proved in the rotation part: Theorem 15.1 gives the $\mathbb{R}^3 \rtimes O(3)$-invariance of King's printed state $Z_\infty^K(\cdot, 0)$, and by Corollary 4.4(iii), with the fixed-cube limits of Theorem 3.3, every J-limit of subsequential fixed-cube limits of the comparison family exists and equals $Z_\infty^K(\cdot, 0)$, which is the statement as posed. King's cubes are compared with periodic tori carrying a smooth mass wall, the walled tori on two lattices whose orientations differ by $\arctan(3/4)$ are compared (Theorem 17.1), and King's state is identified as the lower envelope of the walled tori (Theorem 18.4) through a heavy-shell decoupling estimate on King's block-spin flow (Theorem 20.2).

Remark 3.15 (on Theorem 3.14). (i) The theorem is posed for the comparison family and for the J-limit of its subsequential fixed-cube limits; its proof goes through King's printed state, whose invariance Theorem 15.1 proves directly, and Corollary 4.4 identifies $Z_\infty^{\mathrm{LIM}}(\cdot, 0)$ with King's printed state $Z_\infty^K(\cdot, 0)$, which is thereby $\mathbb{R}^3 \rtimes O(3)$-invariant (Theorem 8.7). (ii) *Where it is used.* The $O(3)$ clause of the curvature sector (Theorem 8.7), the Euclidean clause of the measure's ergodic components (Theorem 11.1(c)), and the $O(3)$ form of the LIM theorem (Theorem 14.9).

Remark 3.16 (the Wick tube on King's convention). Theorem 13.10 is proved for the comparison convention $\mathrm{ND}^{(a)}$. Its transfer to King's convention — the (k, J)-uniform tube bound (31) for $Z_{k,J}^{a,K}$ and the fixed-cube limits on the tube — is Theorem 21.3, with the kill step of Remark 4.8,

the cell statement of Lemma 4.7 and the direction check for the negative leg as its convention-sensitive steps; Form (B) of the main theorem then holds on King's convention (Proposition 21.4).

3.5. **The optional extension.** The extension of the Wick leg to arbitrary amplitudes (Form (A), Theorem 23.7) is stated in the final section, with the three statements it needs: (W2) (Hypothesis 23.1; the fixed-cube limits and the positive-source bound at every amplitude), (R_W) (Hypothesis 23.3; its Euclidean clause) and (FA^K) (Hypothesis 23.5; its King-convention form). No clause of Theorem 21.1 uses them. What this paper proves of (W2) is its cone instance below the threshold $\sigma_1(M_1)$ (Corollary 13.3), which is all that Form (B) needs.

3.6. **Auxiliary statements of the section on continuum subsequences.** The statements on continuum subsequences (Theorem 14.9) are formulated in terms of three named properties proved in that section: the cell bound ($\mathrm{B}_{\mathrm{cell}}$) (Lemma 14.4; the boundedness form of Theorem 3.6 at one cube and one negative Wick source), the scheme bound (SCH) (Lemma 14.5; $a = z$ or $\sup_k |z_k - a_k| < \infty$) and the independence of the subsequence (UNIQ) (Proposition 14.6; any two continuum subsequences give the same LIM state). The first two hold for every admissible scheme under (T3), and the third follows from the rotation theorem; the three are collected in Proposition 14.12. The limit of vanishing photon mass is not treated in this paper (Section 23).

3.7. **Results from the literature used as inputs.**

Remark 3.17 (results from the literature used as inputs). The printed results used as inputs are the following, each applied at the place indicated after its hypotheses are checked there. Of [7], Theorems 6.1 and 6.2(1),(2) are cited as printed; items (3),(4) of its Theorem 6.2, a sketch in print, are a lemma of this paper (Lemma 2.8). The chessboard estimate is Theorem 4.1/4.2 of [9], applied after its hypotheses are checked (Proposition 12.2). The reconstruction in a coordinate direction follows [19, 20] and [10, Thm. 6.1.3], and is proved in the full version on the field spaces used here; the Lusin–Souslin theorem is quoted from [13] (see the full version). Bałaban's torus analysis [1, 2, 3, 4] enters through the torus inputs Proposition 3.9 and Lemma 3.10, established in the full version, in the existence part through what is stated in Section 5, and in the rotation part also through its re-runs for the hybrid flow (Proposition 17.18 of the full version, the full version, for the contours, Proposition 17.73(a) of the full version for the covariances, and for the blocks and the lines the properties (i)–(iv) in the proof of Lemma K.495 of the full version and the replacement of its cube step (v), whose sufficiency is the statement the proof of [3, Prop. 1] and Bałaban's lower-bound induction [1, §§3–4] use the block-spin flow, beyond the contour family (Proposition 17.18 of the full version), only through the properties (i)–(iv) listed in the proof of Lemma K.495 of the full version (the slice decomposition, the slice bounds, the degree count and the renormalisation at $B = 0$) and through the cube step (v) listed there; and with (v) replaced by the constant-background reduction of Lemma 17.41 of the full version in its universal-cover form, with the element bounds of Proposition 17.73(a) of the full version and with the degree bound of the hybrid lines in the form of Proposition K.537(e) of the full version, King's multiscale induction [15, (3.66)–(3.70)] run for the hybrid lines yields [3, (1.33)] with the factor $C(1+k)^p$ for the hybrid flow of X_k on the tori $T \in \{T_1, T_2\}$, and the lower-bound induction yields the lower bound [1, (1.14)] for that flow in the form of Proposition 17.117, with $C(1+k)^p$ in the exponent, uniformly in the level $1 \le j \le r$, with constants depending on the composite blocks $B^{(j)}(y)$ of Lemma 17.20(iv) of the full version only through the tiling, the nesting, the diameter bound $\le C5^j\varepsilon_k$, the blocks of a level being the translates of one set by the lattice Y_{k-j}, and the identities $\hat{Q}\hat{Q}^* = \mathbf{1}$, $Q(A)Q(A)^* = \mathbf{1}$ of Lemma 17.20(iii) of the full version; the induction is run for the hybrid flow in the full version). The general results on the measure, its ergodic decomposition, reconstruction and clustering and the two-loop majorants of the lattice mass counterterm behind Proposition 22.14 are proved in the full version. Nothing of [15, 16] is used as a result: King's existence theorem is proved in Sections 5 and 6, and his reflection-positivity argument for the modified box and his passage to subsequences are replaced by proofs in this paper.

4. The two cube conventions: interlacing and margins

King's printed state lives on King's convention $\mathrm{ND}^{K}_{k,J}$ (Definition 2.2); the rotation part and the Wick-tube estimates of this paper use the comparison convention $\mathrm{ND}^{(a)}_{k,J}$ (Definition 2.1). We show that the two families are interlaced in J (Lemma 4.2), draw the consequences for the limits (Corollaries 4.3 and 4.4), and show that the comparisons of a cube with a reflected cube, a larger box and a tiling by cells carry over to King's convention with the printed margins (Lemmas 4.5 to 4.7).

4.1. **The two conventions and the support condition.** Throughout, $\varepsilon = \varepsilon_k = 5^{-k}$, $X_J = C_J \cap \Lambda_k = \{-MJ, \dots, MJ-\varepsilon\}^3$, $T_J = \{x \in X_J : \ x_\mu = MJ - \varepsilon \text{ for some } \mu\}$ and $\mathrm{alive}^K_J = X_J \setminus T_J = \{-MJ, \dots, MJ-2\varepsilon\}^3$. Both states carry the same photon variables $A_\alpha(x)$, $x \in X_J$, with the Neumann form of X_J, so their photon Gaussians are identical and every plaquette observable is the same function of the shared variables. They differ exactly by the matter on T_J: alive in the comparison convention (scalar Dirichlet planes $x_\mu = -MJ-\varepsilon$, $x_\mu = MJ$), killed in King's (planes $x_\mu = -MJ-\varepsilon$, $x_\mu = MJ-\varepsilon$, $2MJ$ apart, a multiple of every block scale; cf. Remark 3.5). A bond with one end alive and the other killed or outside the site set contributes its diagonal part $\frac{\varepsilon}{2} r_x^2$ at the alive end, so an alive site carries six diagonals in either convention. In the proof of his Theorem 3.2 and in his box form (3.38), King [16] works in the comparison convention: every site of the region is alive there.

For (f, g) with f a real 2-form and $g \geq 0$ a scalar, both in C^∞_c, we write as in Definition 2.4 $\Theta(\mathrm{i}f, g) = e^{\mathrm{i}F(f) + W(g)}$ (class $\mathcal{I}$), $\Theta(f, -g) = e^{F(f) - W(g)}$ (class $\mathcal{D}$), $W(g) = \sum_x \varepsilon^3 g(x)(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon)$ (free Wick constant), and $Z^a_{k,J}(\mathrm{i}f, g) = e^{-a_k \mathsf{S}(g)} \langle \Theta(\mathrm{i}f, g) \rangle$, $Z^a_{k,J}(f, -g) = e^{a_k \mathsf{S}(g)} \langle \Theta(f, -g) \rangle$, $\mathsf{S}(g) = \sum_x \varepsilon^3 g(x)$; a superscript K ($Z^{a,K}_{k,J}$) marks King's convention. All these expectations are real. The sources lie inside X_J if

(S) every plaquette $p = (x; \alpha, \beta)$ of Λ_k with $f_{c(p)}(e_\alpha, e_\beta) \neq 0$ has its four corners in X_J, and every site $x \in \Lambda_k$ with $g(x) \neq 0$ lies in $\mathrm{alive}^K_J = X_J \setminus T_J$.

If $\operatorname{supp} f \cup \operatorname{supp} g \subset \{y : \operatorname{dist}_\infty(y, \mathbb{R}^3 \setminus C_J) > 1\}$, as every hypothesis of this paper demands of its sources, then (S) holds: a site with $g(x) \neq 0$ has $-MJ + 1 < x_\mu < MJ - 1 \leq MJ - \varepsilon$, so $x \in \mathrm{alive}^K_J$; a plaquette with $f_{c(p)} \neq 0$ has its centre at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ and its corners within $\varepsilon/2 \leq \frac{1}{2}$ of it, hence in X_J. Also (S) for X_{J_0} implies (S) for every $J \geq J_0$.

4.2. **Cut-outs and the interlacing lemma.**

Lemma 4.1 (cut-outs from a box with killed sites)**.** *Let $\langle \cdot \rangle_Y$ be the state of Definition 2.1 on a finite set $V \subset \Lambda_k$ of sites with the matter killed on $T \subset V$, and $X \subset V$ any subset. Let (f, g), $g \geq 0$, satisfy* (S_X)*: every plaquette with $f \neq 0$ has its four corners in X and every site with $g \neq 0$ lies in X (g on $X \cap T$ is immaterial). Then deleting from $\langle \cdot \rangle_Y$ the photon gradient terms coupling X to $V \setminus X$ and the cosine parts of the hopping terms between X and $V \setminus X$ yields a product whose X-factor is $\langle \cdot \rangle^{(a), X \cap T}_X$, the state of Definition 2.1 on X with the matter killed on $X \cap T$; consequently*

$$\langle \Theta(\mathrm{i}f, g) \rangle^{(a), X \cap T}_X \ \leq \ \langle \Theta(\mathrm{i}f, g) \rangle_Y, \qquad \langle \Theta(f, -g) \rangle^{(a), X \cap T}_X \ \geq \ \langle \Theta(f, -g) \rangle_Y,$$

and the same for Z^a. In particular, (a) *if $X \cap T = \emptyset$ the factor is $\langle \cdot \rangle^{(a)}_X$;* (b) *if $X = \prod_\mu \{a_\mu, \dots, b_\mu\}$ is a box and $X \cap T = \{x \in X : \ x_\mu = b_\mu \text{ for some } \mu\}$, the factor is the state of King's convention on X.*

Proof. Each deletion is in the cone direction: deleting a photon gradient term does not increase $\langle \Theta_{F,g} \rangle$ and does not decrease $\langle \Theta^-_{F,g} \rangle$ (Lemma 2.8(a),(d)), and setting a hopping coefficient c_{xy} to 0 likewise (Lemma 2.8(b),(d)); every intermediate measure is a BFS measure. After all deletions nothing couples the X-variables to the rest, so the measure is a product (Lemma 2.8(e)). In the X-factor the photon is the Neumann photon of X; a site of $X \cap T$ carries its point mass; an alive $x \in X \setminus T$ carries the single-site weight times $\frac{\varepsilon}{2} r_x^2$ for each of its six bonds, since in $\langle \cdot \rangle_Y$

every bond at an alive x contributes $\frac{\varepsilon}{2}r_x^2$ at x whether its other end is alive in V (the diagonal part of $\frac{\varepsilon}{2}|e^{\mathrm{i}e\varepsilon A}\phi_y-\phi_x|^2$), killed, or outside V ($\phi=0$ there), and the cut keeps all diagonals; the cosines are those of the bonds within X between alive sites. That is the state of Definition 2.1 on X killed on $X\cap T$. By (S_X) the observable is a function of the X-factor's variables (at $x\in X\cap T$ the factor $e^{\varepsilon^3 g(x)r_x^2}$ is 1 in both states), and the c-numbers $e^{\mp 2\mathsf{W}_\varepsilon \mathsf{S}(g)}$, $e^{\mp a_k \mathsf{S}(g)}$ are the same positive constants on both sides (Lemma 2.12(a); the counterterm is site-independent). Hence $\langle\Theta\rangle_X^{(a),X\cap T}$ is the expectation of Θ in the fully deleted measure, which is $\le\langle\Theta\rangle_Y$ on $\mathcal{I}$ and $\ge$ on $\mathcal{D}$. □

The clause of Lemma 2.11 that X and its neighbours lie in one fundamental cell is a torus clause; for a box it has no content, since a bond leaving V already carries its Dirichlet diagonal in $\langle\cdot\rangle_Y$. Every sandwich $C_{J-1}\subset\cdot\subset C_{J+1}$ of this paper, read in King's convention, is an instance of Lemma 4.1.

Lemma 4.2 (interlacing of the two cube families)**.** *Fix $k\ge 0$, $J\ge 1$ and (f,g), $g\ge 0$, satisfying (S) for X_J. Then*

$$\langle\Theta(\mathrm{i}f,g)\rangle_{k,J}^K \le \langle\Theta(\mathrm{i}f,g)\rangle_{k,J}^{(a)} \le \langle\Theta(\mathrm{i}f,g)\rangle_{k,J+1}^K, \qquad \langle\Theta(f,-g)\rangle_{k,J}^K \ge \langle\Theta(f,-g)\rangle_{k,J}^{(a)} \ge \langle\Theta(f,-g)\rangle_{k,J+1}^K,$$

and the same for the scheme functionals $Z_{k,J}^a(\mathrm{i}f,g)$, $Z_{k,J}^a(f,-g)$ of the three states.

Proof. Step 0 (one observable, one multiplier). By (S) the plaquettes with non-zero coefficient have all corners in X_J, whose photon variables are present in all three states, so $F(f)$ is one real linear functional F of the shared variables; the sites with $g\neq 0$ lie in alive_J^K, alive in all three states; W_ε, δm_ε^2, $\mathsf{S}(g)$ are the same for the three regions. Hence, with $g'_x:=\varepsilon^3 g(x)\ge 0$, $\Theta(\mathrm{i}f,g)=e^{-2\mathsf{W}_\varepsilon\mathsf{S}(g)}\Theta_{F,g'}$ and $\Theta(f,-g)=e^{2\mathsf{W}_\varepsilon\mathsf{S}(g)}\Theta^-_{F,g'}$, where $\Theta_{F,g'}=e^{\mathrm{i}F}e^{\sum_x g'_x r_x^2}$, $\Theta^-_{F,g'}=e^F e^{-\sum_x g'_x r_x^2}$ are the observables of Lemma 2.8, and the scheme multiplier $e^{\mp a_k\mathsf{S}(g)}$ is one positive constant for all three states (Lemma 2.12(a)). It suffices to treat $\langle\Theta_{F,g'}\rangle$, $\langle\Theta^-_{F,g'}\rangle$.

Step 1 (kill step on $\mathcal{I}$). For $s\ge 0$ let $\langle\cdot\rangle_s$ be $\langle\cdot\rangle_{k,J}^{(a)}$ with the single-site measure at each $x\in T_J$ multiplied by $e^{-s\varepsilon^3 r_x^2}$, a BFS measure, and $\varphi(s):=\langle\Theta_{F,g'}\rangle_s=N(s)/D(s)$, $N(s)=\int\Theta\, e^{-s\varepsilon^3\sum_{T_J}r_x^2}d\mu_0$, $D(s)=\int e^{-s\varepsilon^3\sum_{T_J}r_x^2}d\mu_0$, μ_0 the unnormalised measure at $s=0$. The s-derivative of either integrand is bounded in modulus by $\varepsilon^3\sum_{T_J}r_x^2$ times the integrand at $s=0$, integrable against the quartic weight $e^{-\lambda\varepsilon^3 r^4}$ and the Gaussian; so differentiation under the integral sign is justified, $D(s)>0$, and

$$\varphi'(s)=-\varepsilon^3\sum_{x\in T_J}\big(\langle\Theta\, r_x^2\rangle_s-\langle\Theta\rangle_s\langle r_x^2\rangle_s\big)=-\varepsilon^3\sum_{x\in T_J}\langle\Theta;r_x^2\rangle_s \le 0$$

by Lemma 2.8(c). So $\varphi(s)\le\varphi(0)=\langle\Theta\rangle_{k,J}^{(a)}$. Since $\Theta_{F,g'}$ is continuous with $|\Theta_{F,g'}|=e^{\sum_x g'_x r_x^2}$ and $\langle\cdot\rangle_{k,J}^{(a)}$ is of King type, Lemma 2.10 gives $\varphi(s)\to\langle\Theta\rangle_{T_J}$, the state with the matter killed on T_J, which is $\langle\cdot\rangle_{k,J}^K$. Hence $\langle\Theta\rangle_{k,J}^K\le\langle\Theta\rangle_{k,J}^{(a)}$.

Step 2 (cut step). Apply Lemma 4.1 with $V:=X_{J+1}$, $T:=T_{J+1}$ (so $\langle\cdot\rangle_Y=\langle\cdot\rangle_{k,J+1}^K$), $X:=X_J$: $X_J\subset X_{J+1}$, and $X_J\cap T_{J+1}=\emptyset$ since every site of T_{J+1} has a coordinate $M(J+1)-\varepsilon\ge MJ>MJ-\varepsilon$. (At $k=0$, $M=1$ the outward neighbours of X_J lie on T_{J+1} and are killed in $\langle\cdot\rangle_Y$; harmless, since the bond from an alive $x\in X_J$ to such a neighbour already carries $\frac{\varepsilon}{2}r_x^2$ at x, the Dirichlet diagonal of the comparison cube.) By Lemma 4.1(a) the X_J-factor is $\langle\cdot\rangle_{k,J}^{(a)}$, and (S) is (S_{X_J}): $\langle\Theta(\mathrm{i}f,g)\rangle_{k,J}^{(a)}\le\langle\Theta(\mathrm{i}f,g)\rangle_{k,J+1}^K$, $\langle\Theta(f,-g)\rangle_{k,J}^{(a)}\ge\langle\Theta(f,-g)\rangle_{k,J+1}^K$.

Step 3 (kill step on $\mathcal{D}$). With $\langle\cdot\rangle_s$ as in Step 1 and $\varphi^-(s):=\langle\Theta^-_{F,g'}\rangle_s$ (differentiation justified as before, now with $|\Theta^-|\le e^{F(A)}$ integrable against the Gaussian), $(\varphi^-)'(s)=-\varepsilon^3\sum_{x\in T_J}\langle\Theta^-;r_x^2\rangle_s\ge 0$ by Lemma 2.8(d), so $\varphi^-(s)\ge\varphi^-(0)=\langle\Theta^-\rangle_{k,J}^{(a)}$; as $|\Theta^-_{F,g'}|\le e^{F(A)}$ with F real linear, Lemma 2.10 (with $\ell=F$, $C=0$) gives $\varphi^-(s)\to\langle\Theta^-\rangle_{k,J}^K$. Hence $\langle\Theta(f,-g)\rangle_{k,J}^K\ge\langle\Theta(f,-g)\rangle_{k,J}^{(a)}$.

Step 4 (Z^a). The multiplier of Step 0 is one positive constant. □

Nothing in the proof uses a reflection symmetry of either cube (neither has one), the counterterm beyond its site-independence, or the renormalisation group. Since the observable vanishes at the killed sites in every kill step of this section, each such step also follows with no limit: set $c_{xy} = 0$ on every bond at the set U to be killed, one bond at a time (Lemma 2.8(b),(d)); the measure is then a product (Lemma 2.8(e)) whose U-factor cancels, and the other factor is the state with U killed.

4.3. **Monotonicity of the K family and the limits.**

Corollary 4.3 (the K family is monotone in J)**.** *Under* (S) *for* X_J*:* $\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J} \le \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J+1}$ *and* $\langle\Theta(f,-g)\rangle^K_{k,J} \ge \langle\Theta(f,-g)\rangle^K_{k,J+1}$*, and the same for* $Z^{a,K}$*; hence for all* $J \ge J_0$ *whenever the sources satisfy* (S) *for* X_{J_0}*.*

Proof. Chain the two inequalities of Lemma 4.2. (For the comparison family this is Lemma 4.1(a) with $V = X_{J+1}$, $X = X_J$, i.e. the first inequality of Lemma 2.11.) □

The proof of Theorem 3.2 of [16] first replaces the J-dependent Dirichlet Wick subtraction (2.13) by the free, J-independent one and then proves the monotonicity in J in the comparison convention (Lemma 4.1(a)); by Corollary 4.3 that statement holds between cubes of King's convention too, with the kill step (Step 1) not printed there. The form with the Dirichlet constant follows in neither convention and is not used here; at $g = 0$ no Wick constant enters.

Corollary 4.4 (the limits)**.** *Let* $f = (f_1, \dots, f_N)$ *be real 2-forms in* C^∞_c *with* $\operatorname{supp} f_j \subset \{\operatorname{dist}_\infty(\cdot, \mathbb{R}^3 \setminus C_{J_0}) > 1\}$*; for* $J \ge J_0$*,* $z \in \mathbb{C}^N$ *put* $a^K_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$*,* $a^{(a)}_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$*,* $g(z) := \sum_j (\operatorname{Re} z_j) f_j$*. At the cone points* $z = \mathrm{i}t$*,* $t \in \mathbb{R}^N$*, both are real with modulus* ≤ 1*, and* (S) *holds for* X_J*.*

(i) (One J-limit at fixed k; no hypothesis.) *For every* k *and* t*,* $J \mapsto a^K_{k,J}(\mathrm{i}t)$ *and* $J \mapsto a^{(a)}_{k,J}(\mathrm{i}t)$ *are nondecreasing on* $J \ge J_0$*, bounded by* 1 *and interlaced; hence* $\lim_J a^K_{k,J}(\mathrm{i}t) = \lim_J a^{(a)}_{k,J}(\mathrm{i}t) \in [-1,1]$*. The same holds for* $\langle\Theta(\mathrm{i}f,g)\rangle$*,* $g \ge 0$*, with a common limit in* $(-\infty,\infty]$*.*

(ii) (King's cube limits.) *By Theorem 3.3, posed on King's convention, for every such family, every* $J \ge J_0$ *and every* $t \in \mathbb{R}^N$*,* $\lim_k a^K_{k,J}(t)$ *exists in* $(0,\infty)$*. Then:* (α) $a^K_{k,J} \to Z^K_J$ *locally uniformly on* $\mathbb{C}^N$*,* Z^K_J *is entire and* $|Z^K_J(z)| \le e^{\|g(z)\|_2^2}$*;* ($\beta$) $Z^K_J \to Z^K_\infty$ *locally uniformly on* $\mathbb{C}^N$*,* Z^K_∞ *is entire and* $|Z^K_\infty(z)| \le e^{\|g(z)\|_2^2}$*;* ($\gamma$) *for all* $J \ge J_0$ *and* $t \in \mathbb{R}^N$*,*

$$Z^K_J(\mathrm{i}t) \;\le\; \liminf_k a^{(a)}_{k,J}(\mathrm{i}t) \;\le\; \limsup_k a^{(a)}_{k,J}(\mathrm{i}t) \;\le\; Z^K_{J+1}(\mathrm{i}t),$$

hence $\lim_J \liminf_k a^{(a)}_{k,J}(\mathrm{i}t) = \lim_J \limsup_k a^{(a)}_{k,J}(\mathrm{i}t) = Z^K_\infty(\mathrm{i}t)$*.*

(iii) (The subsequential limits of the comparison family.) *Let* $\mathcal{S} = (k_m)$ *be an increasing sequence of spacings along which* $a^{(a)}_{k_m,J} \to Z^{\mathcal{S}}_J$ *locally uniformly on* $\mathbb{C}^N$ *for every* $J \ge J_0$ *and every such family (a continuum subsequence of the comparison family, Definition 14.1; they exist with no hypothesis, Lemma 14.7). Then* $Z^{\mathcal{S}}_J \to Z^K_\infty$ *locally uniformly on* $\mathbb{C}^N$ *as* $J \to \infty$*: every subsequential fixed-cube limit of the comparison family, followed by* $J \to \infty$*, is King's printed state. In particular, if the fixed-cube limits of the comparison family exist along the full sequence,* $\lim_J \lim_k a^{(a)}_{k,J} = Z^K_\infty$*.*

Proof. (i) Monotonicity is Corollary 4.3 for the K family and Lemma 4.1(a) for the comparison family; Lemma 4.2 interlaces them; two nondecreasing sequences with $a_J \le b_J \le a_{J+1}$ have the same limit, finite or $+\infty$.

(ii)(α) This is the first step of the proof of Theorem 7.4(b) run on the K family. (1) $a^K_{k,J}$ is entire (Gaussian decay in A, $e^{-\lambda\varepsilon^3 r^4}$ decay in r at the alive sites). (2) The local bound of Lemma 7.3 holds for the K family with the same constant: $\operatorname{Re}\sum_j z_j F(f_j) = F(g(z))$, so $|a^K_{k,J}(z)| \le \langle e^{F(g(z))}\rangle^K_{k,J}$; Lemma 2.9(a) with $s = 1$ and $v := d^*_k g(z)$ (defined by $F_{\Lambda_k}(g(z)) =$

$\langle v, A\rangle_k$) applies to any BFS measure whose Gaussian has covariance C and gives $\langle e^{\langle v,A\rangle}\rangle \le e^{\langle v,Cv\rangle/2}$; the K cube is a BFS measure whose Gaussian is the Neumann photon Gaussian of X_J, the same as the comparison cube's, so its covariance is the C of Lemma 7.1, a statement about this C and the plaquettes with corners in X_J: $\langle v, Cv\rangle_k \le 2\|g(z)\|^2_{2,k}$. Hence $|a^K_{k,J}(z)| \le e^{\|g(z)\|^2_{2,k}} \le e^{3C_0|g(z)|^2_4}$ uniformly in $k \ge 0$, $J \ge J_0$, by (E3) of Lemma 2.7: the plaquette centres of each of the three orientations form a translate of $\varepsilon_k\mathbb{Z}^3$, each contributing at most $C_0|g|^2_4$ to $\|g\|^2_{2,k}$ by (E1), and the three orientations give $3C_0$. Nothing in this chain sees the matter beyond the state being a BFS measure. (3) Theorem 3.3 gives convergence at every real point. (4) $\mathbb{R}^N$ is a set of uniqueness for entire functions on $\mathbb{C}^N$; by Montel's theorem every subsequence has a locally uniformly convergent sub-subsequence, all limits are entire and agree on $\mathbb{R}^N$, hence everywhere [11]. So $a^K_{k,J} \to Z^K_J$ locally uniformly, Z^K_J is entire, and the bound passes to the limit, $|Z^K_J(z)| \le \lim_k e^{\|g(z)\|^2_{2,k}} = e^{\|g(z)\|^2_2}$ (Lemma 2.7), a J-free bound.

(β) By Corollary 4.3 at every k and (α), $J \mapsto Z^K_J(\mathrm{i}t)$ is nondecreasing and ≤ 1, so $Z^K_\infty(\mathrm{i}t) := \lim_J Z^K_J(\mathrm{i}t)$ exists; $\{Z^K_J\}_{J\ge J_0}$ is locally uniformly bounded by the J-free bound of (α); $(\mathrm{i}\mathbb{R})^N$ is a maximal totally real subspace, hence a set of uniqueness for entire functions; Montel's theorem and the identity theorem give $Z^K_J \to Z^K_\infty$ locally uniformly, Z^K_∞ entire, $|Z^K_\infty(z)| \le e^{\|g(z)\|^2_2}$. (This is the proof of Theorem 7.4(b) with Lemma 2.11 replaced by Corollary 4.3.)

(γ) Lemma 4.2 at every k, with $k \to \infty$ at the two ends (limits by (α)) and lim inf, lim sup in the middle, gives the sandwich; (β) gives the limits.

(iii) Lemma 4.2 along $\mathcal{S}$, with the full-sequence limits of (α) at the ends, gives $Z^K_J(\mathrm{i}t) \le Z^{\mathcal{S}}_J(\mathrm{i}t) \le Z^K_{J+1}(\mathrm{i}t)$, so $\lim_J Z^{\mathcal{S}}_J(\mathrm{i}t) = Z^K_\infty(\mathrm{i}t)$ by (β). $\{Z^{\mathcal{S}}_J\}_J$ is locally bounded by the J-free bound $e^{\|g(z)\|^2_2}$ (Lemma 7.3 passed to the limit), so by Montel's theorem and the uniqueness set $(\mathrm{i}\mathbb{R})^N$ it converges locally uniformly to an entire function (Lemma 14.8), which agrees with Z^K_∞ on $(\mathrm{i}\mathbb{R})^N$, hence on $\mathbb{C}^N$. If the fixed-cube limits exist along the full sequence, that sequence is a continuum subsequence. □

Statements (ii)–(iii) also hold with the two conventions exchanged, for $J \ge J_0 + 1$, when the comparison family is convergent at fixed cubes (Definition 3.1; read Lemma 4.2 as $a^{(a)}_{k,J-1} \le a^K_{k,J} \le a^{(a)}_{k,J}$); the exchanged (iii) needs Lemmas 14.7 and 14.8 for the K family, whose proofs use only the local bound and the photon-only bound $\langle F(u-u')^2\rangle \le 2\|u-u'\|^2_{2,k}$, both transferred by the covariance identity in (α).

What does not follow. At fixed J, Lemma 4.2 yields exactly (γ) and nothing on whether $\lim_k a^{(a)}_{k,J}(\mathrm{i}t)$ exists: the bounds $Z^K_J(\mathrm{i}t)$, $Z^K_{J+1}(\mathrm{i}t)$ differ in general (the cubes differ by M, so the gap does not shrink with k), and nothing constrains the oscillation of $a^{(a)}_{k,J}(\mathrm{i}t)$ in k between them. So the existence of the fixed-cube limits on one convention does not imply it on the other by these inputs, and likewise with scalar sources: the existence theorem must be proved on the family on which the renormalisation-group construction is carried out; the other is a comparison family with the same infinite-volume state by (ii)–(iii). No counterexample is claimed.

4.4. **Three margins in King's convention.** Let $\sigma_3 = \mathrm{diag}(1,1,-1)$, $X := X_J$, $X_+ := \sigma_3 X = (C_J + \varepsilon e_3)\cap\Lambda_k$ (third coordinates $\{-MJ+\varepsilon, \dots, MJ\}$), $\sigma_3 T_J = \{x \in X_+ :\ x_1 = MJ-\varepsilon$ or $x_2 = MJ-\varepsilon$ or $x_3 = -MJ+\varepsilon\}$ and $A_\sigma := \sigma_3(\mathrm{alive}^K_J) = \{x :\ x_1, x_2 \in \{-MJ, \dots, MJ-2\varepsilon\},\ x_3 \in \{-MJ+2\varepsilon, \dots, MJ\}\}$.

Lemma 4.5 (the reflection sandwich)**.** *Let $(f,g) \in \mathcal{K}$ be supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$.* (a) *If $M \ge 2\varepsilon_k$ (i.e. $k \ge 1$, or $M \ge 2$), then for every $J \ge J_0 + 1$*

$$Z^{a,K}_{k,J-1}(\mathrm{i}f, g) \ \le\ Z^{a,K}_{k,J}(\mathrm{i}\,\sigma_3\cdot f, \sigma_3\cdot g) \ \le\ Z^{a,K}_{k,J+1}(\mathrm{i}f, g).$$

(b) *For every $k \ge 0$, $M \ge 1$ and $J \ge J_0 + 2$ the same holds with $J \mp 2$ in place of $J \mp 1$.* (c) *For $k = 0$, $M = 1$ the one-step sandwich* (a) *is not obtainable by deletions and kills: A_σ contains alive sites ($x_3 = MJ$) killed in King's cube of index $J+1$, and* alive^K_{J-1} *contains sites*

($x_3 = -MJ + \varepsilon$) killed in the reflected state. Whether the inequality itself holds there is left open.

Proof. Step 1 (transport). By Lemma 8.4, $Z^{a,K}_{k,J}(\mathrm{i}\sigma_3\cdot f, \sigma_3\cdot g) = e^{-a_k \mathsf{S}(g)}\langle\Theta(\mathrm{i}f,g)\rangle_R$, with R the state on X_+ for the frame $(e_1, e_2, -e_3)$ killed on $\sigma_3 T_J$ ($\mathsf{S}(\sigma_3\cdot g) = \mathsf{S}(g)$). In the frame (e_α), as in Lemma 2.14, R is the BFS measure with A_1, A_2 on X_+ and A_3 on $X_+ - \varepsilon e_3 = X$ (gradient pairs within these sets, masses on them), matter alive exactly on A_σ with point masses on $\sigma_3 T_J$ and $\phi = 0$ outside X_+, all six diagonals at each alive site, and the standard cosine on every bond with both ends in A_σ (the flipped bond with $A_3'(x) = -A_3(x - \varepsilon e_3)$ gives the standard cosine of $\langle x - \varepsilon e_3, x\rangle$); $F(f)$, $W(g)$ are the standard ones (the plaquettes $(x;\alpha,3)$ pick up two sign changes and keep their centres). R is of King type. The sources satisfy (S) for X_{J-1} (resp. X_{J-2}), so by the inclusions below Step 0 of Lemma 4.2 applies to all states compared.

Step 2 (upper end). If $A_\sigma \subset \mathrm{alive}^K_{J'}$ and $X_+ \cup X \subset X_{J'}$: from $\langle\cdot\rangle^K_{k,J'}$ kill every site of $\mathrm{alive}^K_{J'} \setminus A_\sigma$ (Lemma 2.8(c), Lemma 2.10, which allows the already killed $T_{J'}$), leaving exactly the cosines of bonds within A_σ, i.e. R's; then delete every photon gradient term except the pairs within X_+ for A_1, A_2 and within X for A_3 (Lemma 2.8(a); these are gradient pairs of $X_{J'}$). The result is a product (Lemma 2.8(e)) whose factor on R's variables is R (photon, alive set with all six diagonals by the bookkeeping of Lemma 4.1, cosines); the other factor cancels. Hence $\langle\Theta(\mathrm{i}f,g)\rangle_R \le \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J'}$. The inclusions hold iff $MJ \le MJ' - 2\varepsilon$ (direction 3; the rest is trivial): for $J' = J+1$ iff $M \ge 2\varepsilon$, for $J' = J+2$ always.

Step 3 (lower end). If $\mathrm{alive}^K_{J''} \subset A_\sigma$ and $X_{J''} \subset X \cap X_+$: from R kill every site of $A_\sigma \setminus \mathrm{alive}^K_{J''}$ (Lemma 2.8(c), Lemma 2.10) and delete every photon gradient term coupling an $X_{J''}$-variable to another variable of R (Lemma 2.8(a)). The $X_{J''}$-factor has the Neumann photon of $X_{J''}$ (gradient pairs within $X_+ \cap X_{J''} = X \cap X_{J''} = X_{J''}$) and the matter alive on $\mathrm{alive}^K_{J''}$ with all diagonals, killed on $T_{J''}$: it is $\langle\cdot\rangle^K_{k,J''}$; the rest cancels. Hence $\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J''} \le \langle\Theta(\mathrm{i}f,g)\rangle_R$. The inclusions hold iff $-MJ'' \ge -MJ + 2\varepsilon$: for $J'' = J-1$ iff $M \ge 2\varepsilon$, for $J'' = J-2$ always.

Step 4. Steps 2–3 with $J\pm1$ give (a), with $J\pm2$ give (b); the multipliers agree (Lemma 2.12(a)). For (c): at $k=0$, $M=1$ the top of A_σ in direction 3 is J while that of alive^K_{J+1} is $J-1$, and the bottom of alive^K_{J-1} is $-J+1 = -MJ+\varepsilon$, the killed layer of R; deletions and kills never revive a killed site. □

In the curvature sector ($g=0$) the limit sentences of Theorem 8.5 run on (a) for $k \ge 1$ or on (b) for all k, with $k \to \infty$ by Corollary 4.4(ii) and $J \to \infty$ by Corollary 4.3; with scalar sources they need the K-family tube limits, supplied by Theorem 21.3 (Remark 4.9).

Lemma 4.6 (the upper box needs no change)**.** *Let $\Omega \subset B$ be lattice boxes at spacing ε such that B contains every site within ℓ^∞-distance N lattice steps of Ω, $N \ge 1$ (the hypothesis of Lemma 9.10, whose mass condition forces $N \ge 1$), and let $\langle\cdot\rangle^K_B$ be the state of King's convention on B (killed on its top face layers T_B). Then* (a) *$\Omega \cap T_B = \emptyset$;* (b) *the second part of Lemma 9.10 holds with $\langle\cdot\rangle^K_B$ in place of the comparison state on B, with the same proof, and its first part is a photon statement, unchanged;* (c) *consequently the sandwich in the proof of Theorem 10.6 holds for King's cubes with the printed cubes and margins (20): for every k and every ζ admitted there, $Z^{a,K}_{k,J}(\zeta) \le \Phi_k(\zeta) \le Z^{a,K}_{k,J+2n}(\zeta)$, Φ_k the functional of the box state of Definition 9.2 on $\Omega_{J,k}$; the lower end is Lemma 9.9 with Step 1 of Lemma 4.2 prepended. (By Corollary 4.3 the cube of index $J+2n+1$ would also do; it is not needed.)*

Proof. (a) For $x \in \Omega$ and each μ, $x + N\varepsilon e_\mu \in B$, so $x_\mu < \max_B x_\mu$ and $x \notin T_B$. (b) By (a) every site of Ω is alive, so once the hopping cosines touching $B \setminus \Omega$ are removed the matter on Ω is the Dirichlet matter of Ω, and the photon law of B is the same in both conventions; the complete proof is given in Lemma K.3 of the full version. (c) Lower end: Lemma 9.9 gives $\langle Y\rangle^{(a)}_{C_J} \le \langle Y\rangle_{\Omega_{J,k}}$ on $\mathcal{I}$ (its proof lands on the Dirichlet matter of C_J, the comparison convention), and Step 1 of Lemma 4.2 gives $\langle Y\rangle^K_{C_J} \le \langle Y\rangle^{(a)}_{C_J}$, (S) holding since the sources of the enlarged family and their reflections are supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $C_{J_0} \subset C_J$. Upper end: (b) with

$B = X_{J+2n}$, whose hypothesis is the second margin of (20), a statement about photon sites. The multipliers agree (Lemma 2.12(a)). □

For the cell bound we use the cells of Definition 13.4: $\ell_c = 2ML_c \geq 10$; the cells $c = \prod_\mu [w_\mu, w_\mu + \ell_c)$ of the tiling $\mathcal{T}_s$, $s \in \{0,1\}^3$, $w_\mu \in W_\mu(s_\mu)$, $W_\mu(0) = \ell_c\mathbb{Z}$, $W_\mu(1) = ML_c + \ell_c\mathbb{Z}$ (so $w_\mu, w_\mu + \ell_c, MJ \in \mathbb{Z}$); the good sets $G_s = \{x : \operatorname{dist}(x_\mu, W_\mu(s_\mu)) > 2\ \forall \mu\}$ and cutoffs χ_s with $\operatorname{supp}\chi_s \subset G_s$; $X_c := c \cap X_J$; $J_1(R) = \min\{J : MJ \geq R + \ell_c\}$ for sources supported in $Q_R = [-R, R]^3$; B_{cell}, defined on the comparison cell C_{L_c}. N^K_{k,C_J} is the normalised scalar-source functional of $\langle\cdot\rangle^K_{k,J}$ and $N^{(a),T}_{k,X}$ that of the comparison state on X killed on $T \subset X$.

Lemma 4.7 (the cell lemmas hold as printed)**.** (a) *For a tiling $\mathcal{T}_s$ and any lattice function $q \geq 0$ on X_J: $N^K_{k,C_J}(0,-q) \leq \prod_i N^{(a),X_i\cap T_J}_{k,X_i}(0,-q|_{X_i})$, over the site sets $X_i = c \cap X_J$ of the cells meeting C_J.* (b) *For a full cell $c \subset C_J$: $X_c \cap T_J \neq \emptyset$ iff $w_\mu + \ell_c = MJ$ for some μ.* (c) *Let $q \geq 0$ be any lattice function supported in Q_R, $J \geq J_1(R)$, $s \in \{0,1\}^3$ and $c \in \mathcal{T}_s$ with $(\chi_s q)|_{X_c} \not\equiv 0$. Then $-MJ < w_\mu$ and $w_\mu + \ell_c < MJ - 2$ for every μ: c is a full cell inside C_J with $X_c \cap T_J = \emptyset$, and $(\chi_s q)|_{X_c}$ is supported at ℓ^∞-distance > 2 from $\mathbb{R}^3 \setminus c$.* (d) *Hence Lemmas 13.6 to 13.8 are unchanged and Proposition 13.9 holds for $\langle\cdot\rangle^K_{k,J}$ as printed, under its hypotheses and with its constant K_-: $\ln N^K_{k,C_J}(0,-q) \leq K_-|q|_w$ for every k, every $J \geq J_1(R)$ and every lattice $0 \leq q \leq H$ supported in Q_R.* (e) *(Direction.) For a cell c and $\emptyset \neq T \subset X_c$: $N^{(a),T}_{k,c}(0,-\tilde q) \geq N^{(a)}_{k,c}(0,-\tilde q)$ for $\tilde q \geq 0$ vanishing on T. So B_{cell}, a bound on a comparison cell, does not bound a source-carrying cell of King's type, and* (c) *is essential.*

Proof. (a) The proof of Lemma 13.5 deletes the cross-cell photon gradient terms and hopping cosines, each in the $\mathcal{D}$ direction (Lemma 2.8(d)); by Lemma 2.8(e) and the factor identification of Lemma 4.1 with $V = X_J$, $T = T_J$, $X = X_i$, the product has the factors $N^{(a),X_i\cap T_J}_{k,X_i}$; the observable $e^{-\sum \varepsilon^3 q r^2}$ and the multiplier $e^{a_k \mathsf{S}(q)} = \prod_i e^{a_k \mathsf{S}(q|_{X_i})}$ factorise, and at the sites of T_J the observable factor is 1 in every state. (b) The sites of a full cell have $x_\mu \in \{w_\mu, \dots, w_\mu + \ell_c - \varepsilon\}$, and $x \in T_J$ iff $w_\mu + \ell_c - \varepsilon = MJ - \varepsilon$ for some μ. (c) Let $x \in X_c$ with $(\chi_s q)(x) \neq 0$. Then $x \in G_s$; since $w_\mu, w_\mu + \ell_c \in W_\mu(s_\mu)$ and $w_\mu \leq x_\mu < w_\mu + \ell_c$, this gives $w_\mu + 2 < x_\mu < w_\mu + \ell_c - 2$; and $q(x) \neq 0$ gives $|x_\mu| \leq R$. Hence $w_\mu + \ell_c < R + \ell_c - 2 \leq MJ - 2$ (as $MJ \geq R + \ell_c$) and $w_\mu > -R - \ell_c + 2 \geq -MJ + 2$. So $c \subset (-MJ, MJ)^3$, $w_\mu + \ell_c \neq MJ$, hence $X_c \cap T_J = \emptyset$ by (b), and $\operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus c) = \min_\mu \min(x_\mu - w_\mu, w_\mu + \ell_c - x_\mu) > 2$. (d) In the chain of Proposition 13.9 the Hölder step of Lemma 13.8 is an inequality within one state, convention-free; then (a) with the tiling $\mathcal{T}_s$ and $q_s := 8\chi_s q$. A factor with $q_s|_{X_i} \equiv 0$ equals 1 whatever is killed on X_i (this covers the clipped cells at the faces of C_J and any cell flush with them); a factor with $q_s|_{X_i} \not\equiv 0$ is, by (c), the comparison-cell functional $N^{(a)}_{k,c}(0,-q_s|_c)$ of a full cell $c \subset C_J$ with $q_s|_c$ supported at distance > 2 from $\mathbb{R}^3 \setminus c$ and $0 \leq q_s \leq 8H = H'$: exactly the hypotheses of Lemma 13.7 (hence of Lemma 13.6: translation to C_{L_c}, B_{cell}, Proposition 13.2 on the comparison box), as printed; Lemma 13.8 sums the cells. (A cell flush with a face, $w_\mu + \ell_c = MJ$, can meet $\operatorname{supp} q$ but never $\operatorname{supp}(\chi_s q)$; the chain feeds the cutoff sources, not q, to (a).) (e) Step 3 of Lemma 4.2 applied to the comparison cell with $U = T$: $s \mapsto \langle\Theta^-\rangle_s$ is nondecreasing (Lemma 2.8(d)) and tends to the killed state (Lemma 2.10). □

Remark 4.8 (one-line kill steps). Proposition 13.1 (cube $\leq$ torus on $\mathcal{I}$), Proposition 13.2 (box $\leq$ cube $\leq$ torus), the lower end of Lemma 9.9, and the comparison of the cube with the box state of Definition 9.2 on the cone (Remark 9.8) are each proved for the comparison cube; for King's cube prepend Step 1 of Lemma 4.2 ($\langle\cdot\rangle^K_{k,J} \leq \langle\cdot\rangle^{(a)}_{k,J}$ on $\mathcal{I}$), and for the $\mathcal{D}$ versions (cube $\geq$ torus for negative sources) Step 3. The same applies to a box of King's type (a killed face layer) in Proposition 13.2. Nothing beyond Lemma 4.2 is needed.

Remark 4.9 (the Wick sector). With scalar sources $g \geq 0$, Corollary 4.4(i) holds as stated. The analogues of Corollary 4.4(ii)–(iii) on the Wick tube need the fixed-cube limits and the (k, J)-uniform bound (31) for both families. For King's family the bound is the chain of Theorem 13.10(i) run on ND^K, whose convention-sensitive steps are the cube-versus-torus comparison

(Corollary 13.3, via Remark 4.8) and the cell bound (Lemma 4.7); the assembly of that chain and the existence of the K-family tube limits are carried out in Theorem 21.3, after which the tube statements hold on King's convention (Remark 13.12, Proposition 21.4).

The existence part

5. King's existence theorem on the cube

The existence theorem for King's cube limits (Theorem 3.3), and its form with scalar sources that the Wick leg consumes (Theorem 3.6), are proved in this section and the next from estimates on King's renormalisation-group integrals on the cube C_J^K. The argument is the one King gives for his Theorem 2.1 [16, Sect. 3.1] and [15, Sect. 3.2]: the generating functional is approximated, up to a small-field remainder, by an integral of the effective action after j renormalisation steps, and two spacings whose effective actions live on the same unit lattice are compared. We write it as a Cauchy property of the *normalised* functional, in a ratio form in which every field-independent constant of the effective action cancels; this removes from King's route the convergence of the vacuum energies, which his printed argument uses and which is not available on the cube.

The inputs are of two kinds. Estimates taken from Bałaban's block-factor-5 analysis [1, 2, 3, 4] on tori, and their transfer to King's cube, are stated as propositions, each followed by a description of its content; their proofs are given in the full version. Three of these statements are used at several places and are therefore stated once, with a precise formulation, in Section 5.2 ((BAL_5)[int], $(\mathrm{REN}_{\mathrm{orb}})$) and Section 6.1 ($(\mathrm{K}_h)$). Everything else in the two sections is proved in place.

5.1. **Renormalisation-group integrals on King's cube.** Fix the couplings, a cube C_J and King's RG cube C_J^K of Definition 2.2. In this section and the next the index k is the spacing index ($\varepsilon_k = 5^{-k}$, as throughout), and j counts renormalisation steps, $0 \le j \le k - J_*$ for a fixed J_* (King's "J" of [15, Thm. 3.1]); the block scale after j steps is $\eta := 5^j\varepsilon_k = 5^{j-k} \le 1$, and the unit lattice after j steps is $\eta^{-1}(C_J \cap \eta\mathbb{Z}^3)$. Since the side $2MJ$ of C_J is an integer, the faces of C_J lie on block planes at every scale.

Sources. A real source is $s = (u, h)$ with $u = u_x := \sum_i x_i f_i$ a real combination of 2-forms $f_i \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ sampled at the plaquette centres in the lattice's own frame, and $h \in C_c^\infty(\mathbb{R}^3)$ real; all supports lie at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ (so $J \ge J_0$). Put $X(s) := F(u) + {:}\phi^2{:}(h)$, with the free Wick constant of Definition 2.4; X is real and linear in s. The unnormalised functional is

$$Z_k(s) \;:=\; \int e^{-S}\, e^{X(s)} \prod dA \prod d^2\phi$$

over the variables of C_J^K at spacing ε_k (any fixed, s-independent normalisation of the measure; it cancels everywhere below), and $N_k(s) := Z_k(s)/Z_k(0) = \langle e^{X(s)}\rangle_{k,J}^K$. At $h = 0$ we write $a_k(x) := N_k(u_x, 0)$, which is the $a_{k,J}^K(x)$ of Corollary 4.4.

King's flow on the cube. The block-spin transformations of [15, Sect. 2] are run on C_J^K with block factor 5: the photon field is averaged over the $5^j\varepsilon_k$-blocks that tile X_J, with the Neumann condition of Definition 2.2; the scalar field is averaged with King's Dirichlet weight [16, (3.8)–(3.9)] (in the corrected form in which the averaging blocks have their top layers on the Dirichlet planes: the weight is $(5/4)^3$ at the sites of each finest block whose three coordinates have residues in $S = \{0, 1, 2, 3\}$ relative to the block's corner and 0 at the sites with a residue 4 in some coordinate (for a block whose top face lies on a Dirichlet plane, the sites with residue 4 in the coordinate normal to the plane form the layer on that plane), Definition D.7 of the full version) at the finest step and with the uniform weight above it. Explicitly, at $A = 0$ and zero block fields, with ι the extension by zero from the alive set alive_J^K to $\varepsilon_k\mathbb{Z}^3$ and Q_j^X the composite average Q_j restricted to the alive set and to the blocks of C_J: the scalar fluctuation covariance after j steps is $G_j(C_J^K) = (H^X)^{-1}$ with $H^X := \iota^*(-\Delta_{\varepsilon_k} + m^2)\iota + \mathfrak{a}_j\eta^{-2}(Q_j^X)^*Q_j^X$, the Dirichlet Laplacian of Definition 2.2 with the block-spin term; the photon covariance after $i \ge 1$ steps is $D_i(C_J^K) = (-\Delta_{X_J}^N + \mu_0^2 + \mathfrak{a}_i\ell_i^{-2}\mathsf{P}_{X_J,\ell_i})^{-1}$ componentwise, with the Neumann

Laplacian of Definition 2.2, $\mathsf{P}_{X_J,\ell}$ the uniform average over the ℓ-blocks of X_J, $D_0(C_J^K) := 0$ and the slices $G^{\mathrm{ph}}_{(i)}(C_J^K) = D_{i+1}(C_J^K) - D_i(C_J^K)$; and the background-field profile and the effective quadratic form after j steps are $H_j(C_J^K) = \mathfrak{a}_j\eta^{-2}G_j(C_J^K)(Q_j^X)^*$ and $\Delta^{(j)}(C_J^K) = \mathfrak{a}_j\eta^{-2}\big(\mathbf{1} - \mathfrak{a}_j\eta^{-2}Q_j^X G_j(C_J^K)(Q_j^X)^*\big)$ on the block lattice of C_J, with the stiffness $\mathfrak{a}_j$ of Definition D.7 of the full version. The Dirichlet planes of C_J^K are $2MJ$ apart, a multiple of every block side, which is what makes this flow possible on C_J^K and not on the comparison cube (Remark 5.3). Let $S_k^{(j)}(A_j, \phi_j; s)$ be King's effective action after j steps [15, (3.14)], [16, (3.12)], with the source vertices [15, (3.20)–(3.21)]; let χ_j be King's small-field indicator, [15, (3.2)] times the face factor [16, (5.41)], a function of (A_j, ϕ_j) alone; and

$$I_k^{(j)}(s) \;:=\; \int dA_j\, d\phi_j\; \chi_j\; e^{-S_k^{(j)}(A_j,\phi_j;s)}.$$

The *vacuum constant* $v_{k,j} := S_k^{(j)}(0,0;0)$ is finite; it contains King's E_0, E_1, the normalisation constants $\ln Z_j$, $\ln Z_j(0)$ of the block-spin kernels and every field- and source-free graph. The *normalised action* is $S_{N,k}^{(j)}(\cdot; s) := S_k^{(j)}(\cdot; s) - v_{k,j}$ for every s (the field-free, source-dependent part is kept), and $\hat{I}_k^{(j)}(s) := \int \chi_j e^{-S_{N,k}^{(j)}(\cdot;s)} = e^{v_{k,j}} I_k^{(j)}(s)$. The flow is a device of the proof: the state $\langle\cdot\rangle_{k,J}^K$ does not depend on it.

Diagonal pairs. Fix $\alpha \in (0, \frac{1}{2})$ and put $j(k) := \lfloor (1-\alpha)k \rfloor$, $\eta_k := 5^{j(k)-k}$; then $\eta_k \le 5^{-\alpha k}$ and $k - j(k) \le \alpha k + 1$. The *diagonal pair* at k consists of the k-model after $j(k)$ steps and the $(k+1)$-model after $j(k)+1$ steps. Both have the block scale η_k ($5^{j+1}\varepsilon_{k+1} = 5^j\varepsilon_k$), hence the same unit lattice and the same indicator $\chi_{j(k)}$; this is the situation of [15, p. 656] ("the fields A_k, ϕ_k have the same bounds in each integral"). No relation between $j(k+1)$ and $j(k)+1$ is used anywhere.

Bałaban's decomposition. In Bałaban's block-spin representation [1, 2], $Z_k(s) = \int d\mu_{k,s}$ with $d\mu_{k,s} = e^{X(s)} d\mu_{k,0}$ a positive measure on the fields and on the block variables of the scales $\le j$, and $P = P_j \in [0,1]$ is the product of all small-field factors of those scales, a function of the block and fluctuation variables alone. The identity $Z_k(s) = \int P\, d\mu_{k,s} + \int (1-P)\, d\mu_{k,s}$ is exact; the first term is the *small-field part*, the second the *large-field remainder*.

5.2. **Two statements from the renormalisation-group analysis.**

Lemma 5.1 ((BAL$_5$)[int]: Bałaban's interaction-term bound at a nonzero background)**.** *In the setting of* [2, Sect. 2] *at block factor $L = 5$, in unit-lattice form at the step k, $t = 5^k\varepsilon$: let $\Omega_A = B^k(\Lambda_2^{(k-1)\prime}) \supset \Omega_C = B^k(\Lambda_2^{(k)})$ be the two propagator domains of* [2, Prop. 2.6]*, let $\tilde{B}$ be the background field of* [2, p. 579]*, regular on Ω_A, let $G_k(\Omega, \tilde{B})$ be the scalar propagator of the step on Ω in the background $\tilde{B}$, and write the interaction terms of the stage of* [2, p. 579]*, localised in $B^k(\Lambda_7^{(k)})$, as a sum of per-label full renormalised classes with scalar lines $G_k(\Omega_A, \tilde{B})$, external scalar fields $E_A = a_k G_k(\Omega_A, \tilde{B})f$ with $\|f\|_\infty \le Cp\lambda(t)^{-1/4}$, and other external fields of norms $\le Cp^a\lambda(t)^{-b}$ (see the full version). Replace $G_k(\Omega_A, \tilde{B})$ by $G_k(\Omega_C, B^{(k+1),\eta})$ in every scalar line and in every E_A. For every $\kappa > 0$ there are $t_0(\kappa) > 0$ and $C_\kappa < \infty$, depending on κ, on Bałaban's parameters $(R, \varrho, b_0, \tilde{p}, \bar{n}, \alpha_0)$ and on the couplings only, such that for $t \le t_0(\kappa)$ the sum over all localisations of the absolute values of all terms containing at least one difference of propagators or of external fields is at most $C_\kappa\, t^\kappa\, |\Lambda_7^{(k)}|$. The constants do not depend on k, on the domains or on $\tilde{B}$, and $\tilde{B} = 0$ is included.*

The proof is given in the full version (Theorem G.16 there). This is the interaction-term half of Bałaban's domain-change estimate [2, Prop. 2.6], at every background. The terms with exactly one difference line, grouped line by line, carry their counterterm partners but not the complete renormalised classes of the cut graph, and a lone divergent two-photon-leg subgraph is not small per graph; the proof expands every line and every external field in the difference and regroups the terms exactly into contractions of *full* classes of the cut components, to which the renormalisation bound [3, Prop. 1, (1.33)] applies with constants free of the domains and of

the background. Every use of the upper stability bound [2, (1.3)] and of [15, Thm. 3.1, (3.4)] at block factor 5 passes through this statement.

Lemma 5.2 (($\mathrm{REN}_{\mathrm{orb}}$): orbit regrouping of cut renormalised families)**.** *Let $\mathcal{R} = \sum_{\Gamma\in\mathcal{G}} c_\Gamma E(\Gamma)$ be a renormalised family of* [3, Sects. 1–3] *with $\mathcal{G}$ a finite union of renormalised classes (Definition G.1 of the full version), in which the objects (lines or external legs) of certain kinds are decomposed as $o = o' - \Delta o$ with symmetric kernels Δo, and expand multilinearly, $\mathcal{R} = \sum_\tau 2^{-|S_L|} c_\Gamma \prod_{o\in S}(-\Delta o)\prod_{o\notin S} o'$ over the oriented cut terms $\tau = (\Gamma, S, \theta)$ of Definition G.2 of the full version. Call two cut terms equivalent when one arises from the other by finitely many of Bałaban's three completions of* [3, pp. 420, 429–432] *(the counterterm partner; the two-vector-leg class; the permutation of the external scalar legs of a divergent $\phi\phi A$ subgraph) applied at divergent subgraphs containing no line of S, the objects of S carried along to the image slots. Then:* (i) *the equivalence classes (orbits) partition the cut terms, and every move keeps the label multiset and the number and kinds of the objects of S;* (ii) *a connected line set of Γ avoiding S is a subgraph of the cut graph $\Gamma\setminus S$ with the same vertices, slots, slot types, order and degree* [3, (2.1)–(2.3)]*, and cutting commutes with the moves, so that the cut graphs of one orbit form exactly one renormalised class of the cut graph, the product of the classes of its connected components;* (iii) *at fixed labels, after the sum over the separable indices of the cut kernels, the sum of the cut terms of an orbit O is $(-1)^{|S|}2^{-|S_L|}r_O^{-1}c_*\prod_K E(\mathrm{Cl}(K);$ its fields$)$, the product over the components K of the class sums with the cut kernels as external fields, c_* a constant and $r_O \geq 1$ the number of relabellings of identical S-objects preserving the class;* (iv) *the sum over all orbits is the full expansion;* (v) *if $\mathcal{G}$ is a per-label full class, the cut terms with a given cut datum (components with labels and orders, stubs and fields at each component, incidence of the S-objects) correspond s_δ-to-one to the tuples of members of the per-label full classes of the components, members without internal line and of order 0 included; each such group is a union of orbits and is closed under the exchange of the scalar legs of every two-scalar-leg or $\phi\phi A$ subgraph.*

The proof is given in the full version (Lemmas G.3 and G.5 there). This is Bałaban's regrouping step in the proof of [3, Prop. 1, p. 432] ("if we remove a line in any graph of the class, then for the obtained graph we can find in this class all graphs necessary to form a new renormalized class"), made explicit for decomposed objects, with the sharper conclusion that one orbit gives exactly one class of the cut graph. It is needed wherever the renormalisation bound is applied to a family some of whose lines have been cut, in particular in the comparison of two propagator domains and in the telescoping of the cube's large-field remainders; at a nonzero background, where the bound of [3, Prop. 1] is available for per-label full classes, it is used in the form (v).

Remark 5.3 (the cube). The propositions below are stated for King's RG cube C_J^K, on which King's Dirichlet weight is compatible with the image group of the two Dirichlet planes (they are $2MJ$ apart). On the comparison cube $\mathrm{ND}^{(a)}$ the planes are $2MJ + \varepsilon_k$ apart, no partition into consecutive blocks is invariant under the image group, and the flow does not exist; this is why Theorem 3.3 is posed on King's convention, and why the fixed-cube limits of the comparison family are reached only through Corollary 4.4. The cube of the existence theorem is C_J^K itself (Remark 3.5).

5.3. **The inputs at zero scalar source.**

Proposition 5.4 (($\mathrm{H0rel}$)$_C$: the small-field representation on King's cube, zero scalar source)**.** *Fix $J \geq J_0$, a family $f_1, \dots, f_N$ and $x \in \mathbb{R}^N$, and let $s \in \{u_x, 0\}$. There are $\eta_1 > 0$, k_0 and $C, c, \sigma > 0$, depending on the couplings, on J, on f and x, and on nothing else (not on k, j), such that for all $k \geq k_0$ and all j with $J_* \leq j \leq k - J_*$ and $\eta = 5^{j-k} \leq \eta_1$:*

$$e^{-C\eta^\sigma} I_k^{(j)}(s) \;\leq\; Z_k(s) \;\leq\; e^{C\eta^\sigma} I_k^{(j)}(s) + \rho(\eta)\, Z_k(0), \qquad \rho(\eta) := e^{-c\,p(\eta)^2 + C}, \tag{2}$$

where $p(\eta) = b_0(1 + \ln \eta^{-1})^p$, with $p = 6$, compatible with the conditions [1, (3.54)]*, is Bałaban's small-field threshold. Moreover, at $s = 0$ the two terms of the upper line are the two parts of Bałaban's decomposition: $\int P\, d\mu_{k,0} \leq e^{C\eta^\sigma} I_k^{(j)}(0)$ and $\int (1-P)\, d\mu_{k,0} \leq \rho(\eta) Z_k(0)$ ((R_0)).*

The proof is given in the full version. This is [15, Thm. 3.1] on the cube, which [16, Sect. 5.2] asserts ("the proof follows by modifying suitably the material in [Ba 1] and [Ba 2]"), in the form in which the remainder is measured against $Z_k(0)$ rather than against an absolute constant, and with constants uniform in the step j (the form the diagonal pairs need, since $\eta_k \to 0$). The lower line is the weaker of the two readings of the sign in [15, (3.3)]. No lower bound on $Z_k(0)$ appears. Of the statements of Section 5.2, (BAL_5)[int] (through Bałaban's torus bounds) and $(\mathrm{REN}_{\mathrm{orb}})$ (through the comparison of propagator domains and the telescoping of the large-field remainders) enter.

Proposition 5.5 ($(\text{K-pos})_N$: one-step bound on the normalised action, zero scalar source)**.** *With J, f, x and $s \in \{u_x, 0\}$ as above there are $\gamma \in (0,1)$, $\beta, \sigma > 0$ and $C_3 < \infty$, depending on the couplings, J, f, x only, such that for all $k \geq k_0$, all $J_* \leq j \leq k - J_*$ with $\eta = 5^{j-k} \leq \eta_1$, and all (A_j, ϕ_j) in the support of χ_j,*

$$\big|S^{(j)}_{N,k}(A_j,\phi_j;s) - S^{(j+1)}_{N,k+1}(A_j,\phi_j;s)\big| \;\leq\; C_3\big(5^{-\gamma j}\eta^{-\beta} + \eta^{\sigma}\big). \tag{3}$$

The proof is given in the full version. This is [15, Thm. 3.4] at $n = 1$ on the cube, for the field- and source-dependent part of the effective action: the two members of the pair (k, j), $(k+1, j+1)$ live on one unit lattice with one indicator χ_j, and the statement compares them there. It is assembled from six pieces: the photon quadratic form (an even-restriction identity reduces the cube to the doubled torus, where [15, Prop. 3.10] applies); the scalar quadratic form at zero background (the Dirichlet symbol) and with background (a two-leg tree graph); the background determinant (one-loop graphs with a remainder, [15, (3.94)–(3.98)], of which the middle line of (3.98) is false as printed and is replaced); the graphs with curvature vertices; the source-free field-dependent graphs away from the faces and near them (the photon face terms and the scalar boundary remainders of [16, pp. 331–332]); and the block-vertex classes. For the curvature-vertex and the source-free field-dependent classes, the localisations whose tree length R exceeds the small-field threshold p are bounded by cutting the localised graph into clusters of mutual distance $> 3\sqrt{d}$ and reducing each cluster in its own box of side $O(p)$, with the cut lines as external lines of both clusters (Corollary B.40(b) and Lemma B.51; the zone-B tails by Theorem B.60(a) and Theorem B.67(d), all in the full version). The vacuum constant $v_{k,j}$ is not part of the statement.

Lemma 5.6 (bounds for free)**.** *Put $\beta_f := C_0 \sum_i |f_i|_4^2$ with C_0 as in Lemma 2.7. For all $k \geq 0$, $J \geq J_0$ and $x \in \mathbb{R}^N$,*

$$e^{-\beta_f |x|^2} \;\leq\; a_k(x) \;\leq\; e^{\beta_f |x|^2}.$$

Proof. By Theorem 7.6(a), $a_k(x) \leq \exp(\|u_x\|^2_{2,k})$, and $\|u_x\|_{2,k} \leq \sum_i |x_i| \|f_i\|_{2,k} \leq |x| (\sum_i \|f_i\|^2_{2,k})^{1/2} \leq |x|\, \beta_f^{1/2}$ by Lemma 2.7(E3). The lower bound is $a_k(x) a_k(-x) \geq 1$, Lemma 2.12(c). □

5.4. **The Cauchy property in the spacing.**

Theorem 5.7 (the Cauchy property, ratio form)**.** *Fix $J \geq J_0$, f, $x \in \mathbb{R}^N$, and assume (2) and (3) for $s \in \{u_x, 0\}$ with the constants of Propositions 5.4 and 5.5. Choose $\alpha \in (0, \min\{\frac{1}{2}, \gamma/(\gamma+\beta)\})$ and the diagonal pairs $j(k) = \lfloor (1-\alpha)k \rfloor$, $\eta_k = 5^{j(k)-k}$. Put*

$$b_k := C\eta_k^{\sigma}, \qquad \theta_k := C_3\big(5^{-\gamma j(k)}\eta_k^{-\beta} + \eta_k^{\sigma}\big), \qquad \rho_k := \rho(\eta_k), \qquad r_k(0) := 2\rho_k, \qquad r_k(u_x) := 2\rho_k e^{\beta_f |x|^2}.$$

Then there is k_1 such that for all $k \geq k_1$

$$\big|\ln a_{k+1}(x) - \ln a_k(x)\big| \;\leq\; 4b_k + 2\theta_k + r_k(u_x) + r_k(0). \tag{4}$$

The right side is summable in k, and its tail from k on is at most $C' 5^{-\kappa k}$ for some $\kappa > 0$. Hence $a_k(x)$ converges, and its limit lies in $[e^{-\beta_f |x|^2}, e^{\beta_f |x|^2}]$.

Proof. Fix $k \geq k_0$ so large that $j(k) \geq J_*$, $\eta_k \leq \eta_1$ and $r_k(u_x) \leq 1$ (possible by Step 6), and put $j := j(k)$, $b := b_k$, $\theta := \theta_k$. For $s \in \{u_x, 0\}$ write $Z := Z_k(s)$, $I := I_k^{(j)}(s)$, $\hat{I} := \hat{I}_k^{(j)}(s)$ for the k-model after j steps, and $Z' := Z_{k+1}(s)$, $I' := I_{k+1}^{(j+1)}(s)$, $\hat{I}' := \hat{I}_{k+1}^{(j+1)}(s)$ for the $(k+1)$-model

after $j+1$ steps. Both members have the block scale η_k and the indicator χ_j, and (2) applies to both, since $j+1 \le (k+1) - J_*$; their remainders are $R := \rho_k Z_k(0)$ and $R' := \rho_k Z_{k+1}(0)$.

Step 1 (the remainders relative to Z). For $s=0$, $R/Z = \rho_k = r_k(0)/2$ and $R'/Z' = \rho_k$. For $s = u_x$, $Z = a_k(x) Z_k(0) \ge e^{-\beta_f |x|^2} Z_k(0)$ by Lemma 5.6, so $R/Z \le \rho_k e^{\beta_f |x|^2} = r_k(u_x)/2$, and likewise $R'/Z' \le r_k(u_x)/2$. In both cases $R/Z,\ R'/Z' \le r_k(s)/2 \le \frac{1}{2}$. No lower bound on $Z_k(0)$ and no bound on $Z_k(u_x)$ beyond Lemma 5.6 is used.

Step 2 (one model). From $e^{-b} I \le Z \le e^b I + R$: $I \le e^b Z$ and $I \ge e^{-b}(Z - R) = e^{-b} Z(1 - R/Z)$. So $I \in [e^{-b} Z(1 - R/Z),\, e^b Z]$, and in the same way $I' \in [e^{-b} Z'(1 - R'/Z'),\, e^b Z']$. In particular $I, I' > 0$.

Step 3 (across the pair). On the support of χ_j, (3) gives $|S^{(j)}_{N,k}(\cdot; s) - S^{(j+1)}_{N,k+1}(\cdot; s)| \le \theta$. The two integrals $\hat I$, $\hat I'$ run over the same variables with the same χ_j and have non-negative integrands, so $\hat I'/\hat I \in [e^{-\theta}, e^{\theta}]$. Since $I = e^{-v_{k,j}} \hat I$ and $I' = e^{-v_{k+1,j+1}} \hat I'$,

$$I'/I \ \in\ e^{\Delta}\,[e^{-\theta}, e^{\theta}], \qquad \Delta := v_{k,j} - v_{k+1,j+1},$$

and Δ does not depend on s: the vacuum constant is the same number in the numerator and in the denominator of a_k.

Step 4 (ratio of partition functions). By Steps 2 and 3,

$$\frac{Z'}{Z} = \frac{Z'}{I'} \cdot \frac{I'}{I} \cdot \frac{I}{Z} \ \in\ e^{\Delta}\Big[e^{-2b-\theta}\big(1 - \tfrac{R}{Z}\big),\ e^{2b+\theta}\big(1 - \tfrac{R'}{Z'}\big)^{-1}\Big].$$

With $|\ln(1-t)| \le 2t$ for $0 \le t \le \frac{1}{2}$ and Step 1, $|\ln(1 - R/Z)|,\ |\ln(1 - R'/Z')| \le r_k(s)$, hence

$$\big|\ln(Z'/Z) - \Delta\big| \ \le\ 2b + \theta + r_k(s).$$

Step 5 (ratio of ratios). $\ln \dfrac{a_{k+1}(x)}{a_k(x)} = \Big[\ln \dfrac{Z_{k+1}(u_x)}{Z_k(u_x)} - \Delta\Big] - \Big[\ln \dfrac{Z_{k+1}(0)}{Z_k(0)} - \Delta\Big]$, and Step 4 at $s = u_x$ and at $s = 0$ gives (4). (If the constants of (2), (3) differ between the two sources, take the larger ones.)

Step 6 (summability). Since $k - j(k) \ge \alpha k$, $\eta_k \le 5^{-\alpha k}$, so b_k and $C_3 \eta_k^{\sigma}$ decay geometrically. Since $j(k) \ge (1-\alpha)k - 1$ and $k - j(k) \le \alpha k + 1$,

$$5^{-\gamma j(k)} \eta_k^{-\beta} \ \le\ 5^{\gamma+\beta}\, 5^{-(\gamma(1-\alpha) - \beta\alpha)k},$$

which decays geometrically because $\alpha < \gamma/(\gamma+\beta)$. Finally $p(\eta_k)^2 \ge b_0^2 (1 + \alpha k \ln 5)^2 \ge c_1 k^2$ for $p \ge 1$, so $\rho_k \le e^{-c c_1 k^2 + C}$; thus $r_k(u_x) \le 1$ for large k, $\sum_k r_k(s) < \infty$, and the tail of $\sum_k \rho_k$ is eventually below any geometric tail. The series in (4) converges with a geometric tail, so $(\ln a_k(x))_k$ is Cauchy, $a_k(x)$ converges, and the limit lies in the closed interval of Lemma 5.6. □

Remark 5.8 (what the argument does not use). (i) No convergence, and no bound, of the vacuum constants $v_{k,j}$ (King's E_0, E_1, the kernel normalisations, the face vacuum energies) enters: they cancel in Step 3. King's printed route [16, p. 329] goes through the convergence of Z_k itself, which requires exactly those constants to converge. (ii) No upper stability bound, no lower bound on $Z_k(0)$ and no bound on $Z_k(u_x)$ beyond Lemma 5.6 is used; the remainder of (2) is compared with $Z_k(0)$ only. (iii) No reflection positivity, no Ward identity, and no subsequence. (iv) The all-n form of [15, (3.13)] (comparing the k-model with the $(k+n)$-model at a fixed stopping scale η for every n) also gives the Cauchy property, provided the constants of (2) and (3) are taken at fixed η with no growth in k: then $|\ln a_{k+n} - \ln a_k| \le 4b(\eta) + 2\theta_k(\eta) + 4\rho(\eta)(e^{\beta_f |x|^2} + 1)$ for every n, and one lets $k \to \infty$ and then $\eta \to 0$. The diagonal form above uses $n = 1$ only, at the price of constants uniform in j; the two forms need different readings of the inputs, and neither implies the other.

Theorem 5.9 (King's existence theorem on the cube: the curvature field)**.** *For every finite family $f_1, \dots, f_N$ of real 2-forms in $C_c^\infty(\mathbb{R}^3; \Lambda^2)$, every $J \ge J_0(f)$ and every $x \in \mathbb{R}^N$ the limit*

$$Z_J^K(x) = \lim_{k \to \infty} \langle e^{\sum_i x_i F(f_i)} \rangle^K_{k,J}$$

exists and lies in $[e^{-\beta_f|x|^2}, e^{\beta_f|x|^2}] \subset (0,\infty)$*, with* β_f *as in Lemma 5.6; the convergence is geometric. That is, Theorem 3.3 holds. Consequently* Z_J^K *and* Z_∞^K *exist locally uniformly on* $\mathbb{C}^N$ *and are entire, with the bounds and the monotonicity of Theorem 7.6(b).*

Proof. Propositions 5.4 and 5.5 supply the hypotheses of Theorem 5.7 at every x; the theorem gives the limit and its bounds; Theorem 7.6(b) (that is, Corollary 4.4(ii)) gives the rest. The constants k_0, η_1 and the rates may depend on x and on f; existence needs no uniformity in x. □

Remark 5.10 (the comparison family). Theorem 5.9 says nothing about the fixed-cube limits of the comparison family $\mathrm{ND}^{(a)}$ (Remark 5.3); for that family Corollary 4.4(ii)(γ) squeezes the limits inferior and superior at fixed J between Z_J^K and Z_{J+1}^K, and Corollary 4.4(iii) identifies every subsequential limit, followed by $J\to\infty$, with Z_∞^K. At fixed J every inequality available for the comparison family bounds its oscillation in k only by the difference of two distinct limits, which tends to zero only as $J\to\infty$; a Cauchy statement at fixed J for that family would need the small-field analysis at its faces, where no flow of the kind used above exists.

6. King's existence theorem on the cube with scalar sources

This section proves Theorem 3.6, the existence theorem with scalar sources in the form the Wick leg consumes: in the reference scheme z, the fixed-cube limits of the generating functional with curvature and scalar sources exist, locally uniformly and holomorphically, on the domain of coefficient vectors whose positive scalar part is below a threshold, and they are positive at real points; the negative scalar part is unrestricted. The route is the Cauchy argument of Section 5 with the scheme c-number carried along (Theorem 6.14), applied at real points with a small non-negative scalar source, followed by a normal-family argument on the domain. The domain is bounded by Hölder's inequality from three one-sided estimates: the curvature bound of Lemma 5.6, a positive-source bound (Proposition 6.10), and a negative-source bound (Theorem 6.16), the last obtained from positive-source statements at a heavier scalar mass by an exact mass shift and a correlation inequality. The notation is that of Section 5.1.

6.1. **Setting; the source-dependent graphs.** Throughout, the scheme is the reference scheme $z = a^\sharp$ of Definition 2.4; write z_k for its k-th term. Put $\mathsf{S}_k(h) := \sum_{x\in\Lambda_k\cap C_J} \varepsilon_k^3 h(x)$ and $c_k(h) := z_k \mathsf{S}_k(h)$, which is linear in h. For a real source $s=(u,h)$ the normalised functional in the scheme is

$$n_k(s) \;:=\; e^{-c_k(h)}\, N_k(s) \;=\; e^{-z_k \mathsf{S}_k(h)}\, \langle e^{F(u)+:\phi^2:(h)}\rangle_{k,J}^K \;=\; Z_{k,J}^{z,K}(s),$$

the object of Theorem 3.6. For a finite family $f_1,\dots,f_n$ of real 2-forms and $h_1,\dots,h_p$ of real functions, all supported at ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_J$, and $\zeta=(z,w)\in\mathbb{C}^{n+p}$, $n_k(\zeta) := e^{-\sum_j w_j c_k(h_j)} \langle e^{\sum_i z_i F(f_i)+\sum_j w_j:\phi^2:(h_j)}\rangle_{k,J}^K$ (here z_i are coefficients and z_k the scheme term; no confusion will arise). Every n_k is entire in ζ, $|n_k(\zeta)|\le n_k(\operatorname{Re}\zeta)$, and $\ln n_k$ is convex on real points (Lemma 2.12(b)): a log-Laplace transform minus a linear function.

A second scalar mass $M>0$ will be used with the *same* e, λ, μ_0, lattice, cube and boundary conditions; $\langle\cdot\rangle^{(M)}$ denotes King's cube state at mass M, $\alpha_k(M) := M^2+\delta m^2_{\varepsilon_k}(M)$ King's free-boundary bare coefficient at mass M (as in Lemma 3.10), $u_k(M) := \frac12(\alpha_k(M)-\alpha_k(m))$, and $s_-(M) := \liminf_k u_k(M)$. Every statement of Section 5 and of this section holds at every point of King's range with point-dependent constants, in particular at the point (e,λ,M,μ_0); we use this at $M=m$ and at one heavier M_2.

Lemma 6.1 ((K_h): one-step convergence of the source-dependent graphs on King's cube)**.** *Let* $P_k^{(j),1}$ *be the interaction part of King's effective action after* j *steps on* C_J^K [15, (3.17)]*, written as a sum over connected renormalised graphs* H*, and let* $\mathcal{N}_h$ *be the family of its members that depend on the scalar source* h*: the field-dependent part of every graph with an* h*-vertex, and every field-independent* h*-dependent graph except the* h*-linear ones of the three lowest orders* $(0,0)$*,* $(0,1)$*,* $(2,0)$ *in* (e^2,λ)*. Then for every fixed real* $h\in C_c^\infty$ *supported at distance* >1 *from* $\mathbb{R}^3\setminus C_J$ *and every member of* $\mathcal{N}_h$*, for* $J_*\le j\le k-J_*$*: the localisation bound* [15, (3.38)]*,*

with the factor $(1+k-j)^{p_0}$ *(King's power of* $\ln(L^k\varepsilon)^{-1}$*, [15, p. 666], a choice of his* n_1*), holds for its localisations at distance from the faces larger than the small-field scale; the bound* [15, (3.38)] *with the restricted degree holds with the factor* $(1+k-j)^{p_0}$*, summed over the localisations sharing the cube of an h-vertex, and for each single localisation at distance* >1 *from the faces. Moreover, there are exponents* $\gamma',\beta',\sigma'>0$*, depending neither on h nor on the member, such that for every k and every j with* $J_*\le j\le k-J_*$ *the sum over positions of the difference between the member at* (k,j) *and the member at* $(k+1,j+1)$*, on the common unit lattice and on the support of* χ_j*, is bounded by* $C\big[5^{-\gamma' j}\eta^{-\beta'}+\eta^{\sigma'}\big]\,|C_J|$*,* $\eta=5^{j-k}$*, with C independent of j and k.*

This is the cube form, on C_J^K with King's Dirichlet weight, of the corresponding statement on King's tori; the h-free graphs, those with a curvature vertex and no h-vertex and the source-free ones, are covered on the cube by Proposition 5.5. The localisations of a member of $\mathcal{N}_h$ with tree length $R\le p$ lie at distance >1 from the faces, where the bounds reduce to those of the free flow on the torus (Proposition 6.4); a localisation with $R>p$ carries the factor $e^{-\delta R}$ once the localised graph is cut into clusters of mutual distance $>3\sqrt{d}$, each reduced in its own box of side $O(p)$, the clusters containing the source vertex lying away from the faces and the clusters meeting the Dirichlet faces being source-free (Proposition 6.5). The third clause holds at every step of the range; Lemma D.3 of the full version uses it along the diagonal pairs of Section 5.1, in the proof of Proposition 6.7. The proof is given in Section 6.2.

6.2. **Proof of Lemma 6.1.** We work in the units of depth j of the proof of Proposition 5.5 in the full version: the live sites are T_η, F denotes the scalar Dirichlet planes, $t(x):=\operatorname{dist}(x,F)$, and a localised graph H carries King's boxes $\square\subset\square''\subset\square'$ and $\Omega=\square'\cap T_\eta$, in zone I if $\square'\cap F=\emptyset$ and in zone B otherwise. The localisation is by centres, as in the proof of Proposition 5.5: a tuple τ has the point set $P(\tau)$, $N:=|P(\tau)|\le N_*$ with N_* depending on the order $\bar n$ of the expansion only, and the tree length $R=R(\tau):=\operatorname{dist}(P(\tau))$. The cut by clusters is that of Lemma B.36 of the full version: $c_0:=\frac{13}{3}\sqrt{3}$, $R_0:=3\sqrt{3}$, the clusters K are the components of the c_0-proximity graph on $P(\tau)$, every line whose ends lie in different clusters is cut and kept whole as an external line of both cluster graphs H_K, and H_K is reduced with its own boxes $\square_K\subset\square''_K\subset\square'_K$ and gauge $B_K=A_j(x_0^K)$; put $\Omega_K:=\square'_K\cap T_\eta$. A cluster is in zone I if $\Omega_K\cap F=\emptyset$ (then $\square'_K$ is not truncated and $\Omega_K=\square'_K$) and in zone B otherwise. The two members of the pair, the k-model at depth j and the $(k+1)$-model at depth $j+1$, live on the same unit lattice; every bound below holds for each of them separately with the same constants, and the two are compared only in zone I.

The h-vertex is King's vertex [15, (3.21)]: the smooth function h sampled at the points of the lattice's own frame, the coupling $\theta\eta^2$ in the units of depth j, and at most two fields $\phi^{(j)}$. In King's count of the coupling power $\rho(H)=\sum_v\rho_v$ of a graph [15, p. 665], every vertex has $\rho_v-s_v/4\ge 0$, s_v its number of external scalar fields, and the h-vertex has $\rho_h-s_h/4\ge 2-\frac12=\frac32$; hence

$$\rho(H)-\tfrac14 s(H)\ \ge\ \tfrac32\qquad\text{for every member of }\mathcal{N}_h\text{ with external fields,}\tag{5}$$

$s(H)$ its number of external scalar legs, and $\rho(H)\ge 0$ for the field-independent members. The Wick constant of [15, (3.21)] is field-free and h-linear of the lowest order, so it enters only the excluded family $\mathcal{T}_h$ of Definition D.1 of the full version and no member of $\mathcal{N}_h$ depends on it.

The range. Let $\eta_2\le\eta_1$ be so small that for $\eta\le\eta_2$, with $p=p(\eta)$ and M the block integer of C_J,

$$\eta^{-1}\ \ge\ 4p+8M+2c_0N_*^2+8,\qquad p\ \ge\ c_0N_*^2,\qquad C_\flat\, e\,\eta^{1/2}\,p^2\ \le\ \tfrac12,\tag{6}$$

where $C_\flat$ is the constant C_{box} of Lemma 6.2(e) below times $\sqrt{3}$ and the constants of the difference bound [15, (3.45)] and of the operator bound in Lemma K.27(b) of the full version. All three hold for η small because $p(\eta)$ grows slower than every power of η^{-1} and faster than every logarithm. The first condition contains the corresponding condition of that proof. Since $C_{\mathrm{box}}p$ bounds the side of every cluster box $\square'_K$ and also the side $\le 5p+4M+2$ of the one box $\square'$ of a localisation with $R\le p$ (Lemma 6.2(e)), the third condition contains the convergence condition $Ce\eta^{1/2}p\sqrt{3}\,(5p+4M+2)\le\frac12$ of the expansion in $\tilde A$ on that one box (Lemma K.27

of the full version with $\Omega_K \to \Omega$), as well as the one on every cluster box. J_* is taken so large that $\eta = 5^{-(k-j)} \le \eta_2$ and $C_{\mathrm{PW}} e \eta^{1/2} p(\eta) \le \theta_{\mathrm{PW}}$ (Lemma K.25 of the full version) whenever $k-j \ge J_*$; the least such integer is the largest of the thresholds of the two geometric conditions of (6), of the convergence conditions of the expansions in the cluster fields and on the one box, and of the smallness of the bounds at the background.

Lemma 6.2 (the source vertices and the clusters)**.** *Let $\eta \le \eta_2$, let H be a member of $\mathcal{N}_h$, s an h-vertex of H with centre z_s, and τ a tuple of H. Then:* (a) $t(z_s) \ge \eta^{-1} - 3$. (b) *If τ is in zone B, then $R(\tau) \ge \eta^{-1}/2$.* (c) *If τ is in zone B, then τ has at least two clusters.* (d) *Every cluster K containing an h-vertex has $\operatorname{dist}(\square_K, F) \ge 2p + M + 1$, $\Omega_K = \square'_K$ and $\operatorname{dist}(\square_K, \partial\Omega_K) \ge 2p$: it is a zone-I cluster. Hence every zone-B cluster is source-free.* (e) *Every $\square'_K$ has side $\le c_0 N_* + 4p + 6M$, and the one box $\square'$ of a localisation with $R \le p$ has side $\le R + 4p + 4M + 2 \le 5p + 4M + 2$; both are $\le C_{\mathrm{box}}\, p$, where $C_{\mathrm{box}} := \max(c_0 N_* + 6M + 4,\, 4M + 7)$. The same holds with a curvature vertex in place of the h-vertex.*

Proof. (a) $h(x) \ne 0$ forces ℓ^∞-$\operatorname{dist}(x, \mathbb{R}^3 \setminus C_J) > 1$ in physical units, that is $> \eta^{-1}$ unit lengths; the planes F lie within one fine spacing $5^{-j} \le 1$ of ∂C_J, and the vertex lies in $\operatorname{supp} \chi_{z_s}$, within $\frac{2}{3}\sqrt{3} < 2$ of its centre. (b) A tree of total length R joins the points of $P(\tau)$, so all of them lie within R of z_s; the vertices lie within $\frac{2}{3}$ of these points in each coordinate, so $\square$ lies in the sup-norm cube of half-side $R+M+1$ about z_s, and $\square'$, which adds at most $2p+M$ on each face (as in (e)), in that of half-side $R+2p+2M+1 \le R+2p+3M$. If $\square'$ meets F, then $t(z_s) \le R+2p+3M$, and with (a) and (6) $R \ge \eta^{-1} - 2p - 3M - 3 \ge \eta^{-1}/2$. (c) With one cluster $\operatorname{diam} P(\tau) \le c_0(N-1)$ (Lemma B.36(c) of the full version), so $R(\tau) \le (N-1) \operatorname{diam} P(\tau) \le c_0 (N_* - 1)^2 < \eta^{-1}/2$ by (6), against (b). (d) If $s \in K$, every point of P_K lies within $c_0(N-1)$ of z_s, so $\square_K$ lies in the cube of half-side $c_0 N + M$ about z_s and $\square'_K$ in that of half-side $c_0 N + 2p + 3M$. By (a) and (6), $t(z_s) - (c_0 N + 2p + 3M) \ge \eta^{-1} - 3 - c_0 N - 2p - 3M > 2p + M + 1$. So $\square'_K$ misses F, it is not truncated, it contains every large block within $2p$ of $\square_K$, and $\operatorname{dist}(\square_K, F) \ge t(z_s) - c_0 N - M \ge 2p + M + 1$. (e) $\square_K$ has side $\le \operatorname{diam} P_K + \frac{4}{3} + 2M \le c_0 N_* + 2M$ (the vertices lie within $\frac{2}{3}$ of the points in each coordinate, $\operatorname{diam} P_K \le c_0(N_* - 1)$ by Lemma B.36(c) of the full version, and $c_0 > \frac{4}{3}$) and $\square'_K$ adds at most $2p + M \le 2p + 2M$ per side, as for the one box below; truncation at ∂T_η only shrinks it. For the one box: the points of $P(\tau)$ lie within R of one another, and every vertex lies in the support of its localisation weight, within $\frac{2}{3}$ of its centre in each coordinate; so $\square$ has side $\le R + \frac{4}{3} + 2M \le R + 2M + 2$, and $\square'$ adds on each face at most the large blocks within $2p$ of $\square$, that is at most $2p + M$; hence its side is $\le R + 4p + 4M + 2$. Since $p \ge 1$, $c_0 N_* + 4p + 6M \le (c_0 N_* + 6M + 4)p$ and $5p + 4M + 2 \le (4M + 7)p$. The curvature source f is supported at the same distance from $\mathbb{R}^3 \setminus C_J$, so the same arithmetic applies to its vertex. $\square$

Lemma 6.3 (cut lines vanish on the Dirichlet planes)**.** *Let ℓ be a scalar, dressed or external cut line of Lemma B.36 of the full version with its end u in a cluster K, in either model, at the background $A^{(j)}$ or at zero background. Put $m(u') := \min(1, t(u'))$ for a scalar or an external line and $m(u') := m_\alpha(u'; 1) = \min(1, t(u'))^\alpha$ for a dressed line (Lemma B.58 of the full version); in every case $m(u') \le m_\alpha(u'; 1)$.* (a) *For every evaluation point u' of the u-end with $t(u') \le \frac{1}{4}$ and every operation O_v at the far end (a difference or Hölder quotient of order $\le 1 + \alpha$, or the identity) whose evaluation set E_v lies at distance $\ge \frac{1}{2}$ from u',*

$$|O_v \ell(u', \cdot)| \;\le\; C\, m(u')\, e^{-\delta \operatorname{dist}(u', E_v)}.$$

(b) *For a scalar or dressed cut line and evaluation points u', v' of its two ends, both undifferentiated, with $|u' - v'| \ge 1$ and $t(u'), t(v') \le \frac{1}{4}$,*

$$|\ell(u', v')| \;\le\; C\, m(u') \min(1, t(v'))\, e^{-\delta |u' - v'|} \;\le\; C\, m_\alpha(u'; 1)\, m_\alpha(v'; 1)\, e^{-\delta |u' - v'|}.$$

The constants depend on those of Lemma B.37(b) and, for dressed lines, also on those of Lemma B.58(i), Lemma K.25 and (123) *of the full version.*

Proof. Each of the three kinds of line vanishes when its u-end lies on F: a scalar line by the Dirichlet condition of the cube propagator, a dressed line and an external line through their first factor, which is such a propagator.

(a) for scalar and external lines. Let $u'_F \in F$ be the foot of u' and telescope along the $t(u')/5^{-j}$ bonds of the normal segment from u'_F to u'; the parallel transporters along a bond are unimodular, so $|O_v\ell(u',\cdot)| \le t(u')\max_b |O_v D_b\ell(\cdot,\cdot)|$, the maximum over the bond differences D_b on that segment. Every such bond lies within $\frac14$ of u', hence at distance $\ge \frac14$ from E_v, and the mixed bound of Lemma B.37(b) of the full version ($C^{1,\alpha}\otimes C^{1,\alpha}$ at unit scale in both arguments, decay $e^{-\delta_1|u-v|}$, no distance condition) bounds each by $Ce^{-\delta\,\mathrm{dist}(u',E_v)}$.

(a) for dressed lines. Write the line as $\sum_x \vartheta^3 G_A(u',x)\,c_{\ge j}\,G_A(x,\cdot)$, with ϑ the fine spacing (chains iterated likewise). Its first factor is not cut off near u', so the face factor is taken from the slices of the first factor at all scales. Each slice $G_{(i)}$ obeys the face clause of Lemma B.58(i) of the full version with $\mathsf{q}=0$, $|G_{(i)}(u',x)| \le C\rho_i^{-1} m_\alpha(u';\rho_i)e^{-\delta|u'-x|/\rho_i}$: its proof uses only the factorisation $G_{(i)} = a_i^2\rho_i^{-4}(G_iQ_i^*)C^{(i)}(Q_iG_i)$ [15, (2.17)], the Hölder clause of the outer factor $G_iQ_i^*$ up to the faces and its vanishing on F, and the exponential decay of the coarse factor $C^{(i)}$. At the background $A^{(j)}$ the outer-factor clauses hold at every level i by [16, (5.2)–(5.3)], the background being regular at level i. The coarse factor $C^{(i)}(A) = K_i(A)^{-1}$ (defined in the full version) is the lower right block of the inverse of the block operator with diagonal entries $M(A)$ and $D_i := a_i + a5^{-2}Q^*Q$ and off-diagonal entries $-a_iQ_i^*$ and $-a_iQ_i$, which is positive, its quadratic form being $\langle g_1, M_0(A)g_1\rangle + a_i\|g_2 - Q_ig_1\|^2 + a5^{-2}\|Qg_2\|^2$. The Schur complement of D_i, with $QQ^*=I$, $QQ_i = Q_{i+1}$ and $a_{i+1}^{-1} = a^{-1} + a_i^{-1}5^{-2}$, gives $C^{(i)} = D_i^{-1} + a_i^2 D_i^{-1}Q_iG'_{i+1}Q_i^*D_i^{-1}$ with $G'_{i+1} := (M_0(A) + a_{i+1}5^{-2}Q_{i+1}^*Q_{i+1})^{-1}$, the propagator of level $i+1$ in the units of level i. Since $D_i^{-1} = a_i^{-1}(I-Q^*Q) + (a_i + a5^{-2})^{-1}Q^*Q$ is block-local and bounded, the decay of $C^{(i)}$ is the outer-factor bound at level $i+1$, applied to the block-local bounded sources $Q_i^*D_i^{-1}\delta_y$. Since $m_\alpha(u';\rho_i) \le \rho_i^{-\alpha} m_\alpha(u';1)$ for $\rho_i \le 1$, the sum over the slices gives

$$|G_A(u',x)| \;\le\; C\,m_\alpha(u';1)\,(|u'-x|+\vartheta)^{2-d-\alpha}\,e^{-\delta|u'-x|/2},$$

a kernel that is locally summable in $d=3$. The sum over x against the constant $c_{\ge j}$ and the last factor $O_vG_A(x,\cdot)$, which (PW1) of Lemma K.25 of the full version bounds, is the convolution of two kernels of the form of (PW1), and (123) of the full version bounds it by $C\,m_\alpha(u';1)\,e^{-\delta\,\mathrm{dist}(u',E_v)}$, since $\mathrm{dist}(u',E_v) \ge \frac12$.

(b). Put $f(w) := \ell^\flat(u',w) = \bar\psi(u'-w)\ell(u',w)$; then $f(v') = \ell(u',v')$, since $|u'-v'| \ge 1$. Let $v'_F \in F$ be the foot of v'; $f(v'_F) = 0$ by the Dirichlet condition in the second variable (for a dressed line through its last factor). Since $|u'-v'| \ge 1$ (on the supports even $|u'-v'| \ge R_0 - 4\cdot 5^{-j} > 1$, Lemma B.37(a) of the full version) and $t(v') \le \frac14$, every point of the segment from v'_F to v' lies at distance $\ge \frac34$ from u'. Telescope along its $t(v')/5^{-j}$ bonds, with unimodular transporters: $|f(v')| \le t(v')\max_b |D_bf|$. On each bond, D_bf is $\bar\psi(u'-\cdot)D_b\ell(u',\cdot)$ plus the difference of the fixed smooth cutoff $\bar\psi$, of size $\le \|\nabla\bar\psi\|_\infty$, times $\ell(u',\cdot)$ at an end of the bond; $\bar\psi$ is differentiated once only. Every point involved lies at distance $\ge \frac34 - 5^{-j} \ge \frac12$ from u' and within $\frac14 + 5^{-j}$ of v', so (a), with $O_v := D_b$ and with O_v the identity, bounds both terms by $C\,m(u')e^{-\delta|u'-v'|}$. Hence $|\ell(u',v')| \le C\,m(u')\min(1,t(v'))e^{-\delta|u'-v'|}$, and the second inequality of (b) is $m(u') \le m_\alpha(u';1)$ and $\min(1,t) \le \min(1,t)^\alpha$. Where $t > \frac14$ at an end, the factor of that end is $\ge \frac14$: if $t(v') > \frac14$, (a) with O_v the identity gives the bound; if $t(u') > \frac14$, the telescoping at v' above, with $D_bf = D_b\ell^\flat(u',\cdot)$ bounded by Lemma B.37(b) of the full version, gives it; if both, Lemma B.37(b) alone. □

Proposition 6.4 (the branch $R \le p$: zone I)**.** *Let H be a member of $\mathcal{N}_h$ and τ a localisation of H in zone I; by Lemma B.23(i) of the full version this covers every localisation with $R \le p$. Then, for each model, on* $\mathrm{supp}\,\chi_j$*,*

$$|E^{(j)}(H;\tau)| \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta R}, \tag{7}$$

which is [15, (3.38)] *for the renormalised family at the tuple, the factor $(1+k-j)^{p_0}$ being King's power of* $\ln(L^k\varepsilon)^{-1}$ [15, p. 666]*, and across the pair (k,j), $(k+1,j+1)$, on the common unit lattice,*

$$|E'^{(j+1)}(H;\tau) - E^{(j)}(H;\tau)| \;\le\; C\,\eta^{\rho(H)-1/2}\,p^{n_1}\big[5^{-\gamma j} + \eta^{3+\sigma+1/2}\big]\,e^{-\delta R}. \tag{8}$$

The constants depend on the point of King's range, J, $\bar{n}$, $\|h\|_{C^1}$ and, for members with curvature vertices, $\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$*; not on j, k, τ.*

Proof. The complete proof is given in the full version, where the statement is restated as Proposition K.36. □

Proposition 6.5 (the branch $R > p$: the cut by clusters)**.** *Let $\eta \le \eta_2$, let H be a member of $\mathcal{N}_h$ and τ a tuple of H with at least two clusters. By Lemma 6.2(c) this covers every zone-B tuple, and by Lemma B.36(c) of the full version and (6) every tuple with $R > p$. Then, for each model separately, on* $\operatorname{supp}\chi_j$*,*

$$|E_N^{(j)}(H;\tau)| \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta\,\operatorname{dist}(P(\tau))}, \tag{9}$$

E_N the field-dependent part (the whole value for a field-independent member); this is [15, (3.38)] *at the tuple, with the same logarithm. Moreover, for every unit cube y_* and $0 < \delta' \le \delta/2$,*

$$\sum_{\substack{\tau \text{ in zone } B\\ \text{with an } h\text{-vertex in } y_*}} |E_N^{(j)}(H;\tau)|\,e^{\delta' R(\tau)} \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta/(8\eta)}. \tag{10}$$

The constants depend on the point of King's range, J, $\bar{n}$, $\|h\|_\infty$ and, for members with curvature vertices, $\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$*.*

Proof. The complete proof is given in the full version; in outline, at a tuple with at least two clusters every connected component of every cluster graph carries a cut-line leg (Corollary B.40(c) of the full version), so the vacuum members and the component-wise cancellation of [15, p. 667] do not arise, and $E(H;A^{(j)};\tau)$ and $E(H;0;\tau)$ are bounded separately; at zero background the cut lines are the cube propagators at $A = 0$ and $B_K = 0$. Fix one of the two terms.

Step 0: regularised cut lines. Replace every cut line ℓ by $\ell^\flat$; by Lemma B.37(a) of the full version this changes nothing.

Step 1: reduction per cluster. Each H_K is reduced by [15, (3.43)–(3.55)] with its own box Ω_K, centre x_0^K, gauge B_K and field $\tilde{A}_K = A^{(j)} - B_K$ on Ω_K (Lemma B.36 of the full version), its cut lines keeping the background and receiving the unitary factors [15, (3.46)] at their ends. The expansion of the reduced propagator in $\tilde{A}_K$ converges on Ω_K: $e_j \sup_{\Omega_K}|\tilde{A}_K| \le Ce_j p \operatorname{diam}\Omega_K \le C_\flat e\eta^{1/2}p^2 \le \frac{1}{2}$ by the difference bound [15, (3.45)], Lemma 6.2(e) and (6): this is the reduction of the branch $R \le p$ on a box of side $O(p)$, run inside each cluster. By Lemma B.36(a) of the full version the partition, the boxes, the gauges and the cut status of every line are the same for every member of every family met in the nesting and for every position, so renormalisation families are formed inside each H_K. A zone-I cluster, in particular every cluster containing a source vertex (Lemma 6.2(d)), satisfies $\operatorname{dist}(\square_K, \partial\Omega_K) \ge 2p$ and $\Omega_K \cap F = \emptyset$, so Lemma B.24 of the full version applies to H_K: it is its free f_D-flow version plus terms carrying an image line of any prescribed degree, with the cut lines as external legs; the vertices of the expansion in $\tilde{A}_K$ that lie in $\square'_K \setminus \square_K$ are joined to a localised vertex of K by a path of lines one of which is long, and that line supplies a credit of any order, as in step (2) of the proof of Theorem B.33 run inside H_K (Theorem B.39 of the full version). A zone-B cluster is source-free and is rewritten by Definition B.53 of the full version within Ω_K.

Step 2: the cut-line factors. Every evaluation point of a cut-line end in an output term is joined to a localised anchor of its own cluster by at most $C(\bar{n})$ lines of that cluster's rewritten units (Theorem B.39 of the full version), and by Lemma B.37(b) of the full version the phase-dressed cut line satisfies $|O_u O_v \ell^{\flat,\mathrm{ph}}| \le Ce^{-\delta_2 d(K,K')} \times$ (transfer factors on the lines of those paths), for all operations O_u, O_v of order $\le 1+\alpha$ at its ends, and $\le C$ crudely. The quotient graph (clusters as nodes, cut lines as edges) is connected because H is (Lemma B.36(d) of the full version); bound the cut lines of one spanning tree T of it by the first bound and the others by C.

Step 3: factorisation. The output term is bounded by $\prod_K \mathfrak{E}_K \times \prod_{\ell\in T} Ce^{-\delta_2 d(K,K')} \times C^{\#\text{cut lines}}$, where $\mathfrak{E}_K$ depends on the positions of K only; the transfer factors cost one extra decay draw per line of a cluster, within the budget of Theorem B.39 of the full version.

Step 4: the cluster factors. Zone-I clusters. The cluster graph is a free f_D-flow graph with its cut lines as undifferentiated external legs of unit scale (Lemma B.37(b) of the full version).

For the cluster containing the h-vertex, the census is that of Proposition 6.4; since the cluster carries a cut leg, every component has an external leg and the excepted field-independent family does not occur. For the source-free zone-I clusters, the degree count holds at every tuple by Theorems B.27 and B.39, and block points by Lemma B.48 of the full version. The multiscale bound Theorem B.44(a) of the full version with the root held, summed over the root in Ω_K (volume $\le Cp^3$ by Lemma B.36(c) of the full version), gives $\sum_{\text{positions of } K} \mathfrak{E}_K \le C\eta^{\rho_K} p^{n_K}(1+k-j)^{p_{0,K}}$, with ρ_K the coupling power of the vertices of K. *Zone-B clusters.* The cluster is source-free (Lemma 6.2(d)); apply Theorem B.60(a) and Theorem B.67 of the full version with no vertex held to the renormalised family of H_K, localised at P_K with the box Ω_K of side $O(p)$; the hypothesis "at least one external field" is the cut leg. The proof of Theorem B.60 of the full version uses an external scalar leg only in the following ways, each of which a cut line satisfies: through its sup and $(1,\alpha)$-norms on Ω_K up to the faces, in the Taylor and transport steps, the Hölder splits and the ordered induction (Lemma B.37(b) of the full version); and, when the charging rule of Lemma B.59 of the full version charges the leg at a $D' = -1$ remainder vertex v, through the vanishing factor $m_\alpha(v;1)$ of Lemma B.58(ii) of the full version, which a cut line supplies by Lemma 6.3(a); only scalar, dressed and external cut lines are ever charged (Lemma K.46 of the full version, (T2)). A cut line joining two zone-B clusters may be charged at both ends, by the charging rules of the two clusters; then Lemma 6.3(b) supplies both factors. After the borrowing of Lemma B.59 of the full version the cut-line factor left in the output term is $\ell/[m_\beta(u';1)m_\beta(v';1)]$ (or with one factor), an undifferentiated evaluation bounded by $C\, m_{\alpha-\beta}(u';1)m_{\alpha-\beta}(v';1)e^{-\delta\,\mathrm{dist}} \le Ce^{-\delta\,\mathrm{dist}}$ by Lemma 6.3, since its face factors are $\le m_\alpha$ and $m_\alpha/m_\beta = m_{\alpha-\beta} \le 1$, which Steps 2–3 treat like every other cut-line factor. The resulting bound is again $\le C\eta^{\rho_K}p^{n_K}(1+k-j)^{p_{0,K}}$, the logarithm coming from the infrared sums inside Ω_K.

Step 5: the tree. By Lemma B.36(d) of the full version, $\prod_{\ell\in T} e^{-\delta_2 d(K,K')} \le e^{\delta_2 c_0 N} e^{-\delta_2\,\mathrm{dist}(P(\tau))}$.

Step 6: exponents. $\rho(H) = \sum_K \rho_K$, the count being vertex-local (an unlocalised mass-counterterm vertex on a cut dressed line keeps its own coupling power); $n_1 := \sum_K n_K$ and $p_0 := \sum_K p_{0,K}$ plus the dressed-line factor of Lemma B.37(b) of the full version; the number of clusters is $\le N_*$ and $\delta := \delta_2$. This is (9).

The root sum. On the tuples of (10), $R(\tau) \ge \eta^{-1}/2$ by Lemma 6.2(b). The number of tuples of H with a point in y_* and $R \in [n, n+1)$ is $\le C(n+1)^{3N_*}$, and $e^{\delta' R}e^{-\delta R} \le e^{-\delta R/2}$. So the sum is $\le C(1+k-j)^{p_0}\eta^{\rho(H)}p^{n_1}\sum_{n\ge\eta^{-1}/2-1}(n+1)^{3N_*}e^{-\delta n/2} \le C'(1+k-j)^{p_0}\eta^{\rho(H)}p^{n_1}e^{-\delta/(8\eta)}$, since $(n+1)^{3N_*}e^{-\delta n/4} \le C$ and $\eta^{-1} \ge 8$. Members with curvature vertices: the curvature vertex lies in a zone-I cluster (Lemma 6.2(d)), is invariant under the reduction (Lemma B.25(b)) and is bounded through Lemma B.25(c) of the full version. □

Proof of Lemma 6.1. The complete proof, with the two propositions above proved in full, is given in the full version; in outline, for a member H of $\mathcal{N}_h$ and $J_* \le j \le k - J_*$ (so that $\eta \le \eta_2$), it runs as follows. *First clause.* (7) is [15, (3.38)] with the factor $(1+k-j)^{p_0}$ for the localisations in zone I, which are those at distance from the faces larger than the small-field scale. *Second clause.* A tuple with one cluster has $R \le c_0(N_*-1)^2 \le p$, lies in zone I by Lemma B.23(i) of the full version, and obeys (7); a tuple with at least two clusters obeys (9). Summed over the localisations sharing the cube y_* of an h-vertex: the zone-I tuples by $\sum_n (n+1)^{3N_*}e^{-(\delta-\delta')n} \le C$, the zone-B tuples by (10). This is [15, (3.38)] with the factor $(1+k-j)^{p_0}$. *Third clause.* Here $J_* \le j \le k - J_*$ and $\eta = 5^{j-k}$. Split the localisations at $R = p$. Those with $R \le p$ are in zone I and are reduced on one box of side $\le 5p + 4M + 2$; there (8) is the comparison of [15, pp. 663–665] made element by element, in which the only element depending on h is the coefficient of the h-vertex, replaced by sampling at the cost $C\varepsilon_k\|\nabla h\|_\infty$, the points of the block products being never source vertices. Sum (8) by Lemma B.51 of the full version (branch $R \le p$): the result is $\le C\eta^{\rho(H)-1/2}p^{n_1}(1+k-j)^{p_0}\eta^{-3}|C_J|[5^{-\gamma j} + \eta^{3+\sigma+1/2}]$, whose first term is $\le C5^{-\gamma' j}\eta^{-\beta'}|C_J|$ for every $\gamma' \le \gamma$ and $\beta' > \frac{7}{2} - \rho(H)$, and whose second term is $\le C\eta^{\rho(H)+\sigma}p^{n_1}(1+k-j)^{p_0}|C_J| \le C\eta^{\sigma'}|C_J|$ for every $\sigma' < \sigma$, since $p^{n_1}(1+k-j)^{p_0}$ is bounded by a constant multiple of a power of $1 + \ln\eta^{-1}$, hence $\le C_\epsilon\eta^{-\epsilon}$ for every $\epsilon > 0$ (Lemma B.51 of the full version). Those with $R > p$ have at least two clusters; bound the two

members of the pair separately by (9), with the roots at the h-vertex, $\le C\eta^{-3}|C_J|$ of them. For the zone-I tuples among them the factor $e^{-(\delta-\delta')p/2}$ of Lemma B.51 of the full version (branch $R > p$) is $\le C_N\eta^N$ for every N, in particular $\le C\eta^{3+\sigma}$, since the exponent of the logarithm in $p(\eta)$ is $6 \ge 2$, and the sum is $\le C\eta^{-3}|C_J|\eta^{\rho(H)}p^{n_1}(1+k-j)^{p_0}\eta^{3+\sigma} \le C\eta^{\sigma'}|C_J|$; for the zone-B tuples (10) gives $\le C\eta^{-3}|C_J|\eta^{\rho(H)}p^{n_1}(1+k-j)^{p_0}e^{-\delta/(8\eta)} \le C_N\eta^N|C_J|$ for every N. In both cases $\rho(H) \ge 0$, and the external fields on $\operatorname{supp}\chi_j$ are accounted for in $\rho(H)$ by (5). Adding the three parts gives the third clause, with C independent of j and k. The constants depend on the point of King's range, so the lemma holds at every point of the range, in particular at the second mass M_2 of Section 6.1. □

6.3. **The inputs with scalar sources.** For a source $s = (u, h)$ put $\mathsf{S}(s) := \sum_i |x_i| \max_{\mu\nu}\|f_{i,\mu\nu}\|_{C^{2,\alpha}} + \|h\|_\infty$, a size; a compact family of sources is contained in $\{\mathsf{S}(s) \le S_0\}$.

Proposition 6.6 ((U^+): the small-field lines at a non-negative scalar source)**.** *Fix $J \ge J_0$, $\alpha \in (0, \frac{1}{2})$ and $S_0 > 0$. There are k_0 and $C, \sigma > 0$, depending on the couplings, J, α, S_0 and on nothing else, such that for every real source $s = (u, h)$ with $h \ge 0$ and $\mathsf{S}(s) \le S_0$, every $k \ge k_0$ and both members $(k', j') \in \{(k, j(k)), (k+1, j(k)+1)\}$ of the diagonal pair,*

$$Z_{k'}(s) \;\ge\; e^{-C\eta_k^\sigma}\, I_{k'}^{(j')}(s) \qquad \text{and} \qquad \int P\, d\mu_{k',s} \;\le\; e^{C\eta_k^\sigma}\, I_{k'}^{(j')}(s). \tag{11}$$

The proof is given in the full version. These are the two small-field lines of (2) with the scalar source inserted into the small-field analysis through the vertex [15, (3.21)]; the large-field remainder at $s \ne 0$ is not part of the statement, and is supplied instead by Lemma 6.12. The source enters the small-field steps only through its vertex, and the sign $h \ge 0$ is used only through the sign of the Wick-ordered source term in the large-field region, which is not examined here.

Proposition 6.7 ((B^h): one-step bound with the scheme c-number)**.** *Fix $J \ge J_0$ and a compact family Σ of real sources. For $\sigma \in \Sigma \cup \{0\}$ and h_σ its scalar part, there is a sequence $\theta_k(\sigma) \le \theta_k$, with θ_k independent of $\sigma \in \Sigma$ and decaying geometrically, such that for $k \ge k_0$, on the support of $\chi_{j(k)}$,*

$$\big|S_{N,k}^{(j(k))}(\cdot;\sigma) + c_k(h_\sigma) - S_{N,k+1}^{(j(k)+1)}(\cdot;\sigma) - c_{k+1}(h_\sigma)\big| \;\le\; \theta_k(\sigma). \tag{12}$$

The proof is given in the full version. The field-dependent and the h-nonlinear parts are Proposition 5.5, the graphs with curvature vertices at $s = (u, h)$, and (K_h). The field-independent h-linear part of the three lowest orders is a c-number $\bar\beta_k^{(j)} \int h$ up to a localisation error, and $c_k(h)$ is exactly the c-number that makes $\bar\beta_k^{(j)} + z_k$ converge along the diagonal pairs at the rate $5^{-\gamma' j}\eta^{-\beta'} + \eta^{\sigma'}$; this is the sense in which z is the scheme adapted to King's flow. At $\sigma = 0$, (12) is (3) with $j = j(k)$.

Lemma 6.8 (c-numbers)**.** (a) $|z_k| \le C(1+k)$. (b) *For every $M > 0$, $\Delta_k(M) := 2(\mathsf{W}_{\varepsilon_k}(m) - \mathsf{W}_{\varepsilon_k}(M)) + z_k(m) - z_k(M)$ is bounded in k, where $\mathsf{W}_\varepsilon(M) = (-\Delta_{\varepsilon\mathbb{Z}^3} + M^2)^{-1}(0,0)$ and $z_k(M)$ is the reference scheme at mass M.*

The proof is given in the full version. Both are statements about lattice Feynman integrals: (a) is the logarithmic growth of the order-e^2 coefficient of $\langle|\phi|^2\rangle$ with the counterterm inserted; in (b) the Wick difference is a convergent lattice integral in three dimensions, and the scheme difference converges.

Proposition 6.9 ((NEG_∞): the bare coefficient grows with the mass)**.** *For every point of King's range ($e, \lambda \ge 0$, $m, \mu_0 > 0$) and every $M > m$, $u_\infty(M) := \lim_k u_k(M)$ exists and*

$$2u_\infty(M) \;\ge\; (M-m)\Big(M + m + \frac{4\lambda}{\pi} - \frac{e^2}{4\pi}\Big) - \Big(\frac{e^4}{4\pi^2} + \frac{8\lambda^2}{\pi^2}\Big)\ln\frac{M}{m};$$

in particular $s_-(M) = u_\infty(M) \to \infty$ as $M \to \infty$, and $s_-(M_2) \ge \frac{1}{2}$ for $M_2 := \max\{m,\ \frac{e^2}{4\pi} + (\frac{e^4}{4\pi^2} + \frac{8\lambda^2}{\pi^2})/m\} + 1$.

The proof is given in the full version. The content is the exact continuum limit of King's free-boundary counterterm at all five of its orders and the k-uniform bounds on its derivatives in the mass; only $\sup_M s_-(M) = \infty$ is used below.

Proposition 6.10 ((W$^\sharp$)[s_W]: the positive-source bound)**.** *Assume* (I2) *and* (I3) *(Propositions 12.2 and 12.6;* (I2) *follows from* (T)*). Then with $s_W := s_1$ (the amplitude of Proposition 13.2) and $\kappa = C_U/s_1$: for every k, J and every lattice function $0 \le q \le s_W$ on X_J,*

$$\ln n_k(0,q) \;\le\; \kappa\,\|q\|_{1,k}.$$

In particular $\sup_k n_k(0,q) < \infty$ for every $0 \le q \in C_c^\infty$ with $\sup q \le s_W$ and $\operatorname{supp} q$ at distance > 1 from $\mathbb{R}^3 \setminus C_J$.

Proof. $e^{\sum_x \varepsilon^3 q(x) r_x^2}$ is a class-$\mathcal{I}$ observable, so by Step 1 of Lemma 4.2 (killing the matter on T_J does not increase it; Remark 4.8) $n_k(0,q) \le N^z_{k,C_J}(0,q)$, the comparison-cube functional, which Proposition 13.2 at $X = X_J$ bounds by $e^{\kappa\|q\|_{1,k}}$; the scheme multiplier is common (Lemma 2.12(a)). For $q \in C_c^\infty$, $\|q\|_{1,k} \le C_0'|q|_w$ by Lemma 2.7(E3). □

Any other threshold with the same property serves below in place of s_1; the full version contains a second instance, obtained from Bałaban's torus bounds through the comparison of the cube with King's tori.

6.4. **Three lemmas on positive and negative scalar sources.** For a mass M let $Y_k^{(M)}(u,h) := e^{-c_k(h)}\langle e^{F(u)+:\phi^2:(h)}\rangle^{(M)}$ denote the functional of the mass-M state *in the conventions of the mass* m (free Wick constant $\mathsf{W}_{\varepsilon_k}(m)$ and scheme $z_k = z_k(m)$), so that $Y_k^{(m)} = n_k$ and $Y_k^{(M)}(u,h) = e^{-\Delta_k(M)\mathsf{S}_k(h)} n_k^{(M)}(u,h)$, with $n_k^{(M)}$ the functional in the mass-M conventions and $\Delta_k(M)$ as in Lemma 6.8(b). Write $r^2(q) := \sum_x \varepsilon_k^3 q(x) r_x^2$ over the live sites.

Lemma 6.11 (mass shift and the correlation inequality)**.** *Let $M > 0$, $k \ge 0$, and let $0 \le q$ be a lattice function on the live sites with $q(x) \le u_k(M)$ for all x. Then*

$$n_k(0,-q) \;\le\; \frac{1}{Y_k^{(M)}(0,q)}, \qquad \text{and} \qquad \frac{1}{n_k(0,q)} \;\le\; n_k(0,-q) \;\le\; \frac{1}{Y_k^{(M)}(0,q)}, \quad \text{hence} \quad Y_k^{(M)}(0,q) \le n_k(0,q).$$

Proof. Mass shift. In King's action the scalar mass enters only through $\sum_x \varepsilon^3 \tfrac12 (M^2 + \delta m_\varepsilon^2(M)) r_x^2$: δm_ε^2 is one number per spacing and mass (Definition 2.4), the Dirichlet terms $\tfrac{\varepsilon}{2} r_x^2$ do not depend on M, and the photon does not see M. Hence, with $u := u_k(M)$ and $r^2(1) := \sum_x \varepsilon_k^3 r_x^2$, $e^{-S^{(m)}} = e^{-S^{(M)}} e^{u\, r^2(1)}$, and for every Φ integrable against both measures $\langle\Phi\rangle^{(m)} = \langle \Phi e^{u r^2(1)}\rangle^{(M)} / \langle e^{u r^2(1)}\rangle^{(M)}$; the quartic term makes $e^{\pm r^2(\cdot)}$ integrable. *Correlation inequality.* Put $g_1 := u\mathbf{1} - q \ge 0$ and $g_2 := q \ge 0$. The mass-M state is a BFS measure, and so is the state with its single-site measures multiplied by $e^{t\, r^2(g_1)}$, $t \in [0,1]$ (Definition 2.1 and the paragraph on BFS measures following it: the exponential moments stay finite because $\lambda > 0$). By Lemma 2.8(c), multiplying the single-site measures by $e^{-t r^2(g_1)}$ does not increase $\langle e^{r^2(g_2)}\rangle$; read from $t = 1$ down to $t = 0$, $\langle e^{r^2(g_2)}\rangle_{t=1}^{(M)} \ge \langle e^{r^2(g_2)}\rangle_{t=0}^{(M)}$, that is

$$\langle e^{r^2(g_1)} e^{r^2(g_2)}\rangle^{(M)} \;\ge\; \langle e^{r^2(g_1)}\rangle^{(M)} \langle e^{r^2(g_2)}\rangle^{(M)}.$$

Conclusion. With $\Phi = e^{-r^2(q)}$: $\langle e^{-r^2(q)}\rangle^{(m)} = \langle e^{r^2(g_1)}\rangle^{(M)} / \langle e^{r^2(g_1+g_2)}\rangle^{(M)} \le 1/\langle e^{r^2(q)}\rangle^{(M)}$. Multiply both sides by $e^{(2\mathsf{W}_{\varepsilon_k}(m) + z_k)\mathsf{S}_k(q)}$: the left side becomes $n_k(0,-q)$ and the right side $1/Y_k^{(M)}(0,q)$, both in the conventions of the mass m. The left inequality of the second display is Lemma 2.12(c), and the last inequality follows. □

Lemma 6.12 (the large-field remainder at a source, from the one at zero source)**.** *Let σ be a real source and $p > 1$, and assume* (11) *at σ and* (R_0) *of Proposition 5.4 for the member (k,j), with $a := C\eta^\sigma$ and $\rho := \rho(\eta)$. Then*

$$e^{-a}\, I_k^{(j)}(\sigma) \;\le\; Z_k(\sigma) \;\le\; e^{a}\, I_k^{(j)}(\sigma) + \rho^{1-1/p}\, \langle e^{X(p\sigma)}\rangle_k^{1/p}\, Z_k(0).$$

Proof. The lower line is the first inequality of (11). For the upper line, $Z_k(\sigma) = \int P\, d\mu_{k,\sigma} + \int (1-P) e^{X(\sigma)} d\mu_{k,0}$. The first term is at most $e^a I_k^{(j)}(\sigma)$ by (11). For the second, Hölder's inequality for the positive measure $(1-P)d\mu_{k,0}$ with the functions 1 and $e^{X(\sigma)}$, then $1-P \le 1$ in the second factor and $pX(\sigma) = X(p\sigma)$ (linearity), give

$$\int (1-P) e^{X(\sigma)} d\mu_{k,0} \;\le\; \Big[\int (1-P) d\mu_{k,0}\Big]^{1-1/p} \Big[\int e^{X(p\sigma)} d\mu_{k,0}\Big]^{1/p} \;\le\; \big[\rho Z_k(0)\big]^{1-1/p} \big[Z_k(0) \langle e^{X(p\sigma)}\rangle_k\big]^{1/p},$$

using (R_0) for the first bracket. □

At a negative scalar source the same step would need a bound on $\langle e^{X(p\sigma)}\rangle$ at a negative source of larger amplitude, which is what is to be proved; Theorem 6.16 avoids this through the mass shift.

Lemma 6.13 (positivity of the limit at a positive source)**.** *Fix a mass M, a function $0 \le q \in C_c^\infty$ as in Proposition 6.10, and $\bar w \in (0,1]$. Assume that at mass M, in its own conventions, the hypotheses of Theorem 6.14 below hold at $\sigma = (0, wq)$ for every $w \in [0, \bar w]$ and at $\sigma = 0$, with sequences b_k, $\theta_k(wq)$, $r_k(0)$ and $e^{-c_k(wq)} r_k(wq)$ dominated by w-independent summable sequences, for all $k \ge k_0$ with k_0 independent of w. Then $\inf_{k \ge k_0} n_k^{(M)}(0,q) > 0$.*

Proof. Steps 1–4 of the proof of Theorem 6.14 use the hypotheses at $\{wq, 0\}$ only and give the lower half of (15) with w-independent summable $\bar\delta_k$, $\bar\epsilon_k$: $n_{k+1}(wq) \ge (1-\bar\delta_k) n_k(wq) - \bar\epsilon_k$. By induction, $n_k(wq) \ge \prod_{i=k_1}^{k-1} (1-\bar\delta_i)\, n_{k_1}(wq) - \sum_{i \ge k_1} \bar\epsilon_i$ for $k \ge k_1$. Choose $k_1 \ge k_0$ with $\prod_{i \ge k_1} (1-\bar\delta_i) \ge \frac12$ and $\sum_{i \ge k_1} \bar\epsilon_i \le \frac18$. At this fixed k_1, $w \mapsto n_{k_1}(wq)$ is continuous (a finite-dimensional integral) and equals 1 at $w = 0$; choose $w_0 \in (0, \bar w]$ with $n_{k_1}(w_0 q) \ge \frac12$. Then $n_k(w_0 q) \ge \frac18$ for all $k \ge k_1$. Since $\ln n_k(wq)$ is convex in w and vanishes at $w = 0$, $n_k(w_0 q) \le n_k(q)^{w_0}$, so $n_k(q) \ge 8^{-1/w_0}$ for $k \ge k_1$; the finitely many $k_0 \le k < k_1$ have $n_k(q) > 0$. □

6.5. The Cauchy argument with a c-number.

Theorem 6.14 (the Cauchy property with the scheme c-number)**.** *Fix a real source $s = (u, h)$ and let σ range over $\{s, 0\}$. Suppose there are k_0 and non-negative sequences b_k, $r_k(\sigma)$, $\theta_k(\sigma)$ such that for all $k \ge k_0$:*

(U) *for every σ and both members (k', j') of the diagonal pair at k, $e^{-b_k} I_{k'}^{(j')}(\sigma) \le Z_{k'}(\sigma) \le e^{b_k} I_{k'}^{(j')}(\sigma) + r_k(\sigma) Z_{k'}(0)$, with $\sum_k b_k < \infty$, $r_k(0) \le \frac12$, $\sum_k r_k(0) < \infty$ and $\sum_k e^{C_c(h)(k+2)} r_k(s) < \infty$, where $|c_k(h)| \le C_c(h)(1+k)$ (Lemma 6.8(a));*

(B) *on the support of $\chi_{j(k)}$, (12) holds at σ with $\sum_k \theta_k(\sigma) < \infty$.*

Put $\Theta_k := \theta_k(s) + \theta_k(0)$, $\delta_k := (4b_k + \Theta_k + 2r_k(0))\, e^{4b_k + \Theta_k + 2r_k(0)}$ and $\epsilon_k := (e^{-c_k(h)} + e^{-c_{k+1}(h)})\, r_k(s)$. Then for all $k \ge k_0$:

(i) $n_k(s) \le \mathsf{M}(s) := \exp\big(\sum_{i \ge k_0} \delta_i\big)\big(n_{k_0}(s) + \sum_{i \ge k_0} \epsilon_i\big) < \infty$;

(ii) $|n_{k+1}(s) - n_k(s)| \le \delta_k \mathsf{M}(s) + \epsilon_k$, *so $n_k(s)$ converges to a limit $n(s) \in [0, \mathsf{M}(s)]$;*

(iii) *if* (U) *and* (B) *hold at $-s$ as well, $n_k(s) \ge 1/\mathsf{M}(-s)$ and $n(s) \in [1/\mathsf{M}(-s), \mathsf{M}(s)] \subset (0, \infty)$.*

If b_k, Θ_k decay geometrically and $r_k(\sigma) \le e^{-ck^2 + Ck}$, then $|n_k(s) - n(s)| \le C' 5^{-\kappa k}$ for some $\kappa > 0$.

Proof. Fix $k \ge k_0$, $j := j(k)$, and let primes denote the $(k+1)$-model after $j+1$ steps.

Step 1 (one model, (U) *only).* For one member write $Z(\sigma)$, $I(\sigma)$, $b = b_k$, $r(\sigma) = r_k(\sigma)$. From (U) at $\sigma = 0$, $(1 - r(0)) Z(0) \le e^b I(0)$ and $Z(0) \ge e^{-b} I(0)$, so $I(0) > 0$. With $\varrho(\sigma) := I(\sigma)/I(0)$ and $N(\sigma) = Z(\sigma)/Z(0)$,

$$e^{-2b}(1 - r(0))\, \varrho(\sigma) \;\le\; N(\sigma) \;\le\; e^{2b} \varrho(\sigma) + r(\sigma)\,: \tag{13}$$

the upper bound from $Z(\sigma) \le e^b I(\sigma) + r(\sigma) Z(0)$ and $1/Z(0) \le e^b / I(0)$, the lower one from $Z(\sigma) \ge e^{-b} I(\sigma)$ and $1/Z(0) \ge e^{-b}(1 - r(0))/I(0)$. No lower bound on $Z(\sigma)$ for $\sigma \ne 0$ is used: the remainder enters relative to $Z(0)$ only.

Step 2 (the vacuum constant). $\varrho(\sigma) = \hat I(\sigma)/\hat I(0)$, since $I = e^{-v} \hat I$ with one v for all σ; likewise for the primed member.

Step 3 (across the pair, (B) *only).* On the support of χ_j, (B) at $\sigma = s$ reads $S'_N(\cdot;s) = S_N(\cdot;s) + c_k(h) - c_{k+1}(h) + E$ with $|E| \le \theta_k(s)$; the integrals $\hat I(s)$, $\hat I'(s)$ run over the same variables with the same χ_j, so $\hat I'(s) \in e^{c_{k+1}(h)-c_k(h)}[e^{-\theta_k(s)}, e^{\theta_k(s)}]\,\hat I(s)$. At $\sigma = 0$ ($c_k(0) = 0$), $\hat I'(0) \in [e^{-\theta_k(0)}, e^{\theta_k(0)}]\,\hat I(0)$. With $\varrho := \varrho(s)$ for the k-model and ϱ' for the $(k+1)$-model,

$$e^{-c_{k+1}(h)}\varrho' \;\in\; [e^{-\Theta_k}, e^{\Theta_k}]\; e^{-c_k(h)}\varrho. \tag{14}$$

Step 4 (two-sided recursion). Write $n := n_k(s)$, $n' := n_{k+1}(s)$, $r_0 := r_k(0)$, $r_s := r_k(s)$, $\Theta := \Theta_k$. Apply (13) to the primed member, multiply by $e^{-c_{k+1}(h)}$, use (14), and then (13) for the k-model in the forms $\varrho \le e^{2b}(1-r_0)^{-1}N(s)$ and $\varrho \ge e^{-2b}(N(s) - r_s)$:

$$n' \le e^{2b}\,e^{-c_{k+1}}\varrho' + e^{-c_{k+1}}r_s \;\le\; e^{2b+\Theta}e^{-c_k}\varrho + e^{-c_{k+1}}r_s \;\le\; e^{4b+\Theta}(1-r_0)^{-1}\,n + e^{-c_{k+1}}r_s,$$

$$n' \ge e^{-2b}(1-r_0)\,e^{-c_{k+1}}\varrho' \;\ge\; e^{-2b-\Theta}(1-r_0)\,e^{-c_k}\varrho \;\ge\; e^{-4b-\Theta}(1-r_0)\big(n - e^{-c_k}r_s\big).$$

For $t \le \frac12$, $(1-t)^{-1} \le e^{2t}$ and $1 - t \ge e^{-2t}$; with $e^y - 1 \le ye^y$ and $1 - e^{-y} \le y$ this gives $e^{4b+\Theta}(1-r_0)^{-1} - 1 \le \delta_k$ and $1 - e^{-4b-\Theta}(1-r_0) \le \delta_k$, hence

$$(1-\delta_k)\,n_k(s) - \epsilon_k \;\le\; n_{k+1}(s) \;\le\; (1+\delta_k)\,n_k(s) + \epsilon_k. \tag{15}$$

Step 5 (boundedness and convergence). By induction from the upper half of (15), $n_k \le \prod_{i=k_0}^{k-1}(1+\delta_i)\big(n_{k_0} + \sum_{i<k}\epsilon_i\big) \le \mathsf M(s)$. Here $\sum_i \delta_i < \infty$ because b_i, Θ_i, $r_i(0)$ are summable (so bounded, and $\delta_i \le C(4b_i + \Theta_i + 2r_i(0))$), and $\sum_i \epsilon_i \le 2\sum_i e^{C_c(h)(i+2)}r_i(s) < \infty$ by (U) and $e^{-c_i(h)} \le e^{C_c(h)(1+i)}$. Then (15) gives $|n_{k+1} - n_k| \le \delta_k n_k + \epsilon_k \le \delta_k \mathsf M(s) + \epsilon_k$, summable; so $(n_k(s))$ is Cauchy. This is (i), (ii).

Step 6 (positivity). $X(-s) = -X(s)$ and $c_k(-h) = -c_k(h)$, so $n_k(s)n_k(-s) = \langle e^X\rangle\langle e^{-X}\rangle \ge 1$ (Lemma 2.12(c)). Steps 1–5 at $-s$ give $n_k(-s) \le \mathsf M(-s)$, hence $n_k(s) \ge 1/\mathsf M(-s)$, and the limit inherits both bounds. The rate follows by summing the bound of (ii) from k on. □

Remark 6.15 (why the ratio is not taken). At $h = 0$ and $c_k \equiv 0$ this is Theorem 5.7 with Lemma 5.6 replaced by Step 6. The logarithmic form of Theorem 5.7 needs $r_k(\sigma)Z(0)/Z(\sigma)$ small, that is a lower bound on $Z_k(s)$ uniform enough in k; at $h \ne 0$ the only candidates are Lemma 2.12(c), which needs the upper bound at $-s$ the theorem is to produce, and Jensen's inequality, which needs a renormalised one-point bound that is not available. The additive comparison (13) avoids both: the remainder is compared with $Z(0)$, and the lower bound on $n_k(s)$ is produced at the end. Nothing in the proof uses the cube or any relation between $j(k+1)$ and $j(k)+1$.

6.6. **The negative-source bound.**

Theorem 6.16 ((UB$^-$): uniform bound at a negative scalar source)**.** *Assume* (I2) *and* (I3). *Let* $0 \le q \in C_c^\infty(\mathbb R^3)$ *be supported at* ℓ^∞*-distance* > 1 *from* $\mathbb R^3\backslash C_J$. *Then* $\sup_k n_k(0,-q) < \infty$. *Moreover, for every real* 2*-form* u *with the same support condition,* $n_k(u,-q) \le n_k(2u,0)^{1/2}n_k(0,-2q)^{1/2}$, *so* $\sup_k n_k(u,-q) < \infty$.

Proof. By Proposition 6.9 choose $M_2 > m$ with $s_-(M_2) > \sup q$. Propositions 5.4, 6.6 and 6.7 and Lemma 6.8 hold at the point (e, λ, M_2, μ_0) as well; $\Delta := \sup_k |\Delta_k(M_2)| < \infty$.

Step 1. Choose $\bar w \in (0, \frac12]$ with $2\bar w \sup q < s_W$, and $k_1 \ge k_0$ with $u_k(M_2) \ge \sup q$ for $k \ge k_1$ (possible since $\sup q < \liminf_k u_k(M_2)$). Then for $k \ge k_1$ and $w \in [0,\bar w]$, both $q \le u_k(M_2)$ and $2wq \le u_k(M_2)$, so Lemma 6.11 applies at q and at $2wq$.

Step 2 (an a priori bound at M_2*).* For $k \ge k_1$, $w \in [0,\bar w]$: by Lemma 6.11, $Y_k^{(M_2)}(0,2wq) \le n_k(0,2wq)$; by convexity of $t \mapsto \ln n_k(0,tq)$ with value 0 at $t = 0$, $n_k(0,2wq) \le \max\{1, n_k(0,2\bar w q)\}$; and by Proposition 6.10, $n_k(0,2\bar w q) \le e^{\kappa\|2\bar w q\|_{1,k}} \le e^{\kappa C_0'|2\bar w q|_w}$. Hence, in the conventions of the mass M_2,

$$n_k^{(M_2)}(0,2wq) = e^{\Delta_k(M_2)\mathsf S_k(2wq)}\,Y_k^{(M_2)}(0,2wq) \;\le\; e^{\Delta\,\mathsf S_k(2q)}\max\{1, e^{\kappa C_0'|2\bar w q|_w}\} \;=:\; \bar{\mathsf M},$$

uniformly in w and k ($\mathsf S_k(2q) \le C_0'|2q|_w$).

Step 3 (the hypotheses of Lemma 6.13 at M_2*).* The family $\{wq :\ 0 \le w \le \bar w\}$ is compact. At $\sigma = (0,wq)$: (U) holds by Lemma 6.12 at mass M_2 with $p = 2$, from (11) (Proposition 6.6 at

M_2; $wq \geq 0$) and (R_0) (Proposition 5.4 at M_2), with $b_k = C\eta_k^\sigma$ and, in the conventions of M_2, $e^{-c_k(wq)} r_k(wq) = \rho_k^{1/2}\, n_k^{(M_2)}(0, 2wq)^{1/2} \leq \rho_k^{1/2} \bar{\mathsf{M}}^{1/2}$ by Step 2, a w-independent sequence; since $\rho_k \leq e^{-ck^2+C}$ and $e^{c_k(wq)} \leq e^{C_c(1+k)}$ with C_c uniform on the family, $\sum_k e^{C_c(k+2)} r_k(wq) < \infty$ uniformly in w. At $\sigma = 0$, (U) is (2) with $r_k(0) = \rho_k \leq \frac{1}{2}$ for $k \geq k_0$. (B) at $\sigma \in \{wq, 0\}$ is Proposition 6.7 at M_2 on the compact family, with w-independent θ_k.

Step 4 (positivity at M_2). Lemma 6.13 gives $c := \inf_{k \geq k_1} n_k^{(M_2)}(0, q) > 0$, hence $Y_k^{(M_2)}(0, q) \geq e^{-\Delta \mathsf{S}_k(q)} c$ for $k \geq k_1$.

Step 5 (the mass shift). By Lemma 6.11 at q, $n_k(0, -q) \leq 1/Y_k^{(M_2)}(0, q) \leq e^{\Delta \mathsf{S}_k(q)}/c \leq e^{\Delta C_0' |q|_w}/c$ for $k \geq k_1$; the finitely many $k < k_1$ are finite numbers. The mixed bound is Cauchy–Schwarz, $\langle e^{F(u) - :\phi^2:(q)} \rangle \leq \langle e^{2F(u)} \rangle^{1/2} \langle e^{-2:\phi^2:(q)} \rangle^{1/2}$, with the c-numbers distributed ($e^{c_k(q)} = (e^{c_k(2q)})^{1/2}$); the first factor is bounded by Lemma 5.6. □

6.7. **Existence on the cube with scalar sources.**

Theorem 6.17 (King's existence theorem on the cube with scalar sources, reference scheme)**.** *Assume* (I2) *and* (I3) *(so that Proposition 6.10 holds with $s_W = s_1$). Let $f_1, \dots, f_n$ be real 2-forms and $h_1, \dots, h_p \geq 0$ non-negative real functions, all in C_c^∞ and supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, $J \geq J_0$. For $\zeta = (z, w) \in \mathbb{C}^{n+p}$ put $k_1(\zeta) := \sum_j (\operatorname{Re} w_j)_+ h_j$, $k_2(\zeta) := \sum_j (\operatorname{Re} w_j)_- h_j$ (both ≥ 0, smooth, with $\sum_j (\operatorname{Re} w_j) h_j = k_1 - k_2$), and*

$$D_J(s_W) \;:=\; \big\{ \zeta \in \mathbb{C}^{n+p} : \; \sup_x 2k_1(\zeta)(x) < s_W \big\}.$$

Then $D_J(s_W)$ is an open convex tube, containing every ζ with $\operatorname{Re} w_j \leq 0$ for all j and every ζ with w purely imaginary, and

(i) *$n_k \to F_J$ locally uniformly on $D_J(s_W)$, and F_J is holomorphic there;*
(ii) *$F_J(\zeta) \in (0, \infty)$ for every real $\zeta \in D_J(s_W)$.*

Proof. The domain. For each x, $\operatorname{Re} w \mapsto 2k_1(\zeta)(x) = \sum_j 2(\operatorname{Re} w_j)_+ h_j(x)$ is convex, since $h_j(x) \geq 0$ and $t \mapsto t_+$ is convex; so $\operatorname{Re} w \mapsto \sup_x 2k_1(\zeta)(x)$ is a finite convex function on $\mathbb{R}^p$, hence continuous, and $D_J(s_W)$ is the preimage under $\zeta \mapsto \operatorname{Re} w$ of an open convex set: open, convex, connected. Write $D := D_J(s_W)$ and $D_\mathbb{R} := D \cap \mathbb{R}^{n+p}$, an open convex subset of $\mathbb{R}^{n+p}$.

Step 1 (local boundedness). Let $Q \subset D$ be compact. $\operatorname{Re} Q$ is a compact subset of the open set $D_\mathbb{R}$, so it is covered by finitely many closed coordinate boxes $B \subset D_\mathbb{R}$. For $\zeta \in Q$ with $\operatorname{Re} \zeta \in B$, $|n_k(\zeta)| \leq n_k(\operatorname{Re} \zeta) \leq \max_{v \in \mathrm{vert}(B)} n_k(v)$, by Lemma 2.12(b) and the convexity of $\ln n_k$ on B. At a vertex $v = (x, y) \in D_\mathbb{R}$ put $k_1 := k_1(v)$, $k_2 := k_2(v)$; then $X(v) = F(u_x) + :\phi^2:(k_1) - :\phi^2:(k_2)$ and the c-number is $e^{-c_k(k_1) + c_k(k_2)}$. Hölder's inequality with exponents $(4, 2, 4)$, with the c-number distributed as $e^{-c_k(k_1)} = (e^{-c_k(2k_1)})^{1/2}$, $e^{c_k(k_2)} = (e^{-c_k(-4k_2)})^{1/4}$, gives

$$n_k(v) \;\leq\; a_k(4x)^{1/4}\, n_k(0, 2k_1)^{1/2}\, n_k(0, -4k_2)^{1/4}.$$

The three factors are bounded uniformly in k: the first by Lemma 5.6; the second by Proposition 6.10, since $2k_1$ is a non-negative smooth function with $\sup 2k_1 < s_W$ ($v \in D_\mathbb{R}$) and the required support; the third by Theorem 6.16 at $q = 4k_2$. Hence $\sup_k \sup_Q |n_k| < \infty$.

Step 2 (convergence on a real open set). Choose $\delta > 0$ with $4\delta \sup_x \sum_j h_j(x) < s_W$ and put $U := \{\zeta = (x, y) \in \mathbb{R}^{n+p} : \; 0 < y_j < \delta \; \forall j\}$, a non-empty open subset of $\mathbb{R}^{n+p}$ contained in $D_\mathbb{R}$. At $\zeta \in U$, $k_2(\zeta) = 0$ and $s := (u_x, k_1(\zeta))$ has a non-negative scalar part; the sources $\{k_1(\zeta) : \zeta \in \bar{U}\}$ form a compact family. Apply Theorem 6.14(i),(ii) at s. Hypothesis (U) at $\sigma = 0$ is (2) on both members, with $r_k(0) = \rho_k$. Hypothesis (U) at $\sigma = s$ is Lemma 6.12 with $p = 2$, from (11) (Proposition 6.6) and (R_0), with $r_k(s) = \rho_k^{1/2} \langle e^{X(2s)} \rangle_k^{1/2}$; here $\langle e^{X(2s)} \rangle_k = e^{c_k(2k_1)} n_k(2u_x, 2k_1) \leq e^{c_k(2k_1)} a_k(4x)^{1/2} n_k(0, 4k_1)^{1/2}$ by Cauchy–Schwarz, bounded by $e^{C_c(1+k)}$ times a constant (Lemma 5.6; Proposition 6.10, as $\sup 4k_1 < s_W$), so $\sum_k e^{C_c(k+2)} r_k(s) \leq C \sum_k e^{2C_c(k+2)} \rho_k^{1/2} < \infty$ since $\rho_k \leq e^{-ck^2+C}$. Hypothesis (B) at $\sigma \in \{s, 0\}$ is Proposition 6.7. Hence $n_k(\zeta)$ converges for every $\zeta \in U$.

Step 3 (Vitali). By Step 1 the family $\{n_k\}$ is locally bounded on the connected domain D; by Montel's theorem every subsequence has a further subsequence converging locally uniformly to a

holomorphic function; any two such limits agree on U (Step 2), hence on D, since a holomorphic function vanishing on a non-empty open subset of $\mathbb{R}^{n+p}$ has all its derivatives equal to zero at a point of it and vanishes on the connected D. So the whole sequence converges locally uniformly on D to a holomorphic F_J: (i).

Step 4 (positivity at real points). On $D_{\mathbb{R}}$ every n_k is positive and $\ln n_k$ is convex, so $F_J \geq 0$ there and $g := \ln F_J : D_{\mathbb{R}} \to [-\infty, \infty)$ satisfies $g(\lambda a + (1-\lambda)b) \leq \lambda g(a) + (1-\lambda)g(b)$ (the limit of the inequalities for $\ln n_k$; all values are $< \infty$). For $\zeta \in U$ the point $-\zeta$ lies in $D_{\mathbb{R}}$ ($k_1(-\zeta) = 0$), both $n_k(\zeta)$ and $n_k(-\zeta)$ converge by (i), and $n_k(\zeta)n_k(-\zeta) \geq 1$ (Lemma 2.12(c)), so $F_J(\zeta)F_J(-\zeta) \geq 1$ and $g > -\infty$ on U. If $g(t) = -\infty$ for some $t \in D_{\mathbb{R}}$, pick $t' \in U$ and, $D_{\mathbb{R}}$ being open, a point $t'' \in D_{\mathbb{R}}$ beyond t' on the ray from t through t', so that $t' = \lambda t + (1-\lambda)t''$ with $\lambda \in (0,1)$; then $g(t') \leq \lambda g(t) + (1-\lambda)g(t'') = -\infty$, a contradiction. Hence $F_J > 0$ on $D_{\mathbb{R}}$: (ii). □

Corollary 6.18 (King's cube limits with scalar sources in the reference scheme, where they are consumed)**.** *Under the hypotheses of Theorem 6.17:*

(a) *For every finite family of real 2-forms f_i and real functions h_j (of either sign) in C_c^∞ with the support condition, and every real coefficient vector (x, y) with $2\|(\sum_j y_j h_j)_+\|_\infty < s_W$, the limit $\lim_k Z^{z,K}_{k,J}(x, y)$ exists and lies in $(0, \infty)$. In particular Theorem 3.6 for the reference scheme holds at every real point of the tube $\mathcal{D}_{\sigma_+,H}$ with $\sigma_+ = s_1$ (there $2\|k_+\|_\infty < \sigma_+ = s_W$), and at every real coefficient vector whose scalar source $\sum_j y_j h_j$ is non-positive, with no restriction on its amplitude.*

(b) *For $0 \leq q$ as in Theorem 6.16, the comparison-convention functional satisfies $\sup_k N^z_{k,C_J}(0, -q) \leq \sup_k n_k(0, -q) < \infty$.*

Proof. (a) Put $\mathsf{k} := \sum_j y_j h_j$ and let k_+ be its pointwise positive part. Choose $\chi \in C_c^\infty$ with $0 \leq \chi \leq 1$, $\chi = 1$ on a neighbourhood of $\operatorname{supp} \mathsf{k}$ and $\operatorname{supp} \chi$ at distance > 1 from $\mathbb{R}^3 \setminus C_J$, and $\delta > 0$ with $2(\|\mathsf{k}_+\|_\infty + \delta) < s_W$; put $h^+ := (\|\mathsf{k}_+\|_\infty + \delta)\chi$ and $h^- := h^+ - \mathsf{k}$. Both are smooth, non-negative ($h^+ \geq \mathsf{k}_+ \geq \mathsf{k}$ on $\operatorname{supp} \mathsf{k}$, and $h^+ \geq 0 = \mathsf{k}$ off it), compactly supported with the support condition, and $\mathsf{k} = h^+ - h^-$. Then $Z^{z,K}_{k,J}(x, y)$ for the given family equals $n_k(\zeta')$ for the family $(f_1, \dots, f_n; h^+, h^-)$ at $\zeta' = (x; 1, -1)$, for which $k_1(\zeta') = h^+$ and $\sup 2k_1(\zeta') = 2(\|\mathsf{k}_+\|_\infty + \delta) < s_W$; so $\zeta' \in D_J(s_W)$ and Theorem 6.17 applies. (b) $e^{-:\phi^2:(q)}$ is a class-$\mathcal{D}$ observable; killing the matter on T_J does not decrease its expectation (Step 3 of Lemma 4.2, the $\mathcal{D}$ direction), and the scheme multiplier is common. □

Remark 6.19 (other schemes). For a scheme a, $Z^{a,K}_{k,J}(x, y) = e^{-(a_k - z_k)\sum_j y_j \mathsf{S}_k(h_j)} Z^{z,K}_{k,J}(x, y)$. At a coefficient vector with $\sum_j y_j \int h_j \neq 0$ the limit for a therefore exists if and only if $a_k - z_k$ converges: on the domain of Corollary 6.18(a), Theorem 3.6 for a is the convergence of the scheme difference, in agreement with Remark 2.5, and Corollary 6.18(b) holds for a whenever $\sup_k |a_k - z_k| < \infty$.

Remark 6.20 (the cell bound). ($\mathrm{B}_{\mathrm{cell}}$) (Lemma 14.4) is the boundedness of $N^a_{k,C_{L_c}}(0, -2H'\chi_0)$ on one comparison-convention cell. When the cell is a King cube, that is $\ell_c = 2MJ'$ for some J' up to a translation by a lattice vector (Lemma 2.13), and χ_0 is supported at ℓ^∞-distance > 1 from the cell's complement, Corollary 6.18(b) and Remark 6.19 give ($\mathrm{B}_{\mathrm{cell}}$) for every scheme with $\sup_k |a_k - z_k| < \infty$, and Corollary 6.18(a) gives the convergence of the King-convention cell functional $Z^{z,K}_{k,C_{L_c}}(0, -2H'\chi_0)$ as well; for the comparison-convention cell functional only the boundedness is asserted (Remark 5.10).

Remark 6.21 (what is and is not claimed). Beyond the statements of Section 5.2 and Lemma 6.1, the only input of this section is the threshold s_W of Proposition 6.10, supplied by (I2) and (I3). Nothing is claimed at a non-negative scalar source of amplitude $\geq s_W/2$: there the positive-source bound is the content of (W2) above the threshold (Remark 23.2). The negative scalar sources are unrestricted, because the mass shift of Lemma 6.11 turns a negative source of the mass-m model into a positive source of a heavier model, where the small-field analysis applies

with no threshold on the amplitude, and Proposition 6.9 supplies a heavy enough mass for every amplitude.

The axioms

7. Regularity in the curvature sector

This section proves the regularity axiom E0″ for the curvature field, for the comparison convention $\mathrm{ND}^{(a)}$ (Definition 2.1) and for King's convention ND^{K} (Definition 2.2), on which King's printed state Z^{K}_{∞} lives. Gaussian domination (Lemma 2.9) and a bound on the Neumann photon form of the cube (Lemma 7.1) give a bound on every lattice generating functional, uniform in the spacing, the cube and the scalar sector, with absolute constants (Remark 7.5); under the existence hypothesis it passes to the limits by Montel's theorem, and Cauchy's estimates turn it into E0″ with $s = 4$. The section closes with the curvature-sector properties of the fixed-cube limits of the comparison family (locally uniform convergence, monotonicity in the cube, positivity, continuity and translations), proved with cone-direction correlation inequalities only (Lemma 7.7).

Throughout, $f_1, \dots, f_n$ are real 2-forms in $C^{\infty}_{c}(\mathbb{R}^3;\Lambda^2)$ with supports at ℓ^{∞}-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $J \geq J_0$, $z \in \mathbb{C}^n$, and $g(z) := \sum_j (\mathrm{Re}\, z_j) f_j$, so that $\mathrm{Re}\sum_j z_j F(f_j) = F(g(z))$ (Lemma 2.12(b)). For a box X, $\langle A, B\rangle_k$ and $d^*_k g$ are as in (1): $d^*_k g$ is the lattice 1-form with $F_{\Lambda_k}(g) = \langle d^*_k g, A\rangle_k$, defined whenever every plaquette at which $g \neq 0$ has its four corners in X. The support condition implies this for $X = X_J$: the plaquette centre is at ℓ^{∞}-distance > 1 from $\mathbb{R}^3 \setminus C_J$ and the corners are within $\varepsilon/2 \leq \frac{1}{2}$ of it.

7.1. The Neumann cube photon form.

Lemma 7.1 (the Neumann cube photon form)**.** *Let $X = C_J \cap \Lambda_k$, let C be the photon covariance of the cube states on X (the same for $\mathrm{ND}^{(a)}_{k,J}$ and $\mathrm{ND}^{K}_{k,J}$) with respect to $\langle\cdot,\cdot\rangle_k$, let g be a real 2-form on the plaquettes $p = (x;\alpha<\beta)$ all four of whose corners lie in X, and $v := d^*_k g$. Then*

$$\langle v, Cv\rangle_k \;\leq\; \min\big\{\, 2\|g\|^2_{2,k},\ \mu_0^{-2}\|v\|^2_{2,k}\,\big\}.$$

Proof. The Gaussian density of the photon field is $e^{-Q(A)/2}$ with

$$Q(A) = \langle A, C^{-1}A\rangle_k = \sum_{\alpha,\beta}\ \sum_{x,\, x+\varepsilon e_\beta \in X} \varepsilon\,(\partial_\beta A_\alpha(x))^2 + \mu_0^2\|A\|^2_{2,k}, \qquad \partial_\beta A_\alpha(x) := A_\alpha(x+\varepsilon e_\beta) - A_\alpha(x)\,:$$

the Neumann rule keeps exactly the gradient terms of pairs inside X. By Legendre duality, $\langle v, Cv\rangle = \sup_A \big[2\langle v, A\rangle - Q(A)\big]$, attained at $A = Cv$. *Second bound.* $Q(A) \geq \mu_0^2\|A\|^2$, so $C \leq \mu_0^{-2}$ as forms. *First bound.* Let P_g be the set of plaquettes on which $g \neq 0$. For $p = (x;\alpha<\beta) \in P_g$ both gradient terms $(\partial_\alpha A_\beta(x))^2$ (pair $(x, x+\varepsilon e_\alpha)$, component β) and $(\partial_\beta A_\alpha(x))^2$ (pair $(x, x+\varepsilon e_\beta)$, component α) are present in Q, because the pairs lie in X. With $F(p) = \varepsilon^{-1}(\partial_\alpha A_\beta(x) - \partial_\beta A_\alpha(x))$,

$$\|dA\|^2_{P_g} := \sum_{p\in P_g} \varepsilon^3 F(p)^2 = \sum_{p\in P_g} \varepsilon\,(\partial_\alpha A_\beta(x) - \partial_\beta A_\alpha(x))^2 \leq 2\sum_{p\in P_g} \varepsilon\big[(\partial_\alpha A_\beta(x))^2 + (\partial_\beta A_\alpha(x))^2\big] \leq 2Q(A),$$

where the last step is a counting statement: the ordered triple (site x, component β, direction $\alpha \neq \beta$) determines the plaquette $(x;\{\alpha,\beta\})$, so distinct plaquettes use distinct gradient terms of Q and no term is counted twice. Also $\langle v, A\rangle_k = F_{\Lambda_k}(g) = \sum_{p\in P_g} \varepsilon^3 g_p F(p) =: \langle g, dA\rangle_{P_g}$. Hence

$$\langle v, Cv\rangle \leq \sup_A \big[2\langle g, dA\rangle_{P_g} - \tfrac{1}{2}\|dA\|^2_{P_g}\big] \leq \sup_{B\in\mathbb{R}^{P_g}} \big[2\langle g, B\rangle_{P_g} - \tfrac{1}{2}\|B\|^2_{P_g}\big] = 2\|g\|^2_{2,k},$$

the last supremum being attained at $B = 2g$. The inequality $Q(A) - \frac{1}{2}\|dA\|^2_{P_g} \geq 0$ used in the middle is in fact the identity

$$Q(A) - \tfrac{1}{2}\|dA\|^2_{P_g} = \tfrac{1}{2}\sum_{p\in P_g} \varepsilon\,(\partial_\alpha A_\beta(x) + \partial_\beta A_\alpha(x))^2 + \big(\text{gradient terms of } Q \text{ used by no } p \in P_g\big) + \mu_0^2\|A\|^2_{2,k}.$$

□

Remark 7.2 (on the constant 2). On a King torus (Definition 12.1) the Hodge identity of the periodic lattice gives the constant 1 in place of 2 (Lemma L.2 and Corollary L.3 of the full version); on the Neumann cube no Hodge argument is available, and whether 2 is the best constant for the exact quantity $\rho := \sup_g \operatorname{Var} F(g)/\|g\|_{2,k}^2$ (which depends on ε and μ_0 only through $\mu_0\varepsilon$) is not known; nothing below depends on it. The second bound is the cube form of the two-point bound $0 \le S_2(f) \le \mu_0^{-2}\|d^*f\|_{L^2}^2$; it is the worse of the two at small μ_0.

7.2. **The comparison convention.**

Lemma 7.3 (standalone local bound)**.** *For either convention and every compact $K \subset \mathbb{C}^n$,*

$$\sup_{k\ge 0}\ \sup_{J\ge J_0}\ \sup_{z\in K} \left|\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)}\rangle_{k,J}\right| \ \le\ \exp\Big(3C_0 \sup_{z\in K}|g(z)|_4^2\Big) < \infty,$$

with C_0 the constant of Lemma 2.7(E1)*. No hypothesis and no correlation inequality is used.*

Proof. $|\langle e^{\sum z_j F(f_j)}\rangle| \le \langle e^{F(g(z))}\rangle$. Both cube states are BFS measures whose Gaussian is the Neumann photon Gaussian of X_J with covariance C (the matter on T_J does not touch the photon variables), so Lemma 2.9 with $s = 1$ and $v = d_k^* g(z)$ gives $\langle e^{F(g(z))}\rangle \le e^{\langle v, Cv\rangle_k/2} \le e^{\|g(z)\|_{2,k}^2}$ by Lemma 7.1. The plaquette centres of each of the three orientations form a translate of $\varepsilon_k\mathbb{Z}^3$, and $|g_{c(p)}(e_\alpha, e_\beta)| \le (1+|c(p)|^2)^{-2}|g|_4$, so by (E1) each orientation contributes at most $C_0|g|_4^2$ to $\|g\|_{2,k}^2$ and $\|g(z)\|_{2,k}^2 \le 3C_0|g(z)|_4^2$ for every k (this is the count behind (E3) of Lemma 2.7). Finally $|g(z)|_4 \le \sum_j |\operatorname{Re} z_j||f_j|_4$ is bounded on K. □

The existence theorem Theorem 3.3 is posed on King's convention. For the comparison family the analogous clause is its convergence at fixed cubes (Definition 3.1): for every such family, every $J \ge J_0$ and every real z, $\lim_k \langle e^{\sum_j z_j F(f_j)}\rangle_{k,J}^{(a)}$ exists in $(0,\infty)$. It is not needed for the main theorem (the two clauses do not imply one another by the inputs of this paper, Corollary 4.4); Theorem 7.4 is stated under it because the rotation part and the tube estimates are written for the comparison family, and Theorem 7.6 is its form for King's printed state, where the limits exist by Theorem 3.3.

Theorem 7.4 (regularity, comparison convention)**.** *Let $f_1, \dots, f_n$, J_0, $g(z)$ be as above.*

(a) (Finite volume; no hypothesis.) *For all $k \ge 0$, $J \ge J_0$, $\mu_0 > 0$, $z \in \mathbb{C}^n$, and in the presence of any real scalar source $h \in C_c^\infty$,*

$$\left|\frac{\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)} e^{:\phi^2:(h)}\rangle_{k,J}^{(a)}}{\langle e^{:\phi^2:(h)}\rangle_{k,J}^{(a)}}\right| \ \le\ \frac{\langle e^{F_{\Lambda_k}(g(z))} e^{:\phi^2:(h)}\rangle_{k,J}^{(a)}}{\langle e^{:\phi^2:(h)}\rangle_{k,J}^{(a)}} \ \le\ \exp\Big(\min\big\{\|g(z)\|_{2,k}^2,\ \tfrac12\mu_0^{-2}\|d_k^* g(z)\|_{2,k}^2\big\}\Big).$$

(b) (Limits; if the comparison family is convergent at fixed cubes.) *$Z_J(\sum_j z_j f_j, 0) := \lim_k \langle e^{\sum_j z_j F(f_j)}\rangle_{k,J}^{(a)}$ exists locally uniformly on $\mathbb{C}^n$ and is entire; $Z_\infty(\sum_j z_j f_j, 0) := \lim_J Z_J(\sum_j z_j f_j, 0)$ exists locally uniformly on $\mathbb{C}^n$ and is entire; and for $J \ge J_0$*

$$|Z_J(\textstyle\sum_j z_j f_j, 0)|,\ |Z_\infty(\sum_j z_j f_j, 0)| \ \le\ \exp\Big(\min\big\{\|g(z)\|_2^2,\ \tfrac12\mu_0^{-2}\|d^* g(z)\|_{L^2}^2\big\}\Big).$$

(c) (Schwinger functions; if the comparison family is convergent at fixed cubes.) *$S_n(f_1, \dots, f_n) := \partial_{z_1}\cdots\partial_{z_n} Z_\infty(\sum_j z_j f_j, 0)\big|_{z=0}$ satisfies, for real f_i,*

$$|S_n(f_1,\dots,f_n)| \ \le\ (\sqrt{2}\,e)^n (n!)^{1/2} \prod_{i=1}^n \|f_i\|_2 \ \le\ (e\pi/2)^n (n!)^{1/2} \prod_{i=1}^n |f_i|_4.$$

S_n extends to a continuous symmetric multilinear form on $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n$ and defines a tempered distribution; for complex test functions the bound carries the factor 2^n. This is the axiom E0″ *of* [20, (4.2), p. 287] *with $s = 4$ and $\sigma_n := (e\pi)^n (n!)^{1/2} \le \alpha_\beta (n!)^\beta$ for every $\beta > \frac12$ (and for no α at $\beta = \frac12$).*

Proof. (a) The first inequality is $\operatorname{Re}\sum_j z_j F(f_j) = F(g(z))$. The measure $e^{:\phi^2:(h)} d\langle\cdot\rangle_{k,J}^{(a)}$, normalised, is a BFS measure whose single-site weights carry the extra factor $e^{\varepsilon^3 h(x) r_x^2}$ and whose

photon covariance is the same C; Lemma 2.9 with $s = 1$ and $v = d_k^* g(z)$ bounds the middle expression by $e^{\langle v, Cv\rangle_k/2}$, and Lemma 7.1 gives both terms of the minimum.

(b) *Step 1: $k \to \infty$ at fixed $J \geq J_0$.* The functions $\Phi_k(z) := \langle e^{\sum_j z_j F(f_j)}\rangle_{k,J}^{(a)}$ are entire on $\mathbb{C}^n$: finite-dimensional integrals with Gaussian decay in A ($\mu_0 > 0$) and $e^{-\lambda\varepsilon^3 r^4}$ decay in r, converging absolutely and locally uniformly in z. By Lemma 7.3 they are locally uniformly bounded; by the convergence at fixed cubes they converge at every real point. $\mathbb{R}^n$ is a set of uniqueness for entire functions on $\mathbb{C}^n$ (a holomorphic function vanishing on $\mathbb{R}^n$ has all its Taylor coefficients at 0 equal to 0). By Montel's theorem every subsequence has a locally uniformly convergent sub-subsequence; all subsequential limits are entire and agree on $\mathbb{R}^n$, hence everywhere [11]; so $\Phi_k \to Z_J(\cdot, 0)$ locally uniformly on $\mathbb{C}^n$ and $Z_J(\cdot, 0)$ is entire. The bound of (a) at $h = 0$ passes to the limit: $|Z_J(\sum z_j f_j, 0)| \leq \lim_k e^{\|g(z)\|_{2,k}^2} = e^{\|g(z)\|_2^2}$ by (E3) of Lemma 2.7, and likewise $|Z_J(\sum z_j f_j, 0)| \leq \lim_k \exp(\frac{1}{2}\mu_0^{-2}\|d_k^* g(z)\|_{2,k}^2) = \exp(\frac{1}{2}\mu_0^{-2}\|d^* g(z)\|_{L^2}^2)$ by (1). Both bounds are free of J.

Step 2: monotonicity in J on the imaginary axes. For $t \in \mathbb{R}^n$ and $f := \sum_j t_j f_j$, $\langle e^{\mathrm{i}F(f)}\rangle_{k,J}^{(a)}$ is real (the state is invariant under $(A, \vartheta) \mapsto (-A, -\vartheta)$, which maps $F(f)$ to $-F(f)$) and of modulus ≤ 1, and $J \mapsto \langle e^{\mathrm{i}F(f)}\rangle_{k,J}^{(a)}$ is nondecreasing on $J \geq J_0$ at fixed k: this is the first inequality of Lemma 2.11 with $Y = C_{J+1} \cap \Lambda_k$, $X = C_J \cap \Lambda_k$ and the class-$\mathcal{I}$ observable $\Theta(\mathrm{i}f, 0)$, whose plaquettes lie in X by the support condition (its proof uses Lemma 2.8(a),(b) only, and the site-independence of the counterterm). Letting $k \to \infty$ (Step 1), $J \mapsto Z_J(\mathrm{i}f, 0)$ is nondecreasing and bounded by 1, hence convergent, for every $t \in \mathbb{R}^n$.

Step 3: $J \to \infty$. $\{Z_J(\cdot, 0)\}_{J \geq J_0}$ is locally uniformly bounded on $\mathbb{C}^n$ by the J-free bound of Step 1, and converges on $(\mathrm{i}\mathbb{R})^n$ (Step 2), a totally real subspace of maximal dimension, hence a set of uniqueness for entire functions (at a point $\mathrm{i}t_0$ the restriction to $\mathrm{i}t_0 + \mathrm{i}\mathbb{R}^n$ determines the Taylor series). Montel's theorem and the identity theorem as in Step 1 give $Z_J(\cdot, 0) \to Z_\infty(\cdot, 0)$ locally uniformly on $\mathbb{C}^n$, $Z_\infty(\cdot, 0)$ entire, and both bounds of Step 1 pass to the limit.

(c) By multilinearity normalise $\|f_i\|_2 = 1$. On the polydisc $|z_i| \leq r$, $\|g(z)\|_2 \leq \sum_i |\operatorname{Re} z_i| \leq nr$, so $|Z_\infty(\sum z_j f_j, 0)| \leq e^{n^2 r^2}$ by (b). Cauchy's estimate for the mixed first derivative in the n distinct variables gives $|S_n| \leq r^{-n} \sup_{|z_i|=r} |Z_\infty| \leq r^{-n} e^{n^2 r^2}$; at $r^2 = 1/(2n)$ this is $(2en)^{n/2} = (2e)^{n/2} n^{n/2} \leq (\sqrt{2}\, e)^n (n!)^{1/2}$ by (E4) of Lemma 2.7. Then (E2), $\|f_i\|_2 \leq \frac{\pi}{2\sqrt{2}} |f_i|_4$, gives the second bound. S_n is symmetric and multilinear (derivatives of an entire function of a linear combination) and continuous for the L^2 norms, so it extends by density from C_c^∞ to $\mathcal{S}$; for complex $f_i = f_i' + \mathrm{i}f_i''$ with real parts, multilinearity writes $S_n(f_1, \dots, f_n)$ as a sum of 2^n terms $\mathrm{i}^m S_n(f_1^\sharp, \dots, f_n^\sharp)$, each bounded by $(e\pi/2)^n (n!)^{1/2} \prod |f_i^\sharp|_4 \leq (e\pi/2)^n (n!)^{1/2} \prod |f_i|_4$, since $|\operatorname{Re} f|_4, |\operatorname{Im} f|_4 \leq |f|_4$; the kernel theorem gives the tempered distribution. The axiom E0″ of [20, p. 287] asks for $|S_n(f_1, \dots, f_n)| \leq \sigma_n \prod_i |f_i|_s$ on $\mathcal{S}$ with $\sigma_n \leq \alpha (n!)^\beta$; here $s = 4$: on scalar test functions $|\cdot|_4$ is the norm $|\cdot|_4$ of [20, (2.1), p. 284] (Definition 2.6), and a 2-form f whose only non-zero components are $f_{\alpha\beta} = -f_{\beta\alpha} = q$ ($\alpha < \beta$) has $|f|_4 = |q|_4$ by the definition of $|\cdot|_4$ on 2-forms, so the bound, read on such test functions, is E0″ for the Schwinger functions of the three component fields $F_{\alpha\beta}(q) := F(f)$, $\alpha < \beta$, mixed ones included (each slot may carry a different component); and $\sigma_n = (e\pi)^n (n!)^{1/2} \leq \alpha_\beta (n!)^\beta$ exactly when $(e\pi)^n / (n!)^{\beta - 1/2}$ is bounded, that is for every $\beta > \frac{1}{2}$. □

Remark 7.5 (the constants are absolute). The bound $e^{\|g\|_2^2}$ of Theorem 7.4 and the constants $(\sqrt{2}\, e)^n (n!)^{1/2}$, $(e\pi/2)^n (n!)^{1/2}$, σ_n of E0″ depend on none of e, λ, m, μ_0 and M: the only inputs are Gaussian domination, the Neumann form's constant 2 and the elementary facts (E1)–(E4). The second bound of (a) and (b) is the one that carries μ_0^{-2}. The proof of (b) uses, besides the existence clause, only [7, Thm. 6.2(1),(2)] through Lemma 2.11, Gaussian domination and Lemma 7.1; it uses neither [7, Thm. 6.2(3),(4)] (Lemma 2.8(d)) nor any local bound resting on that reversed inequality.

7.3. **King's convention.** For the K family write $a_{k,J}^K(z) := \langle e^{\sum_j z_j F(f_j)}\rangle_{k,J}^K$ as in Corollary 4.4, and Z_J^K, Z_∞^K for its limits.

Theorem 7.6 (regularity of King's printed state)**.** *Let $f = (f_1, \dots, f_n)$, J_0, $g(z)$ be as above.*

(a) (No hypothesis.) *For all $k \geq 0$, $J \geq J_0$, $z \in \mathbb{C}^n$,*

$$|a^K_{k,J}(z)| \;\leq\; \langle e^{F(g(z))}\rangle^K_{k,J} \;\leq\; \exp\Big(\min\big\{\|g(z)\|^2_{2,k},\; \tfrac{1}{2}\mu_0^{-2}\|d_k^* g(z)\|^2_{2,k}\big\}\Big),$$

and for every real 2-form u whose plaquettes have their corners in X_J: $\langle F(u)\rangle^K_{k,J} = 0$ and $\langle F(u)^2\rangle^K_{k,J} \leq \langle v, Cv\rangle_k \leq 2\|u\|^2_{2,k}$, $v = d_k^* u$.

(b) (Limits.) *$Z^K_J(z) := \lim_k a^K_{k,J}(z)$ exists locally uniformly on $\mathbb{C}^n$ and is entire; $Z^K_\infty(z) := \lim_J Z^K_J(z)$ exists locally uniformly on $\mathbb{C}^n$ and is entire; $|Z^K_J(z)|, |Z^K_\infty(z)| \leq e^{\|g(z)\|_2^2}$ and also $\leq \exp(\frac{1}{2}\mu_0^{-2}\|d^* g(z)\|^2_{L^2})$; $J \mapsto Z^K_J(\mathrm{i}t)$ is nondecreasing on $J \geq J_0$ with values in $[0,1]$ for $t \in \mathbb{R}^n$; and $Z^K_\infty(x) \geq 1$ at every real $x \in \mathbb{R}^n$.*

(c) (Schwinger functions.) *The Schwinger functions $S^K_n(f_1, \dots, f_n) := \partial_{z_1} \cdots \partial_{z_n} Z^K_\infty(0)$ satisfy $|S^K_n| \leq (\sqrt{2}\, e)^n (n!)^{1/2} \prod_i \|f_i\|_2 \leq (e\pi/2)^n (n!)^{1/2} \prod_i |f_i|_4$ for real f_i; the lattice Schwinger functions $S^{K,(k,J)}_n := \partial_{z_1} \cdots \partial_{z_n} a^K_{k,J}(0)$ satisfy the same bound at every (k, J) with $\|f_i\|_{2,k}$ in place of $\|f_i\|_2$, and $S^{K,(k,J)}_n \to S^K_n$ as $k \to \infty$ and then $J \to \infty$. Each S^K_n extends to a continuous symmetric multilinear form on $\mathcal{S}(\mathbb{R}^3; \Lambda^2_{\mathbb{C}})^n$ with the same bound, carrying the factor 2^n for complex test functions: E0″ holds with $s = 4$ and $\sigma_n = (e\pi)^n (n!)^{1/2} \leq \alpha_\beta (n!)^\beta$ for every $\beta > \frac{1}{2}$.*

Proof. (a) The two inequalities are the bound of Lemma 7.3 for the K family, with Lemma 7.1 kept in both forms: $\mathrm{Re}\sum z_j F(f_j) = F(g(z))$; $\langle\cdot\rangle^K_{k,J}$ is a BFS measure (the killed sites carry point masses, which Lemma 2.9 allows: at a killed site $I_q(0) = \delta_{q0}$ and the currents vanish on its bonds), and its Gaussian is the Neumann photon Gaussian of X_J, the same as the comparison cube's, so Lemma 2.9 with $s = 1$, $v = d_k^* g(z)$ and Lemma 7.1 give $\langle e^{F(g(z))}\rangle^K_{k,J} \leq e^{\langle v, Cv\rangle_k/2} \leq \exp(\frac{1}{2}\min\{2\|g\|^2_{2,k}, \mu_0^{-2}\|v\|^2_{2,k}\})$. *Second-moment clause.* $\langle F(u)\rangle^K_{k,J} = 0$ because the state is invariant under $(A, \vartheta) \mapsto (-A, -\vartheta)$ and $F(u)$ is odd. For real s, $e^x + e^{-x} \geq 2 + x^2$ pointwise and $e^{\pm sF(u)}$ are integrable, so $2 + s^2\langle F(u)^2\rangle^K_{k,J} \leq \langle e^{sF(u)}\rangle^K_{k,J} + \langle e^{-sF(u)}\rangle^K_{k,J} \leq 2e^{s^2\langle v, Cv\rangle_k/2}$ (Lemma 2.9 at $\pm s$; $\pm v$ give the same quadratic form). Hence $\langle F(u)^2\rangle^K_{k,J} \leq 2(e^{s^2\langle v, Cv\rangle_k/2} - 1)/s^2 \to \langle v, Cv\rangle_k$ as $s \downarrow 0$, and Lemma 7.1 applies since the corners lie in X_J.

(b) The two limits, their entireness and the J-free bound $e^{\|g(z)\|_2^2}$ are Corollary 4.4(ii)(α),(β): the proof of Theorem 7.4(b) run on the K family, with the local bound of (a) (uniform in k, J by $\|g\|^2_{2,k} \leq 3C_0|g|^2_4$), Theorem 3.3 for the convergence at real points, Montel's theorem with the uniqueness set $\mathbb{R}^n$, and then, for $J \to \infty$, the monotonicity of Corollary 4.3 on the cone in place of Lemma 2.11 and the uniqueness set $(\mathrm{i}\mathbb{R})^n$. Three additions. (1) *Values in* $[0,1]$. $a^K_{k,J}(\mathrm{i}t)$ is real by the symmetry used in (a) and of modulus ≤ 1; it is ≥ 0 by the current representation of Lemma 2.9: with $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q), A\rangle$, $w(q) \geq 0$, and the Gaussian identity at $a = \mathrm{i}v \pm \mathrm{i}\kappa$,

$$\langle e^{\mathrm{i}\langle v, A\rangle}\rangle^K_{k,J} = e^{-\langle v, Cv\rangle_k/2}\, \mathbb{E}^\sharp \cosh\langle Cv, \kappa\rangle \;\geq\; 0, \qquad v = d_k^* \textstyle\sum_j t_j f_j,$$

the killed sites being covered as in (a). Monotonicity in J is Corollary 4.3 at every k (condition (S) holds for X_{J_0} by the support condition), passed to the limit $k \to \infty$. (2) *The μ_0^{-2} bound.* $\|d_k^* g\|_{2,k} \to \|d^* g\|_{L^2}$ by (1), so the second bound of (a) passes to Z^K_J and then to Z^K_∞ exactly as the first. (3) $Z^K_\infty(x) \geq 1$. For real x, $a^K_{k,J}(x) = \langle e^{F(g(x))}\rangle^K_{k,J} \geq e^{\langle F(g(x))\rangle^K_{k,J}} = 1$ by Jensen's inequality and (a); the inequality is closed under both limits.

(c) Let $\|f_j\|_{2,k} > 0$ for all j (otherwise $F_{\Lambda_k}(f_j) = 0$ and $S^{K,(k,J)}_n = 0$). Cauchy's estimates on the polydisc $|z_j| \leq \rho_j := \rho/(n\|f_j\|_{2,k})$ with the bound (a): there $\|g(z)\|_{2,k} \leq \sum_j |z_j|\|f_j\|_{2,k} \leq \rho$, so $|a^K_{k,J}| \leq e^{\rho^2}$ and $|S^{K,(k,J)}_n| \leq e^{\rho^2} \prod_j \rho_j^{-1} = e^{\rho^2} n^n \rho^{-n} \prod_j \|f_j\|_{2,k}$; at $\rho^2 = n/2$ the prefactor is $e^{n/2} 2^{n/2} n^{n/2} \leq e^{n/2} 2^{n/2} e^{n/2} (n!)^{1/2} = (\sqrt{2}\, e)^n (n!)^{1/2}$ by (E4). The Taylor coefficients at 0 converge under the locally uniform convergences of (b) (Weierstrass), and $\|f_j\|_{2,k} \to \|f_j\|_2$ by (E3), which gives the bound for S^K_n; (E2) gives the $|\cdot|_4$ form. The 2^n factor for complex test functions, the extension to $\mathcal{S}$ and the growth class σ_n are as in Theorem 7.4(c). □

7.4. **The curvature sector of the comparison family's limits.** For the comparison family and an admissible scheme a, the general-source functionals $Z^a_J(\zeta) = \lim_k Z^a_{k,J}(\zeta)$, $\zeta \in \mathbb{C}^{2N}$, have, under (W2) (the optional extension), the following properties: (i) locally uniform convergence to an entire Z^a_J; (ii) on the cone, $J \mapsto Z^a_J(\mathrm{i}f, g)$ nondecreasing and $J \mapsto Z^a_J(f, -g)$ nonincreasing; (iii) the limit $Z^a_\infty = \lim_J Z^a_J$, locally uniform and entire, positive at real sources, with $Z^a_J \le Z^a_\infty$ on the cone; (iv) $\Phi^a = \ln Z^a_\infty$ finite, convex and continuous on the Fréchet space $E_\rho = C^\infty_c(\bar B_\rho; \Lambda^2) \times C^\infty_c(\bar B_\rho)$; (v) translation invariance; (vi) the change of scheme. At $h = 0$ these conclusions hold under the existence clause alone; the natural local bound, $|Z_J(\zeta)| \le Z_J(\mathrm{Re}\,\zeta) \le Z_{J_0}(\mathrm{Re}\,\zeta)$, would use the nonincreasing clause of (ii), that is Lemma 2.8(d). The following lemma states the curvature sector with that use removed; the Wick sector on the tube is Lemma 13.11.

Lemma 7.7 (the comparison family's limits, curvature sector, with cone-direction inequalities only)**.** *Assume that the comparison family is convergent at fixed cubes (Definition 3.1). For every finite family $f_1, \dots, f_N$ of real 2-forms in C^∞_c supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, with $Z_{k,J}(\sum_j z_j f_j, 0) := \langle e^{\sum_j z_j F(f_j)} \rangle^{(a)}_{k,J}$ and $g(z) = \sum_j (\mathrm{Re}\, z_j) f_j$:*

(i) *$Z_{k,J}(\cdot, 0) \to Z_J(\cdot, 0)$ locally uniformly on $\mathbb{C}^N$ as $k \to \infty$, and $Z_J(\cdot, 0)$ is entire, for every $J \ge J_0$;*

(ii) *$J \mapsto Z_J(\mathrm{i}f, 0)$ is nondecreasing on $J \ge J_0$ for every real $f = \sum_j t_j f_j$;*

(iii) *for all $k \ge 0$, $J \ge J_0$ and $z \in \mathbb{C}^N$,*

$$|Z_{k,J}(\textstyle\sum_j z_j f_j, 0)| \le \langle e^{F(g(z))} \rangle^{(a)}_{k,J} \le e^{\frac12 \langle v, Cv\rangle_k} \le \exp\left(\tfrac12 \mu_0^{-2} \|d_k^* g(z)\|^2_{2,k}\right) \le \exp\left(\tfrac12 \mu_0^{-2} \sup_{k'} \|d_{k'}^* g(z)\|^2_{2,k'}\right), \quad v = c$$

a bound uniform in k and J and locally uniform in z; consequently $Z_\infty(\cdot, 0) := \lim_J Z_J(\cdot, 0)$ exists locally uniformly on $\mathbb{C}^N$, is entire, satisfies $Z_\infty(f, 0) > 0$ for real f, and $Z_J(\mathrm{i}f, 0) \le Z_\infty(\mathrm{i}f, 0)$ for real f;

(iv) *for every $\rho > 0$, $\Phi(f) := \ln Z_\infty(f, 0)$ is finite, convex and continuous on the Fréchet space $C^\infty_c(\bar B_\rho; \Lambda^2)$;*

(v) *$Z_\infty(\tau_b f, 0) = Z_\infty(f, 0)$ for all $b \in \mathbb{R}^3$ and all complex coefficient vectors, τ_b acting on every f_j.*

Clauses (i), (iii)–(v) *use, besides the convergence at fixed cubes, only* [7, Thm. 6.2(1),(2)] *through Lemma 2.11, Lemma 2.9, Lemma 2.12*(b), (1), *Lemma 2.15, Montel's theorem and the identity theorem. The nonincreasing clause* (ii) *at $h = 0$, $J \mapsto Z_J(f, 0)$ nonincreasing for real f, holds as well and uses Lemma 2.8*(d)*; it is not consumed by the curvature sector of this paper. Clause* (vi) *is void at $h = 0$, no scheme entering.*

Proof. (i) is Step 1 of the proof of Theorem 7.4(b), which uses Lemma 7.3, the convergence at fixed cubes and Montel's theorem. (ii) is its Step 2. (iii) The first inequality is Lemma 2.12(b), the second is Lemma 2.9 with $s = 1$ (the cube state is a BFS measure), the third is $C \le \mu_0^{-2}$ as forms (the photon form is bounded below by $\mu_0^2 \|A\|^2_{2,k}$, the second bound of Lemma 7.1), and the supremum over k' is finite by (1). The bound passes to Z_J by (i) and makes $\{Z_J(\cdot, 0)\}_{J \ge J_0}$ locally bounded on $\mathbb{C}^N$; with the convergence on $(\mathrm{i}\mathbb{R})^N$ from (ii) and the bound $|Z_J(\mathrm{i}f, 0)| \le 1$, Montel's theorem and the identity theorem give the locally uniform convergence to an entire $Z_\infty(\cdot, 0)$ as in Step 3 of Theorem 7.4(b). Positivity: $Z_{k,J}(f, 0) \ge e^{\langle F(f) \rangle^{(a)}_{k,J}} = 1$ for real f by Jensen's inequality and $\langle F(f) \rangle^{(a)}_{k,J} = 0$ (the symmetry $(A, \vartheta) \mapsto (-A, -\vartheta)$), closed under both limits. The last clause is (ii) in the limit.

(iv) Φ is finite by (iii). Convexity: for $f, f' \in C^\infty_c(\bar B_\rho; \Lambda^2)$ and $t \in [0, 1]$ apply (i) and (iii) to the family (f, f') at the real coefficients $(t, 1 - t)$; at fixed k, J, $\ln Z_{k,J}(\cdot, 0)$ is convex in the coefficients by Hölder's inequality, and convexity survives the two pointwise limits. Borel measurability: fix J with $MJ > \rho + 1$, so that every $f \in C^\infty_c(\bar B_\rho; \Lambda^2)$ is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$; at each k, $\ln Z_{k,J}(f, 0)$ depends on f only through the finitely many sampled values $f_{c(p)}(e_\alpha, e_\beta)$, continuous functionals of f in the C^∞ topology, and the finite-dimensional integral is continuous in them (dominated convergence); so $\Phi = \lim_J \lim_k$ of continuous functions is Borel (of Baire class ≤ 2). $C^\infty_c(\bar B_\rho; \Lambda^2)$ is a closed subspace of $C^\infty(\bar B_{\rho+1}; \Lambda^2)$, hence Fréchet. Lemma 2.15 gives the continuity.

(v) is Proposition 8.1, whose proof uses the lattice sandwich between C_{J-1} and C_{J+1} (first inequality of Lemma 2.11), $k \to \infty$ by (i) (the convergence at fixed cubes covers the translated family), $J \to \infty$ by (iii), the identity theorem with the uniqueness set $(\mathrm{i}\mathbb{R})^N$, the continuity of (iv) and the density of $\bigcup_k 5^{-k}\mathbb{Z}^3$. □

For King's convention the same clauses hold by Theorem 3.3: (i) and (iii) are Theorem 7.6(b), with the local bound of Theorem 7.6(a); (ii) is Corollary 4.3; (iv) follows from Theorem 7.6(b) by the argument above, word for word; (v) is Corollary 8.3. In every clause the inequalities used are the cone directions of Lemma 2.8(a),(b) and of Lemma 4.2; the reversed inequalities of Lemma 2.8(d) enter the curvature sector of this paper nowhere.

8. Translations, the cubic group, and rotations

This section proves the clause OS1 of King's printed state as far as the lattice allows: invariance under all translations and under the cubic group O_h about every point; full $O(3)$ invariance comes from the rotation theorem (Theorem 3.14, proved in the rotation part, Theorem 15.1) and is stated at the end. No cube is invariant under a non-trivial translation or a reflection; what replaces exactness is a sandwich between a smaller and a larger cube of the same family, in the direction of the cone inequalities of Lemmas 2.11 and 4.1, followed by the limits $k \to \infty$, $J \to \infty$. Coordinate permutations alone are exact (Lemma 8.4).

Conventions. We use the notation of Definition 2.4 and Corollary 4.4: for a finite family $f = (f_1, \dots, f_N)$ of real 2-forms in $C_c^\infty(\mathbb{R}^3; \Lambda^2)$ and $z \in \mathbb{C}^N$, $a^K_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)} \rangle^K_{k,J}$ and $a^{(a)}_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)} \rangle^{(a)}_{k,J}$ are the lattice functionals, Z^K_J, Z^K_∞ and Z_J, Z_∞ their limits where these exist, and $g(z) = \sum_j (\operatorname{Re} z_j) f_j$. Since F_{Λ_k} is linear in the source (extended $\mathbb{C}$-linearly), a lattice functional depends on a family only through its total source, and we use the same symbol for a coefficient vector and for its total source: $a^K_{k,J}(z) = Z^K_{k,J}(\sum_j z_j f_j)$ with $Z^K_{k,J}(u) := \langle e^{F(u)} \rangle^K_{k,J}$ for a complex-valued 2-form u; in particular, for a real 2-form u, $Z^K_{k,J}(\mathrm{i}u)$ is the one-member functional at the coefficient i, and the same convention is carried to Z^K_J, Z^K_∞ (so $Z^K_\infty(\mathrm{i}u_t) = Z^{K,f}_\infty(\mathrm{i}t)$ for $u_t = \sum_j t_j f_j$). When several families are in play we write $a^{K,f}_{k,J}$, $Z^{K,f}_J$, $Z^{K,f}_\infty$. $J_0(f)$ is the least J such that every $\operatorname{supp} f_j$ lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$; for $J \geq J_0(f)$ the support condition (S) holds for X_J. The transport $(O \cdot f)_x(u, v) = f_{O^{-1}x}(O^{-1}u, O^{-1}v)$ and the translation $(\tau_b f)_x = f_{x-b}$ are those of Definition 2.4; $O \cdot (O' \cdot f) = (OO') \cdot f$, and both are linear in f. Translations and the cubic group act on the Schwinger functions by this signed push-forward of 2-forms: for an improper O no extra sign appears.

8.1. Translations on the comparison convention.

Proposition 8.1 (translations, comparison convention)**.** *Assume that the comparison family is convergent at fixed cubes (Definition 3.1), so that by Lemma 7.7(i),(iii) the limits $Z_J(z) = \lim_k a^{(a)}_{k,J}(z)$ and $Z_\infty(z) = \lim_J Z_J(z)$ exist locally uniformly on $\mathbb{C}^N$ for every family and are entire. Then for every finite family f of real 2-forms, every $b \in \mathbb{R}^3$ and every $z \in \mathbb{C}^N$,*

$$Z_\infty\Big(\sum_j z_j \tau_b f_j\Big) = Z_\infty\Big(\sum_j z_j f_j\Big).$$

Proof. Step 1 (lattice covariance). Let $k_0 \geq 0$, $b \in \Lambda_{k_0}$ and $k \geq k_0$. Then τ_b maps Λ_k onto itself, and the substitution $A'_\alpha(x) := A_\alpha(x + b)$, $\phi'(x) := \phi(x + b)$ is a bijection of the configurations of the comparison state on X_J onto those of the comparison state on $X_J - b$ (Definition 2.1 on the translated box) which carries the Lebesgue measure and the action to themselves: the action is a sum of local terms with site-independent coefficients (δm^2_ε, W_ε depend on ε only), and the boundary conditions are transported. The plaquette p corresponds to $p - b$ with the same orientation and $F(p)(A) = F(p - b)(A')$, so $F_{\Lambda_k}(\tau_b f)(A) = F_{\Lambda_k}(f)(A')$. Hence, for all coefficients,

$$Z_{k,X_J}\Big(\sum_j z_j \tau_b f_j\Big) = Z_{k,X_J - b}\Big(\sum_j z_j f_j\Big) \qquad (b \in \Lambda_{k_0},\ k \geq k_0), \tag{16}$$

the right side being the functional of the comparison state on $X_J - b$.

Step 2 (interlacing of the translated box). For $|b|_\infty \le M$: $X_{J-1} \subset X_J - b \subset X_{J+1}$, since $-MJ - b_\mu \le -M(J-1)$, $MJ - b_\mu \ge M(J-1)$, $-MJ - b_\mu \ge -M(J+1)$ and $MJ - b_\mu \le M(J+1)$ for every μ.

Step 3 (the sandwich on the imaginary axes). Let $J \ge J_0(f) + 1$, $t \in \mathbb{R}^N$ and $u_t = \sum_j t_j f_j$, so that $(u_t, 0) \in \mathcal{K}$ is supported at distance > 1 from $\mathbb{R}^3 \setminus C_{J-1}$. Apply the first inequality of Lemma 2.11 twice: with $Y = X_J - b$ and $X = X_{J-1}$, and with $Y = X_{J+1}$ and $X = X_J - b$. The support condition of the lemma holds because the plaquettes carrying u_t have their corners in X_{J-1}; no site is killed, so each cut-out factor is the comparison state on the smaller box (the Dirichlet terms of the smaller state are the diagonal parts of the cut bonds, and where the boundaries of two of the boxes touch they coincide). With (16),

$$a^{(a)}_{k,J-1}(\mathrm{i}t) \;\le\; a^{(a),\tau_b f}_{k,J}(\mathrm{i}t) \;\le\; a^{(a)}_{k,J+1}(\mathrm{i}t) \qquad (k \ge k_0,\ J \ge J_0(f) + 1).$$

Let $k \to \infty$: the two ends converge by Lemma 7.7(i) for the family f at $J \mp 1 \ge J_0(f)$, and the middle term by Lemma 7.7(i) for the family $\tau_b f$, which is admissible at J because $\operatorname{supp} f_j + b$ lies at distance > 1 from $\mathbb{R}^3 \setminus C_J$ (as $C_{J-1} + b \subset C_J$). Then let $J \to \infty$ by Lemma 7.7(iii): $Z^{\tau_b f}_\infty(\mathrm{i}t) = Z^f_\infty(\mathrm{i}t)$ for all $t \in \mathbb{R}^N$.

Step 4 (all coefficients). $D(z) := Z^{\tau_b f}_\infty(z) - Z^f_\infty(z)$ is entire on $\mathbb{C}^N$ and vanishes on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal real dimension N; every partial derivative of D at 0 vanishes ($\partial_{y_j} D(\mathrm{i}y) = \mathrm{i}\,\partial_{z_j} D$), so $D \equiv 0$. Thus the identity holds for $b \in \Lambda_{k_0}$, $|b|_\infty \le M$, and all $z \in \mathbb{C}^N$.

Step 5 (all lattice translations). A vector $b \in \bigcup_k \Lambda_k$ lies in some Λ_{k_0} and is a finite sum $b_1 + \cdots + b_n$ of vectors $b_i \in \Lambda_{k_0}$ with $|b_i|_\infty \le M$ (each coordinate splits into steps $\pm M$ and a remainder in $\varepsilon_{k_0}\mathbb{Z}$ of modulus $< M$). Steps 3–4 hold for every family, in particular for the translated families $\tau_{b_1 + \cdots + b_{i-1}} f$; composing gives the identity for b.

Step 6 (all $b \in \mathbb{R}^3$ at real coefficients). Fix $\rho > 0$ and let $E_\rho := C^\infty_c(\bar{B}_\rho; \Lambda^2)$ with its C^∞ topology, a Fréchet space. On E_ρ the function $\Phi(u) := \ln Z_\infty(u)$ at real u is finite ($Z_\infty(u) \ge 1$ by Jensen's inequality, as $\langle F(u)\rangle = 0$ by charge conjugation), convex (a pointwise limit of the convex functions $u \mapsto \ln\langle e^{F(u)}\rangle^{(a)}_{k,J}$) and Borel (each lattice functional depends continuously on finitely many values of u), hence continuous by Lemma 2.15; this is Lemma 7.7(iv). For $|b| \le \rho' - \rho$ the map $b \mapsto \tau_b u \in E_{\rho'}$ is continuous, $\bigcup_k \Lambda_k$ is dense, and Φ on $E_{\rho'}$ restricted to E_ρ is the same pointwise limit; so Step 5 gives $Z_\infty(\tau_b u) = Z_\infty(u)$ for all $b \in \mathbb{R}^3$ and real $u \in E_\rho$, and exhausting over ρ for all real $u \in C^\infty_c$.

Step 7 (complex coefficients). For a family f and $b \in \mathbb{R}^3$, both sides of the asserted identity are entire in z and agree on $\mathbb{R}^N$ by Step 6 (applied to $u = \sum_j x_j f_j$), a set of uniqueness; hence they agree on $\mathbb{C}^N$. □

Only the first inequality of Lemma 2.11 enters: the sandwich is taken on the cone, and the passage off it is the identity theorem. The translation clause for the joint field on the tube, by the same sandwich at the cone points of the tube, is Lemma 13.11(v).

8.2. **The translation sandwich on King's convention.** On King's convention the translated cube has its killed sites on $T_J - b$, which neither contains nor is contained in $T_{J\pm1}$, so Lemma 2.11 does not compare it with $C^K_{J\pm1}$ directly; one kill step (Lemma 2.8(c) through Lemma 2.10) and one cut-out with killed sites (Lemma 4.1) at each end do.

Lattice covariance on King's convention. For $b \in \Lambda_{k_0}$ and $k \ge k_0$ let $\langle\cdot\rangle^{K,-b}_{k,J}$ denote the comparison state on $X_J - b$ with the matter killed on $T_J - b$ (King's cube translated by $-b$). The substitution of Step 1 above is a bijection of the configurations on X_J killed on T_J onto those on $X_J - b$ killed on $T_J - b$, carrying the point masses to the point masses and the action to the action (the kill set translates with the box). Hence, for every finite family and all coefficients,

$$Z^{a,K}_{k,J}\Big(\sum_j z_j \tau_b f_j, \sum_j w_j \tau_b h_j\Big) = Z^{a,K,-b}_{k,J}\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big) \qquad (b \in \Lambda_{k_0},\ k \ge k_0), \tag{17}$$

the right side being the scheme functional of $\langle\cdot\rangle^{K,-b}_{k,J}$ with the same multiplier ($\mathsf{S}_{\Lambda_k}(\tau_b h) = \mathsf{S}_{\Lambda_k}(h)$).

Lemma 8.2 (translation sandwich, King's convention)**.** *Let $(f,g) \in \mathcal{K}$ be supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and let $b \in \Lambda_{k_0}$ with $|b|_\infty \le M$. Then for every $k \ge k_0$, every $M \ge 1$ and every $J \ge J_0 + 1$,*

$$Z^{a,K}_{k,J-1}(\mathrm{i}f,g) \;\le\; Z^{a,K}_{k,J}(\mathrm{i}\,\tau_b f, \tau_b g) \;\le\; Z^{a,K}_{k,J+1}(\mathrm{i}f,g),$$

and the reversed inequalities hold for $(f,-g)$.

Proof. By (17) the middle term is $Z^{a,K,-b}_{k,J}(\mathrm{i}f,g)$; the multipliers of the three terms are the same positive constant (Lemma 2.12(a); S is translation invariant), so it suffices to prove the inequalities for $\langle\Theta(\mathrm{i}f,g)\rangle$, and the reversed ones for $\Theta(f,-g)$; all these expectations are real.

Geometry. Here $b_\mu \in \varepsilon_{k_0}\mathbb{Z}$, $|b_\mu| \le M$, and all sets are subsets of Λ_k. (G1) $\mathrm{alive}^K_J - b = \prod_\mu\{-MJ - b_\mu, \dots, MJ - 2\varepsilon - b_\mu\} \subset \mathrm{alive}^K_{J+1} = \{-MJ - M, \dots, MJ + M - 2\varepsilon\}^3$, since $-MJ - b_\mu \ge -MJ - M$ and $MJ - 2\varepsilon - b_\mu \le MJ + M - 2\varepsilon$. (G2) $X_J - b \subset X_{J+1}$: the same arithmetic with ε in place of 2ε. (G3) $\mathrm{alive}^K_{J-1} = \{-MJ + M, \dots, MJ - M - 2\varepsilon\}^3 \subset \mathrm{alive}^K_J - b$, since $-MJ + M \ge -MJ - b_\mu$ and $MJ - M - 2\varepsilon \le MJ - 2\varepsilon - b_\mu$. (G4) $X_{J-1} \subset X_J - b$, likewise. All four inclusions hold with margin $M - |b_\mu| \ge 0$ on every face, at every spacing, including $k = 0$, $M = 1$, $|b_\mu| = M$; non-strict inclusions are all that Lemma 4.1 needs.

Sources. By (S) for X_{J-1}, every plaquette carrying f has its corners in X_{J-1} and every site carrying g lies in alive^K_{J-1}; by (G1)–(G4) these lie in the site sets and alive sets of all three states. So the observable is one function of variables present and alive in all three states, and its scalar factor equals 1 at every site killed below.

Upper end. Start from $\langle\cdot\rangle^K_{k,J+1}$, the comparison state on X_{J+1} killed on T_{J+1}, a BFS measure of King type (Definition 2.2). (i) Kill $U_+ := \mathrm{alive}^K_{J+1} \setminus (\mathrm{alive}^K_J - b)$. These are alive sites of the state, $g = 0$ on U_+ (by (G3), $U_+ \cap \mathrm{alive}^K_{J-1} = \emptyset$), and $\Theta(\mathrm{i}f,g)$ is continuous with $|\Theta| = e^{\sum_x g_x r_x^2}$; so by Lemma 2.8(c), through Lemma 2.10 (which allows the already killed sites T_{J+1}), killing U_+ does not increase $\langle\Theta(\mathrm{i}f,g)\rangle$, and by Lemma 2.8(d) with Lemma 2.10 it does not decrease $\langle\Theta(f,-g)\rangle$. The result is the comparison state on X_{J+1} killed on $T' := T_{J+1} \cup U_+ = X_{J+1} \setminus (\mathrm{alive}^K_J - b)$ (by (G1); nothing is revived). (ii) Apply Lemma 4.1 with $V = X_{J+1}$, $T = T'$ and $X = X_J - b \subset V$ (by (G2)); the lemma allows any $T \subset V$ and any $X \subset V$. The cut-out factor is the comparison state on $X_J - b$ killed on $(X_J - b) \cap T' = (X_J - b) \setminus (\mathrm{alive}^K_J - b) = T_J - b$, that is $\langle\cdot\rangle^{K,-b}_{k,J}$; the support condition (S_X) holds for $X = X_J - b$ by *Sources*. Hence $\langle\Theta(\mathrm{i}f,g)\rangle^{K,-b}_{k,J} \le \langle\Theta(\mathrm{i}f,g)\rangle^{(a),T'}_{X_{J+1}} \le \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J+1}$, and $\ge$ for class $\mathcal{D}$.

Lower end. Start from $\langle\cdot\rangle^{K,-b}_{k,J}$, the comparison state on $X_J - b$ killed on $T_J - b$ (of King type, a translate of a state of Definition 2.2). (i) Kill $U_- := (\mathrm{alive}^K_J - b) \setminus \mathrm{alive}^K_{J-1}$, alive sites at which $g = 0$ by *Sources*; as above, $\langle\Theta(\mathrm{i}f,g)\rangle$ does not increase and $\langle\Theta(f,-g)\rangle$ does not decrease, and the result is the comparison state on $X_J - b$ killed on $T'' := (X_J - b) \setminus \mathrm{alive}^K_{J-1}$ (by (G3)). (ii) Lemma 4.1 with $V = X_J - b$, $T = T''$ and $X = X_{J-1} \subset V$ (by (G4)): the factor is the comparison state on X_{J-1} killed on $X_{J-1} \cap T'' = X_{J-1} \setminus \mathrm{alive}^K_{J-1} = T_{J-1}$, that is $\langle\cdot\rangle^K_{k,J-1}$; ($\mathrm{S}_X$) for X_{J-1} is the hypothesis. Hence $\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J-1} \le \langle\Theta(\mathrm{i}f,g)\rangle^{K,-b}_{k,J}$, and $\ge$ for class $\mathcal{D}$.

There is no edge case because a translation by $b \in \Lambda_{k_0}$, $|b|_\infty \le M$, moves King's cube rigidly, with the alive sets nested at margins $M - |b_\mu| \ge 0$ at every spacing; the reflection of Lemma 4.5 shifts the cube by one lattice layer instead, which is why its one-step form needs $M \ge 2\varepsilon_k$. □

The class-$\mathcal{I}$ clause uses Lemma 2.8(a),(b),(c),(e) and Lemma 2.10 only; the class-$\mathcal{D}$ clause uses Lemma 2.8(d) in addition. Only the class-$\mathcal{I}$ clause is used in this section.

Corollary 8.3 (translations of King's printed state)**.** *For every finite family f of real 2-forms, every $b \in \mathbb{R}^3$ and every $z \in \mathbb{C}^N$, $Z^K_\infty(\sum_j z_j \tau_b f_j) = Z^K_\infty(\sum_j z_j f_j)$. Moreover, for all real 2-forms $u, u' \in C^\infty_c$,*

$$|Z^K_\infty(\mathrm{i}u) - Z^K_\infty(\mathrm{i}u')| \le \sqrt{2}\,\|u - u'\|_2,$$

the L^2 norm of a 2-form being computed in a basis (e_c) of Λ^2 orthonormal for the norm formula.

Proof. Write $u_t = \sum_j t_j f_j$ for $t \in \mathbb{R}^N$. (1) *$b \in \Lambda_{k_0}$, $|b|_\infty \le M$, on the imaginary axes.* For $J \ge J_0(f)+1$ and $k \ge k_0$, Lemma 8.2 at $(u_t, 0) \in \mathcal{K}$ gives $a^K_{k,J-1}(\mathrm{i}t) \le a^{K,\tau_b f}_{k,J}(\mathrm{i}t) \le a^K_{k,J+1}(\mathrm{i}t)$. Let $k \to \infty$: the ends converge by Theorem 7.6(b) for f at $J \mp 1 \ge J_0(f)$; the middle term by Theorem 7.6(b) for the family $\tau_b f$, which is admissible at J since $\operatorname{supp} f_j + b$ lies at distance > 1 from $\mathbb{R}^3 \setminus C_J$ ($C_{J-1} + b \subset C_J$ for $|b|_\infty \le M$), so that Theorem 3.3, posed for every finite family, applies to it. Hence $Z^K_{J-1}(\mathrm{i}t) \le Z^{K,\tau_b f}_J(\mathrm{i}t) \le Z^K_{J+1}(\mathrm{i}t)$, and $J \to \infty$ (Corollary 4.4(ii)(β) at both ends) gives $Z^{K,\tau_b f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for all $t \in \mathbb{R}^N$. (2) *All coefficients, same b.* $D(z) := Z^{K,\tau_b f}_\infty(z) - Z^{K,f}_\infty(z)$ is entire on $\mathbb{C}^N$ (Theorem 7.6(b) for both families) and vanishes on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal real dimension, so every partial derivative of D at 0 vanishes and $D \equiv 0$. (3) *All $b \in \bigcup_k \Lambda_k$:* as in Step 5 of Proposition 8.1, by composition over the translated families. (4) *The Lipschitz bound.* For real 2-forms u, u' whose plaquettes have corners in X_J, at every (k, J), $|\langle e^{\mathrm{i}F(u)}\rangle^K_{k,J} - \langle e^{\mathrm{i}F(u')}\rangle^K_{k,J}| \le \langle |F(u-u')|\rangle^K_{k,J} \le \langle F(u-u')^2\rangle^{K\,1/2}_{k,J} \le \sqrt{2}\|u-u'\|_{2,k}$, by $|e^{\mathrm{i}x} - e^{\mathrm{i}y}| \le |x-y|$, the linearity of F_{Λ_k} and the second-moment clause of Theorem 7.6(a). Both sides have limits as $k \to \infty$ and $J \to \infty$ (Theorem 7.6(b) for the one-member families at the coefficient i; $\|u-u'\|_{2,k} \to \|u-u'\|_2$ by (1)). (5) *All $b_0 \in \mathbb{R}^3$.* Pick $b_n \to b_0$ with $b_n \in \bigcup_k \Lambda_k$. For real $u \in C^\infty_c$, $\tau_{b_n} u \to \tau_{b_0} u$ in L^2 (uniform continuity and compact support), so by (4) and (3), $Z^K_\infty(\mathrm{i}\tau_{b_0} u) = \lim_n Z^K_\infty(\mathrm{i}\tau_{b_n} u) = Z^K_\infty(\mathrm{i}u)$. With $u = u_t$ this is $Z^{K,\tau_{b_0} f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for all t, and (2) extends it to all $z \in \mathbb{C}^N$. □

Nothing here uses a reflection or any property of the limit beyond Theorem 7.6; King's derivation of translation invariance from the uniqueness of his limit [16, Thm. 2.3(i)] is not used.

8.3. **The cubic group.**

Lemma 8.4 (passive covariance with killed sites)**.** *Let O, Λ, (f_α) and X be as in Lemma 2.13, and let $U \subset X$ be a set of killed sites. Then the identities of Lemma 2.13 hold between the state on X with the matter killed on U and the state on OX (frame (Of_α), transported boundary conditions and wall) with the matter killed on OU:*

$$\big\langle \Theta(z\,O\cdot f, w\,O\cdot h)\big\rangle_{O\Lambda, OX; OU} = \big\langle \Theta(zf, wh)\big\rangle_{\Lambda, X; U}, \qquad \mathsf{S}_{O\Lambda}(O\cdot h) = \mathsf{S}_\Lambda(h).$$

Proof. The configuration-space bijection $\phi'(Ox) = \phi(x)$, $A'_\alpha(Ox) = A_\alpha(x)$ of Lemma 2.13 carries $\prod_{x\in U} \delta_0(d\phi(x))\, d\nu$ to $\prod_{x\in OU} \delta_0(d\phi'(Ox))\, d\nu$ (the point mass at $\phi(x) = 0$ goes to the point mass at $\phi'(Ox) = 0$), and the diagonal parts $\frac{\varepsilon}{2} r^2$ at the alive ends of the bonds into U are frame-free; the rest of that proof is unchanged. □

Theorem 8.5 (cubic symmetry of King's printed state)**.** *For all real 2-forms $f_1, \dots, f_N \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$, all $z \in \mathbb{C}^N$, all $O \in O_h$ (the signed permutation matrices) and all $b \in \mathbb{R}^3$,*

$$Z^K_\infty\Big(\sum_j z_j\, \tau_b O\cdot f_j\Big) = Z^K_\infty\Big(\sum_j z_j f_j\Big).$$

Hence the Schwinger functions S^K_n of Theorem 7.6(c) are $\mathbb{R}^3 \rtimes O_h$-covariant, F transforming as a 2-form.

Proof. "(k, J) exact" means an identity of the lattice functionals for all complex coefficients and all k, J for which both families have their plaquettes in X_J.

Step 1 (permutations are exact). Let P be a permutation matrix, $Pe_\alpha = e_{\pi(\alpha)}$. Then $P\Lambda_k = \Lambda_k$, $PX_J = X_J$ and $PT_J = T_J$ (the condition "$x_\mu = MJ - \varepsilon$ for some μ" is permutation invariant), hence P $\mathrm{alive}^K_J = \mathrm{alive}^K_J$. By Lemma 8.4 with $O = P$, $X = X_J$, $U = T_J$, $\langle \Theta(zP\cdot f)\rangle_{P\Lambda_k, PX_J; PT_J}$ in the frame $(e_{\pi(\alpha)})$ equals $\langle \Theta(zf)\rangle^K_{k,J}$, and the state on the left is $\langle\cdot\rangle^K_{k,J}$ itself: the Dirichlet rule is a condition on the site set X_J, the kill set is T_J, the Neumann rule keeps the same set of (bond, direction) pairs relabelled, the mass term runs over the same bonds, and the hopping and single-site terms are frame-free. The plaquette set and the centres are the same; for a transposed pair $(x; \beta, \alpha)$ both $F(p)$ and $f_{c(p)}(e_\beta, e_\alpha)$ change sign by antisymmetry, so

their product is unchanged. Thus $a^{K,P\cdot f}_{k,J}(z) = a^{K,f}_{k,J}(z)$ is (k,J) exact. The distance->1 interior $(-MJ+1, MJ-1)^3$ of C_J is P-invariant, so $J_0(P\cdot f) = J_0(f)$, and Theorem 7.6(b) for both families gives $Z^{K,P\cdot f}_\infty = Z^{K,f}_\infty$ on $\mathbb{C}^N$.

Step 2 (the reflection σ_3 on the imaginary axes). Let $\sigma_3 = \mathrm{diag}(1,1,-1)$, $t \in \mathbb{R}^N$, $u_t = \sum_j t_j f_j$, $J_0 = J_0(f)$. Lemma 4.5(b), for every $k \geq 0$, every $M \geq 1$ and every $J \geq J_0 + 2$, gives

$$a^{K,f}_{k,J-2}(\mathrm{i}t) \;\leq\; a^{K,\sigma_3\cdot f}_{k,J}(\mathrm{i}t) \;\leq\; a^{K,f}_{k,J+2}(\mathrm{i}t),$$

since $\sigma_3 \cdot u_t = \sum_j t_j \sigma_3 \cdot f_j$. The family $\sigma_3 \cdot f$ is admissible at J, the interior being σ_3-symmetric, so $J_0(\sigma_3 \cdot f) = J_0(f)$. Let $k \to \infty$ (Theorem 7.6(b) for f at $J \mp 2$ and for $\sigma_3 \cdot f$ at J), then $J \to \infty$ (Corollary 4.4(ii)(β) at both ends): $Z^{K,\sigma_3\cdot f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for all $t \in \mathbb{R}^N$. The one-step sandwich of Lemma 4.5(a) would serve whenever $M \geq 2\varepsilon_k$, that is for all but finitely many k, which the k-limit ignores; the two-step form is used because it needs no case distinction, and the J-limit absorbs the extra step. This is the use King makes of the correlation inequalities in the proof of his Theorem 3.2 [16], with the kill steps added; no reversed inequality enters.

Step 3 (σ_1, σ_2). $\sigma_\nu = P\sigma_3 P^{-1}$ with P the transposition of the axes ν and 3. Since the transport is a left action, for every real 2-form u,

$$Z^K_\infty(\mathrm{i}\sigma_\nu\cdot u) = Z^K_\infty(\mathrm{i}P\cdot(\sigma_3 P^{-1}\cdot u)) = Z^K_\infty(\mathrm{i}\sigma_3\cdot(P^{-1}\cdot u)) = Z^K_\infty(\mathrm{i}P^{-1}\cdot u) = Z^K_\infty(\mathrm{i}u),$$

by Step 1 for the family $\sigma_3 P^{-1}\cdot u$, Step 2 for the family $P^{-1}\cdot u$, and Step 1 again; each step is a statement about every family.

Step 4 (the group, on the imaginary axes). Every $O \in O_h$ is a product of a permutation matrix and some of $\sigma_1, \sigma_2, \sigma_3$, and if $Z^K_\infty(\mathrm{i}O\cdot u) = Z^K_\infty(\mathrm{i}u)$ for all real u holds for O_1 and O_2, it holds for $O_1 O_2$. By Steps 1–3, $Z^{K,O\cdot f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for every $O \in O_h$, every family f and every $t \in \mathbb{R}^N$.

Step 5 (all coefficients). $D(z) := Z^{K,O\cdot f}_\infty(z) - Z^{K,f}_\infty(z)$ is entire on $\mathbb{C}^N$ (Theorem 7.6(b); $J_0(O\cdot f) = J_0(f)$) and vanishes on $(\mathrm{i}\mathbb{R})^N$, a set of uniqueness; so $D \equiv 0$. No splitting of sources is needed, there being no scalar source.

Step 6 (about every point). O_h about b is $\tau_b O \tau_{-b}$, and translations are Corollary 8.3 (every family, every $b \in \mathbb{R}^3$, all coefficients): $Z^K_\infty(\sum_j z_j \tau_b O\cdot f_j) = Z^K_\infty(\sum_j z_j O\cdot f_j) = Z^K_\infty(\sum_j z_j f_j)$.

Step 7 (Schwinger functions). Differentiating the identity of entire functions at $z = 0$ gives $S^K_n(\tau_b O\cdot f_1, \dots, \tau_b O\cdot f_n) = S^K_n(f_1, \dots, f_n)$ for all n and $(b, O) \in \mathbb{R}^3 \rtimes O_h$. □

Remark 8.6 (what cubic symmetry needs). King's cube is exactly symmetric under the coordinate permutations (Step 1) but not under the reflections; there the sandwich of Lemma 4.5 replaces exactness, as that of Lemma 2.14 does on the comparison convention. So O_h-invariance of King's printed state follows from the uniqueness of the monotone cube limit (Corollary 4.3 with Theorem 7.6(b)) together with the cut sandwich, and does not need the uniqueness of the infinite-volume state. Only class-$\mathcal{I}$ clauses enter, and no scheme or Wick constant.

8.4. **Rotations.**

Theorem 8.7 (rotations)**.** *Z^K_∞ is $\mathbb{R}^3 \rtimes O(3)$-invariant: for every finite family f, every $z \in \mathbb{C}^N$ and every $(b, R) \in \mathbb{R}^3 \rtimes O(3)$, $Z^K_\infty(\sum_j z_j \tau_b R\cdot f_j) = Z^K_\infty(\sum_j z_j f_j)$, and the Schwinger functions S^K_n are Euclidean covariant, F transforming as a 2-form.*

Proof. By Theorem 15.1 (the rotation theorem Theorem 3.14 on King's convention, at every point of King's range), $Z^K_\infty(\cdot, 0)$ is $O(3)$-invariant for all complex coefficients; translations are Corollary 8.3; the Schwinger functions follow by differentiation at $z = 0$. By Corollary 4.4(iii), with the fixed-cube limits of Theorem 3.3, the limit state $Z^{\mathrm{LIM}}_\infty(\cdot, 0)$ of the comparison family equals $Z^K_\infty(\cdot, 0)$ and is therefore $\mathbb{R}^3 \rtimes O(3)$-invariant as well. □

The rotation theorem is proved in the rotation part (Section 15, Theorem 15.1). Nothing in Corollary 8.3 and Theorem 8.5 depends on it. Its consequences for reflection positivity about every plane and for the components of the measure are drawn in Theorem 11.1(c).

9. Reflection positivity I: the modified box

King's Neumann–Dirichlet cube is not reflection symmetric about any plane, and the natural symmetrisation of its photon form is not reflection positive on gauge-invariant functionals (Remark 9.7). Reflection positivity is therefore proved first for the modified box state of [16, (3.38)] and then carried to the cube states by a two-sided comparison on the cone: the cube C_J sits inside the box (Lemma 9.9), and the box sits inside a larger cube once a margin proportional to μ_0^{-1} is allowed (Lemma 9.10). The passage to the limits is in Theorem 10.6. Throughout, $\varepsilon = \varepsilon_k$, the lattice is $\Lambda_k \subset \mathbb{R}^d$ with $d = 3$ (the certificates of Lemma 9.6 are also given for $d = 2$), a *lattice box* is a product $\Omega = \prod_\mu \{a_\mu, a_\mu + \varepsilon, \dots, b_\mu\} \subset \Lambda_k$, Ω^* is the set of its internal bonds (both ends in Ω) and Ω^{**} the set of its plaquettes (all four corners in Ω).

9.1. The lattice reflection and the modified box.

Definition 9.1 (mid-planes and the lattice reflection)**.** The *mid-planes* are the planes $\{x_\nu = c\}$ with $c \in D := \bigcup_{k\ge 0} \varepsilon_k(\mathbb{Z} + \frac12)$. If $c \in \varepsilon_{k_0}(\mathbb{Z} + \frac12)$ then $c \in \varepsilon_k(\mathbb{Z} + \frac12)$ for every $k \ge k_0$, because $5^j(m + \frac12) = 5^j m + \frac{5^j-1}{2} + \frac12$ and 5^j is odd; the same set D is obtained from any tail $k \ge k_1$ of the spacings. D is dense in $\mathbb{R}$ but not all of $\mathbb{R}$.

Fix ν, a mid-plane $c \in \varepsilon_k(\mathbb{Z}+\frac12)$ and a lattice box $\Omega \subset \Lambda_k$ with $\rho\Omega = \Omega$, $\rho = \rho_{\nu,c}$ the reflection of Definition 2.4. On the fields put $(\hat\rho\phi)(x) := \phi(\rho x)$ and $(\hat\rho A)(b) := A(\rho b)$, the image bond taken with its orientation: $(\hat\rho A)_\alpha(x) = A_\alpha(\rho x)$ for $\alpha \ne \nu$ and $(\hat\rho A)_\nu(x) = -A_\nu(\rho x - \varepsilon e_\nu)$. Then $\hat\rho$ is a signed permutation of the variables attached to Ω and Ω^* (it preserves Lebesgue measure and maps Ω^*, Ω^{**} to themselves); the *crossing bonds* $B_0 := \{\langle x^-, x^- + \varepsilon e_\nu\rangle : \ x^-_\nu = c - \varepsilon/2\}$ are mapped to their reversals, $(\hat\rho A)(b) = -A(b)$ for $b \in B_0$. Let $\Omega_\pm := \{x \in \Omega : \ \pm(x_\nu - c) > 0\}$ (no site lies on the plane), B_+ the bonds with both ends in Ω_+, $\mathfrak{B}_+$ the Borel functions of $(\phi|_{\Omega_+}, A|_{B_+})$, and

$$\theta G := \overline{G \circ \hat\rho} \qquad (G \in \mathfrak{B}_+).$$

On the smeared fields, $F_{\Lambda_k}(f)\circ\hat\rho = F_{\Lambda_k}(\rho^* f)$ and $W^a_{\Lambda_k}(h)\circ\hat\rho = W^a_{\Lambda_k}(h\circ\rho)$. Indeed, $F(p)\circ\hat\rho = \sigma_p F(\rho p)$ with $\sigma_p = -1$ exactly when $\nu \in \{\alpha, \beta\}$ (the orientation of the image plaquette, whose base point is shifted back one layer), the centres satisfy $c(\rho p) = \rho c(p)$, and $(\rho^* f)_{c(\rho p)}(e_\alpha, e_\beta) = f_{c(p)}(\rho e_\alpha, \rho e_\beta) = \sigma_p f_{c(p)}(e_\alpha, e_\beta)$; reindexing $p \mapsto \rho p$ gives the first identity. The scalar case is a relabelling of sites, and the c-numbers $2\mathsf{W}_\varepsilon\mathsf{S}(h)$ and $a_k\mathsf{S}(h)$ are invariant because ρ permutes Λ_k ($2(x_\nu - c) \in \varepsilon\mathbb{Z}$). Consequently, for families ζ_i whose lattice observables $\Theta(\zeta_i)$ lie in $\mathfrak{B}_+$,

$$\theta(\Theta(\zeta_i))\,\Theta(\zeta_j) = \Theta(\theta\zeta_i^* \sqcup \zeta_j), \qquad m(\theta\zeta_i^* \sqcup \zeta_j) = \overline{m(\zeta_i)}\, m(\zeta_j) \tag{18}$$

for the scheme multiplier $m(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}(h_j)}$ (a_k real), so that for any state $\langle\cdot\rangle$ on the box

$$\sum_{i,j} \bar\gamma_i\gamma_j\, m(\theta\zeta_i^* \sqcup \zeta_j)\langle\Theta(\theta\zeta_i^* \sqcup \zeta_j)\rangle = \sum_{i,j} \overline{(\gamma_i m(\zeta_i))}\,(\gamma_j m(\zeta_j))\,\big\langle\theta(\Theta(\zeta_i))\,\Theta(\zeta_j)\big\rangle. \tag{19}$$

Support. If the sources of ζ_i are supported in $\{x_\nu \ge c+\delta\}$ with $\delta > 0$, then for $\varepsilon_k < 2\delta$ every site carrying a sampled h and every corner of every plaquette carrying a sampled f (centre in the support, corners at $x_\nu \ge c+\delta-\varepsilon_k/2 > c$) lies in Ω_+, so $\Theta(\zeta_i) \in \mathfrak{B}_+$. The splitting $h = h^+ - h^-$ of Definition 10.1 is taken with cutoffs supported in $\{x_\nu \ge c + \delta/2\}$, so that $h^\pm$ are supported in $\{x_\nu > c\}$ as well.

Definition 9.2 (the modified box state)**.** Let $\Omega \subset \Lambda_k$ be a lattice box, $F(p) = \varepsilon^{-1}\sum_{b\in\partial p} A(b)$, $\Omega(y)$ the indicator of Ω, and x^* the nearest neighbours of x in Λ_k. King's modified action [16, (3.38)] on Ω is

$$\begin{aligned} S^\varepsilon_\Omega(A,\phi) = {}& \tfrac12 \sum_{p\in\Omega^{**}} \varepsilon^d |F(p)|^2 + \tfrac12\mu_0^2 \sum_{b\in\Omega^*} \varepsilon^d |A(b)|^2 + \tfrac12 \sum_{x\in\Omega} \varepsilon^d \Big| \sum_{y\in x^*} \varepsilon^{-1}\Omega(y) A(\langle x,y\rangle)\Big|^2 \\ & + \tfrac12 \sum_{\langle x,y\rangle} \varepsilon^{d-2}\big[(\Omega r)(x)^2 + (\Omega r)(y)^2 - 2(\Omega r)(x)(\Omega r)(y)\cos(\vartheta_x - e\varepsilon A(\langle x,y\rangle) - \vartheta_y)\big] \end{aligned}$$

$$+\sum_{x\in\Omega}\varepsilon^d\big[\tfrac12 m^2 r_x^2+\lambda r_x^4+\tfrac12\delta m_\varepsilon^2\, r_x^2\big],$$

with the *constant* free-boundary counterterm δm_ε^2 of Definition 2.2 at every site, and with the free Wick constant W_ε in the Wick observables. (King prints a site-dependent counterterm in the last term; it is not used here, and the statements below are made for the constant one.) The photon variables are the internal bonds Ω^* only. The matter part is exactly the Dirichlet matter of the Neumann–Dirichlet state on $X=\Omega$: cosines on the internal bonds, and the diagonal $\frac12\varepsilon^{d-2}r_x^2$ for every bond leaving Ω. The *modified box state* $\langle\cdot\rangle_{k,\Omega}^{(3.38)}$ is the normalised measure $e^{-S_\Omega^\varepsilon}\prod_{x\in\Omega}d^2\phi_x\prod_{b\in\Omega^*}dA_b$; it is a BFS measure (the Gaussian is non-degenerate since $\mu_0>0$, the single-site measures have all exponential moments since $\lambda>0$). We write $Z_{k,\Omega}^{a,(3.38)}(\zeta):=m(\zeta)\langle\Theta(\zeta)\rangle_{k,\Omega}^{(3.38)}$ with the same observables and multiplier as in Definition 2.4.

The third term is the graph divergence of (Ω,Ω^*): with $d_0:\mathbb{R}^\Omega\to\mathbb{R}^{\Omega^*}$, $(d_0\chi)(\langle x,x+\varepsilon e_\alpha\rangle)=\varepsilon^{-1}(\chi(x+\varepsilon e_\alpha)-\chi(x))$, one has $\sum_{y\in x^*}\Omega(y)A(\langle x,y\rangle)=-\varepsilon(d_0^TA)(x)$ (only internal bonds contribute). With $d_1A:=(\varepsilon F(p))_{p\in\Omega^{**}}$, the *unitary form* on the box and the modified form are

$$Q_\Omega^u(A):=\tfrac12\sum_{p\in\Omega^{**}}\varepsilon^d|F(p)|^2+\tfrac12\mu_0^2\sum_{b\in\Omega^*}\varepsilon^d|A(b)|^2,\qquad Q_\Omega^{(3.38)}=Q_\Omega^u+\tfrac12\varepsilon^d\|d_0^TA\|^2,$$

i.e. the modified form is the unitary form plus a gauge-fixing term depending on A only through d_0^TA. The modified action is invariant under $\hat\rho$ for every mid-plane c with $\rho\Omega=\Omega$: every term is written through $A(\langle x,y\rangle)$ and nearest-neighbour pairs, which ρ permutes, $(d_0^T\hat\rho A)(x)=(d_0^TA)(\rho x)$, and δm_ε^2 is site independent.

The boxes. Fix $c\in D$, $k\ge k_0(c)$, $J\ge J_0$ such that every source of the enlarged family of Definition 10.1 and its reflection is supported at ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_{J_0}$, an integer $n\ge1$, and put $a_k:=c+\varepsilon_k/2\in\varepsilon_k\mathbb{Z}$ and

$$\Omega_{J,k}:=(C_{J+n}+a_ke_\nu)\cap\Lambda_k,\qquad Mn\ \ge\ |c|+2+R(\mu_0),\qquad R(\mu_0):=\frac{\pi\sqrt3}{2\mu_0}.\tag{20}$$

$C_{J+n}\cap\Lambda_k$ is symmetric in direction ν about $-\varepsilon_k/2$ (sites $x_\nu\in\{-M(J+n),\dots,M(J+n)-\varepsilon_k\}$, and $M(J+n)\in\varepsilon_k\mathbb{Z}$), so $\Omega_{J,k}$ is a lattice box symmetric about $\{x_\nu=c\}$ with an even number $2M(J+n)5^k$ of layers. Under the margin condition of (20), for every k: (i) C_J plus one layer is contained in $\Omega_{J,k}$ (this needs $Mn\ge|c|+\frac32\varepsilon_k$); (ii) C_{J+2n} contains every site at ℓ^∞-distance $\le R(\mu_0)+\varepsilon_k$ from $\Omega_{J,k}$ (this needs $Mn\ge|c|+R(\mu_0)+\frac32\varepsilon_k$). We fix such an $n=n(\mu_0,c)$, independent of k and J. The two margins are exactly the hypotheses of Lemmas 9.9 and 9.10 below; by Lemma 4.6 they serve King's convention as they stand.

9.2. **Reflection positivity of the modified box.**

Lemma 9.3 (lattice reflection positivity of the modified box)**.** *Let $\mu_0>0$, c a mid-plane and Ω a lattice box with $\rho\Omega=\Omega$, $\rho=\rho_{\nu,c}$.*

(a) *For every gauge-invariant $G\in\mathfrak{B}_+$ that is bounded, or of the form $\Theta(\zeta)$ with sources supported in Ω_+, or more generally with $\langle|G|^2\rangle_{k,\Omega}^{(3.38)}<\infty$: $\langle\theta(G)G\rangle_{k,\Omega}^{(3.38)}\ge0$.*

(b) *$(G,G')\mapsto\langle\theta(G')G\rangle_{k,\Omega}^{(3.38)}$ is a Hermitian positive semidefinite form on such G; hence $\sum_{i,j}\bar\gamma_i\gamma_j\langle\theta(G_i)G_j\rangle_{k,\Omega}^{(3.38)}\ge0$, and by (19) and the support clause of Definition 9.1, $\sum_{i,j}\bar\gamma_i\gamma_jZ_{k,\Omega}^{a,(3.38)}(\theta\zeta_i^*\sqcup\zeta_j)\ge0$ for families ζ_i with sources supported in $\{x_\nu\ge c+\delta\}$, $\varepsilon_k<2\delta$.*

Proof. Step 1: gauge-invariant expectations of the modified form equal those of the unitary form. The graph (Ω,Ω^*) is connected, $\ker d_0$ consists of the constants, $\mathbb{R}^{\Omega^*}=\operatorname{ran}d_0\oplus\ker d_0^T$ orthogonally, and $d_1d_0=0$. Write $A=A^T+d_0\chi$ with $d_0^TA^T=0$ and $\chi\perp1$; $(A^T,\chi)\mapsto A$ is a linear bijection with constant Jacobian. Then $Q_\Omega^u(A)=Q_\Omega^u(A^T)+\frac{\mu_0^2\varepsilon^d}{2}\|d_0\chi\|^2$ (orthogonality and $d_1d_0=0$) and $Q_\Omega^{(3.38)}(A)=Q_\Omega^u(A^T)+q(\chi)$ with $q(\chi)=\frac{\mu_0^2\varepsilon^d}{2}\|d_0\chi\|^2+\frac{\varepsilon^d}{2}\|d_0^Td_0\chi\|^2$;

both χ-forms are positive definite on $1^\perp$ because $\mu_0 > 0$. A gauge transformation is $A \mapsto A + d_0\chi$ together with $\vartheta \mapsto \vartheta - e\chi$: the hopping phase $\vartheta_x - e\varepsilon A_b - e(\chi(y) - \chi(x)) - \vartheta_y$ equals $(\vartheta_x + e\chi_x) - e\varepsilon A_b - (\vartheta_y + e\chi_y)$ and the Dirichlet diagonals carry no A, so for gauge-invariant G the matter integral $\Psi(A) := \int \prod_x d^2\phi_x\, (Ge^{-S_m})(A,\phi)$ satisfies $\Psi(A^T + d_0\chi) = \Psi(A^T)$ (rotate ϕ site by site; $\prod d^2\phi$ is invariant). Hence $\int dA\, e^{-Q_\Omega^{(3.38)}}\Psi = \int dA^T e^{-Q_\Omega^u(A^T)}\Psi(A^T) \int d\chi\, e^{-q(\chi)}$, and the same with Q_Ω^u in place of $Q_\Omega^{(3.38)}$ and a different finite χ-factor; the factors cancel in the normalised expectation. Fubini applies, since $|G| \le Ce^{a\sum_b |A(b)|} e^{C\sum_x r_x^2}$ for $G = \Theta(\zeta)$ (trivially for bounded G) is dominated by the Gaussian and by $e^{-\lambda\varepsilon^d r^4}$. This is the mechanism of [7, Thm. 5.1, p. 270], carried to the box graph.

Step 2: the unitary box measure is reflection positive on all of $\mathfrak{B}_+$. Let $S_\Omega^u := Q_\Omega^u + S_m$, write $\omega = (\omega_+, \omega_-, a)$ with $a = (a_b)_{b\in B_0}$ the crossing-bond variables, and put $\mathsf{M} := \mu_0^2\varepsilon^d$. The terms of S_Ω^u are of four kinds. Single-site terms (local matter and Dirichlet diagonals), hopping cosines on bonds of $B_\pm$, masses of bonds of $B_\pm$, and plaquettes with all corners in Ω_+ or all in Ω_- involve one side only; by $\hat\rho$-invariance the minus-side terms are the $\hat\rho$-transport of the plus-side terms. The crossing masses are $\frac{\mathsf{M}}{2}a_b^2$, $b \in B_0$. A crossing plaquette $p = (x^-; \alpha, \nu)$, $\alpha \ne \nu$, is present iff $x^-, x^- + \varepsilon e_\alpha \in \Omega$ (then all four corners are in Ω by symmetry); with $x^+ := x^- + \varepsilon e_\nu$, $b = \langle x^-, x^+\rangle$ and $b' = \langle x^- + \varepsilon e_\alpha, x^+ + \varepsilon e_\alpha\rangle$ it reads $\varepsilon F(p) = \pm[A_\alpha(x^-) + a_{b'} - A_\alpha(x^+) - a_b]$, the overall sign immaterial after squaring. A crossing hopping term is $-\varepsilon^{d-2} r_{x^-} r_{x^+} \cos(\vartheta_{x^-} - e\varepsilon a_b - \vartheta_{x^+})$, $b \in B_0$.

Split each crossing variable: $\int_{\mathbb{R}} g(a_b) e^{-\frac{\mathsf{M}}{2}a_b^2} da_b = \sqrt{2\pi/\mathsf{M}}\, \mathbb{E}\, g(\alpha_b - \alpha_b')$ with α_b, α_b' independent $N(0, \frac{1}{2\mathsf{M}})$, whose difference is $N(0, \frac{1}{\mathsf{M}})$. Put $\xi := (\omega_+, \alpha)$ and $\xi' := (\omega_-^\rho, \alpha')$, where ω_-^ρ is the $\hat\rho$-image of the minus data transported to the plus side ($(\omega_-^\rho)_\alpha(x^+) = A_\alpha(x^-)$ for the tangential bonds, $\phi^\rho(x^+) = \phi(x^-)$, the ν-bonds of B_- carried with the sign of Definition 9.1); by $\hat\rho$-invariance the minus-side one-sided terms are the plus-side terms evaluated on ω_-^ρ, and the a priori measures of ξ and ξ' coincide. For a crossing plaquette put $X_p(\xi) := A_\alpha(x^+) - \alpha_{b'} + \alpha_b$, a function of plus variables and α (note $A_\alpha(x^+) \in B_+$ since $x^+, x^+ + \varepsilon e_\alpha \in \Omega_+$), and the same function on ξ', $X_p(\xi') = A_\alpha(x^-) - \alpha_{b'}' + \alpha_b'$; then $\varepsilon F(p) = \pm[X_p(\xi') - X_p(\xi)]$ and

$$\tfrac{\varepsilon^d}{2} F(p)^2 = \tfrac{\varepsilon^{d-2}}{2} X_p(\xi)^2 + \tfrac{\varepsilon^{d-2}}{2} X_p(\xi')^2 - \varepsilon^{d-2} X_p(\xi) X_p(\xi').$$

For a crossing hopping term put $\psi_b(\xi) := \phi(x^+) e^{\mathrm{i}e\varepsilon\alpha_b}$, $\psi_b(\xi') = \phi(x^-) e^{\mathrm{i}e\varepsilon\alpha_b'}$; then $r_{x^-} r_{x^+} \cos(\vartheta_{x^-} - e\varepsilon(\alpha_b - \alpha_b') - \vartheta_{x^+}) = \mathrm{Re}[\psi_b(\xi')\overline{\psi_b(\xi)}] = \sum_{i=1,2} \psi_b^i(\xi)\psi_b^i(\xi')$. Hence, with S' the action minus the crossing masses,

$$e^{-S'} = \mathcal{G}(\xi)\, \mathcal{G}(\xi') \exp\Big(\sum_\iota \gamma_\iota Y_\iota(\xi) Y_\iota(\xi') \Big), \qquad \mathcal{G}(\xi) := e^{-S_+(\xi) - \frac{\varepsilon^{d-2}}{2}\sum_p X_p(\xi)^2},$$

where S_+ collects the one-sided plus terms and $\{(\gamma_\iota, Y_\iota)\} = \{(\varepsilon^{d-2}, X_p)\}_p \cup \{(\varepsilon^{d-2}, \psi_b^i)\}_{b,i}$: all $\gamma_\iota > 0$, all Y_ι real. For $G \in \mathfrak{B}_+$ we have $G(\omega) = G(\xi)$ and $\theta G(\omega) = \overline{G(\xi')}$, the same function on the transported data (G depends neither on a nor on α, α'). Expanding the exponential,

$$Z\, \langle \theta(G) G\rangle_{k,\Omega}^u = K \sum_{n\in\mathbb{N}^I} \prod_\iota \frac{\gamma_\iota^{n_\iota}}{n_\iota!} \Big| \int d\xi\, G(\xi)\mathcal{G}(\xi) \prod_\iota Y_\iota(\xi)^{n_\iota} \Big|^2 \ge 0, \qquad K > 0,$$

the two integrals of each term being complex conjugates because measures and functions coincide on the two copies. The interchange of sum and integral is justified by $\sum_n \prod_\iota \gamma_\iota^{n_\iota} |Y_\iota Y_\iota'|^{n_\iota}/n_\iota! \le \exp(\frac{1}{2}\sum_\iota \gamma_\iota(Y_\iota^2 + Y_\iota'^2))$: the plaquette parts cancel against $\mathcal{G}$ exactly, $|\psi_b| = r_{x^+}$, and $\int |G| e^{-S_+(\xi) + \frac{\varepsilon^{d-2}}{2}\sum_b r_{x^+}^2} d\xi <$ ∞ by $\lambda > 0$ (a quartic term at every plus site), $\mu_0 > 0$ (a mass on every B_+ bond and the α-Gaussians) and the growth bound on G. The argument is the positive cross-term criterion of [10, 9], with the crossing bond variables split into two independent Gaussians and the Dirichlet diagonals as single-site terms. The Hermitian property $\overline{\langle\theta(G')G\rangle} = \langle (G'\circ\hat\rho)\bar G\rangle = \langle G'\, \overline{G\circ\hat\rho}\rangle = \langle\theta(G)G'\rangle$ follows from the $\hat\rho$-invariance of the measure.

Step 3: combination and L^2 extension. For gauge-invariant $G \in \mathfrak{B}_+$, bounded or equal to $\Theta(\zeta)$, the product $\theta(G)G$ is gauge invariant ($\hat\rho$ maps gauge transformations to gauge transformations: $\hat\rho\, d_0\chi = d_0(\chi\circ\rho)$), so $\langle\theta(G)G\rangle^{(3.38)}_{k,\Omega} = \langle\theta(G)G\rangle^{u}_{k,\Omega} \geq 0$ by Steps 1–2. For gauge-invariant $G \in \mathfrak{B}_+$ with $\langle|G|^2\rangle < \infty$ (for instance polynomials in the plus-side smeared fields, Proposition 10.10), $G_n := G\,\mathbf{1}\{|G| \leq n\}$ is bounded, gauge invariant and in $\mathfrak{B}_+$, so $\langle\theta(G_n)G_n\rangle \geq 0$; moreover $|\theta(G_n)G_n| \leq |\theta G||G|$ and $\langle|\theta G||G|\rangle \leq \langle|G|^2\rangle^{1/2}\langle|\theta G|^2\rangle^{1/2} = \langle|G|^2\rangle$ by Cauchy–Schwarz and $\hat\rho$-invariance, so dominated convergence gives (a). The modified measure is itself $\hat\rho$-invariant (Definition 9.2), so the computation of the Hermitian property in Step 2 holds verbatim for it on L^2; the form is sesquilinear, Hermitian and nonnegative on the diagonal, hence positive semidefinite (apply (a) to $G = \sum_i \gamma_i G_i$). With (19) and the support clause this is (b). □

Remark 9.4 (on King's Proposition 3.3). Lemma 9.3 is the statement of [16, Prop. 3.3], which King derives from [7, Cor. 5.4]. That corollary (with [7, Lemma 5.2, Thm. 5.3]) concerns the lattice actions of [7, §5], whose photon Gaussian is translation invariant with a reflection-positive boundary term; King's finite-box Gaussian with its masked divergence is not one of them, the obstruction being the box geometry of the gauge-fixing term. So his Proposition 3.3 is true, and it is proved here rather than derived from [7, Cor. 5.4]. The gauge-fixing term $\|d_0^T A\|^2$ is not itself compatible with the mechanism of Step 2 (at a site next to the plane it produces cross terms $Y(\xi)Z(\xi') + Z(\xi)Y(\xi')$ of indefinite sign); on the torus it drops out for gauge-invariant observables, on King's cube it does not reduce to a divergence at the boundary (Lemma 9.5), and this is why the photon form has to be changed before reflecting (Remark 9.7).

9.3. **The same box: the boundary identity and the reversed inequality.** In this subsection we use lattice units, $\varepsilon = 1$ (every term of the three photon forms carries the same weight $\frac{1}{2}\varepsilon^{d-2}$), and write $\mu := \mu_0\varepsilon$. For a lattice box Ω and $A \in \mathbb{R}^{\Omega^*}$ let $K^{\mathrm{int}}_\Omega(A) := \sum_\mu \sum_\nu \sum |A_\mu(x + e_\nu) - A_\mu(x)|^2$, the sum over pairs with both bonds $(x,\mu),(x+e_\nu,\mu)$ in Ω^* (the componentwise Neumann gradient form on the internal bonds), and $N_\partial(A) := \sum |A_b|^2$, the sum over the normal crossing pairs: for $b = (x,\mu) \in \Omega^*$ one term for each of $(x \pm e_\mu, \mu)$ that is not in Ω^*. Split $N_\partial = N^{\mathrm{up}}_\partial + N^{\mathrm{low}}_\partial$ according to whether the missing partner is $(x+e_\mu,\mu)$ (i.e. $x + 2e_\mu \notin \Omega$) or $(x-e_\mu,\mu)$ (i.e. $x - e_\mu \notin \Omega$).

Lemma 9.5 (the boundary identity)**.** *For every lattice box Ω and every $A \in \mathbb{R}^{\Omega^*}$,*

$$\|d_1 A\|^2_{\Omega^{**}} + \|d_0^T A\|^2 = K^{\mathrm{int}}_\Omega(A) + N_\partial(A).$$

Consequently $Q^{(3.38)}_\Omega = \frac{1}{2}\big[K^{\mathrm{int}}_\Omega + N_\partial + \mu^2\|\cdot\|^2_{\Omega^*}\big]$: the modified photon form is the internal Neumann–Feynman form plus the positive boundary term N_∂. (By the proof of Lemma 9.6(a), the marginal of the Neumann–Dirichlet Gaussian of Ω on the internal bonds has the precision $M^{(3.38)} - N^{\mathrm{low}}_\partial - P_{12}P_{22}^{-1}P_{21}$: the modified form minus the squares of the near-normal bonds and minus the Schur term of the far-face variables.)

Proof. Extend A by 0 to a 1-form a on $\mathbb{Z}^d$. For finitely supported 1-forms, $\sum_p |da(p)|^2 + \sum_x |d^*a(x)|^2 = \sum_\mu \sum_\nu \sum_x |a_\mu(x+e_\nu) - a_\mu(x)|^2$ (the Hodge Laplacian on $\mathbb{Z}^d$ is the componentwise Laplacian; summation by parts). On the left, $d^*a(x)$ is the masked divergence for $x \in \Omega$ and 0 outside, since no internal bond has an end outside Ω; a plaquette $p \notin \Omega^{**}$ meeting Ω^* has exactly one internal bond b (two parallel internal edges put all four corners in Ω; two adjacent ones put three corners in Ω, and a product of intervals then contains the fourth), so $|da(p)|^2 = |A_b|^2$. On the right, pairs with both bonds internal give K^{int}_Ω; a pair $(x,\mu),(x+e_\nu,\mu)$ with exactly one internal bond contributes $|A_b|^2$; for $\nu \neq \mu$ the partner is non-internal iff $x + e_\nu \notin \Omega$ (as $x, x+e_\mu \in \Omega$, and in a box $x + e_\nu \in \Omega \iff x + e_\mu + e_\nu \in \Omega$), i.e. iff the plaquette spanned by b and e_ν lies outside Ω^{**}; these pairs are in bijection with the plaquettes counted on the left and cancel; for $\nu = \mu$ they are the terms of N_∂. □

Lemma 9.6 (the same box: the second inequality of King's (3.42) is reversed)**.** *Let Ω be a lattice box, $\langle\cdot\rangle^{\mathrm{ND}}_\Omega$ the Neumann–Dirichlet state on $X = \Omega$ (variables $A_\mu(x)$, $x \in \Omega$, all outgoing bonds; photon form $Q^{\mathrm{ND}}_\Omega = \frac{1}{2}[K_\Omega + \mu^2\|\cdot\|^2]$ of Definition 2.1) and $\langle\cdot\rangle^{(3.38)}_\Omega$ the modified box*

state of Definition 9.2 (variables Ω^*); both carry the same Dirichlet matter on Ω (constant δm_ε^2, free Wick constant), coupled to A through the internal bonds only. Let $\Sigma_\Omega^{\mathrm{ND}}$ and Σ_Ω^u be the covariances of $F = (d_1 A(p))_{p\in\Omega^{**}}$ under the two photon Gaussians. Then:

(a) $\Sigma_\Omega^u = \Sigma_\Omega^{(3.38)} \le \Sigma_\Omega^{\mathrm{ND}}$ *on* $\mathbb{R}^{\Omega^{**}}$;

(b) *for every gauge-invariant $Y \in \mathcal{I}$ with sources in Ω and every $e \ge 0$: $\langle Y\rangle_\Omega^{\mathrm{ND}} \le \langle Y\rangle_\Omega^{(3.38)}$, and $\langle X\rangle_\Omega^{\mathrm{ND}} \ge \langle X\rangle_\Omega^{(3.38)}$ for $X \in \mathcal{D}$. This is the reverse of the second inequality of* [16, (3.42)], *whose middle term is the modified box state on C_{J+1}; it holds weakly, for the modified state with the constant δm_ε^2 (nothing is asserted for King's printed site-dependent counterterm);*

(c) *King's inequality fails strictly in the Gaussian sector: at $e = 0$ and $Y = e^{\mathrm{i}t\langle v,F\rangle}$ it reads $v^T\Sigma_\Omega^{\mathrm{ND}} v \le v^T\Sigma_\Omega^u v$, while $\Delta_\Omega := \Sigma_\Omega^{\mathrm{ND}} - \Sigma_\Omega^u \ge 0$ is nonzero. In exact rational arithmetic the inertia (negative, zero, positive) of Δ_Ω is $(0,1,8)$ on the 4×4 box ($d=2$) for $\mu^2 \in \{\frac{1}{20}, \frac{1}{4}, 1\}$ and $(0,3,13)$ on the $4\times2\times2$ box ($d=3$) for $\mu^2 \in \{\frac{1}{20}, \frac{1}{4}\}$; single-plaquette certificates are $\Delta_{pp} = \mathrm{Var}^{\mathrm{ND}} F(p) - \mathrm{Var}^{(3.38)} F(p) = \frac{14285248}{44314767}$ at $p = ((0,0);0,1)$ on 4×4, $\mu^2 = \frac{1}{4}$, and $\frac{121897601600}{395580222961}$ at $p = ((0,0,0);1,2)$ on $4\times2\times2$, $\mu^2 = \frac{1}{20}$.*

Proof. (a) Write the Neumann–Dirichlet precision $P := K_\Omega + \mu^2$ in blocks, the internal variables Ω^* first and the outgoing boundary variables second. Its internal block is $P_{11} = K_\Omega^{\mathrm{int}} + N_\partial^{\mathrm{up}} + \mu^2$: pairs of K_Ω with both bonds internal give K_Ω^{int}; a pair $(x,\mu),(x+e_\nu,\mu)$ with exactly one internal bond is impossible for $\nu \ne \mu$ (box convexity, as in Lemma 9.5) and for $\nu = \mu$ is (x,μ) internal with $(x+e_\mu,\mu)$ outgoing, i.e. $x + 2e_\mu \notin \Omega$ (the other assignment would need $x + e_\mu \notin \Omega$ and $x + e_\mu \in \Omega$ at once); its diagonal contribution to the internal block is $|A_\mu(x)|^2$, which is N_∂^{up}. Hence, by Lemma 9.5, $M^{(3.38)} := K_\Omega^{\mathrm{int}} + N_\partial + \mu^2 = P_{11} + N_\partial^{\mathrm{low}} \ge P_{11}$. The marginal covariance of the internal variables under the Neumann–Dirichlet Gaussian is the inverse of the Schur complement, $(P^{-1})_{11} = (P_{11} - P_{12}P_{22}^{-1}P_{21})^{-1} \ge P_{11}^{-1} \ge (M^{(3.38)})^{-1}$, by operator monotonicity of the inverse on positive definite matrices and $P_{12}P_{22}^{-1}P_{21} \ge 0$. Since $F = d_1 A$ on Ω^{**} uses internal bonds only, $\Sigma_\Omega^{\mathrm{ND}} = D_1(P^{-1})_{11}D_1^T \ge D_1(M^{(3.38)})^{-1}D_1^T = \Sigma_\Omega^{(3.38)}$, and $\Sigma_\Omega^{(3.38)} = \Sigma_\Omega^u$ by Step 0 in the proof of Lemma 9.10 (the field strength depends only on the $\ker d_0^T$-component, whose law is the same under both Gaussians, Step 1 of Lemma 9.3). (b) Both states carry the Dirichlet matter of Ω coupled to Ω^*, and their field-strength laws on Ω^{**} are ordered by (a); the complete proof is given in Lemma K.2 of the full version. (c) At $e = 0$ the matter decouples from A and is identical in the two states, so for $Y = e^{\mathrm{i}t\langle v,F\rangle}$ both sides equal $\exp(-\frac{t^2}{2}v^T\Sigma v)$ with the respective Σ; King's $\le$ becomes $v^T\Sigma_\Omega^{\mathrm{ND}} v \le v^T\Sigma_\Omega^u v$, which is false for every v with $v^T\Delta_\Omega v > 0$, for instance $v = e_p$ at the plaquettes listed. The inertias and the certificates are obtained from the two covariance matrices in exact rational arithmetic (symmetric elimination with diagonal pivoting over $\mathbb{Q}$). □

Scope. (i) King's passage from the box states to the cube limit on [16, p. 336] needs the sandwich for the box functionals with purely imaginary curvature coefficients, i.e. for the whole class $\mathcal{I}$; (c) refutes the second inequality on that class exactly. (ii) For the scalar observable $\Theta(0, 2h_+)$ of the printed display, (b) shows that the inequality can hold only with equality; at $e = 0$ it does (identical decoupled matter), and at $e > 0$ a strict failure is to be expected (the Wick square couples to F at order e^2) but is not certified here. Both clauses are for the modified state with the constant δm_ε^2: if King's site-dependent counterterm differs from the free-boundary number, the two matter measures differ, the mechanism of (b) does not apply, and neither clause is proved for his printed objects; (c) alone is independent of the counterterm, the matter decoupling from A at $e = 0$.

Remark 9.7 (what is false in the printed route). (i) King's Neumann–Dirichlet cube state is not reflection symmetric about any plane: the site set $C_J \cap \Lambda_k$ is symmetric in direction ν about $x_\nu = -\varepsilon_k/2$, the set of ν-bonds about $x_\nu = 0$, and no reflection maps the variable set to itself (Lemma 2.14 accordingly obtains the reflection symmetry of the limit from a cut sandwich, not from a lattice symmetry). (ii) The natural symmetrisation, the Feynman gradient form on the internal bonds $K_\Omega^{\mathrm{int}} + \mu^2$ (the modified form without N_∂; reflection symmetric), is

not reflection positive on gauge-invariant functionals. This is exact at $e = 0$: on the 4×4 box at $\mu^2 = \frac{1}{20}$ the reflection kernel $R_{pq} = \sigma_p \Sigma[\rho p, q]$ of the plus-side plaquettes is negative definite (inertia $(3,0,0)$), and on $4 \times 2 \times 2$ at $\mu^2 = \frac{1}{20}$ it has inertia $(2,2,2)$ with $R_{pp} = \mathrm{Cov}(\theta F(p), F(p)) < 0$ for $p = ((2,0,0);0,1)$. Indeed, at $e = 0$ the matter decouples, and the gauge-invariant plus-side functionals $G_1 = e^{\mathrm{i}t\langle v, F_+\rangle}$, $G_2 = 1$ give the matrix $(\langle \theta(G_i) G_j\rangle)$ with determinant $e^{-t^2 v^T \Sigma_{++} v}(e^{t^2 v^T R v} - 1) < 0$ for every $t > 0$ when $v^T R v < 0$, since $\langle \theta(G_1) G_1\rangle = \exp(-t^2 v^T \Sigma_{++} v + t^2 v^T R v)$ and $\langle \theta(G_1)\rangle = \langle G_1\rangle = \exp(-\frac{1}{2} t^2 v^T \Sigma_{++} v)$ by the reflection symmetry $\Sigma_{--} = \Sigma_{++}$. By continuity at fixed spacing the symmetrised cube state is not reflection positive for small $e > 0$ either. A proof must work at every coupling, so reflection positivity of King's own cube form, or of this symmetrisation, at fixed spacing is not available as a route: any route through cube states at fixed spacing has to change the photon form, as the modified box does (a reflection-symmetric, reflection-positive modification whose values converge to the same limit, Theorem 10.6). Nothing is claimed about the symmetrised form at a given larger e or about other proofs; the route taken here needs the comparison of Lemmas 9.9 and 9.10, and its reflecting plane must be a lattice mid-plane for the half-bond mechanism of Lemma 9.3 (King's cubes have an even number of layers). (iii) Accordingly, the second inequality of [16, (3.42)] is false as printed in the following sense: false for the class of observables that the argument on [16, p. 336] needs (exactly, Lemma 9.6(c)), and never strictly true for the modified state with the constant counterterm (Lemma 9.6(b)). Its valid first half is Lemma 9.9; the invalid half is replaced by Lemma 9.10 with the margin $R(\mu_0)$ of (20), $n(\mu_0, c)$ cube steps instead of one.

Remark 9.8 (the box form is the larger one). King describes the modified box action as obtained from the cube action by omitting some terms. By Lemma 9.5 the relation is the reverse: on one box the modified photon form is the internal Neumann–Dirichlet form P_{11} plus the positive term $N_\partial^{\mathrm{low}}$ and without the outgoing boundary variables, so the modified state is the one with the *larger* form, the smaller field-strength covariance, and the larger $\mathcal{I}$-expectations: $\langle Y\rangle_C^{\mathrm{ND}} \le \langle Y\rangle_C^{(3.38)}$ on the cone for every box C (Lemma 9.6(b)), the direction that Lemma 2.11 requires for consistency with Lemmas 9.9 and 9.10. For King's convention the same comparison holds with Step 1 of Lemma 4.2 prepended (Remark 4.8): $\langle Y\rangle_{k,J}^{K} \le \langle Y\rangle_{k,J}^{(a)} \le \langle Y\rangle_{k,X_J}^{(3.38)}$ on $\mathcal{I}$.

9.4. **The sandwich.**

Lemma 9.9 (the lower half)**.** *Let $C \subset \Omega$ be lattice boxes with C plus one layer contained in Ω, so that the variables $A_\mu(x)$, $x \in C$, of the Neumann–Dirichlet state on C (the outgoing bonds) are internal bonds of Ω. Then*

(a) $Q_\Omega^{(3.38)} \ge Q_C^{\mathrm{ND}} \oplus \frac{\mu^2}{2} I_{\mathrm{rest}}$ *as quadratic forms on* $\mathbb{R}^{\Omega^*} = \mathbb{R}^{C\text{-variables}} \oplus \mathbb{R}^{\mathrm{rest}}$;

(b) *for sources such that every plaquette and every site at which they do not vanish lies in C,*

$$\langle Y\rangle_C^{\mathrm{ND}} \le \langle Y\rangle_\Omega^{(3.38)} \quad (Y \in \mathcal{I}), \qquad \langle X\rangle_C^{\mathrm{ND}} \ge \langle X\rangle_\Omega^{(3.38)} \quad (X \in \mathcal{D}),$$

and the same for Z^a (equal positive multipliers, Lemma 2.12(a)). The $\mathcal{D}$ half uses Lemma 2.8(d); the $\mathcal{I}$ half only (a), (b), (e).

Proof. (a) K_Ω^{int} contains every gradient pair of K_C (both base points in C, both bonds internal to Ω); the remaining pairs and N_∂ are nonnegative; the mass term splits. (b) Start from $\langle\cdot\rangle_\Omega^{(3.38)}$ and remove the hopping cosines across ∂C and all cosines not internal to C; by Lemma 2.8(b),(d) $\langle Y\rangle$ does not increase and $\langle X\rangle$ does not decrease, every intermediate measure being a BFS measure. The matter is now the Dirichlet matter of C: the diagonal parts $\frac{1}{2} r^2$ of the cut bonds stay and are exactly the Dirichlet terms, as in the proof of Lemma 2.11; it is coupled to A only through bonds internal to C, which are C-variables, and the matter of $\Omega \setminus C$ is decoupled and cancels (Lemma 2.8(e)). Then pass from the form $Q_\Omega^{(3.38)}$ to the smaller form $Q_C^{\mathrm{ND}} \oplus \frac{\mu^2}{2} I_{\mathrm{rest}}$: by (a) the difference is positive semidefinite, a finite sum $\sum_i s_i |\ell_i(A)|^2$ with $s_i \ge 0$, and adding each term does not decrease $\langle Y\rangle$ and does not increase $\langle X\rangle$ (Lemma 2.8(a),(d)). So the state with the larger form, the modified one, has the larger $\langle Y\rangle$ and the smaller $\langle X\rangle$. The state with the smaller form factorises as $\langle\cdot\rangle_C^{\mathrm{ND}}$ times an independent Gaussian on the rest, which cancels. □

For King's convention the lower end is used in the form $\langle Y\rangle^{K}_{k,J} \le \langle Y\rangle^{(a)}_{k,J} \le \langle Y\rangle^{(3.38)}_{\Omega}$ on $\mathcal{I}$ (Step 1 of Lemma 4.2, then Lemma 9.9; Remark 4.8), and the upper end needs no change (Lemma 4.6).

Lemma 9.10 (the upper half, with a margin)**.** *Let $\Omega \subset B$ be lattice boxes at spacing ε such that B contains every site within ℓ^∞-distance N lattice steps of Ω, and let $\mu = \mu_0\varepsilon$ satisfy $d(\pi/2N)^2 \le \mu^2$, i.e.*

$$N\varepsilon \;\ge\; \frac{\pi\sqrt{d}}{2\mu_0} \qquad \big(= R(\mu_0) = \tfrac{\pi\sqrt{3}}{2\mu_0} \text{ in } d=3\big).$$

Let Σ_X denote the covariance of $(\varepsilon F(p))_{p\in\Omega^{**}}$ under the photon Gaussian of the state on X. Then

(a) $\Sigma^{\mathrm{ND}}_B \le \Sigma^{(3.38)}_{\Omega} = \Sigma^{u}_{\Omega}$ *on* $\mathbb{R}^{\Omega^{**}}$;

(b) *for sources supported in Ω as in Lemma 9.9,*

$$\langle Y\rangle^{(3.38)}_{\Omega} \le \langle Y\rangle^{\mathrm{ND}}_{B} \quad (Y\in\mathcal{I}), \qquad \langle X\rangle^{(3.38)}_{\Omega} \ge \langle X\rangle^{\mathrm{ND}}_{B} \quad (X\in\mathcal{D}),$$

and the same for Z^a. Only the $\mathcal{I}$ half is used in the sequel.

Proof. (a), in lattice units. Step 0: $\Sigma^{(3.38)}_{\Omega} = \Sigma^{u}_{\Omega}$, because $d_1A = d_1A^T$ depends only on the $\ker d_0^T$-component, whose law $\propto e^{-Q^u_\Omega(A^T)}$ is the same under both Gaussians (Step 1 of Lemma 9.3). Step 1 (variational identity): for a Gaussian with density $\propto e^{-\frac12 A^T M_X A}$, $\langle g,\Sigma_X g\rangle = \mathrm{Var}\langle g,d_1A\rangle = \sup_A[2\langle g,d_1A\rangle - A^TM_XA]$, the supremum attained at $A = M_X^{-1}d_1^Tg$. So it suffices to show that for every A on the B-variables there is $A'\in\mathbb{R}^{\Omega^*}$ with $d_1A' = d_1A$ on Ω^{**} and $\|d_1A'\|^2_{\Omega^{**}} + \mu^2\|A'\|^2_{\Omega^*} \le K_B(A) + \mu^2\|A\|^2_B$; take $A' := A|_{\Omega^*}$ (the plaquettes of Ω^{**} use only Ω^*-bonds). Step 2 (cutoff): let $\psi(x) = \prod_\nu \phi_\nu(x_\nu)$ with $\phi_\nu = 1$ on the ν-range of Ω and on the first layer outside, $\phi_\nu(t) = \cos(\pi(t-1)/2N)$ at the t-th layer outside for $1\le t\le N+1$ (so $\phi_\nu(N+1) = 0$), and 0 beyond; then $0\le\psi\le1$, $\psi = 1$ on Ω and its neighbours, $\mathrm{supp}\,\psi\subset B$. Put $a := \psi A$ on $\mathbb{Z}^d$ (the bond $\langle x, x+e_\mu\rangle$ weighted by $\psi(x)$; $A := 0$ off the B-variables). Step 3 (Hodge on $\mathbb{Z}^d$): $\|d_1A\|^2_{\Omega^{**}} = \|d_1A'\|^2_{\Omega^{**}} \le \sum_p |da(p)|^2 \le \sum_p |da|^2 + \sum_x |d^*a|^2 = \sum_\mu\sum_\nu\sum_x |a_\mu(x+e_\nu) - a_\mu(x)|^2$, since the bonds of plaquettes in Ω^{**} have base points in Ω, where $\psi = 1$. Step 4 (discrete IMS identity): for finitely supported ψf and a direction e,

$$\sum_x |\psi(x+e)f(x+e) - \psi(x)f(x)|^2 = \sum_x \psi(x)\psi(x+e)|f(x+e)-f(x)|^2 + \sum_x \psi(x)\big(2\psi(x) - \psi(x+e) - \psi(x-e)\big)f(x)^2,$$

exactly (per bond the difference of the two sides is $\psi'(\psi'-\psi)f'^2 + \psi(\psi-\psi')f^2$; collect by site). With $f = A_\mu$ and $e = e_\nu$: $\psi\psi' \le 1$, and $\psi\psi' > 0$ only for pairs with both base points in B, which are gradient pairs of K_B; so the first sums, over μ,ν, total at most $K_B(A)$. The potential is $V(x) = \sum_\nu \big(\prod_{\kappa\ne\nu}\phi_\kappa^2\big)\phi_\nu(-\Delta\phi_\nu)(x_\nu)$. Step 5 (the profile): with $\vartheta := \pi/2N$, $\phi(-\Delta\phi) = 0$ on the range of Ω (all neighbours equal 1), $= 1-\cos\vartheta \le \vartheta^2/2$ at $t = 1$, $= 2\cos^2(\vartheta(t-1))(1-\cos\vartheta) \le \vartheta^2$ for $2\le t\le N$ (using $\phi(N+1) = \cos(\vartheta N) = 0$ at $t = N$), and $= 0$ for $t\ge N+1$; hence $V = 0$ on Ω and $V \le d\vartheta^2$ everywhere. Step 6: $\|d_1A'\|^2_{\Omega^{**}} \le K_B(A) + d\vartheta^2\sum_{x\in B\setminus\Omega}|A(x)|^2$; the variables with base point in $B\setminus\Omega$ and those of Ω^* (base point in Ω) are disjoint subsets of the B-variables, so $\|d_1A'\|^2_{\Omega^{**}} + \mu^2\|A'\|^2_{\Omega^*} \le K_B(A) + \mu^2\|A\|^2_B$ as soon as $d\vartheta^2 \le \mu^2$.

(b). Start from $\langle\cdot\rangle^{\mathrm{ND}}_B$ and remove all hopping cosines touching $B\setminus\Omega$ (Lemma 2.8(b): $\mathcal{I}$ does not increase; (d): $\mathcal{D}$ does not decrease). The matter is now the Dirichlet matter of Ω (the diagonals of the bonds leaving Ω kept; every such bond ends in B since $N\ge1$), coupled to A on Ω^*, and the sites of $B\setminus\Omega$ decouple. This matter is identical to that of the modified box state: the same cosines on Ω^*, the same diagonals $\frac12 r_x^2$ for every bond leaving Ω, the same single-site weights with the constant δm_ε^2 and the same Wick constant. For a gauge-invariant observable with sources in Ω the matter integral at fixed A, $\Psi_Y(A) = \int\prod_{x\in\Omega} d^2\phi_x\, Y\, e^{-S_m}$, is invariant under $A\mapsto A + d_0\chi$, $\chi\in\mathbb{R}^\Omega$ (as in Step 1 of Lemma 9.3), hence a function of the class of $A|_{\Omega^*}$ modulo $\mathrm{ran}\,d_0 = \ker d_1$ (a box has $H^1 = 0$: the cycle space of a grid graph is generated by its 4-cycles), hence a function $\tilde\Psi_Y(d_1A|_{\Omega^{**}})$ of the field strengths on Ω^{**}. So $\langle Y\rangle = \mathbb{E}[\tilde\Psi_Y(F)]/\mathbb{E}[\tilde\Psi_1(F)]$ depends on the photon law only through the centred Gaussian law of $F = (d_1A(p))_{p\in\Omega^{**}}$, with

covariance Σ_B^{ND} on the Neumann–Dirichlet side after the removal and Σ_Ω^u on the modified side (Step 0). Both laws live on $\operatorname{ran} d_1$. Let $T : \mathbb{R}^{\Omega^{**}} \to \mathbb{R}^{\Omega^*}$ be a linear map with $d_1 T = \mathrm{id}$ on $\operatorname{ran} d_1$, extended linearly to all of $\mathbb{R}^{\Omega^{**}}$ (any extension serves, both laws being supported on $\operatorname{ran} d_1$), and realise each law as the photon Gaussian of $A'' := TF \in \mathbb{R}^{\Omega^*}$, covariance $T\Sigma T^T$: then $d_1 A'' = F$ almost surely, the gauge-invariant expectations are reproduced, and $T\Sigma_B^{\mathrm{ND}} T^T \le T\Sigma_\Omega^u T^T$ by (a). These covariances are degenerate; regularise to $T\Sigma T^T + \delta I$, $\delta > 0$, which are non-degenerate, so the regularised states are BFS measures, the ordering of the covariances is preserved, and the law of $F'' = d_1 A''$ is $N(0, \Sigma + \delta d_1 d_1^T)$, still ordered. Lemma 2.8(a) by interpolation (the difference of the precision matrices is positive semidefinite by operator monotonicity of the inverse, hence a finite sum of terms $s|\ell(A)|^2$, $s \ge 0$) gives $\langle Y\rangle_{\Sigma^u+\delta} \le \langle Y\rangle_{\Sigma^{\mathrm{ND}}+\delta}$ for $Y \in \mathcal{I}$, and the reverse for $\mathcal{D}$ by Lemma 2.8(d). *The limit* $\delta \downarrow 0$. For the $\mathcal{I}$ half, $\tilde\Psi_Y$ is bounded and continuous: $|\Psi_Y(A)| \le \int e^{r^2(g)} e^{\sum \varepsilon^{d-2} r_x r_y} \prod_x d\rho_x < \infty$ independently of A ($|e^{\mathrm{i}F}| = 1$; the hopping bound $e^{\varepsilon^{d-2} r_x r_y} \le e^{\frac12 \varepsilon^{d-2}(r_x^2 + r_y^2)}$ cancels the diagonals and $\lambda > 0$ does the rest), and so is $\tilde\Psi_1$, with $\tilde\Psi_1 > 0$; the Gaussians $N(0, \Sigma + \delta d_1 d_1^T)$ converge weakly to $N(0,\Sigma)$, so numerator and denominator converge and the inequality passes to the limit. For the $\mathcal{D}$ half (not used below), $X = \Theta(f, -g)$ gives $\tilde\Psi_X(F) = e^{\langle v, F\rangle} \tilde\Psi_X^0(F)$ with v the plaquette vector of f and $0 < \tilde\Psi_X^0 \le e^{2\mathsf{W}_\varepsilon \mathsf{S}(g)} \tilde\Psi_1 \le C$ bounded and continuous, so $\tilde\Psi_X$ is continuous but unbounded; under $N(0, \Sigma + \delta d_1 d_1^T)$ one has $\mathbb{E}[e^{2\langle v, F''\rangle}] = e^{2v^T(\Sigma + \delta d_1 d_1^T)v} \le e^{2v^T \Sigma v + 2\|d_1^T v\|^2}$ for $\delta \le 1$, so $\{\tilde\Psi_X(F'')\}_{0<\delta\le 1}$ is bounded in L^2, hence uniformly integrable, and weak convergence with uniform integrability and continuity gives $\mathbb{E}[\tilde\Psi_X(F'')] \to \mathbb{E}[\tilde\Psi_X(F)]$. Altogether $\langle Y\rangle_\Omega^{(3.38)} \le \langle Y\rangle^{\text{after removal}} \le \langle Y\rangle_B^{\mathrm{ND}}$, and the reverse for $\mathcal{D}$. □

Remark 9.11 (why a margin, and why a fixed photon mass). Part (a) of Lemma 9.10 fails without a margin: for $B = \Omega$ the comparison is reversed (Lemma 9.6), and with a single layer ($N = 1$) the difference $\Sigma_\Omega^u - \Sigma_B^{\mathrm{ND}}$ has a negative eigenvalue already at moderate μ. No fixed margin works at $\mu_0 = 0$: there Σ_Ω^u is the projector onto exact 2-forms, while $\mathrm{Var}_B\langle g, dA\rangle > \|g\|^2$ for a Coulomb potential whose gradient leaks out of B. So Lemma 9.10 is a statement at fixed $\mu_0 > 0$ with $R(\mu_0) \to \infty$ as $\mu_0 \to 0$; a limit of vanishing photon mass inherits reflection positivity, if it exists, as a pointwise limit of positive semidefinite matrices. On one box the three states are ordered on $\mathcal{I}$ as $\langle Y\rangle_\Omega^{\mathrm{ND}} \le \langle Y\rangle_\Omega^{(3.38)} \le \langle Y\rangle_B^{\mathrm{ND}}$ (Lemmas 9.6 and 9.10), as the monotonicity of Lemma 2.11 requires. The sandwich with the boxes (20) is assembled in the proof of Theorem 10.6; for King's convention its lower end carries the kill step of Remark 4.8 and its upper end is unchanged (Lemma 4.6).

10. Reflection positivity II: the limit, OS3 and the polynomial form

The lattice statements of the previous section are now carried to the limits $k \to \infty$ and $J \to \infty$. The scheme of proof is the same for the curvature sector and for the Wick tube: uniform local bounds for the modified-box functionals at complex coefficients (Lemma 10.3), Montel's theorem in k at fixed J, the two-sided sandwich of Lemmas 9.9 and 9.10 on the cone as $J \to \infty$ with the identity theorem off it, and a translation for the planes that are not mid-planes. The sector-specific input is the convergence of the cube functionals themselves: the convergence at fixed cubes (Definition 3.1; on King's convention Theorem 3.3) in the curvature sector, and on the Wick tube the positive-leg bound of Corollary 13.3 together with the convergence at fixed cubes on the tube (Definition 3.2; on King's convention Theorem 3.6) (Proposition 10.5). The arbitrary-amplitude form of the Wick sector, which needs (W2), is the optional extension of Theorem 23.8; the one lemma written for it here, Lemma 10.2, is marked as such and is not used by the main theorem.

10.1. **The enlarged family.**

Definition 10.1 (the enlarged family, its tube and its bounded domain)**.** Fix $\nu \in \{1,2,3\}$, a mid-plane $c \in D$ (Definition 9.1), and finite families $\zeta_1, \dots, \zeta_N$ of real sources $f_{il} \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, $h_{il} \in C_c^\infty(\mathbb{R}^3)$ supported in $\{x_\nu \ge c + \delta\}$ for some $\delta > 0$, with complex coefficients z_{il}, w_{il}. Split each scalar source as $h_{il} = h_{il}^+ - h_{il}^-$ with $h_{il}^- := \|h_{il}\|_\infty \chi_{il}$, where $\chi_{il} \in C_c^\infty$, $0 \le \chi_{il} \le 1$, $\chi_{il} = 1$

on $\operatorname{supp} h_{il}$ and $\operatorname{supp}\chi_{il} \subset \{x_\nu \geq c+\delta/2\}$, and $h^+_{il} := h_{il} + h^-_{il} \geq 0$. The *enlarged family* is the family of real sources

$$\mathcal{F} := \{\rho^* f_{il}\} \cup \{f_{il}\} \cup \{h^\pm_{il} \circ \rho\} \cup \{h^\pm_{il}\}, \qquad \text{coefficient space } \mathbb{C}^m,\ m = m_F + m_h,$$

whose scalar members $\tilde{h}_j \geq 0$ are all nonnegative. For a real coefficient vector $\xi \in \mathbb{R}^m$ the total source is $(f_\xi, k_1(\xi) - k_2(\xi))$ with $k_1(\xi) := \sum_j (\xi_j)_+ \tilde{h}_j$, $k_2(\xi) := \sum_j (\xi_j)_- \tilde{h}_j$, both smooth and nonnegative ($x_\pm := \max(\pm x, 0)$). Let J_0 be the least J such that every member of $\mathcal{F}$ is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$; for $J \geq J_0$, $n = n(\mu_0, c)$ and the boxes $\Omega_{J,k}$ of (20) put

$$\Phi_k(\zeta) := Z^{a,(3.38)}_{k,\Omega_{J,k}}(\zeta), \qquad \zeta \in \mathbb{C}^m,$$

an entire function (a finite-dimensional integral with Gaussian decay in A and $e^{-\lambda\varepsilon^3 r^4}$ decay in r). The *matrix points* $\zeta^*_{ij} \in \mathbb{C}^m$ carry $\bar{z}_{il}$ on $\rho^* f_{il}$, z_{jl} on f_{jl}, $\pm\bar{w}_{il}$ on $h^\pm_{il} \circ \rho$, $\pm w_{jl}$ on $h^\pm_{jl}$ and 0 elsewhere; their total source is that of $\theta\zeta^*_i \sqcup \zeta_j$, so $\Phi_k(\zeta^*_{ij})$ is the (i,j) entry of Lemma 9.3(b). Since reflected and unreflected members have disjoint supports, $\|h^+\|_\infty \leq 2\|h\|_\infty$ and $\|h^-\|_\infty = \|h\|_\infty$,

$$\|k_1(\operatorname{Re}\zeta^*_{ij})\|_\infty \ \leq\ 2\max\big(\mathsf{a}(\zeta_i), \mathsf{a}(\zeta_j)\big). \tag{21}$$

(i) The *Wick tube* of $\mathcal{F}$ is $T_\mathcal{F} := \{\zeta \in \mathbb{C}^m :\ 2\|k_1(\operatorname{Re}\zeta)\|_\infty < \sigma_+/2\}$, with $\sigma_+ = s_1$ the amplitude of (I2) (Definition 12.5, (30)). It is open, convex ($\xi \mapsto \|k_1(\xi)\|_\infty$ is a supremum of convex functions), connected, invariant under $z \mapsto -z$ on the curvature coordinates, contains 0, and by (21) contains every ζ^*_{ij} as soon as $\mathsf{a}(\zeta_i) < \sigma_+/8 =: \sigma_W$ for every i.

(ii) The *bounded domain*: let $B_1 \subset \mathbb{R}^m$ be the open convex hull of the unit ℓ^∞-balls about 0 and about the real parts $\operatorname{Re}\zeta^*_{ij}$; let $\beta_j := 2\sup_{\xi\in B_1}|\xi_j|$ for the scalar coordinates (so $\beta_j = 2(|\operatorname{Re} w_{il}| + 1)$ for the coordinate of $h^\pm_{il}$ or of $h^\pm_{il} \circ \rho$), $B'' := \prod_j [0, \beta_j]$ over the scalar coordinates, and $q_v := \sum_j v_j \tilde{h}_j$ for $v \in \operatorname{vert}(B'')$, the *vertex sources*. All q_v are pointwise $\leq q_\zeta := \sum_{i,l} 2(|\operatorname{Re} w_{il}| + 1)(h^+_{il} + h^-_{il} + h^+_{il} \circ \rho + h^-_{il} \circ \rho)$. Put $\mathcal{T}_1 := \{\zeta \in \mathbb{C}^m : \operatorname{Re}\zeta \in B_1\}$: open, convex, connected, containing every ζ^*_{ij} and the unit polydisc about 0, so that $\mathcal{T}_1 \cap T_\mathcal{F}$ is a non-empty open set. For $\xi \in B_1$ both $2\xi_+$ and ξ_+ lie in B''.

(iii) The *uniqueness set* $U \subset \mathbb{C}^m$ is the set of coefficient vectors whose curvature coefficients are purely imaginary and whose scalar coefficients are positive reals. It is a non-empty, relatively open subset of the totally real subspace $(\mathrm{i}\mathbb{R})^{m_F} \times \mathbb{R}^{m_h}$ of maximal real dimension m (it is not open in $\mathbb{C}^m$), hence a set of uniqueness for holomorphic functions on any connected open set meeting it: a holomorphic function vanishing on a relatively open piece of a maximal totally real subspace has all derivatives zero there [11]. $U \cap \mathcal{T}_1 \neq \emptyset$ and $U \cap T_\mathcal{F} \neq \emptyset$ (small positive scalar coefficients). On U the total source is $(\mathrm{i}f_\zeta, g_\zeta) \in \mathcal{K}$, so $\Theta(\zeta) \in \mathcal{I}$; the scheme multipliers are positive reals, equal for the Neumann–Dirichlet and the modified states; and every lattice value is real (charge conjugation, Definition 2.4: $\overline{Z^a(\mathrm{i}f, g)} = Z^a(-\mathrm{i}f, g) = Z^a(\mathrm{i}f, g)$).

Every lattice functional, and hence every limit below, depends on a family only through its total sources; the splitting is therefore immaterial at the matrix points and at the Taylor coefficients at 0.

10.2. **Uniform local bounds.** The following local bound is used only in the optional extension (Theorem 23.8); it is stated here because its domain $\mathcal{T}_1$ is what makes "finitely many instances of (W2)" literally true there. The fixed-cube limits $Z^a_J := \lim_k Z^a_{k,J}$ exist for all complex coefficients and are entire under (W2)(a) alone: it gives convergence at the real points, the vertex argument of Lemma 10.3 with Lemma 2.12(b) gives local boundedness in k, and Montel's theorem with the uniqueness set $\mathbb{R}^m$ does the rest.

Lemma 10.2 (uniform local bound; optional extension)**.** *Assume* (W2)*(a) and* (W2)*(b) at the finitely many sources q_v, $v \in \operatorname{vert}(B'')$. Then $\{Z^a_J\}_{J\geq J_0}$ is uniformly bounded on $\mathcal{T}_1$, $Z^a_J \to Z^a_\infty$ locally uniformly on $\mathcal{T}_1$, Z^a_∞ is holomorphic on $\mathcal{T}_1$, and $Z^a_J \leq Z^a_\infty$ on $U \cap \mathcal{T}_1$.*

Proof. For real $\xi \in B_1$, Lemma 2.12(d) gives $Z^a_J(\xi) \leq Z^a_J(2f_\xi, -2k_2(\xi))^{1/2} Z^a_J(0, 2k_1(\xi))^{1/2}$. The first factor is nonincreasing in J: at fixed k this is the second inequality of Lemma 2.11 with

Lemma 2.12(a), and it passes to $k \to \infty$ by (W2)(a); (W2)(b) is not used there. Hence it is $\le Z^a_{J_0}(2f_\xi, -2k_2(\xi))$, and $\xi \mapsto \ln Z^a_{J_0}(2f_\xi, -2k_2(\xi))$ is bounded on the compact $\bar B_1$: the coefficient map is piecewise linear, so the image of $\bar B_1$ lies in a real box B', on which $\ln Z^a_{J_0}$ is convex (Lemma 2.12(b)) and so bounded by its maximum over the vertices of B', each finite by (W2)(a). For the second factor, $2\xi_+ \in B''$ and $\ln Z^a_J(0,\cdot)$ is convex in the coefficients, so $Z^a_J(0, 2k_1(\xi)) \le \max_{v\in\mathrm{vert}(B'')} Z^a_J(0,q_v) \le \max_v \sup_{J'\ge J_{q_v}} Z^a_{J'}(0,q_v) =: C_{B''} < \infty$ by (W2) at the q_v (note $J \ge J_0 \ge J_{q_v}$). By Lemma 2.12(b), $|Z^a_J(\zeta)| \le Z^a_J(\mathrm{Re}\,\zeta)$, so $\sup_{J\ge J_0}\sup_{\mathcal{T}_1}|Z^a_J| < \infty$. On $U \cap \mathcal{T}_1$ the sources lie in $\mathcal{K}$, so $J \mapsto Z^a_J(\zeta)$ is nondecreasing (first inequality of Lemma 2.11 at fixed k, then $k \to \infty$) and bounded, hence convergent. $U \cap \mathcal{T}_1$ is a set of uniqueness for holomorphic functions on the connected $\mathcal{T}_1$ (Definition 10.1(iii)); Montel's theorem gives locally uniform convergence on $\mathcal{T}_1$ to a holomorphic limit, which is Z^a_∞ wherever the latter is defined (the same J-limit), and the monotone limit on $U \cap \mathcal{T}_1$ gives $Z^a_J \le Z^a_\infty$ there. Under (W2)(b) for every $0 \le q \in C^\infty_c$ the same holds on all of $\mathbb{C}^m$. □

Lemma 10.3 (uniform local bounds for the modified-box functionals)**.** *Let $\mathcal{F}$, Φ_k, $\mathcal{T}_1$, $T_\mathcal{F}$ be as in Definition 10.1.*

(A) (Optional extension.) *Under* (W2)*(a) and* (W2)*(b) at the q_v, $v \in \mathrm{vert}(B'')$: there is $C < \infty$, independent of k and of $J \ge J_0$, with $\limsup_k \sup_{\mathcal{T}_1} |\Phi_k| \le C$; and at fixed J, $\sup_k \sup_Q |\Phi_k| < \infty$ for every compact $Q \subset \mathbb{C}^m$ (this part uses* (W2)*(a) only).*

(A_∞) (Optional extension.) *Under* (W2) *for every $0 \le q \in C^\infty_c$: the same with $\mathcal{T}_1$ replaced by any compact $Q \subset \mathbb{C}^m$, with $C = C_Q$.*

(T) (The tube.) *Under Definition 3.2 (on King's convention Theorem 3.6),* (I2), (I3) *and the correlation inequalities of Lemma 2.8(a)–(e) only (no* (W2)*): the same for every compact $Q \subset T_\mathcal{F}$ and all $J \ge J_0$, with no further threshold on J.*

(C) (Curvature sector, $h \equiv 0$; no hypothesis.) $|\Phi_k(z)| \le \langle e^{F(f_{\mathrm{Re}\,z})}\rangle^{(3.38)}_{k,\Omega_{J,k}} \le \exp(\frac12 \mathrm{Var}^{(3.38)} F(f_{\mathrm{Re}\,z}))$ *and* $\mathrm{Var}^{(3.38)} F(f) \le \sum_p \varepsilon^3 f^2_{c(p)} \to \|f\|^2_{L^2}$*, uniformly in k and J.*

Proof. $|\Phi_k(\zeta)| \le \Phi_k(\mathrm{Re}\,\zeta)$ by the proof of Lemma 2.12(b) ($|\Theta(\zeta)| = \Theta(\mathrm{Re}\,\zeta)$ and the modulus of the multiplier), which is valid for any probability measure with the same real multiplier, in particular for the modified box state, which is not one of the states of Definitions 2.1 and 2.2. At a real ξ the proof of Lemma 2.12(d) (Cauchy–Schwarz with the multiplier identity, again valid for any such measure) gives

$$\Phi_k(\xi) \;\le\; Z^{a,(3.38)}_{k,\Omega_{J,k}}(2f_\xi, -2k_2(\xi))^{1/2}\; Z^{a,(3.38)}_{k,\Omega_{J,k}}(0, 2k_1(\xi))^{1/2}. \tag{22}$$

First factor (class $\mathcal{D}$). By the $\mathcal{D}$ half of Lemma 9.9 with $C = C_J \cap \Lambda_k$ (its hypothesis is (20)(i)) it is $\le Z^a_{k,J}(2f_\xi, -2k_2) \le Z^a_{k,J_0}(2f_\xi, -2k_2)$, the second step being the second inequality of Lemma 2.11 at fixed k. The map $\xi \mapsto \eta(\xi) :=$ (coefficient vector of $(2f_\xi, -2k_2(\xi))$) is continuous, so $\eta(\mathrm{Re}\,Q)$, resp. $\eta(\bar B_1)$, lies in a real box B'; $\eta \mapsto \ln Z^a_{k,J_0}(\eta)$ is convex (Lemma 2.12(b)), so $\sup_\xi Z^a_{k,J_0}(\eta(\xi)) \le \max_{v\in\mathrm{vert}(B')} Z^a_{k,J_0}(v)$, finitely many values, each convergent in k by Definition 3.2 (the sources here have negative Wick coefficients, on which the clause has no amplitude restriction; in versions (A), (A_∞) by (W2)(a)), and convergence in k implies boundedness in k. So the first factor is bounded uniformly in k, ξ and $J \ge J_0$. *Second factor* (class $\mathcal{I}$). By the $\mathcal{I}$ half of Lemma 9.10 with $B = C_{J+2n} \cap \Lambda_k$ and $N := \lceil R(\mu_0)/\varepsilon_k \rceil$ (hypothesis (20)(ii)) it is $\le Z^a_{k,J+2n}(0, 2k_1(\xi))$. (A): for $\xi \in B_1$, $2\xi_+ \in B''$, so by convexity it is $\le \max_{v\in\mathrm{vert}(B'')} Z^a_{k,J+2n}(0,q_v)$; each of these finitely many values converges in k by (W2)(a) (hence is bounded in k, which is the fixed-J bound Montel's theorem needs, and the $\limsup_k$ bound passes to any subsequential limit) with limit $Z^a_{J+2n}(0,q_v) \le \sup_{J'\ge J_{q_v}} Z^a_{J'}(0,q_v) < \infty$: this is (W2) at q_v, used directly ($J+2n \ge J_0 \ge J_{q_v}$), uniformly in J; the finitely many instances q_v are all of (W2) that version (A) uses. (A_∞): the same with a box $B'' \supset 2(\mathrm{Re}\,Q)_+$ depending on Q. (T): for $\xi \in \mathrm{Re}\,Q$, $Q \subset T_\mathcal{F}$, one has $0 \le 2k_1(\xi) \le \sigma_+/2 \le s_1$ on the lattice (the sampled supremum is at most the continuum one), so Corollary 13.3, $\ln N^a_{k,C_J}(0,q) \le \kappa\|q\|_{1,k}$ for $0 \le q \le s_1$ on $C_J \cap \Lambda_k$, uniformly in k and J, applied with $N^a_{k,X}(0,q) = Z^a_{k,X}(0,q)$ (the same functional with the same scheme multiplier, so no J-dependent factor intervenes), gives

$Z^a_{k,J+2n}(0,2k_1(\xi)) \le \exp(\kappa\|2k_1(\xi)\|_{1,k}) \le \exp\big(2\kappa\sum_j|\xi_j|\sup_{k'}\|\tilde h_j\|_{1,k'}\big)$, bounded uniformly in k, J and $\xi \in \operatorname{Re} Q$ because $\|\tilde h_j\|_{1,k'} \to \|\tilde h_j\|_{L^1}$ (Riemann sums, (1)). No lower bound on J beyond J_0 enters: Corollary 13.3 holds for every k and J. *(C).* At $h \equiv 0$ the first inequality is Lemma 2.12(b) as above and the second is Lemma 2.9 with $s = 1$, the modified box state being a BFS measure (Definition 9.2). Its photon Gaussian has the unitary form for gauge-invariant observables (Step 1 of Lemma 9.3), and $\operatorname{Var}^u\langle v, d_1A\rangle = \varepsilon^{-1}\langle d_1^T v, (d_1^T d_1 + \mu_0^2\varepsilon^2)^{-1} d_1^T v\rangle \le \varepsilon^{-1}\|v\|^2$; with $v_p = \varepsilon^2 f_{c(p)}$ this is $\operatorname{Var} F(f) \le \sum_p \varepsilon^3 f^2_{c(p)}$. □

Remark 10.4 (one instance of (W2) in place of the vertex set). For $v \in \operatorname{vert}(B'')$ one has $0 \le q_v \le q_\zeta$ pointwise, and for $p = 1+\eta > 1$, $p' = p/(p-1)$, $q_v = \frac{1}{p}(pq_\zeta) - \frac{1}{p'}\big(p'(q_\zeta - q_v)\big)$. Hölder's inequality with the multiplier bookkeeping $e^{-a_k \mathsf{S}(q_v)} = (e^{-a_k\mathsf{S}(pq_\zeta)})^{1/p}(e^{a_k\mathsf{S}(p'(q_\zeta-q_v))})^{1/p'}$ gives, for every state, $Z^a(0,q_v) \le Z^a(0,pq_\zeta)^{1/p} Z^a(0,-p'(q_\zeta-q_v))^{1/p'}$. The second factor is of class $\mathcal{D}$, nonincreasing in J (Lemma 2.11, then $k \to \infty$ by (W2)(a)), hence $\le Z^a_{J_0}(0,-p'(q_\zeta-q_v)) < \infty$, one real point of Theorem 3.6; the first is bounded uniformly in J by (W2)(b) at the single source $(1+\eta)q_\zeta$. So (W2) at $(1+\eta)q_\zeta$, for any one $\eta > 0$, implies the finitely many instances used in Lemmas 10.2 and 10.3(A). Only this direction is proved; no converse is claimed. Pointwise domination $q_v \le q_\zeta$ alone gives nothing, since $Z^a(0,\cdot)$ is not monotone in q: the Wick square has no sign, and the scheme multiplier $e^{-a_k\mathsf{S}(q)}$ has no sign control.

10.3. **King's state on the Wick tube.**

Proposition 10.5 (the Wick-tube object: existence and translations)**.** *Let a be an admissible scheme (Definition 3.7), assume* (I2) *and* (I3)*, and assume that the comparison family is convergent at fixed cubes on the tube (Definition 3.2); the form on King's convention, where Theorem 3.6 supplies the limits, is Proposition 21.4(c). For a finite family $\mathcal{F}'$ of real sources with nonnegative scalar members $\tilde h_j \ge 0$ (any family is brought to this form by the splitting of Definition 10.1) let $T_{\mathcal{F}'}$ be its Wick tube, as in Definition 10.1(i).*

(a) *$Z^a_J \to Z^a_{\infty,T}$ locally uniformly on $T_{\mathcal{F}'}$ as $J \to \infty$; $Z^a_{\infty,T}$ is holomorphic on $T_{\mathcal{F}'}$; $Z^a_{\infty,T} = G_\infty$ on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ and $Z^a_{\infty,T} = Z^a_\infty$ wherever the latter exists (the same J-limit of the same Z^a_J); $|Z^a_{\infty,T}(\zeta)| \le Z^a_{\infty,T}(\operatorname{Re}\zeta)$; $Z^a_{\infty,T}(\xi) > 0$ for real $\xi \in T_{\mathcal{F}'}$ with $-\xi \in T_{\mathcal{F}'}$; and $Z^a_{\infty,T}(-z,w) = Z^a_{\infty,T}(z,w)$.*

(b) (Translations.) *$Z^a_{\infty,T}(\tau_b\cdot\zeta) = Z^a_{\infty,T}(\zeta)$ for all $b \in \mathbb{R}^3$ and $\zeta \in T_{\mathcal{F}'}$, where τ_b acts on all sources; the translated family has the same tube.*

The proof uses, besides Definition 3.2, only Corollary 13.3 (hence (I2)*,* (I3) *and* [7, Thm. 6.2(1),(2)]*), the second inequality of Lemma 2.11 (hence Lemma 2.8(d)), Lemmas 2.12 and 2.15, Montel's theorem and the identity theorem;* (W2) *is not used.*

Proof. (a) *Local boundedness* of $\{Z^a_J\}_{J\ge J_0}$ on $T_{\mathcal{F}'}$: $|Z^a_J(\zeta)| \le Z^a_J(\operatorname{Re}\zeta) \le Z^a_J(2f,-2k_2)^{1/2} Z^a_J(0,2k_1)^{1/2} \le Z^a_{J_0}(2f,-2k_2)^{1/2}\exp(\kappa\|k_1\|_{L^1})$, by Lemma 2.12(b),(d), the nonincreasing $\mathcal{D}$-factor (second inequality of Lemma 2.11 at fixed k, then $k \to \infty$), and Corollary 13.3 in the limit $k \to \infty$ for the $\mathcal{I}$-factor, valid since the source $2k_1$ has $2\sup(2k_1) = 4\|k_1\|_\infty < \sigma_+ = s_1$; the first factor is bounded on compacts by the vertex argument of Lemma 10.3 with Definition 3.2 at C_{J_0}. *Convergence on $U \cap T_{\mathcal{F}'}$*: there the values are real, nondecreasing in J (first inequality of Lemma 2.11, then $k \to \infty$) and bounded, since $Z^a_J(\mathrm{i}f_\zeta, g_\zeta) \le Z^a_J(0,g_\zeta)$ by Lemma 2.12(b) and $Z^a_J(0,g_\zeta) \le \exp(\kappa\|g_\zeta\|_{L^1})$ by Corollary 13.3 in the limit $k \to \infty$ ($0 \le g_\zeta \le \sigma_+/4$ there, so $2\sup g_\zeta < s_1$). Montel's theorem on the connected open $T_{\mathcal{F}'}$ with the uniqueness set $U \cap T_{\mathcal{F}'}$ gives locally uniform convergence to a holomorphic $Z^a_{\infty,T}$; on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ this is the limit G_∞ of Theorem 13.10(i), the same argument on the other tube. The bound $|Z^a_{\infty,T}(\zeta)| \le Z^a_{\infty,T}(\operatorname{Re}\zeta)$ and the evenness in z pass from the lattice (Lemma 2.12(b); the invariance $(A,\vartheta) \mapsto (-A,-\vartheta)$ of every state (charge conjugation, Definition 2.4), which maps $F(f) \mapsto -F(f)$ and fixes r; $T_{\mathcal{F}'}$ is invariant under $z \mapsto -z$). Positivity at symmetric real points is Lemma 2.12(c), $Z^a(\xi)Z^a(-\xi) \ge 1$, in the limit.

(b) *Lattice translations.* Let $b \in 5^{-k_1}\mathbb{Z}^3$, $|b|_\infty \le M$, $k \ge k_1$. τ_b preserves Λ_k and S_{Λ_k}, so $Z^a_{k,J}(\tau_b\cdot\zeta) = Z^a_{k,(C_J-b)\cap\Lambda_k}(\zeta)$ (the action is translation covariant on Λ_k). As sets of sites,

$C_{J-1}\cap\Lambda_k\subset(C_J-b)\cap\Lambda_k\subset C_{J+1}\cap\Lambda_k$, so the first inequality of Lemma 2.11 (stated for any finite site set) gives, for $\zeta\in U$ and $J\ge J_0+1$ (so that every plaquette and site at which the sources do not vanish lies in C_{J-1}), $Z^a_{k,J-1}(\zeta)\le Z^a_{k,J}(\tau_b\cdot\zeta)\le Z^a_{k,J+1}(\zeta)$. Let $k\to\infty$ and then $J\to\infty$ on $U\cap T_{\mathcal{F}'}$, where the outer limits converge by (a): $Z^a_{\infty,T}(\tau_b\cdot\zeta)=Z^a_{\infty,T}(\zeta)$ there, hence on all of $T_{\mathcal{F}'}$ by the identity theorem (both sides are holomorphic on the connected $T_{\mathcal{F}'}$, the translated family having the same supremum norms). Iterating, the identity holds for all $b\in\bigcup_k 5^{-k}\mathbb{Z}^3$. *All* $b\in\mathbb{R}^3$. For $\rho>0$ let $E_\rho:=C_c^\infty(\bar B_\rho;\Lambda^2)\times C_c^\infty(\bar B_\rho)$ (a Fréchet space) and $O_\rho:=\{(f,h)\in E_\rho:\ 4\|h\|_\infty<\sigma_+/2\}$, open, convex and symmetric under $s\mapsto -s$. For $s=(f,h)\in O_\rho$ the one-member family with its splitting has the coefficient vector $(1,1,-1)$ in its tube ($2\|k_1\|_\infty=2\|h^+\|_\infty\le 4\|h\|_\infty<\sigma_+/2$), so $\Phi_T(s):=\ln Z^a_{\infty,T}(f,h)$ is defined and finite on O_ρ by (a). Φ_T is convex (a pointwise limit of the functions $\ln Z^a_{k,J}$, convex by Hölder's inequality) and Borel (a pointwise limit of continuous functions of finitely many sampled values). By Lemma 2.15, Φ_T is continuous on O_ρ. For $s\in O_\rho$ and $b\in\mathbb{R}^3$ pick $b_n\to b$ in $\bigcup_k 5^{-k}\mathbb{Z}^3$; then $\tau_{b_n}s,\tau_b s\in O_{\rho+|b|+1}$, $\tau_{b_n}s\to\tau_b s$ in $E_{\rho+|b|+1}$ (translations act continuously on C_c^∞), and $\Phi_T(\tau_{b_n}s)=\Phi_T(s)$; so $\Phi_T(\tau_b s)=\Phi_T(s)$. *Complex coefficients.* For a family $\mathcal{F}'$ the two holomorphic functions $\zeta\mapsto Z^a_{\infty,T}(\tau_b\cdot\zeta)$ and $\zeta\mapsto Z^a_{\infty,T}(\zeta)$ on $T_{\mathcal{F}'}$ agree at the real points of $\{\xi\in\mathbb{R}^m:4\|h_\xi\|_\infty<\sigma_+/2\}\cap T_{\mathcal{F}'}$, where both limits exist and both equal the one-member value $\exp\Phi_T(f_\xi,h_\xi)$ (the same J-limit of the same Z^a_J). This intersection is open in $\mathbb{R}^m$ and contains 0 (the first set alone need not lie in the tube: with $\tilde h_1=\tilde h_2$ and $\xi=(1,-1)$, $h_\xi=0$ while $k_1=\tilde h_1$), hence is a set of uniqueness in the connected $T_{\mathcal{F}'}$; so the two functions agree on $T_{\mathcal{F}'}$. □

10.4. **The two limits and the passage to every plane.** Throughout this subsection $\mathcal{F}$, Φ_k, ζ^*_{ij}, $\mathcal{T}_1$, $T_{\mathcal{F}}$, U are those of Definition 10.1 for a mid-plane $c\in D$, and "version" refers to Lemma 10.3: (C) is the curvature sector when the comparison family is convergent at fixed cubes (the uniform local bound being Theorem 7.4(b)), (T) the tube under Theorem 3.6, (I2), (I3), and (A), (A_∞) the optional extension under (W2).

Step 5 ($k\to\infty$ at fixed J). By the fixed-J bound of Lemma 10.3 and Montel's theorem there is a subsequence $k_m\to\infty$ along which $\Phi_{k_m}\to\Phi^J$ locally uniformly on $\mathbb{C}^m$, with Φ^J entire. No convergence of the full sequence is claimed or needed. By Lemma 9.3(b), $\sum_{ij}\bar\gamma_i\gamma_j\Phi_k(\zeta^*_{ij})\ge 0$ for every $k\ge k_0(c,\delta)$ and $\gamma\in\mathbb{C}^N$; positive semidefiniteness of a Hermitian matrix is closed under entrywise limits, so $\sum_{ij}\bar\gamma_i\gamma_j\Phi^J(\zeta^*_{ij})\ge 0$. The uniform bounds of Lemma 10.3 pass to Φ^J: $\sup_{\mathcal{T}_1}|\Phi^J|\le C$ in (A), $\sup_Q|\Phi^J|\le C_Q$ in (A_∞), (T) and (C), uniformly in $J\ge J_0$, whatever subsequence was taken. (The box $\Omega_{J,k}$ depends on J, so Φ^J does; the subsequence may depend on J.)

Step 6 ($J\to\infty$: sandwich on the cone, identity theorem off it). By the $\mathcal{I}$ halves of Lemmas 9.9 and 9.10, whose hypotheses are (20)(i),(ii), for every k and every $\zeta\in U$,

$$Z^a_{k,J}(\zeta)\ \le\ \Phi_k(\zeta)\ \le\ Z^a_{k,J+2n}(\zeta). \tag{23}$$

Let $k\to\infty$ along the subsequence of Step 5. The outer terms converge along the full sequence (version (C): Theorem 7.4(b); (T): Definition 3.2; (A), (A_∞): (W2)(a); each with Montel's theorem), so $Z^a_J\le\Phi^J\le Z^a_{J+2n}$ on U for every subsequential Φ^J. As $J\to\infty$ the outer terms converge on the respective uniqueness set: in (C) on $U=(\mathrm{i}\mathbb{R})^{m_F}$ to $Z_\infty(\cdot,0)$ (Theorem 7.4(b)); in (T) on $U\cap T_{\mathcal{F}}$ to $Z^a_{\infty,T}$ (Proposition 10.5(a); there $J\mapsto Z^a_J(\zeta)$ is real, nondecreasing and bounded by $Z^a_J(0,g_\zeta)\le\exp(\kappa\|g_\zeta\|_{L^1})$, since $0\le g_\zeta\le\sigma_+/4$, so $2\sup g_\zeta<s_1$); in (A) on $U\cap\mathcal{T}_1$ to Z^a_∞ (Lemma 10.2); in (A_∞) on U to Z^a_∞ (Lemma 10.2 with (W2)(b) for every q). Hence $\Phi^J\to Z^a_{\infty(,T)}$ pointwise on that set. By Step 5 the family $\{\Phi^J\}_J$ is locally bounded on the corresponding connected domain ($\mathbb{C}^m$, $T_{\mathcal{F}}$, $\mathcal{T}_1$, $\mathbb{C}^m$), on which the limit functional is holomorphic (Theorem 7.4(b), Proposition 10.5(a), Lemma 10.2, Lemma 10.2 on $\mathbb{C}^m$ respectively); by Montel's theorem every subsequence of (Φ^J) has a further subsequence converging locally uniformly to a holomorphic function, which agrees with the limit functional on the uniqueness set and hence on the whole domain [11]. So $\Phi^J\to Z^a_{\infty(,T)}$ locally uniformly on the domain, in particular at the finitely many matrix points ζ^*_{ij} (which lie in $\mathcal{T}_1$ by construction and in $T_{\mathcal{F}}$ by (21) when

every $\mathsf{a}(\zeta_i) < \sigma_W$), where $\Phi^J(\zeta^*_{ij}) \to Z^a_{\infty(,T)}(\theta\zeta^*_i \sqcup \zeta_j)$ because the functionals depend on the family only through the total sources. Positivity passes to the limit:

$$\sum_{i,j} \bar\gamma_i\gamma_j\, Z^a_{\infty(,T)}(\theta\zeta^*_i \sqcup \zeta_j) \;\geq\; 0 \qquad \text{for the plane } \{x_\nu = c\},\ c \in D. \tag{24}$$

A side result: the modified-box values converge to the cube state, $\lim_J \lim_{k_m} Z^{a,(3.38)}_{k_m,\Omega_{J,k_m}}(\zeta) = Z^a_{\infty(,T)}(\zeta)$, on the respective domain and along any k-subsequence on which the fixed-J limits exist; this is the corrected form of King's passage from the modified boxes to the cube limit [16, (3.42)], with the margin $2n$ in place of one cube step (Lemma 9.6 and Remark 9.8). The reversal of King's second inequality is proved, weakly at every coupling and strictly at $e = 0$ (Lemma 9.6); no step of the passage to the limit uses that inequality in King's direction.

Step 7 (every plane). Let $c \in \mathbb{R}$ be arbitrary and the sources supported in $\{x_\nu > c\}$. If $c \in D$, (24) is the result. Otherwise pick $c' \in D$ and $b := (c' - c)e_\nu$. The translated sources $\tau_b\zeta_i$ are supported in $\{x_\nu > c'\}$, and $\theta_{c'}(\tau_b\zeta)^* = \tau_b(\theta_c\zeta^*)$, because $\rho_{c'}(x + b) = \rho_c(x) + b$ for this b and the linear parts of $\rho_c, \rho_{c'}$ agree. By (24) for the plane c' and the translation invariance of the limit functional,

$$\sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}(\theta_c\zeta^*_i \sqcup \zeta_j) = \sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}\big(\tau_b(\theta_c\zeta^*_i \sqcup \zeta_j)\big) = \sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}\big(\theta_{c'}(\tau_b\zeta_i)^* \sqcup \tau_b\zeta_j\big) \;\geq\; 0.$$

The translation invariance is supplied in (C) by Proposition 8.1, and in (T) by Proposition 10.5(b) on $T_{\mathcal{F}}$ (the amplitude a is translation invariant, so the matrix points stay in the tube). In the optional extension a plane $c \notin D$ needs a non-lattice translation, which finitely many instances of (W2)(b) do not supply; there the step is carried by Proposition 10.5(b) together with Lemma 10.2 for $\mathcal{F}$ and for $\tau_b\mathcal{F}$ and the identity theorem on $\mathcal{T}_1$ (Theorem 23.8).

10.5. **The theorems.**

Theorem 10.6 (reflection positivity, curvature sector, comparison convention)**.** *Assume that the comparison family is convergent at fixed cubes (Definition 3.1). Let $\mu_0 > 0$, $\nu \in \{1, 2, 3\}$, $c \in \mathbb{R}$, and let $\zeta_1, \dots, \zeta_N$ be finite families of real 2-form sources $f_{il} \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$ supported in the open half-space $\{x_\nu > c\}$, with arbitrary complex coefficients z_{il} and no scalar sources. Then for all $\gamma \in \mathbb{C}^N$*

$$\sum_{i,j=1}^{N} \bar\gamma_i\gamma_j\, Z_\infty\big(\theta\zeta^*_i \sqcup \zeta_j,\, 0\big) \;\geq\; 0.$$

Besides the convergence at fixed cubes the proof uses only [7, Thm. 6.2(1),(2)] *(through the first inequality of Lemma 2.11 and the $\mathcal{I}$ halves of Lemmas 9.9 and 9.10), Theorem 7.4(b) with Lemma 7.3, and Lemma 2.9, Lemma 2.12(b),(c), Lemma 2.8(e), Lemma 2.15; no inequality of class $\mathcal{D}$ enters.*

Proof. At $h \equiv 0$ there is no Wick constant and no scheme, and $\mathcal{F}$ consists of the f_{il} and $\rho^* f_{il}$ with $U = (\mathrm{i}\mathbb{R})^m$. What the argument of Steps 5–7 needs of the cube functionals is: the existence of $Z_J(\cdot, 0) = \lim_k Z_{k,J}(\cdot, 0)$ locally uniformly on $\mathbb{C}^m$, entire; the monotonicity of $J \mapsto Z_J(\mathrm{i}f, 0)$ on the cone (first inequality of Lemma 2.11); the existence of $Z_\infty(\cdot, 0) = \lim_J Z_J(\cdot, 0)$, entire, locally uniformly on $\mathbb{C}^m$; and translation invariance. The first three are Theorem 7.4(b), whose local bound is the μ_0-free bound of Theorem 7.4(a) and Lemma 7.3 rather than a local bound resting on the nonincreasing clause of Lemma 7.7 and so on [7, Thm. 6.2(3),(4)]; the fourth is Proposition 8.1, which uses Lemma 2.15 with $\ln Z_\infty$ finite on all of E_ρ (upper bound from Theorem 7.4(b), lower bound Lemma 2.12(c)) and the identity theorem. The lattice steps Lemmas 9.3, 9.9 and 9.10 include $g = 0$, and only their $\mathcal{I}$ halves are used; Lemma 10.3(C) is the fixed-J and J-uniform bound; Steps 5–7 in version (C) give the claim. □

Theorem 10.7 (reflection positivity on the Wick tube, comparison convention)**.** *Let a be an admissible scheme, and assume* (I2) *and* (I3) *(the hypotheses of Corollary 13.3) and, as in Proposition 10.5, that the comparison family is convergent at fixed cubes on the tube (Definition 3.2); the $\mathcal{D}$-class steps use in addition Lemma 2.8(d). Let $\sigma_+ = s_1$ be the amplitude of*

(I2) *and* $\sigma_W := \sigma_+/8$. *Let* $\mu_0 > 0$, ν, $c \in \mathbb{R}$, *and let* $\zeta_1, \dots, \zeta_N$ *be finite families of real sources* $f_{il} \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, $h_{il} \in C_c^\infty(\mathbb{R}^3)$ *supported in* $\{x_\nu > c\}$, *with complex coefficients* z_{il}, w_{il} *such that* $\mathsf{a}(\zeta_i) < \sigma_W$ *for every* i; *in particular, all purely imaginary* w_{il} *with arbitrary* z_{il} *are admitted. Then the values* $Z^a_{\infty,T}(\theta\zeta_i^* \sqcup \zeta_j) = \lim_J Z^a_J(\theta\zeta_i^* \sqcup \zeta_j)$ *exist (Proposition 10.5) and*

$$\sum_{i,j=1}^{N} \bar\gamma_i \gamma_j \, Z^a_{\infty,T}\big(\theta\zeta_i^* \sqcup \zeta_j\big) \;\geq\; 0 \qquad \text{for all } \gamma \in \mathbb{C}^N.$$

Where Z^a_∞ *is defined,* $Z^a_{\infty,T} = Z^a_\infty$ *there. No lower bound on* J *beyond* J_0 *enters, and* (W2) *is not used.*

Proof. With the splitting of Definition 10.1 the matrix points lie in $T_{\mathcal{F}}$ by (21). Lemma 10.3(T) gives the bounds; Steps 5 and 6 in version (T), with Proposition 10.5(a) for the limit on $U \cap T_{\mathcal{F}}$ and its holomorphy on $T_{\mathcal{F}}$, give (24) for $c \in D$; Step 7 with Proposition 10.5(b) gives every plane. The $\mathcal{D}$-class steps are the first factor of (22) (the $\mathcal{D}$ half of Lemma 9.9 and the second inequality of Lemma 2.11) and the nonincreasing factor in Proposition 10.5(a). □

The arbitrary-amplitude form, reflection positivity of Z^a_∞ at all complex coefficients under (W2), is Theorem 23.8 in the optional extension; its proof is Steps 5–7 in version (A) for $c \in D$, where (W2) enters only at the finitely many q_v, and in version (A) or (A′) for $c \notin D$.

Lemma 10.8 (reflection positivity of King's printed state)**.** *Theorem 10.6 holds for* Z^K_∞*: for* ν, c, $\zeta_1, \dots, \zeta_N$ *as there,* $\sum_{i,j} \bar\gamma_i \gamma_j Z^K_\infty(\theta\zeta_i^* \sqcup \zeta_j, 0) \geq 0$ *for all* $\gamma \in \mathbb{C}^N$*; and Propositions 10.9 and 10.10 below hold for* $Z^K_\infty(\cdot, 0)$. *The inputs are Theorem 3.3,* [7, Thm. 6.2(1),(2)]*, Theorem 7.6(a),(b) in place of Theorem 7.4, Lemma 4.6(c) and Remark 4.8 for the two places where the cube convention is seen, and Corollary 8.3 for the translations; no inequality of class* $\mathcal{D}$ *enters.*

Proof. Lemmas 9.3 and 9.5 concern the modified box state alone and the lattice reflection, and Lemma 10.3(C) bounds Φ_k on the modified photon form: these are convention-free, as is Step 5 (Montel in k; positivity passes). The two places where the Neumann–Dirichlet cube states enter are the ends of the sandwich (23), i.e. the $\mathcal{I}$ halves of Lemmas 9.9 and 9.10; on King's convention they read $Z^K_{k,J}(\zeta) \leq \Phi_k(\zeta) \leq Z^K_{k,J+2n}(\zeta)$ for every k and $\zeta \in U = (\mathrm{i}\mathbb{R})^m$: the lower end is Lemma 9.9 with Step 1 of Lemma 4.2 prepended (Remark 4.8: $\langle\cdot\rangle^K_{k,J} \leq \langle\cdot\rangle^{(a)}_{k,J} \leq \langle\cdot\rangle^{(3.38)}_{k,\Omega_{J,k}}$ on $\mathcal{I}$), the upper end is Lemma 9.10 with $\langle\cdot\rangle^K_{k,J}$ on X_{J+2n} in place of the comparison state (Lemma 4.6(b),(c): the box $\Omega_{J,k}$ does not meet the killed layers T_{J+2n}, and part (a) of Lemma 9.10 is photon-only), under the hypotheses of (20). Here J_0 is the J_0 of the enlarged family (the sources and their reflections), which exceeds that of the sources alone when $c \neq 0$; Theorem 7.6(b) is applied to the enlarged family at J and $J + 2n$, $J \geq J_0$. Along Step 5's subsequence the ends converge along the full sequence by Theorem 7.6(b), so $Z^K_J \leq \Phi^J \leq Z^K_{J+2n}$ on U for every subsequential Φ^J; as $J \to \infty$ both ends tend to Z^K_∞ on U, $\{\Phi^J\}_J$ is locally bounded on $\mathbb{C}^m$ by Lemma 10.3(C), and U is a set of uniqueness, so $\Phi^J \to Z^K_\infty$ locally uniformly on $\mathbb{C}^m$ as in Step 6, in particular at the matrix points; positivity passes to the limit. Step 7 uses Corollary 8.3 in place of Proposition 8.1. Propositions 10.9 and 10.10 use only the existence, holomorphy near 0 and lattice symmetries of the limit, and the locally uniform convergences just established; their proofs below apply verbatim with Z^K_∞, the charge-conjugation invariance being exact for $\langle\cdot\rangle^K_{k,J}$ as well. □

10.6. OS3 and the polynomial form.

Proposition 10.9 (symmetry, realness and charge conjugation)**.** *Let* Z *be one of:* $Z_\infty(\cdot, 0)$ *if the comparison family is convergent at fixed cubes (resp.* $Z^K_\infty(\cdot, 0)$*);* $Z^a_{\infty,T}$ *under the hypotheses of Proposition 10.5;* Z^a_∞ *under* (W2) *(optional extension). The Schwinger functions*

$$S_{n,p}(f_1, \dots, f_n; h_1, \dots, h_p) := \partial_{z_1} \cdots \partial_{z_n} \partial_{w_1} \cdots \partial_{w_p} \, Z\big(\textstyle\sum_i z_i f_i, \sum_l w_l h_l\big)\big|_0$$

(with $p = 0$ *and* S_n *in the curvature sector) are (a) symmetric multilinear in the* f*'s and in the* h*'s; (b) real for real test functions; (c) zero for odd* n. *The proof uses only that* Z *exists and*

is holomorphic near 0*, with the stated symmetries of the lattice functionals; the existence is the input of the respective sector.*

Proof. (a) Z depends on a family only through its total sources (so does every lattice functional, hence both limits), so its Taylor coefficients at 0 are symmetric multilinear forms in the f's and in the h's; $Z^a_{\infty,T}$ is holomorphic on a tube containing 0. The definition of $S_{n,p}$ does not depend on the family used to compute it (restrict a larger family to the given sources). (b) For real coefficients every lattice functional is real (real integrand and multiplier, Lemma 2.12(a)), so Z is real at the real points near 0 (for $Z^a_{\infty,T}$ on $\mathbb{R}^m \cap T$, an open piece of $\mathbb{R}^m$ containing 0); a holomorphic function real on the real points near 0 has real Taylor coefficients at 0. (c) Every cube state and every modified box state is invariant under $(A,\vartheta)\mapsto(-A,-\vartheta)$ (the photon terms are even in A, the masked divergence included; $\cos(\vartheta_x - e\varepsilon A - \vartheta_y)$ is invariant; W^a depends on r only; Lebesgue measure is preserved), which maps $F_{\Lambda_k}(f)\mapsto -F_{\Lambda_k}(f)$ and fixes $W^a_{\Lambda_k}(h)$. Hence $Z_{k,X}(-z,w) = Z_{k,X}(z,w)$ for all k, X, and in both limits $Z(-z,w) = Z(z,w)$ (on the $z\mapsto -z$ invariant tube for $Z^a_{\infty,T}$). So $\partial_z^{(n)}\partial_w^{(p)} Z(0) = 0$ for odd n. □

Proposition 10.10 (the polynomial form of reflection positivity)**.** *Let $P = \sum_\alpha c_\alpha \prod_i F(f_{\alpha i}) \prod_l W^a(h_{\alpha l})$ be a polynomial in the smeared fields with real sources supported in $\{x_\nu > c\}$, and $\theta P := \sum_\alpha \bar c_\alpha \prod_i F(\rho^* f_{\alpha i}) \prod_l W^a(h_{\alpha l}\circ\rho)$. Then $S(\theta(P)P)\ge 0$, where S assigns to a monomial the corresponding Schwinger function of Proposition 10.9: the Taylor coefficient at 0 of $Z^a_{\infty,T}$ under the hypotheses of Theorem 10.7, resp. of $Z_\infty(\cdot,0)$ (of $Z^K_\infty(\cdot,0)$), where at $h\equiv 0$ the W^a factors are absent.*

Proof. Let all sources of P be supported in $\{x_\nu \ge c+\delta\}$ and let $\mathcal{F}'$ be the family of all sources of P and their reflections. Its scalar members have no sign, so Steps 4–6 are applied to the split family $\mathcal{F}'_{\text{split}}$ (members $h^\pm_{\alpha l}$, $h^\pm_{\alpha l}\circ\rho$, all ≥ 0); its tube contains 0 ($k_1(0)=0$), and the unsplit functional is the composition of the split one with the linear map $w_{\alpha l}\mapsto (w^+_{\alpha l}, w^-_{\alpha l}) = (w_{\alpha l}, -w_{\alpha l})$, so by the chain rule the derivatives at 0 of the unsplit Φ_k, Φ^J, $Z^a_{\infty(,T)}$ are finite linear combinations of those of the split ones, and locally uniform convergence near 0 passes through the linear map. (i) *Lattice.* For $k\ge k_0$ the sampled P is a gauge-invariant polynomial in the plus-side smeared fields, lies in L^2 of the modified box state (Gaussian moments; $\lambda>0$) and in $\mathfrak{B}_+$ (Definition 9.1), and $\theta(P) = \overline{P\circ\hat\rho}$ by (18). By Lemma 9.3(a) (its L^2 extension), $\langle\theta(P)P\rangle^{(3.38)}_{k,\Omega_{J,k}} \ge 0$. (ii) *Moments as derivatives.* $\langle\theta(P)P\rangle^{(3.38)} = \sum_{\alpha,\beta}\bar c_\alpha c_\beta \langle\prod_i F(\rho^* f_{\alpha i})\prod_l W^a(h_{\alpha l}\circ\rho)\prod_i F(f_{\beta i})\prod_l W^a(h_{\beta l})\rangle^{(3.38)}$, and each mixed moment is a mixed partial derivative at 0 of Φ_k for the family $\mathcal{F}'$ (differentiate under the integral; the scheme multiplier $e^{-wa_k\mathsf{S}(h)}$ is the $-a_k$ inside W^a). (iii) *Limits.* By Steps 5–6 for $\mathcal{F}'_{\text{split}}$, $\Phi_{k_m}\to\Phi^J$ and $\Phi^J\to Z^a_{\infty(,T)}$ locally uniformly on a neighbourhood of 0, so by Weierstrass' theorem all derivatives at 0 converge, and $\sum \bar c_\alpha c_\beta\, \partial^{(\alpha,\beta)} Z^a_{\infty(,T)}(0)\ge 0$; the left side is $S(\theta(P)P)$ by the family-independence of the Schwinger functions (Proposition 10.9(a)). Every plane: Step 7 (translation of the sources; S is translation invariant by Proposition 10.5(b), resp. Proposition 8.1 or Corollary 8.3). The curvature sector is the case $h\equiv 0$ with no splitting. This is the analogue, for the curvature and the renormalised Wick square with reflection in the planes $\{x_\nu = c\}$, of the positivity condition [19, Axiom E2, p. 88] in its form for finite sums of tensor products of test functions supported in a half-space. □

Remark 10.11 (which form is used, and the amplitude constants). The main theorem (Theorem 21.1) states reflection positivity of the Wick sector in the tube form of Theorem 10.7, with Propositions 10.9 and 10.10 for the Schwinger functions; its inputs for this clause are Theorem 3.6, (I2), (I3) and [7, Thm. 6.2(1)–(4)], without (W2). This suffices for the consumer: the weak form of OS4 and the reconstruction of Theorem 11.1 use reflection positivity for exponentials $e^{\mathrm{i}\Phi(f)}$ of the smeared fields, whose Wick coefficients are purely imaginary, so amplitude 0 only. Three positive, ε-free amplitude constants occur in the Wick sector: $\sigma_W = \sigma_+/8$ here; $H/2 = \sigma_+/32$, the negative-part radius of the tube $\mathcal{D}_{\sigma_+,H}$ of Theorem 13.10 ($H = s_1/16$); and $\sigma_0 = H/4$ in the bound E0″ of Theorem 13.10(ii). On $T_{\mathcal{F}}\cap\mathcal{D}_{\sigma_+,H}$ the objects $Z^a_{\infty,T}$ and G_∞ coincide (Proposition 10.5(a)). On King's convention the tube statements of this section are

read under (TW^K) (Remark 13.12, Proposition 21.4); the curvature-sector statement needs no such hypothesis (Lemma 10.8). The margin $R(\mu_0)$ of Lemma 9.10 enters the proof, not the conclusion: reflection positivity is proved at every fixed $\mu_0 > 0$, and in a limit of vanishing photon mass it passes to the limit functionals as a pointwise limit of positive semidefinite matrices, the polynomial form by Weierstrass' theorem.

11. The measure and weak clustering

This section builds the measure of King's printed state in the curvature sector and derives from it, with the general results on the measure, its ergodic decomposition, reconstruction and clustering proved in the full version, the consequences of reflection positivity: the ergodic decomposition, clustering, the unique joint vacuum on a component, the growth of the components' moments, and weak OS4 on almost every component. The joint field with the Wick square is the subject of the tube sections. The measure is built from the characteristic functional, not from the lattice laws, which converge only locally (Remark 11.2).

Conventions. $\mathcal{V} := \Lambda^2\mathbb{R}^3$ with its Euclidean fibre norm. For a real $u \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ put $\mathcal{Z}^K(u) := Z_\infty^K(iu)$, the one-member functional at the coefficient i (Theorem 7.6(b), with the total-source convention of Theorem 8.5). Fix an even $r \geq 14$. $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ is the Hermite–Sobolev space, the dual of the completion of $\mathcal{S}(\mathbb{R}^3;\Lambda^2)$ under $\|u\|_r^2 = \sum_{\alpha,c}\lambda_\alpha^r\langle h_{\alpha,c},u\rangle^2$, where $(h_{\alpha,c})_{\alpha\in\mathbb{N}^3,\,c\leq 3}$ is the Hermite basis of $L^2(\mathbb{R}^3;\Lambda^2)$ and $\lambda_\alpha = 2|\alpha| + 4$ are the eigenvalues of $-\Delta+|x|^2+1$; $|\cdot|_4$ is the weighted C^4 norm of Lemma 2.7, and $|u|_4 \leq C\|u\|_{10}$ on $\mathcal{S}$ (Lemma P.3(b) of the full version). $X := H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2)$ carries the actions $(\tau_a\Phi)(u) = \Phi(\tau_{-a}u)$, $(g\Phi)(u) = \Phi(g^{-1}\cdot u)$ for $g \in O_h$ (or $O(3)$), $(\theta_j\Phi)(u) = \Phi(\sigma_j\cdot u)$ for the coordinate reflections σ_j, with the signed transport of 2-forms of Definition 2.4, and the local σ-algebras $\Sigma(U)$ generated by $\Phi \mapsto \Phi(u)$, $u \in C_c^\infty(U;\Lambda^2)$ real; $\theta_{j,c} := \tau_{ce_j}\theta_j\tau_{-ce_j}$ is the reflection in the plane $x_j = c$ and $\rho_{j,c*}u$ the reflected source (Definition 9.1). These are the space, actions and σ-algebras of Definition P.2 of the full version with the fibre $\mathcal{V} = \Lambda^2$. The *lattice curvature field* of King's cube is the random distribution $\Phi^K_{k,J} : u \mapsto F_{\Lambda_k}(u)$ under $\langle\cdot\rangle^K_{k,J}$. $J_0(u)$ is the least J with $\operatorname{supp} u$ at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$.

Theorem 11.1 (the measure, the local lattice laws, weak clustering)**.** *The limits below exist by Theorem 3.3.*

(a) (The measure; uses Theorem 7.6 and Corollary 8.3 only.) *For every even $r \geq 14$ there is exactly one Borel probability measure $\mu^{K,H}$ on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ whose mixed moments on C_c^∞ are the Schwinger functions S_n^K of King's printed state. Its characteristic functional on C_c^∞, indeed on $\mathcal{S}$, is $\mathcal{Z}^K$ (extended by continuity); its moments on $\mathcal{S}$ are the continuous extensions of the S_n^K and satisfy the bounds of Theorem 7.6(c). Its image μ_S^K under the restriction map $\iota : \mathcal{H}^{-r} \to X$ is the unique Borel probability measure on X with characteristic functional $\mathcal{Z}^K$ on C_c^∞, and also the unique one whose mixed moments on C_c^∞ are the S_n^K.* Local convergence of the lattice laws: *for every real $\chi \in C_c^\infty(\mathbb{R}^3)$ and every box Q containing $\operatorname{supp}\chi$ in its interior, the laws $\nu^K_{k,J,\chi}$ of the localised lattice fields $\chi\Phi^K_{k,J}$ on $H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$ converge weakly, along the full sequence $k \to \infty$ at every fixed $J \geq J_0(\chi)$, to a law $\nu^K_{J,\chi}$, and $\nu^K_{J,\chi} \Rightarrow \nu^K_\chi$ as $J \to \infty$, where ν^K_χ is the law of $\chi\Phi$ under μ_S^K. Nothing is asserted about the laws of the unlocalised lattice fields at fixed J (Remark 11.2).*

(b) (Invariance and weak clustering; uses also Theorem 8.5 and Lemma 10.8.) *μ_S^K satisfies the hypotheses (H1)–(H3) of the full version, is invariant under $\mathbb{R}^3 \rtimes O_h$ about every point, and the results of the full version apply to it: $\mu_S^K = \int \mu_\xi\, d\nu(\xi)$ with components that are translation-invariant and ergodic, θ_j-invariant, reflection positive about every coordinate plane and satisfy (H2) (Theorem M.6 of the full version); clustering in the mean (Theorem N.8 of the full version) and under diagonal separation (Theorem N.10 of the full version); the Osterwalder–Schrader reconstruction in each coordinate direction (Proposition N.5 of the full version) with a unique joint vacuum on every component (Corollary N.12(iii) of the full version); the growth bound $|S_N^\xi(u_1,\dots,u_N)| \leq \sigma_N^\xi \prod_i \|u_i\|_{r+2}$, $\sigma_N^\xi \leq \alpha_\beta(\xi)(N!)^\beta$ for every $\beta > 1$, for the components' Schwartz moments (Theorem M.12*

and Corollary M.13 of the full version); and the cubic-invariant decomposition into mixtures of $k(\zeta) \in \{1,2,3,6\}$ ergodic measures (Theorem M.9 of the full version). The same holds on $\mathcal{H}^{-r}$ for $\mu^{K,H}$ (Lemma P.10 and Remark P.11 of the full version). ν-almost every component satisfies OS4 *in the weak form, ergodicity under the translation group $\mathbb{R}^3$ (Definition N.2(iii) of the full version, Corollary N.12(i) of the full version).*

(c) (The rotation corollary.) *$\mathcal{Z}^K$ is $O(3)$-invariant, $\mathcal{Z}^K(R\cdot u) = \mathcal{Z}^K(u)$ for all $R \in O(3)$ and real u, by Theorem 8.7 (the rotation theorem Theorem 3.14); the clauses below use only this invariance. Then for ν-a.e. ξ: μ_ξ is invariant under $\mathbb{R}^3 \rtimes O(3)$; $\mathbb{E}_{\mu_\xi}[\overline{F\circ\theta_{n,c}}\,F] \geq 0$ for every unit vector n, every $c \in \mathbb{R}$ and every $F \in L^2(\Sigma(H^+_{n,c}),\mu_\xi)$, $H^+_{n,c} = \{x\cdot n > c\}$; and $\mathbb{E}_{\mu_\xi}[G'\,U(a)G] \to \mathbb{E}_{\mu_\xi}[G']\,\mathbb{E}_{\mu_\xi}[G]$ as $|a| \to \infty$ in any manner, for all $G, G' \in L^2(\mu_\xi)$. Consequently μ_ξ satisfies* OS4 *in each of the forms of Definitions N.1 and N.2 of the full version and Definition O.6 of the full version (the cluster property on the positive-time algebra, ergodicity and clustering of the translations along the time direction, ergodicity under $\mathbb{R}^3$) for every choice of the time direction, coordinate or not, and in the reconstruction from μ_ξ in any direction n the joint fixed space is* $\ker H_n = \mathbb{C}\Omega_n$ *(Corollary O.7 of the full version, applied in Step 5 of the proof). The moments of μ_ξ are real and symmetric; the vanishing of its odd curvature moments is* not *claimed for a component.*

11.1. **Proof of (a).** *Step 1 (moment bounds, and where they hold).* For real 2-forms $f_1,\dots,f_n$ at distance >1 from $\mathbb{R}^3\setminus C_{J_0}$, Theorem 7.6(c) gives, at every k and $J \geq J_0$, $|S_n^{K,(k,J)}| \leq (\sqrt{2}\,e)^n (n!)^{1/2}\prod_j \|f_j\|_{2,k}$ with $\|f\|^2_{2,k} \leq 3C_0|f|_4^2$ uniformly in k (Lemma 2.7), and in the limit the bound of Theorem 7.6(c); by Theorem 7.6(a), $\langle F(u)^2\rangle^K_{k,J} \leq 2\|u\|^2_{2,k}$ and hence $S_2^K(u,u) \leq 2\|u\|_2^2$. The lattice Schwinger functions are the Taylor coefficients at 0 of $a^K_{k,J}$, that is the moments of $\Phi^K_{k,J}$, and converge to the S_n^K. These bounds are uniform in k and $J \geq J_0$ for sources at distance >1 inside C_{J_0}; at a fixed J they are not available for sources reaching ∂C_J, and neither is the convergence in k: Theorem 3.3 and Theorem 7.6(a) are stated for sources whose plaquettes have their corners in X_J.

Step 2 (the measure). (i) *The functional on $\mathcal{S}$.* For real $u_1,\dots,u_N \in C_c^\infty$ and $c \in \mathbb{C}^N$, at every (k,J) with $J \geq J_0(u)$, $\sum_{l,m}\bar c_l c_m\, Z^K_{k,J}(\mathrm{i}(u_m - u_l)) = \langle|\sum_m c_m e^{\mathrm{i}F(u_m)}|^2\rangle^K_{k,J} \geq 0$ by the linearity of F_{Λ_k} in the source. The limits $k\to\infty$, then $J\to\infty$, exist at these points by Theorem 7.6(b) for the family u, so $\mathcal{Z}^K$ is positive definite on C_c^∞ with $\mathcal{Z}^K(0)=1$. By Corollary 8.3's Lipschitz bound and (E2), $|\mathcal{Z}^K(u)-\mathcal{Z}^K(u')| \leq \sqrt{2}\|u-u'\|_2 \leq \frac{\pi}{2}|u-u'|_4$. Since $|\cdot|_4$ is a continuous seminorm on $\mathcal{S}(\mathbb{R}^3;\Lambda^2)$ and C_c^∞ is dense in $\mathcal{S}$, $\mathcal{Z}^K$ extends uniquely to a continuous function on $\mathcal{S}$, still positive definite with $\mathcal{Z}^K(0)=1$. By the Bochner–Minlos theorem [10, 12] ($\mathcal{S}$ is nuclear) there is exactly one Borel probability measure μ' on $\mathcal{S}'(\mathbb{R}^3;\Lambda^2)$, with its cylinder σ-algebra, such that $\int e^{\mathrm{i}\Phi(u)}\,d\mu' = \mathcal{Z}^K(u)$ for all real $u\in\mathcal{S}$.

(ii) *Moments on C_c^∞, and on $\mathcal{S}$ by uniform integrability.* For real $u_1,\dots,u_N \in C_c^\infty$ the characteristic function of $(\Phi(u_l))_l$ under μ' is $\varphi(t) = \mathcal{Z}^K(\sum_l t_l u_l) = Z^{K,u}_\infty(\mathrm{i}t)$, the restriction to $\mathrm{i}\mathbb{R}^N$ of the entire function $Z^{K,u}_\infty$ (Theorem 7.6(b)). A characteristic function real-analytic near 0 has moments of all orders, with $\int\prod_l\Phi(u_l)\,d\mu' = (-\mathrm{i})^N\partial_{t_1}\cdots\partial_{t_N}\varphi(0)$ [18]; the chain rule gives $\partial_{t_1}\cdots\partial_{t_N}\varphi(0) = \mathrm{i}^N\partial_{z_1}\cdots\partial_{z_N}Z^{K,u}_\infty(0)$, so the factors cancel and

(25)
$$\int\prod_{l=1}^N\Phi(u_l)\,d\mu' = S_N^K(u_1,\dots,u_N);\qquad \int\Phi(u)^2\,d\mu' = S_2^K(u,u) \leq 2\|u\|_2^2 \leq \tfrac{\pi^2}{4}|u|_4^2 \leq C_2''\|u\|_{10}^2$$

by Step 1, (E2) and $|u|_4 \leq C\|u\|_{10}$. *Extension to $\mathcal{S}$.* By Theorem 7.6(c) each S_N^K extends to a continuous symmetric multilinear form on $\mathcal{S}^N$ with $|S_N^K| \leq \sigma_N\prod_l|u_l|_4$. For $u_l\in\mathcal{S}$ take cut-offs $\chi_R = \chi(\cdot/R)$, $\chi\in C_c^\infty$, $\chi=1$ on the unit ball, $0\leq\chi\leq1$, $R\geq1$: $\chi_R u_l \to u_l$ in $\mathcal{S}$, so $\Phi(\chi_R u_l)\to\Phi(u_l)$ for every $\Phi\in\mathcal{S}'$; and $|\chi_R u|_4 \leq C_\chi|u|_4$ uniformly in $R\geq1$ with $C_\chi = 2^4\|\chi\|_{C^4}$, by Leibniz with $|R^{-|\beta|}(D^\beta\chi)(\cdot/R)| \leq \|\chi\|_{C^4}$ (the norm $|\cdot|_4$ carries derivatives up to order 4, so the cut-off does not pass through it with constant 1). Hence $\int\prod_l\Phi(\chi_R u_l)^2\,d\mu' = S^K_{2N}(\chi_R u_1,\chi_R u_1,\dots) \leq \sigma_{2N}C_\chi^{2N}\prod_l|u_l|_4^2$ uniformly in R: the products

$\prod_l \Phi(\chi_R u_l)$ are bounded in $L^2(\mu')$, hence uniformly integrable, and converge pointwise, so $\int \prod_l \Phi(u_l)\, d\mu' = \lim_R S_N^K(\chi_R u_1, \dots) = S_N^K(u_1, \dots, u_N)$ by the continuity of S_N^K on $\mathcal{S}^N$.

(iii) *The Hermite carrier.* Put $M(\Phi) := \sum_{\alpha,c} \lambda_\alpha^{-r} \Phi(h_{\alpha,c})^2$. By (25) with $\|h_{\alpha,c}\|_{10}^2 = \lambda_\alpha^{10}$, $\int M\, d\mu' \le 3C_2'' \sum_\alpha \lambda_\alpha^{10-r} < \infty$ as soon as $r - 10 > 3$ (there are $(n+1)(n+2)/2$ multi-indices with $|\alpha| = n$), that is for every even $r \ge 14$; so $\mu'(M < \infty) = 1$. For $\Phi \in \mathcal{S}'$ and $u \in \mathcal{S}$ the Hermite expansion $u = \sum_{\alpha,c} \langle h_{\alpha,c}, u\rangle\, h_{\alpha,c}$ converges in $\mathcal{S}$ (Lemma P.3(c) of the full version), so $\Phi(u) = \sum_{\alpha,c} \langle h_{\alpha,c}, u\rangle\, \Phi(h_{\alpha,c})$ and, on $\{M < \infty\}$, $|\Phi(u)| \le M(\Phi)^{1/2} \|u\|_r$ by Cauchy–Schwarz: Φ extends to $\Psi(\Phi) \in \mathcal{H}^{-r}$ with $\|\Psi(\Phi)\|_{-r}^2 = M(\Phi)$ and $\Psi(\Phi) = \Phi$ on $\mathcal{S}$. The map $\Phi \mapsto \Psi(\Phi)$ ($:= 0$ off $\{M < \infty\}$) is measurable from the cylinder σ-algebra to the Borel σ-algebra of the separable Hilbert space $\mathcal{H}^{-r}$, its coordinates being evaluations. Put $\mu^{K,H} := \Psi_* \mu'$. Then $\int e^{\mathrm{i}\Psi(u)}\, d\mu^{K,H} = \mathcal{Z}^K(u)$ and $\int \prod_l \Psi(u_l)\, d\mu^{K,H} = S_N^K(u_1, \dots, u_N)$ for all $u, u_l \in \mathcal{S}$, by (ii). This is Lemma P.7 of the full version, applied with the two-point bound (25) on $\mathcal{S}$.

(iv) *Uniqueness on $\mathcal{H}^{-r}$.* By Proposition P.8(b) of the full version on $\mathcal{H}^{-r}$, $\mu^{K,H}$ is the only Borel probability measure on $\mathcal{H}^{-r}$ whose mixed moments on C_c^∞ exist and equal the S_n^K, provided the moments m_n of $Z = \sum_l t_l \Psi(u_l)$ satisfy $|m_n| \le c_S (n+1) K^n n!$ for constants c_S, K depending on the u_l and t. Our moments give this with $c_S = 1$: $m_n = S_n^K(g, \dots, g)$ for $g = \sum_l t_l u_l$, and $|m_n| \le (\sqrt{2}\, e)^n (n!)^{1/2} \|g\|_2^n \le (n+1) K^n n!$ with $K = \sqrt{2}\, e \|g\|_2$.

(v) *The law on X and its two uniqueness statements.* The map $\iota : \mathcal{H}^{-r} \to X$, $\iota\Psi := \Psi|_{C_c^\infty}$, is continuous, linear and injective (Lemma P.4(d) of the full version): $|\Psi(\chi f)| \le \|\Psi\|_{-r} \|\chi f\|_r \le C_\chi \|\Psi\|_{-r} \|f\|_{H^r}$ for a cut-off χ, and C_c^∞ is dense in $\mathcal{S}$, which is dense in $\mathcal{H}^r$. Put $\mu_S^K := \iota_* \mu^{K,H}$; its characteristic functional on real C_c^∞ is $\mathcal{Z}^K$. By Proposition P.8(a) of the full version on X (for the countable $(f_n) \subset C_c^\infty$ of Lemma P.4(c) of the full version the sets $\{\Phi : (\Phi(f_1), \dots, \Phi(f_N)) \in B\}$ form a π-system generating the Borel σ-algebra, and the law of each such vector is determined by $t \mapsto \mathcal{Z}^K(\sum_k t_k f_k)$), μ_S^K is the only Borel probability measure on X with this characteristic functional. *Moment uniqueness on X:* if μ'' is any Borel probability measure on X whose mixed moments on C_c^∞ are the S_n^K, Proposition P.8(b) of the full version on X applies with the moment bound of (iv), so $\mu'' = \mu_S^K$. And $\mu^{K,H}$ is the only measure on $\mathcal{H}^{-r}$ with $\iota_* \mu^{K,H} = \mu_S^K$, since ι is a Borel isomorphism onto the Borel set $\iota\mathcal{H}^{-r}$. Nothing in Step 2 sees the lattice: the measure clauses of (a) depend on King's printed state only through $\mathcal{Z}^K$ and the S_n^K.

Step 2′ (local convergence of the lattice laws). Fix a real $\chi \in C_c^\infty(\mathbb{R}^3)$ and a box Q with $\operatorname{supp}\chi$ compact in the interior of Q, and let $(e_{\alpha,c})_{\alpha \in \mathbb{Z}^3, c \le 3}$ be the Fourier basis of $L^2(Q; \Lambda^2)$, $|e_{\alpha,c}(x)| = |Q|^{-1/2}$. For a real Q-periodic smooth 2-form ψ put $\langle \chi \Phi_{k,J}^K, \psi\rangle := \Phi_{k,J}^K(\chi\psi) = F_{\Lambda_k}(\chi\psi)$, where $\chi\psi \in C_c^\infty$ is supported in $\operatorname{supp}\chi$ (for the complex basis vectors, used only in (1), take real and imaginary parts); this identifies $\chi\Phi_{k,J}^K$ with an element of the real Hilbert space $H_\chi := H_{\mathrm{per}}^{-r}(Q; \Lambda^2)$, and $\nu_{k,J,\chi}^K$ denotes its law.

(1) *Second moments, uniformly.* For $J \ge J_0(\chi)$ and every k, every plaquette carrying $\chi e_{\alpha,c}$ has its centre in $\operatorname{supp}\chi$, hence its corners in X_J, so the second-moment clause of Theorem 7.6(a) applies to the real and imaginary parts: $\mathbb{E}_{k,J}^K |F_{\Lambda_k}(\chi e_{\alpha,c})|^2 \le 2\|\chi \operatorname{Re} e_{\alpha,c}\|_{2,k}^2 + 2\|\chi \operatorname{Im} e_{\alpha,c}\|_{2,k}^2 \le 4\|\chi e_{\alpha,c}\|_{2,k}^2$. *Riemann-sum bound:* the plaquettes of one orientation have their centres on a translate of $\varepsilon_k \mathbb{Z}^3$; those carrying $\chi e_{\alpha,c}$ have centres in $\operatorname{supp}\chi$; the cubes of side $\varepsilon_k \le 1$ about these centres are disjoint and lie in the $\frac{1}{2}$-neighbourhood $N_{1/2}(\operatorname{supp}\chi)$; so $\|\chi e_{\alpha,c}\|_{2,k}^2 \le 3\,\mathrm{vol}(N_{1/2}(\operatorname{supp}\chi))\, \|\chi\|_\infty^2 / |Q| =: K_\chi$, uniformly in α, c, k and $J \ge J_0(\chi)$ (no derivative of χ enters). Hence, for $s_0 > \frac{3}{2}$,

(26)
$$\mathbb{E}_{k,J}^K \|\chi \Phi_{k,J}^K\|_{H_{\mathrm{per}}^{-s_0}(Q)}^2 = \sum_{\alpha,c} (1+|\alpha|^2)^{-s_0}\, \mathbb{E}_{k,J}^K |F_{\Lambda_k}(\chi e_{\alpha,c})|^2 \;\le\; 12 K_\chi \sum_{\alpha \in \mathbb{Z}^3} (1+|\alpha|^2)^{-s_0} =: C_\chi' < \infty,$$

uniformly in k and $J \ge J_0(\chi)$ (the constants 4 and $12K_\chi$ are valid but not sharp).

(2) *Tightness.* The embedding $H_{\mathrm{per}}^{-s_0}(Q) \hookrightarrow H_{\mathrm{per}}^{-r}(Q)$ is compact for $r > s_0$ (Rellich on the torus); by Chebyshev, $\nu_{k,J,\chi}^K(\|\cdot\|_{H^{-s_0}} > \rho) \le C_\chi'/\rho^2$, and the H^{-s_0}-ball of radius ρ is compact in H_χ; so $(\nu_{k,J,\chi}^K)_{k \ge 0,\, J \ge J_0(\chi)}$ is tight on the separable Hilbert space H_χ, and Prokhorov's theorem [8] gives weak limit points along k at fixed J.

(3) *Identification at fixed J: the full sequence, and where Theorem 3.3 enters.* For a real trigonometric polynomial ψ, $\mathbb{E}^K_{k,J} e^{\mathrm{i}\langle \chi\Phi^K_{k,J},\psi\rangle} = Z^K_{k,J}(\mathrm{i}\chi\psi) \to Z^K_J(\mathrm{i}\chi\psi)$ as $k\to\infty$, along the full sequence, by Theorem 7.6(b) (that is Corollary 4.4(ii)) for the one-member family $(\chi\psi)$ at $J \ge J_0(\chi) \ge J_0(\chi\psi)$: this is Theorem 3.3 for that family on King's cube, together with the local bound of Theorem 7.6(a) and Montel's theorem. A Borel probability measure on H_χ is determined by its characteristic functional on the dual $H^r_{\mathrm{per}}(Q;\Lambda^2)$, which is continuous there (dominated convergence), hence by its values on the dense set of real trigonometric polynomials. So every weak limit point of $(\nu^K_{k,J,\chi})_k$ is the one probability measure $\nu^K_{J,\chi}$ on H_χ with characteristic functional $\psi\mapsto Z^K_J(\mathrm{i}\chi\psi)$, and $\nu^K_{k,J,\chi}\Rightarrow\nu^K_{J,\chi}$ along the whole sequence $k\to\infty$ by tightness and the uniqueness of the limit point. Without Theorem 3.3 the same argument gives the convergence along every continuum subsequence (Theorem 14.9 and Corollary 14.14), not along the full sequence; the full-sequence clause is what Theorem 3.3 adds for King's printed state.

(4) $J\to\infty$. $\|\cdot\|^2_{H^{-s_0}_{\mathrm{per}}} = \sum_{\alpha,c}(1+|\alpha|^2)^{-s_0}|\langle\cdot,e_{\alpha,c}\rangle|^2$ is lower semicontinuous on H_χ (a supremum of finite sums of continuous functions), so by the portmanteau theorem [8] the bound (26) passes to $\nu^K_{J,\chi}$: $\int\|\cdot\|^2_{H^{-s_0}}\,d\nu^K_{J,\chi}\le C'_\chi$ for every $J\ge J_0(\chi)$; hence $(\nu^K_{J,\chi})_J$ is tight on H_χ by the argument of (2), and $Z^K_J(\mathrm{i}\chi\psi)\to Z^K_\infty(\mathrm{i}\chi\psi)=\mathcal{Z}^K(\chi\psi)$ (Theorem 7.6(b)) identifies every limit point: $\nu^K_{J,\chi}\Rightarrow\nu^K_\chi$, the measure with characteristic functional $\psi\mapsto\mathcal{Z}^K(\chi\psi)$.

(5) *ν^K_χ is the law of $\chi\Phi$ under μ^K_S.* For $\Phi\in X$, $\chi\Phi\in H^{-r}(\mathbb{R}^3;\Lambda^2)$ is supported in $\operatorname{supp}\chi$, compact in the interior of Q, so its periodisation is an element of H_χ with $\langle\chi\Phi,\psi\rangle=\Phi(\chi\psi)$ for Q-periodic smooth ψ; the map $\Phi\mapsto\chi\Phi$ is continuous from X to H_χ, since $|\Phi(\chi\psi)|\le\|\tilde\chi\Phi\|_{H^{-r}}\|\chi\psi\|_{H^r}\le c_\chi\|\tilde\chi\Phi\|_{H^{-r}}\|\psi\|_{H^r_{\mathrm{per}}(Q)}$ for a cut-off $\tilde\chi=1$ on Q. And $\int e^{\mathrm{i}\langle\chi\Phi,\psi\rangle}\,d\mu^K_S = \int e^{\mathrm{i}\Phi(\chi\psi)}\,d\mu^K_S=\mathcal{Z}^K(\chi\psi)$ by Step 2(v).

(6) *Moments.* For every trigonometric polynomial ψ (indeed every $\psi\in H^r_{\mathrm{per}}(Q;\Lambda^2)$, by the uniform-integrability argument of Step 2(ii) with $\chi\psi_n\to\chi\psi$), all moments of $\langle\chi\Phi^K_{k,J},\psi\rangle = F_{\Lambda_k}(\chi\psi)$ are bounded uniformly in k and $J\ge J_0(\chi)$ (Step 1; $|\chi\psi|_4\le d_\chi\|\psi\|_{C^4}$ by Leibniz), so by uniform integrability the moments of $\nu^K_{J,\chi}$ are the Taylor coefficients of Z^K_J at $\chi\psi$, and those of ν^K_χ are the $S^K_n(\chi\psi,\dots,\chi\psi)$, consistently with (25). This proves (a). □

Remark 11.2 (why the convergence clause is local). The convergence $Z^K_{k,J}(\mathrm{i}u)\to Z^K_J(\mathrm{i}u)$ is available only for sources u whose plaquettes have their corners in X_J, that is for u at distance >1 inside C_J, where Theorem 3.3 speaks. For a u reaching ∂C_J the lattice functional is defined only after a convention for the plaquettes with a corner outside X_J (for instance, drop them), and for it no bound uniform in k and no convergence is asserted: the second-moment bound of Theorem 7.6(a) carries the corner hypothesis, and the crude bound $\langle e^{F(\tilde u)}\rangle^K_{k,J}\le e^{\frac12\mu_0^{-2}\|d^*_k\tilde u\|^2_{2,k}}$ (Lemma 2.9 with $C\le\mu_0^{-2}$ as forms) is not uniform in k for a source $\tilde u$ truncated at ∂C_J: $d^*_k\tilde u$ is of order $\|u\|_\infty/\varepsilon_k$ on the $O(\varepsilon_k^{-2}|\partial C_J|)$ bonds of the truncation layer, so $\|d^*_k\tilde u\|^2_{2,k}=O(\varepsilon_k^{-1})$. In particular the fixed-J laws of the unlocalised lattice field on X are not asserted to be tight; nothing below depends on them.

11.2. **Proof of (b) and (c).** *Step 3 ((H1)–(H3)).* The hypotheses (H1)–(H3) are those stated in the full version. On X with the actions of the Conventions, the structural parts of (H1) (a Polish space with a jointly continuous $\mathbb{R}^3$-action and involutive homeomorphisms θ_j with $\theta_j\tau_a=\tau_{\rho_j a}\theta_j$) and all of (H2) hold by Lemma P.5 of the full version, which uses only the space and the push-forward actions. *Invariance of μ^K_S:* by the uniqueness in Step 2(v) (Proposition P.9(a) of the full version) it suffices that $\mathcal{Z}^K(\tau_a u)=\mathcal{Z}^K(u)$ and $\mathcal{Z}^K(\sigma_j\cdot u)=\mathcal{Z}^K(u)$ for real u; the first is Corollary 8.3 at the coefficient i, the second Theorem 8.5 at $z=\mathrm{i}$ with $O=\sigma_j$, and the same theorem with $O=\tau_b g\tau_{-b}$ gives the O_h-invariance about every point. *(H3):* for $E_a=e^{\mathrm{i}\Phi(u_a)}$ with u_a real in $C^\infty_c(\{x_j>c\};\Lambda^2)$, $\overline{E_a\circ\theta_{j,c}}\,E_b=e^{\mathrm{i}\Phi(u_b-\rho_{j,c*}u_a)}$ and

$$\mathbb{E}_{\mu^K_S}\big[\overline{E_a\circ\theta_{j,c}}\,E_b\big]=\mathcal{Z}^K(u_b-\rho_{j,c*}u_a)=Z^K_\infty(\theta\zeta^*_a\sqcup\zeta_b),\qquad \zeta_a=(\mathrm{i},u_a),$$

the conjugated coefficient $-\mathrm{i}$ sitting on the reflected source as in Definition 9.1. By Lemma 10.8, for the plane $x_j=c$ and every $c\in\mathbb{R}$, this Gram matrix is positive semidefinite. The exponential

algebra is dense in $L^2(\Sigma(H^+_{j,c}), \mu^K_S)$ (Lemma P.6 of the full version: an algebra of bounded complex functions, closed under conjugation and containing the constants, is dense in L^2 of the σ-algebra it generates), and $\theta_{j,c}$ preserves μ^K_S, so (H3) follows by Proposition P.9(c) of the full version. Finally Lemma P.10 of the full version carries (H1)–(H3) and the O_h-invariance about every point from (X, μ^K_S) to $(\mathcal{H}^{-r}, \mu^{K,H})$.

Step 4 (weak clustering). The results on the ergodic decomposition, reconstruction and clustering in the full version assume only that μ satisfies (H1)–(H3) there; so Theorem M.6 of the full version gives the ergodic decomposition $\mu^K_S = \int \mu_\xi \, d\nu$ with components translation-invariant and ergodic, θ_j-invariant, satisfying (H3) and (H2); Theorems N.8 and N.10 of the full version give clustering in the mean and under diagonal separation; Proposition N.5 of the full version is the reconstruction in direction j, and on an ergodic component the joint fixed space is $\mathbb{C}\Omega_j$ (Proposition N.6(iii) of the full version, Corollary N.12(iii) of the full version). The hypothesis (312) of the full version of Theorem M.12 of the full version for $\mu^{K,H}$ is supplied by Theorem 7.6(c) through (25): $\int \Psi(h_{\alpha,c})^{2n} \, d\mu^{K,H} = S^K_{2n}(h_{\alpha,c}, \dots, h_{\alpha,c}) \le (e\pi/2)^{2n}((2n)!)^{1/2} |h_{\alpha,c}|^{2n}_4 \le (e\pi/2)^{2n}(2n)! \, |h_{\alpha,c}|^{2n}_4$, and by Remark M.14 of the full version, with the inequality $|u|_4 \le C\|u\|_{10}$ of the Conventions, (312) of the full version holds with $c_S = 1$ and $C_0 = Ce\pi/2$. So Theorem M.12 of the full version gives the growth bound of (b) for the components of $\mu^{K,H}$, and Corollary M.13 of the full version, with the restriction map ι of Step 2(v), for those of μ^K_S; the translation invariance of $\mu^{K,H}$ and the joint continuity of the translations on $\mathcal{H}^{-r}$ assumed there are (H1) for $(\mathcal{H}^{-r}, \mu^{K,H})$, from Step 3. The norm $\|u_i\|_{r+2}$ of (b) dominates $\|u_i\|_r$. Theorem M.9 of the full version assumes in addition that O_h acts on X by homeomorphisms with $\rho_j = \theta_j$, $g\tau_a g^{-1} = \tau_{ga}$, μ invariant under every g, and $G \mapsto G \circ g^{-1}$ mapping $\Sigma(U)$-measurable functions to $\Sigma(gU)$-measurable ones: structural for the push-forward action ($(g\Phi)(u) = \Phi(g^{-1} \cdot u)$ is a linear homeomorphism of X, the relations are identities on test functions, $\operatorname{supp} g \cdot u = g \operatorname{supp} u$) together with the invariance of Step 3; it gives the cubic-invariant decomposition with $k(\zeta) \in \{1, 2, 3, 6\}$. Weak OS4 is Definition N.2(iii) of the full version, ergodicity for the action of $\mathbb{R}^3$, the ergodicity clause of the decomposition for ν-a.e. component.

Step 5 (the rotation corollary). Corollary O.7 of the full version assumes (H1)–(H3) and an admissible action of $O(3)$ on X (Definition O.1 of the full version: homeomorphisms $x \mapsto Rx$, continuous in R for each x, with $\rho_j x = \theta_j x$ for the coordinate reflections, $R\tau_a R^{-1} = \tau_{Ra}$, and $G \mapsto G \circ R^{-1}$ mapping $\Sigma(U)$-measurable functions to $\Sigma(RU)$-measurable ones), under which μ is invariant. (H1)–(H3) hold by Step 3. The push-forward $(R\Phi)(u) = \Phi(R^{-1} \cdot u)$ is an admissible action on $X = H^{-r}_{\mathrm{loc}}(\mathbb{R}^3; \Lambda^2)$ by Lemma O.2(i) of the full version, with $\mathcal{V} = \Lambda^2$. By Lemma O.2(ii) of the full version, the invariance of μ^K_S under every $R \in O(3)$ is the identity $\mathcal{Z}^K(R \cdot u) = \mathcal{Z}^K(u)$ for real u, the hypothesis of (c), that is the $O(3)$-invariance of Z^K_∞ at imaginary coefficients, supplied by Theorem 8.7; this section does not derive it. So Corollary O.7 of the full version applies to μ^K_S and its ergodic decomposition of Step 4, and gives the conclusions quoted in (c): the Euclidean invariance and the reflection positivity about every plane of almost every component (Corollary O.7(a) of the full version,(b)), clustering as $|a| \to \infty$ in any manner (Corollary O.7(c) of the full version), every form of OS4 for every time direction, coordinate or not, and $\ker H_n = \mathbb{C}\Omega_n$ in every direction n (the final clause of Corollary O.7 of the full version). They contain nothing on OS3: the vanishing of the odd curvature moments of a component is not among them, and is not known for a single component.

Step 6 (curvature alone). Everything above runs on $X = H^{-r}_{\mathrm{loc}}(\mathbb{R}^3; \Lambda^2)$ for the curvature field alone: the regularity input is Theorem 7.6, the reflection positivity input Lemma 10.8, the invariance input Theorem 8.5, the $O(3)$ input that of (c). The result is an ergodic decomposition of the law of the curvature field; nothing is claimed about the components of a joint law with the Wick square. □

Remark 11.3 (the hypotheses of the results of the full version, matched). For each result of the full version used above, the step that supplies its hypotheses. (H1)–(H3) of the full version: Step 3 (structure from Lemma P.5 of the full version; invariance from Corollary 8.3 and Theorem 8.5 through Proposition P.9(a) of the full version; positivity from Lemma 10.8, Lemma P.6 of the full version and Proposition P.9(c) of the full version). Lemma P.3 of the full version: no

hypothesis; Step 2(iii). The two-point bound $\mathbb{E}\Phi(f)^2 \le C\|f\|_{10}^2$ of Lemma P.7 of the full version: Step 2(ii). The moment bound of Proposition P.8(b) of the full version, $r \ge 14$ even: Step 2(iv), with $K = \sqrt{2}\,e\|u\|_{L^2}$ and $c = 1$. The uniqueness of a measure on X with a given characteristic functional (Proposition P.8(a) of the full version, with the Borel structure of Lemma P.4 of the full version): Step 2(v). The transfer between X and $\mathcal{H}^{-r}$ (Lemma P.10 of the full version): Steps 2(v) and 3. The standing hypothesis "μ satisfies (H1)–(H3)" of Theorems M.6, N.8 and N.10 and Propositions N.5 and N.6 of the full version: Step 3; the ergodicity needed in Proposition N.6(iii) of the full version holds on a component by Theorem M.6 of the full version. Theorem M.9 of the full version (the O_h action and invariance): Step 4 and Theorem 8.5. The bound (312) of the full version of Theorem M.12 of the full version: Theorem 7.6(c) through Step 2(ii), with $c_S = 1$ and $C_0 = Ce\pi/2$ (Step 4). Corollary O.7 of the full version (the admissible $O(3)$ action and the invariance): Step 5, the structure from Lemma O.2(i) of the full version, the invariance from the hypothesis of (c) through Lemma O.2(ii) of the full version. The forms of OS4 (Definitions N.1 and N.2 of the full version): definitions, Steps 4 and 5; Definition N.2(iii) of the full version is the ergodicity clause of Theorem M.6 of the full version. The same matching for the joint field with the Wick square and for the limit states of the comparison family is Remark P.11(ii) of the full version,(iii), Remark N.13 of the full version and Remark O.9 of the full version. The vanishing of the odd moments is not transferred to the components (clause (c)).

Remark 11.4 (the comparison family). The same conclusions hold for the limit states of the comparison family $\mathrm{ND}^{(a)}$: for every continuum subsequence $\mathcal{K}$ (Definition 14.2) the limit state $Z_\infty^{\mathcal{K}}$ carries a measure with the properties of Theorem 11.1(a) and (b), the local lattice laws converging along $\mathcal{K}$ instead of the full sequence, and (c) (Theorem 14.9, clauses (v)–(vii)). By Theorem 3.3 every such limit state equals Z_∞^K (Corollary 4.4(iii)), so for King's printed state the two routes to the measure agree. For the iterated limit of the comparison family itself, when it is convergent at fixed cubes (Definition 3.1), see Proposition 14.12(b); nothing beyond Corollary 4.4(iii) is asserted about it here, the two existence clauses not being interchangeable by the inputs of this paper (Corollary 4.4(iv)).

12. The Wick tube I: torus inputs and the extensive bound

The Wick leg controls the joint functional of the curvature and the Wick square on the tube $\mathcal{D}_{\sigma_+,H}$ of Definition 2.6. Its positive leg (sources $q \ge 0$) and negative leg (sources $-q$) are proved in the next section from three inputs on *King's tori*: lattice reflection positivity (Proposition 12.2), the chessboard estimate it yields (Lemma 12.4), and an *extensive bound* (I2) for constant positive scalar sources in the fixed-spacing volume limit (Definition 12.5). This section proves the first two and derives the third from the torus inputs Proposition 3.9 and Lemma 3.10 and the sign clause (T3) (Proposition 12.6); a variant with the auxiliary mass on either side of m (Proposition 12.11) serves the discussion of the coupling range; no continuum limit is taken here.

12.1. **King's tori, the two conventions and mid-plane pairs.**

Definition 12.1 (King's tori)**.** A *King torus* is $T = \mathbb{R}^3/(\ell\mathbb{Z})^3$ with $\ell \in 2M\mathbb{N}$; $|T| := \ell^3$ is its volume, and $T_J := \mathbb{R}^3/(2MJ\mathbb{Z})^3$, $J \in \mathbb{N}$, is the torus with fundamental cell C_J. Since $\ell\mathbb{Z}^3 \subset \mathbb{Z}^3 \subset \Lambda_k$, the lattice Λ_k descends to T, with $\ell\,5^k$ sites per side, an even number, and $|T|/\varepsilon_k^3$ sites in all. The *periodic state* $\langle\cdot\rangle_{k,T}$ is King's action of Section 2 summed over the finite graph $\Lambda_k/(\ell\mathbb{Z})^3$, with periodic conditions for both fields [16, (4.1)], [15, (2.21)]: Feynman-form gradient terms for every pair of parallel neighbouring bonds, the photon mass μ_0^2, hopping terms on every bond, the single-site weights $e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m^2)r^2+\lambda r^4]}$. Two normalisations of the counterterm are used:

(i) the *free convention* of Section 2 ($\delta m^2 = \delta m_\varepsilon^2$, Wick constant W_ε; the convention of every cube state of this paper), with $N_{k,T}^a(0,q) := e^{-(a_k+2\mathsf{W}_\varepsilon)\mathsf{S}(q)}\langle e^{\sum_x \varepsilon^3 q(x) r_x^2}\rangle_{k,T}$ for a real

function q on the sites of T, exactly as on boxes (Definition 2.4), and

$$\alpha^{a}_{k,T}(u) := |T|^{-1} \ln N^{a}_{k,T}(0, u\mathbf{1}_T) \qquad (u \in \mathbb{R}),$$

the *free energy density* of a constant source;

(ii) *King's torus convention* [15, (2.4), (3.33), (3.35)]: the counterterm $\delta m^2_{k,T}(M)$ and the Wick constant $c_{k,T}(M)$ computed with the periodic propagators of T at scalar mass M, together with the field-independent constants $E(M) = E_0(M) + E_1(M)$ of King's torus action. $Z^{(M)}_{k,T} := \int e^{-S}$ is the source-free partition function of this model at scalar mass M (bare coefficient $M^2 + \delta m^2_{k,T}(M)$; Proposition 3.9 is a statement about these numbers), and $N^{z,\mathrm{tor}}_{k,T}(0,q)$ the normalised source functional of the torus model at mass m in the reference scheme z, with $c_{k,T} = c_{k,T}(m)$ in place of W_ε.

In either convention the coefficients are the same at every site of T. A *mid-plane pair* is $\pi = \{x_\nu = c\} \cup \{x_\nu = c + \ell/2\}$ with $\nu \in \{1,2,3\}$ and $c \in \varepsilon_k(\mathbb{Z} + \frac{1}{2})$ (both planes lie between lattice planes, since $\ell 5^k$ is even); θ denotes the reflection of T in π, which exchanges the two slabs of $T \setminus \pi$ ("the two sides of π"), acting on the fields as in Definition 9.1.

12.2. **Reflection positivity of King's tori.**

Proposition 12.2 (lattice reflection positivity of King's tori)**.** *Let T be a King torus, $k \geq 0$, π a mid-plane pair and θ the reflection in π. For every finite family $G_1, \dots, G_N$ of gauge-invariant Borel functions of the fields on one side of π with $\langle |G_i|^2 \rangle_{k,T} < \infty$ and every $\gamma \in \mathbb{C}^N$,*

$$\sum_{i,l=1}^{N} \bar\gamma_i \gamma_l \, \langle \theta(G_i)\, G_l \rangle_{k,T} \;\geq\; 0.$$

The periodic state is invariant under θ and under all lattice translations. This holds in both conventions of Definition 12.1: reflection positivity does not see the value of the (site-independent) counterterm.

Proof. The proof is the mechanism of Lemma 9.3 with a pair of planes in place of one and without Dirichlet diagonals, in three steps.

Step 1: from the Feynman form to the Proca form. On the torus write $A = Z + d_0\chi$ with $d_0^T Z = 0$, $\chi \perp 1$ (Hodge decomposition on the finite periodic graph). The Feynman form equals $\frac{1}{2}\langle Z, (d_1^T d_1 + \mu_0^2) Z\rangle$ plus a quadratic form in χ, and the Proca form $\frac{1}{2}\langle A, (d_1^T d_1 + \mu_0^2) A\rangle$ equals the same Z-form plus another χ-form; both χ-forms are positive definite on $1^\perp$ because $\mu_0 > 0$. For gauge-invariant G the matter integral at fixed A is invariant under $A \mapsto A + d_0\chi$ (rotate ϕ site by site), so the χ-integrals factor and cancel in the normalised expectation: gauge-invariant expectations in the Feynman form equal those in the Proca form. Fubini's theorem applies by $\mu_0 > 0$, $\lambda > 0$ and a growth bound $|G| \leq C e^{a\sum_b |A(b)|} e^{C\sum_x r_x^2}$, satisfied by bounded G and by the observables $\Theta(\zeta)$; so this step uses $\mu_0 > 0$, gauge invariance, $\lambda > 0$ and the growth bound, and nothing about the value of the counterterm.

Step 2: the Proca torus measure is reflection positive about π. In the Proca form the action couples the two sides of π only through the plaquettes and hopping terms crossing one of the two planes of π (no gradient terms between parallel bonds across a plane; the mass term is local). Write each crossing bond variable as $a_b = \alpha_b - \alpha'_b$ with α_b, α'_b independent centred Gaussians of variance $(2\mu_0^2\varepsilon^3)^{-1}$, attached to the two sides. Then θ becomes the swap of the two data sets $\xi = (\omega_+, \alpha)$, $\xi' = (\omega_-^\theta, \alpha')$, whose a priori measures coincide by the θ-invariance of the one-sided terms, and exactly as in Step 2 of Lemma 9.3 every crossing plaquette term is $\frac{\varepsilon}{2}X_p(\xi)^2 + \frac{\varepsilon}{2}X_p(\xi')^2 - \varepsilon X_p(\xi)X_p(\xi')$ and every crossing hopping term is $-\varepsilon\sum_{i=1,2}\psi^i_b(\xi)\psi^i_b(\xi')$, with the same real functions on both copies. So the Boltzmann factor is $\mathcal{G}(\xi)\mathcal{G}(\xi')\exp(\sum_\iota \gamma_\iota Y_\iota(\xi)Y_\iota(\xi'))$ with all $\gamma_\iota = \varepsilon > 0$ and all Y_ι real, and for G a function of one side $\theta(G)G = \overline{G(\xi')}G(\xi)$. Expanding the exponential, $Z\langle\theta(G)G\rangle$ is a sum with nonnegative coefficients of squared moduli $|\int d\xi\, G(\xi)\mathcal{G}(\xi)\prod_\iota Y_\iota(\xi)^{n_\iota}|^2$, the interchange of sum and integral being justified as in Lemma 9.3 ($\lambda > 0$, $\mu_0 > 0$ and the growth bound). The Hermitian property $\overline{\langle\theta(G')G\rangle} = \langle\theta(G)G'\rangle$ follows

from the θ-invariance of the measure, so $(G, G') \mapsto \langle \theta(G')G\rangle$ is a positive semidefinite Hermitian form on bounded gauge-invariant one-sided G.

Step 3: L^2 observables. For gauge-invariant one-sided G with $\langle |G|^2\rangle < \infty$ apply Step 2 to $G_n := G\mathbf{1}\{|G| \le n\}$; since $|\theta(G_n)G_n| \le |\theta G||G|$ and $\langle |\theta G||G|\rangle \le \langle |G|^2\rangle$ by Cauchy–Schwarz and θ-invariance, dominated convergence gives the inequality for $G = \sum_i \gamma_i G_i$, hence the form statement.

Invariance. Under θ and under every lattice translation each term of the periodic action, in either convention, goes to a term of the same kind with the same coefficient, and the a priori measure is invariant; the value of the site-independent bare coefficient enters nowhere above. □

Remark 12.3 (reflection positivity in every gauge, and the chessboard estimate). (i) [7, Cor. 5.4] (reflection positivity "in every gauge", for covariances $C_F = C_u + dFd^* \ge 0$ with F of either sign) is the mechanism of the proof above: on the torus the Feynman covariance has this form with $F = -\mu_0^{-2}(\Delta_0 + \mu_0^2)^{-1} \le 0$ on functions orthogonal to the constants. It is not a literal instance (the region of [7, §5] is a bounded subset of the lattice with a boundary condition, and reflection positivity of periodic conditions is asserted there in a remark), and Proposition 12.2 does not rest on it. (ii) The chessboard estimate uses Proposition 12.2 for functions of the r_x only; nothing in this section uses reflection positivity of a cube or of a limit state.

12.3. **The chessboard estimate on King's tori.**

Lemma 12.4 (chessboard estimate)**.** *Let T be a King torus at spacing $\varepsilon = \varepsilon_k$, a a scheme, and q a real function on the sites of T. In the free convention,*

$$\ln N^a_{k,T}(0,q) \;\le\; \sum_x \varepsilon^3\, \alpha^a_{k,T}(q(x)), \tag{27}$$

and $u \mapsto \alpha^a_{k,T}(u)$ is convex on $\mathbb{R}$ with $\alpha^a_{k,T}(0) = 0$. In King's torus convention $u \mapsto |T|^{-1} \ln N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1}_T)$ is likewise convex with value 0 at $u = 0$.

Proof. We apply the chessboard estimate [9, Thm. 4.2] (lattice-gauge form: [7, Thm. 2.10]), whose hypotheses we check: the site set of T is a rectangular lattice box with sides of even length $\ell 5^k$; the periodic state is invariant under translations modulo the sides and reflection positive with respect to the reflections in all planes perpendicular to the coordinate axes running midway between neighbouring sites (Proposition 12.2; on the torus such a reflection is the reflection in a mid-plane pair, the antipodal plane $c + \ell/2$ being again a mid-plane). Take as the "spin at x" the gauge-invariant variable r_x and

$$G_x := \exp\big(\varepsilon^3 q(x)(r_x^2 - 2\mathsf{W}_\varepsilon - a_k)\big), \qquad \text{so that} \qquad \Big\langle \prod_x G_x \Big\rangle_{k,T} = N^a_{k,T}(0,q).$$

The G_x are positive with all moments ($\lambda > 0$), and the Schwarz inequality for such unbounded gauge-invariant functions, all that [9, Thm. 4.1] uses beyond cyclic invariance, follows from the bounded case by truncation and monotone convergence. The conclusion is $|\langle \prod_x G_x\rangle| \le \prod_x \langle \prod_y G_x^{(y)}\rangle^{1/|\Lambda|}$, $G_x^{(y)}$ the copy of G_x carried to y by reflections, $|\Lambda| = |T|/\varepsilon^3$ the number of sites. Because W_ε, a_k and δm_ε^2 are the same at every site, $\prod_y G_x^{(y)} = \exp(q(x)\sum_y \varepsilon^3(r_y^2 - 2\mathsf{W}_\varepsilon - a_k))$ has expectation $N^a_{k,T}(0, q(x)\mathbf{1}_T) = e^{|T|\alpha^a_{k,T}(q(x))}$ in the same state; taking logarithms gives (27). Convexity: $|T|\alpha^a_{k,T}(u)$ is the logarithm of a Laplace transform in the linear parameter u plus a linear term, and vanishes at $u = 0$; the same computation gives the convexity statement in the torus convention. □

12.4. **Constant sources on large tori: the extensive bound.**

Definition 12.5 (the extensive bound (I2))**.** We say that (I2) (one-sided, in the volume limit at fixed spacing; also written (UVS$^+_\infty$)) holds for the scheme a in the free convention if there are $s_1 > 0$ and $C_U \in [0, \infty)$ such that

$$\sup_{k \ge 0} \limsup_{J \to \infty} \alpha^a_{k,T_J}(s_1) \;\le\; C_U. \tag{28}$$

The positive leg of the next section consumes exactly this, with $\kappa := C_U/s_1$ (by convexity and $\alpha(0) = 0$, $\limsup_J \alpha^a_{k,T_J}(u) \le \kappa u$ for $0 \le u \le s_1$). Throughout, $M_1 \in (0, m)$ is an auxiliary scalar mass, $\alpha^\infty_k(M) = M^2 + \delta m^2_k(M)$ is King's free-boundary bare coefficient on Λ_k at scalar mass M (Lemma 3.10; it is the fixed-k volume limit of the torus coefficient $M^2 + \delta m^2_{k,T_J}(M)$, Lemma 12.9), and, in the reference scheme z,

$$s^\infty_k(M_1) := \tfrac{1}{2}\big(\alpha^\infty_k(m) - \alpha^\infty_k(M_1)\big), \qquad \sigma_1 = \sigma_1(M_1) := \liminf_{k\to\infty} s^\infty_k(M_1),$$

$$B^z_{k,T}(M) := E(M) - E(m) - \tfrac{1}{2}\big(2c_{k,T} + z_k\big)\big(\alpha^{\mathrm{tor}}_{k,T}(m) - \alpha^{\mathrm{tor}}_{k,T}(M)\big)|T|,$$

$$b^\infty_k(M_1) := \lim_{J\to\infty} |T_J|^{-1} B^z_{k,T_J}(M_1), \qquad \beta_1 = \beta_1(M_1) := \limsup_{k\to\infty} b^\infty_k(M_1),$$

where $\alpha^{\mathrm{tor}}_{k,T}(M) := M^2 + \delta m^2_{k,T}(M)$ is the bare coefficient of King's torus convention and the limit defining b^∞_k exists by Lemma 12.9. The constant $\beta_1(M_1)$ is the second constant of Lemma 3.10, which says that $\sigma_1(M_1)$ and $\beta_1(M_1)$ are finite; (T3) says that $\sigma_1(M_1) > 0$.

Proposition 12.6 ((I2) from the torus inputs)**.** *Assume* (T3) *(Hypothesis 3.12); the torus inputs are Proposition 3.9 at the two masses m and M_1 and Lemma 3.10. Then* (I2) *holds in the reference scheme z for every $s_1 \in (0, \sigma_1)$, with a constant $C_U = C_U(s_1) < \infty$ depending on $(e, \lambda, m, \mu_0, M)$, on M_1 and on s_1. For a scheme $a \ne z$ with $\sup_k |z_k - a_k| < \infty$ (an admissible scheme, Definition 3.7) it holds with the constant $C_U + s_1 \sup_k |z_k - a_k|$.*

The proof assembles four lemmas on King's torus model at fixed spacing, stated in the form used and with their short proofs where elementary. Notation at fixed k (spacing ε) on $T = T_J$: $f^{\mathrm{tor}}(u) := \ln N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1})$ (King's torus model at mass m, Wick constant $c_{k,T}$, scheme z) and $f^F(u) := \ln N^z_{k,T}(0, u\mathbf{1})$ (free convention); both are convex with $f(0) = 0$ (Lemma 12.4). Write $\alpha^{\mathrm{tor}} := m^2 + \delta m^2_{k,T}(m)$, $\alpha^F := m^2 + \delta m^2_\varepsilon$, $\sigma_{k,T} := \frac{1}{2}(\alpha^{\mathrm{tor}} - \alpha^F)$ and $\Delta c_T := \mathsf{W}_\varepsilon - c_{k,T}$ (the two *convention offsets*), and $s_{k,T}(M_1) := \frac{1}{2}(\alpha^{\mathrm{tor}}_{k,T}(m) - \alpha^{\mathrm{tor}}_{k,T}(M_1))$.

Lemma 12.7 (mass shift)**.** *If $M'' > 0$ satisfies $\alpha^{\mathrm{tor}}_{k,T}(M'') = \alpha^{\mathrm{tor}}_{k,T}(m) - 2u$, then*

$$N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1}) = e^B\, Z^{(M'')}_{k,T} / Z^{(m)}_{k,T}, \qquad B = E(M'') - E(m) - (2c_{k,T} + z_k)\, u\, |T|.$$

Proof. The source $u\sum_x \varepsilon^3 r_x^2$ shifts the bare coefficient of the torus model at mass m by $-2u$, that is to the bare coefficient of the model at mass M''; beyond the fixed terms, King's torus action [15, (2.4)] contains only the counterterm and the field-independent constants E_0, E_1, so the two Boltzmann factors differ by the constant $e^{E(M'')-E(m)}$, and the Wick and scheme multipliers $e^{-(2c_{k,T}+z_k)u|T|}$ of $N^{z,\mathrm{tor}}$ are bookkeeping. □

Lemma 12.8 (chord)**.** *If f is convex on $\mathbb{R}$ with $f(0) = 0$, then $f(u) \le (u/s) f(s)$ for $0 \le u \le s$, $s > 0$.*

Proof. $u = (1 - u/s)\cdot 0 + (u/s)\cdot s$ and $f(0) = 0$. □

Lemma 12.9 (the volume limit at fixed spacing)**.** *Fix k. As $J \to \infty$: $\alpha^{\mathrm{tor}}_{k,T_J}(M) \to \alpha^\infty_k(M)$ uniformly for $M \in [M_1, m]$; $s_{k,T_J}(M_1) \to s^\infty_k(M_1)$; $|T_J|^{-1} B^z_{k,T_J}(M_1) \to b^\infty_k(M_1)$, a finite limit; and the offsets vanish, $\sigma_{k,T_J} \to 0$ and $\Delta c_{T_J} \to 0$. Moreover (the* boundedness clause*) $\sup_{J\ge1} \sup_{M\in[M_1,m]} \big(|\delta m^2_{k,T_J}(M)| + |c_{k,T_J}(M)|\big) < \infty$; in particular $\delta m^2_{k,T_J}(m)$ and c_{k,T_J} are bounded in $J \ge 1$.*

The complete proof, including the boundedness clause, is given in the full version. No rate and no uniformity in the source are claimed or used. Of the clauses listed, Step 2 of the proof of Proposition 12.6 uses the convergence of $s_{k,T_J}(M_1)$ and the $\limsup_J$ of $|T_J|^{-1} B^z_{k,T_J}(M_1)$, and Step 3 the two offset clauses.

Lemma 12.10 (the fixed-spacing extensive bound)**.** *Fix k and $v > 0$. There is $C_k(v) < \infty$ such that for every King torus $T = T_J$, $J \ge 1$, in either convention, $f(\pm v) \le C_k(v)|T|$; hence, by convexity, $|f(w)| \le C_k(v)\, |w|\, |T|/v$ for $|w| \le v$.*

Proof. Let $I(b) := \int_{\mathbb{C}} e^{-\varepsilon^3[\frac{1}{2}b|\phi|^2+\lambda|\phi|^4]}d^2\phi < \infty$ for real b ($\lambda > 0$), α the bare coefficient and c the Wick constant of the convention in use, and Z_γ the photon Gaussian integral, finite because $\mu_0 > 0$ (at $\mu_0 = 0$ the constant mode is in the kernel of the Feynman form). The hopping term is ≥ 0, so dropping it bounds the numerator: $\int e^{-S+w\sum\varepsilon^3 r^2} \leq Z_\gamma I(\alpha - 2w)^{|T|/\varepsilon^3}$. The hopping term of a bond is $\leq \varepsilon(r_x^2 + r_y^2)$, and each site lies on six bonds, so the hopping part of the action is $\leq 6\varepsilon^{-2}\sum_x \varepsilon^3 r_x^2 = \sum_x \varepsilon^3 \cdot \frac{1}{2}(12\varepsilon^{-2})r_x^2$ and $\int e^{-S} \geq Z_\gamma I(\alpha + 12\varepsilon^{-2})^{|T|/\varepsilon^3}$. Hence $\ln\langle e^{w\sum\varepsilon^3 r^2}\rangle \leq \varepsilon^{-3}|T|[\ln I(\alpha - 2w) - \ln I(\alpha + 12\varepsilon^{-2})]$ and

$$f(w) \leq \big\{\varepsilon^{-3}[\ln I(\alpha - 2w) - \ln I(\alpha + 12\varepsilon^{-2})] + (2|c| + |z_k|)|w|\big\}|T|.$$

In the free convention α and c do not depend on T; in the torus convention they converge as $J \to \infty$ and are bounded in $J \geq 1$ (the boundedness clause of Lemma 12.9). So $C_k(v)$, the supremum over the sign and over $J \geq 1$ of the brace at $w = \pm v$, is finite. The last claim is convexity with $f(0) = 0$: $f(w) \leq (|w|/v)f(\pm v)$ and $f(w) \geq -f(-w) \geq -(|w|/v)f(\mp v)$. □

Proof of Proposition 12.6. Step 1: the exact convention bridge. On the same torus the free-convention action differs from the torus-convention action by $\frac{1}{2}(\alpha^F - \alpha^{\mathrm{tor}})\sum_x \varepsilon^3 r_x^2 = -\sigma_{k,T}\sum_x \varepsilon^3 r_x^2$ plus field-independent constants. Hence $\langle e^{u\sum\varepsilon^3 r^2}\rangle^F = \langle e^{(u+\sigma)\sum\varepsilon^3 r^2}\rangle^{\mathrm{tor}} / \langle e^{\sigma\sum\varepsilon^3 r^2}\rangle^{\mathrm{tor}}$ with $\sigma = \sigma_{k,T}$, and inserting the Wick and scheme multipliers,

$$f^F(u) = f^{\mathrm{tor}}(u + \sigma) - f^{\mathrm{tor}}(\sigma) - 2\,\Delta c_T\, u\,|T|. \tag{29}$$

Step 2: the bound in the torus convention. Let $0 \leq u < \sigma_1$ and $\eta > 0$. By the definitions of σ_1 and β_1 there is $K_1 = K_1(u, \eta)$, which we take at least as large as the tail of Proposition 3.9, such that $s_k^\infty(M_1) > u$ and $b_k^\infty(M_1) < \beta_1 + \eta$ for all $k \geq K_1$. Fix such a k. By Lemma 12.9 there is $J_* \geq J_*(k)$ with $s := s_{k,T_J}(M_1) > u$ for $J \geq J_*$. The chord (Lemma 12.8) at this s gives $f^{\mathrm{tor}}(u) \leq (u/s)f^{\mathrm{tor}}(s) \leq [f^{\mathrm{tor}}(s)]_+$. Lemma 12.7 with $M'' = M_1$ applies exactly, since $\alpha^{\mathrm{tor}}_{k,T_J}(M_1) = \alpha^{\mathrm{tor}}_{k,T_J}(m) - 2s$ by the definition of s:

$$f^{\mathrm{tor}}(s) = B^z_{k,T_J}(M_1) + \ln Z^{(M_1)}_{k,T_J} - \ln Z^{(m)}_{k,T_J} \leq B^z_{k,T_J}(M_1) + \big(E_+(M_1) + E_-(m)\big)|T_J|$$

by Proposition 3.9, used as an upper bound at M_1 and as a lower bound at m. Divide by $|T_J|$ and let $J \to \infty$ (Lemma 12.9 for the first term):

$$\limsup_{J\to\infty} |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) \leq \big[\beta_1 + \eta + E_+(M_1) + E_-(m)\big]_+ \qquad (k \geq K_1).$$

For each of the finitely many $k < K_1$, Lemma 12.10 with any $v \geq u$ gives $\limsup_J |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) \leq C_k(v)u/v < \infty$. Fix $\eta := 1$ (any $\eta > 0$ serves); then

$$C^{\mathrm{tor}}(u) := \sup_{k\geq 0} \limsup_{J\to\infty} |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) < \infty \qquad \text{for every } u \in [0, \sigma_1).$$

Step 3: transfer to the free convention. Fix $u \in [0, \sigma_1)$ and $p > 1$ with $pu < \sigma_1$, and let $p' = p/(p-1)$. Since $u + \sigma = \frac{1}{p}(pu) + \frac{1}{p'}(p'\sigma)$, convexity gives $f^{\mathrm{tor}}(u+\sigma) \leq \frac{1}{p}f^{\mathrm{tor}}(pu) + \frac{1}{p'}f^{\mathrm{tor}}(p'\sigma)$, whatever the sign of $\sigma = \sigma_{k,T_J}$. Once $p'|\sigma_{k,T_J}| \leq v$, which holds for all large J by Lemma 12.9, Lemma 12.10 gives $|f^{\mathrm{tor}}(p'\sigma)|, |f^{\mathrm{tor}}(\sigma)| \leq C_k(v)p'|\sigma_{k,T_J}||T_J|/v$. Inserting this into (29),

$$|T_J|^{-1} f^F_{k,T_J}(u) \;\leq\; \tfrac{1}{p}|T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(pu) + \tfrac{2C_k(v)p'}{v}|\sigma_{k,T_J}| + 2|\Delta c_{T_J}|\,u \;\xrightarrow[J\to\infty]{\limsup}\; \tfrac{1}{p}C^{\mathrm{tor}}(pu),$$

because $\sigma_{k,T_J} \to 0$ and $\Delta c_{T_J} \to 0$ at fixed k (Lemma 12.9; the factor $\frac{1}{p'} \leq 1$ in the middle term has been dropped). Hence $\sup_k \limsup_J \alpha^z_{k,T_J}(u) \leq \frac{1}{p}C^{\mathrm{tor}}(pu) < \infty$ for every $u \in [0, \sigma_1)$. Take $u = s_1 \in (0, \sigma_1)$ and $C_U := [\inf_{p>1:\, ps_1<\sigma_1} \frac{1}{p}C^{\mathrm{tor}}(ps_1)]_+$; this is (28) in the scheme z. For a scheme $a \neq z$, $N^a_{k,T}(0,q) = e^{-(a_k - z_k)\mathsf{S}(q)}N^z_{k,T}(0,q)$ gives $\alpha^a_{k,T}(u) = \alpha^z_{k,T}(u) + (z_k - a_k)u \leq \alpha^z_{k,T}(u) + |z_k - a_k|\,u$, so (28) holds with $C_U + s_1 \sup_k |z_k - a_k|$. □

Proposition 12.11 (the auxiliary mass on either side of m)**.** *Let $M_1 > 0$, $M_1 \neq m$. Assume Proposition 3.9 at m and at M_1, and Lemma 3.10 and* (T3) *read at this M_1: $s_k^\infty(M_1) = \frac{1}{2}(\alpha_k^\infty(m) - \alpha_k^\infty(M_1))$, of either sign, has $\sigma_1(M_1) = \liminf_k s_k^\infty(M_1) > 0$, and $\beta_1(M_1) < \infty$.*

Then (I2) *holds in the scheme z for every $s_1 \in (0, \sigma_1(M_1))$, and for $a \neq z$ with the shifted constant, exactly as in Proposition 12.6.*

Proof. The proof of Proposition 12.6 is repeated with Lemma 12.9 stated on the interval $I := [\min(M_1, m), \max(M_1, m)]$, and we check that no step uses $M_1 < m$. Lemma 12.7 is an identity for any M'' with $\alpha^{\mathrm{tor}}(M'') = \alpha^{\mathrm{tor}}(m) - 2u$, whatever the order of M'' and m. Lemma 12.8 needs $0 \leq u \leq s$ only. Lemma 12.9: its proof uses only that the propagators are translation-invariant massive lattice covariances at fixed ε with masses in $I \cup \{\mu_0\}$, bounded below by $\min(M_1, m, \mu_0) > 0$, so it holds on I word for word (stated so in the full version); Step 2 uses it at the two masses m and M_1 only. Lemma 12.10 carries no mass order, and (29) is an identity. In Step 2 the sign clause is $s_k^{\infty}(M_1) > u \geq 0$, the chord is applied at $s = s_{k,T_J}(M_1) > u \geq 0$, Lemma 12.7 at $M'' = M_1$ is exact by the definition of s, and Proposition 3.9 is used as an upper bound at M_1 and a lower bound at m; none of this uses $M_1 < m$. Step 3 carries no mass. □

The two-sided variant matters exactly on the set $E_W^{<} \setminus E_W$ of couplings at which m minimises α_∞ on $(0, m]$ but not on $(0, \infty)$ (Lemma 22.1; for instance when $\alpha'_\infty(m) < 0$ and m minimises α_∞ on $(0, m]$): there (T3) fails for every $M_1 < m$ while some $M_1 > m$ has $\sigma_1(M_1) > 0$. The main results use Proposition 12.6 with $M_1 \in (0, m)$, as (T3) is posed.

Remark 12.12 (the convention of Proposition 3.9 and how much of it is used). (i) Proposition 3.9 is posed in King's torus convention (counterterm and Wick constant computed with the periodic propagators of the torus itself), not in the normalisation of Bałaban's own papers. This is what the proof needs: the offsets σ_{k,T_J}, Δc_{T_J} between King's torus convention and the free convention vanish in the volume limit at fixed spacing (Lemma 12.9), and Step 3 of the proof of Proposition 12.6 transfers the bound through them; between other normalisations they need not vanish. (ii) Only a tail of Proposition 3.9 is used: Step 2 of that proof invokes it for $k \geq K_1$ and $J \geq J_*(k)$, the finitely many $k < K_1$ being handled by Lemma 12.10 at fixed spacing, and the constants $E_\pm(M)$ are independent of k and of the torus. (iii) In the scheme z, C_U depends on the couplings, on M and on the auxiliary choices M_1, s_1, and on nothing else; neither it nor any other bound of this section is uniform in μ_0.

13. The Wick tube II: the uniform bound and its limits

This section proves the (k, J)-uniform bound (31) on the general-source functionals of the comparison cubes on the tube $\mathcal{D}_{\sigma_+, H}$ from the extensive bound (I2) (Definition 12.5): the positive part of the Wick source is controlled through King's tori, the negative part through a tiling into cells of fixed size, each controlled by one instance of Theorem 3.6 and the positive bound. Then come the limits on the tube, E0″, and the continuity and translation invariance of the tube object. States are those of the comparison convention $\mathrm{ND}^{(a)}$ (Definition 2.1) unless King's convention is named; $N^a_{k,X}(0, q)$ is the functional of Definition 2.4 on a finite $X \subset \Lambda_k$ or a King torus (Definition 12.1), the torus with fundamental cell C_J being written $\mathbb{T}_J$ here (T_J is the top-layer set of King's RG cube), and $\alpha^a_{k,\mathbb{T}}(u) = |\mathbb{T}|^{-1} \ln N^a_{k,\mathbb{T}}(0, u\mathbf{1})$. $s_1 \in (0, \sigma_1)$ and C_U are the amplitude and constant of (I2), $\kappa := C_U / s_1$; the sources $f_1, \dots, f_n$, $h_1, \dots, h_p$ are in C_c^∞ with supports in $Q_R = [-R, R]^3$; $g(\zeta)$, $k(\zeta) = k_+ - k_-$ and the norms are as in Definition 2.6. The scheme is the reference scheme z, or an admissible scheme a (Definition 3.7, so that $\sup_k |z_k - a_k| < \infty$); then C_U stands for $C_U + s_1 \sup_k |z_k - a_k|$ throughout (Proposition 12.6).

13.1. **The positive leg on cubes.**

Proposition 13.1 (cube ≤ torus at one bare mass)**.** *For every k, J and every lattice function $q \geq 0$ on X_J: $N^a_{k,C_J}(0, q) \leq N^a_{k,\mathbb{T}_J}(0, q)$, both in the free convention.*

Proof. The periodic state on $\mathbb{T}_J$ has the sites of X_J with opposite faces identified; its action is the Neumann–Dirichlet action on C_J plus, for every wrap-around pair $\{x, y\}$, the photon gradient terms $\varepsilon(A_\alpha(y) - A_\alpha(x))^2$ and the hopping term $\frac{\varepsilon}{2}(r_x^2 + r_y^2) - \varepsilon r_x r_y \cos(\cdot)$, whose diagonal part is exactly the pair of Dirichlet terms the cube action carries at x, y for their outward bonds; so the cube state is the torus state with the wrap-around gradient terms and cosine parts deleted.

For the class-$\mathcal{I}$ observable $e^{\sum \varepsilon^3 q r^2}$ ($\Theta_{F,g}$, $F=0$, $g=\varepsilon^3 q \geq 0$) deleting a gradient term, a quadratic $|\ell(A)|^2$, does not increase the expectation (Lemma 2.8(a)), nor does deleting a cosine part (Lemma 2.8(b)); each intermediate measure is a BFS measure, and δm_ε^2, W_ε, a_k, $\mathsf{S}(q)$ agree on both sides. □

Proposition 13.2 (the positive leg in L^1 form, on every Neumann–Dirichlet box)**.** *Assume* (I2) *and* (I3) *(Proposition 12.2, entering through the chessboard estimate Lemma 12.4). Let $X \subset \Lambda_k$ be finite, $\langle\cdot\rangle_{k,X}^{\mathrm{ND}}$ the Neumann–Dirichlet state of Definition 2.1 on X, and $0 \leq q \leq s_1$ a lattice function on X. Then*

$$\ln N_{k,X}^a(0,q) \;\leq\; \kappa\,\|q\|_{1,k}, \qquad \kappa = C_U/s_1.$$

Proof. Choose J' with X and all its nearest neighbours in $C_{J'}$. (1) *Box ≤ cube:* the first inequality of Lemma 2.11 with $Y = X_{J'}$ and $(f,g) = (0,q)$ (Lemma 2.8(a),(b)); its X-state is the state of Definition 2.1 on X, since all six bonds at each site of X are present in Y and no site is killed; the scheme multiplier is common (Lemma 2.12(a)). (2) *Cube ≤ torus:* Proposition 13.1 on $C_{J'}$, q extended by 0. (3) *Chessboard:* Lemma 12.4 on $\mathbb{T}_{J'}$, $\ln N_{k,\mathbb{T}_{J'}}^a(0,q) \leq \sum_x \varepsilon^3 \alpha_{k,\mathbb{T}_{J'}}^a(q(x))$. (4) *Chord:* $\alpha = \alpha_{k,\mathbb{T}_{J'}}^a$ is convex with $\alpha(0)=0$, so $\alpha(u) \leq (u/s_1)[\alpha(s_1)]_+$ for $0 \leq u \leq s_1$, and (3) is at most $s_1^{-1}[\alpha(s_1)]_+\|q\|_{1,k}$ (the ℓ^1 norm of q is the same on X, $C_{J'}$, $\mathbb{T}_{J'}$). (5) The left side does not depend on J'; let $J' \to \infty$ at fixed k and use (I2): $\limsup_{J'} \alpha_{k,\mathbb{T}_{J'}}^a(s_1) \leq C_U$ for every k. □

Corollary 13.3 (the positive leg on the cubes)**.** *Under the hypotheses of Proposition 13.2, for $0 \leq q \leq s_1$ on X_J: $\ln N_{k,C_J}^a(0,q) \leq \kappa\|q\|_{1,k}$, uniformly in k and J. Consequently, for every $0 \leq q \in C_c^\infty$ with $\sup q \leq s_1$ and every J with $\operatorname{supp} q$ at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, $\limsup_k Z_{k,J}^a(0,q) \leq e^{\kappa\|q\|_{L^1}}$, uniformly in J. If moreover the comparison family is convergent at fixed cubes on the tube (Definition 3.2), then* (W2)*(b) holds for $2\sup q < s_1$: for every $0 \leq q \in C_c^\infty$ with $2\sup q < s_1$ and every J as above, $\lim_k Z_{k,J}^a(0,q)$ exists and $\sup_J \lim_k Z_{k,J}^a(0,q) \leq e^{\kappa\|q\|_{L^1}}$.*

Proof. Proposition 13.2 at $X = X_J$. The sampled source satisfies $0 \leq q \leq s_1$ (Lemma 2.7(E3)) and $\|q\|_{1,k} \to \|q\|_{L^1}$, which gives the lim sup bound. The source q is the family $h_1 = q$ at the coefficient $w = 1$, whose positive scalar part is q; Definition 3.2 gives the limit when $2\|q\|_\infty < s_W = s_1$, and there the limit equals the lim sup. □

13.2. **Cells and the constants of the negative leg.**

Definition 13.4 (cells, tilings, cutoffs)**.** Let $L_c := \min\{L \in \mathbb{N} : ML \geq 5\}$, $\ell_c := 2ML_c \geq 10$; the *cell* $C_{L_c} = [-ML_c, ML_c)^3$ is a King cube. For $s \in \{0,1\}^3$ let $W_\mu(0) := \ell_c\mathbb{Z}$, $W_\mu(1) := ML_c + \ell_c\mathbb{Z}$, and let the tiling $\mathcal{T}_s$ consist of the cells $\prod_\mu [w_\mu, w_\mu + \ell_c)$, $w_\mu \in W_\mu(s_\mu)$; every cell is $C_{L_c} + b$ with $b \in ML_c\mathbb{Z}^3 \subset \Lambda_k$. The good sets $G_s := \{x : \operatorname{dist}(x_\mu, W_\mu(s_\mu)) > 2\ \forall \mu\}$ are open and cover $\mathbb{R}^3$ (the 2-neighbourhoods of $W_\mu(0)$, $W_\mu(1)$ are disjoint as $ML_c \geq 5 > 4$). Fix a smooth partition of unity $0 \leq \chi_s \leq 1$, $\sum_s \chi_s = 1$, $\operatorname{supp}\chi_s \subset G_s$; so χ_s vanishes within distance 2 of every wall of $\mathcal{T}_s$. Fix $\chi_0 \in C_c^\infty$, $0 \leq \chi_0 \leq 1$, $\chi_0 = 1$ on $\{\operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus C_{L_c}) \geq 2\}$, $\operatorname{supp}\chi_0 \subset \{\operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus C_{L_c}) > 1\}$, and put $\chi_c := \tau_b\chi_0$ for the cell $c = C_{L_c} + b$. For sources supported in Q_R let $J_1(R) := \min\{J \in \mathbb{N} : MJ \geq R + \ell_c\}$; a cell meeting Q_R has $w_\mu \in [-R - \ell_c, R]$, hence lies in $[-R-\ell_c, R+\ell_c)^3 \subset C_J$ for $J \geq J_1(R)$, so every cell meeting Q_R is a full cell inside C_J and the cells clipped by the faces of C_J carry no source.

The constants of the Wick leg are

(30)

$$\sigma_+ := s_1, \qquad H' := \frac{s_1}{2}, \qquad H := \frac{H'}{8} = \frac{s_1}{16}, \qquad \kappa := \frac{C_U}{s_1}, \qquad B_{\mathrm{cell}} := \sup_k \ln N_{k,C_{L_c}}^a(0, -2H'\chi_0),$$

$$C_{\mathrm{cell}} := \tfrac{1}{2}B_{\mathrm{cell}} + \kappa H'\ell_c^3, \qquad C_\Sigma := \frac{(2+6\ell_c^2)^2\pi^2}{\ell_c^3}, \qquad K_- := \frac{8\,[C_{\mathrm{cell}}]_+\,C_\Sigma}{H'}.$$

B_{cell} is finite by Lemma 14.4, that is by Corollary 6.18(b): at the single cube C_{L_c} and the single real source $-2H'\chi_0$ the comparison-convention functional is bounded by the King-convention

one (the $\mathcal{D}$ direction of Lemma 4.2), which converges by Theorem 3.6 ($\operatorname{supp}\chi_0$ is at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{L_c}$ as Theorem 3.6 requires, and the theorem puts no amplitude restriction on a negative Wick source, a position-dependent increase of the scalar mass squared). This boundedness statement is ($\mathrm{B}_{\mathrm{cell}}$) (Lemma 14.4) and the only place where the existence part enters the bound (31); no fixed-cube limit of the comparison family is used.

13.3. **The negative leg on cubes.**

Lemma 13.5 (wall insertion)**.** *Let $X_1, \dots, X_N$ be the site sets $c \cap X_J$ of the cells c of one tiling $\mathcal{T}_s$ that meet C_J, and $q \geq 0$ a lattice function on X_J. Then $N^a_{k,C_J}(0,-q) \leq \prod_{i=1}^N N^a_{k,X_i}(0,-q|_{X_i})$, with N^a_{k,X_i} the functional of the Neumann–Dirichlet state of Definition 2.1 on X_i.*

Proof. $e^{-\sum \varepsilon^3 q r^2}$ is the class-$\mathcal{D}$ observable $\Theta^-_{F,g}$, $F = 0$, $g = \varepsilon^3 q \geq 0$. Delete, for every nearest-neighbour pair with ends in different cells, the three photon gradient terms and the cosine part of the hopping term, one at a time: by Lemma 2.8(d) (the inequalities [7, Thm. 6.2(3),(4)], proved in Section 2) neither deletion decreases $\langle \Theta^- \rangle$, and each intermediate measure is a BFS measure. The result is a product over the cells (Lemma 2.8(e)) whose factors are the Neumann–Dirichlet states on the X_i: the kept diagonal parts of the deleted bonds are their Dirichlet terms, the remaining gradient terms their Neumann terms, and δm^2_ε, W_ε, a_k are site-independent. The observable and the multiplier $e^{a_k \mathsf{S}(q)} = \prod_i e^{a_k \mathsf{S}(q|_{X_i})}$ factorise. □

Lemma 13.6 (one cell)**.** *Assume* (I2) *and* (I3). *Let $c = C_{L_c} + b$ be a cell ($b \in ML_c\mathbb{Z}^3$) and $0 \leq \tilde{q} \leq H'$ a lattice function supported in $\{x : \operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus c) \geq 2\}$. Then $\ln N^a_{k,c}(0,-\tilde{q}) \leq C_{\mathrm{cell}}$ for every k.*

Proof. $\chi_c = 1$ on $\operatorname{supp}\tilde{q}$, so $-\tilde{q} = k_1 - k_2$ with $k_1 := H'\chi_c - \tilde{q} \geq 0$, $k_2 := H'\chi_c$, and Lemma 2.12(d) at $f = 0$ gives $N^a_{k,c}(0,-\tilde{q}) \leq N^a_{k,c}(0,-2H'\chi_c)^{1/2} N^a_{k,c}(0,2k_1)^{1/2}$. *First factor:* exact lattice translation covariance ($b \in \Lambda_k$; the change of variables $x \mapsto x - b$ transports the boundary conditions, and the action, δm^2_ε, W_ε, a_k are site-independent) gives $N^a_{k,c}(0,-2H'\chi_c) = N^a_{k,C_{L_c}}(0,-2H'\chi_0) \leq e^{B_{\mathrm{cell}}}$. *Second factor:* Proposition 13.2 on the box $c \cap \Lambda_k$ with $q' := 2k_1$, $0 \leq q' \leq 2H' = s_1$, gives at most $\exp(\kappa \|q'\|_{1,k}) \leq \exp(2\kappa H' \ell_c^3)$, the cell having volume ℓ_c^3. Hence $\ln N^a_{k,c}(0,-\tilde{q}) \leq \frac{1}{2} B_{\mathrm{cell}} + \kappa H' \ell_c^3$. □

Lemma 13.7 (chord on the cell)**.** *Under the hypotheses of Lemma 13.6:* $\ln N^a_{k,c}(0,-\tilde{q}) \leq ([C_{\mathrm{cell}}]_+ / H') \sup \tilde{q}$.

Proof. Let $t := \sup \tilde{q}/H' \in (0,1]$, $\hat{q} := \tilde{q}/t$ ($0 \leq \hat{q} \leq H'$, same support). $\varphi(u) := \ln N^a_{k,c}(0,-u\hat{q})$ is convex (Lemma 2.12(b)), $\varphi(0) = 0$, $\varphi(1) \leq C_{\mathrm{cell}}$ (Lemma 13.6); so $\varphi(t) \leq t\varphi(1) \leq t[C_{\mathrm{cell}}]_+$. □

Lemma 13.8 (eight families and the sum over cells)**.** (a) *For a lattice function $q \geq 0$ on X_J: $N^a_{k,C_J}(0,-q) \leq \prod_{s \in \{0,1\}^3} N^a_{k,C_J}(0,-8\chi_s q)^{1/8}$. If q is supported in Q_R and $J \geq J_1(R)$, then for each s every cell of $\mathcal{T}_s$ meeting $\operatorname{supp}(\chi_s q)$ is a full cell inside C_J, on which $\chi_s q$ is supported at ℓ^∞-distance > 2 from the complement of the cell.* (b) *For any tiling of $\mathbb{R}^3$ by cubes of side ℓ_c: $\sum_c \sup_{x \in c} (1 + |x|^2)^{-2} \leq C_\Sigma$.*

Proof. (a) Hölder's inequality with eight exponents 8 in the one state $\langle \cdot \rangle^{\mathrm{ND}}_{k,C_J}$ for $e^{-\sum \varepsilon^3 q r^2} = \prod_s e^{-\sum \varepsilon^3 \chi_s q r^2}$; the multipliers match since $\mathsf{S}(q) = \sum_s \mathsf{S}(\chi_s q)$ at lattice points. Support: every cell meeting Q_R is a full cell inside C_J for $J \geq J_1(R)$ (Definition 13.4), and a point of G_s in the cell $\prod_\mu [w_\mu, w_\mu + \ell_c)$ has $\operatorname{dist}(x_\mu, \{w_\mu, w_\mu + \ell_c\}) > 2$ for every μ. (b) For x, y in one cell $|y| \leq |x| + \sqrt{3}\ell_c$, so $1 + |y|^2 \leq (2 + 6\ell_c^2)(1 + |x|^2)$ and $\sup_{x \in c}(1 + |x|^2)^{-2} \leq (2 + 6\ell_c^2)^2 \ell_c^{-3} \int_c (1 + |y|^2)^{-2} dy$; sum over the disjoint cells, with $\int_{\mathbb{R}^3} (1 + |y|^2)^{-2} dy = \pi^2$. □

Proposition 13.9 (the negative leg)**.** *Assume* (I2) *and* (I3). *For every k, every $J \geq J_1(R)$ and every lattice function $q \geq 0$ supported in Q_R with $\sup q \leq H$:*

$$\ln N^a_{k,C_J}(0,-q) \;\leq\; K_- \, |q|_w .$$

Proof. By Lemma 13.8(a), then Lemma 13.5 for each tiling $\mathcal{T}_s$ (the clipped cells carry no source and contribute $\ln N^a_{k,X_i}(0,0)=0$), then Lemma 13.7, applicable since $8\chi_s q \le 8\sup q \le H'$ and by the support clause of Lemma 13.8(a):

$$\ln N^a_{k,C_J}(0,-q) \le \frac{1}{8}\sum_s \sum_{c\in\mathcal{T}_s} \ln N^a_{k,c}(0,-8\chi_s q|_c) \le \frac{1}{8}\sum_s\sum_c \frac{[C_{\mathrm{cell}}]_+}{H'}\, 8\sup_c q \le \frac{[C_{\mathrm{cell}}]_+}{H'}\,|q|_w \sum_s\sum_c \sup_{x\in c}(1+|x|^2)^{-2}$$

using $\sup_c q \le |q|_w \sup_{x\in c}(1+|x|^2)^{-2}$ (a sampled supremum is at most the continuum one, Lemma 2.7(E3)) and Lemma 13.8(b) for each of the eight tilings. □

The output norm is $|q|_w$: the chord bound on a cell sees only $\sup_c q$, so no L^1 information inside a cell survives, and the tube bound carries $K_-|k_-|_w$ rather than an L^1 norm of k_-.

13.4. **The uniform bound on the tube and its limits.** For a family with supports in Q_R and $\zeta=(z,w)\in\mathbb{C}^{n+p}$ put $G^{(k,C_J)}(\zeta) := Z^a_{k,J}(\sum_i z_i f_i, \sum_j w_j h_j)$, and let $\mathcal{D}_{\sigma_+,H}$ be the tube of Definition 2.6 with $\sigma_+ = s_1$, $H = s_1/16$. Two facts from several complex variables [11] are used: (T2) a holomorphic function on a connected open $D\subset\mathbb{C}^m$ vanishing on a non-empty relatively open subset of $D\cap\Sigma$, Σ an affine totally real subspace of real dimension m (such as $\mathbb{R}^m$ or $(\mathrm{i}\mathbb{R})^{m_1}\times\mathbb{R}^{m_2}$), vanishes on D (all its derivatives vanish at such a point); (T3) a locally bounded family of holomorphic functions on D is normal, and if it converges on a set as in (T2) it converges locally uniformly on D to a holomorphic limit (Montel, Vitali).

Theorem 13.10 (the tube bound)**.** *Assume* (I2) *(which Proposition 12.6 derives from* (T3) *and the torus inputs Proposition 3.9 and Lemma 3.10), for the reference scheme z or for an admissible scheme a (Definition 3.7), with* (I3) = *Proposition 12.2 and* (I4) = *Lemma 2.8. The constants σ_+, H, κ, K_- of (30) depend on (e,λ,m,μ_0,M,a) and on the fixed auxiliary choices M_1, χ_0, $s_1\in(0,\sigma_1)$, and on nothing else; no uniformity in μ_0 is claimed. For every family with supports in Q_R, every $k\ge 0$, $J\ge J_1(R)$ and $\zeta\in\mathcal{D}_{\sigma_+,H}$:*

$$|G^{(k,C_J)}(\zeta)| \;\le\; G^{(k,C_J)}(\mathrm{Re}\,\zeta) \;\le\; \exp\Big(\|g(\zeta)\|^2_{2,k} + \kappa\|k_+(\zeta)\|_{1,k} + K_-|k_-(\zeta)|_w\Big). \tag{31}$$

Consequently, if moreover the comparison family is convergent at fixed cubes on the tube (Definition 3.2; its form on King's convention, where Theorem 3.6 supplies the limits, is Theorem 21.3):

(i) (Limits on the whole tube.) *$G_J := \lim_k G^{(k,C_J)}$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ for every $J\ge J_1(R)$, $G_\infty := \lim_J G_J$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ (the limit along $J\ge J_1(R+\frac12)$, hence along any tail), both are holomorphic there, and*

$$|G_\infty(\zeta)| \le \exp\big(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w\big) \qquad (\zeta\in\mathcal{D}_{\sigma_+,H}).$$

(W2) is not used; where the full object Z^a_∞ of the optional extension exists, $G_\infty = Z^a_\infty$ on $\mathcal{D}_{\sigma_+,H}$.

(ii) (E0″.) *With $\sigma_0 := H/4$, $K' := \kappa\pi^2 + K_-$, the mixed Schwinger functions $S_{n,p}(f;h) := \partial_z^{(n)}\partial_w^{(p)} G_\infty(0)$ satisfy, for real test functions,*

$$|S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p)| \le e^{K'\sigma_0}(\sqrt{2}e)^n (n!)^{1/2}(e/\sigma_0)^p\, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w \le \sigma_{n+p}\prod_i |f_i|_4 \prod_j |h_j|_4,$$

$\sigma_N := e^{K'\sigma_0} C_1^N N!$, $C_1 := 2\max(e\pi/2, e/\sigma_0)$; for complex test functions the first bound carries the factor $2^{n+p}\le 2^N$, $N=n+p$, absorbed in C_1. This is E0″ *with $s=4$ and every $\beta>1$; the $S_{n,p}$ are tempered distributions.*

σ_1, C_U depend on (e,λ,m,μ_0,M_1) through the torus stability constants of Proposition 3.9 at m, M_1, the c-number β_1 of Lemma 3.10 and finitely many fixed-spacing constants; B_{cell} depends on everything, including μ_0, M, a. The curvature term of (31) rests on the constant 2 of Lemma 7.1, which is not known to be sharp (Remark 7.2).

Proof. The bound (31). The first inequality is Lemma 2.12(b). Write $g=g(\zeta)$, $k=k(\zeta)$ (real). *Step 1* (curvature off): Theorem 7.4(a) in the presence of the real scalar source k gives $N^a(g,k) \le e^{\|g\|^2_{2,k}} N^a(0,k)$; the supports of the f_i are at distance >1 from $\mathbb{R}^3\setminus C_J$ since

$MJ \geq R+\ell_c > R+1$. *Step 2*: $N^a(0,k_+-k_-) \leq N^a(0,2k_+)^{1/2}N^a(0,-2k_-)^{1/2}$ (Lemma 2.12(d) at $f=0$). *Step 3* (positive leg): $\zeta \in \mathcal{D}_{\sigma_+,H}$ and (E3) give $0 \leq 2k_+ \leq s_1$ on the lattice, so $\frac{1}{2}\ln N^a(0,2k_+) \leq \kappa\|k_+\|_{1,k}$ by Corollary 13.3. *Step 4* (negative leg): $0 \leq 2k_- \leq H$, supported in Q_R, $J \geq J_1(R)$, so $\frac{1}{2}\ln N^a(0,-2k_-) \leq \frac{1}{2}K_-|2k_-|_w = K_-|k_-|_w$ by Proposition 13.9.

(i), $k\to\infty$ at fixed $J \geq J_1(R)$. Each $G^{(k,C_J)}$ is entire on $\mathbb{C}^{n+p}$. On a compact $K \subset \mathcal{D}_{\sigma_+,H}$ the family $\{G^{(k,C_J)}\}_k$ is uniformly bounded by (31) and Lemma 2.7(E1)–(E3) ($\|g\|_{2,k}^2 \leq C_0|g|_4^2$, $\|k_+\|_{1,k} \leq C_0'|k_+|_w$, $|k_\pm|_w \leq \sum_j |\operatorname{Re} w_j||h_j|_w$). At the real points of $\mathcal{D}_{\sigma_+,H}$, a non-empty open subset of $\mathbb{R}^{n+p}$, it converges by Definition 3.2 (the positive scalar part is below $\sigma_+/2 < s_W$). By (T3) with $\Sigma = \mathbb{R}^{n+p}$, $G^{(k,C_J)} \to G_J$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G_J is holomorphic, and (31) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (k_+ continuous with compact support) and the sampled $|k_-|_w$ at most the continuum one: $|G_J|$ is bounded on $\mathcal{D}_{\sigma_+,H}$ by the right side of the display in (i), uniformly in $J \geq J_1(R)$.

(i), $J\to\infty$ on a sub-tube. Split $h_j = h_j^+ - h_j^-$, $h_j^- := \|h_j\|_\infty \chi_j$, $h_j^+ := h_j + h_j^- \geq 0$, with $\chi_j \in C_c^\infty$, $0 \leq \chi_j \leq 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset Q_{R+1/2}$. Put $F_{k,J}(z,w^+,w^-) := Z^a_{k,J}(\sum_i z_i f_i, \sum_j (w_j^+ h_j^+ + w_j^- h_j^-))$ and $\mathcal{D}'' := \{(z,w^+,w^-) \in \mathbb{C}^{n+2p} : |\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j \delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) < H/2$. On $\mathcal{D}''$ the split family's scalar source has $\|k_\pm^{\mathrm{sp}}\|_\infty \leq \|k^{\mathrm{sp}}\|_\infty < H/2 \leq \sigma_+/2$, so (31) holds for it uniformly in k and $J \geq J_1(R+\frac{1}{2})$, and the previous paragraph gives $F_{k,J} \to F_J$ locally uniformly on $\mathcal{D}''$ with the limit bound. On $\Omega_+'' := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the sources lie in the cone $\mathcal{K}$ (imaginary curvature coefficients, nonnegative total scalar source), so $J \mapsto F_J$ is nondecreasing there (first inequality of Lemma 2.11 with Lemma 2.12(a) at fixed k, then $k\to\infty$) and bounded, hence convergent. Ω_+'' is a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$ and $\mathcal{D}''$ is convex, so (T3) gives $F_J \to F_\infty$ locally uniformly on $\mathcal{D}''$. Restricting to $w_j^+ = w_j$, $w_j^- = -w_j$ gives $G_J \to G_\infty$ locally uniformly on $\mathcal{D}' := \{(z,w) : |\operatorname{Re} w_j| < \delta_j\} \subset \mathcal{D}_{\sigma_+,H}$ (there $\|k_\pm\|_\infty \leq \|k\|_\infty < \sum_j \delta_j\|h_j\|_\infty < H/2$).

(i), the whole tube. $\{G_J\}_{J \geq J_1(R+1/2)}$ is locally uniformly bounded on $\mathcal{D}_{\sigma_+,H}$ by the J-free limit bound; every Montel subsequential limit is holomorphic on $\mathcal{D}_{\sigma_+,H}$ and agrees with G_∞ on the open set $\mathcal{D}'$; $\mathcal{D}_{\sigma_+,H}$ is connected, so all subsequential limits coincide: $G_J \to G_\infty$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, and the bound passes to G_∞. Where Z^a_∞ exists it is the same iterated limit of the same Z^a_J.

(ii). Normalise $\|f_i\|_2 = 1$ and let $\rho_j := \sigma_0/(p|h_j|_w)$. On the polydisc $|w_j| \leq \rho_j$, $z \in \mathbb{C}^n$: $\|k\|_\infty \leq \sum_j |\operatorname{Re} w_j| \|h_j\|_\infty \leq \sum_j \rho_j |h_j|_w = \sigma_0 < H/2 \leq \sigma_+/2$, so the polydisc lies in $\mathcal{D}_{\sigma_+,H}$, where G_∞ exists with its bound by (i); the exponent there is $\leq \kappa\pi^2|k_+|_w + K_-|k_-|_w \leq K'|k|_w \leq K'\sum_j |\operatorname{Re} w_j||h_j|_w \leq K'\sigma_0$ by (E2). Cauchy's estimate for the mixed first w-derivatives at $w=0$, z fixed: $|\partial_w^{(p)} G_\infty(z,0)| \leq \prod_j \rho_j^{-1} \sup|G_\infty| \leq (p/\sigma_0)^p \prod_j |h_j|_w\, e^{K'\sigma_0} e^{\|g(z)\|_2^2}$, and $p^p \leq e^p p!$. The entire function $z \mapsto \partial_w^{(p)} G_\infty(z,0)$ is thus bounded by $Ae^{\|g(z)\|_2^2}$, and Cauchy's estimate in the z's with radius $(2n)^{-1/2}$, as in the proof of Theorem 7.4(c), gives $(\sqrt{2}e)^n (n!)^{1/2}$. Then (E2) and $(n!)^{1/2}p! \leq (n+p)!$ give the second bound; the extension to $\mathcal{S}$, to complex test functions (the factor $2^{n+p} \leq 2^N$) and to tempered distributions is as in Theorem 7.4(c), and $C_1^N/(N!)^{\beta-1} \to 0$ for every $\beta > 1$. □

Lemma 13.11 (the tube object: limit, continuity, translations)**.** *Assume the bound* (31)*, which Theorem 13.10 derives from* (T3) *and the torus inputs (through* (I2)*), and, as in Theorem 13.10(i), that the comparison family is convergent at fixed cubes on the tube (Definition 3.2). Write $\mathcal{D} = \mathcal{D}_{\sigma_+,H}$ for the tube of a family with supports in Q_R.*

(iii) (The J-limit on the tube.) *For $J \geq J_1(R+\frac{1}{2})$, which exceeds the least J_0 with the supports at ℓ^∞-distance >1 from $\mathbb{R}^3 \setminus C_J$ (as $MJ_1(R') - R' \geq \ell_c \geq 10 > 1$), $G_\infty = \lim_J G_J$ exists locally uniformly on $\mathcal{D}$, is holomorphic on $\mathcal{D}$, satisfies the bound of Theorem 13.10*(i)*, is >0 at every real point of $\mathcal{D}$, and $G_J \leq G_\infty$ at the cone points of $\mathcal{D}$ ($z \in (\mathrm{i}\mathbb{R})^n$, $w \in \mathbb{R}^p$, $k(\zeta) \geq 0$: the real scalar source $k(\zeta) \geq 0$ sits in the slot of the cone).*

(iv) (Continuity on the real tube.) *Let $E_\rho := C_c^\infty(\bar{B}_\rho;\Lambda^2) \times C_c^\infty(\bar{B}_\rho)$ with its C^∞ topology, a Fréchet space, and $U_\rho := \{(f,h) \in E_\rho : \|h_+\|_\infty < \sigma_+/2,\ \|h_-\|_\infty < H/2\}$, open and convex*

($h \mapsto \|h_\pm\|_\infty$ is continuous and convex). Then $\Phi^a(f,h) := \ln G_\infty(f,h)$ is finite, convex and continuous on U_ρ.

(v) (Translations on the tube.) *For every family and every $b \in \mathbb{R}^3$: $G_\infty(\tau_b \cdot \zeta) = G_\infty(\zeta)$ for all $\zeta \in \mathcal{D}$; the tube of the translated family is that of the family, since $k(\tau_b \cdot \zeta) = \tau_b k(\zeta)$ has the same supremum norms.*

Besides Definition 3.2 and (31) the proofs use only the first inequality of Lemma 2.11, Lemmas 2.12 and 2.15, lattice covariance and (T2), (T3): not (W2)*, and Lemma 2.8(d) only inside* (31).

Proof. (iii) Existence, holomorphy and the bound are the last three paragraphs of the proof of Theorem 13.10(i), which use (31), Definition 3.2 for the k-limits and the first inequality of Lemma 2.11. *Positivity at real points.* For real $\zeta \in \mathcal{D}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}$ ($\mathcal{D}$ is an open neighbourhood of 0). $\varphi_J(t) := \ln G_J(t\zeta)$ is convex on $[-\epsilon, 1]$ (Lemma 2.12(b) in the limit) with $\varphi_J(0) = 0$, so $0 \le \frac{\epsilon}{1+\epsilon}\varphi_J(1) + \frac{1}{1+\epsilon}\varphi_J(-\epsilon)$, i.e. $\varphi_J(1) \ge -\varphi_J(-\epsilon)/\epsilon \ge -C_\epsilon/\epsilon$, C_ϵ the limit bound of (31) at $-\epsilon\zeta$; hence $G_J(\zeta) \ge e^{-C_\epsilon/\epsilon}$ uniformly in J and $G_\infty(\zeta) > 0$. *Cone points.* There $G_J \le G_{J+1}$ (first inequality of Lemma 2.11 with Lemma 2.12(a) at fixed k, then $k \to \infty$; it needs $J \ge J_0$, which the range of (iii) ensures), so $G_J \le G_\infty$.

(iv) *Finite:* (iii) for the two-source family $(f; h)$ at $\zeta = (1,1)$, in $\mathcal{D}$ by the definition of U_ρ; the value is the pointwise iterated limit and does not depend on the family used. *Convex:* for $s, s' \in U_\rho$, $t \in [0,1]$, use the family $(f, f'; h, h')$ at the real point $(t, 1-t, t, 1-t)$, in $\mathcal{D}$ because $(th + (1-t)h')_\pm \le th_\pm + (1-t)h'_\pm$ pointwise; the convexity of $\ln Z^a_{k,J}$ in the real coefficients (Lemma 2.12(b)) passes to the limits. *Borel:* a pointwise limit of functions each depending continuously on finitely many values of the sources. *Continuous:* Lemma 2.15 on the open convex U_ρ.

(v) *Lattice translations at cone points.* Let $b \in 5^{-k_0}\mathbb{Z}^3$, $|b|_\infty \le M$, and ζ a cone point of $\mathcal{D}$ ($k_- = 0$, $q := k(\zeta) \ge 0$, $\|q\|_\infty < \sigma_+/2$). Steps 1–3 of the proof of Proposition 8.1 run unchanged with the real scalar source q in the slot of the cone: lattice covariance (τ_b preserves Λ_k and S_{Λ_k} for $k \ge k_0$) gives $Z^a_{k,C_J}(\tau_b \cdot \zeta) = Z^a_{k,C_J - b}(\zeta)$, the interlacing $X_{J-1} \subset (C_J - b) \cap \Lambda_k \subset X_{J+1}$ and the first inequality of Lemma 2.11 twice (no killed site) give $Z^a_{k,C_{J-1}}(\zeta) \le Z^a_{k,C_J}(\tau_b \cdot \zeta) \le Z^a_{k,C_{J+1}}(\zeta)$ for $J \ge J_0 + 1$. Let $k \to \infty$ by Theorem 13.10(i) for the three families (the translated family is supported at distance > 1 from $\mathbb{R}^3 \setminus C_{J_0+1}$), then $J \to \infty$ by (iii), the three terms being cone points of their tubes: $G_\infty(\tau_b \cdot \zeta) = G_\infty(\zeta)$ at the cone points of $\mathcal{D}$. *All $\zeta \in \mathcal{D}$.* For the split family of the proof of Theorem 13.10(i) and its translate, $F^{\tau_b}_\infty - F_\infty$ is holomorphic on the connected $\mathcal{D}''$ and vanishes on Ω''_+, whose points are cone points of the split family's tube and which is an open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$; by (T2) it vanishes on $\mathcal{D}''$, and the diagonal gives $G^{\tau_b}_\infty = G_\infty$ on $\mathcal{D}'$. Then $G^{\tau_b}_\infty - G_\infty$ is holomorphic on the connected $\mathcal{D}$ ((iii) for both families) and vanishes on the open set $\mathcal{D}'$, hence on $\mathcal{D}$. *All $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$:* iterate; the tube is the same at every step. *All $b \in \mathbb{R}^3$ at real points.* For real $\zeta \in \mathcal{D}$ the source pair $s := (\sum_i z_i f_i, \sum_j w_j h_j)$ lies in some U_ρ, $\tau_b s \in U_{\rho'}$ for $|b| \le \rho' - \rho$, and $b \mapsto \tau_b s$ is continuous into $E_{\rho'}$; by (iv) and the density of $\bigcup_k 5^{-k}\mathbb{Z}^3$, $\Phi^a(\tau_b s) = \Phi^a(s)$. *Complex $\zeta \in \mathcal{D}$.* $G^{\tau_b}_\infty - G_\infty$ is holomorphic on $\mathcal{D}$ and vanishes on the real points of $\mathcal{D}$, a non-empty open subset of $\mathbb{R}^{n+p}$; (T2) gives the claim. □

Remark 13.12 (King's convention). On King's RG cube $\mathrm{ND}^K_{k,J}$ the negative leg holds as printed: by Lemma 4.7, the wall insertion of Lemma 13.5 runs on $\mathrm{ND}^K_{k,J}$ with factors that are comparison box states killed on $X_i \cap T_J$, and every cell on which a cutoff source $8\chi_s q$ does not vanish is a full cell inside C_J meeting no killed site, because χ_s vanishes within distance 2 of every cell wall; so Lemmas 13.6 to 13.8 and Proposition 13.9 apply with the printed K_-. The direction matters: killing sites of a cell increases a class-$\mathcal{D}$ functional (Lemma 4.7(e)), so B_{cell} does not control a source-carrying cell with killed sites. For the positive leg, Remark 4.8 prepends the interlacing inequality $\langle\cdot\rangle^K_{k,J} \le \langle\cdot\rangle^{(a)}_{k,J}$ on class $\mathcal{I}$ (Step 1 of Lemma 4.2) to Propositions 13.1 and 13.2, so Corollary 13.3 holds for $N^K_{k,C_J}(0,q)$ at every (k,J) and, for the fixed-cube limits, in the form $\limsup_k Z^{a,K}_{k,J}(0,q) \le e^{\kappa\|q\|_{L^1}}$. The assembly of (31) on ND^K from these pieces, with the check

that every step of the proof of Theorem 13.10 runs in the right direction on King's convention, and the existence of the fixed-cube limits of $Z^{a,K}_{k,J}$ on the tube (Theorem 13.10(i) on ND^K) are carried out in Theorem 21.3; the limit statements of this section and of the Wick sector of Sections 8, 9 and 12 then hold on King's convention and the Wick sector of the main theorem is stated for King's printed state (Proposition 21.4).

14. Continuum subsequences and LIM states

The axioms of the preceding sections rest on the existence of the fixed-cube limits $\lim_k$: on Theorem 3.3 in the curvature sector, on Theorem 3.6 on the Wick tube, both proved in the existence part on King's convention (for the comparison family, on which the tube statements are written, the corresponding clauses are the template hypotheses Definitions 3.1 and 3.2). This section shows that the axioms hold without that input, for the limit state along any continuum subsequence of spacings, and identifies every such state with the iterated limit. The k-uniform bounds of Lemma 7.3 and (31) make the fixed-cube functionals normal families, so one subsequence of spacings serves every cube and every family at once; along it the cube limit $J \to \infty$ exists because the cone functionals are monotone in the cube at fixed k (Lemma 2.11). The limit state of such a subsequence (a *LIM state*) satisfies every axiom proved above, with no hypothesis in the curvature sector and, on the Wick tube, under (T3) for every admissible scheme (through two properties of the scheme, the cell bound and the scheme bound, proved below); Theorem 3.3 (resp. Theorem 3.6) adds exactly the identification with the iterated limit and the independence of the subsequence. The section is written on the comparison convention $\mathrm{ND}^{(a)}$ (Definition 2.1), in which the tube lemmas and the rotation part are written; King's convention is Corollary 14.14. Notation is that of Definitions 2.4, 2.6 and 13.4 and (30): $a^{(a)}_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)} \rangle^{(a)}_{k,J}$ for a family $f = (f_1, \dots, f_N)$ of real 2-forms; $Z^a_{k,J}(\zeta)$ for $\mathcal{F} = (f_1, \dots, f_n; h_1, \dots, h_p)$ and $\zeta = (z, w) \in \mathbb{C}^{n+p}$ in the scheme a; $g(\zeta)$, $k(\zeta) = k_+ - k_-$, $\mathsf{a}(\zeta)$; $U_J := \{x : \mathrm{dist}_\infty(x, \mathbb{R}^3 \setminus C_J) > 1\}$, $J_0(f)$ the least J with all supports in U_J, and $R(\mathcal{F})$ the least R with all supports in $Q_R = [-R, R]^3$. Two standard facts [11] are used by name: the *identity theorem* (a holomorphic function on a connected open $D \subset \mathbb{C}^m$ vanishing on a non-empty relatively open subset of $D \cap \Sigma$, Σ a maximal totally real affine subspace such as $\mathbb{R}^m$ or $(\mathrm{i}\mathbb{R})^{m_1} \times \mathbb{R}^{m_2}$, vanishes on D) and *Montel–Vitali* (a locally bounded family of holomorphic functions on D is normal, and converges locally uniformly on D if it converges on such a set).

14.1. Definitions, the cell bound and the scheme bound.

Definition 14.1 (continuum subsequences)**.** An increasing sequence $\mathcal{K} = (k_m)_{m \geq 1} \subset \mathbb{N}$ of spacings is a *continuum subsequence* (of the comparison family) if for every finite family f of real 2-forms and every $J \geq J_0(f)$ the sequence $(a^{(a)}_{k_m,J})_m$ converges locally uniformly on $\mathbb{C}^N$; the limit is written $Z^{\mathcal{K}}_J(\sum_j z_j f_j, 0)$ and depends on the family only through the total complex source, as each $a^{(a)}_{k,J}$ does. Fix a scheme a. $\mathcal{K}$ is a *continuum subsequence for the Wick sector* (Form (B)) if it is a continuum subsequence and, for every finite family $\mathcal{F}$ of real 2-forms and scalars and every J with $MJ > R(\mathcal{F}) + \ell_c$, the sequence $(Z^a_{k_m,J})_m$ converges locally uniformly on the tube $\mathcal{D}_{\sigma_+,H}$ of $\mathcal{F}$; the limit is $Z^{a,\mathcal{K}}_J$. (The strict inequality replaces the threshold $J_1(R)$ of Theorem 13.10 by at most one cube step; only tails in J are used.) Form (A), convergence on all of $\mathbb{C}^{n+p}$ for the same J, is the optional extension of Theorem 23.7 and is not used here.

Definition 14.2 (LIM states)**.** The *LIM state* of a continuum subsequence $\mathcal{K}$ is $Z^{\mathcal{K}}_\infty := \lim_{J\to\infty} Z^{\mathcal{K}}_J$ on $\mathbb{C}^N$ (curvature sector), resp. $Z^{a,\mathcal{K}}_\infty := \lim_J Z^{a,\mathcal{K}}_J$ on the tube (Wick sector); Lemma 14.8 gives existence. A subsequence of a continuum subsequence is a continuum subsequence with the same fixed-cube limits, hence the same LIM state.

Definition 14.3 (the limit state of the rotation theorem)**.** For a family f and $J \geq J_0(f)$ let L^f_J be the set of locally uniform limits on $\mathbb{C}^N$ of subsequences of $(a^{(a)}_{k,J})_k$. The rotation theorem (Theorem 3.14) gives an entire $Z^{\mathrm{LIM}}_\infty(\cdot, 0)$ to which *every* choice $\zeta_J \in L^f_J$ converges

locally uniformly as $J \to \infty$, the subsequence being allowed to depend on J. With J-dependent subsequences one only has $\liminf_k a^{(a)}_{k,J}(iu) \le \zeta_J(iu) \le \limsup_k a^{(a)}_{k,J}(iu)$, and the J-limit is pinned only if the J-limits of the two bounds agree, which is the content of that theorem. A LIM state in the sense of Definition 14.2 uses one $\mathcal{K}$ common to all J and all families, along which the fixed-cube limits are monotone in J (Lemma 14.8), and needs no hypothesis; Proposition 14.12(a) relates the two.

The curvature sector needs no hypothesis. On the Wick tube the bound (31) of Theorem 13.10 is used; it rests on (T3) and the torus inputs (Proposition 3.9, Lemma 3.10, and Hypothesis 3.12, which give (I2) with amplitude $s_1 \in (0, \sigma_1)$ and constant C_U through Proposition 12.6; (I3) is Proposition 12.2 and (I4) is Lemma 2.8) and on one instance of Theorem 3.6, isolated as follows.

We say that the scheme a has the *cell bound* ($\mathrm{B}_{\mathrm{cell}}$) if $B_{\mathrm{cell}} := \sup_k \ln N^a_{k,C_{L_c}}(0, -2H'\chi_0) < \infty$, for the one cell C_{L_c} (side $\ell_c \ge 10$) and the one negative Wick source $-2H'\chi_0$ of Definition 13.4, and that it has the *scheme bound* (SCH) if either $a = z$, the reference scheme in which Lemma 3.10 is computed, or $\sup_k |z_k - a_k| < \infty$. The cell bound is the boundedness form of Theorem 3.6 at one cube and one source, and the only place where that theorem enters the proof of (31) (through Lemmas 13.6 and 13.7 and Proposition 13.9; K_- is built from B_{cell}). Both properties hold for the schemes of the main theorem:

Lemma 14.4 (($\mathrm{B}_{\mathrm{cell}}$), the cell bound)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme (Definition 3.7). Then a has the cell bound: $B_{\mathrm{cell}} = \sup_k \ln N^a_{k,C_{L_c}}(0, -2H'\chi_0) < \infty$.*

Proof. By Definition 13.4 the cell $C_{L_c} = [-ML_c, ML_c)^3$ is a King cube and $\operatorname{supp}\chi_0$ lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{L_c}$; the source $q := 2H'\chi_0$ is a non-negative function of $C^\infty_c(\mathbb{R}^3)$ with this support condition. Under (T3) the inputs (I2) and (I3) of the existence theorem with scalar sources hold (Proposition 12.6, Proposition 12.2), so Corollary 6.18(b) applies and gives $\sup_k N^z_{k,C_{L_c}}(0, -q) < \infty$ for the comparison-convention functional in the reference scheme z. For a scheme a with $\sup_k |a_k - z_k| < \infty$ the two functionals differ by the factor $e^{-(a_k - z_k)\mathsf{S}_k(-q)}$ (Remark 2.5), bounded uniformly in k since $\mathsf{S}_k(q) \to \int q$; this is Remark 6.19, and an admissible scheme has $\sup_k |z_k - a_k| < \infty$ by Definition 3.7. Hence $\sup_k N^a_{k,C_{L_c}}(0, -2H'\chi_0) < \infty$, and $B_{\mathrm{cell}} < \infty$ because each term is positive (Remark 6.20). □

Lemma 14.5 ((SCH), the scheme bound)**.** *An admissible scheme has the scheme bound. If $a \ne z$ has the scheme bound, the constants of* (31) *hold with C_U replaced by $C_U + s_1 \sup_k |z_k - a_k|$.*

Proof. For an admissible scheme either $a = z$ or $\lim_k (z_k - a_k)$ exists, so $\sup_k |z_k - a_k| < \infty$ (Definition 3.7). The replacement of the constant is the last clause of Proposition 12.6, carried through Theorem 13.10. □

Proposition 14.6 ((UNIQ), independence of the subsequence)**.** *Any two continuum subsequences $\mathcal{K}, \mathcal{K}'$ have the same LIM state: $Z^{\mathcal{K}}_\infty(\cdot, 0) = Z^{\mathcal{K}'}_\infty(\cdot, 0)$ for every family.*

Proof. This is Proposition 14.12(a) below, from the rotation theorem Theorem 3.14: every continuum subsequence's LIM state is the limit state $Z^{\mathrm{LIM}}_\infty(\cdot, 0)$ of that theorem. It also follows from Theorem 3.3 through Corollary 4.4(iii) (Corollary 14.14(b)): every LIM state of the comparison family is King's printed state Z^K_∞. □

The proofs of the axioms below hold for each LIM state separately and do not use (UNIQ); it enters only in the Euclidean clause (vii) of Theorem 14.9, through Proposition 14.12(a). Without the rotation theorem and the existence theorem one would only know that every LIM state lies on $(i\mathbb{R})^N$ between $\lim_J \liminf_k a^{(a)}_{k,J}$ and $\lim_J \limsup_k a^{(a)}_{k,J}$ (both exist, by monotonicity in J), with equality of the LIM states if these coincide at every real argument, an entire function being determined on $(i\mathbb{R})^N$.

14.2. **Existence of continuum subsequences and the cube limit.**

Lemma 14.7 (continuum subsequences exist)**.** (a) (Curvature sector; no hypothesis.) *Every increasing sequence of spacings has a continuum subsequence. For every continuum subsequence $\mathcal{K}$, family f and $J \ge J_0(f)$: $Z_J^{\mathcal{K}}(\cdot,0)$ is entire, $|Z_J^{\mathcal{K}}(\sum_j z_j f_j,0)| \le e^{\|g(z)\|_2^2}$, $Z_J^{\mathcal{K}}(x,0)$ is real and ≥ 1 at real x, and $0 \le Z_J^{\mathcal{K}}(\mathrm{i}t,0) \le 1$ at imaginary* it. Several models at once: *for every countable set $\mathcal{M}$ of lattice models of King's type (Neumann–Dirichlet cube states with a site-independent real bare scalar coefficient, photon mass $\mu > 0$, $\lambda > 0$; for instance the models at the photon masses of a countable set, with the counterterm fixed) one $\mathcal{K}$ is a continuum subsequence for every member of $\mathcal{M}$.*

(b) (Wick sector, Form (B).) *Let the couplings satisfy* (T3) *and let the scheme a have the cell bound* ($\mathrm{B}_{\mathrm{cell}}$) *and the scheme bound* (SCH) *(both hold for every admissible scheme, Lemmas 14.4 and 14.5). Then every increasing sequence of spacings has a continuum subsequence for the Wick sector, and for every such $\mathcal{K}$, family $\mathcal{F}$ and J with $MJ > R(\mathcal{F}) + \ell_c$, $Z_J^{a,\mathcal{K}}$ is holomorphic on $\mathcal{D}_{\sigma_+,H}$ and*

$$|Z_J^{a,\mathcal{K}}(\zeta)| \le \exp\big(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w\big) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H}), \tag{32}$$

with the constants of (30). Several models at once: *for a countable set of models all satisfying* (31) *with one set of constants, one $\mathcal{K}$ serves all.*

Form (A), which needs in addition the existence of the limits at every real coefficient vector ((W2)(a)), is Theorem 23.7. The subsequence is chosen in Step 2 below and nowhere else (Remark 14.15).

Proof. (a). Inputs: Lemma 7.3 ($\sup_{k\ge0,\,J\ge J_0(f)} \sup_K |a^{(a)}_{k,J}| < \infty$ for every family and compact $K \subset \mathbb{C}^N$); Lemma 7.1 at second order ($\langle F(u)^2\rangle^{(a)}_{k,J} \le 2\|u\|^2_{2,k}$ for real u with plaquettes in X_J); charge conjugation $(A,\vartheta) \mapsto (-A,-\vartheta)$, a symmetry of every Neumann–Dirichlet state, so $\langle F(u)\rangle^{(a)}_{k,J} = 0$ and $\langle e^{\mathrm{i}F(u)}\rangle^{(a)}_{k,J}$ is real. *Step 1 (a countable core).* Let $\mathcal{B}$ be the countable set of finite unions of closed boxes with rational corners, for $B \in \mathcal{B}$ let $S_0(B)$ be a countable $\|\cdot\|_\infty$-dense subset of $C_c^\infty(\operatorname{int} B;\Lambda^2)$, and let $\mathcal{P}$ be the countable set of pairs (J,u'), u' a finite family from $\bigcup_B S_0(B)$, $J \ge J_0(u')$. *Step 2 (Montel and the diagonal).* For $(J,u') \in \mathcal{P}$, $(a^{u'}_{k,J})_k$ is locally bounded on $\mathbb{C}^N$ uniformly in k (Lemma 7.3), hence normal; enumerate $\mathcal{P}$, extract successively, and let $\mathcal{K} = (k_m)$ be the diagonal sequence: $(a^{u'}_{k_m,J})_m$ converges locally uniformly for every $(J,u') \in \mathcal{P}$. *Step 3 (every family, imaginary coefficients).* Let f be a family, $J \ge J_0(f)$, and $B \in \mathcal{B}$ with $\bigcup_j \operatorname{supp} f_j \subset \operatorname{int} B \subset B \subset U_{J_0(f)}$ (so every $u' \in S_0(B)$ is admissible at every $J \ge J_0(f)$). For real 2-forms u, u', g supported in B and every k with $\varepsilon_k \le 1$,

$$|\langle e^{\mathrm{i}F(u)}\rangle^{(a)}_{k,J} - \langle e^{\mathrm{i}F(u')}\rangle^{(a)}_{k,J}| \le \langle F(u-u')^2\rangle^{(a)}_{k,J}{}^{1/2} \le \sqrt{2}\|u-u'\|_{2,k}, \qquad \|g\|^2_{2,k} \le V_B\|g\|^2_\infty, \quad V_B := 3\,\mathrm{vol}(N_{1/2}(B)),$$

the first by $|e^{\mathrm{i}x}-1| \le |x|$ and the second-order bound, the second because the cubes of side ε_k about the centres of the plaquettes of one orientation on which $g \ne 0$ are disjoint and lie in the closed ℓ^∞-neighbourhood $N_{1/2}(B)$. Given $t \in \mathbb{R}^N$ and $\eta > 0$ pick $u' \in S_0(B)$ with $\|\sum_j t_j f_j - u'\|_\infty \le \eta$; then $|a^f_{k_m,J}(\mathrm{i}t) - a^f_{k_{m'},J}(\mathrm{i}t)| \le 2\sqrt{2V_B}\,\eta + |\langle e^{\mathrm{i}F(u')}\rangle^{(a)}_{k,J_{k_m}} - \langle e^{\mathrm{i}F(u')}\rangle^{(a)}_{k,J_{k_{m'}}}|$, and the last term tends to 0 by Step 2. So $(a^f_{k_m,J}(\mathrm{i}t))_m$ converges for every t. *Step 4 (all complex coefficients).* $(a^f_{k_m,J})_m$ is locally bounded on $\mathbb{C}^N$ and converges on the maximal totally real $(\mathrm{i}\mathbb{R})^N$; Montel–Vitali gives locally uniform convergence to an entire $Z_J^{\mathcal{K}}(\cdot,0)$. The bound $|a^{(a)}_{k,J}(z)| \le e^{\|g(z)\|^2_{2,k}}$ (Theorem 7.4(a)) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$ (1); realness at real points passes; $a^{(a)}_{k,J}(x) = \langle e^{F(u_x)}\rangle^{(a)}_{k,J} \ge e^{\langle F(u_x)\rangle^{(a)}_{k,J}} = 1$ by Jensen, and $[1,\infty)$ is closed; $0 \le a^{(a)}_{k,J}(\mathrm{i}t) \le 1$ by the current representation of Lemma 2.9(b), $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle^{(a)}_{k,J} = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp \cosh\langle Cv,\kappa\rangle \ge 0$, and $|e^{\mathrm{i}F}| = 1$. Dependence only on the total source holds at every (k,J), hence in the limit. *Step 5 (several models).* The constant of Lemma 7.3, the bound $e^{\|g\|^2_{2,k}}$ and the Lipschitz constant $\sqrt{2}$ are model-free: Theorem 7.4(a) is Lemma 2.9(a) for a measure of the class of Lemma 2.8 plus Lemma 7.1 for the Neumann–Dirichlet photon covariance, valid for every real site-independent

bare coefficient (the matter weight does not enter) and every $\mu > 0$. For countable $\mathcal{M}$ run Step 2 over $\mathcal{M} \times \mathcal{P}$ and Steps 3–4 member by member.

(b). Inputs: under (I2)–(I4) and $B_{\mathrm{cell}} < \infty$, for every family with supports in Q_R, every k, $J \geq J_1(R)$ and $\zeta \in \mathcal{D}_{\sigma_+,H}$, (31) gives $|Z^a_{k,J}(\zeta)| \leq Z^a_{k,J}(\operatorname{Re}\zeta) \leq \exp(\|g\|^2_{2,k} + \kappa\|k_+\|_{1,k} + K_-|k_-|_w)$, with (I2) from (T3) by Proposition 12.6 and (SCH) for the scheme; on compacts of the tube the right side is bounded uniformly in k (Lemma 2.7). *Steps 1–2.* As in (a), with $S_0^{\mathsf{W}}(B)$ a countable $\|\cdot\|_\infty$-dense subset of $C_c^\infty(\operatorname{int} B; \Lambda^2) \times C_c^\infty(\operatorname{int} B)$ and $\mathcal{P}^{\mathsf{W}}$ the countable set of pairs $(J, \mathcal{F}')$, $\mathcal{F}'$ a finite family from $\bigcup_B S_0^{\mathsf{W}}(B)$, $MJ > R(\mathcal{F}') + \ell_c$. For such a pair $J \geq J_1(R)$ for every $R \in (R(\mathcal{F}'), MJ - \ell_c]$, so (31) holds for $\mathcal{F}'$ at every k and $(Z^a_{k,J})_k$ is normal on the tube of $\mathcal{F}'$; the diagonal over $\mathcal{P}^{\mathsf{W}}$, together with that of (a), gives $\mathcal{K}$. *Step 3 (every family, imaginary coefficients).* Let $\mathcal{F} = (f_i; h_j)$, $MJ > R(\mathcal{F}) + \ell_c$, $R_J := MJ - \ell_c > R(\mathcal{F})$, so that all supports lie in the open cube $Q^\circ_{R_J}$ and $J \geq J_1(R_J)$; choose $B \in \mathcal{B}$ with the supports in $\operatorname{int} B \subset B \subset Q^\circ_{R_J}$, so that $(J, \mathcal{F}') \in \mathcal{P}^{\mathsf{W}}$ for every $\mathcal{F}'$ from $S_0^{\mathsf{W}}(B)$. For $(s, s') \in \mathbb{R}^{n+p}$ put $\sigma := (\sum_i s_i f_i, \sum_j s'_j h_j)$, so that $Z^a_{k,J}(\mathrm{i}s, \mathrm{i}s') = \langle e^{\mathrm{i}\Phi^a_k(\sigma)}\rangle^{(a)}_{k,J}$ with $\Phi^a_k(g,h) := F_{\Lambda_k}(g) + W^a_{\Lambda_k}(h)$. *Lipschitz bound.* For a real pair $q = (g_q, h_q)$ supported in Q_{R_J}, $(t_1, t_2) \mapsto Z^a_{k,J}(t_1 g_q, t_2 h_q)$ is holomorphic, and on the polydisc $|t_1| \leq \|g_q\|^{-1}_{2,k}$, $|t_2| \leq \sigma_0/|h_q|_w$ one has $\|k_\pm\|_\infty \leq |\operatorname{Re} t_2|\,\|h_q\|_\infty \leq \sigma_0 < H/2 < \sigma_+/2$, so the polydisc lies in the tube and (31) gives $|Z^a_{k,J}| \leq e^{1 + K'_{\mathrm{lat}}\sigma_0}$ there, $K'_{\mathrm{lat}} := \kappa C'_0 + K_-$ ($\|k_+\|_{1,k} \leq C'_0|k_+|_w$ by Lemma 2.7). The Taylor coefficients at 0 are the second moments of $F(g_q)$ and $W^a(h_q)$, and Cauchy's estimates give

$$\langle \Phi^a_k(q)^2\rangle^{(a)\,1/2}_{k,J} \leq C\,(\|g_q\|_{2,k} + |h_q|_w), \qquad C := \sqrt{2}\, e^{(1+K'_{\mathrm{lat}}\sigma_0)/2} \max(1, \sigma_0^{-1}),$$

uniformly in k, J and R_J; hence $|Z^a_{k,J}(\mathrm{i}\sigma) - Z^a_{k,J}(\mathrm{i}\sigma')| \leq C(\|g_{\sigma-\sigma'}\|_{2,k} + |h_{\sigma-\sigma'}|_w)$ for real pairs supported in Q_{R_J}. *Density.* Given $\eta > 0$ pick $\sigma' \in S_0^{\mathsf{W}}(B)$ with $\|g_\sigma - g_{\sigma'}\|_\infty, \|h_\sigma - h_{\sigma'}\|_\infty \leq \eta$; with V_B and $|h|_w \leq (1+3R_J^2)^2\|h\|_\infty$ for h supported in Q_{R_J}, $|Z^a_{k,J}(\mathrm{i}\sigma) - Z^a_{k,J}(\mathrm{i}\sigma')| \leq C(\sqrt{V_B} + (1+3R_J^2)^2)\eta$ for every k, while $Z^a_{k_m,J}(\mathrm{i}\sigma')$, the core functional at the point (i,i) of its tube, converges by Step 2. So $(Z^a_{k_m,J})_m$ converges on $(\mathrm{i}\mathbb{R})^{n+p}$. *Step 4 (the tube).* $(Z^a_{k_m,J})_m$ is locally bounded on the connected open $\mathcal{D}_{\sigma_+,H}$ and converges on the maximal totally real $(\mathrm{i}\mathbb{R})^{n+p} \subset \mathcal{D}_{\sigma_+,H}$; Montel–Vitali gives locally uniform convergence to a holomorphic $Z^{a,\mathcal{K}}_J$, and (31) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (Riemann sums; k_+ is continuous with compact support) and sampled $|k_-|_w \leq$ continuum $|k_-|_w$: this is (32). *Step 5 (several models).* With one set of constants in (31) for all members, C is model-free and Step 2 runs over the product of the model set with $\mathcal{P}^{\mathsf{W}}$. □

Lemma 14.8 (the cube limit along $\mathcal{K}$ is a full limit)**.** *Let $\mathcal{K}$ be a continuum subsequence.* (Curvature sector.) *For every family, $Z^{\mathcal{K}}_J(\cdot, 0) \to Z^{\mathcal{K}}_\infty(\cdot, 0)$ locally uniformly on $\mathbb{C}^N$ as $J \to \infty$; the limit is entire, $|Z^{\mathcal{K}}_\infty(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|^2_2}$, real and ≥ 1 at real points, in $[0,1]$ at imaginary points; $J \mapsto Z^{\mathcal{K}}_J(\mathrm{i}t, 0)$ is nondecreasing, so $Z^{\mathcal{K}}_J \leq Z^{\mathcal{K}}_\infty$ on $(\mathrm{i}\mathbb{R})^N$.* (Wick sector, Form (B).) *Under the hypotheses of Lemma 14.7(b), $Z^{a,\mathcal{K}}_J \to Z^{a,\mathcal{K}}_\infty$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$; the limit is holomorphic, obeys (32), is > 0 at the real points of the tube, and $Z^{a,\mathcal{K}}_J \leq Z^{a,\mathcal{K}}_\infty$ at the cone points ($z \in (\mathrm{i}\mathbb{R})^n$, w real, $k(\zeta) \geq 0$).*

Proof. Curvature sector. For $t \in \mathbb{R}^N$, $J \geq J_0(f)$ and every k: $a^{(a)}_{k,J}(\mathrm{i}t) \leq a^{(a)}_{k,J+1}(\mathrm{i}t) \leq 1$ (first inequality of Lemma 2.11 with $Y = C_{J+1} \cap \Lambda_k$, $X = X_J$; the cut-out factor is the Neumann–Dirichlet state on C_J because δm^2_ε is site-independent; realness by charge conjugation). Both sides converge along $\mathcal{K}$, so $J \mapsto Z^{\mathcal{K}}_J(\mathrm{i}t, 0)$ is nondecreasing, bounded by 1, convergent. $\{Z^{\mathcal{K}}_J\}_J$ is locally bounded on $\mathbb{C}^N$ by the J-free bound $e^{\|g(z)\|^2_2}$ and converges on $(\mathrm{i}\mathbb{R})^N$; Montel–Vitali gives the entire limit with the same bound; realness, ≥ 1 at real points and $[0,1]$ at imaginary points pass to the limit.

Wick sector. (1) *Split family.* Write $h_j = h_j^+ - h_j^-$, $h_j^- := \|h_j\|_\infty \chi_j$, $h_j^+ := h_j + h_j^- \geq 0$, with $\chi_j \in C_c^\infty$, $0 \leq \chi_j \leq 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset N_{1/2}(\operatorname{supp} h_j)$; put $F_{k,J}(z, w^+, w^-) := Z^a_{k,J}(\sum_i z_i f_i, \sum_j (w_j^+ h_j^+ + w_j^- h_j^-))$ and $\mathcal{D}'' := \{|\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j \delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) <$

$H/2$, a convex subset of the split family's tube. By Lemma 14.7(b) for the split family, $F_{k_m,J} \to F_J^{\mathcal{K}}$ locally uniformly on $\mathcal{D}''$ for $MJ > R(\mathcal{F}) + \frac{1}{2} + \ell_c$, with (32). (2) On $\Omega_+ := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the observable has imaginary curvature coefficients and nonnegative total scalar source, so $J \mapsto F_{k,J}$ is nondecreasing at every k (first inequality of Lemma 2.11 with Lemma 2.12(a), the scheme multiplier being common); along $\mathcal{K}$, $J \mapsto F_J^{\mathcal{K}}$ is nondecreasing and bounded on Ω_+, hence convergent. (3) Ω_+ is a non-empty open subset of the maximal totally real $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$ and $\mathcal{D}''$ is connected: $F_J^{\mathcal{K}} \to F_\infty^{\mathcal{K}}$ locally uniformly on $\mathcal{D}''$. (4) At $w^+ = w$, $w^- = -w$ the lattice functionals equal $Z^a_{k,J}(\zeta)$ exactly (equal total sources), so $Z_J^{a,\mathcal{K}}$ converges locally uniformly on $\mathcal{D}' := \{|\operatorname{Re} w_j| < \delta_j\} \times \mathbb{C}^n \subset \mathcal{D}_{\sigma_+,H}$; $\{Z_J^{a,\mathcal{K}}\}_J$ is locally bounded on $\mathcal{D}_{\sigma_+,H}$ by the J-free (32) and converges on the open $\mathcal{D}'$; Montel–Vitali on the connected tube gives $Z_J^{a,\mathcal{K}} \to Z_\infty^{a,\mathcal{K}}$, holomorphic, with (32). (5) *Positivity at real tube points.* For real $\zeta \in \mathcal{D}_{\sigma_+,H}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}_{\sigma_+,H}$ and let $C_\epsilon := \sup_k(\|g\|^2_{2,k} + \kappa\|k_+\|_{1,k} + K_-|k_-|_w)|_{-\epsilon\zeta} < \infty$, the exponent of (31) at $-\epsilon\zeta$ (bounded in k by Lemma 2.7, free of J). At every k, $t \mapsto \ln Z^a_{k,J}(t\zeta)$ is convex on $\mathbb{R}$ (Hölder, Lemma 2.12(b)) and vanishes at $t = 0$, so $1 \le Z^a_{k,J}(\zeta)^{\epsilon/(1+\epsilon)} Z^a_{k,J}(-\epsilon\zeta)^{1/(1+\epsilon)} \le Z^a_{k,J}(\zeta)^{\epsilon/(1+\epsilon)} e^{C_\epsilon/(1+\epsilon)}$, i.e. $Z^a_{k,J}(\zeta) \ge e^{-C_\epsilon/\epsilon}$ for every k and J; the bound passes along $\mathcal{K}$ and then in J: $Z_\infty^{a,\mathcal{K}}(\zeta) \ge e^{-C_\epsilon/\epsilon} > 0$. (The product inequality of Lemma 2.12(c) is not used: the tube is asymmetric, $H = \sigma_+/16$, and for real $\xi \in \mathcal{D}_{\sigma_+,H}$ with $\|k_+(\xi)\|_\infty \ge H/2$ the point $-\xi$ lies outside it, where Definition 14.1 gives no limit.) (6) At a cone point of the original family $J \mapsto Z^a_{k,J}(\zeta)$ is nondecreasing at every k (first inequality of Lemma 2.11 with Lemma 2.12(a)); pass along $\mathcal{K}$. □

14.3. **The axioms for LIM states.**

Theorem 14.9 (LIM states of the comparison family at fixed photon mass)**.** *Let $e > 0$, $\lambda, m, \mu_0 > 0$, $\varepsilon_k = 5^{-k}$, $C_J = [-MJ, MJ)^3$, and the states $\mathrm{ND}^{(a)}_{k,J}$ of Definition 2.1.*

Curvature sector (no hypothesis). *Every increasing sequence of spacings has a continuum subsequence. For every continuum subsequence $\mathcal{K}$ and every finite family of real 2-forms,*

$$Z_\infty^{\mathcal{K}}\Big(\sum_j z_j f_j, 0\Big) = \lim_{J\to\infty}\lim_{m\to\infty} \Big\langle e^{\sum_j z_j F_{\Lambda_{k_m}}(f_j)} \Big\rangle^{(a)}_{k_m,J}$$

exists locally uniformly in $z \in \mathbb{C}^N$, is entire, and depends on the family only through the total source. Its Schwinger functions $S_n^{\mathcal{K}}(f_1,\dots,f_n) := \partial_{z_1}\cdots\partial_{z_n} Z_\infty^{\mathcal{K}}(\sum_j z_j f_j, 0)|_{z=0}$ satisfy:

(i) (OS0.) $|Z_\infty^{\mathcal{K}}(\sum_j z_j f_j, 0)| \le e^{\|g(z)\|_2^2}$ *(no dependence on μ_0, e, λ, m) and* $\le \exp(\frac{1}{2}\mu_0^{-2}\|\sum_j (\operatorname{Re} z_j) d^* f_j\|_2^2)$; $|S_n^{\mathcal{K}}| \le (\sqrt{2}e)^n (n!)^{1/2} \prod_j \|f_j\|_2 \le (e\pi/2)^n (n!)^{1/2} \prod_j |f_j|_4$; *the $S_n^{\mathcal{K}}$ are tempered distributions, and* E0″ *holds with $s = 4$ and every $\beta > \frac{1}{2}$.*

(ii) (OS1.) *$S_n^{\mathcal{K}}$ is invariant under all translations and under O_h about every point, F transforming as a 2-form.*

(iii) (OS2.) *Reflection positivity about every coordinate plane $\{x_\nu = c\}$: for exponentials $E_a = e^{\sum_l z_{al} F(f_{al})}$ with arbitrary complex z_{al} and f_{al} supported in $\{x_\nu > c\}$, $(S(\overline{\theta_c E_a} E_b))_{ab}$ is positive semidefinite, and $S(\theta_c(P)P) \ge 0$ for polynomials P in such fields.*

(iv) (OS3.) *$S_n^{\mathcal{K}}$ is real, symmetric, and zero for odd n.*

(v) (The measure.) *For every even $r \ge 14$ there is exactly one Borel probability measure $\mu_F^{\mathcal{K}}$ on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ whose moments on C_c^∞ are the $S_n^{\mathcal{K}}$; its characteristic functional is $Z_\infty^{\mathcal{K}}(\mathrm{i}\,\cdot, 0)$ extended by continuity to $\mathcal{S}$, and its image on H^{-r}_{loc} is the unique measure there with this characteristic functional and the unique one with these moments. For every $\chi \in C_c^\infty$ and box Q containing* $\operatorname{supp}\chi$ *in its interior, the laws of $\chi F_{\Lambda_{k_m}}$ under $\mathrm{ND}^{(a)}_{k_m,J}$ converge weakly on $H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$,* first as $m \to \infty$ along $\mathcal{K}$ *at every fixed $J \ge J_0(\chi)$ and then as $J \to \infty$, to the law of $\chi\Phi$ under $\mu_F^{\mathcal{K}}$.*

(vi) (Weak OS4.) *$\mu_F^{\mathcal{K}}$ is $\mathbb{R}^3 \rtimes O_h$-invariant and satisfies the clauses of Theorem 11.1(b) with $\mu_F^{\mathcal{K}}$ in place of μ_S^K: it is an integral of translation-ergodic, reflection-positive, θ_j-invariant components clustering in the mean and under diagonal separation, with a unique joint vacuum*

per coordinate direction and the component growth bound; almost every component satisfies OS4 *in the weak form; the O_h-invariant decomposition into mixtures of $k \in \{1,2,3,6\}$ ergodic measures exists.*

(vii) (Euclidean clause.) *$Z^{\mathcal{K}}_\infty$ does not depend on $\mathcal{K}$, equals the limit state Z^{LIM}_∞ of Definition 14.3, is $\mathbb{R}^3 \rtimes O(3)$-invariant for all complex coefficients (Theorem 3.14), and the Euclidean forms of (ii) (full $\mathbb{R}^3 \rtimes O(3)$), (iii) (reflection positivity about every plane) and (vi) (almost every component $\mathbb{R}^3 \rtimes O(3)$-invariant, reflection positive about every plane, clustering in any manner, with a unique vacuum in every direction; the vanishing of a component's odd moments is not claimed) hold.*

Wick sector, Form (B). *Let the couplings satisfy* (T3) *and let the scheme a have the cell bound* ($\mathrm{B}_{\mathrm{cell}}$) *and the scheme bound* (SCH) *(Lemmas 14.4 and 14.5; every admissible scheme does). Every increasing sequence of spacings has a continuum subsequence for the Wick sector, and for every such $\mathcal{K}$ and family the LIM state $Z^{a,\mathcal{K}}_\infty$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$, is holomorphic there, obeys (32), and (i)–(vi) hold for the joint Schwinger functions $S^{\mathcal{K}}_{n,p}(f;h) = \partial_z^{(n)}\partial_w^{(p)} Z^{a,\mathcal{K}}_\infty(0)$ with: in (i) the mixed bounds $|S^{\mathcal{K}}_{n,p}| \le e^{K'\sigma_0}(\sqrt{2}e)^n (n!)^{1/2}(e/\sigma_0)^p\, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w$, $\sigma_0 = H/4$, $K' = \kappa\pi^2 + K_-$, and* E0″ *with $s = 4$ and every $\beta > 1$; in (ii) W^a a scalar, the identity holding on the tube; in (iii) exponentials with arbitrary complex curvature coefficients and Wick coefficient vectors $\zeta_1, \dots, \zeta_N$ of* amplitude $\mathsf{a}(\zeta_i) < H/2$ *(in particular all purely imaginary Wick coefficients), and the polynomial form; in (iv) $S^{\mathcal{K}}_{n,p} = 0$ for odd n; in (v) the measures on $\mathcal{H}^{-r}(\mathbb{R}^3; \Lambda^2 \oplus \mathbb{R})$ and H^{-r}_{loc}, the localised lattice fields $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ converging along $\mathcal{K}$ and then in J; (vi) unchanged; and* no Euclidean clause. Schemes: *if a, a' both satisfy the hypotheses, every continuum subsequence has a further subsequence $\mathcal{K}'$ along which $a'_k - a_k \to \kappa^{\mathcal{K}'}$, and $Z^{a',\mathcal{K}'}_\infty(\zeta) = e^{-\kappa^{\mathcal{K}'}\sum_j w_j \int h_j} Z^{a,\mathcal{K}'}_\infty(\zeta)$: the two LIM states differ by the shift $W \mapsto W - \kappa^{\mathcal{K}'}\int(\cdot)$ of the Wick component, and joint cumulants of order ≥ 2 are scheme-free.*

Proof. The complete proof is given in the full version; in summary, each clause is the corresponding proof for the iterated limit (Theorems 7.4, 8.5, 10.6, 10.7 and 11.1 and Propositions 8.1 and 10.9) with exactly one of three replacements: (α) a fixed-(k, J) identity or inequality between Neumann–Dirichlet functionals of explicit finite families is passed along $\mathcal{K}$ (both sides converge by Lemma 14.7, which covers *every* family); (β) Montel on a k-uniform bound, the identification of the limit by convergence at real points being replaced by the choice of $\mathcal{K}$; (γ) a step about the limit object only, unchanged. The only uniqueness used is that of the monotone cube limit along $\mathcal{K}$ (Lemma 14.8).

(i). Lemmas 14.7 and 14.8 give existence, entireness and the first bound; the second, $\exp(\frac{1}{2}\mu_0^{-2}\|d^*_k g\|^2_{2,k})$ at every (k, J) (Theorem 7.4(a)), passes along $\mathcal{K}$ and J with $\|d^*_k g\|_{2,k} \to \|d^* g\|_2$. The Schwinger bounds, the extension to $\mathcal{S}$, the kernel theorem and E0″ are the arguments of Theorem 7.4 (Wick sector: Theorem 13.10(ii), on the polydisc inside the tube) run on the limit function with its bound (γ).

(ii), translations. For $b \in 5^{-k_0}\mathbb{Z}^3$, $|b|_\infty \le M$, $k \ge k_0$: $Z^a_{k,C_J}(\tau_b \cdot \zeta) = Z^a_{k,C_J - b}(\zeta)$ (exact lattice covariance) and $C_{J-1} \cap \Lambda_k \subset (C_J - b) \cap \Lambda_k \subset C_{J+1} \cap \Lambda_k$, so for a cone point s supported in U_{J_0} and $J \ge J_0 + 1$, $Z^a_{k,C_{J-1}}(s) \le Z^a_{k,C_J}(\tau_b \cdot s) \le Z^a_{k,C_{J+1}}(s)$ (first inequality of Lemma 2.11, twice). The three terms are functionals of the families s and $\tau_b \cdot s$ (supports in Q_{R+M}, the same tube), so they converge along $\mathcal{K}$ (α), and Lemma 14.8 sends both ends to $Z^{\mathcal{K}}_\infty(s)$: $Z^{\mathcal{K}}_\infty(\tau_b \cdot s) = Z^{\mathcal{K}}_\infty(s)$ at cone points, with no uniqueness of the k-limit. All coefficients (γ): the difference of the two limit functions is entire and vanishes on $(\mathrm{i}\mathbb{R})^N$ (curvature), resp. for the split family is holomorphic on the connected $\mathcal{D}''$ and vanishes on Ω_+, so the identity theorem, the diagonal (4) of Lemma 14.8 and the identity theorem on the connected tube give it on $\mathcal{D}_{\sigma_+,H}$. All $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$ by iteration; all $b \in \mathbb{R}^3$ at imaginary coefficients because the Lipschitz bounds of Lemma 14.7, Step 3, pass to the limits ($|Z^{\mathcal{K}}_\infty(\mathrm{i}u, 0) - Z^{\mathcal{K}}_\infty(\mathrm{i}u', 0)| \le \sqrt{2}\|u - u'\|_2$; $|Z^{a,\mathcal{K}}_\infty(\mathrm{i}\sigma) - Z^{a,\mathcal{K}}_\infty(\mathrm{i}\sigma')| \le C(\|g_{\sigma-\sigma'}\|_2 + |h_{\sigma-\sigma'}|_w)$) and $\tau_{b_n} s \to \tau_{b_0} s$ in L^2 and in $|\cdot|_w$ for $s \in C^\infty_c$; complex coefficients by the identity theorem.

(ii), the cubic group. Coordinate permutations are exact at every (k, J) (Lemma 2.13); pass along $\mathcal{K}$ and J (α). For the reflection σ_3 and a cone point (f, g) supported in U_{J-1}, at every k, $Z^a_{k,C_{J-1}}(\mathrm{i}f, g) \le Z^a_{k,C_J}(\mathrm{i}\sigma_3 \cdot f, \sigma_3 \cdot g) \le Z^a_{k,C_{J+1}}(\mathrm{i}f, g)$ (the frame flip of Lemma 2.13 with the one-site shift $X_+ = (C_J + \varepsilon_k e_3) \cap \Lambda_k$ and two deletions in the cone direction, Lemma 2.8(a),(b)); the middle term is the functional of the reflected family (same tube), so (α) and Lemma 14.8 give $Z^{\mathcal{K}}_\infty(\mathrm{i}\sigma_3 \cdot f, \sigma_3 \cdot g) = Z^{\mathcal{K}}_\infty(\mathrm{i}f, g)$ on the cone (all of it at $g = 0$; in Form (B) at cone points with $\|g\|_\infty < \sigma_+/2$): the reflections are not exact lattice symmetries; the sandwich replaces exactness and survives along $\mathcal{K}$. Conjugation $\sigma_\nu = P\sigma_3 P^{-1}$, group closure on the cone, the identity theorem off the cone (on $\mathbb{C}^N$ with $(\mathrm{i}\mathbb{R})^N$; on $\mathcal{D}''$ with Ω_+), invariance about every point by the translations, and differentiation are (γ), as in Theorem 8.5.

(iii), reflection positivity. Fix ν, a mid-plane $c \in D$ (Definition 9.1) and the enlarged family $\mathcal{F}$ of Definition 10.1 with the modified-box functionals $\Phi_k(\zeta) = Z^{a,(3.38)}_{k,\Omega_{J,k}}(\zeta)$, $\zeta \in \mathbb{C}^m$, and support radius $R_{\mathcal{F}}$. The fixed-k steps (reflection positivity of the modified box, Lemma 9.3; the sandwich lemmas Lemmas 9.9 and 9.10) concern the photon form and the inequalities of Lemma 2.8 only and are unchanged. *Curvature sector.* At $h = 0$, $|\Phi_k(z)| \le e^{\|f_{\mathrm{Re}\, z}\|^2_{2,k}}$ by Lemma 2.9(a) and the modified photon form, uniformly in k and J. (β) Extract from $\mathcal{K}$ a further $\mathcal{K}'$ along which $\Phi_k \to \Phi^J$ locally uniformly on $\mathbb{C}^m$; positive semidefiniteness of the Gram matrices at every $k \ge k_0(c, \delta)$ passes to Φ^J. (α) On U (imaginary curvature coefficients) $a^{(a)}_{k,C_J}(\zeta) \le \Phi_k(\zeta) \le a^{(a)}_{k,C_{J+2n}}(\zeta)$ at every k (Lemmas 9.9 and 9.10, cone halves); along $\mathcal{K}'$ the ends converge to $Z^{\mathcal{K}}_J, Z^{\mathcal{K}}_{J+2n}$, so $Z^{\mathcal{K}}_J \le \Phi^J \le Z^{\mathcal{K}}_{J+2n}$ on U, and Lemma 14.8 sends both ends to $Z^{\mathcal{K}}_\infty$. $\{\Phi^J\}_J$ is locally bounded on $\mathbb{C}^m$ (J-free bound) and U is a set of uniqueness, so $\Phi^J \to Z^{\mathcal{K}}_\infty$ locally uniformly, in particular at the matrix points, where $\Phi^J(\zeta^*_{ij}) \to Z^{\mathcal{K}}_\infty(\theta\zeta^*_i \sqcup \zeta_j)$ (dependence through the total source only); positivity passes. $\mathcal{K}'$ may depend on J; this is harmless, since the sandwich ends converge along all of $\mathcal{K}$ and the bound on Φ^J is J-free, so neither $\lim_J \Phi^J$ nor the positivity depends on the choice. Every plane: translate by $(c' - c)e_\nu$ and use (ii). The polynomial form: Weierstrass along $\mathcal{K}'$ and J (Proposition 10.10). *Wick sector, amplitude $< H/2$, without Theorem 3.6.* Let $\zeta_1, \dots, \zeta_N$ be families of real sources supported in $\{x_\nu > c\}$ with $\mathsf{a}(\zeta_i) < H/2$; the (i, j) Gram entry is $\Phi_k(\zeta^*_{ij})$, with total scalar source $\sum_l \bar{w}_{il}\, h_{il} \circ \rho + \sum_l w_{jl} h_{jl}$. The bound of Lemma 10.3, which depends on Theorem 3.6, is replaced by Lemma 14.10 below, after which the argument is completed.

(iv). Symmetry: Taylor coefficients of a function of the total source. Realness: real at real points at every (k, J), closed under limits (Form (B): at the real points of the tube). Charge conjugation gives $Z^a_{k,J}(-z, w) = Z^a_{k,J}(z, w)$ at every (k, J), hence for $Z^{\mathcal{K}}_J$ and $Z^{\mathcal{K}}_\infty$ (the tube is invariant under $z \mapsto -z$); odd z-derivatives at 0 vanish.

(v), (vi). The moment bounds at (k, J) (Theorem 7.4(a), resp. (31), with Cauchy's estimates) are uniform in k and J above the thresholds, and the lattice Schwinger functions converge along $\mathcal{K}$ then J (Weierstrass). The measure is built from the limit functional (γ): $\mathcal{Z}^{\mathcal{K}}(f) := Z^{\mathcal{K}}_\infty(\mathrm{i}f)$ on real $f \in C^\infty_c$ is normalised, positive definite (at every (k, J) the matrix $(Z^a_{k,J}(\mathrm{i}(f_m - f_l)))_{lm}$ is a Gram matrix; the entries converge along $\mathcal{K}$ then J) and Lipschitz for $\sqrt{2}\|\cdot\|_2$, resp. $C(\|\cdot\|_2 + |\cdot|_w)$, hence continuous on $\mathcal{S}$; Bochner–Minlos, moments by Lukacs [18] and uniform integrability, the Hermite carrier on $\mathcal{H}^{-r}$, $r \ge 14$ even, and the two uniqueness statements are the steps of Theorem 11.1(a) with Lemmas P.4 and P.7 and Proposition P.8 of the full version. *Local laws along $\mathcal{K}$* ((β); the one weakened clause): tightness of the laws of $\chi\Phi^a_{k,J}$ on $H^{-r}_{\mathrm{per}}(Q; \mathcal{V})$ is uniform in k and $J \ge J(\chi)$ from the second-moment bounds (Prokhorov [8]); the characteristic functionals $Z^a_{k,J}(\mathrm{i}\chi\psi)$ converge *along* $\mathcal{K}$ to $Z^{a,\mathcal{K}}_J(\mathrm{i}\chi\psi)$; a probability measure on H^{-r}_{per} is determined by its characteristic functional on trigonometric polynomials; so the laws converge along $\mathcal{K}$, then in J (Lemma 14.8), to the law of $\chi\Phi$ under the measure just built (the periodisation identity of Theorem 11.1). (vi): $\mathbb{R}^3 \rtimes O_h$-invariance of the measure is (ii) at imaginary coefficients plus uniqueness of the characteristic functional; (H3) of the full version is (iii) at purely imaginary Wick coefficients ($\mathsf{a} = 0 < H/2$) with the density of the exponential algebra (Lemma P.6 of the full version) and Proposition P.9(c) of the full version;

then Theorems M.6, M.9, M.12, N.8 and N.10, Proposition N.5, and Corollaries M.13 and N.12 of the full version and Definition N.2(iii) of the full version, stated for a measure satisfying (H1)–(H3), apply.

Schemes. $\sup_k |a'_k - a_k| \le \sup_k |a'_k - z_k| + \sup_k |z_k - a_k| < \infty$ by (SCH) for both. For a continuum subsequence $\mathcal{K}$ for a, Bolzano–Weierstrass gives $\mathcal{K}' \subset \mathcal{K}$ with $a'_k - a_k \to \kappa^{\mathcal{K}'} \in \mathbb{R}$. At every (k, J), $Z^{a'}_{k,J}(\zeta) = e^{-\sum_j w_j (a'_k - a_k) \mathsf{S}_{\Lambda_k}(h_j)} Z^{a}_{k,J}(\zeta)$ (Remark 2.5) with $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$; so along $\mathcal{K}'$ the a'-functionals converge wherever the a-functionals do, $\mathcal{K}'$ is a continuum subsequence for a' with $Z^{a',\mathcal{K}'}_J = e^{-\kappa^{\mathcal{K}'} \sum_j w_j \int h_j} Z^{a,\mathcal{K}'}_J$, and then $J \to \infty$. (For admissible a, a', $a'_k - a_k$ converges along the full sequence and $\kappa^{\mathcal{K}'} = \kappa$.) The cumulant statement is that of Theorem 21.1. ($\mathrm{B}_{\mathrm{cell}}$) transfers between schemes with bounded difference: $B^{a'}_{\mathrm{cell}} \le B^{a}_{\mathrm{cell}} + \sup_k |a'_k - a_k| \, \|2H' \chi_0\|_{1,k}$.

(vii). By Proposition 14.12(a), $Z^{\mathcal{K}}_\infty = Z^{\mathrm{LIM}}_\infty$, which is $\mathbb{R}^3 \rtimes O(3)$-invariant for all $z \in \mathbb{C}^N$ by Theorem 3.14. (ii) in Euclidean form follows by differentiation; for (iii), a plane P is $E\{x_\nu = c\}$ for a Euclidean motion E, $\theta_P = E\theta_{\nu,c}E^{-1}$, and $Z^{\mathcal{K}}_\infty(\theta_P \zeta^*_i \sqcup \zeta_j) = Z^{\mathcal{K}}_\infty(\theta_{\nu,c}(E^{-1}\zeta_i)^* \sqcup E^{-1}\zeta_j)$ by invariance at complex coefficients, the $E^{-1}\zeta_i$ being supported in $\{x_\nu > c\}$; (vi) is Corollary O.7 of the full version, with the admissible push-forward action of Lemma O.2(i) of the full version and the $O(3)$-invariance of the measure, which by Lemma O.2(ii) of the full version is that of $\mathcal{Z}^{\mathcal{K}}$, as in Theorem 11.1(c), with its proviso on the odd moments of a component. □

Lemma 14.10 (a k- and J-uniform bound for the modified-box functionals on the tube of the enlarged family)**.** *Under* (T3), ($\mathrm{B}_{\mathrm{cell}}$), (SCH), *for every k, every $J \ge \max(J_0, J_1(R_{\mathcal{F}}))$ and every $\zeta \in \mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$ (the tube of $\mathcal{F}$ in the sense of Definition 2.6),*

$$|\Phi_k(\zeta)| \le \exp\big(2\|g(\zeta)\|^2_{2,k} + \kappa \|k_+(\zeta)\|_{1,k} + K_- |k_-(\zeta)|_w\big).$$

The curvature constant is 2, not the 1 of (31), because the Cauchy–Schwarz split below doubles the curvature source and Theorem 7.4(a) is quadratic in it; nothing depends on its value.

Proof. $|\Phi_k(\zeta)| \le \Phi_k(\xi)$, $\xi := \mathrm{Re}\,\zeta$ (Lemma 2.12(b)). Let $k(\xi)$ be the total real scalar source at ξ and $k_1 := k(\xi)_+$, $k_2 := k(\xi)_-$ its *pointwise* parts, sampled on the lattice. Cauchy–Schwarz at the lattice level (Lemma 2.12(d); the multiplier $e^{-a_k \mathsf{S}(k_1 - k_2)}$ splits by linearity of S): $\Phi_k(\xi) \le Z^{a,(3.38)}(2f_\xi, -2k_2)^{1/2} Z^{a,(3.38)}(0, 2k_1)^{1/2}$. *First factor* (class $\mathcal{D}$): the $\mathcal{D}$-half of Lemma 9.9, which needs only that the sources vanish off X_J ($J \ge J_0$), gives $\le Z^a_{k,C_J}(2f_\xi, -2k_2)$; Theorem 7.4(a) with the real scalar source $-2k_2$ gives $\le e^{\|2f_\xi\|^2_{2,k}} N^a_{k,C_J}(0, -2k_2)$ (the same scheme multiplier on both sides); Proposition 13.9 with $q := 2k_2 \ge 0$, supported in $Q_{R_{\mathcal{F}}}$, $J \ge J_1(R_{\mathcal{F}})$ and $\sup q = 2\|k(\xi)_-\|_\infty < H$ (this is $\zeta \in \mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$; the proposition allows $\sup q \le H$) gives $\ln N^a_{k,C_J}(0, -2k_2) \le K_- |2k_2|_w$. Hence $\frac{1}{2}\ln(\text{first factor}) \le 2\|g(\zeta)\|^2_{2,k} + K_- |k_-(\zeta)|_w$, as $f_\xi = g(\zeta)$ and $k_2 = k_-(\zeta)$. *Second factor* (class $\mathcal{I}$): the $\mathcal{I}$-half of Lemma 9.10 with $B = C_{J+2n} \cap \Lambda_k$ gives $\le Z^a_{k,C_{J+2n}}(0, 2k_1)$, and Corollary 13.3, uniform in k and in the cube, with $0 \le 2k_1 \le s_1$ on the lattice ($\|k_+(\zeta)\|_\infty < \sigma_+/2$, sampled $\le$ continuum) gives $\ln Z^a_{k,C_{J+2n}}(0, 2k_1) \le \kappa \|2k_1\|_{1,k}$, so $\frac{1}{2}\ln(\text{second factor}) \le \kappa \|k_+(\zeta)\|_{1,k}$. Add. On compacts of the tube the right side is bounded uniformly in k (Lemma 2.7). □

The lemma is (31) (with curvature constant 2) for the modified-box functional, transported through Lemmas 9.9 and 9.10; the two factors live on C_J and C_{J+2n}, which is harmless as Proposition 13.9 and Corollary 13.3 are uniform in the cube, and the amplitude $H/2$ is exactly the tube condition $\|k(\xi)_-\|_\infty \le H/2$ on the total source. *The matrix points lie in the tube:* at $\xi^*_{ij} = \mathrm{Re}\,\zeta^*_{ij}$ the two sums of the total scalar source have disjoint supports, so $\|k(\xi^*)_\pm\|_\infty \le \max(\mathsf{a}(\zeta_i), \mathsf{a}(\zeta_j)) < H/2 \le \sigma_+/2$, and by continuity a neighbourhood of the finitely many matrix points lies in $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$. *End of the proof of (iii), Wick sector.* (β) By Lemma 14.10, $(\Phi_k)_k$ is locally bounded on the open connected $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$; extract $\mathcal{K}' \subset \mathcal{K}$ with $\Phi_k \to \Phi^J$ locally uniformly there; Gram positivity passes ($\mathcal{K}'$ may depend on J; harmless as in the curvature sector). (α) Let U be the set of coefficient vectors with purely imaginary curvature coefficients and positive real scalar coefficients on the nonnegative $\tilde{h}_j$; $U \cap \mathcal{D}^{\mathcal{F}} \ne \emptyset$ ($k_- = 0$, k_+ small), and it is a relatively open subset of the maximal totally real $(\mathrm{i}\mathbb{R})^{m_F} \times \mathbb{R}^{m_h}$, a set of uniqueness in $\mathcal{D}^{\mathcal{F}}$.

On U, $Z^a_{k,C_J}(\zeta) \le \Phi_k(\zeta) \le Z^a_{k,C_{J+2n}}(\zeta)$ at every k (Lemmas 9.9 and 9.10, $\mathcal{I}$-halves); the ends are functionals of $\mathcal{F}$ at points of its tube, converging along $\mathcal{K}'$ (Lemma 14.7(b), J above the threshold of $\mathcal{F}$) to $Z^{a,\mathcal{K}}_J, Z^{a,\mathcal{K}}_{J+2n}$, which tend to $Z^{a,\mathcal{K}}_\infty$ (Lemma 14.8, locally uniformly on the tube of $\mathcal{F}$, in particular on $U \cap \mathcal{D}^{\mathcal{F}}$). $\{\Phi^J\}_J$ is locally bounded on $\mathcal{D}^{\mathcal{F}}$ (J-free), so $\Phi^J \to Z^{a,\mathcal{K}}_\infty$ locally uniformly there, in particular at the matrix points, where the limit is $Z^{a,\mathcal{K}}_\infty(\theta\zeta_i^* \sqcup \zeta_j)$; positivity passes. Every plane by (ii) (the amplitude is translation invariant); the polynomial form by Weierstrass.

Remark 14.11 (amplitudes in $[H/2, \sigma_+/8)$: the route through the fixed-cube limits, for a further subsequence). The proof of Lemma 10.3 bounds the first factor on the Wick tube $T_{\mathcal{F}'}$ of Definition 10.1 by the vertex argument at the cube C_{J_0}, which needs the fixed-cube limits in their boundedness form only. For amplitudes in $[H/2, \sigma_+/8)$ the matrix points lie in $T_{\mathcal{F}'}$ but in general outside $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$, the only domain on which Definition 14.1 gives convergence along $\mathcal{K}$; what follows is therefore stated here only for a further subsequence. Assume the boundedness in k of the functionals of the class-$\mathcal{D}$ families $(2f_\xi, -2k_2(\xi))$ at the one cube C_{J_0} (with the coefficient-sign split of Definition 10.1). The vertex argument then gives local boundedness of $(Z^a_{k,J})_k$ and $(\Phi_k)_k$ on $T_{\mathcal{F}'}$ (first factors at C_{J_0} by that boundedness, second factors by Corollary 13.3); Montel, with a diagonal over the countably many J, yields a further subsequence $\mathcal{K}' \subset \mathcal{K}$, depending on the family, along which $Z^a_{k,J} \to Z^{a,\mathcal{K}'}_J$ and $\Phi_k \to \Phi^J$ locally uniformly on $T_{\mathcal{F}'}$ (on $T_{\mathcal{F}'} \cap \mathcal{D}^{\mathcal{F}}$ the first limit is $Z^{a,\mathcal{K}}_J$). The sandwich step runs on $T_{\mathcal{F}'}$ with the uniqueness set $U \cap T_{\mathcal{F}'}$, the J-free local bound of Proposition 10.5(a) on $T_{\mathcal{F}'}$ serving the Vitali step in J, and Proposition 10.5(b) along $\mathcal{K}'$ for the remaining planes; so OS2 holds up to $\mathsf{a}(\zeta_i) < \sigma_W = \sigma_+/8$ for the LIM state of such a $\mathcal{K}'$ on $T_{\mathcal{F}'}$. Whether the convergence holds along $\mathcal{K}$ itself on $T_{\mathcal{F}'}$ is not claimed here; under the hypotheses of Form (A), Theorem 23.7 gives OS2 along $\mathcal{K}$ with no amplitude restriction. The remark is used nowhere else.

14.4. **Identification, and King's printed state.**

Proposition 14.12 (identification)**.** (a) *For every continuum subsequence $\mathcal{K}$ and every family, $Z^{\mathcal{K}}_\infty(\cdot,0) = Z^{\mathrm{LIM}}_\infty(\cdot,0)$, the limit state of the rotation theorem (Theorem 3.14); in particular all LIM states coincide,* (UNIQ) *holds, and they are $\mathbb{R}^3 \rtimes O(3)$-invariant. Theorem 3.14 does* not *give Theorem 3.3.*

(b) *The following are equivalent: (i) the comparison family is convergent at fixed cubes (Definition 3.1), that is the existence in $(0,\infty)$ of the fixed-cube limits of the comparison family at real coefficients; (ii) for every J and family, $a^{(a)}_{k,J} \to Z_J$ locally uniformly on $\mathbb{C}^N$ along the full sequence; (iii) every increasing sequence of spacings is a continuum subsequence. Under them $Z^{\mathcal{K}}_J = Z_J$ and $Z^{\mathcal{K}}_\infty = Z_\infty := \lim_J Z_J$ for every $\mathcal{K}$: every LIM state is the iterated limit of the comparison family.*

(c) *Assume* (T3) *and let a be admissible. Then* (SCH) *and* ($\mathrm{B}_{\mathrm{cell}}$) *hold (Lemmas 14.4 and 14.5). If moreover the comparison family is convergent at fixed cubes on the tube (Definition 3.2), every increasing sequence of spacings is a continuum subsequence for the Wick sector, and for every $\mathcal{K}$, $Z^{a,\mathcal{K}}_\infty = G_\infty$ of Theorem 13.10 on $\mathcal{D}_{\sigma_+,H}$, which equals $Z^a_{\infty,T}$ of Proposition 10.5 on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ (Proposition 10.5 lives on the Wick tube $T_{\mathcal{F}'}$, a different domain). Once* (SCH) *and* ($\mathrm{B}_{\mathrm{cell}}$) *are given, the argument uses Definition 3.2 for a alone; on King's convention, where Theorem 3.6 supplies the limits, the same argument is Step 7 of Theorem 21.3.*

Proof. (a) Let $\mathcal{K}$ be a continuum subsequence, f a family, $J \ge J_0(f)$. By Definition 14.1, $Z^{\mathcal{K}}_J(\cdot,0)$ is the locally uniform limit of $(a^{(a)}_{k_m,J})_m$, so $Z^{\mathcal{K}}_J(\cdot,0) \in L^f_J$. By Theorem 3.14 every choice $\zeta_J \in L^f_J$, $J \ge J_0(f)$, converges locally uniformly to $Z^{\mathrm{LIM}}_\infty(\cdot,0)$; take $\zeta_J := Z^{\mathcal{K}}_J(\cdot,0)$, which by Lemma 14.8 converges to $Z^{\mathcal{K}}_\infty(\cdot,0)$, and limits are unique. Hence any two continuum subsequences have the same LIM state, and invariance is the second clause of Theorem 3.14. That theorem pins the J-limit of the subsequential limits, not the fixed-cube limits themselves, so it does not give Theorem 3.3.

(b) (i)⇒(ii): $(a^{(a)}_{k,J})_k$ is locally bounded on $\mathbb{C}^N$ (Lemma 7.3) and converges at every real point; $\mathbb{R}^N$ is a set of uniqueness for entire functions, and Montel–Vitali gives locally uniform convergence of the whole sequence to an entire Z_J. (ii)⇒(iii): every subsequence converges to the same limit. (iii)⇒(i): the full sequence is itself an increasing sequence, hence a continuum subsequence, so $a^{(a)}_{k,J}$ converges locally uniformly, and at real x the limit lies in $[1,\infty) \subset (0,\infty)$ (Lemma 14.7, Step 4). The last clause is Lemma 14.8.

(c) (SCH) and ($\mathrm{B}_{\mathrm{cell}}$) are Lemmas 14.4 and 14.5; so (31) holds. $(Z^a_{k,J})_k$ is locally bounded on the tube by (31) and converges at the real points of the tube by Definition 3.2, a non-empty open subset of $\mathbb{R}^{n+p}$ and a set of uniqueness in the connected tube; Montel–Vitali: the whole sequence converges locally uniformly on the tube (Theorem 13.10(i)), so every increasing sequence is a continuum subsequence for the Wick sector with $Z^{a,\mathcal{K}}_J = G_J$, and $Z^{a,\mathcal{K}}_\infty = G_\infty$ by Lemma 14.8; on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ this is the same J-limit of the same G_J as $Z^a_{\infty,T}$ (Proposition 10.5(a)). □

Corollary 14.13 (the iterated limits of the comparison family)**.** *Assume that the comparison family is convergent at fixed cubes (Definition 3.1). Then every increasing sequence of spacings is a continuum subsequence, all LIM states coincide with the iterated limit* $Z_\infty(f,0) = \lim_J \lim_k \langle e^{F(f)}\rangle^{(a)}_{k,J}$ *of the comparison family, and the curvature sector (i)–(vi) of Theorem 14.9 holds for it with the lattice laws of (v) converging along the* full *sequence of spacings; (vii) holds as well. For general sources assume* (T3) *and let a be admissible; then every increasing sequence is a continuum subsequence for the Wick sector, every LIM state is* G_∞ *(*$= Z^a_{\infty,T}$ *on* $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$*),* $\kappa^{\mathcal{K}} = \kappa$ *is independent of* $\mathcal{K}$*, and Form (B) of Theorem 14.9 holds for* G_∞ *with the full-sequence lattice clause.*

Proof. Curvature sector: Proposition 14.12(b) makes every increasing sequence a continuum subsequence with $Z^{\mathcal{K}}_\infty = Z_\infty$; (i)–(vi) are those of Theorem 14.9, and in (v) the local laws converge along the full sequence because the characteristic functionals now do (step (β) with the full sequence); (vii) is Proposition 14.12(a). Wick sector: Proposition 14.12(c); the clauses are Form (B) of Theorem 14.9 for G_∞, with OS2 at amplitude $< H/2$ by the route of Lemma 14.10, which does not use Theorem 3.6, and at amplitude $< \sigma_W = \sigma_+/8$ by the route of Theorem 10.7; both hold ($H/2 = \sigma_+/32 < \sigma_+/8$: the route without the fixed-cube limits gives the smaller amplitude, and what it improves is the list of inputs); $\kappa^{\mathcal{K}} = \kappa$ by the scheme clause. □

Theorem 3.6 is the scheme-a analogue, with scalar sources, of the existence clause of King's theorem; it is not that theorem, which is false as printed with scalar sources in King's Wick ordering. With (W2) in addition one obtains Form (A) for the full object Z^a_∞ and, under (R_W), its Euclidean clause; this is the optional extension of Theorem 23.7.

Corollary 14.14 (King's convention)**.** *Define continuum subsequences and LIM states of the K-family* $a^K_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ *(Definition 2.2) by Definitions 14.1 and 14.2 with* $\langle\cdot\rangle^K_{k,J}$ *in place of* $\langle\cdot\rangle^{(a)}_{k,J}$*.*

(a) (No hypothesis.) *Every increasing sequence of spacings has a continuum subsequence of the K-family; for every such* $\mathcal{K}$*, family and* $J \geq J_0(f)$ *the fixed-cube limits* $Z^{K,\mathcal{K}}_J$ *and the LIM state* $Z^{K,\mathcal{K}}_\infty = \lim_J Z^{K,\mathcal{K}}_J$ *exist, are entire and obey* $|Z^{K,\mathcal{K}}_J|, |Z^{K,\mathcal{K}}_\infty| \leq e^{\|g(z)\|^2_2}$*, with the other clauses of Lemma 14.7(a) and Lemma 14.8. Nothing beyond this is asserted for a K-family LIM state from the bounds alone.*

(b) (With Theorem 3.3.) *Every increasing sequence of spacings is a continuum subsequence of the K-family, every K-family LIM state is King's printed state* Z^K_∞*, every LIM state of the comparison family is* Z^K_∞ *as well (Corollary 4.4(iii)), and the curvature-sector axioms of Theorem 21.1 hold for it, with the lattice laws of Theorem 11.1(a) converging along the full sequence.*

With scalar sources, the corresponding statements on King's convention are supplied by Theorem 21.3 and Proposition 21.4.

Proof. (a) The proof of Lemma 14.7(a) uses only the k- and J-uniform local bound of Lemma 7.3 (Step 2), the second-order bound $\langle F(u)^2\rangle \leq 2\|u\|^2_{2,k}$ of Lemma 7.1 (Step 3), and the current

representation and Jensen's inequality (Step 4); that of Lemma 14.8 uses in addition the first inequality of Lemma 2.11. All of these hold on King's convention: the bounds by Theorem 7.6 (the second-order bound is photon-only and transfers by the covariance identity of Lemma 4.2), the cut inequality by Corollary 4.3. So the two lemmas hold for the K-family by the same proofs; the argument makes no use of Theorem 3.3. (b) By Theorem 3.3, Corollary 4.4(ii)(α) gives, for every $J \geq J_0$ and family, locally uniform convergence of $a^K_{k,J}$ along the full sequence, so every increasing sequence is a continuum subsequence of the K-family, and by (β) every K-family LIM state is Z^K_∞; Corollary 4.4(iii) is the statement for the comparison family; the axioms are the curvature sector of Theorem 21.1, and the full-sequence lattice laws are Theorem 11.1(a). □

Remark 14.15 (on the construction of the subsequences). Every subsequence of spacings used in this paper is constructed in Step 2 of the proof of Lemma 14.7 from the k-uniform bounds of Lemma 7.3 and (31) and Montel's theorem, or is a further extraction from one so constructed, and the identification of its limits uses only the monotone cube limit of Lemma 14.8; no subsequence statement is cited from [16].

Rotation invariance

15. The rotation part: lattices, walled tori and the route

This and the following six sections form the rotation part of the paper. Its object is King's printed state on curvature sources, $z \mapsto Z^K_\infty(\sum_j z_j f_j, 0)$, the entire function of the coefficients of real 2-forms $f_j \in C^\infty_c$ whose derivatives at $z = 0$ are the curvature Schwinger functions (Theorem 3.3 and the curvature-sector section); its result, Theorem 15.1, is that this function is invariant under every Euclidean motion.

Rotation invariance is not automatic, since all lattices in King's construction have the same orientation. Theorem 2.4 of [16] states that on the tori D_J the lattices $\varepsilon_k\mathbb{Z}^3$ and $R(\varepsilon_k\mathbb{Z}^3)$, with R the rotation by $\theta_0 = \arctan(3/4)$ in a coordinate plane, have the same continuum limit; since θ_0/π is irrational these rotations generate a dense subgroup of $SO(3)$. As printed the statement is false in two ways (Proposition H.2 of the full version, Remark 16.1): the source it pairs with the plaquettes is one scalar function on both lattices, which couples the field to two different 2-forms, and the Wick constant it uses misses an order-e^2 logarithm. We therefore call the printed statement *King's torus conjecture*, and we use instead the corrected statement, for 2-form sources sampled in each lattice's own frame and for the two lattices at the same spacing; it is the torus comparison (K_0) of Theorem 17.1, proved in Section 17.3. Moreover, even granting the torus comparison, two steps remain between it and rotation invariance of King's state: the infinite-volume limit along tori, where correlation inequalities give no monotonicity, and its identification with the limit along cubes, which King writes can be expected from a cluster expansion [16, (2.24)]. The ultraviolet analysis of [15, 16] needs boundaries that are lattice hyperplanes [16, p. 326], and no bounded region has that property for both lattices (Proposition 20.21). Where to place the two-lattice comparison is therefore the whole question. The rotation part places it on periodic tori carrying a smooth mass wall, and proves $O(3)$ invariance of King's printed state in the curvature sector from three inputs, each stated in one line and proved where it is stated: the existence theorem (E_C) of Theorem 3.3, the torus comparison (K_0) (Theorem 17.1) and the heavy-shell estimates (HS), (DS) (Theorem 20.2 and Proposition 20.1).

15.1. **The torus route.** On each lattice separately King's cubes are compared with tori carrying a *smooth mass wall*. Fix R in the generating set $\mathcal{R}_0 = \{R_{\mu\nu}(\pm\theta_0)\}$ of Definition 15.3, a pair $(f, g) \in \mathcal{K}$ (a real 2-form f and a function $g \geq 0$, both smooth with compact support) and a scheme a; at $g = 0$ nothing below depends on the scheme. For a walled torus $\mathfrak{c} = (J', w)$, the periodic model on $D_{J'}$ in which the scalar mass squared is raised by $2w$, with $w \geq 0$ smooth and vanishing near the supports (Definition 15.7, Definition 18.1), the chain is

(33)
$$Z^a_{k,L}(\mathrm{i}\,R\cdot f, R\cdot g) \;=\; Z^a_{R^{-1}\Lambda_k,\,R^{-1}X_L}(\mathrm{i}f, g) \;\leq\; T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f,g) \;\xrightarrow[k\to\infty]{}\; \mathcal{W}^a_{\mathfrak{c}}(f,g) \;\xleftarrow[k\to\infty]{}\; T^a_{\mathfrak{c},\Lambda_k}(f,g).$$

Here $Z^a_{k,L}$ is the functional of the comparison cube $X_L = C_L \cap \Lambda_k$ on the axis lattice (Definition 2.4); the first equality is a passive rotation (Lemma 15.10(b)); the inequality is a correlation inequality on the rotated lattice $R^{-1}\Lambda_k$ (Lemma 15.9), for which $R^{-1}C_L$ is a cube with faces on lattice hyperplanes; $T^a_{\mathfrak{c},\Lambda}$ is the walled ratio of Definition 15.7; and the two continuum limits exist and agree by the torus comparison (K_0) (Theorem 17.1; at $g \neq 0$ its scheme form $(\mathrm{K})_a$, Definition 17.2). The comparison cubes on the left are squeezed, as $k \to \infty$ and then $L \to \infty$, to King's printed state by the interlacing of the two cube families (the section on the interlacing). What is missing is an upper bound for walled tori on the axis lattice by King's printed state:

(34) for every $\eta > 0$ there is an admissible $\mathfrak{c}$ with $\mathcal{W}_{\mathfrak{c}}(f,0) \le Z^K_\infty(\mathrm{i}f,0) + \eta$.

This is the closing estimate (TIGHT_0) of Theorem 18.4. It involves one lattice and one orientation, and the reverse inequality holds in the limit (Lemma 18.2), so (TIGHT_0) says that King's printed state is the lower envelope of the walled tori. With it, letting $L \to \infty$ in (33) gives $Z^K_\infty(\mathrm{i}\,R\cdot f,0) \le Z^K_\infty(\mathrm{i}f,0)$, and applying this to $R^{-1} \in \mathcal{R}_0$ gives equality. The passage from the dense subgroup to $O(3)$ uses the Lipschitz bound of Corollary 8.3 at imaginary coefficients and the cubic symmetry of King's printed state (Theorem 18.12), so no regularity of the Schwinger functions beyond that corollary is needed.

The closing estimate (TIGHT_0) does not follow from the correlation inequalities: Remark 18.8 gives a family of Gaussian functionals with all the other properties used in the proof whose analogue of King's state is not rotation invariant. We therefore isolate the local estimate that implies (TIGHT_0). In the walled torus let the sources sit in a box H and let U be a thick shell around H inside the heavy plateau, where the wall is at least 1. Raise the scalar mass squared on the shell further by $2K$, through a second smooth profile, and let $\Psi_k(n,K)$ be the resulting expectation at lattice spacing ε_k, with $\Psi_k(n,\infty)$ the expectation with the matter removed on the shell (Definition 19.3). The decoupling property (RES_0) of Definition 19.8 says that for one such configuration and one finite K

$$\liminf_{k\to\infty} \big[\Psi_k(n,K) - \Psi_k(n,\infty)\big] \le \eta,$$

with the width of the shell chosen so that a photon boundary term is below η as well. At every fixed spacing the bracket tends to 0 as $K \to \infty$ (Proposition 20.20); the content of (RES_0) is that the heavy-shell limit and the continuum limit can be interchanged, at one geometry and with no rate. A two-line sandwich (Lemma 19.7) shows that the heavy shell is bounded below by a higher wall, and once the matter on the shell is dead the photon on the far side can be pinned and then replaced by King's Neumann photon on a larger cube at a cost exponentially small in the width of the shell, uniformly in the spacing (Lemma 19.1); so (RES_0) implies (TIGHT_0) (Theorem 19.10); in the exterior-shell geometry the same chain is (DS) (Proposition 20.1). Under (K_0) the limit $\lim_k \Psi_k(n,K)$ exists for every finite K, and (RES_0) is equivalent to the statement that the monotone limit $\lim_K \lim_k \Psi_k(n,K)$ lies within η of $\limsup_k \Psi_k(n,\infty)$ (Theorem 20.18); when the dead-shell limit $\lim_k \Psi_k(n,\infty)$ exists, which is a further existence statement not needed for (RES_0), this is the continuum form (RES^0), in which no lattice spacing appears. In the curvature sector (RES_0) is proved at one configuration (Theorem 20.14): the shell mass $\varkappa$ is put inside the matter covariance of King's block-spin flow, and two estimates on that flow, a two-sided bound on the stopping-scale integral uniformly in $\varkappa$ and a $\varkappa$-Cauchy property of the reduced small-field action on one test set, and a disc bound for the heavy shell, which is established there, give the tail bound at one configuration. The two estimates form (HS) (Theorem 20.2); they come from a renormalisation-group analysis of King's walled torus model with the shell mass inside the matter covariance (Propositions 20.3 and 20.13).

The torus comparison is reduced to one-sided bounds at a single lattice spacing (Theorem 17.6). The passive map $S = R^{-1}\sigma_\nu$ is an involution that exchanges the two lattices and maps the torus D_J to the torus $R^{-1}D_J$. A forward bound of King's type on each of the two tori therefore gives both directions of the comparison on D_J. Theorem 17.6 reduces (K_0) in the curvature sector to the one-sided bound (F_0) of Definition 17.5 together with the clause (C_0) attached to it (a bound at one spacing, with a condition on its constants) together with the existence of the axis-lattice limits (Remark 17.9); these are what the proof of (K_0) along

King's lines produces, with the large-field clause of King's upper bound, in the form of Proposition 17.100 at the wall, in place of King's torus bound [16, (4.2)], which is vacuous at a non-zero wall (Remark 17.9, Remark 17.10). An alternative placement of the comparison, on a Neumann–Dirichlet box aligned with one lattice and a staircase for the other, is possible in principle and is not used here.

15.2. **The main theorem of the rotation part.** For $R \in O(3)$ and $b \in \mathbb{R}^3$, sources are transported as 2-forms, $R \cdot f$ and $\tau_b f$ (Definition 2.4).

Theorem 15.1 ($O(3)$ invariance of the curvature sector)**.** *King's printed state on curvature sources is $O(3)$-invariant: for every $R \in O(3)$, all real 2-forms $f_1, \dots, f_N \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ and all $z \in \mathbb{C}^N$,*

$$Z_\infty^K\Big(\sum_j z_j\, R \cdot f_j,\, 0\Big) = Z_\infty^K\Big(\sum_j z_j f_j,\, 0\Big),$$

so all curvature Schwinger functions are $O(3)$-covariant. With the translations of Corollary 8.3 the same holds for every Euclidean motion $x \mapsto Rx + b$, with $\tau_b R \cdot f_j$ in place of $R \cdot f_j$: the curvature sector is invariant under $\mathbb{R}^3 \rtimes O(3)$, and the curvature Schwinger functions are Euclidean covariant.

The proof is assembled at the end of the heavy-shell sections (after Theorem 20.14): the heavy-shell estimates (HS) (Theorem 20.2) give (RES_0) at one configuration (Theorem 20.14), (RES_0) gives the closing estimate (TIGHT_0) (Theorem 19.10, with the torus comparison (K_0) of Theorem 17.1 and the dead-shell chain (DS) of Proposition 20.1), and Theorem 18.12 gives the invariance from (E_C) (Theorem 3.3), (K_0) and (TIGHT_0) at every point of King's range off the set E_W of Remark 15.2; at the points of E_W the invariance, with its Euclidean clause, follows from the statement at the nearby scalar masses by Theorem 20.15 (Remark 15.2). For the curvature two-point function the weaker form (TIGHT_2) of the closing estimate suffices (Corollary 18.13). The rotation part is organised as follows: this section sets up the lattices with a frame, the walled tori and the three lemmas (cutting, passive covariance, second moment) that every later argument uses; the next section treats Wick schemes and the full object on the comparison convention, which the scalar-source forms of the theorems need; then come the torus comparison, the closing hypothesis with the rotation theorems, the two heavy-shell sections, and the corrections to [15, 16] that the rotation part relies on.

Remark 15.2 (the exceptional set E_W and the threshold J_{pos}). Throughout the rotation part, E_W denotes the set (58) of the section on the coupling range: the couplings (e, λ, m, μ_0) at which the scalar mass m is a global minimiser on $(0, \infty)$ of the function α_∞ of the scalar mass, the continuum limit of the relative bare mass coefficient, $\alpha_\infty(M) - \alpha_\infty(m) = \lim_k [\alpha_k^\infty(M) - \alpha_k^\infty(m)]$ (Theorem F.168(b) of the full version); and J_{pos} denotes the threshold of Theorem 17.98 of the full version, a number depending on the couplings and on the large-block integer M. Off E_W, Theorem 17.98 of the full version supplies the positivity margin that the large-field clause of King's upper bound needs at a wall, and the chain (HS) $\Rightarrow$ (RES_0) $\Rightarrow$ (TIGHT_0) $\Rightarrow$ invariance of Theorem 15.1 is run there; Theorem 17.1, Theorem 18.4 and Theorem 18.12 are stated at every point of King's range off E_W. At a point of E_W the chain is not run: Theorem 20.15 gives the statement of Theorem 15.1 from its statement at the nearby scalar masses $M \downarrow m$, which lie off E_W, since α_∞ is real-analytic and non-constant in M (Lemma K.560 of the full version), so that $M = m$ is an isolated minimiser. The set E_W is contained in $\{\alpha_\infty'(m) = 0\}$, $' = d/dM^2$, and meets each line of fixed (λ, m, μ_0) in at most two values of e^2 whenever the polynomial $e^2 \mapsto \alpha_\infty'(m)$, of degree ≤ 2, is not identically zero (Lemma 22.1(b)(iii)). In the Wick sector the one-sided clause (T3) of Theorem 21.1 excludes the larger set $E_W^< \supseteq E_W$ (Lemma 22.1(b)(i)).

15.3. **Geometry.**

Definition 15.3. Let $\theta_0 \in (0, \pi/2)$ with $\cos\theta_0 = \frac{4}{5}$, $\sin\theta_0 = \frac{3}{5}$. For $1 \le \mu < \nu \le 3$ let $R_{\mu\nu}(\theta)$ be the rotation with $Re_\mu = \cos\theta\, e_\mu + \sin\theta\, e_\nu$, $Re_\nu = -\sin\theta\, e_\mu + \cos\theta\, e_\nu$, $Re_\rho = e_\rho$ for $\rho \ne \mu, \nu$. Put $\mathcal{R}_0 := \{R_{\mu\nu}(\pm\theta_0) : 1 \le \mu < \nu \le 3\}$ and let G_0 be the group generated by $\mathcal{R}_0$. The spacings $\varepsilon_k = 5^{-k}$, the axis lattices $\Lambda_k = \varepsilon_k \mathbb{Z}^3$, King's large-block integer M [15, §2.3] (here $M \in 50\mathbb{N}$,

$M \geq M_{\min}$, Remark 2.3) and King's cubes $C_J = [-MJ, MJ)^3$, $J \in \mathbb{N}$ [16, (2.7)], are those of the first sections. The *King tori* of the rotation part are $D_J := \mathbb{R}^3/(50MJ\,\mathbb{Z})^3$ with fundamental cell $\Gamma_J := [-25MJ, 25MJ)^3$ [16, (4.1)]; D_J is a King torus in the sense of the section on the model ($\ell = 50MJ \in 2M\mathbb{N}$), so Λ_k descends to it. We write $Q_b := [-b, b]^3$ (written Q_R in the first sections), $N_r(X) := \{y : \mathrm{dist}_\infty(y, X) \leq r\}$, and $\bar{B}_\rho$ for the closed Euclidean ball of radius ρ about 0.

Lemma 15.4. (a) *For $R \in \mathcal{R}_0$ the matrix $5R$ has integer entries, $R^{-1} \in \mathcal{R}_0$, and $\sigma_\nu R_{\mu\nu}(\theta)\sigma_\nu = R_{\mu\nu}(-\theta)$, where σ_ν is the reflection $x_\nu \mapsto -x_\nu$.*
(b) *For $R \in \mathcal{R}_0$, $k \geq 0$ and $J \geq 1$ the lattices $R\Lambda_k$ and $R^{-1}\Lambda_k$ are invariant under translations by $50MJ\,\mathbb{Z}^3$, so they descend to D_J.*
(c) *For $R \in \mathcal{R}_0$, $|Rx|_\infty \leq \frac{7}{5}|x|_\infty$, and $Q_{5b/7} \subset R^{-1}Q_b \subset Q_{7b/5}$ for $b > 0$.*
(d) *θ_0/π is irrational, and G_0 is dense in $SO(3)$.*

Proof. (a) The entries of R are $0, 1, \pm\frac{3}{5}, \pm\frac{4}{5}$, and $R_{\mu\nu}(\theta)^{-1} = R_{\mu\nu}(-\theta)$; conjugation by σ_ν reverses the orientation of the (x_μ, x_ν)-plane and fixes the third axis. (b) For $p \in 50MJ\,\mathbb{Z}^3$ we have $R^{\pm1}p = \frac{1}{5}(5R^{\pm1})p \in 10MJ\,\mathbb{Z}^3 \subset \Lambda_k$. Hence $R^{-1}\Lambda_k + p = R^{-1}(\Lambda_k + Rp) = R^{-1}\Lambda_k$, and likewise for $R\Lambda_k$. (c) The sums of the absolute values of the entries in each row of R are at most $\frac{7}{5}$. If $Rx \in Q_b$ then $|x|_\infty = |R^{-1}(Rx)|_\infty \leq \frac{7}{5}b$, since $R^{-1} \in \mathcal{R}_0$; if $|x|_\infty \leq \frac{5}{7}b$ then $|Rx|_\infty \leq b$. (d) If $\theta_0 \in \pi\mathbb{Q}$, then $e^{\mathrm{i}\theta_0} = (4 + 3\mathrm{i})/5$ is a root of unity, hence an algebraic integer. Its minimal polynomial over $\mathbb{Q}$ is $x^2 - \frac{8}{5}x + 1$, which does not have integer coefficients; this is a contradiction (compare [16, pp. 337–338]). Hence $\{n\theta_0\}_{n\in\mathbb{Z}}$ is dense modulo 2π, and the closure of $\{R_{\mu\nu}(\theta_0)^n\}_{n\in\mathbb{Z}}$ is the group of all rotations in the (x_μ, x_ν)-plane. Every element of $SO(3)$ is a product $R_{12}(\alpha)R_{23}(\beta)R_{12}(\gamma)$ (Euler angles), so $\overline{G_0} = SO(3)$. □

15.4. **Lattices with a frame.**

Definition 15.5 (lattices with a frame, plaquettes, sources)**.** A *lattice of spacing* ε_k is $\Lambda = O\Lambda_k$ with $O \in O(3)$, equipped with the frame $f_\alpha := Oe_\alpha$, $\alpha = 1, 2, 3$, and with the nearest-neighbour bonds $\langle x, x + \varepsilon_k f_\alpha\rangle$. (The frame vectors f_α and the 2-form sources f share a letter; the index tells them apart.) A field configuration on a set X of sites consists of $\phi(x) = r_x e^{\mathrm{i}\vartheta_x} \in \mathbb{C}$ and $A_\alpha(x) \in \mathbb{R}$, $x \in X$; the bond $\langle x, x + \varepsilon f_\alpha\rangle$ carries $A_{\langle x, x+\varepsilon f_\alpha\rangle} := A_\alpha(x)$ and the reversed bond carries $-A_\alpha(x)$ (Feynman form of the vector field, [16, (2.5)]). The action on Λ is King's action of the section on the model written in the frame, e_α replaced by f_α throughout, with the same free-boundary counterterm δm_ε^2 and the same free Wick constant W_ε, which depend on ε only. A plaquette is $p = (x; \alpha, \beta)$ with $\alpha < \beta$; its centre is $c(p) := x + \frac{\varepsilon}{2}(f_\alpha + f_\beta)$ and its field strength is

$$F(p) := \varepsilon^{-1}\big(A_\alpha(x) + A_\beta(x + \varepsilon f_\alpha) - A_\alpha(x + \varepsilon f_\beta) - A_\beta(x)\big) \qquad [16, (2.11)].$$

For a real 2-form f and a real function h, smooth and compactly supported (on a torus: smooth and periodic; a compactly supported source whose support lies in the interior of a fundamental cell is periodised from it), put

$$F_\Lambda(f) := \sum_p \varepsilon^3\, f_{c(p)}(f_\alpha, f_\beta)\, F(p), \qquad \mathsf{S}_\Lambda(h) := \sum_x \varepsilon^3 h(x), \qquad W_\Lambda(h) := \sum_x \varepsilon^3 h(x)\big(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon\big),$$

the sums running over the plaquettes $p = (x; \alpha, \beta)$, respectively the sites, of the region (of one fundamental cell on a torus): the 2-form is *sampled in the lattice's own frame*. On Λ_k with the standard frame these are $F_{\Lambda_k}(f)$, $\mathsf{S}_{\Lambda_k}(h)$ and ${:}\phi^2{:}(h)$ of Definition 2.4. For a finite family $\zeta = (z_j, f_j; w_j, h_j)_j$ with complex coefficients, $\Theta(\zeta) := \exp(\sum_j z_j F_\Lambda(f_j) + \sum_j w_j W_\Lambda(h_j))$ as there, and $O \cdot \zeta := (z_j, O \cdot f_j; w_j, O \cdot h_j)_j$ with the transport of Definition 2.4. The cone $\mathcal{K}$ of Definition 2.4 is invariant under $O(3)$, under translations, under $(f, g) \mapsto (tf, tg)$, $t \geq 0$, and under sums.

Remark 15.6 (the frame convention). In [15, 16] the curvature source is written $F(g) = \sum_p \varepsilon^3 g(p)F(p)$ with one function g on all plaquettes [16, (2.12)]. On one lattice a 2-form sampled as above is a function on plaquettes, and King's proofs treat the source terms componentwise ("the sources

$g_{\mu\nu}(x'), h(x') \dots$ can be replaced by $g_{\mu\nu}(x), h(x)$", [15, p. 665]; the sentence is stated there without an argument). King's single-lattice statements at $h = 0$ are used in this sense throughout: the existence theorem (E_C) is stated, in Theorem 3.3, for real smooth 2-form sources sampled at plaquette centres in the lattice's own frame, and this existence clause of King's Theorem 2.1 of [16] is proved in Section 5; the statements of [16, Thms. 2.2, 2.3, (3.40)–(3.44)] that the rotation part uses, all at $h = 0$, are established in the first sections from (E_C) and the correlation inequalities, [16, (3.42)–(3.43)] are not used, and the torus statement [15, Thm. 2.1] at a wall is part of the torus comparison (K_0) (Theorem 17.1). The two-lattice statement is different: Proposition H.2 of the full version shows that it fails for a single scalar function on all plaquettes, and (K_0) is formulated for 2-forms sampled in each lattice's own frame.

15.5. **Tori and walled tori.**

Definition 15.7 (periodic and walled torus states)**.** Let Λ be a lattice of spacing $\varepsilon = \varepsilon_k$ with a frame, and $d\nu := \prod_x \prod_\alpha dA_\alpha(x)\, d^2\phi(x)$ over the sites in question.

(a) *Boxes.* For a finite set X of sites of Λ, $\langle\cdot\rangle^{\mathrm{ND}}_{\Lambda,X}$ is the Neumann–Dirichlet state of Definition 2.1 written in the frame of Λ (Dirichlet condition for the scalar field outside X, Neumann condition for the vector field), possibly with the matter killed on a set $U \subset X$ in the sense of that definition. On Λ_k these are the states $\langle\cdot\rangle^{\mathrm{ND}}_{k,X}$ of Definition 2.1, among them the comparison cube $\langle\cdot\rangle^{(a)}_{k,J}$ and King's RG cube $\langle\cdot\rangle^{K}_{k,J}$ (Definition 2.2).

(b) *Tori.* If Λ descends to D_J, $\langle\cdot\rangle_{\Lambda,D_J}$ is the normalised measure $Z^{-1}e^{-S}d\nu$ with the action S summed over the sites and bonds of the finite graph $\Lambda/(50MJ\,\mathbb{Z})^3$ (periodic conditions for both fields). On Λ_k this is the periodic state $\langle\cdot\rangle_{k,T}$ of the section on the model for the King torus $T = D_J$.

(c) *Walled tori.* A *walled configuration* is $\mathfrak{c} = (J', w)$ with $J' \in \mathbb{N}$ and $w \in C^\infty(D_{J'}; [0,\infty))$, the *wall*. If Λ descends to $D_{J'}$, $\langle\cdot\rangle_{\mathfrak{c},\Lambda}$ is the torus state of (b) with the density multiplied by $\prod_x \exp(-\varepsilon^3 w(x)|\phi(x)|^2)$. The wall raises the scalar mass squared by $2w$ and acts in the single-site measures. We write $(J', \psi, n) := (J', n\psi)$ for $n \geq 0$ and a fixed profile ψ.

(d) *Killed sites.* For any of the states (b), (c) and a set U of sites, the state *with the matter killed on* U is defined as in Definition 2.1: the single-site measure at $x \in U$ is replaced by the point mass at $\phi(x) = 0$, every hopping term with an endpoint in U reduces to its diagonal part at the other endpoint, and the photon variables are unchanged.

For every state $\langle\cdot\rangle_{\Lambda,X}$ above and a scheme a (Definition 2.4, Remark 2.5) the scheme functional is defined by the formula of Definition 2.4 with F_Λ, W_Λ, S_Λ in place of their axis-lattice versions,

$$Z^a_{\Lambda,X}(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)} \big\langle \Theta(\zeta) \big\rangle_{\Lambda,X};$$

on Λ_k we keep the notation $Z^a_{k,X}$ of Definition 2.4, and on the rotated lattice we write $Z^a_{R^{-1}\Lambda_k,X}$. All these states are invariant under the charge conjugation $(A, \vartheta) \mapsto (-A, -\vartheta)$, which maps $F_\Lambda(f)$ to $-F_\Lambda(f)$ and fixes r; hence $\langle \Theta(\mathrm{i}f, g)\rangle = \langle \cos F_\Lambda(f)\, e^{W_\Lambda(g)}\rangle$ is real for $(f, g) \in \mathcal{K}$.

Definition 15.8 (the unnormalised torus functional and the walled ratio)**.** For a lattice Λ of spacing ε_k descending to a torus T, and real smooth periodic sources G (a 2-form) and h (a function), the *unnormalised* generating functional is

$$\mathsf{Z}_{\Lambda,T}(G, h) := \int e^{-S(A,\phi)+F_\Lambda(G)+W_\Lambda(h)}\, d\nu,$$

with no field-independent constant in the action; for a pair $s = (G, h)$ we also write $\mathsf{Z}_{\Lambda,T}(s)$. Two lattices of the same spacing descending to T have the same number of sites in a fundamental cell, so any field-independent constant per site multiplies both by the same factor. For a walled configuration $\mathfrak{c} = (J', w)$, a lattice Λ descending to $D_{J'}$, a scheme a and $(f, g) \in \mathcal{K}$ with $w = 0$ on the support of g, the *walled ratio* is

$$T^a_{\mathfrak{c},\Lambda}(f, g) := e^{-a_k \mathsf{S}_\Lambda(g)} \big\langle \Theta(\mathrm{i}f, g) \big\rangle_{\mathfrak{c},\Lambda} = \frac{Z^a_{\Lambda,D_{J'}}(\mathrm{i}f, g - w)}{Z^a_{\Lambda,D_{J'}}(0, -w)}.$$

The second expression agrees with the first because

$$e^{-W_\Lambda(w)} = e^{2\mathsf{W}_\varepsilon \mathsf{S}_\Lambda(w)} \prod_x e^{-\varepsilon^3 w(x)|\phi(x)|^2},$$

so that the wall is the scalar source $-w$ up to the constant $e^{2\mathsf{W}_\varepsilon \mathsf{S}_\Lambda(w)}$, which cancels between numerator and denominator, as do the multipliers $e^{a_k \mathsf{S}_\Lambda(w)}$ of both. The same formulas define the walled ratio for the periodic model of Λ on any torus $T = \mathbb{R}^3/P$ to which Λ descends (defined as in Definition 15.7(b), with the finite graph Λ/P) and any smooth wall $w \geq 0$ on T vanishing on the support of g; this is used only for the transported tori of Lemma 15.10(b).

Every state of Definition 15.7, with or without killed sites, is a BFS measure of King type in the sense of the section on the model: $c_{xy} = \varepsilon$ for the bonds present, the single-site measures contain $e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]}$, the diagonal parts $\frac{\varepsilon}{2} r_x^2$ of all bonds at x (including, for boxes, those leading out of X, and those leading into a killed site) and the wall factor $e^{-\varepsilon^3 w(x) r_x^2}$ with $w \geq 0$; the photon form is positive definite because $\mu_0 > 0$, and the exponential moments are finite because $\lambda > 0$. The same holds for every measure obtained from such a state by multiplying some c_{xy} by $t \in [0,1]$, multiplying photon gradient terms by $t \in [0,1]$, adding $s|\ell(A)|^2$ ($s \geq 0$) to the action, multiplying single-site measures by e^{-sr^2} ($s \geq 0$), or killing sites. Curvature sources $F_\Lambda(f)$ are real linear functionals of A, and $e^{W_\Lambda(g)}$ is a positive constant times $e^{\sum_x \varepsilon^3 g(x) r_x^2}$; so Lemma 2.8, Lemma 2.9 and Lemma 2.10 apply to all of them, on every lattice with a frame.

15.6. **Cutting, passive covariance and the second moment.**

Lemma 15.9 (cutting, walled tori)**.** *Let $\langle\cdot\rangle_Y$ be a box, torus or walled-torus state of Definition 15.7 on a lattice Λ with a frame, possibly with killed sites, and let X be a set of sites of Λ such that X and its nearest neighbours lie in one fundamental cell. Let $\langle\cdot\rangle_X^{\mathrm{ND}}$ be the Neumann–Dirichlet state on X whose single-site measures (wall, killed sites and diagonal parts of the bonds leaving X included) are those of $\langle\cdot\rangle_Y$. Let $(f,g) \in \mathcal{K}$ be such that every plaquette and every site at which f, respectively g, does not vanish lies in X. Then, at every lattice spacing,*

$$\langle \Theta(\mathrm{i}f, g)\rangle_X^{\mathrm{ND}} \leq \langle \Theta(\mathrm{i}f, g)\rangle_Y, \qquad \langle \Theta(f, -g)\rangle_X^{\mathrm{ND}} \geq \langle \Theta(f, -g)\rangle_Y,$$

and the same for Z^a. If the wall of Y vanishes on X and no site of X is killed, $\langle\cdot\rangle_X^{\mathrm{ND}} = \langle\cdot\rangle_{\Lambda,X}^{\mathrm{ND}}$ is the box state of Definition 15.7(a).

Proof. The wall and the killed sites are part of the single-site measures, and the frame plays no role, so the proof of Lemma 2.11 applies word for word. Delete from Y every photon gradient term coupling a site of X to a site outside X, and the cosine parts of all hopping terms between X and its complement, keeping their diagonal parts $\frac{\varepsilon}{2}(r_x^2 + r_y^2)$; since X and its neighbours lie in one fundamental cell, these are bonds between distinct sites of the finite graph. By Lemma 2.8(a),(b) this does not increase the first expectation, by (d) it does not decrease the second, and every intermediate measure is a BFS measure. The resulting measure is a product (Lemma 2.8(e)); its X-factor is $\langle\cdot\rangle_X^{\mathrm{ND}}$, since the diagonal parts of the cut bonds are exactly the Dirichlet terms and the remaining photon terms are the Neumann ones, and the observable depends on the X-variables only. The free Wick constant, δm_ε^2 and the scheme multiplier $e^{\mp a_k \mathsf{S}_\Lambda(g)}$ are the same in both states. □

With $Y = X_{J+1}$ and $X = X_J$ on Λ_k this is King's monotonicity in the volume [16, Thm. 3.2, (3.31)–(3.33)] in the form his proof proves, with one Wick constant for all J, as discussed after Lemma 2.11; with Y a walled torus and X a cube of Λ inside the cell, it is the inequality of (33).

Lemma 15.10 (passive covariance and the algebra of the walled ratio)**.** *Let $\mathfrak{a}$ be a scheme.*

(a) *For real coefficients the multiplier $e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)}$ is a positive constant depending only on k, the coefficients and the numbers $\mathsf{S}_\Lambda(h_j)$. Hence the inequalities of Lemma 2.8 and Lemma 15.9 between two states on the same lattice with the same sources hold for Z^a, and for $T^a_{\mathfrak{c},\Lambda}$, in the same direction.*

(b) *Let $O \in O(3)$, let Λ be a lattice of spacing ε with frame (f_α), and let X be a box, a torus or a walled torus for Λ, possibly with the matter killed on U. Equip $O\Lambda$ with the frame (Of_α) and OX with the transported boundary conditions, the wall $O\cdot w$ and the killed set OU; for a torus $X = \mathbb{R}^3/P$ this means the torus $OX = \mathbb{R}^3/OP$, to which $O\Lambda$ descends. Then for every finite family ζ*

$$\big\langle \Theta(O\cdot\zeta)\big\rangle_{O\Lambda,OX} = \big\langle \Theta(\zeta)\big\rangle_{\Lambda,X}, \qquad \mathsf{S}_{O\Lambda}(O\cdot h) = \mathsf{S}_\Lambda(h), \qquad Z^a_{O\Lambda,OX}(O\cdot\zeta) = Z^a_{\Lambda,X}(\zeta).$$

In particular, for a walled configuration $\mathfrak{c} = (J', w)$ and (f, g) as in Definition 15.8, $T^a_{O\cdot\mathfrak{c},O\Lambda}(O\cdot f, O\cdot g) = T^a_{\mathfrak{c},\Lambda}(f,g)$, where $O\cdot\mathfrak{c}$ denotes the transported walled torus, the torus $OD_{J'}$ with the wall $O\cdot w$, and the left side is the walled ratio on that torus in the sense of Definition 15.8; $O\cdot\mathfrak{c}$ is a walled configuration $(J', O\cdot w)$ of Definition 15.7(c) when $OD_{J'} = D_{J'}$, as for a signed permutation matrix O. Moreover, on every torus the periodic model on Λ does not change if the frame (f_α) is replaced by $(\pm f_{\pi(\alpha)})$, for a permutation π and any choice of signs.

(c) $T^a_{\mathfrak{c},\Lambda}(f,g) = e^{-a_k \mathsf{S}_\Lambda(g)}\langle\Theta(\mathrm{i}f,g)\rangle_{\mathfrak{c},\Lambda}$ *is the normalised expectation of the walled torus state up to the multiplier of* (a)*, and at $g = 0$ the walled ratio $T^a_{\mathfrak{c},\Lambda}(f,0) = \langle e^{\mathrm{i}F_\Lambda(f)}\rangle_{\mathfrak{c},\Lambda}$ does not depend on the scheme.*

Proof. (a) The first statement is immediate (compare the remark in [7, p. 281] that the inequalities are not affected by Wick ordering); for $T^a_{\mathfrak{c},\Lambda}$ the multiplier is $e^{-a_k \mathsf{S}_\Lambda(g)}$. (b) The map $\phi'(Ox) = \phi(x)$, $A'_\alpha(Ox) = A_\alpha(x)$, with components taken in the frames (f_α) and (Of_α), is a bijection of the configuration spaces preserving $d\nu$; this is the proof of Lemma 2.13, which covers boxes and tori, and for a torus $X = \mathbb{R}^3/P$ it carries the finite graph Λ/P onto $O\Lambda/OP$. The action is written in the frame, and δm^2_ε, W_ε and a_k depend on ε only; the wall term $\sum_x \varepsilon^3 w(x)|\phi(x)|^2$ becomes $\sum_x \varepsilon^3 (O\cdot w)(Ox)|\phi'(Ox)|^2$ since $(O\cdot w)(Ox) = w(x)$, and a point mass at $\phi(x) = 0$ becomes a point mass at $\phi'(Ox) = 0$. Plaquettes, their centres and field strengths are transported, and $(O\cdot f)_{Oc}(Of_\alpha, Of_\beta) = f_c(f_\alpha, f_\beta)$, $\sum_{Ox} \varepsilon^3 (O\cdot h)(Ox) = \sum_x \varepsilon^3 h(x)$. The clause on the walled ratio is the first identity for $\zeta = (\mathrm{i}, f; 1, g)$ and the walled torus $X = D_{J'}$, together with $\mathsf{S}_{O\Lambda}(O\cdot g) = \mathsf{S}_\Lambda(g)$ and the first expression of Definition 15.8 read on $OD_{J'}$. For the last statement, a permutation of the frame relabels the components; reversing f_α replaces $A_\alpha(x)$ by $-A_\alpha(x - \varepsilon f_\alpha)$, a bijection of the torus variables that preserves the action, and changes the sign of both $F(p)$ and $f_{c(p)}(f_\alpha, f_\beta)$ while preserving the set of plaquette centres. (c) The expression for $T^a_{\mathfrak{c},\Lambda}$ is its definition, and at $g = 0$ the multiplier is 1 and $W_\Lambda(0) = 0$. □

Lemma 15.11 (the characteristic function and the second moment of a photon functional)**.** *Let $\langle\cdot\rangle$ be a BFS measure whose Gaussian has covariance C, and let $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q), A\rangle$ and $\mathbb{E}^\sharp$ be the current representation and the current measure of Lemma 2.9. Then for $v \in \mathbb{R}^m$:*

(a) $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp\cosh\langle Cv,\kappa\rangle$*; in particular $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle \in [0,1]$ is real;*

(b) $\langle\langle v,A\rangle\rangle = 0$ *and* $\langle\langle v,A\rangle^2\rangle = \langle v,Cv\rangle - \mathbb{E}^\sharp\langle Cv,\kappa\rangle^2 \in [0, \langle v,Cv\rangle]$.

Proof. (a) For a centred Gaussian γ with covariance C and $a \in \mathbb{C}^m$, $\int e^{\langle a,A\rangle} d\gamma = e^{\langle a,Ca\rangle/2}$. With $a = \mathrm{i}(v \pm \kappa)$ and the average of the two signs,

$$\int e^{\mathrm{i}\langle v,A\rangle}\cos\langle\kappa,A\rangle\, d\gamma = e^{-\langle v,Cv\rangle/2 - \langle\kappa,C\kappa\rangle/2}\cosh\langle v,C\kappa\rangle,$$

since C is symmetric. Multiply by $w(q) \geq 0$, sum over the currents q (the series converges absolutely: $\cosh\langle v,C\kappa\rangle \leq e^{|\langle v,C\kappa\rangle|} \leq e^{\langle v,Cv\rangle/2 + \langle\kappa,C\kappa\rangle/2}$ by the Cauchy–Schwarz inequality for the form C, so the terms are dominated by $w(q)$ and $\sum_q w(q) < \infty$), and divide by the same sum at $v = 0$, which is the positive normalisation $\sum_q w(q) e^{-\langle\kappa,C\kappa\rangle/2}$ of $\mathbb{E}^\sharp$; this is the formula. Its right side lies in $[0,\infty)$, and $|\langle e^{\mathrm{i}\langle v,A\rangle}\rangle| \leq 1$. (b) Replace v by tv, $t \in \mathbb{R}$, in (a). The left side is the characteristic function of the real random variable $\langle v,A\rangle$, which has all moments because $\langle e^{s\langle v,A\rangle}\rangle \leq e^{s^2\langle v,Cv\rangle/2}$ for every real s (Lemma 2.9); so it equals $1 + \mathrm{i}t\langle\langle v,A\rangle\rangle - \frac{t^2}{2}\langle\langle v,A\rangle^2\rangle + O(t^3)$. The right side is $e^{-t^2\langle v,Cv\rangle/2}\mathbb{E}^\sharp\cosh(t\langle Cv,\kappa\rangle)$, and $\mathbb{E}^\sharp\cosh(t\langle Cv,\kappa\rangle) = 1 + \frac{t^2}{2}\mathbb{E}^\sharp\langle Cv,\kappa\rangle^2 + O(t^4)$, the expansion under $\mathbb{E}^\sharp$ being justified by the domination $\cosh(t\langle Cv,\kappa\rangle) \leq e^{t^2\langle v,Cv\rangle/2 + \langle\kappa,C\kappa\rangle/2}$

of (a), which shows that $\langle Cv,\kappa\rangle$ has all moments under $\mathbb{E}^\sharp$. The right side is real and even in t, so $\langle\langle v,A\rangle\rangle = 0$; comparing the terms of order t^2 gives $\langle\langle v,A\rangle^2\rangle = \langle v,Cv\rangle - \mathbb{E}^\sharp\langle Cv,\kappa\rangle^2$. The bounds follow since both $\langle\langle v,A\rangle^2\rangle$ and $\mathbb{E}^\sharp\langle Cv,\kappa\rangle^2$ are non-negative. □

The lemma applies to every state of Definition 15.7 and to every state of the first sections, with v the vector $d^*_\Lambda f$ defined by $F_\Lambda(f) = \langle d^*_\Lambda f, A\rangle$ for a real 2-form f whose plaquettes carry variables of the state; the photon Gaussian of a torus or walled torus is the periodic Feynman form with mass μ_0, and killed sites are covered as in Lemma 2.9.

16. Schemes, admissibility and the full object

This section fixes the scheme in which generating functionals with scalar sources are taken, defines W-admissibility of a scheme, and proves the full-object theorem (Theorem 16.6): the iterated limit of the comparison cubes in a W-admissible scheme exists for all curvature and scalar sources, is entire and translation invariant, and depends on the scheme only through one explicit factor. The curvature sector $h = 0$ of all this is scheme-free and is already in place (Theorem 7.6, Lemma 7.7); the rotation statements of the Wick sector (Theorem 18.9, Theorem 18.11) are stated for the object defined here.

16.1. **The Wick square.**

Remark 16.1 (a logarithm in the Wick square). With the free Wick constant W_ε alone, the expectation of the Wick square $:\phi^2:(h)$ of Definition 2.4 diverges at order e^2 in perturbation theory: for the model on $\varepsilon\mathbb{Z}^3$, at order $e^2\lambda^0$,

$$\big\langle :\phi^2:(h)\big\rangle = \frac{e^2}{4\pi^2}\,\ln(1/\varepsilon)\int h + O(1),$$

with a coefficient that depends neither on m nor on μ_0. We indicate the computation, in formal perturbation theory. In the normalisation of the action each real component of ϕ has propagator $(p^2+m^2)^{-1}$ at $e=0$. Write the one-loop correction to the inverse propagator as $\Sigma(p)$; the mass counterterm removes $\Sigma(0)$. The photon-exchange part is $-e^2\int \frac{d^3k}{(2\pi)^3}\frac{(2p-k)^2}{((p-k)^2+m^2)(k^2+\mu_0^2)}$; writing $(2p-k)^2 = 2p^2 + 2(p-k)^2 - k^2$ and using $\int \frac{d^3k}{(2\pi)^3}\frac{1}{k^2(p-k)^2} = \frac{1}{8|p|}$, one finds $\Sigma(p)-\Sigma(0) = -\frac{e^2}{4}|p| + O(m+\mu_0)$ for $1 \ll |p| \ll \varepsilon^{-1}$, the lattice regularisation changing only momentum-independent terms and terms of order εp^2. Inserting,

$$\langle|\phi(0)|^2\rangle - 2\mathsf{W}_\varepsilon = -2\int\frac{d^3p}{(2\pi)^3}\frac{\Sigma(p)-\Sigma(0)}{(p^2+m^2)^2} + O(e^4) = \frac{e^2}{2}\cdot\frac{1}{2\pi^2}\ln(1/\varepsilon) + O(1)$$

at order e^2. Two consequences.

(a) For h with $\int h \neq 0$ the generating functionals with scalar sources in King's printed Wick ordering have no finite continuum limit beyond the curvature sector, at least in perturbation theory: for $h \geq 0$, $h \not\equiv 0$, Jensen's inequality gives $\langle e^{:\phi^2:(h)}\rangle \geq e^{\langle:\phi^2:(h)\rangle} \to \infty$. This affects [15, Thm. 2.1] and [16, Thms. 2.1–2.3, (3.43), (4.2)] for scalar sources with $\int h \neq 0$. In the cube King subtracts the Dirichlet constant [16, (2.13)] and, at distance > 1 from the boundary, the free one, which differ by a convergent amount only. The statement of [15, p. 677] that the sources are proportional to $(5^k\varepsilon)^\alpha$, $\alpha > 0$, in the small-field region does not hold for the marginal graph that produces the logarithm. We therefore allow an additional vacuum counterterm, linear in h and depending only on the lattice spacing: the schemes of Definition 16.2. Walled ratios (Definition 15.7) and the curvature sector $h = 0$ are unaffected, since the h-linear c-numbers cancel in them.

(b) King's proof of [16, Thm. 2.4] compares the effective action generated from spacing ε by a first block-spin step with rotated blocks with the effective action generated from the rotated lattice of spacing 5ε [16, (4.3)–(4.9)]. In a scheme a the two generating functionals carry the multipliers $e^{-wa_\varepsilon\mathsf{S}(h)}$ and $e^{-wa_{5\varepsilon}\mathsf{S}(h)}$. If a removes the logarithm, then $a_\varepsilon - a_{5\varepsilon} \to \frac{e^2}{4\pi^2}\ln 5$, so the comparison [16, (4.6)] must carry the c-number $\frac{e^2}{4\pi^2}\ln 5\int h$, and as printed it cannot hold for $\int h \neq 0$. Since a walled ratio contains the wall source $-w$ with $\int w > 0$, this

enters the proof of walled comparisons even though their statement is scheme-free. In the equal-spacing formulation of the torus comparison (Theorem 17.1) the two lattices carry the same a_k, and the issue does not arise.

16.2. **Schemes.**

Definition 16.2 (schemes)**.** A *scheme* is a real sequence $a = (a_k)_{k\ge 0}$ such that a_k depends only on ε_k and on (e, λ, m, μ_0), and $Z^a_{k,X}(\zeta) = e^{-\sum_j w_j a_k \mathsf{S}_{\Lambda_k}(h_j)} \langle \Theta(\zeta)\rangle^{\mathrm{ND}}_{k,X}$ is the functional of Definition 2.4, the expectation of $\exp(\sum_j z_j F(f_j) + \sum_j w_j \sum_x \varepsilon_k^3 h_j(x)[|\phi(x)|^2 - 2\mathsf{W}_{\varepsilon_k} - a_k])$; the same multiplier defines $Z^a_{\Lambda,T}(\zeta)$ for the periodic and walled states of Definition 15.7 on a lattice Λ of spacing ε_k. Three schemes are used.

(i) The *printed scheme* $a \equiv 0$.

(ii) The *reference scheme* $z = a^\sharp$ of Definition 2.4: $a^\sharp_k = e^2 X_{\varepsilon_k}$, where X_ε is the coefficient of e^2 in the perturbative expansion of $\langle|\phi(0)|^2\rangle$ for the model on $\varepsilon\mathbb{Z}^3$ at $\lambda = 0$, with the order-e^2 part of δm^2_ε inserted.

(iii) The *self-normalised scheme*: fix J_0 and $0 \le h_0 \in C^\infty_c(\mathbb{R}^3)$, $h_0 \not\equiv 0$, supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and put $a^{sn}_k := \mathsf{S}_{\Lambda_k}(h_0)^{-1} \ln Z^0_{k,J_0}(0, h_0)$ when $\mathsf{S}_{\Lambda_k}(h_0) > 0$, and $a^{sn}_k := 0$ otherwise; here $Z^0_{k,J_0}(0,h_0) = \langle e^{:\phi^2:(h_0)}\rangle^{(a)}_{k,J}$ at $J = J_0$ is the functional of the comparison cube in the printed scheme.

At order e^2 the reference scheme removes the logarithm of Remark 16.1 identically on $\varepsilon\mathbb{Z}^3$: by translation invariance the order-$e^2\lambda^0$ part of $\langle :\phi^2:(h)\rangle$ there equals $e^2 X_\varepsilon \mathsf{S}(h)$.

16.3. **W-admissible schemes.**

Definition 16.3 ($(\mathrm{W})_a$: W-admissibility of the scheme a)**.** Let a be a scheme. $(\mathrm{W})_a$ is the statement (W2) of Hypothesis 23.1 for the scheme a, posed on the comparison convention $\mathrm{ND}^{(a)}$, in the following two clauses:

$(\mathrm{W1})^{(a)}$ For every King cube C_J, every finite family of curvature sources f_j and scalar sources h_j supported at ℓ^∞-distance > 1 from $\mathbb{R}^3\setminus C_J$, and every $(z,w) \in \mathbb{R}^{2N}$, the limit $\lim_{k\to\infty} Z^a_{k,J}(\sum_j z_j f_j, \sum_j w_j h_j)$ exists in $(0,\infty)$.

$(\mathrm{W2})^{(a)}$ For every $0 \le q \in C^\infty_c(\mathbb{R}^3)$, $\sup_{J\ge J_q} \lim_k Z^a_{k,J}(0,q) < \infty$, where J_q is the least J such that $\operatorname{supp} q$ has ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$.

A scheme with $(\mathrm{W})_a$ is called W-*admissible*. Below the tube threshold, $(\mathrm{W1})^{(a)}$ is the existence theorem with scalar sources, Theorem 3.6; at arbitrary amplitudes (W2) is a hypothesis of the optional extension of the Wick leg (Theorem 23.7), and so are the Wick-sector statements of this part that assume $(\mathrm{W})_a$. Two W-admissible schemes are compared in Lemma 16.4; by its part (b) they differ by a convergent sequence, so a W-admissible scheme is admissible in the sense of Definition 3.7 as soon as the reference scheme is.

$(\mathrm{W1})^{(a)}$ is the analogue in the scheme a of [16, Thm. 2.1], and $(\mathrm{W2})^{(a)}$ of the bound [16, (3.43)], which King asserts for the Neumann–Dirichlet cubes from (3.42), reflection positivity and ultra-violet stability by "a standard argument (see [GJ1])" (not printed) and which, in King's Wick ordering, fails for $\int h_+ > 0$ at order e^2 in perturbation theory (Remark 16.1). At $h = 0$ no multiplier enters: $(\mathrm{W1})^{(a)}$ at zero Wick coefficient is $(\mathrm{E}_C)^{(a)}$, the existence theorem (E_C) of Theorem 3.3 read for the comparison family, with 2-form sources read as in Remark 15.6, and $(\mathrm{W2})^{(a)}$ is not needed, the curvature sector being Lemma 7.7 on the comparison convention and Theorem 7.6 on King's.

Lemma 16.4 (comparison of schemes)**.** (a) *If a is* W*-admissible and $a'_k - a_k \to \kappa_{a'a} \in \mathbb{R}$, then a' is* W*-admissible, and for every family as in* $(\mathrm{W1})^{(a)}$, $\lim_k Z^{a'}_{k,J}(\zeta) = e^{-\kappa_{a'a}\sum_j w_j \int h_j} \lim_k Z^a_{k,J}(\zeta)$.

(b) *If a and a' are* W*-admissible, then $a'_k - a_k$ converges.*

(c) *If some scheme is* W*-admissible, then a^{sn} is* W*-admissible.*

Proof. (a) $Z^{a'}_{k,J}(\zeta) = e^{-\sum_j w_j (a'_k - a_k)\mathsf{S}_{\Lambda_k}(h_j)} Z^a_{k,J}(\zeta)$, and $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$ (a Riemann sum of a smooth compactly supported function). (b) For $0 \le h \in C^\infty_c$ with $\int h > 0$, supported at

distance > 1 from $\mathbb{R}^3 \setminus C_J$, the ratio $Z^{a'}_{k,J}(0,h)/Z^{a}_{k,J}(0,h) = e^{-(a'_k - a_k)\mathsf{S}_{\Lambda_k}(h)}$ converges to a number in $(0,\infty)$ by $(\mathrm{W1})^{(a)}$ and $(\mathrm{W1})^{(a')}$, and $\mathsf{S}_{\Lambda_k}(h) \to \int h > 0$; this is the argument of Remark 2.5. (c) If a is W-admissible, $a^{sn}_k - a_k = \mathsf{S}_{\Lambda_k}(h_0)^{-1} \ln Z^{a}_{k,J_0}(0,h_0)$ for all large k, which converges by $(\mathrm{W1})^{(a)}$ and $\mathsf{S}_{\Lambda_k}(h_0) \to \int h_0 > 0$; apply (a). □

Remark 16.5 (the class of W-admissible schemes). By Lemma 16.4, $(\mathrm{W})_a$ holds for some scheme if and only if it holds for a^{sn}, and W-admissible schemes form a single class modulo convergent sequences; the reference scheme $a^\sharp$ is W-admissible if and only if some scheme is and $a^\sharp - a^{sn}$ converges. If the logarithm of Remark 16.1 is present beyond perturbation theory, the printed scheme $a \equiv 0$ is not W-admissible. A proof of $(\mathrm{W})_a$ would carry the counterterm $a_k\mathsf{S}(h)$ as an additional h-linear vacuum vertex through King's renormalisation group. $(\mathrm{W})_a$ enters none of the curvature-sector results of this part: at $g = 0$ every walled quantity is scheme-free (Lemma 15.10), and King's printed state on curvature sources is Z^K_∞ of Theorem 7.6, under (E_C) alone.

16.4. **The full object in a W-admissible scheme.** For $\rho > 0$ let $\mathcal{E}_\rho := C^\infty_c(\bar B_\rho; \Lambda^2) \times C^\infty_c(\bar B_\rho)$ be the space of pairs $s = (f,h)$ of smooth real curvature and scalar sources supported in the closed ball $\bar B_\rho$, with the C^∞ topology. It is a closed subspace of $C^\infty(\bar B_{\rho+1}; \Lambda^2) \times C^\infty(\bar B_{\rho+1})$, hence a Fréchet space, and it is invariant under $O(3)$ and under the translations τ_b in the sense that τ_b maps $\mathcal{E}_\rho$ into $\mathcal{E}_{\rho+|b|}$. (The 2-form space $E_\rho = C^\infty_c(\bar B_\rho; \Lambda^2)$ of the translation section is the first factor.) For a W-admissible scheme a and a finite family $\zeta = (z_j, f_j; w_j, h_j)_j$ with all supports at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ write

$$Z^a_J(\zeta) := \lim_{k\to\infty} Z^a_{k,J}(\zeta),$$

the limit of the comparison cubes at fixed J, which exists at real coefficients by $(\mathrm{W1})^{(a)}$. We use the total-source convention of the translation section: a functional depends on a family only through its total sources $(\sum_j z_j f_j, \sum_j w_j h_j)$, and we write $Z^a_J(s)$, $Z^a_\infty(s)$ for $s \in \mathcal{E}_\rho$.

Theorem 16.6 (the full-object theorem)**.** *Let a be a* W*-admissible scheme,* $(\mathrm{W})_a$.

(i) *For C_J and families as in* $(\mathrm{W1})^{(a)}$, *$Z^a_{k,J} \to Z^a_J$ locally uniformly on $\mathbb{C}^{2N}$ as $k \to \infty$, and Z^a_J is entire.*

(ii) *For $(f,g) \in \mathcal{K}$ supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $J \mapsto Z^a_J(\mathrm{i}f, g)$ is nondecreasing and $J \mapsto Z^a_J(f, -g)$ is nonincreasing on $J \geq J_0$.*

(iii) *For every finite family of curvature and scalar sources the limit $Z^a_\infty(\zeta) := \lim_{J\to\infty} Z^a_J(\zeta)$ exists locally uniformly on $\mathbb{C}^{2N}$ and is entire; $Z^a_\infty(f,h) > 0$ for real sources; and $Z^a_J \leq Z^a_\infty$ on $\mathcal{K}$.*

(iv) *For every $\rho > 0$, $\Phi^a(s) := \ln Z^a_\infty(s)$ is finite, convex and continuous on $\mathcal{E}_\rho$.*

(v) *$Z^a_\infty(\tau_b \cdot \zeta) = Z^a_\infty(\zeta)$ for all $b \in \mathbb{R}^3$, where τ_b acts on all sources of the family.*

(vi) *If a' is another* W*-admissible scheme, then $a'_k - a_k \to \kappa_{a'a} \in \mathbb{R}$ and $Z^{a'}_\infty(\zeta) = e^{-\kappa_{a'a}\sum_j w_j \int h_j} Z^a_\infty(\zeta)$.*

Proof. (i) At fixed k the functionals are entire in the coefficients: a finite-dimensional integral with Gaussian decay in A and decay $e^{-\lambda\varepsilon^3 r^4}$ in r. Fix a compact $K \subset \mathbb{C}^{2N}$ and a real box $Q \supset \operatorname{Re} K$. By Lemma 2.12(b), $\sup_K |Z^a_{k,J}| \leq \sup_Q Z^a_{k,J}$, and since $\ln Z^a_{k,J}$ is convex in the real coefficients the latter is the maximum over the vertices of Q, at which the sequences converge by $(\mathrm{W1})^{(a)}$, hence are bounded. So the family $\{Z^a_{k,J}\}_k$ is locally bounded; by Montel's theorem and since $\mathbb{R}^{2N}$ is a set of uniqueness for entire functions on $\mathbb{C}^{2N}$, it converges locally uniformly, and the limit is entire [11]. (Compare [16, (3.40)–(3.44)]: Lemma 2.12(b) plays the role of (3.40); King's (3.42)–(3.43) are not used, $(\mathrm{W2})^{(a)}$ replacing (3.43).)

(ii) The sources of (f,g) lie inside X_J for $J \geq J_0$: the corners of a plaquette lie within ℓ^∞-distance $\varepsilon/2 \leq \frac12$ of its centre, so every plaquette and site at which (f,g) does not vanish lies in C_{J_0}. Lemma 2.11 with $Y = X_{J+1}$, $X = X_J$ (no killed sites, so the cut-out factor is the comparison cube) and Lemma 2.12(a) give both inequalities at fixed k for $Z^a_{k,J}$; let $k \to \infty$ by (i).

(iii) Split each $h_j = h_j^+ - h_j^-$ with $h_j^- := \|h_j\|_\infty \chi_j$ and $h_j^+ := h_j + h_j^-$, where $\chi_j \in C_c^\infty$ takes values in $[0,1]$, equals 1 on $\operatorname{supp} h_j$ and is supported in $N_{1/2}(\operatorname{supp} h_j)$; both $h_j^\pm$ are smooth and non-negative. Let J_0 be such that all supports (those of the f_j, $h_j^\pm$) have ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and for $J \geq J_0$ and $(z, w^+, w^-) \in \mathbb{C}^{3N}$ let

$$F_J(z, w^+, w^-) := Z_J^a\Big(\sum_j z_j f_j,\ \sum_j (w_j^+ h_j^+ + w_j^- h_j^-)\Big),$$

entire by (i). *Local bound.* For real $\zeta = (z, w^+, w^-)$ let $k_1 := \sum_j [(w_j^+)_+ h_j^+ + (w_j^-)_+ h_j^-]$ and $k_2 := \sum_j [(w_j^+)_- h_j^+ + (w_j^-)_- h_j^-]$, where $x_\pm = \max(\pm x, 0)$; both are smooth and non-negative, and the total scalar source is $k_1 - k_2$. By Lemma 2.12(d), passed to the limit $k \to \infty$ by (i),

$$F_J(\zeta) \leq Z_J^a\Big(2\sum_j z_j f_j, -2k_2\Big)^{1/2} Z_J^a(0, 2k_1)^{1/2}.$$

The first factor is nonincreasing in J by (ii), hence at most its value at J_0; the second is bounded uniformly in J by $(\mathrm{W2})^{(a)}$ applied to $q = 2k_1$ (whose support lies at distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$ by the choice of J_0). By the convexity of $\ln F_J$ in the real coefficients and Lemma 2.12(b) (both inherited from $Z_{k,J}^a$ through (i)), the family $\{F_J\}_{J \geq J_0}$ is locally bounded on $\mathbb{C}^{3N}$. *Convergence on a set of uniqueness.* On $\Omega_+ := (\mathrm{i}\mathbb{R})^N \times (0,\infty)^{2N}$ the total sources are $(\mathrm{i}f, g)$ with f real and $g = \sum_j (w_j^+ h_j^+ + w_j^- h_j^-) \geq 0$, that is, they lie in $\mathcal{K}$; so $J \mapsto F_J$ is nondecreasing on Ω_+ by (ii) and bounded, hence convergent. Ω_+ is a non-empty open subset of the totally real subspace $(\mathrm{i}\mathbb{R})^N \times \mathbb{R}^{2N}$ of maximal real dimension, hence a set of uniqueness for entire functions on $\mathbb{C}^{3N}$. By Montel's theorem, $F_J \to F_\infty$ locally uniformly on $\mathbb{C}^{3N}$, with F_∞ entire [11]. Setting $w^+ = w$ and $w^- = -w$ gives the claim for the original family, since $\sum_j w_j h_j = \sum_j (w_j h_j^+ - w_j h_j^-)$; the limit is $\lim_J Z_J^a(\sum_j z_j f_j, \sum_j w_j h_j)$ and does not depend on the splitting. Positivity at real sources follows from Lemma 2.12(c), $Z_J^a(f,h) Z_J^a(-f,-h) \geq 1$, in the limit $J \to \infty$ (both factors are finite, so neither vanishes), and the last claim from (ii).

(iv) Φ^a is finite by the positivity in (iii), and convex as a pointwise limit of the convex functions $s \mapsto \ln Z_{k,J}^a(s)$. For J so large that $N_1(\bar{B}_\rho) \subset C_J$, and each k, the function $s \mapsto \ln Z_{k,J}^a(s)$ on $\mathcal{E}_\rho$ depends on s only through the finitely many sampled values $f_{c(p)}(e_\alpha, e_\beta)$, $h(x)$, continuous functionals of s in the C^∞ topology, and the finite-dimensional integral depends continuously on them (dominated convergence); so it is continuous, and $\Phi^a = \lim_J \lim_k$ of continuous functions is Borel. Lemma 2.15, applied on the open convex set $\mathcal{E}_\rho$ itself, gives the continuity.

(v) Let $b \in 5^{-k_0}\mathbb{Z}^3$ with $|b|_\infty \leq M$. For $k \geq k_0$, τ_b preserves Λ_k and S_{Λ_k}, and the substitution $A'_\alpha(x) = A_\alpha(x+b)$, $\phi'(x) = \phi(x+b)$ of Step 1 of the proof of Proposition 8.1 carries the comparison state on X_J to the comparison state on $(C_J - b) \cap \Lambda_k$ together with both sources, the single-site weights, W_ε, δm_ε^2 and a_k being site-independent; so $Z_{k,X_J}^a(\tau_b \cdot s) = Z_{k,(C_J-b)\cap\Lambda_k}^a(s)$, the right side being the functional of the comparison state on the translated box. As sets of sites, $C_{J-1} \cap \Lambda_k \subset (C_J - b) \cap \Lambda_k \subset C_{J+1} \cap \Lambda_k$. Let $s \in \mathcal{K}$ have all sources supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$ and let $J \geq J_0 + 1$; then, as in (ii), every plaquette and site at which s does not vanish lies in C_{J-1}. The first inequality of Lemma 2.11 (with $X = C_{J-1} \cap \Lambda_k$ in the state on $(C_J - b) \cap \Lambda_k$, then $X = (C_J - b) \cap \Lambda_k$ in the state on $C_{J+1} \cap \Lambda_k$; no site is killed, and the Dirichlet terms of each smaller state are the diagonal parts of the cut bonds) and Lemma 2.12(a) give

$$Z_{k,J-1}^a(s) \leq Z_{k,J}^a(\tau_b \cdot s) \leq Z_{k,J+1}^a(s) \qquad (k \geq k_0,\ J \geq J_0 + 1).$$

Let $k \to \infty$ by (i) (the translated family is as in $(\mathrm{W1})^{(a)}$ at J, since $C_{J-1} + b \subset C_J$ for $|b|_\infty \leq M$) and then $J \to \infty$ by (iii): $Z_\infty^a(\tau_b \cdot s) = Z_\infty^a(s)$ on $\mathcal{K}$. The identity extends to all complex coefficients of a family by the argument of (iii) (both sides are entire in the split coefficients and agree on the set of uniqueness Ω_+), to all $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$ by iteration over the translated families (each $b \in 5^{-k_0}\mathbb{Z}^3$ is a finite sum of vectors of $5^{-k_0}\mathbb{Z}^3$ with $|\cdot|_\infty \leq M$), and to all $b \in \mathbb{R}^3$ at real coefficients by (iv), since $b \mapsto \tau_b \cdot s \in \mathcal{E}_{\rho'}$ is continuous for $|b| \leq \rho' - \rho$ and $\bigcup_k 5^{-k}\mathbb{Z}^3$ is dense; then to complex coefficients by the identity theorem, $\mathbb{R}^{2N}$ being a set of

uniqueness. (King states translation invariance as [16, Thm. 2.3(i)]; his argument is not used.) At $h = 0$, where $(\mathrm{W1})^{(a)}$ is $(\mathrm{E}_C)^{(a)}$, this is Proposition 8.1, whose proof is the one just given with the scalar sources absent.

(vi) Lemma 16.4(a),(b) and (iii). □

Definition 16.7 (the full object in the scheme a)**.** Let the couplings e, λ, m, μ_0 be fixed in King's range, together with the block factor 5 and the large-block integer M; let C_J be the King cubes, $\Lambda_k = \varepsilon_k \mathbb{Z}^3$, $\varepsilon_k = 5^{-k}$, and let $\langle\cdot\rangle^{(a)}_{k,J}$ be the comparison cube of Definition 2.1, with the free-boundary mass counterterm $\delta m^2_{\varepsilon_k}$ of the setting section, one number per spacing. Let a be a W-admissible scheme ($(\mathrm{W})_a$). For a finite family ζ of curvature and scalar sources (Definition 2.4) the *full object in the scheme* a is

$$Z^a_\infty(\zeta) := \lim_{J\to\infty} \lim_{k\to\infty} Z^a_{k,J}(\zeta) :$$

the inner limit is the cube limit along the spacings 5^{-k} at fixed μ_0 and fixed J, taken for the J at which the sources have ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and the outer limit runs along the cube family $(C_J)_J$. Both limits exist, locally uniformly in the coefficients, by Theorem 16.6(i),(iii), and Z^a_∞ is the functional so named among the objects of the setting section, on the comparison convention. It is the object of the Wick-sector theorems of this part (Theorems 18.9 and 18.11) and of Form (A). At $h = 0$ the multiplier of Definition 16.2 is 1, so the object does not depend on the scheme: it is the iterated limit $\lim_J \lim_k Z_{k,J}(\sum_j z_j f_j, 0)$ of the comparison family, which exists under $(\mathrm{E}_C)^{(a)}$ (Lemma 7.7) and then equals King's printed state Z^K_∞ by Corollary 4.4(iii). The existence theorem (E_C) is posed on King's convention, and nothing is asserted about the fixed-cube limits of the comparison family; Corollary 4.4(ii)(γ) gives, at the cone points $z = \mathrm{i}t$,

$$Z^K_J(\mathrm{i}t) \le \liminf_k Z_{k,J}(\mathrm{i}t) \le \limsup_k Z_{k,J}(\mathrm{i}t) \le Z^K_{J+1}(\mathrm{i}t),$$

and both ends tend to $Z^K_\infty(\mathrm{i}t)$ as $J \to \infty$. Throughout this part, *King's printed state* means Z^K_∞ of Theorem 7.6, on King's convention ((E_C)); the curvature-sector statements of the part are posed for it.

Remark 16.8 (scope). The scheme changes Z^a_∞ only by the factor in Theorem 16.6(vi), which affects the one-point function of the Wick square; joint cumulants of order ≥ 2 do not depend on the scheme. The relation of the full object to infinite-volume limits along periodic boxes, King's expected identity [16, (2.24)], is never used.

16.5. **The general-source extension.** For scalar sources with $\int h \neq 0$ King's printed Wick ordering has no finite continuum limit beyond the curvature sector (Remark 16.1), and the object is the full object in a W-admissible scheme (Definition 16.7). For the reference scheme $z = a^\sharp$ the form of $(\mathrm{W})_a$ that carries the extension to such sources is the following.

Remark 16.9 (the reference scheme on the cone). $(\mathrm{W})_{a^\sharp}$ on the cone consists of the two clauses

($\mathrm{W}^\sharp$1) For every King cube C_J, every finite family of real 2-forms $f_j \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$, sampled at the plaquette centres as in Definition 2.4, and of scalar sources $h_j \in C^\infty_c(\mathbb{R}^3)$, all supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and every real (z, w), the limit $\lim_k Z^{a^\sharp}_{k,J}(\sum_j z_j f_j, \sum_j w_j h_j)$ exists in $(0, \infty)$, locally uniformly in complex (z, w).

($\mathrm{W}^\sharp$2) For every real $0 \le q \in C^\infty_c(\mathbb{R}^3)$ with $\sup q < s_*(e)$, $\sup_{J \ge J_q} \limsup_k Z^{a^\sharp}_{k,J}(0, q) < \infty$; here $s_*(e) > 0$ is a threshold fixed by the bulk bounds of the scheme, positive for small λ outside an exceptional set of at most two values of e^2 for given (λ, μ_0, m).

The first clause is Theorem 3.6 with the threshold on the positive scalar part removed, that is clause (a) of (W2) for the reference scheme; the second is the bound of clause (b) of (W2) below a threshold, in the form proved on the tube in Proposition 6.10 and Corollary 13.3. With these two clauses in place of $(\mathrm{W})_a$, and with $(\mathrm{K})_a$ and $(\mathrm{TIGHT})_a$ for scalar sources (Definitions 17.2 and 18.5), the arguments of this and the following two sections apply on the corresponding cone of sources. None of this enters Theorem 15.1.

17. The torus comparison

The rotation enters the rotation part through a single statement, the torus comparison, posed on King's tori for the two lattices Λ_k and $R^{-1}\Lambda_k$. This section states it, reduces it by an exact symmetry to one-sided bounds at one lattice spacing, and proves it along the lines of [16].

17.1. **The torus comparison.** Throughout, for a lattice Λ of spacing ε_k descending to a King torus D_J (Definitions 15.5 and 15.7), $Z^a_{\Lambda,D_J}(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)}\langle\Theta(\zeta)\rangle_{\Lambda,D_J}$ is the functional of the periodic state in the scheme a (Definition 2.4), $\mathsf{Z}_{\Lambda,D_J}(s)$ the unnormalised functional of Definition 15.8, and $\langle\cdot\rangle_{\mathfrak{c},\Lambda}$ the walled state of the configuration $\mathfrak{c} = (J', w)$.

Theorem 17.1 ((K_0): the torus comparison, curvature sector)**.** *Let the couplings be a point of King's range off E_W (Remark 15.2). For every $R \in \mathcal{R}_0$, every walled configuration $\mathfrak{c} = (J', w)$ of the class of Remark 18.6 (the conditions of Theorem 17.13, $J' \geq J_{\mathrm{pos}}$ among them) and every real smooth $D_{J'}$-periodic 2-form G, the limits $\lim_k \langle e^{F(G)}\rangle_{\mathfrak{c},\Lambda_k}$ and $\lim_k \langle e^{F(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ exist and are equal.*

The proof is given at the end of the next subsection, which reduces the statement to one-sided bounds at a single lattice spacing; the bounds themselves are proved in Section 17.3, and Remark 17.10 lists the inputs of that proof from the literature. With scalar sources the comparison is a property of a scheme a, which is not proved here (it is listed among the open problems, in the final section) and is used only by the optional extension of the Wick leg (Theorem 23.7):

Definition 17.2 ($(\mathrm{K})_a$: the torus comparison with scalar sources)**.** For every $R \in \mathcal{R}_0$, every King torus D_J, every finite family of real smooth D_J-periodic 2-forms G_j and functions h_j, and every $(z, w) \in \mathbb{R}^{2N}$, the limits

$$\lim_{k\to\infty} Z^a_{\Lambda_k,D_J}\Big(\sum_j z_j G_j, \sum_j w_j h_j\Big) \qquad \text{and} \qquad \lim_{k\to\infty} Z^a_{R^{-1}\Lambda_k,D_J}\Big(\sum_j z_j G_j, \sum_j w_j h_j\Big)$$

exist in $(0,\infty)$ and are equal.

(K_0) is stated on the sub-class of walled configurations described in Remark 18.6, which is the one on which the results that consume it use it, and for couplings off E_W (Remark 15.2), at which the large-field clause of King's upper bound at a wall is proved (Proposition 17.100 with Theorem 17.98 of the full version); the points of E_W are reached by right-continuity in the scalar mass (Theorem 20.15). Here $R^{-1}\Lambda_k$ carries the frame $(R^{-1}e_\alpha)$ and descends to D_J by Lemma 15.4(b); both lattices have spacing ε_k and carry the same a_k, $\delta m^2_{\varepsilon_k}$ and $\mathsf{W}_{\varepsilon_k}$. Since $\mathcal{R}_0 = \mathcal{R}_0^{-1}$ (Lemma 15.4(a)), the statement for all $R \in \mathcal{R}_0$ is the same as the statement for the lattices $R\Lambda_k$ used in [16]. Read with a single scalar function on all plaquettes, as printed in [16, (2.12), Thm. 2.4], the comparison is false already at $e = 0$ (Proposition H.2 of the full version); the 2-form reading, in which each lattice samples the source in its own frame, is consistent with King's proof, which treats the source components separately (Remark 15.6).

Lemma 17.3. (a) *Under $(\mathrm{K})_a$ the convergence holds locally uniformly on $\mathbb{C}^{2N}$, the two limits are the same entire function, and for every walled configuration $\mathfrak{c}$ and every (f, g) supported in one fundamental cell with $w = 0$ on $\operatorname{supp} g$, $\lim_k T^a_{\mathfrak{c},\Lambda_k}(f, g)$ and $\lim_k T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f, g)$ exist and are equal. Under (K_0) the same holds, for the configurations of its class, for $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ and $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ at complex z.*

(b) *$(\mathrm{K})_a$ implies (K_0), and $(\mathrm{K})_a$ implies $(\mathrm{K})_{a'}$ if $a'_k - a_k$ converges.*

(c) *If the statement of $(\mathrm{K})_a$, respectively (K_0), holds for $R = R_{12}(\theta_0)$, it holds for every $R \in \mathcal{R}_0$.*

(d) *Under $(\mathrm{K})_a$, for every King torus D_J and every $0 \leq q \in C^\infty_c$ supported in one fundamental cell, $C_{D_J}(q) := \sup_k Z^a_{\Lambda_k,D_J}(0, q) < \infty$.*

Proof. (a) As in the proof of Theorem 16.6(i): at fixed k the functionals are entire; by Lemma 2.12(b), whose proof applies verbatim to periodic states, $|Z^a(\zeta)| \leq Z^a(\operatorname{Re}\zeta)$ and $\ln Z^a$ is convex in

the real coefficients, so the convergence on $\mathbb{R}^{2N}$ gives local boundedness of the sequence on $\mathbb{C}^{2N}$; Montel's theorem gives convergent subsequences, whose limits agree on $\mathbb{R}^{2N}$, a set of uniqueness for entire functions on $\mathbb{C}^{2N}$ [11]; hence both sequences converge locally uniformly, to the same entire function. The walled ratios are quotients of such limits: with the family $(f; g, w)$ and the coefficients $(\mathrm{i}; 1, -1)$ in the numerator and $(0; 0, -1)$ in the denominator, $T^a_{\mathfrak{c},\Lambda}(f, g) = Z^a_{\Lambda,D_{J'}}(\mathrm{i}f, g - w)/Z^a_{\Lambda,D_{J'}}(0, -w)$ (Definition 15.7), and the denominators have positive limits. For (K_0) at complex z use $|\langle e^{zF(G)}\rangle| \le \langle e^{(\mathrm{Re}\, z)F(G)}\rangle$ and that (K_0) for all G includes all real multiples tG; the same argument applies. (b) $\langle e^{F(G)}\rangle_{\mathfrak{c},\Lambda} = Z^a_{\Lambda,D}(G, -w)/Z^a_{\Lambda,D}(0, -w)$, the wall being the scalar source $-w$ (Definition 15.7; the scheme multiplier cancels). For the second claim, the factors $e^{-\sum_j w_j(a'_k - a_k)\mathsf{S}_\Lambda(h_j)}$ have the same limit on both lattices, since $\mathsf{S}_{\Lambda_k}(h)$ and $\mathsf{S}_{R^{-1}\Lambda_k}(h)$ are Riemann sums of the smooth periodic function h over lattices descending to D_J and both tend to $\int_{\Gamma_J} h$. (c) Let $\sigma_2 = \mathrm{diag}(1, -1, 1)$. It preserves Λ_k and D_J, and changes the frame only by a sign, which does not change the periodic model (Lemma 2.13); moreover $\sigma_2 R_{12}(\theta_0)^{-1}\sigma_2 = R_{12}(\theta_0)$ (Lemma 15.4(a)). By Lemma 2.13 with $O = \sigma_2$ (for walled configurations Lemma 15.10(b), the wall becoming $\sigma_2 \cdot w$) the statement for $(R_{12}(\theta_0), \text{sources } s)$ is equivalent to the statement for $(R_{12}(-\theta_0), \sigma_2 \cdot s)$; as s ranges over all sources, so does $\sigma_2 \cdot s$. For the other planes, $R_{\mu\nu}(\pm\theta_0) = \Pi R_{12}(\pm\theta_0)\Pi^{-1}$ for a permutation matrix Π mapping e_1, e_2 to e_μ, e_ν; apply Lemma 2.13 with $O = \Pi$, which preserves Λ_k and D_J and permutes frames. (d) A convergent sequence with a finite limit is bounded. □

17.2. **Reduction to one-sided bounds at one lattice spacing.** King's proof of his Theorem 2.4 compares the lattice of spacing ε with the rotated lattice of spacing 5ε, and obtains the reverse inequality by "switching around the argument" [16, p. 337]. We compare the two lattices at the *same* spacing, and obtain the reverse inequality from an exact symmetry.

Lemma 17.4 (the involution)**.** *Let $R = R_{\mu\nu}(\pm\theta_0) \in \mathcal{R}_0$, let $\sigma := \sigma_\nu$ be the reflection $x_\nu \mapsto -x_\nu$, and put $S := R^{-1}\sigma$. Then $S^2 = 1$, $S\Lambda_k = R^{-1}\Lambda_k$ and $S(R^{-1}\Lambda_k) = \Lambda_k$, and S maps $D_J = \mathbb{R}^3/P_J$, $P_J := (50MJ\mathbb{Z})^3$, onto the torus $R^{-1}D_J := \mathbb{R}^3/R^{-1}P_J$. Both lattices descend to both tori, and for all real smooth D_J-periodic sources $s = (G, h)$,*

$$\mathsf{Z}_{R^{-1}\Lambda_k,D_J}(s) = \mathsf{Z}_{\Lambda_k,R^{-1}D_J}(S \cdot s), \qquad \mathsf{Z}_{\Lambda_k,D_J}(s) = \mathsf{Z}_{R^{-1}\Lambda_k,R^{-1}D_J}(S \cdot s),$$
$$\mathsf{S}_{R^{-1}\Lambda_k,D_J}(h) = \mathsf{S}_{\Lambda_k,R^{-1}D_J}(S \cdot h).$$

Proof. By Lemma 15.4(a), $\sigma R\sigma = R^{-1}$, so $S^2 = R^{-1}(\sigma R^{-1}\sigma) = R^{-1}R = 1$; $\sigma\Lambda_k = \Lambda_k$ gives $S\Lambda_k = R^{-1}\Lambda_k$, and then $S(R^{-1}\Lambda_k) = S^2\Lambda_k = \Lambda_k$. Also $\sigma P_J = P_J$, so $SP_J = R^{-1}P_J$. The lattice $R^{-1}\Lambda_k$ is invariant under $R^{-1}P_J$ trivially, and Λ_k is invariant under $R^{-1}P_J$ because $R^{-1}p \in 10MJ\,\mathbb{Z}^3$ for $p \in P_J$ (Lemma 15.4(b)). The map S carries the frame $(R^{-1}e_\alpha)$ to $(SR^{-1}e_\alpha) = (\sigma e_\alpha)$ and the frame (e_α) to $(R^{-1}\sigma e_\alpha)$; these differ from the frames of Λ_k and $R^{-1}\Lambda_k$ only by signs, which do not change the periodic models (Lemma 2.13). Lemma 2.13 with $O = S$, applied to the unnormalised integrals (the bijection of configurations preserves $d\nu$), gives the identities. □

Definition 17.5 ((F), (V), (C) and (F_0), (C_0): one-sided bounds at one spacing)**.** Let $R \in \mathcal{R}_0$ and $J \in \mathbb{N}$. Write $T_1 := D_J$, $T_2 := R^{-1}D_J$. The following are properties of the pair (R, J). (F): for $i = 1, 2$ there are real numbers $v_k(T_i)$ and $c_k(T_i)$, $k \ge k_0(J)$, independent of the fields and the sources, with $|c_k(T_i)| \le C\varepsilon_k^{-q}$ for some $q < \infty$ and $v_k(T_i)$ bounded, such that for every real smooth T_i-periodic source $s = (G, h)$,

$$\mathsf{Z}_{R^{-1}\Lambda_k,T_i}(s) \le \big(1 + \delta_k(T_i, s)\big)\exp\big(-v_k(T_i) - c_k(T_i)\,\mathsf{S}_{\Lambda_k,T_i}(h)\big)\,\mathsf{Z}_{\Lambda_k,T_i}(s), \qquad \lim_{k\to\infty}\delta_k(T_i, s) = 0.$$

(V): $v_k(T_1) + v_k(T_2) \to 0$. (C): $c_k(T_1) - c_k(T_2) \to 0$. (F_0) is (F) restricted to the sources with $h = 0$ and to the sources (G, h_0), where $h_0 = -w \le 0$ is a wall, in total form ($v_k \equiv 0$), on T_1 for these sources and on T_2 for their images under the involution S of Lemma 17.4, together with (C_0): $\limsup_k c_k(T_1) \le 0$ and $c_k(T_2) - c_k(T_1) \to 0$; it involves no Wick scheme. The sources $(G, -h_0)$, with a mass-decreasing wall, are not needed.

On each torus, (F) bounds the functional of the rotated lattice from above by that of the axis lattice. The rotated lattice is the one whose flow is King's standard block-spin flow in its own frame, and the axis lattice is the one run through King's hybrid flow; this is the direction of [16, (4.7)–(4.9)]. Since smooth periodic functions have rapidly convergent Riemann sums, $|\mathsf{S}_{\Lambda,T}(h) - \int_\Gamma h| \le C_N(h)\varepsilon_k^N$ for every N and every lattice Λ of spacing ε_k descending to T, Γ a fundamental cell; with $N > q$ the error $c_k(\mathsf{S}_{\Lambda,T}(h) - \int h)$ tends to 0.

Theorem 17.6 (closing at one spacing)**.** *Let $R \in \mathcal{R}_0$, $J \in \mathbb{N}$, and assume* (F)*. Write $v = v_k(T_1)$, $v' = v_k(T_2)$, $c = c_k(T_1)$, $c' = c_k(T_2)$, and $\rho_k(s) := \mathsf{Z}_{R^{-1}\Lambda_k,D_J}(s)/\mathsf{Z}_{\Lambda_k,D_J}(s)$.*

(a) *For every real smooth D_J-periodic source $s = (G,h)$ (for this item* (F) *is needed at s and at $S\cdot s$ only), $e^{v'+c'\int h}(1+o(1)) \le \rho_k(s) \le e^{-v-c\int h}(1+o(1))$ as $k\to\infty$.*

(b) *$v + v' \le o(1)$. If* (V) *holds, then $c + c' \to 0$.*

(b′) *If, in place of the standing assumption* (F)*,* (F) *holds in total form ($v = v' = 0$) on T_1 for the sources $s_t = (tG, h_0)$, $t \in \mathbb{R}$, with $\int h_0 < 0$, and on T_2 for their images under the involution S of Lemma 17.4, with c, c' the constants of that hypothesis, and $\limsup_k(c + c') \le 0$, then $c + c' \to 0$ and $\rho_k(s_t)e^{c\int h_0} \to 1$ for every real t; in particular the identity in (c) holds for $s_0 = (0, h_0)$ and $s_1 = (tG, h_0)$.*

(c) *If* (V) *holds, then $\rho_k(s) = e^{-v-c\int h}(1+o(1))$ for every s. Consequently, for sources s_0, s_1 with $\int h_0 = \int h_1$,*

$$\frac{\mathsf{Z}_{R^{-1}\Lambda_k,D_J}(s_1)}{\mathsf{Z}_{R^{-1}\Lambda_k,D_J}(s_0)} = \frac{\mathsf{Z}_{\Lambda_k,D_J}(s_1)}{\mathsf{Z}_{\Lambda_k,D_J}(s_0)}\,(1+o(1)).$$

(d) *If* (V) *and* (C) *hold, then $c, c' \to 0$, and the identity in (c) holds for all s_0, s_1.*

No convergence of either family and no Wick scheme is assumed.

Proof. (a) The upper bound is (F) on T_1, with $c\,\mathsf{S}_{\Lambda_k,D_J}(h) = c\int h + o(1)$. Apply (F) on T_2 to the T_2-periodic source $S\cdot s$ and use Lemma 17.4: $\mathsf{Z}_{R^{-1}\Lambda_k,T_2}(S\cdot s) = \mathsf{Z}_{\Lambda_k,D_J}(s)$, $\mathsf{Z}_{\Lambda_k,T_2}(S\cdot s) = \mathsf{Z}_{R^{-1}\Lambda_k,D_J}(s)$ and $\mathsf{S}_{\Lambda_k,T_2}(S\cdot h) = \mathsf{S}_{R^{-1}\Lambda_k,D_J}(h) = \int h + O(\varepsilon_k^N)$. This is the lower bound. (b) The interval in (a) is non-empty, so $(v+v') + (c+c')\int h \le o(1)$ for every s. Apply this to (G,h) and to $(G,-h)$ and add: $v + v' \le o(1)$. The same two inequalities give $|c+c'|\,|\int h| \le o(1) - (v+v')$; under (V) the right side tends to 0, and choosing h with $\int h \ne 0$ gives $c + c' \to 0$. (b′) By (a) at s_t the interval is non-empty, so $(c + c')\int h_0 \le o(1)$; as $\int h_0 < 0$, $c + c' \ge -o(1)$, and with $\limsup_k(c + c') \le 0$ this gives $c + c' \to 0$. The two ends of the interval in (a) then differ by a factor $e^{o(1)}$, so $\rho_k(s_t)e^{c\int h_0} \to 1$; dividing the statement for s_1 by the one for s_0 cancels the prefactor. (c) Under (V) and (b), the ratio of the two ends of the interval in (a) tends to 1. Dividing the statement for s_1 by the one for s_0, the prefactors cancel when $\int h_1 = \int h_0$. (d) Now $c + c' \to 0$ and $c - c' \to 0$. □

The proof of (b) uses (F) at the images under $h \mapsto -h$; (b′) does not. In particular, under (F_0) the conclusions (a) and (b′) and the identity of (c) hold for the sources (tG, h_0) with $\int h_0 < 0$ (every wall $w \ne 0$), and (a) in total form and the identity of (c) hold, with no condition on c, c', for the sources with $h = 0$ (both ends of the interval in (a) then tend to 1, so $\rho_k(s) \to 1$).

Corollary 17.7 (walled curvature ratios)**.** *Assume* (F_0) *for R and J'. Then for every walled configuration $\mathfrak{c} = (J', w)$ and every real smooth periodic 2-form G, $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k} - \langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k} \to 0$ locally uniformly in $z \in \mathbb{C}$. If moreover $\lim_k \langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ exists for real t, then* (K_0) *holds for $\mathfrak{c}$ and G.*

Proof. With $h_0 = -w$, $\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda} = \mathsf{Z}_\Lambda(tG, h_0)/\mathsf{Z}_\Lambda(0, h_0)$ (the Wick constants cancel; Lemma 15.10(c)), a ratio of two sources with the same $\int h$. By Theorem 17.6(b′) (at $w = 0$ by (a) in total form), $u_k(t) := \langle e^{tF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}/\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k} \to 1$ for real t. The walled states are BFS measures (Definition 15.7), so by Lemma 2.9(a) and, for the lower bound, Jensen's inequality with $\langle F(G)\rangle = 0$ by the symmetry $(A,\vartheta) \mapsto (-A,-\vartheta)$, $1 \le \langle e^{tF(G)}\rangle \le e^{t^2\sigma_\Lambda/2}$ for real t, where $\sigma_\Lambda = \langle v, Cv\rangle$ is the variance of $F_\Lambda(G) = \langle v, A\rangle$ under the free periodic photon on the lattice; since the periodic photon form is at least μ_0^2 times the ℓ^2 norm, $\sigma_\Lambda \le \mu_0^{-2}\|\delta_\varepsilon G\|^2$, δ_ε the lattice codifferential,

and this is bounded uniformly in k (a Riemann sum for $\|d^*G\|^2_{L^2(\Gamma_{J'})}$). Hence the difference of the two functions tends to 0 on $\mathbb{R}$, and it is bounded on every strip $|\operatorname{Re} z| \le r$ uniformly in k, because $|\langle e^{zF}\rangle| \le \langle e^{(\operatorname{Re} z)F}\rangle$. Vitali's theorem [11] gives locally uniform convergence to 0 on $\mathbb{C}$. The last statement follows by the same argument. □

Corollary 17.8 (the full comparison)**.** *Let a be a scheme with $|a_k| \le C(1+k)$. Assume* (F)*,* (V) *and* (C) *for $R = R_{12}(\theta_0)$ and every J, and that for every King torus and every finite family of real smooth periodic sources the limit $\lim_k Z^a_{\Lambda_k,D_J}$ exists in $(0,\infty)$ for real coefficients. Then* $(\mathrm{K})_a$ *holds.*

Proof. $Z^a_{\Lambda,D_J}(s) = e^{-a_k \mathsf{S}_\Lambda(h)} \mathsf{Z}_\Lambda(s)/\mathsf{Z}_\Lambda(0)$, and the multipliers of the two lattices differ by $e^{-a_k(\mathsf{S}_{R^{-1}\Lambda_k}(h) - \mathsf{S}_{\Lambda_k}(h))} = 1 + O(k\varepsilon_k^N)$. Theorem 17.6(d) with $s_0 = 0$ gives $Z^a_{R^{-1}\Lambda_k,D_J}(s)/Z^a_{\Lambda_k,D_J}(s) \to 1$ for every real source, which with the assumed convergence gives $(\mathrm{K})_a$ for $R_{12}(\theta_0)$; Lemma 17.3(c) gives the other generators. □

Remark 17.9 (what this subsection supplies). Lemma 17.4, Theorem 17.6 and its corollaries are a reduction of the torus comparison to the one-sided bounds (F), (V), (C); the bounds themselves come from the renormalisation-group analysis. In the curvature sector, what Corollary 17.7 needs is (F_0) (with its clause (C_0)) together with the existence of the axis-lattice limits, and these are what the proof of (K_0) along the lines of [16] produces (Remark 17.10). In that proof King's torus bound [16, (4.2)] is of no use, because at a non-zero wall it is vacuous; its place is taken by the large-field clause of King's upper bound [15, Thm. 3.1, (3.4)] at blocking factor 5, in the form of Proposition 17.100 at the wall, and at a mass-increasing wall by a polynomial lower bound for the walled partition function. Corollary 17.8 concerns scalar sources and is not used in the curvature sector (Theorem 15.1).

Proof of Theorem 17.1. Let the couplings be off E_W, as in the statement. By Lemma 17.3(c) it suffices to treat $R = R_{12}(\theta_0)$. Fix a walled configuration $\mathfrak{c} = (J', w)$ of the class of Remark 18.6 and a real smooth periodic 2-form G. By Corollary 17.7 the statement follows from (F_0) for R and J' together with the existence of $\lim_k \langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ for real t. The existence of the axis-lattice limit is King's torus theorem [15, Thm. 2.1] with the wall as scalar source, read on the normalised ratio, in which the vacuum and the h-linear constants cancel (Theorem 17.127). The bounds (F_0) on the two tori T_1, T_2, in total form and with (C_0), come from the comparison, at equal spacing, of the effective actions of King's standard block-spin flow on the rotated lattice and of the hybrid flow on the axis lattice; they are proved in Section 17.3, and the proof is completed there (Theorem 17.13 and the paragraph after Proposition 17.140). □

Remark 17.10 (the inputs of the proof of (K_0)). (K_0) is the corrected form of King's torus theorem [16, Thm. 2.4]: 2-form sources in each lattice's own frame, equal spacing, the free-boundary counterterm, walls allowed. The proof of Section 17.3 follows King's lines. Passive rotations and Lemma 17.4 reduce the comparison to the two lattices on T_1, T_2. The effective actions of two block-spin flows, the standard flow of $Y_k = R^{-1}\Lambda_k$ in its own frame and a hybrid flow of $X_k = \Lambda_k$ whose first step averages over rotated blocks, are compared at equal spacing on King's small-field domain, in total form (field-independent constants included). Inserting the comparison into King's upper bound [15, Thm. 3.1] for the rotated lattice and into the hybrid analogue of [16, Prop. 4.1] for the axis lattice gives a bound of type (F) on each torus, the large-field term absorbed by a polynomial lower bound on the partition function; Theorem 17.6 closes, with (V) automatic because the comparison is in total form. The inputs: the two-flow comparison bound in per-flow normal form (Proposition 17.39, proved as Theorem 17.40), the h-linear field-independent families excepted, which rests on Bałaban's renormalisation bound [3, Prop. 1, (1.33)] for the renormalised families of [15, §3.5] taken on each flow by itself; decay and regularity of the block-spin operators, [15, Thm. 3.3], [4], and of the hybrid kernels; King's upper bound, also on the torus skew in the rotated frame; the large-field clause of that bound at a wall, which uses the lighter scalar mass M_1 of Theorem 17.98 of the full version, off E_W (Remark 15.2); the hybrid lower bound; the polynomial lower bound [1, (1.14)] at zero wall, and at a mass-increasing wall the floor shift Theorem 17.120 from [1, (1.14)] and [2, (1.3)] at

higher King points; the existence of the walled values on the axis lattice, [15, Thm. 2.1] with the wall as scalar source, read on the normalised ratio; Reisz's theorems [22, 23] in a corrected three-dimensional form; the local limit law [17]; the bridge between King's torus counterterm [15, (3.33)] and the free-boundary one [16, p. 330]. The estimates of Bałaban's papers that the proof shares with the proof of (E_C) (Section 5), stated in those papers without a printed argument, are among them [1, (2.33), (2.39)–(2.42), (3.20)–(3.21), (3.25), (3.34)–(3.36)] and the frame proof of its (1.14), [2, Prop. 2.6, Lemma 2.7, (3.5), (3.7)–(3.8)], [3, Prop. 1 with (1.33)], [4, Lemma 2.4, (1.8), (1.15), (1.16), (1.20)], [15, Thm. 3.1, Props. 3.7–3.9, Lemma 4.5], the weakened-element step of the proof of [15, Prop. 3.6] (p. 665), and the sentence of [16] that Bałaban's results apply to both models; King's treatment of the sources ([15, pp. 665, 677]) is proved in the full version, and the mass clause of [2, Prop. 3.1] is cited under it as in Remark 22.2. Of the degree properties of the renormalised families, both flows use the strict positivity of the degrees after Bałaban's rewriting in the componentwise form of [3, (3.5)], in the form stated in the proof of Corollary 17.42 of the full version (Proposition K.537(e) of the full version): for the standard flow of Y_k it is the degree count of Theorem K.167 of the full version for King's lines, and for the hybrid flow of X_k it is the same count for the hybrid lines (for the hybrid flow of X_k, with the lines of the flow at $A = 0$ and the vertices at $B = 0$, every connected source-free point-free component S of a prefix of a non-vanishing graph of a renormalised family of order $\leq \bar{n}$ in per-flow normal form whose plain degree $D(S)$ is ≤ 0 is rewritten, finest first, with the output shapes (R0)–(R2) of Theorem K.167 of the full version (the plain and current degrees D, D' as in the renormalised degree count of the full version, ρ_{j_0} the spacing of the finest slice j_0 of S, $\alpha \in (0,1)$ the Hölder exponent of [15, (3.8)]): the resummed coefficients $c_a(F_S)$ of its inside family vanish exactly, by the counterterm of the flow per scale assignment (Lemma 17.31 of the full version), by charge parity, by the inversion about the coarse site or by the Ward identity of the full fixed-order two-photon function of the flow and of its free-lattice replacement (Proposition K.166(v) of the full version), or are bounded by $C\rho_{j_0}^{D(S)+|a|}$, the coefficients with $D(S)+|a| \leq 0$ vanishing exactly, so that $D'(S) \geq \alpha$ and the current degree of no other component is lowered; hence every non-vanishing source-free point-free ultraviolet prefix component of the hybrid flow has current degree $D' \geq \min\{\alpha, 1\}$ after the rewriting). Statements false as printed are replaced; the sense in which each is false, or is not available in its printed form, and the corrected form proved here are as follows (details in the full version). [15, Thm. 3.5, Prop. 3.6, (3.42)] are stated for every graph of King's expansion; they fail on the order-e^2 members of the h-linear field-free family $\mathcal{T}_h$ of Lemma 17.61 of the full version, whole graphs of degree 0 without a counterterm in King's scheme, whose per-site kernel grows like the number of ultraviolet steps and whose two depths differ by the drift of Remark 16.1 (the full version); and under King's rule of one partition of unity per vertex [15, (3.36)] the per-tuple bound (3.38) is not available for the photon two-point classes, whose members have different numbers of vertices (the full version). On every renormalised family whose non-vanishing ultraviolet ordered subgraphs have positive degree after Bałaban's rewriting and whose infrared lines lie in mass-counterterm units, the displays hold with King's power of $\ln(L^k\varepsilon)^{-1}$ [15, p. 666], written $(1+k-j)^{p_0}$; in this form they are, on the cube with the localisation by centres of Lemma B.31 of the full version, Lemma 6.1, Theorem B.41 of the full version, Corollary B.40(b) of the full version and Theorem B.44 of the full version, and on the torus Corollary 17.42 of the full version and Proposition 17.134 of the full version, the degree-zero family being bounded separately there with the factor $(1+r)$. [15, (4.36)–(4.37)] treat the block average Q^*Q as a Fourier multiplier, which it is not; (4.33) itself holds (Proposition H.3 of the full version). [15, (3.78)] is used only for the mass-counterterm units, attached at one vertex (the full version), where it holds for the units of order two and the sunset in every ordering, and for every mass-counterterm (δm^2) unit of order four and every term of its expansion in slices, after the rewriting of its divergent sub-pieces at infrared indices, in every admitted ordering of the lines, every placement of the derivatives and every assignment of indices, the current degree of the whole graph is at most the current degree of every prefix containing the ultraviolet lines ($D(H_m) \leq D(H_t)$ for every $m_1 \leq t < m$, H_t the prefix of the first t lines, m_1 the number of ultraviolet lines and H_m the whole graph) (the full version,

Remark K.168(a) of the full version); it fails for some vacuum graphs of E_1 whose infrared lines lie outside the units (the full version), field-independent constants which cancel in every normalised functional and enter only the comparison of the vacuum constants of the two flows, Theorem 17.58(e) of the full version, which uses (3.78) on the units only. In [15, (4.21)] a mass term is dropped: the factor $|p'|^2$ reads $|p'|^2 + m^2(L^k\varepsilon)^2$, and the corrected ratio has the decay $C|p' + l + m|^{-2}$ that [15, (4.23)] uses; (4.21) enters this paper only through King's proof of the fourth line of [15, (3.71)], taken on the torus in Lemma K.542 of the full version (the full version). The remaining items are the count (3.45) and the first inequality of (3.46) of [2] and the naturalness clause of Reisz's theorem.

Remark 17.11 (the content of (K_0)). (K_0) is [16, Thm. 2.4] with the corrections of Proposition H.2 of the full version and Remark 16.1, for arbitrary smooth periodic sources, walls included, and with free-boundary counterterms. By Corollaries 17.7 and 17.8 its two-lattice content is, in the curvature sector, (F_0) (with its clause (C_0)), and with scalar sources (F), (V) and (C); its one-lattice content is the existence of the torus limits on the axis lattice. The rotation enters the whole argument of the rotation part only through Theorem 17.1.

Remark 17.12 (what the torus comparison alone does not give). Given (K_0), the walled values of a configuration $\mathfrak{c}$ on D_J agree on the two lattices, and by passive covariance (Lemma 15.10(b)) the walled value of $R\cdot\mathfrak{c}$ on the rotated torus $R\,D_J$ for the axis lattice equals the walled value of $\mathfrak{c}$ on D_J for the rotated lattice. Rotation invariance of a limit along tori would therefore follow from (K_0) if the limit did not depend on the shape and orientation of the tori, which is the content of King's expected identity [16, (2.24)] between torus and cube limits. That identity is a uniqueness statement for the infinite-volume limit, and we do not use it. The walls of the next section replace it by a one-sided statement about one lattice, (TIGHT_0) (Theorem 18.4), which is proved from the heavy-shell estimates (Theorem 20.14, Theorem 19.10).

17.3. **Proof of the torus comparison.** This subsection proves the one-sided bounds (F_0) and (V) of Definition 17.5, in the total form in which the vacuum constants are absent and the source-linear constants tend to zero, for the walled configurations of the class of Remark 18.6; with the existence of the axis-lattice limits (Theorem 17.127) and Corollary 17.7 this completes the proof of Theorem 17.1. Here the principal statements are reproduced with their numbers, each proof is replaced by its idea, and every auxiliary statement is cited by its number in the full version, which contains the complete proofs. Throughout, $R \in \mathcal{R}_0$ is fixed, $5R$ and $5R^{-1}$ are integer matrices (Lemma 15.4(a)), and

$$X_k := \Lambda_k = \varepsilon_k\mathbb{Z}^3 \quad \text{(the axis lattice)}, \qquad Y_k := R^{-1}\Lambda_k \quad \text{(the rotated lattice)}, \qquad Y_{k-j} := R^{-1}(5^j\varepsilon_k\mathbb{Z}^3),$$

so that $Y_{k-j} \subset X_k \cap Y_k$ for $j \geq 1$; the frame of X_k is (e_α) and that of Y_k is $(R^{-1}e_\alpha)$; the tori are $T_1 = D_J = \mathbb{R}^3/P_J$, $P_J = 50MJ\,\mathbb{Z}^3$, and $T_2 = R^{-1}D_J$, and $S = R^{-1}\sigma_\nu$ is the involution of Lemma 17.4. Here r denotes the number of block-spin steps.

Theorem 17.13 (the torus comparison at one configuration)**.** *Let the couplings be a point of King's range off E_W (Remark 15.2) and $\mathfrak{c} = (J, \psi, n)$ a walled configuration with $J \in 5^{n_*}\mathbb{N}$, $J \geq \max(J_0(\delta_r), J_{\mathrm{pos}})$, $n > 0$, the wall condition $n\mathsf{S}_{X_k}(\psi) > \Delta_*|T_1|/2$ for every k (Δ_* the constant of Proposition 17.126(b) of the full version, which does not depend on T) and block alignment (the source box and the torus unions of blocks of the final scale); here n_* is the number of large-field scales of the flow of Y_k, fixed by the couplings, J_{pos} is the threshold of Theorem 17.98 of the full version, and $J_0(\delta_r)$ the non-wrapping threshold of Lemma 17.124 of the full version at the coarsest spacing δ_r of the flow (a condition void for large k, since $\delta_r \to 0$). The class carries in addition the cap $\varepsilon_0^{\mathrm{Ba}} \leq \varepsilon_{\mathrm{sys}}$ of Lemma K.518 of the full version on Bałaban's stopping scale, $\varepsilon_{\mathrm{sys}}$ being the minimum over $R \in \mathcal{R}_0$ of the thresholds defined there (one number, the same for every R); where the proof uses, in Theorem 17.120 on the skew pairs (which Proposition 17.139 uses), the ε-uniform bounds of Theorem K.506 of the full version on a skew torus, which run the flow to Bałaban's final spacing, it needs the non-wrapping condition at that spacing, and under the cap that condition holds with the threshold 1 for every J of the class. Then for every real smooth P_J-periodic 2-form G:*

(i) $\langle e^{zF(G)}\rangle_{\mathfrak{c},Y_k} - \langle e^{zF(G)}\rangle_{\mathfrak{c},X_k} \to 0$ *as* $k\to\infty$, *locally uniformly in* $z\in\mathbb{C}$; *in particular all moments* $\langle F(G)^m\rangle$ *differ by* $o(1)$;
(ii) *both families converge, locally uniformly on* $\mathbb{C}$, *to the same entire function* $\mathcal{W}_{\mathfrak{c}}(zG)$, *and all moments converge to the same limits;*
(iii) *the same holds with* R *replaced by any* $R'\in\mathcal{R}_0$.

At $e=0$ *the statement holds for every walled configuration with no condition:* $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda} = e^{z^2 q_\Lambda(G)/2}$ *exactly (Lemma 17.22), and* $q_{X_k}(G), q_{Y_k}(G)\to q(G)$ *(Lemma 17.23 of the full version).*

The three conditions on $\mathfrak{c}$ are those of Remark 18.6, and the configuration produced by Theorem 19.10 satisfies them (Remark 19.11(iii), Theorem 20.14; on the skew pairs it also meets the condition at Bałaban's final spacing of Theorem K.506 of the full version at the raised wall height). The proof is assembled in Proposition 17.139 and the paragraph after it, from the comparison of the effective actions of two block-spin flows at equal spacing (Theorem 17.63), King's upper bound for the rotated lattice (Proposition 17.100), the hybrid lower bound (Proposition 17.117) and a polynomial lower bound at the wall (Theorem 17.120), closed by Theorem 17.6. Both period lattices lie in every block lattice of both flows (Lemma 17.14 of the full version).

The two block-spin flows.

Definition 17.15 (the standard flow of Y_k and the hybrid flow of X_k)**.** Fix k and $r\le k$; $\delta_r := 5^r\varepsilon_k$ is the physical spacing of the coarse lattice Y_{k-r}. In the unit scale of level r write $\eta := 5^{1-r}$ and $\eta' := \eta/5 = 5^{-r}$. Blocks are centred at every level (Convention 17.17 of the full version).

- *Standard flow on* Y_k (flow Y): Bałaban's and King's block-spin transformation [1, (2.1)–(2.11)], [15, (2.10)–(2.17)] in the frame of Y_k: centred cubes of 125 points, $Y_{k-j+1}\to Y_{k-j}$, with contours made of fine Y_k-bonds in the axial order f_1,f_2,f_3 of the frame inside each block.
- *Hybrid flow on* X_k (flow X): the first step averages over the centred tiles $y+\varepsilon B_0$, $y\in Y_{k-1}$, with
$$B_0 := \mathbb{Z}^3\cap R^{-1}\big(-\tfrac{5}{2},\tfrac{5}{2}\big)^3$$
(125 sites, a complete residue system of $\mathbb{Z}^3/5R^{-1}\mathbb{Z}^3$, Lemma 17.20 of the full version), onto Y_{k-1}; this is [16, (4.3)–(4.4)], with the photon average read frame-covariantly, $(\hat{Q}A)_\alpha(y) = \frac{1}{125}\sum_{x\in B(y)} f_\alpha\cdot A(x)$, $f_\alpha = R^{-1}e_\alpha$ (Remark 17.19 of the full version). Steps $2,\dots,r$ are the standard flow's steps $Y_{k-1}\to Y_{k-r}$ with the same blocks, except that every rotated bond of Y_{k-1} occurring in a contour is replaced by a fixed path of fine X_k-bonds ([16, p. 336]: "succeeding steps of the renormalization transformation in the usual way, that is without rotating the blocks").
- *Contours, rotation-paired.* Let $\Gamma^{\mathrm{st}}_{z,w}$ be King's coordinate staircase from z to w [15, (2.12)] and $\Gamma^{(r)}_{y,x}$ his composite contour [15, (2.11)], which the standard flow uses in Y's frame. The legs of a Y-frame staircase between two centres of Y_{k-1}-blocks are concatenations of the η-bonds $(u,u+5\varepsilon f_\alpha)$, $u\in Y_{k-1}$. The hybrid composite contour from the level-r block centre y to $x'\in X_k$ is the same staircase with every η-bond $(u,u+5\varepsilon f_\alpha)$ replaced by the *fixed* fine axis staircase $u+\varepsilon\gamma_\alpha$ (and the reversed bond by its reverse), where, for $R=R_{12}(\theta_0)$ and $f_\alpha = R^{-1}e_\alpha$ with $5f_1=(4,3,0)$, $5f_2=(-3,4,0)$,
$$\gamma_1:\ e_1,e_2,e_1,e_2,e_1,e_2,e_1\ \text{(to }(4,3,0)\text{)},\qquad \gamma_2:\ e_2,-e_1,e_2,-e_1,e_2,-e_1,e_2\ \text{(to }(-3,4,0)\text{)},\qquad \gamma_3:\ e_3^5,$$
followed, inside the first-step tile, by a fine axis path of at most 5 bonds (Lemma 17.20(ii) of the full version), chosen once as a function of the displacement only and the same in every tile, so that the block and contour system of the flow is invariant under Y_{k-1}, in particular under the periods of both tori; this is the first-step contour of [16, (4.4)]. (For the other generators conjugate by the permutation and sign matrices of Lemma 17.3(c).) The gauge field along a composite contour is read by King's convention [15, (2.19)] (Convention 17.17(ii) of the full version).

Both flows use King's coefficients $a_j = a(1-5^{-2})/(1-5^{-2j})$ [15, p. 653], take r steps and land on the same coarse lattice Y_{k-r}/P with the same unit-scaled fields (A_r, ϕ_r) and the same small-field indicator χ_r [15, (3.2)]. The effective actions are written so that

$$\mathsf{Z}_{X_k,T}(s) = \int (dA_r)(d\phi_r)\, e^{-\hat{s}_T^{(r)}(A_r,\phi_r;s)}, \qquad \mathsf{Z}_{Y_k,T}(s) = \int (dA_r)(d\phi_r)\, e^{-S_T^{(r)}(A_r,\phi_r;s)}$$

exactly, every c-number kept inside: this is [15, (3.14), (3.90)] with E_0 inserted from [15, (3.89)].

Centred blocks, paired contours, the tile (Lemmas 17.16, 17.20 and 17.21, Convention 17.17, and Remark 17.19 of the full version). Blocks centred at the sites of the next coarse lattice make the inversion about any site of the coarsest lattice an exact symmetry of both flows, which removes every first-moment term from both effective actions; one translation of the fine lattice by one of its own vectors carries King's corner-rooted blocks of every level, and with them his construction, its contours and every estimate printed for it, onto the centred cubes with the same constants (Proposition R.54 of the full version), the transported contours being rooted at a corner of the centred cube, not at its centre, so that the contour half of the convention is a separate statement (Hypothesis R.55 of the full version); that the printed estimates use the blocks and the contours only through the tiling, nesting, bounded-diameter and translation properties, which centred blocks share, and through the contours staying inside their blocks with bounded length, is the insensitivity of the fluctuation-integral estimates of [1, 2, 3, 4] to the root of the block, for in-block contour families of bounded length with the central symmetry of the fixed staircases (Convention 17.17(i) of the full version); the gauge field along a composite contour is read in King's convention [15, (2.19)], and the photon average frame-covariantly, which lets a constant background pass through the step. Each fixed staircase lies within 0.6ε of the rotated bond it replaces, so every piece of the hybrid contour stays inside its block and the two composite contours bound, with the matching segment, a surface of area $\le 1.5\eta' + 33\eta'^2$; the tile B_0 is a complete residue system of $\mathbb{Z}^3/5R^{-1}\mathbb{Z}^3$ with 125 points, $B_0 = -B_0$, and $\hat{Q}\hat{Q}^* = \mathbf{1}$, $Q(A)Q(A)^* = \mathbf{1}$ for every bond field A, so the first hybrid step is King's transformation with the tile in place of the cube; and by Stokes' theorem the phases $U(A(\Gamma))$ of the two flows at matched endpoints differ by at most $C\eta'\|dA\|_\infty$. The pairing is a condition, not a convenience: if the enclosed loops did not shrink with the scale, the flux would be of the order of the Wilson-line phase at the coarse momenta at which the comparison is made, and Theorem 17.57 would fail.

The Cameron–Martin transfer of the curvature source.

Lemma 17.22 (Cameron–Martin transfer)**.** *Let Λ be a lattice of spacing $s = \varepsilon_k$ with frame (f_α) descending to a torus T, γ_Λ the normalised Gaussian photon measure of the action, with covariance $C_\Lambda = (-\Delta_\Lambda + \mu_0^2)^{-1} \otimes \mathbf{1}_3$ for the pairing $\langle u, v\rangle_\Lambda = \sum_x s^3 \sum_\alpha u_\alpha(x) v_\alpha(x)$, and write $\mathsf{Z}_{\Lambda,T}(G,h) = \int d\gamma_\Lambda(A)\, e^{F_\Lambda(G)} W_\Lambda(A;h)$ with W_Λ the matter integral, which depends on A only through the bond phases. Then $F_\Lambda(G) = \langle A, \ell_\Lambda\rangle_\Lambda$ with $\ell_{\Lambda,\beta} := \sum_{\alpha\neq\beta} \partial_\alpha^* G_{\alpha\beta}$, $G_{\alpha\beta}(x) := G_{c(x;\alpha\beta)}(f_\alpha, f_\beta)$, $\partial_\alpha^* u(x) := s^{-1}(u(x - sf_\alpha) - u(x))$, and for real z*

$$\mathsf{Z}_{\Lambda,T}(zG,h) = e^{z^2 q_\Lambda(G)/2}\, \mathsf{Z}_\Lambda[z a_\Lambda](T;h), \qquad q_\Lambda(G) := \langle \ell_\Lambda, C_\Lambda \ell_\Lambda\rangle_\Lambda, \quad a_\Lambda := C_\Lambda \ell_\Lambda,$$

where $\mathsf{Z}_\Lambda[b](T;h) := \int d\gamma_\Lambda(A)\, W_\Lambda(A+b;h)$. Counterterms, Wick constant and wall are untouched. At $e = 0$, W_Λ does not depend on A, so $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda} = e^{z^2 q_\Lambda(G)/2}$ exactly.

Idea of the proof, and what the transfer does (Lemmas 17.23, 17.24 and 17.26 and Remark 17.25 of the full version). Summation by parts writes $F_\Lambda(G) = \langle A, \ell_\Lambda\rangle$, and completing the square in the photon Gaussian moves the source into the matter coupling. The Gaussian factor is frame-free, $|q_\Lambda(G) - q(G)| \le C(\mu_0) s^2 \|G\|^2_{H^3(T)} + C_N(G) s^N$, and the shift field a_Λ is the sample of one smooth co-exact 1-form $a^{\mathrm{c}} = (-\Delta + \mu_0^2)^{-1} d^* G$ up to $O(s^2)$, so that in King's unit scaling $z a_\Lambda$ and $A^{(r)} + z a_\Lambda$ are regular in the sense of [15, Def. 3.2]. After the transfer the model is King's with a scalar source and a fixed smooth background in the matter coupling only, which enters exactly where King's coarse background enters ([15, (3.82)–(3.87)]), and no frame-paired source vertex [15, (3.20)] appears. Finally $|\mathsf{S}_\Lambda(h) - \int_T h| \le C_N(h) s^N$ for every lattice $\Lambda = O(s\mathbb{Z}^3) \supset P$, with $C_N(h)$ independent of O.

The number of steps, the effective actions, the matching. Summary (Lemmas 17.27, 17.30 and 17.31 and Definitions 17.28 and 17.29 of the full version). The number of steps is $r = r(k)$ with $k - r = \lfloor \varrho k \rfloor$ for a fixed $0 < \varrho < \min\{\gamma/(\gamma+\beta), \frac{1}{9}\}$, γ the smallest of the exponent of [15, Thm. 3.4], the gain θ_1 of Theorem 17.40, $\frac{1}{2}$ and the exponent γ_0 of the range of Proposition 17.134(i) of the full version, β the larger of the field-norm exponents; then $\delta_r = 5^r \varepsilon_k \to 0$, $5^{-\gamma r}\delta_r^{-\beta}(1+k)^p \to 0$ for every p, $\varepsilon_k^{-q} e^{-c\delta_r^{-1}} \to 0$, and $p(\delta_r)$ grows at least like k^{p}. The effective actions $S^{(r)}$ (standard flow) and $\hat{s}^{(r)}$ (hybrid flow) are [15, (3.14)] rewritten as [15, (3.90)], with the Taylor polynomial $P^{(r)}$ of order $\bar{n} = 13$ of King's E'_r; a wall enters through the h-vertex [15, (3.21)], in the free-boundary convention in which the subtraction is the same number for both lattices. In the unit scale of level r the first-step cells of the two flows, the tile in $\Lambda_X = \eta'\mathbb{Z}^3$ and the centred cube in $\Lambda_Y = R^{-1}(\eta'\mathbb{Z}^3)$ around each $x \in \Gamma_1 = R^{-1}(\eta\mathbb{Z}^3)$, have 125 sites each; fixed bijections between them glue to a bijection $\tau : \Lambda_X \to \Lambda_Y$ with $|\tau x' - x'| \le c_\tau \eta'$, $c_\tau < 7.1$, and points $x'' = \tau x'$ are *matched*. The fluctuation covariance of each flow is sliced exactly, $G_r^F = \sum_{j<r} S_j^F$, slice j having length scale $\ell_j = 5^j \eta'$. Every Y-graph is, exactly, a sum over X-vertices with τ-transported kernels; a smooth vertex function sampled at x' and at $\tau x'$ differs by $\le c_\tau \eta' \|\nabla w\|_\infty$; and a counterterm graph of δm^2 or E_1, expanded in the slices of either flow, sums to the same number for both flows, because $\sum_j S_j^F = C^F$ pointwise and C^Y is the image of C^X under the isometry R^{-1}.

The equal-spacing propagator package. In the unit scale of level $r \ge 2$, on the infinite lattice and at $A = 0$, let u_{r-1} be the common symbol of the $(r-1)$-step rotated centred average, u_X the tile symbol and u_Y the cube symbol, A^F the coarse multiplier of flow F, $A^F \le a$, Δ'_F the fine Laplacian of the flow's lattice plus the running mass, and L_2^F the 125 alias classes.

Proposition 17.32 (free core at equal spacing)**.** $A^F(p)^{-1} = a_{r-1}^{-1} + \sum_{l_1 \in \Gamma_r^*/\Gamma_1^*} u_{r-1}(p+l_1)^2 [\eta^2 a^{-1} + S_F(p+l_1)]$ *with* $S_F(q) := \sum_{l \in L_2^F} u_F(q+l)^2/\Delta'_F(q+l)$, *and* $\sup_{V_\eta} |S_X - S_Y| \le 0.198\,\eta^2$. *Hence for all* $r \ge 2$, *all* p *and all* $m_r \ge 0$,

$$|A^X(p) - A^Y(p)| \le 0.20\,\eta^2 A^X(p) A^Y(p) \le 0.20\,a\,\eta^2 A^Y(p),$$

and the same for the photon (three copies at mass $\mu_0 \delta_r$*).*

Idea of the proof, and the package (Lemmas 17.33 to 17.36 and Corollaries 17.37 and 17.38 of the full version). The formula is the Woodbury identity for a composite average whose symbol is $u_{r-1}u_F$; the main terms differ by at most 0.0924 by the second moments of the tile and of the cube, and the 124 aliases by at most 0.1047. From the Fourier representation of the lifts, the lifts of the two flows differ at matched points by $\le C_E J_1 \eta$, with a decaying form by the geometric mean with the Combes–Thomas decay [27]; the coarse covariances $\Gamma^F = H^F B^F H^{F*}$, with $C^F = G_r^F + \Gamma^F$ exactly, differ at matched pairs by $\le C_\Gamma \eta^{1/2}$ and are $\frac{1}{2}$-Hölder; the slices $j \ge 2$ differ at matched points by $\le C(5\eta'/\ell_j)^{1/2} \ell_j^{-1} e^{-\delta|x'-y'|/\ell_j}$. The *uniform comparison form*: with $D_j := S_j^X - S_j^Y \circ \tau$ and $\theta = \frac{1}{2}$, the fine slices obey $|D_j(x',y')| \le C_D \eta'^\theta \ell_j^{-1-\theta} e^{-\delta|x'-y'|/\ell_j}$ (for $j \in \{0,1\}$ there is no gain and the factor is paid by a degree loss, King's device at the finest slice [15, p. 675]), the coarse slices obey the same with $\eta^{1/2}$, and, given Proposition 17.73(a) of the full version and [15, Thm. 3.3], the derivative, Hölder and contour versions hold, the contour lines [15, (3.27)] along the paired contour systems differing by the kernel difference along the contour plus the flux of the slice kernel through the sliver. Summing the slices, $|C^X(x'-y') - C^Y(x''-y'')| \le C_6 \varepsilon^{1/2} |u|^{-3/2}$ for matched pairs with $|u| \ge 25\varepsilon$: a quantitative rotation estimate for the massive lattice Green's function.

The comparison principle and the two-flow bound. A *renormalised family* is a finite sum of King's renormalised graphs [15, §3.5], Bałaban's classes of [3, §§2–3] (Definition G.1 of the full version), localised as in [3, (1.33)]. It is in *per-flow normal form* when, for each flow separately and before any element is mixed, every counterterm graph is expanded in the lines of the flow [15, (3.76)], every Taylor subtraction [15, (3.83)] and Ward-identity rewriting has been carried out with that flow's exact identities, and the exact cancellations between members with identical element sets are co-switched. Its *elements* are lines, vertex functions and external fields.

Proposition 17.39 (comparison principle)**.** *Let $\mathcal{R}$ be a renormalised family in per-flow normal form other than the h-linear field-free family $\mathcal{T}_h$ of Lemma 17.61 of the full version (excepted as in Theorem 17.40), with $n_{\mathcal{R}}$ elements and $E^F(\mathcal{R})$ its value along flow F. Then, with $\theta = \min(\frac{1}{2}, \theta_1)$ and θ_1 of Theorem 17.40,*

$$|E^X(\mathcal{R}) - E^Y(\mathcal{R})| \le C n_{\mathcal{R}}\, p_r^2 \delta_r^{-1/2}\, \eta'^{\theta} (1+k)^p \cdot [\textit{right side of } [3,\ (1.33)] \textit{ for } \mathcal{R}],$$

the prefactor $p_r^2\delta_r^{-1/2}$ being that of the contour elements at $e > 0$ (absent at $e = 0$).

Idea of the proof. Both values are sums over X-vertices; order the elements consistently across members and telescope, switching one element at a time; in each term the difference element, divided by $C\eta'^{\theta}$, obeys the bound of its type weakened by the degree θ (Corollary 17.37, Proposition 17.73(b) of the full version), and the weakened-element bound for each term is Theorem 17.40.

Theorem 17.40 (the two-flow comparison bound)**.** *Let $\alpha \in (0,1)$ be the Hölder exponent of* [15, (3.8)] *and put $\theta_1 := \alpha^2/(4(1+\alpha)^2)$. There are $p = p(\bar{n})$ and C such that for every renormalised family $\mathcal{R}$ of order $\le \bar{n}$ in per-flow normal form other than the h-linear field-free family $\mathcal{T}_h$ of Lemma 17.61 of the full version, which is excepted (its members are the order-0 tadpole, of degree -1, and the order-λ figure-eight and the order-e^2 exchange and seagull tadpoles, of degree 0, the degree-0 members with their δm^2 partners: whole graphs, the order-e^2 ones carrying no counterterm for their own degree-0 divergence; Proposition 17.134(ii) of the full version, the full version), that is, for the h-families of Lemma 17.61 of the full version other than $\mathcal{T}_h$ and for the families of step (5) of the proof of Theorem 17.57 in the full version at $e > 0$ after the constant-background reduction of Lemma 17.41 of the full version, and for $0 < \theta \le \theta_1$, uniformly in r, k, the torus and the masses, the two-flow difference obeys the bound of Proposition 17.39: in every term of its telescoping, the element replaced by the difference $e_i^X - e_i^Y$, which obeys the bounds of its type weakened by the degree θ,* [3, (1.33)] *holds with the extra factor $C(1+k)^p\eta'^{\theta}$ and the prefactor stated there. The exponent θ_1 does not depend on $\bar{n}$, and any $\theta > 0$ with any polynomial factor suffices for the uses below (Lemma 17.27 of the full version).*

Idea of the proof (Lemmas 17.43 to 17.46, 17.49 and 17.51 to 17.56, Corollary 17.47, Definition 17.48, and Remark 17.50 of the full version). Inputs: Proposition 17.73 of the full version, [15, Thm. 3.3], [3, Prop. 2.1, (2.8)] with the ordering device (2.7) there, the counterterm conditions [15, (3.33)] through order four with [3, (2.23)–(2.28)], the local limit law [17, Thm. 2.5.6], and the h-census of Lemma 17.61 of the full version. After the constant-background reduction every line is an $A = 0$ line, $G_r^F = C^F - \Gamma^F$, where C^F is *one* lattice function $C = (-\Delta_{\eta'} + m_r^2)^{-1}$ evaluated in each flow's own frame, sliced in the time variable of the simple random walk. Degrees are read per ordering of the scale-summed elements, so that Bałaban's multiscale bound tests only the connected components of initial segments, and a distinguished element carrying $(\eta'/\ell_j)^{\theta}$ keeps the factor η'^{θ} whenever every component containing it has degree $> \theta$. The connected all-C sets of non-positive bare degree are enumerated; each unit (self-energy, vacuum polarisation) is bundled with one of its legs into a *compound*, an own-frame second-order Taylor subtraction whose local terms are common numbers of both flows, and one subtraction rule, Bałaban's integration by parts [3, (2.8)] or a first-order subtraction at a terminal with a common zero moment, raises every newly formed component of degree zero to degree one; afterwards every component of every initial segment has degree ≥ 1, or $n + m\alpha$ with $m \ge 1$, or is an explicit-η' component (a finite case analysis, given in full in the full version). Every element type obeys a two-flow difference law with a positive exponent (the local limit law gives $(\eta'/\ell_j)^{1/2^n}$ for a slice line with n derivatives, by rotation covariance of the isotropic continuum slice; gradients of externals and coarse elements gain η^{κ}, $\kappa = \alpha/(2(1+\alpha))$; compounds gain η'^{θ_U}, $\theta_U \ge \min\{\theta, s/8\}$, $s = \alpha/(1+\alpha)$), and no frame vector ever appears. The telescoping switches compounds first and externals last, uses identities only at the two pure endpoints in each flow's own frame, and bounds every term by the multiscale bound with the switched element weakened by θ; $\theta_1 = \alpha^2/(4(1+\alpha)^2)$ is at most every gain exponent and does not depend on $\bar{n}$.

The constant-background reduction and the one-flow instance (Lemma 17.41 and Corollary 17.42; statements and complete proofs in the full version). King's expansion [15, (3.43)–(3.55)] is carried out for each flow separately on its infinite lattice, in universal-cover form; the constant part of the background is removed exactly by a gauge function, leaving King's phases [15, (3.46)]; every family at background $A^{(r)}$ becomes a sum of families at $A = 0$ in per-flow normal form, times common explicit coarse-field factors and the phases, plus terms bounded per flow by $C(\delta_r^{\sigma} + e^{-\delta' p(\delta_r)})|T|$. Corollary 17.42 of the full version is King's weakened-element induction [15, (3.67)–(3.70)] for one flow, standard or hybrid, with zero-degree logarithms $(1+k)^p$, for the renormalised families whose non-vanishing ultraviolet ordered subgraphs have positive degree after Bałaban's rewriting and whose infrared lines, of slice index $\geq r$, lie on the counterterm lines of their mass-counterterm units; it gives the step of King's proof of [15, Prop. 3.6] asserted on [15, p. 665] without a printed proof for both flows, and [15, Prop. 3.6] itself for the standard flow, for every such family, while on the degree-0 members of the order-e^2 h-linear field-free family it fails as printed (Proposition 17.134(ii) of the full version). In the infrared stage the counterterm-attachment subgraphs of [15, (3.78)] with degree ≤ 0 cost a factor $\leq (1+k)$ each; the stage is verified for the units of order two and the sunset, and for the other units of order four. For the hybrid flow, that the properties of its lines and blocks suffice for the proof of [3, Prop. 1] is the generality of the proof of [3, Prop. 1] in the flow, as in the proof of Lemma K.495 of the full version.

The comparison of the effective actions at equal spacing.

Theorem 17.57 (equal-spacing comparison, total form)**.** *Let $T \in \{T_1, T_2\}$, G a real smooth P-periodic 2-form, $z \in \mathbb{R}$, and $r = r(k)$ as in Lemma 17.27 of the full version. Then for $k \geq k_0(zG)$,*

$$\sup_{\operatorname{supp}\chi_r} \left|\hat{s}_T^{(r)}(A_r, \phi_r; zG, 0) - S_T^{(r)}(A_r, \phi_r; zG, 0)\right| \leq \vartheta_k |T|, \qquad \vartheta_k := C\big(5^{-\gamma r}\delta_r^{-\beta}(1+k)^p + \delta_r^{\sigma}\big) \to 0,$$

every c-number included: there is no vacuum constant and no source-linear constant at $h = 0$.

Idea of the proof (with Theorem 17.58, Proposition 17.59, and Remark 17.60 of the full version). Both effective actions are written in the form of [15, (3.90)] and compared term by term on $\operatorname{supp}\chi_r$, per coarse site, times $|T|\delta_r^{-3}$ sites: the quadratic forms by Proposition 17.32; the normalisations by $\leq 12.5\, a\, 5^{3k-5r}|T|$; the background determinants by $\leq \vartheta_k|T|$ (see the full version; King's [15, (3.94)–(3.98)] is not used, the per-p middle line of (3.98) being false as printed); and the interaction $P^{(r)}$, after the constant-background reduction, by Proposition 17.39 family by family, in per-flow normal form with the same combinatorics in both flows. The vacuum constant $v_k(T) = S_T^{(r)}(0,0;0) - \hat{s}_T^{(r)}(0,0;0)$ tends to 0 on each torus (Theorem 17.58 of the full version): E_1 is the same c-number for both lattices, the torus vacuum graphs are up to $O(\varepsilon^{-q}e^{-c\delta_r^{-1}})$ the infinite-lattice densities, the same for both tori, and the vacuum classes of positive degree reduce, with their mass-counterterm units, to families with at least two legs, to which Proposition 17.39 applies; for the classes of non-positive degree the all-ultraviolet pieces of E_1 cancel exactly, the orders λ and λ^2 are computed (Proposition 17.59 of the full version), and the remaining classes, whose pieces of E_1 carry infrared lines outside the units, are compared between the two flows. The comparison must be applied to manifest forms in which every counterterm is expanded in the flow's own lines (Remark 17.60 of the full version).

Walls: the source vertex.

Theorem 17.63 (comparison with a wall)**.** *For $T \in \{T_1, T_2\}$ and $s = (zG, h)$ real, on $\operatorname{supp}\chi_r$: $\left|\hat{s}_T^{(r)}(\cdot; s) - S_T^{(r)}(\cdot; s) + c_k \mathsf{S}_{X_k}(h)\right| \leq \vartheta_k(s)|T| \to 0$, with c_k of Lemma 17.62 of the full version.*

Idea of the proof (Lemmas 17.61 and 17.62 of the full version). In King's degree the h-vertex [15, (3.21)] has $D = 2$, so every non-vanishing field-dependent h-family has $D \geq 1$; the only h-families with $D \leq 0$ are field-independent and linear in h (the free tadpole, the order-λ figure-eight and the order-e^2 tadpoles with their counterterm partners), the family $\mathcal{T}_h$ (Lemma 17.61 of the full version). With f^F the infinite-lattice one-point density of $\mathcal{T}_h$ along the flow F, $\bar{f}^F$ its

cell average and $c_k := -(\bar f^X - \bar f^Y)$, independent of the torus with $|c_k| \le C\varepsilon^{-q}$, the $\mathcal{T}_h$-part of $\hat s_T - S_T$ at zero coarse fields is $-c_k \mathsf{S}_{X_k}(h) + o(1)$ on both tori (Lemma 17.62 of the full version); every other h-family is telescoped with Proposition 17.39 as in the proof of Theorem 17.57.

The source-linear constant vanishes.

Proposition 17.72 (the source-linear constant vanishes)**.** *With c_k as in Lemma 17.62 of the full version and $r = k - \lfloor \varrho k \rfloor$, $\varrho < \frac{1}{9}$, for $k \ge k_0(a, m, \mu_0, \varrho)$*

$$|c_k| \le N C_\Gamma \, \eta^{1/2} \delta_r^{-1} + C_1 \lambda \, \eta^{1/2} + C_2 e^2 \, \eta'^{1/8} (1 + \delta_r^{-1})(1 + \log(1/\eta'))^4 \le C \, 5^{-(1-9\varrho)k/8} (1+k)^4,$$

with constants depending on a, m, μ_0, N, τ only (not on J, ψ, n, the torus or h). Hence $c_k \to 0$.

Idea of the proof (Lemmas 17.64 to 17.71 of the full version). $|\bar f^X - \bar f^Y| \le \sup_{x'} |f^X(x') - f^Y(\tau x')|$, and the three members of $\mathcal{T}_h$ are compared at matched points: the free tadpole leaves the residue $-N\Gamma^F(x,x)$, with difference $\le N C_\Gamma \eta^{1/2}$; the figure-eight is $-4N(N+2)\lambda_r \sum_y G^F(x,y)^2 \Gamma^F(y,y)$, telescoped with the matched-pair bounds of Corollary 17.38 of the full version; and the order-e^2 density is split exactly as $Z^F + A^F(x) + B^F(x)$ with the renormalised one-loop kernel, where the all-C part Z^F cancels exactly between the flows and the parts with external or internal Γ-lines have a positive power-counting margin and gain $\eta'^{1/6}$ up to logarithms. The conversion to physical units costs δ_r^{-1}, and $\varrho < \frac{1}{9}$ makes every term geometric in k; only $c_k \to 0$ is used.

The inputs of the forward bound. Throughout $T \in \{T_1, T_2\}$ and $r = r(k)$ is the choice of Lemma 17.27 of the full version; $p(\delta) = b_0(1 + \log \delta^{-1})^{\mathrm{p}}$ is King's small-field function.

The kernel bounds of the hybrid flow at a non-zero gauge field (Proposition 17.73 of the full version). For $r \ge 2$ and a background A regular in the sense of [15, Def. 3.2] with $e_r^\beta \le c_0(a)$: (a) the operators G_r, $\Delta^{(r)}$, $C^{(r)}$ of the hybrid flow with the paired contours obey the bounds of [15, Thm. 3.3, (3.5)–(3.8)] in the periodic form, on the infinite lattice and on tori fitted by both lattices, each in its own quotient distance, with constants depending on a, M, α only, except the Hölder constant of (3.8), which is $C_\alpha \log^{3\alpha}(1/\eta')$; the differences between two such domains with a common fundamental region on which the backgrounds agree carry the boundary factor of [15, Thm. 3.3] in the smaller of the two quotient metrics; the restricted inverses on unions of big blocks obey [16, (5.54)]; the fine-slice lines obey [15, (3.63)–(3.65)]; and the level-r elements H^X, B^X, Γ^X obey their decay and regularity bounds, all uniformly in r, k and the torus, at every level $j \le r$ and, in the form the hybrid lower bound uses, at the levels 0 and 1. (b) After the constant-background reduction the difference elements $\Gamma^X - \Gamma^Y \circ \tau$, ΔH, $\Delta(A^{-1})$ obey the bounds of Corollary 17.37 of the full version with the factor $\eta'^{1/2}$; the contour vertices [15, (3.24)–(3.26)] differ entrywise by a relative $\eta^{1/2}$ against their size $C e_r p_r^2$; the constant-background phases [15, (3.46)] at matched vertices differ by $\le e\, p_r \mu_0^{-1} c_\tau \delta_r^{-1/2} \eta'$; and the contour lines, one-ended and double, obey the slice-j bounds in both flows with a relative mismatch $(\eta'/\ell_j)^{1/2}$, a one-ended line with one lattice derivative at its lattice end with $(\eta'/\ell_j)^{\theta_\partial}$, $\theta_\partial = \alpha/(2(2+\alpha)) \ge \theta_1$. At $A = 0$ this is the propagator package; at $A \ne 0$ the proof is the programme of [16, §5.3] written out: coercivity at a regular background by a cover with axis cubes carrying a local gauge for the constant part of the field ([16, (5.59)–(5.60)]); Combes–Thomas decay in L^2 form [27]; the geometric resolvent identity for the two-domain clause; Bałaban's random walk [4, §2] with periodic cells fitted by both lattices [16, (5.55)–(5.56)], the Neumann series summed in the sup norm first so that the smallness condition is uniform in r; the cubic-Hermite cardinal interpolation of the Y kernels; and, for (b), King's constant-background reduction applied to both flows on their infinite lattices, after which every difference element is of one of seven types (an $A = 0$ line; a contour line, with the kernel difference along the contour plus the flux of the slice kernel through the sliver; a double contour line; a contour vertex; an external lifted field; a constant-background phase; a hopping vertex). What would break (b) are mis-paired contours or mis-rooted blocks, both excluded by Definition 17.15 and the centred blocks. The walk-sum bounds are written out in the full version.

King's upper bound for the rotated lattice.

Proposition 17.100 (King's upper bound for the rotated lattice)**.** *Let the couplings be off E_W (Remark 15.2). For Y_k on T, every real source $s = (zG, h)$ with $h = -n\psi$ the wall of a configuration (J, ψ, n) of the class of Theorem 17.13 (so that $J \geq J_{\mathrm{pos}}$, a condition of that class), or its transport, the free-boundary convention of Definition 15.7, the centred blocks of Convention 17.17 of the full version, and every k with $\lfloor \varrho k \rfloor \geq J_{\mathrm{K}}$ (King's stopping index, the integer called J in* [15, Thm. 3.1]*; so $r \leq k - J_{\mathrm{K}}$) and $\varepsilon_k \leq s_0$ (s_0 of Proposition 17.126(b) of the full version, the spacing below which its rate clause holds; by Proposition 17.126(b) of the full version, $|u(\varepsilon_k)| \leq \Delta_*/2$ for every $\varepsilon_k \leq s_0$),*

$$\mathsf{Z}_{Y_k,T}(s) \leq e^{C\delta_r^{\sigma}|T|} \int \chi_r \, e^{-S_T^{(r)}} + B, \qquad 0 \leq B \leq \mathfrak{N}\, e^{-c\,p(\delta_r)^2 + C|T|}\, \varepsilon^{-q},$$

where $\mathfrak{N} > 0$ is the field-independent prefactor of Proposition 17.126(c) of the full version at $\Lambda = Y_k$ and spacing ε_k, so that $\mathsf{Z}_{Y_k,T}(s) = \mathfrak{N}\, \mathsf{Z}^{\mathrm{tor}}_{Y_k,T}(zG, h + u(\varepsilon_k)\mathbf{1})$; and $c = 1 - 1/p_H \in (0,1)$, with $p_H > 1$ depending on the torus T and the lattice family only, not on k (Theorem 17.98 of the full version; when $\Delta_T(\varepsilon_k) \geq 0$ at every admitted k no hypothesis on King points enters and $p_H = 2$ is taken), $C = C(s, |T|, p_H)$ and $q = q(s, |T|, p_H) < \infty$ do not depend on k.

Idea of the proof. On T_2, aligned in Y's frame, the passive rotation R maps (Y_k, T_2) to (X_k, D_J) with its standard frame, and with it the blocks, the contours, the small-field indicator and the effective actions of the flow; so the bound with the small-field integral is King's Theorem 3.1 [15, (3.4)] at block factor 5 (Theorem F.2(iii) of the full version with Lemma 5.1), in the torus convention for which it is printed, at the constant-shifted source $(zG, h + u(\varepsilon_k)\mathbf{1})$, carried to the free-boundary convention by Proposition 17.126(b) of the full version,(c), and with centred blocks (Convention 17.17(i) of the full version). King's constant depends on g, h and $|T|$ [15, p. 655]; the shift by $u(\varepsilon_k)\mathbf{1}$ changes $\sup|h|$ by at most $\Delta_*/2$ and no derivative of h, and the constants of [15, Thm. 3.1] are uniform over the family $\{h + u\mathbf{1} : |u| \leq \Delta_*/2\}$, as in Lemma 17.128 of the full version. On T_1 the frame torus of Y_k is skew, and the instance is Proposition 17.121. For a walled source ($\int h \neq 0$) King's additive large-field term carries the constant of the stability bound [15, (2.23)], which fails as printed (Remark 16.1); on both tori the large-field term, in the torus convention at the source $(zG, h + u(\varepsilon_k)\mathbf{1})$ and at depth r, is at most $e^{-c\,p(\delta_r)^2 + C|T|}\varepsilon^{-q}$ with $c = c_* := 1 - 1/p_H \in (0,1)$, $p_H > 1$ fixed, and $q = q(h, |T|, p_H)$: on T_2 by Proposition R.72(a), under the contour half of Convention 17.17(i) stated as Hypothesis R.71 of the full version; on T_1 by Proposition R.76(a), under the display-by-display form of that step, Hypothesis R.75, and for $J \in 5^{n_*}\mathbb{N}$ with $J \geq J_0(e^{-1})$, the mass-free threshold of Definition R.74 of the full version; on either torus for every wall of the class of Theorem 17.13 or its transport, unconditionally when $\Delta_T(\varepsilon_k) \geq 0$ for every admitted k, and otherwise under the hypothesis $(\mathrm{POS})_V$ of Hypothesis R.57 of the full version, with $V > V_+ := \frac{1}{2}\sup_k \max(0, -\Delta_T(\varepsilon_k))$ and $p_H < V/V_+$. In the two remaining cases, at a wall where $\Delta_T < 0$ at some admitted spacing, without $(\mathrm{POS})_V$, and on T_1 for the J admitted by Proposition 17.121 but below $J_0(e^{-1})$, the large-field term is at most $e^{-c\,p(\delta_r)^2 + C|T|}\varepsilon^{-q}$ for some $c \in (0,1]$; the prefactor $\mathfrak{N}$ carries it to B. The coefficient c and the factor ε^{-q} are absorbed by Proposition 17.139.

The cumulant bound for the step Gaussian. Summary (Theorem 17.116 and Lemmas 17.101 to 17.115 of the full version). The fluctuation integral of every block-spin step of both flows has the form $\int \chi_b\, e^V d\mu_P$: μ_P is a centred Gaussian on $(\mathbb{R}^n)^\Lambda$, Λ a finite subset of $\mathbb{Z}^3$, of a cubic torus or of one of the skew tori of Proposition 17.121, whose precision $P \geq \gamma_0 > 0$ has exponentially decaying off-diagonal blocks; $V = \sum_{Y \subset J} V_Y$ is a real polynomial of degree at most D with exponentially decaying coefficients, $\mathsf{A} := \|V\|_\varsigma$; and χ_b is a product over the sites of indicators of a fixed symmetric convex body scaled by b. Theorem 17.116 of the full version is the order-$\bar{n}$ cumulant expansion of such an integral: there are $S < \infty$, $\rho_3 > 0$ and a threshold b_*, depending only on $(n, \bar{n}, D, \varsigma, \gamma_0, c_P, \delta_P, c_K)$, such that for $b \geq b_*$ and $\mathsf{A}b^{D+6} \leq 1$

$$\Big| \log \int \chi_b\, e^V d\mu_P - \sum_{k=1}^{\bar{n}} \frac{\kappa_{\mu_P}(V;k)}{k!} \Big| \leq S|J| \big[(\mathsf{A}b^{D+6})^{\bar{n}+1} + e^{-\rho_3 b^{3/2}} \big] + 4n|\Lambda| e^{-\gamma_0 \gamma_K^8 b^2/(2n)},$$

the last term in the lower bound only; nothing depends on Λ, on J, on the torus or on translation invariance. For the step Gaussians of both flows the hypotheses hold with Bałaban's coercivity constant [1, (2.33)] (for the hybrid flow by Proposition 17.73(a) of the full version), the Combes–Thomas decay of the step precision, $\mathsf{A} \le s_0 O(1)\ell_j^{\beta_0}$ by Corollary 17.42 of the full version, Bałaban's two-component cutoff [1, (3.43), (3.50)] and $b = p_1(\ell_j)$. The printed cluster-expansion lemma of [5], which King cites for this step, uses the Markov property of the nearest-neighbour field in four places; the proof keeps its architecture (cells with corridors, four shifted pavements, the cutoff moved from the corridors into the cells) and replaces those four uses by a Combes–Thomas bound for every principal block, a conditioning lemma, a decoupling lemma across corridors and the comparison of two Gaussian laws.

Proposition 17.117 (the hybrid lower bound)**.** *For the hybrid flow on T and every real smooth periodic source s,*

$$\mathsf{Z}_{X_k,T}(s) \ge e^{-C(1+k)^p \delta_r^\sigma |T|} \int \chi_r \, e^{-\hat{s}_T^{(r)}}.$$

Idea of the proof. Bałaban's lower-bound induction [1, §§3–4] at block factor 5 run for the hybrid flow on both tori, with the hybrid kernel bounds in place of [1, Props. 2.1–2.3] and [16, Prop. 5.6], the cumulant lower bound, the skew instance Proposition 17.121 on T_1 (whence the constant $C(1+k)^p$ of Corollary 17.42 of the full version) and the counterterm bridge; the induction at block factor 5, the kernel bounds, the cumulant lower bound and the bridge are Theorem F.2, Proposition 17.73(a), Theorem 17.116 and Proposition 17.126 of the full version; that Bałaban's argument holds for the hybrid flow is the generality of the proof of [3, Prop. 1] in the flow and of the operations of [1, §3] in the covariances, as in the proofs of Lemma K.495 and Proposition K.517 of the full version. The complete proof is in the full version.

Proposition 17.118 (the polynomial lower bound for the rotated lattice)**.** *For Y_k on T, in the torus convention of Proposition 17.126 of the full version, and every real source $s = (zG, h_0)$ ($z \in \mathbb{R}$) with $h_0 = -n\psi$, $n \ge 0$: $\mathsf{Z}^{\mathrm{tor}}_{Y_k,T}(s) \ge e^{-C_1|T|}\varepsilon^{q(s)}$ for some $q(s) < \infty$ (on T_1 under the cap of Lemma K.518 of the full version, automatic for every J of the class). In the free-boundary convention of Definition 15.7 the same bound holds, for $n = 0$ as for $n > 0$, only up to the field-independent factor $\mathfrak{N}_T(h_0)$ of Proposition K.550 of the full version.*

Idea of the proof. At $h = 0$ this is Bałaban's lower bound [1, (1.14)] (Theorems F.1 and F.2 of the full version) in the torus convention, in which it is proved (Proposition 17.126 of the full version); on T_1 by the skew instance Proposition 17.121, under the condition at Bałaban's final spacing marked in Theorem K.506 of the full version. The source zG is removed by Jensen's inequality, since $\langle F(G)\rangle = 0$ under charge conjugation. At the wall the two-sided ε-uniform bound [16, (4.2)] is false (Remark 16.1) and only the polynomial form is asked for: it is Theorem 17.120 in the torus convention, at every King point. The free-boundary form follows from it at the shifted source by the identity of Proposition K.550 of the full version, with the factor $\mathfrak{N}_T(h_0)$, whose logarithm is of order $\varepsilon^{-1}|T|$ in general (Proposition 17.126(b) of the full version); no proof uses that form, nor the alternative through Proposition 17.119 of the full version.

Theorem 17.120 (floor shift: the polynomial lower bound at the wall at every King point)**.** *Let m^2 be any King point and h a real P_J-periodic function (it may depend on k) with $\|h\|_\infty \le H$ uniformly in k. Write $a_s := s + \delta m^2(s;\varepsilon)$ for the bare scalar coefficient at the King point s and let M^2 be a King point with $\Delta_0(M^2) := \liminf_{\varepsilon\to 0}[a_M - a_m] > 2H$; such an M^2 exists for every coupling and every H, since $\Delta_0(M^2) \to +\infty$ as $M^2 \to \infty$ (the coefficients of $\partial_s \delta m^2$ are non-negative at order λ, $O(s^{-1/2})$ at order e^2 and $O(s^{-1})$ at the two-loop orders; see the full version). Then on $\Lambda \in \{Y_k, X_k\}$, $T \in \{T_1, T_2\}$ (on the pairs (Y_k, T_1) and (X_k, T_2), skew in the frame of the flow and with equal source-free partition functions (Lemma 17.4, in either convention by Proposition 17.126(c) of the full version), under the cap of Lemma K.518 of the full version, automatic for every J of the class), in the torus convention of Proposition 17.126 of the full version, and in the free-boundary convention of Definition 15.7 up to the field-independent*

factor $\mathfrak{N}_T(h)$ of Proposition K.550 of the full version,

$$\ln \mathsf{Z}^{m^2}_{\Lambda,T}(0,h) \geq -C|T| - C(1+k)|T| - qk \ln 5, \qquad \textit{hence} \qquad \mathsf{Z}^{m^2}_{\Lambda,T}(tG,h) \geq e^{-C_1|T|} \varepsilon^{q'} \quad \textit{for every real } t.$$

In the torus convention this is the polynomial form of Proposition 17.118 at the wall $h_0 = -n\psi$ for every $n > 0$; the sign of h is used nowhere.

Idea of the proof. The *floor shift* $e^{\sum \varepsilon^3 h|\phi|^2} = e^{h_{\min} \sum \varepsilon^3 |\phi|^2} e^{\sum \varepsilon^3 (h - h_{\min})|\phi|^2}$ moves the wall into the bare scalar coefficient, Jensen's inequality bounds the remaining factor below by 1, and the monotonicity of $\langle \sum \varepsilon^3 |\phi|^2 \rangle_\gamma$ in the bare coefficient bounds the shifted integral below by the one at the higher King point M^2, where [1, (1.14)] applies; the ε^{-1}-divergent c-numbers collapse exactly by the integrated counterterm identity, and the perturbative remainder is $\geq -C(1+k)|T|$ by Reisz's theorem [22, Thm. 1] with [15, (3.33)]; on the skew pairs (Y_k, T_1) and (X_k, T_2) the two stability bounds are the skew instances of Proposition 17.121. The complete proof is in the full version.

The skew-torus instances. A torus $\mathbb{R}^3/\Pi$ is *aligned* in the frame of a lattice $O(s\mathbb{Z}^3)$ if $\Pi = O(\ell_1\mathbb{Z} \times \ell_2\mathbb{Z} \times \ell_3\mathbb{Z})$, and *skew* otherwise; [1, 2, 3] and [15] are written for a torus aligned in the frame of the blocks. T_1 is aligned for X_k and skew for Y_k; T_2 is aligned for Y_k and skew for X_k. On T_1 every block-spin step performed in Y's frame is a step on a torus skew in the frame of its blocks; on T_2 only the first hybrid step is.

Proposition 17.121 (the skew-torus instances)**.** *King's Theorem 3.1* [15, (3.4)] *(Bałaban's upper bound with the renormalisation of* [3]*) and Bałaban's lower-bound machinery* [1, §§2–4, (1.14)] *hold for the standard flow of Y_k on T_1, the torus RD_J in Y's frame, which is skew, with the constants of the aligned case times at most* 125 *in the linear part; and Bałaban's lower-bound argument holds for steps $2, \dots, r$ of the hybrid flow on T_1 with the constant $C(1+k)^p$ of Corollary 17.42 of the full version, applied with its positivity hypothesis for the hybrid lines (for the hybrid flow of X_k, with the lines of the flow at $A = 0$ and the vertices at $B = 0$, every connected source-free point-free component S of a prefix of a non-vanishing graph of a renormalised family of order $\leq \bar{n}$ in per-flow normal form whose plain degree $D(S)$ is ≤ 0 is rewritten, finest first, with the output shapes (R0)–(R2) of Theorem K.167 of the full version (the plain and current degrees D, D' as in the renormalised degree count of the full version, ρ_{j_0} the spacing of the finest slice j_0 of S, $\alpha \in (0,1)$ the Hölder exponent of* [15, (3.8)]*): the resummed coefficients $c_a(F_S)$ of its inside family vanish exactly, by the counterterm of the flow per scale assignment (Lemma 17.31 of the full version), by charge parity, by the inversion about the coarse site or by the Ward identity of the full fixed-order two-photon function of the flow and of its free-lattice replacement (Proposition K.166(v) of the full version), or are bounded by $C\rho_{j_0}^{D(S)+|a|}$, the coefficients with $D(S) + |a| \leq 0$ vanishing exactly, so that $D'(S) \geq \alpha$ and the current degree of no other component is lowered; hence every non-vanishing source-free point-free ultraviolet prefix component of the hybrid flow has current degree $D' \geq \min\{\alpha, 1\}$ after the rewriting). In both, the torus enters* [1, 2, 3] *only through the embeddings of the block lattices, the external fields and the image terms of the periodised kernels; no torus symmetry and no torus Fourier analysis is used in the proof of* [3, Prop. 1]*. The statements hold under the provisos $J \in 5^{n_*}\mathbb{N}$ and $J \geq J_0(\delta_r)$ of Theorem 17.13; the two ε-uniform bounds, the lower half of* [1, (1.14)] *and the upper bound* [2, (1.3)] *(Theorem K.506(i) of the full version–(ii)), which run the flow to Bałaban's final spacing, hold under the cap of Lemma K.518 of the full version, under which the non-wrapping condition* (259) *of the full version holds at every spacing $t \leq \varepsilon_0^{\mathrm{Ba}}$, the final spacing t_K included, for every $J \geq 1$ (Lemma K.518(iii) of the full version), their absolute clauses with the extensivity of the final-step vacuum c-numbers on the skew torus (the vacuum c-numbers $E^{\pm}_{\mathrm{fin}}$ accumulated by Bałaban's lower-bound induction* [1, §§3–4] *and upper-bound induction* [2, §§2–3]*, run at the spacing $\varepsilon = \varepsilon_k$ on the skew torus $T' = RD_J$ ($|T'| = |D_J|$) through the steps $r < j \leq K$, $r = r(k)$ the depth of Lemma 17.27 of the full version (the step errors of the steps $j \leq r$ are those of the re-runs Propositions K.513 and K.514 of the full version on T', extensive with the aligned constants), to the final step K, the final integral at the spacing $t_K \in (\varepsilon_0^{\mathrm{Ba}}/5, \varepsilon_0^{\mathrm{Ba}}]$ included, at the couplings (e, λ, μ_0) and at every scalar mass $\sqrt{s}$ with s one of the King points in*

use (m^2, the points M^2, M_+^2 of Theorem 17.120 and M_1^2 of Theorem 17.98 of the full version), are extensive: $|E_{\mathrm{fin}}^{\pm}| \le C_F(\varepsilon_0^{\mathrm{Ba}})\,|T'|$ with C_F as in Theorem F.4 of the full version, depending on the scalar mass and the couplings and not on ε, k or J, for every $J \in 5^{n_*}\mathbb{N}$ under the cap $\varepsilon_0^{\mathrm{Ba}} \le \varepsilon_{\mathrm{sys}}$ of Lemma K.518 of the full version).

Idea of the proof (Lemmas 17.122 to 17.124 and the census, all in the full version). The skew torus $T' = RD_J$ in Y's frame is covered by the cube $\tilde{T} = \mathbb{R}^3/250MJ\,\mathbb{Z}^3$, aligned in Y's frame, with a deck group of order 125 that commutes with every block average, so every kernel of a Y-frame flow on T' is the 125-fold periodisation of the kernel on $\tilde{T}$ (*folding*); inequalities between liftable functionals descend from $\tilde{T}$ to T' (*restriction principle*); and every step of [1, §§2–4], [2, §§2–3], [3, §§1–3], [15, §3] that the proofs of this part use, taken in the form in which it is used (the corrections of Remark 17.10 included), is of one of six types, each of which transfers verbatim to T' (*type closure*), the local arguments needing sets of sup-diameter below the systole, the proviso $J \ge J_0(\delta_r)$; the cumulant bound is Theorem 17.116 of the full version on the skew tori. The complete proof is in the full version.

Summary (Proposition 17.126 of the full version; the image-sum estimates are in the full version). [15] and [1, 2, 3] are proved for the model in which the counterterms, the Wick subtraction and the source vertex are defined with torus propagators, while Definition 15.7 uses the free-boundary conventions of [16, p. 330]. With $\Delta_T(s) := \delta m^2(s) - \delta m_T^2(s)$, $\Delta E_T(s) := E_1(s) - E_{1,T}(s)$ and $\Delta c_T(s) := \mathsf{W}_s - c_T(s)$, the free-boundary model is, by exact algebra, the torus-convention model with the bare mass shifted by Δ_T, times a c-number and a shift of the Wick constant, which both block-spin flows carry through unchanged (Lemma F.3 of the full version); the three quantities are bounded uniformly in s, $\Delta_T(0) = \lim_{s\to 0}\Delta_T(s)$ exists and $|\Delta_T(s) - \Delta_T(0)| \le Cs$. Proposition 17.139, the only consumer of Propositions 17.100 and 17.117, is c-number blind, so those bounds, proved for the torus-convention model at the constant-shifted source $h - \frac{1}{2}\Delta_T(s)\mathbf{1}$ with constants uniform over the family $\{h + u\mathbf{1} : |u| \le \Delta_*/2\}$ (for Proposition 17.117 part of its proof; for Proposition 17.100 the uniformity step of its proof), hold for the model of Definition 15.7 as consumed; for Proposition 17.118 no shift is consumed, Proposition 17.139 taking its lower bound at a wall from Theorem 17.120 in the torus convention at the shifted source; and the walled curvature ratio is the same number in both conventions at the shifted wall $h_0 - \Delta_T(\varepsilon_k)/2$.

The existence of the walled axis-lattice limits.

Theorem 17.127 (the walled torus limit on the axis lattice)**.** *Let the couplings be off E_W (Remark 15.2). For every walled configuration $\mathfrak{c} = (J, \psi, n)$ of the class of Theorem 17.13 and every real smooth P_J-periodic 2-form G the limit*

$$\mathcal{W}_{\mathfrak{c}}(tG) := \lim_{k\to\infty} \frac{\mathsf{Z}_{X_k,T_1}(tG, h_0)}{\mathsf{Z}_{X_k,T_1}(0, h_0)} = \lim_{k\to\infty} \langle e^{tF(G)} \rangle_{\mathfrak{c},X_k}$$

exists for every real t; it is the same number in the free-boundary convention of Definition 15.7 and in King's torus convention at the shifted constant source $h_0 - \Delta_T(\varepsilon_k)/2$ (Proposition 17.126(c) of the full version), and it equals the limit of the torus-convention ratios $\mathsf{Z}^{\mathrm{tor}}_{X_k,T_1}(tG, h_\infty)/\mathsf{Z}^{\mathrm{tor}}_{X_k,T_1}(0, h_\infty)$ at the frozen wall $h_\infty := h_0 - \Delta_T(0)/2$. The convergence is locally uniform in $t \in \mathbb{C}$, the limit is entire, and all moments of $F(G)$ converge.

Idea of the proof (Lemmas 17.128, 17.129, 17.132, 17.133 and 17.136 to 17.138 and Propositions 17.130, 17.131, 17.134 and 17.135 of the full version). This is King's torus theorem [15, Thm. 2.1] with the wall as his scalar source, read on the normalised ratio, in the only form in which it is true at a wall; it is a one-lattice statement. With $\mathsf{Z}_k^{\mathrm{tor}}(t; h)$ the torus-convention functional, $h_k := h_0 - \frac{1}{2}\Delta_T(\varepsilon_k)\mathbf{1}$ the *moving wall* and $h_\infty := h_0 - \frac{1}{2}\Delta_T(0)\mathbf{1}$ the *frozen wall*, $|h_k - h_\infty| \le C\varepsilon_k$, the bridge gives $r_k(t) := \langle e^{tF(G)} \rangle_{\mathfrak{c},X_k} = \mathsf{Z}_k^{\mathrm{tor}}(t; h_k)/\mathsf{Z}_k^{\mathrm{tor}}(0; h_k)$, literally the object of [15, Thms. 3.1, 3.4] with the scalar source h_k, whose constants are uniform over the family; all ratios obey $1 \le \mathsf{Z}_k^{\mathrm{tor}}(t; h)/\mathsf{Z}_k^{\mathrm{tor}}(0; h) \le e^{C_G t^2}$, so convergence for real t gives locally uniform convergence on $\mathbb{C}$. King's upper bound, with the large-field term $B_k = e^{-c\,p(\delta)^2 + C|T|}\varepsilon_k^{-q}$

at a wall, $c = c_* = 1 - 1/p_H$ for a fixed $p_H > 1$ (Proposition R.66 of the full version: Hölder's inequality on the large-field remainder reduces it to the source-free large-field term of [15, (3.4)] and to the upper stability bound at a wall; unconditionally when $\Delta_T \geq 0$ at every spacing, and otherwise under the hypothesis $(\mathrm{POS})_V$ of Hypothesis R.57 of the full version; where $\Delta_T < 0$ at some spacing the bound holds without $(\mathrm{POS})_V$ for every wall of the class of Theorem 17.13), and Theorem 17.120 show that each ratio differs from its small-field ratio by $o(1)$ (Proposition R.82(a) of the full version, uniformly in h in the family and in t on compact sets). King's renormalisation theorem [15, Thm. 3.5] is false as printed, in both its clauses, on the degree-0 members of the order-e^2 h-linear field-free family (Remark 16.1; their kernel grows like r at fixed δ_r, and the two depths differ by the drift), but its convergence clause holds for the field- or source-dependent parts of every graph, because every connected ordered subgraph of their non-vanishing derived graphs has strictly positive degree and their infrared lines lie on the counterterm lines of the mass-counterterm units, so that Corollary 17.42 of the full version applies; hence the effective actions at consecutive k on the common unit lattice differ by a field-free, t- and G-independent constant d_k up to $\eta_k|T|$, King's [15, (3.9)] without the field-free h-sector constant, which is exactly what drifts (the photon quadratic form is compared by Lemma R.85 of the full version, and the scalar quadratic forms and the background determinants through their torus forms by Proposition R.101 of the full version, at the fixed photon mass $\mu_0 > 0$ and for $k \geq k_0$, which is why Proposition 17.135 of the full version is stated at fixed μ_0 and from a threshold k_0 independent of t, G and h); d_k cancels in every ratio, and the consecutive differences of the frozen ratios are summable, the exponents being chosen in the window of Lemma 17.27 of the full version (Corollary R.102 of the full version) and the large-field terms decaying like $e^{-c_2 k^2}$ (Proposition R.82(b) of the full version). A constant shift u of the scalar source changes the effective action by at most $C(1+r)|u|p(\delta_r)^{n_1}\delta_r^{-3}|T|$, and the rate $|u_k| \leq C\varepsilon_k$ against δ_r^{-3} is geometric for $\varrho < \frac{1}{3}$. No value of d_k and no Wick scheme is used, and the sign of the wall enters nowhere.

The forward bound and the assembly.

Proposition 17.139 (the forward bound)**.** *Let the couplings be off E_W (Remark 15.2). For $T \in \{T_1, T_2\}$ and every real source $s = (zG, h)$ with $h = -n\psi$ the wall of a configuration (J, ψ, n) of the class of Theorem 17.13, or its transport,*

$$\mathsf{Z}_{Y_k,T}(s) \leq (1+\delta_k)\, e^{-c_k \mathsf{S}_{X_k}(h)}\, \mathsf{Z}_{X_k,T}(s), \qquad \delta_k = \delta_k(T,s) \to 0,$$

with the constant c_k of Lemma 17.62 of the full version, independent of T and of the source, and $|c_k| \leq C\varepsilon_k^{-q}$. No bound on c_k is used.

Idea of the proof. By Proposition 17.100, for k large, $\mathsf{Z}_Y \leq e^{C\delta_r^{\sigma}|T|} \int \chi_r e^{-S^{(r)}} + B$ with the large-field term $B \leq \mathfrak{N}\, e^{-c\,p(\delta_r)^2 + C|T|}\varepsilon^{-q}$, where $\mathsf{Z}_Y = \mathfrak{N}\, \mathsf{Z}^{\mathrm{tor}}_{Y_k,T}(zG, h + u(\varepsilon_k)\mathbf{1})$; by Theorem 17.120 in the torus convention at the shifted source (at a wall this bound takes the place of Proposition 17.118) $B \leq b\, \mathsf{Z}_Y$ with $b \to 0$, since $c\,p(\delta_r)^2$ grows at least quadratically in k and $\ln \varepsilon^{-q-q'}$ linearly (Proposition R.82(c) of the full version), so the large-field term is absorbed on the left, which is why no bound on c_k and no lower bound for the axis lattice is needed; on $\operatorname{supp}\chi_r$, Theorem 17.63 gives $e^{-S} \leq e^{-\hat{s} - c_k \mathsf{S}_{X_k}(h) + \vartheta_k|T|}$, and Proposition 17.117 completes the chain.

Proposition 17.140 (mass-increasing walls only)**.** *Let $\mathcal{S}^- := \{(tG, h_0) : t \in \mathbb{R}\}$ with $h_0 = -n\psi$, $n > 0$, a family not closed under $h \mapsto -h$, and assume the forward bounds of Proposition 17.139 on T_1 for $s \in \mathcal{S}^-$ and on T_2 for their S-images, with the common constant c_k. Then the conclusions* (a) *and* (c) *of Theorem 17.6 hold for the sources of $\mathcal{S}^-$, with $v = v' = 0$, $c = c' = c_k$ and $c_k \to 0$ (Proposition 17.72); this is Theorem 17.6(b′). The sources $(tG, +n\psi)$ do not occur.*

Idea of the proof. Part (a) of Theorem 17.6 uses only the two forward bounds, one source at a time; part (b) is the only place where the pair $(G, \pm h)$ enters, and its conclusion $c + c' \to 0$ is supplied directly by $c = c' = c_k \to 0$; part (c) follows as there.

Proof of Theorem 17.13 (route). Fix $\mathfrak{c} = (J, \psi, n)$ in the class and G, and let $\mathcal{S}^- = \{(tG, h_0) : t \in \mathbb{R}\}$, $h_0 = -n\psi$. *Step 1*: Theorem 17.63 on T_1 for $s \in \mathcal{S}^-$ and on T_2 for $S\cdot s$, with the common constant c_k. *Step 2*: Proposition 17.139 on both tori (inputs: Propositions 17.100 and 17.117 and Theorem 17.120 in the torus convention, hence Theorem F.2 and Proposition 17.126 of the full version and Convention 17.17 of the full version throughout, Proposition 17.121 on T_1; in the torus convention the instance of Proposition 17.118 at the wall is Theorem 17.120). *Step 3*: Proposition 17.140 with Proposition 17.72 gives conclusion (c) of Theorem 17.6 for the balanced pair (tG, h_0), $(0, h_0)$, i.e. $\langle e^{tF(G)}\rangle_{\mathfrak{c},Y_k} - \langle e^{tF(G)}\rangle_{\mathfrak{c},X_k} \to 0$ for every real t. *Step 4*: by Lemma 2.9 and Lemma 17.22 the difference is bounded on strips uniformly in k, so Vitali's theorem gives (i). *Step 5*: by Theorem 17.127 the axis family converges, and by (i) so does the rotated family, to the same entire limit: (ii). *Step 6*: Lemma 17.3(c) gives (iii). No reverse step ("switching around", [16, p. 337]), no comparison of two spacings and no infinite-volume limit is used. □

Completion of the proof of Theorem 17.1. By Lemma 17.3(c) it suffices to treat $R = R_{12}(\theta_0)$, and by Remark 18.6 the configurations to be treated are those of the class of Theorem 17.13. For such $\mathfrak{c}$ and G, Steps 1–2 of the proof of Theorem 17.13 give (F) on both tori for the sources (tG, h_0), in total form, with $v_k(T_1) = v_k(T_2) = 0$ and $c_k(T_1) = c_k(T_2) = c_k \to 0$; Proposition 17.140 supplies conclusion (c) of Theorem 17.6, Corollary 17.7 gives $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k} - \langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k} \to 0$ locally uniformly in z, and with Theorem 17.127 its last clause gives (K_0) for $\mathfrak{c}$ and G. □

18. Walled tori, the closing estimate and rotation invariance

18.1. **Walled tori and the closing estimate.**

Definition 18.1 (admissible walls and walled values)**.** For a bounded set $X \subset \mathbb{R}^3$ let $\mathcal{A}(X)$ be the set of walled configurations $\mathfrak{c} = (J', w)$ of Definition 15.7 with $N_1(X) \subset (-25MJ', 25MJ')^3$ and $w = 0$ on $N_1(X)$. The interior $(-25MJ', 25MJ')^3$ of the cell $\Gamma_{J'}$ is a fundamental domain of $D_{J'}$ for both lattices, $R^{-1}\Lambda_k$ descending to $D_{J'}$ for $R \in \mathcal{R}_0$ by Lemma 15.4(b), and sources supported in X are periodised from it. If $X \subset Y$ then $\mathcal{A}(Y) \subset \mathcal{A}(X)$. Recall the walled ratio $T^a_{\mathfrak{c},\Lambda}(f,g) = e^{-a_k \mathsf{S}_\Lambda(g)}\langle\Theta(\mathrm{i}f, g)\rangle_{\mathfrak{c},\Lambda}$ of a lattice Λ of spacing ε_k descending to $D_{J'}$, defined for $(f,g) \in \mathcal{K}$ with $w = 0$ on $\operatorname{supp} g$ (Lemma 15.10(c)). For $g \neq 0$ assume $(\mathrm{K})_a$ (Definition 17.2). For $(f,g) \in \mathcal{K}$ supported in X and $\mathfrak{c} \in \mathcal{A}(X)$ the periodised sources are smooth, and by Lemma 17.3(a) the *walled value*

$$\mathcal{W}^a_{\mathfrak{c}}(f,g) := \lim_{k\to\infty} T^a_{\mathfrak{c},\Lambda_k}(f,g) = \lim_{k\to\infty} T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f,g) \qquad (R \in \mathcal{R}_0)$$

exists in $(0,\infty)$. For $b > 0$ and $(f,g) \in \mathcal{K}$ supported in Q_b let $I^a_b(f,g) := \inf_{\mathfrak{c}\in\mathcal{A}(Q_b)} \mathcal{W}^a_{\mathfrak{c}}(f,g)$, which is nondecreasing in b, and

$$\mathcal{W}^{a,\mathrm{env}}(f,g) := \lim_{b\to\infty} I^a_b(f,g) \in (0,\infty].$$

At $g = 0$ these definitions rest on (K_0) alone (Theorem 17.1; Lemma 17.3(a) at the complex coefficient i), and they do not depend on the scheme (Lemma 15.10(c)): $\mathcal{W}_{\mathfrak{c}}(f,0) = \lim_k \langle e^{\mathrm{i}F_{\Lambda_k}(f)}\rangle_{\mathfrak{c},\Lambda_k} \in [-1,1]$, and we write $I_b(f,0)$, $\mathcal{W}^{\mathrm{env}}(f,0)$. Where (K_0) is supplied for the walled configurations of the class of Remark 18.6 only, as in Theorem 17.1, the walled value at $g = 0$ is defined for the configurations of the class, $I_b(f,0)$ is the infimum over the configurations of the class in $\mathcal{A}(Q_b)$ and $\mathcal{W}^{\mathrm{env}}(f,0)$ its limit (the class reading of Theorem 18.4); every infimum and supremum over a set $\mathcal{A}(X)$ at $g = 0$ below is read in the same way. King's printed state at a curvature source is written $Z^K_\infty(\mathrm{i}f, 0)$ in this part, the second slot recording the absence of a scalar source; it is $Z^K_\infty(\mathrm{i}f)$ of Theorem 7.6(b), and $Z^K_J(\mathrm{i}f)$ denotes the fixed-cube limit there.

Lemma 18.2 (the free direction)**.** (a) *Assume* $(\mathrm{W})_a$ *and* $(\mathrm{K})_a$*. If* $(f,g) \in \mathcal{K}$ *is supported at* ℓ^∞*-distance* > 1 *from* $\mathbb{R}^3 \setminus C_J$ *and* $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$*, then* $\mathcal{W}^a_{\mathfrak{c}}(f,g) \geq Z^a_J(\mathrm{i}f, g)$*, where* $Z^a_J = \lim_k Z^a_{k,J}$ *is the fixed-cube limit of Theorem 16.6(i).*

(b) *Assume* $(\mathrm{W})_a$ *and* $(\mathrm{K})_a$*. Then* $\mathcal{W}^{a,\mathrm{env}} \geq Z^a_\infty$ *on* $\mathcal{K}$*.*

(c) *If the real 2-form $f \in C_c^\infty$ is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ and $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$ is a configuration of the class of Remark 18.6, then $\mathcal{W}_\mathfrak{c}(f,0) \geq Z_J^K(\mathrm{i}f)$.*
(d) $\mathcal{W}^{\mathrm{env}}(f,0) \geq Z_\infty^K(\mathrm{i}f,0) \geq 0$ *for every real 2-form $f \in C_c^\infty$.*

Proof. (a) Fix k. The set $X_J = C_J \cap \Lambda_k$ and its nearest neighbours lie in $N_1(\bar{C}_J)$, hence in the interior of the cell, and the wall vanishes there; so Lemma 15.9 applied to the walled torus state $\langle\cdot\rangle_{\mathfrak{c},\Lambda_k}$ with $X = X_J$ has as its X-factor the comparison cube $\langle\cdot\rangle_{k,J}^{(a)}$ of Definition 2.1, and gives $\langle\Theta(\mathrm{i}f,g)\rangle_{k,J}^{(a)} \leq \langle\Theta(\mathrm{i}f,g)\rangle_{\mathfrak{c},\Lambda_k}$; every plaquette and site at which (f,g) does not vanish lies in X_J by the support condition. The multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is the same positive constant on both sides (Lemma 15.10(c), Lemma 2.12(a)), so $Z_{k,J}^a(\mathrm{i}f,g) \leq T_{\mathfrak{c},\Lambda_k}^a(f,g)$. Let $k \to \infty$: the left side converges to $Z_J^a(\mathrm{i}f,g)$ by Theorem 16.6(i), the right side to $\mathcal{W}_\mathfrak{c}^a(f,g)$ by Definition 18.1.

(b) If $\bar{C}_J \subset Q_b$ then $\mathcal{A}(Q_b) \subset \mathcal{A}(\bar{C}_J)$, so $I_b^a(f,g) \geq Z_J^a(\mathrm{i}f,g)$ by (a) for every J at which the support condition of (a) holds; let $b \to \infty$ and then $J \to \infty$ (Theorem 16.6(iii)).

(c) At $g = 0$ no multiplier enters. As in (a), Lemma 15.9 gives $Z_{k,J}(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle_{k,J}^{(a)} \leq T_{\mathfrak{c},\Lambda_k}(f,0)$, and the first inequality of Lemma 4.2, whose support condition (S) holds for X_J because f is supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$, gives $Z_{k,J}^K(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle_{k,J}^K \leq Z_{k,J}(\mathrm{i}f)$. Hence $Z_{k,J}^K(\mathrm{i}f) \leq T_{\mathfrak{c},\Lambda_k}(f,0)$ at every k. Let $k \to \infty$: the left side converges to $Z_J^K(\mathrm{i}f)$ by Theorem 7.6(b) under (E_C), the right side to $\mathcal{W}_\mathfrak{c}(f,0)$ by (K_0) (Lemma 17.3(a)).

(d) Fix f and $J \geq J_0(f)$. For $b \geq MJ$ we have $\bar{C}_J = Q_{MJ} \subset Q_b$, so $\mathcal{A}(Q_b) \subset \mathcal{A}(\bar{C}_J)$ and $I_b(f,0) \geq Z_J^K(\mathrm{i}f)$ by (c); letting $b \to \infty$, $\mathcal{W}^{\mathrm{env}}(f,0) \geq Z_J^K(\mathrm{i}f)$ for every $J \geq J_0(f)$. By Corollary 4.3 and Theorem 7.6(b), $J \mapsto Z_J^K(\mathrm{i}f)$ is nondecreasing with values in $[0,1]$, and by Corollary 4.4(ii)(β) it converges to $Z_\infty^K(\mathrm{i}f,0)$; let $J \to \infty$. □

Lemma 18.2(a),(c) compare a walled value with a cube of fixed size. They do not assert $\mathcal{W}_\mathfrak{c}^a \geq Z_\infty^a$ for an individual configuration, and this need not hold: for instance, at $e = \lambda = 0$, $f = 0$ and $g \geq 0$ small, as $n \to \infty$ the walled value of (J', ψ, n) decreases (Lemma 2.8(c)) towards the value of the Dirichlet state on the set where $\psi = 0$, which is smaller than the infinite-volume value, as the Gaussian formula for $\langle e^{:\phi^2:(g)}\rangle$ shows.

Lemma 18.3 (independence of the orientation)**.** *Assume $(\mathrm{K})_a$ if $g \neq 0$. For $R \in \mathcal{R}_0$ and $(f,g) \in \mathcal{K}$,*

$$\lim_{L\to\infty} \inf_{\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}_\mathfrak{c}^a(f,g) = \mathcal{W}^{a,\mathrm{env}}(f,g),$$

the infimum being defined as soon as $\operatorname{supp} f \cup \operatorname{supp} g \subset Q_{5ML/7}$. At $g = 0$ the same holds, with no assumption, for $\mathcal{W}_\mathfrak{c}(f,0)$ and $\mathcal{W}^{\mathrm{env}}(f,0)$, with the infima over the configurations of the class of Remark 18.6 where (K_0) is supplied on that class only (Definition 18.1).

Proof. By Lemma 15.4(c), $Q_{5ML/7} \subset R^{-1}Q_{ML} \subset Q_{7ML/5}$, hence $\mathcal{A}(Q_{7ML/5}) \subset \mathcal{A}(R^{-1}Q_{ML}) \subset \mathcal{A}(Q_{5ML/7})$, and the same inclusions hold between the configurations of the class in these sets; so $I_{5ML/7}^a \leq \inf_{\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}_\mathfrak{c}^a \leq I_{7ML/5}^a$ in either reading. Both bounds tend to $\mathcal{W}^{a,\mathrm{env}}$ as $L \to \infty$. The proof uses only the inclusions between shrinking families, so it holds on the class. □

Theorem 18.4 ((TIGHT_0), (TIGHT_2): the closing estimate)**.** *Let the couplings be a point of King's range off E_W (Remark 15.2). (TIGHT_0): $\mathcal{W}^{\mathrm{env}}(f,0) \leq Z_\infty^K(\mathrm{i}f,0)$ for every real 2-form $f \in C_c^\infty$. Where (K_0) is supplied on a sub-class of the walled configurations only, as in Theorem 17.1 (the class of Remark 18.6), (TIGHT_0) is read with the infimum over that sub-class, whose conditions do not depend on R (J_{pos} being the maximum over the finite set $\mathcal{R}_0$ and the three torus pairs, Theorem 17.98 of the full version; the condition $J \geq J_0(\delta_r)$ being void for $k \geq k_0$, k_0 the number of Remark 18.6, which depends on neither R nor J; n_* being a number of King's flow; the vanishing of the wall near $R^{-1}C_L$ is the membership in $\mathcal{A}(R^{-1}Q_{ML})$ of Definition 18.1, not a condition of the class; the block alignment and the grid offset are conditions on H' and T in the frame of the axis lattice Λ_k and in units of its final block, which do not involve R): equivalently, for every $b > 0$, every real 2-form f supported in Q_b and every $\eta > 0$ there is a configuration $\mathfrak{c}$ of the class in $\mathcal{A}(Q_b)$ with $\mathcal{W}_\mathfrak{c}(f,0) \leq Z_\infty^K(\mathrm{i}f,0) + \eta$. For $b \geq \frac{7}{5}ML + 2$ such a configuration has $\psi = 0$ on the 2-neighbourhood of $R^{-1}C_L$ for every $R \in \mathcal{R}_0$ (Lemma 15.4(c)),*

so it lies in $\mathcal{A}(R^{-1}Q_{ML})$ *and serves the comparison with the cube* C_L *on the pair* $(\Lambda_k, R^{-1}\Lambda_k)$ *for every* $R \in \mathcal{R}_0$. (TIGHT$_2$)*: for every real* 2*-form* $f \in C_c^\infty$,

$$\lim_{b\to\infty} \sup_{\mathfrak{c}\in\mathcal{A}(Q_b)} \sigma_{\mathfrak{c}}^2(f) \;\geq\; S_2^K(f), \qquad \sigma_{\mathfrak{c}}^2(f) := \lim_k \langle F_{\Lambda_k}(f)^2\rangle_{\mathfrak{c},\Lambda_k}, \quad S_2^K(f) := S_2^K(f,f) = \lim_J \lim_k \langle F(f)^2\rangle_{k,J}^K,$$

the supremum being taken, where (K$_0$) *is supplied on the class of Remark 18.6 only, over the configurations of that class in* $\mathcal{A}(Q_b)$*, as the infimum in* (TIGHT$_0$).

(TIGHT$_0$) involves only (K$_0$), which defines its left side (on the class of Remark 18.6, at the couplings off E_W at which (K$_0$) is stated), and King's printed state, which exists by (E$_C$). It is proved in the heavy-shell sections: Theorem 19.10 derives it from the decoupling property (RES$_0$) of Definition 19.8, which Theorem 20.14 establishes at one configuration from the heavy-shell estimates (HS); (TIGHT$_2$) follows from (TIGHT$_0$) by Lemma 18.7(c). In the Wick sector the closing statement is a property of the scheme, not proved here (it is listed among the open problems, in the final section) and used only by the optional extension of the Wick leg (Theorem 23.7):

Definition 18.5 ((TIGHT)$_a$: the closing property, Wick sector)**.** Let a be W-admissible and assume (K)$_a$. (TIGHT)$_a$: $\mathcal{W}^{a,\mathrm{env}} \leq Z_\infty^a$ on $\mathcal{K}$. Equivalently: for every $b > 0$, every $(f,g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}_{\mathfrak{c}}^a(f,g) \leq Z_\infty^a(\mathrm{i}f, g) + \eta$.

The equivalence in (TIGHT)$_a$ holds because I_b^a is nondecreasing in b. By Lemma 18.2, (TIGHT)$_a$ is equivalent to $\mathcal{W}^{a,\mathrm{env}} = Z_\infty^a$ on $\mathcal{K}$, and (TIGHT$_0$) to $\mathcal{W}^{\mathrm{env}}(\cdot,0) = Z_\infty^K(\mathrm{i}\cdot, 0)$: *King's printed state is the limit of the lower envelopes of walled tori, the continuum limit being taken first.* The limit defining $\sigma_{\mathfrak{c}}^2$ exists, for the configurations of the class, by (K$_0$), since $t \mapsto \langle e^{\mathrm{i}tF(f)}\rangle_{\mathfrak{c},\Lambda_k}$ converges locally uniformly as an entire function (Lemma 17.3(a)) and second derivatives of locally uniformly convergent entire functions converge; S_2^K is the two-point Schwinger function of Theorem 7.6(c), with the lattice second moments converging to it. The supremum over the configurations of the class in $\mathcal{A}(Q_b)$ is nonincreasing in b, since $\mathcal{A}(Q_{b'}) \subset \mathcal{A}(Q_b)$ for $b' \geq b$ and the class does not depend on b, so the limit in (TIGHT$_2$) exists.

Remark 18.6 (the configuration class). (K$_0$) is supplied by Theorem 17.13 for the walled configurations of its class; this class, with the grid-offset condition of Theorem 20.2 added so that one class serves every statement of this part, is smaller than $\mathcal{A}(X)$: $\mathfrak{c} = (J, \psi, n)$ with $J \in 5^{n_*}\mathbb{N}$ (n_* the number of large-field scales of the rotated flow), $J \geq \max(J_0(\delta_r), J_{\mathrm{pos}})$ ($J_0(\delta_r)$ the non-wrapping threshold of Lemma 17.124 of the full version at the coarsest spacing of the flow, J_{pos} the threshold of Theorem 17.98 of the full version), $n > 0$ (a mass-increasing wall), the wall condition $n\mathsf{S}_{X_k}(\psi) > \Delta_*|T_1|/2$ for every k (Δ_* the constant of Proposition 17.126(b) of the full version, which does not depend on the torus), block alignment (the source box and the torus unions of blocks of the final scale) and the grid offset of the corners of the source box H' of Theorem 20.2 (every corner coordinate, in units of the final block, congruent to $M'/2$ modulo M', where $M = 5M'$); the threshold J_{pos} belongs to the class because the margin of the positivity condition is proved for $J \geq J_{\mathrm{pos}}$ only. This costs nothing: in the proofs of Theorem 18.12 and Corollary 18.13 below the infimum over $\mathcal{A}(R^{-1}Q_{ML})$ is used only through an upper bound by the walled value of one configuration supplied by (TIGHT$_0$), so the infimum may be taken over any sub-class that contains that configuration, (TIGHT$_0$) is then read with the same sub-class (Theorem 18.4), and Lemma 18.3 holds on it, since its proof uses only the inclusions between shrinking families. The configuration that Theorem 19.10 produces lies in the class: its wall is $(n+K)\psi$ with $n+K \geq n > 0$, mass-increasing; the wall condition is linear in the height and holds for K large at a fixed torus, which Remark 19.11(iii) allows; J (the J' of Theorem 20.14), the block alignment of H' and T and the grid offset of the corners of H' are choices of the torus and of the box that Theorem 20.14 makes ($J' \geq \max(J_0, J_{\mathrm{pos}})$ among them), the last moving H' by less than M' final blocks, since in each direction every interval of M' final blocks contains an admissible corner coordinate (Theorem 20.2). On the skew pairs the large-field clause of King's upper bound at a wall (Proposition 17.100, through the fixed-mass bounds of Theorem K.506 of the full version) takes one step at Bałaban's final spacing,

which needs the non-wrapping condition at that spacing; the class carries the cap $\varepsilon_0^{\mathrm{Ba}} \le \varepsilon_{\mathrm{sys}}$ of Lemma K.518 of the full version on Bałaban's stopping scale (Theorem 17.13), $\varepsilon_{\mathrm{sys}}$ being the minimum over $R \in \mathcal{R}_0$ of the thresholds defined there, which does not depend on J and is closed downward, and under it the non-wrapping threshold equals 1 for every J of the residue class $5^{n_*}\mathbb{N}$ at every spacing $t \le \varepsilon_0^{\mathrm{Ba}}$ (Lemma K.518(iii) of the full version), Bałaban's final spacing included. The threshold $J_0(\delta_r)$ is the least J satisfying (259) of the full version at the coarsest spacing $\delta_r = 5^{-\lfloor \varrho k \rfloor}$ of the flow (Lemma 17.27 of the full version), one and the same function of the spacing for every $R \in \mathcal{R}_0$ (Lemma K.518(ii) of the full version); it depends on k and is imposed at every k at which the comparison is run. Put $k_0 := \min\{k \in \mathbb{N} : 5^{-\lfloor \varrho k \rfloor} \le \varepsilon_0^{\mathrm{Ba}}\}$, a number depending on $\varepsilon_0^{\mathrm{Ba}}$ and ϱ only and neither on R, on J nor on the configuration. For $k \ge k_0$ the spacing δ_r lies below the cap, so $J_0(\delta_r) = 1$ and the condition $J \ge J_0(\delta_r)$ holds for every J of the residue class; for $k \ge k_0$ the class is therefore the fixed set of configurations defined by the remaining conditions, none of which depends on k or on R (n_* is the number of large-field scales of the standard flow of Y_k in its own frame, which is King's flow and does not depend on R), and (TIGHT_0) and Lemma 18.3 are read on this set.

Lemma 18.7. (a) *If a, a' are* W*-admissible and* $(\mathrm{K})_a$ *holds, then* $(\mathrm{TIGHT})_a$ *and* $(\mathrm{TIGHT})_{a'}$ *are equivalent.*

(b) *Assume* $(\mathrm{W})_a$ *and* (E_C)*. Then* $(\mathrm{TIGHT})_a$ *implies* (TIGHT_0)*, both read over all walled configurations (the reading when* (K_0) *holds on all of them, Lemma 17.3(b)).*

(c) *Under* (K_0)*,* (TIGHT_0) *implies* (TIGHT_2)*, in the same reading; and* $\lim_b \sup_{\mathcal{A}(Q_b)} \sigma_{\mathfrak{c}}^2(f) \le S_2^K(f)$ *always, so* (TIGHT_2) *is an equality.*

Proof. (a) By Lemma 16.4(b), $a'_k - a_k \to \kappa_{a'a}$, and $T^{a'}_{\mathfrak{c},\Lambda} = e^{-(a'_k - a_k)\mathsf{S}_\Lambda(g)} T^{a}_{\mathfrak{c},\Lambda}$ by Lemma 15.10(c); since $\mathsf{S}_{\Lambda_k}(g) \to \int g$, $\mathcal{W}^{a'}_{\mathfrak{c}} = e^{-\kappa_{a'a}\int g}\, \mathcal{W}^{a}_{\mathfrak{c}}$ with a factor independent of $\mathfrak{c}$. The same factor relates $Z^{a'}_\infty$ and Z^a_∞ (Theorem 16.6(vi)).

(b) At $g = 0$ the walled values are scheme-free (Lemma 15.10(c)), so the left sides of $(\mathrm{TIGHT})_a$ and (TIGHT_0) agree, and the right side of $(\mathrm{TIGHT})_a$ is $Z^a_\infty(\mathrm{i}f, 0) = \lim_J \lim_k Z_{k,J}(\mathrm{i}f)$, the iterated limit of the comparison family, which exists under $(\mathrm{W})_a$ (Theorem 16.6(iii)) and equals $Z^K_\infty(\mathrm{i}f, 0)$ under (E_C) by Corollary 4.4(ii)(γ).

(c) Fix f, b and $\eta' > 0$. All functions of t below are real and even. Since $\cos x \ge 1 - \frac{x^2}{2}$ and $\cos x \le 1 - \frac{x^2}{2} + \frac{x^4}{24}$ at every lattice level, and the moments converge in the limits (for the walled torus by Lemma 17.3(a), for King's cube by Theorem 7.6(c)),

$$\mathcal{W}_{\mathfrak{c}}(tf, 0) \ge 1 - \tfrac{t^2}{2}\sigma^2_{\mathfrak{c}}(f), \qquad Z^K_\infty(\mathrm{i}tf, 0) \le 1 - \tfrac{t^2}{2} S_2^K(f) + \tfrac{t^4}{24} M_4(f),$$

with $M_4(f) := \partial_t^4 Z^K_\infty(tf, 0)|_{t=0} = \lim_J \lim_k \langle F(f)^4 \rangle^K_{k,J} < \infty$. By (TIGHT_0) applied to tf, for every $t > 0$ and $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ (of the class, in the class reading) with $\mathcal{W}_{\mathfrak{c}}(tf, 0) \le Z^K_\infty(\mathrm{i}tf, 0) + \eta$, hence $\sigma^2_{\mathfrak{c}}(f) \ge S_2^K(f) - \frac{t^2}{12} M_4(f) - \frac{2\eta}{t^2}$. Choosing t with $t^2 M_4(f)/12 < \eta'/2$ and then $\eta = t^2\eta'/4$ gives $\sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \ge S_2^K(f) - \eta'$.

For the second claim, let $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$ (of the class, in the class reading) with f supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$. By Lemma 18.2(c) applied to tf, $\mathcal{W}_{\mathfrak{c}}(tf, 0) \ge Z^K_J(\mathrm{i}tf)$ for all real t, with equality ($= 1$) at $t = 0$; both sides are even and smooth, so comparing second derivatives at $t = 0$ gives $\sigma^2_{\mathfrak{c}}(f) \le v_J(f) := \lim_k \langle F(f)^2 \rangle^K_{k,J}$, the limit existing by Theorem 7.6(c). Hence $\sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \le v_J(f)$ whenever $\bar{C}_J \subset Q_b$. By Corollary 4.3, $\langle e^{\mathrm{i}tF(f)} \rangle^K_{k,J} \le \langle e^{\mathrm{i}tF(f)} \rangle^K_{k,J+1}$ for all real t, with equality at $t = 0$; comparing second derivatives, the cube variances $\langle F(f)^2 \rangle^K_{k,J}$ are nonincreasing in J at every k, hence so are the $v_J(f)$, and they converge to $S_2^K(f)$ (Theorem 7.6(c)). Therefore $\lim_b \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \le \inf_J v_J(f) = S_2^K(f)$. □

Remark 18.8 (the role of the closing estimate). Apart from the closing estimate, the proofs of Theorems 18.9 and 18.12 below use only: monotonicity of the cube functionals in J (on $\mathcal{K}$ and on the decreasing class), the comparison "cube $\le$ walled torus" on $\mathcal{K}$ and "cube $\ge$ walled torus" on the decreasing class on each lattice, the equality of walled values on the two lattices, passive covariance, and the existence of the limits. All of these hold for the following family

of Gaussian functionals, in which the analogue of King's state is not rotation invariant. Let $A = \mathrm{diag}(1,2,3)$, $\mu > 0$, and let C_I, C_R, C_U be the translation-invariant operators on $L^2(\mathbb{R}^3)$ with Fourier symbols $\hat{c}_I(k) = (\langle k, Ak\rangle + \mu^2)^{-1}$, $\hat{c}_R(k) = \hat{c}_I(Rk)$ and $\hat{c}_U = \min(\hat{c}_I, \hat{c}_R)$. Let the "cube of size J" on the lattice $X \in \{I, R\}$ have covariance $C_X + J^{-1}$, let every "walled torus" have covariance C_U on both lattices, and let a covariance C define the functionals $e^{-\frac{1}{2}\langle f, Cf\rangle}$ on the cone and $e^{\frac{1}{2}\langle f, Cf\rangle}$ on the decreasing class. Every relation listed reduces to $C_U \le C_X + J^{-1}$ or to the monotonicity of $C_X + J^{-1}$ in J, and passive covariance holds since $\langle f, C_R f\rangle = \langle R\cdot f, C_I\, R\cdot f\rangle$. But $e^{-\frac{1}{2}\langle R\cdot f, C_I R\cdot f\rangle} \neq e^{-\frac{1}{2}\langle f, C_I f\rangle}$ in general, and the closing estimate fails since $C_U \neq C_I$. Hence the closing estimate is not a consequence of the other properties, and an input asserting "walled torus $\le$ cube + small" is needed. (The same family shows that any placement of the two-lattice comparison on a finite box must also carry an input beyond the correlation inequalities: there the input is the two-lattice comparison itself, which fails for $C_R \neq C_I$.)

18.2. **Rotation invariance.**

Theorem 18.9 (invariance under the generators, Wick sector)**.** *Let a be* W*-admissible (*$(\mathrm{W})_a$*), and assume* $(\mathrm{K})_a$ *and* $(\mathrm{TIGHT})_a$*. Then for every $R \in \mathcal{R}_0$ and every $(f,g) \in \mathcal{K}$,*

$$Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g) = Z^a_\infty(\mathrm{i} f, g),$$

and consequently the same holds for every $R \in G_0$.

Proof. Fix $R \in \mathcal{R}_0$ and $(f,g) \in \mathcal{K}$ supported in Q_ρ.

Step 0 (geometry). Take L with $ML \ge \frac{7}{5}\rho + 2$. By Lemma 15.4(c), $R\cdot(f,g)$ is supported in $Q_{7\rho/5}$, at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$, and (f,g) is supported in $Q_{5ML/7} \subset R^{-1}Q_{ML}$. Let $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$. Then $R^{-1}\Lambda_k$ descends to $D_{J'}$ (Lemma 15.4(b)), and $B'_k := R^{-1}(C_L \cap \Lambda_k)$ is a set of sites of $R^{-1}\Lambda_k$ which, with its nearest neighbours, lies in the interior of the cell, and on which $w = 0$. Since $R\cdot(f,g)$ has distance > 1 from $\mathbb{R}^3 \setminus C_L$, every plaquette and site of $R^{-1}\Lambda_k$ at which (f,g) does not vanish lies in B'_k.

Step 1 (cutting on the rotated lattice). By Lemma 15.9, applied to the walled torus state $\langle\cdot\rangle_{\mathfrak{c}, R^{-1}\Lambda_k}$ with $X = B'_k$, whose X-factor is the Neumann–Dirichlet state on B'_k with the scalar alive on all of B'_k, and by the multiplier being the same constant on both sides (Lemma 15.10(c), Lemma 2.12(a)), at every k,

$$Z^a_{R^{-1}\Lambda_k, B'_k}(\mathrm{i} f, g) \le T^a_{\mathfrak{c}, R^{-1}\Lambda_k}(f,g).$$

Step 2 (passive rotation). By Lemma 15.10(b) with $O = R$, which maps $R^{-1}\Lambda_k$ with its frame $(R^{-1}e_\alpha)$ onto Λ_k with the frame (e_α) and B'_k onto X_L, $Z^a_{R^{-1}\Lambda_k, B'_k}(\mathrm{i} f, g) = Z^a_{k,L}(\mathrm{i}\, R\cdot f, R\cdot g)$.

Step 3 (continuum limits). As $k \to \infty$ the left side converges by Theorem 16.6(i) (the sources $R\cdot(f,g)$ being at distance > 1 from $\mathbb{R}^3 \setminus C_L$) and the right side by $(\mathrm{K})_a$ (Definition 18.1):

$$Z^a_L(\mathrm{i}\, R\cdot f, R\cdot g) \le \mathcal{W}^a_{\mathfrak{c}}(f,g) \qquad \text{for every } \mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML}).$$

Step 4 (infimum, volume limit, closing hypothesis). Take the infimum over $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$ and let $L \to \infty$. By Theorem 16.6(iii), Lemma 18.3 and $(\mathrm{TIGHT})_a$,

$$Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g) = \lim_{L\to\infty} Z^a_L(\mathrm{i}\, R\cdot f, R\cdot g) \le \mathcal{W}^{a,\mathrm{env}}(f,g) \le Z^a_\infty(\mathrm{i} f, g).$$

Step 5 (the inverse). $R^{-1} \in \mathcal{R}_0$ (Lemma 15.4(a)) and $R\cdot(f,g) \in \mathcal{K}$, so Steps 0–4 for $(R^{-1}, R\cdot(f,g))$ give $Z^a_\infty(\mathrm{i} f, g) \le Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g)$. Since $\mathcal{K}$ is G_0-invariant and the transport is a left action, the claim extends to G_0. □

Remark 18.10 (the order of limits). The order of limits is: at fixed R, (f,g), L and $\mathfrak{c}$, first $k \to \infty$; then the infimum over $\mathfrak{c}$; then $L \to \infty$. The continuum limit is always taken first, at fixed volume and fixed wall, as in King's construction [16, p. 336], and the torus comparison is used only as an equality of continuum limits, never at a fixed lattice spacing.

Theorem 18.11 (closure: from the generators to $O(3)$, Wick sector)**.** *Let a be* W*-admissible ($(\mathrm{W})_a$), and suppose $Z^a_\infty(\mathrm{i}\,R\cdot f, R\cdot g) = Z^a_\infty(\mathrm{i}f,g)$ for all $(f,g)\in\mathcal{K}$ and $R\in\mathcal{R}_0$. Then for every $R\in O(3)$, every N, all real 2-forms f_j and real h_j in C^∞_c, and all $(z,w)\in\mathbb{C}^{2N}$,*

$$Z^a_\infty\Big(\sum_j z_j\,R\cdot f_j,\sum_j w_j\,R\cdot h_j\Big)=Z^a_\infty\Big(\sum_j z_j f_j,\sum_j w_j h_j\Big).$$

Proof. Complex coefficients. The hypothesis passes to G_0, since $\mathcal{K}$ is invariant under $O(3)$ and $\mathcal{R}_0=\mathcal{R}_0^{-1}$. For $R\in G_0$ and a finite family, split $h_j=h_j^+-h_j^-$ as in the proof of Theorem 16.6(iii). The difference of the two sides, as a function of $(z,w^+,w^-)\in\mathbb{C}^{3N}$, is entire by Theorem 16.6(iii) and vanishes on $\Omega_+=(\mathrm{i}\mathbb{R})^N\times(0,\infty)^{2N}$, where all sources lie in $\mathcal{K}$; Ω_+ has non-empty interior within a totally real subspace of maximal dimension, hence a set of uniqueness [11], and the difference vanishes identically.

Closure in $SO(3)$. At real coefficients, $\Phi^a(R\cdot s)=\Phi^a(s)$ for $s\in\mathcal{E}_\rho$ (the pairs of real sources supported in $\bar B_\rho$, Theorem 16.6(iv)) and $R\in G_0$; note $R\cdot s\in\mathcal{E}_\rho$ since $R\bar B_\rho=\bar B_\rho$. The map $R\mapsto R\cdot s\in\mathcal{E}_\rho$ is continuous on $O(3)$, and Φ^a is continuous on $\mathcal{E}_\rho$ by Theorem 16.6(iv). Hence $G:=\{R\in O(3):\Phi^a(R\cdot s)=\Phi^a(s)$ for all ρ and $s\in\mathcal{E}_\rho\}$ is a closed subgroup of $O(3)$ containing G_0, so $G\supset SO(3)$ by Lemma 15.4(d).

Reflection. Let $\sigma_3=\mathrm{diag}(1,1,-1)$ and $(f,g)\in\mathcal{K}$ supported at ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_{J-1}$. Lemma 2.14 gives, at every k, $Z^a_{k,J-1}(\mathrm{i}f,g)\le Z^a_{k,J}(\mathrm{i}\,\sigma_3\cdot f,\sigma_3\cdot g)\le Z^a_{k,J+1}(\mathrm{i}f,g)$. The family $\sigma_3\cdot(f,g)$ is supported at distance >1 from $\mathbb{R}^3\setminus C_J$, the region $\{\mathrm{dist}_\infty(\cdot,\mathbb{R}^3\setminus C_{J-1})>1\}$ being σ_3-symmetric and contained in C_J; so all three terms converge as $k\to\infty$ by Theorem 16.6(i), and then as $J\to\infty$ by Theorem 16.6(iii): $Z^a_\infty(\mathrm{i}\,\sigma_3\cdot f,\sigma_3\cdot g)=Z^a_\infty(\mathrm{i}f,g)$ on $\mathcal{K}$. The first step, with σ_3 in place of R, gives $\sigma_3\in G$, so $G=O(3)=SO(3)\cup\sigma_3SO(3)$.

Conclusion. For $R\in O(3)$ both sides of the claim are entire in $(z,w)\in\mathbb{C}^{2N}$ (Theorem 16.6(iii)) and agree on $\mathbb{R}^{2N}$, since $R\in G$; hence they agree on $\mathbb{C}^{2N}$. □

Theorem 18.12 (the curvature sector)**.** *Let the couplings be a point of King's range off E_W (Remark 15.2). Then King's printed state on curvature sources is $O(3)$-invariant: for every $R\in O(3)$, all real 2-forms $f_1,\dots,f_N\in C^\infty_c$ and all $z\in\mathbb{C}^N$,*

$$Z^K_\infty\Big(\sum_j z_j\,R\cdot f_j,0\Big)=Z^K_\infty\Big(\sum_j z_j f_j,0\Big),$$

and the curvature Schwinger functions S^K_n of Theorem 7.6(c) are $O(3)$-covariant, F transforming as a 2-form. Moreover, for every Euclidean motion $x\mapsto Rx+b$, $Z^K_\infty(\sum_j z_j\,\tau_b R\cdot f_j,0)=Z^K_\infty(\sum_j z_j f_j,0)$. At the points of E_W the same conclusions hold: this is Theorem 15.1, whose proof obtains them there from the nearby scalar masses through Theorem 20.15.

Proof. The inputs are the existence theorem (E_C) (Theorem 3.3), the torus comparison (K_0) (Theorem 17.1) and the closing estimate (TIGHT_0) (Theorem 18.4), the last two stated, like this theorem, off E_W and on the class of Remark 18.6.

Invariance under the generators at imaginary points. Fix $R\in\mathcal{R}_0$ and a real 2-form $f\in C^\infty_c$ supported in Q_ρ. At $g=0$ the walled values are those of (K_0) (Definition 18.1). Steps 0–2 of the proof of Theorem 18.9 run unchanged with the source f alone: with L and B'_k as there and $\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})$ a configuration of the class of Remark 18.6 (the infimum below is taken over the class; the configuration that (TIGHT_0) supplies for a source supported in Q_b with $b\ge\frac{7}{5}ML+2$ has $\psi=0$ on the 2-neighbourhood of $R^{-1}C_L$, Lemma 15.4(c), so it is admissible there), Lemma 15.9 on the rotated lattice and Lemma 15.10(b) give, at every k,

$$Z_{k,L}(\mathrm{i}\,R\cdot f)=\langle e^{\mathrm{i}F(R\cdot f)}\rangle^{(a)}_{k,J}\big|_{J=L}=Z_{R^{-1}\Lambda_k,B'_k}(\mathrm{i}f)\le T_{\mathfrak{c},R^{-1}\Lambda_k}(f,0),$$

the middle term being the functional of the comparison cube X_L. *Step 3.* The right side converges to $\mathcal{W}_{\mathfrak{c}}(f,0)$ by (K_0) (Lemma 17.3(a)), so $\limsup_k Z_{k,L}(\mathrm{i}\,R\cdot f)\le\mathcal{W}_{\mathfrak{c}}(f,0)$; and since $R\cdot f$ is supported at distance >1 from $\mathbb{R}^3\setminus C_L$ (Step 0), Corollary 4.4(ii)(γ) under (E_C) gives $Z^K_L(\mathrm{i}\,R\cdot f)\le\liminf_k Z_{k,L}(\mathrm{i}\,R\cdot f)$. Hence

$$Z^K_L(\mathrm{i}\,R\cdot f)\le\mathcal{W}_{\mathfrak{c}}(f,0)\qquad\text{for every }\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})\text{ of the class.}$$

Step 4. Take the infimum over the configurations $\mathfrak{c}$ of the class and let $L \to \infty$: the left side converges to $Z^K_\infty(\mathrm{i}\,R\cdot f,0)$ by Corollary 4.4(ii)(β), the infima converge to $\mathcal{W}^{\mathrm{env}}(f,0)$ by Lemma 18.3 (on the class, Remark 18.6), and (TIGHT$_0$) gives

$$Z^K_\infty(\mathrm{i}\,R\cdot f,0) \le \mathcal{W}^{\mathrm{env}}(f,0) \le Z^K_\infty(\mathrm{i}f,0).$$

Step 5. $R^{-1}\in\mathcal{R}_0$ and $R\cdot f$ is a real 2-form in C^∞_c, so the same for $(R^{-1},R\cdot f)$ gives the reverse inequality: $Z^K_\infty(\mathrm{i}\,R\cdot u,0)=Z^K_\infty(\mathrm{i}u,0)$ for every real 2-form $u\in C^\infty_c$ and $R\in\mathcal{R}_0$, hence for every $R\in G_0$, the transport being a left action.

Closure in $SO(3)$. Fix a real 2-form $u\in C^\infty_c$. The map $R\mapsto R\cdot u$ is continuous from $O(3)$ to $L^2(\mathbb{R}^3;\Lambda^2)$: for $R\to R_0$ the 2-forms $R\cdot u$ converge to $R_0\cdot u$ uniformly, with supports in one compact set. By the Lipschitz bound of Corollary 8.3, $|Z^K_\infty(\mathrm{i}R\cdot u)-Z^K_\infty(\mathrm{i}R_0\cdot u)|\le\sqrt{2}\,\|R\cdot u-R_0\cdot u\|_2$, so $R\mapsto Z^K_\infty(\mathrm{i}\,R\cdot u,0)$ is continuous on $O(3)$. It is constant on G_0, which is dense in $SO(3)$ (Lemma 15.4(d)); hence $Z^K_\infty(\mathrm{i}\,R\cdot u,0)=Z^K_\infty(\mathrm{i}u,0)$ for all $R\in SO(3)$.

The reflection. By Theorem 8.5 with $O=\sigma_3\in O_h$, $Z^K_\infty(\mathrm{i}\,\sigma_3\cdot u,0)=Z^K_\infty(\mathrm{i}u,0)$ for every real u. Since $O(3)=SO(3)\cup\sigma_3SO(3)$ and $(\sigma_3R)\cdot u=\sigma_3\cdot(R\cdot u)$, the identity at imaginary points holds for every $R\in O(3)$.

All complex coefficients. Fix $R\in O(3)$ and the family $f_1,\dots,f_N$. By Theorem 7.6(b) both sides of the claim are entire functions of $z\in\mathbb{C}^N$. At $z=\mathrm{i}t$, $t\in\mathbb{R}^N$, we have $\sum_j \mathrm{i}t_j\,R\cdot f_j=\mathrm{i}\,R\cdot u_t$ with $u_t=\sum_j t_jf_j$, so the two sides agree on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal dimension, hence a set of uniqueness [11]; they agree on $\mathbb{C}^N$. Differentiating at $z=0$ gives $S^K_n(R\cdot f_1,\dots,R\cdot f_n)=S^K_n(f_1,\dots,f_n)$ for every n, the $O(3)$-covariance of the Schwinger functions.

Euclidean motions. By Corollary 8.3, $Z^K_\infty(\sum_j z_j\tau_bR\cdot f_j,0)=Z^K_\infty(\sum_j z_jR\cdot f_j,0)$ for every $b\in\mathbb{R}^3$, and the right side equals $Z^K_\infty(\sum_j z_jf_j,0)$ by the above. □

Corollary 18.13 (the curvature two-point function)**.** *Let the couplings be any point of King's range. The two-point Schwinger function $S^K_2(f_1,f_2)=\lim_J\lim_k\langle F(f_1)F(f_2)\rangle^K_{k,J}$ of Theorem 7.6(c) satisfies $S^K_2(R\cdot f_1,R\cdot f_2)=S^K_2(f_1,f_2)$ for all $R\in O(3)$ and all real 2-forms $f_1,f_2\in C^\infty_c$.*

Proof. Of the closing estimate only (TIGHT$_2$) is used. Let first the couplings be off E_W, where (K$_0$) and (TIGHT$_2$) are stated (Theorems 17.1 and 18.4). Let $R\in\mathcal{R}_0$ and f a real 2-form. Steps 0–3 of the proof of Theorem 18.12, with the source tf, give $Z^K_L(\mathrm{i}t\,R\cdot f)\le\mathcal{W}_{\mathfrak{c}}(tf,0)$ for all real t and every configuration $\mathfrak{c}$ of the class of Remark 18.6 in $\mathcal{A}(R^{-1}Q_{ML})$. Both sides are even and smooth in t and equal 1 at $t=0$; comparing second derivatives, $v_L(R\cdot f)\ge\sigma^2_{\mathfrak{c}}(f)$, where $v_L(u):=\lim_k\langle F(u)^2\rangle^K_{k,J}|_{J=L}$ is the cube variance (Theorem 7.6(c)). Since every configuration of the class in $\mathcal{A}(Q_{7ML/5+2})$ lies in $\mathcal{A}(R^{-1}Q_{ML})$ and has $\psi=0$ on the 2-neighbourhood of $R^{-1}C_L$ (Lemma 15.4(c)), and $b\mapsto\sup_{\mathcal{A}(Q_b)}\sigma^2_{\mathfrak{c}}(f)$, the supremum over the configurations of the class in $\mathcal{A}(Q_b)$ (Theorem 18.4), is nonincreasing,

$$v_L(R\cdot f)\ge\sup_{\mathcal{A}(Q_{7ML/5+2})}\sigma^2_{\mathfrak{c}}(f)\ge\lim_{b\to\infty}\sup_{\mathcal{A}(Q_b)}\sigma^2_{\mathfrak{c}}(f)\ge S^K_2(f)$$

by (TIGHT$_2$). The cube variances converge to $S^K_2(R\cdot f)$ as $L\to\infty$ (Theorem 7.6(c)), so $S^K_2(R\cdot f)\ge S^K_2(f)$, and the inverse generator gives equality; hence $S^K_2(u,u)$ is G_0-invariant.

The bound. By Theorem 7.6(a), $\langle e^{tF(f)}\rangle^K_{k,J}\le\exp(\frac12t^2\mu_0^{-2}\|d^*_kf\|^2_{2,k})$ for real t, with equality at $t=0$; comparing second derivatives, $\langle F(f)^2\rangle^K_{k,J}\le\mu_0^{-2}\|d^*_kf\|^2_{2,k}$, and $\|d^*_kf\|_{2,k}\to\|d^*f\|_{L^2}$ by (1). Letting $k\to\infty$ and then $J\to\infty$, $S^K_2(f)\le\mu_0^{-2}\|d^*f\|^2_{L^2}$; and $S^K_2(f)\ge0$ since $Z^K_\infty(\mathrm{i}tf,0)\le1=Z^K_\infty(0,0)$ (Theorem 7.6(b)). (The continuum bound of Theorem 7.6(b) gives the same at once.)

Closure. S^K_2 is a positive semidefinite symmetric bilinear form, so $|S^K_2(f,f)-S^K_2(f',f')|=|S^K_2(f-f',f+f')|\le S^K_2(f-f')^{1/2}S^K_2(f+f')^{1/2}$, and with the bound, $R\mapsto S^K_2(R\cdot f,R\cdot f)$ is continuous on $O(3)$, because $d^*(R\cdot f)\to d^*(R_0\cdot f)$ in L^2 as $R\to R_0$ (uniform convergence of the first derivatives on one compact set). Density of G_0 gives $SO(3)$; Theorem 8.5 with $O=\sigma_3$, differentiated twice at $z=0$, gives σ_3, hence $O(3)$; and polarisation gives the bilinear form.

The points of E_W. Let the couplings lie in E_W, and write $Z^K_\infty(\,\cdot\,,0;M)$ and $S^K_2(\,\cdot\,,\cdot\,;M)$ for King's printed state and its two-point function at the scalar mass M, the other couplings fixed. By Theorem 20.15 the scalar masses $M > m$ close to m lie off E_W, so the identity $S^K_2(R\cdot f_1, R\cdot f_2; M) = S^K_2(f_1,f_2;M)$ holds at them by the above, and for all real 2-forms $f_1, f_2 \in C^\infty_c$ the entire functions $z \mapsto Z^K_\infty(z_1 f_1 + z_2 f_2, 0; M)$ converge to $Z^K_\infty(z_1 f_1 + z_2 f_2, 0; m)$ locally uniformly on $\mathbb{C}^2$ as $M \downarrow m$. Second derivatives at $z = 0$ of locally uniformly convergent entire functions converge, and $S^K_2(f_1,f_2;M) = \partial_{z_1}\partial_{z_2} Z^K_\infty(z_1 f_1 + z_2 f_2, 0; M)|_{z=0}$ (Theorem 7.6(c)); hence $S^K_2(f_1,f_2;M) \to S^K_2(f_1,f_2;m)$, and likewise for the pair $(R\cdot f_1, R\cdot f_2)$, so the identity passes to the limit $M = m$ for every $R \in O(3)$. □

19. The heavy shell I: the photon switch, two-level walls and the dead shell

This section and the next prove the closing estimate (TIGHT_0) of Theorem 18.4. Here we prove the photon boundary switch (Lemma 19.1), define the two-level walls and the dead shell, and show that each of the two shell-decoupling properties (SH_0) and (RES_0) implies (TIGHT_0) (Theorems 19.6 and 19.10); the next section proves (RES_0) from the heavy-shell estimates.

Throughout this section $\Lambda = \Lambda_k$ is the axis lattice, $\varepsilon = \varepsilon_k \le 1/\mu_0$, and

$$m_\varepsilon := \frac{2}{\varepsilon}\operatorname{arcsinh}\Big(\frac{\varepsilon\mu_0}{2}\Big);$$

since $t \mapsto \operatorname{arcsinh}(t)/t$ is decreasing on $(0,\infty)$ and $\varepsilon\mu_0/2 \le \frac12$, $m_\varepsilon \ge 0.96\,\mu_0$. Let $G_{\mathbb{Z}^3}$ be the Green's function of simple random walk on $\mathbb{Z}^3$ and

$$\mathsf{a} := \sup_{z\in\mathbb{Z}^3\setminus\{0\}} |z|_\infty\, G_{\mathbb{Z}^3}(0,z),$$

which is finite by the asymptotics of the Green's function of simple random walk in three dimensions [17, Chap. 4]. For a set $X \subset \mathbb{R}^3$ put $\mathrm{vol}_\varepsilon(X) := \varepsilon^3 \#(X \cap \Lambda)$, and $\mathrm{vol}(X)$ for the Lebesgue measure. King cubes are $C_L = [-ML, ML)^3$ with $X_L = C_L \cap \Lambda_k$ (Definition 15.3); the comparison cube on X_L is $\langle\cdot\rangle^{(a)}_{k,J}$ with $J = L$ and its functional is $Z^a_{k,L}$ (Definitions 2.1 and 2.4).

19.1. **Photon boundary conditions across a matter-free collar.** The photon in Feynman form consists of three independent scalar Gaussian fields A_α, $\alpha = 1,2,3$, and on a finite graph $\mathcal{G}$ of sites of Λ each has the precision $K_\mathcal{G} = \varepsilon L_\mathcal{G} + \mu_0^2\varepsilon^3$, with $L_\mathcal{G}$ the graph Laplacian of $\mathcal{G}$ with unit conductances and $\langle u, v\rangle = \sum_x u(x)v(x)$; King's Neumann photon on a cube is the case of the induced subgraph [16, (2.5), p. 333], and the periodic photon the case of the torus graph.

Lemma 19.1 (photon boundary switch)**.** *Let $\mathsf{H} = Q_{b'}$ be a closed cube and C_L a King cube with $\operatorname{dist}_\infty(\mathsf{H}, \mathbb{R}^3 \setminus C_L) = D + 3$, where $D \ge \max(1, 3/\mu_0)$ and $\varepsilon \le \min(1/\mu_0, 1/4)$. Let $\langle\cdot\rangle_\natural$ and $\langle\cdot\rangle_N$ be BFS measures with the same matter part: the sites of $\mathsf{H}\cap\Lambda$, BFS single-site measures (arbitrary; wall and killed sites included), and hopping terms only between neighbouring sites of H, coupled to the photon variables of the bonds of H as in King's action. Their photon parts are:*

(♮) the variables $A_\alpha(x)$ with $x, x+\varepsilon e_\alpha \in C_L$, with the law of the periodic photon of a torus whose cell contains $N_1(C_L)$, conditioned on $A_\beta(y) = 0$ for all other (β, y);

(N) King's Neumann photon on $C_L \cap \Lambda$.

Let $\Theta = e^{\mathrm{i}F(f)} e^{\sum_x \varepsilon^3 g(x) r_x^2 - c}$ with f supported on plaquettes of H, $g \ge 0$ supported in H, and $c \in \mathbb{R}$. Then

$$0 \le \langle\Theta\rangle_\natural - \langle\Theta\rangle_N \le \langle|\Theta|^2\rangle_N^{1/2}\big(e^\tau - 1\big)^{1/2}, \qquad \tau \le C_{\mathrm{GB}}\,\mathsf{a}\,\mathrm{vol}_\varepsilon(N_2(\mathsf{H}))\,D^{-1}e^{-\frac32 m_\varepsilon D},$$

with an absolute constant C_{GB}.

Proof. Step 1 (marginals and order). Fix a component α. Let $X_\alpha := \{x \in C_L\cap\Lambda : x+\varepsilon e_\alpha \in C_L\}$; conditioning a Gaussian on the vanishing of some coordinates leaves the principal submatrix of the precision on the others, so in (♮) the variables $A_\alpha(x)$, $x \in X_\alpha$, have precision $K_\natural = \varepsilon(6 - \mathrm{Adj}_{X_\alpha}) + \mu_0^2\varepsilon^3$, where Adj_{X_α} is the adjacency matrix of X_α. In (N) the variables are $A_\alpha(x)$, $x \in C_L \cap \Lambda$, with precision $K_N = \varepsilon L_{C_L} + \mu_0^2\varepsilon^3$. The principal submatrix of K_N on X_α is $\varepsilon(\deg_{C_L} - \mathrm{Adj}_{X_\alpha}) + \mu_0^2\varepsilon^3 \le K_\natural$, and the Schur complement of K_N on X_α is at most its principal

submatrix; hence $C_\natural := K_\natural^{-1} \le C_N|_{X_\alpha}$, $C_N := K_N^{-1}$. Let $\hat X := \{(\alpha,x) : x, x+\varepsilon e_\alpha \in \mathsf{H}\}$. The integrands depend on A only through $A_{\hat X}$, whose laws are $N(0, C_\natural|_{\hat X})$ and $N(0, C_N|_{\hat X})$, with $C_\natural|_{\hat X} \le C_N|_{\hat X}$. Put $M := (C_\natural|_{\hat X})^{-1} - (C_N|_{\hat X})^{-1} \ge 0$. The $\natural$-measure is obtained from the N-measure, after integrating out the photon variables off $\hat X$, by adding $\frac12\langle A, M\rangle A$ to the action; writing $M = \sum_j \lambda_j v_j v_j^T$ with $\lambda_j \ge 0$, this adds $\frac{s}{2}\sum_j \lambda_j \langle v_j, A\rangle^2$ at $s=1$, and by Lemma 2.8(a) along $s \in [0,1]$, $\langle\Theta\rangle_\natural \ge \langle\Theta\rangle_N$.

Step 2 (Cauchy–Schwarz). With $Q := \langle A, M\rangle A \ge 0$ and $\varsigma := e^{-Q/2}$, $\langle\Theta\rangle_\natural = \langle\Theta\varsigma\rangle_N / \langle\varsigma\rangle_N$, so

$$\langle\Theta\rangle_\natural - \langle\Theta\rangle_N = \frac{\langle(\Theta - \langle\Theta\rangle_N)(\varsigma - \langle\varsigma\rangle_N)\rangle_N}{\langle\varsigma\rangle_N} \le \langle|\Theta|^2\rangle_N^{1/2}\Big(\frac{\langle e^{-Q}\rangle_N}{\langle e^{-Q/2}\rangle_N^2} - 1\Big)^{1/2} \le \langle|\Theta|^2\rangle_N^{1/2}\big(e^{\langle Q\rangle_N} - 1\big)^{1/2},$$

by $\langle e^{-Q}\rangle_N \le 1$ and Jensen's inequality, $\langle e^{-Q/2}\rangle_N \ge e^{-\langle Q\rangle_N/2}$. With $M = \sum_j \lambda_j v_j v_j^T$ as in Step 1, Lemma 15.11(b) in the N-measure, whose Gaussian has covariance C_N, gives $\langle Q\rangle_N = \sum_j \lambda_j \langle\langle v_j, A\rangle^2\rangle_N \le \sum_j \lambda_j \langle v_j, C_N v_j\rangle = \operatorname{Tr}(M C_N|_{\hat X}) =: \tau$.

Step 3 (the trace). Let $\Delta := C_N|_{X_\alpha} - C_\natural \ge 0$, for each α. Then $M C_N|_{\hat X} = (C_\natural|_{\hat X})^{-1}\Delta_{\hat X}$, so $\tau = \mathbb{E}\langle Y, (C_\natural|_{\hat X})^{-1} Y\rangle$ with $Y \sim N(0, \Delta_{\hat X})$. Since the inverse of a principal block of the covariance is the minimum of the precision form over extensions, $\langle a, (C_\natural|_{\hat X})^{-1} a\rangle = \min\{\langle u, K_\natural u\rangle : u|_{\hat X} = a\}$. Let $\tilde Y \sim N(0,\Delta)$ on X_α and $\eta(x) := (1 - \operatorname{dist}_\infty(x, \mathsf{H}))_+$; then $\eta\tilde Y$ is an extension of $\tilde Y|_{\hat X}$, so $\tau \le \sum_\alpha \mathbb{E}\langle \eta\tilde Y, K_\natural \eta\tilde Y\rangle$. Let int be the set of $x \in X_\alpha$ whose six neighbours lie in X_α; it contains $N_2(\mathsf{H}) \cap \Lambda$. At $x \in$ int the rows of $K_\natural$, K_N and of the infinite-lattice precision K_∞ coincide, so $(K_\infty\Delta)(x,w) = (K_N C_N)(x,w) - (K_\natural C_\natural)(x,w) = 0$ for all $w \in X_\alpha$, hence $\operatorname{Var}((K_\infty\tilde Y)(x)) = 0$ and $K_\infty\tilde Y = 0$ on int almost surely. For u with $K_\infty u = 0$ on int $\supset \operatorname{supp}\eta$, summation by parts and the identity $(\eta_x u_x - \eta_y u_y)^2 - (u_x - u_y)(\eta_x^2 u_x - \eta_y^2 u_y) = (\eta_x - \eta_y)^2 u_x u_y$ give $\langle \eta u, K_\natural \eta u\rangle = \langle \eta u, K_\infty \eta u\rangle = \varepsilon \sum_{\langle xy\rangle} (\eta_x - \eta_y)^2 u_x u_y$. Since $|\eta_x - \eta_y| \le \varepsilon$ and $|\Delta(x,y)| \le \frac12(\Delta(x,x) + \Delta(y,y))$,

$$\tau \le \sum_\alpha \varepsilon^3 \sum_{\langle xy\rangle \subset N_2(\mathsf{H})} |\Delta(x,y)| \le 9\ \operatorname{vol}_\varepsilon(N_2(\mathsf{H})) \sup_{\alpha,\ x \in N_2(\mathsf{H})} \Delta(x,x).$$

Step 4 (decay of Δ). For $K = \mathcal{D} - \varepsilon \operatorname{Adj}_\mathcal{G}$ with $\mathcal{D}$ diagonal, $K^{-1}(x,y) = \sum_\omega \prod_{(u\to v)\in\omega} (\varepsilon/\mathcal{D}(u)) \cdot \mathcal{D}(y)^{-1}$, the sum over walks ω from x to y in $\mathcal{G}$. For $K_\natural$ and K_N the weights of steps from sites of int coincide. Decomposing walks from $x \in N_2(\mathsf{H})$ at their first visit to $\Sigma := X_\alpha \setminus$ int, $K^{-1}(x,y) = G^{\mathrm{int}}(x,y) + \sum_{z\in\Sigma} \mathsf{h}(x,z) K^{-1}(z,y)$ with G^{int} and the hitting kernel h the same for both precisions. Hence

$$\Delta(x,x) = \sum_{z\in\Sigma} \mathsf{h}(x,z)\big(K_N^{-1}(z,x) - K_\natural^{-1}(z,x)\big) \le \Big(\sum_{z\in\Sigma} \mathsf{h}(x,z)\Big) \sup_{z\in\Sigma} K_N^{-1}(z,x).$$

The first factor is the probability that simple random walk from x, killed with probability $\mu_0^2\varepsilon^2/(6+\mu_0^2\varepsilon^2)$ per step, reaches Σ, which lies at ℓ^∞-distance at least D from x. With $\cosh\lambda = 1 + \mu_0^2\varepsilon^2/2$, that is $\lambda = \varepsilon m_\varepsilon$, $e^{\pm\lambda S_n^\beta}\mathbf{1}_{\{\text{alive at } n\}}$ is a martingale for each coordinate S^β of the walk; optional stopping and a union over the six half-spaces give $\sum_z \mathsf{h}(x,z) \le 6e^{-m_\varepsilon D}$. The same argument, with the decomposition at the first exit from the ℓ^∞-ball of radius $|y|_\infty/2$ and the bound $K^{-1}_{\varepsilon\mathbb{Z}^3,\mu_0} \le K^{-1}_{\varepsilon\mathbb{Z}^3,0} = (6\varepsilon)^{-1} G_{\mathbb{Z}^3}(\cdot/\varepsilon)$ entrywise, gives the free bound

$$G_\varepsilon(y) := K^{-1}_{\varepsilon\mathbb{Z}^3}(0,y) \le 4\mathsf{a}\,|y|_\infty^{-1} e^{-m_\varepsilon|y|_\infty/2} \qquad (|y|_\infty \ge 4\varepsilon).$$

King's Neumann precision on the box $C_L \cap \Lambda$ has the Green's function $K_N^{-1}(z,x) = \sum_{x'} G_\varepsilon(z - x')$, the sum over the images x' of x under the reflections of $\varepsilon\mathbb{Z}^3$ in the planes half a lattice spacing outside the faces of the box (method of images; the mass term is uniform). Every image other than x lies outside C_L at ℓ^∞-distance at least D from every site of C_L, and so does x from $z \in \Sigma$. If $s \ge 2D$ is the side of C_L, at most $8(j+3)^3$ images satisfy $|z - x'|_\infty < D + (j+1)s$; since $m_\varepsilon D \ge 2$,

$$K_N^{-1}(z,x) \le \sum_{j\ge0} 8(j+3)^3\, \frac{4\mathsf{a}\, e^{-m_\varepsilon(D+js)/2}}{D+js} \le C_1\, \mathsf{a}\, D^{-1} e^{-m_\varepsilon D/2}$$

with an absolute constant C_1. Combining, $\Delta(x,x) \le 6C_1 \mathsf{a} D^{-1} e^{-\frac{3}{2}m_\varepsilon D}$, and Step 3 gives the claim with $C_{\mathrm{GB}} = 54C_1$. □

The Gaussian measures in Lemma 19.1 are compared only on the bonds of H, where no live matter crosses the region in which the two photon boundary conditions differ; this is what makes the bound uniform in ε. A comparison in which the photon is pinned or cut on a surface that live matter crosses is not of this type: in formal perturbation theory, at one loop, the order-e^2 difference of two such states diverges logarithmically as $\varepsilon \to 0$, because the free-boundary mass counterterm compensates the photon self-energy of the matter only where the photon has its bulk covariance.

19.2. **Shell configurations, two-level walls, and shell-kill decoupling.**

Definition 19.2 (shell configurations and shell profiles)**.** Let $b > 0$. A *shell configuration* for Q_b is $\mathfrak{s} = (\mathfrak{c}, b', L)$, where $\mathfrak{c} = (J', \psi, n)$ with $\psi \in C^\infty(D_{J'}; [0,\infty))$ and $n \ge 0$ is a walled configuration in $\mathcal{A}(Q_b)$ (so $\psi = 0$ on $N_1(Q_b)$; Definitions 15.7 and 18.1), $\mathsf{H} := Q_{b'}$ with $b' \ge b+3$, and C_L is a King cube with $\mathrm{dist}_\infty(\mathsf{H}, \mathbb{R}^3 \setminus C_L) = D(\mathfrak{s}) + 3$, $D(\mathfrak{s}) \ge \max(1, 3/\mu_0)$, and $N_1(\bar{C}_L) \subset (-25MJ', 25MJ')^3$. Its *shell* is $U := C_L \setminus \mathsf{H}$, and we require the *plateau condition* $\psi \ge 1$ on $N_1(U)$ (compatible with $\psi = 0$ on $N_1(Q_b)$, since $\mathrm{dist}_\infty(U, Q_b) \ge 3$). Put

$$\tau(\mathfrak{s}) := C_{\mathrm{GB}}\, \mathsf{a}\, \mathrm{vol}(N_3(\mathsf{H}))\, D(\mathfrak{s})^{-1} e^{-1.44\,\mu_0 D(\mathfrak{s})}, \tag{44}$$

with the constant C_{GB} of Lemma 19.1. A *shell profile* for $\mathfrak{s}$ is $\chi \in C^\infty(D_{J'}; [0,1])$ with $\chi = 1$ on U and $\mathrm{supp}\,\chi \subset N_1(U)$; then $\chi \le \psi$ automatically, and $\chi = 0$ on $N_1(Q_b)$. Its *dead set* is $D_\chi := \{\chi > 0\} \subset N_1(U)$; every site of C_L within ε of $\mathbb{R}^3 \setminus C_L$, and every site of U, lies in D_χ. We call $(\mathfrak{s}, \chi)$ a *two-level configuration* for Q_b.

Definition 19.3 (two-level states and the tail)**.** Let $(\mathfrak{s}, \chi)$ be a two-level configuration for Q_b, $(f, g) \in \mathcal{K}$ supported in Q_b, a a scheme, and k with $\varepsilon_k \le \min(1/\mu_0, 1/4)$. For $K \in [0,\infty)$ let

$$\Psi_k(n, K) := T^a_{(J',\, n\psi + K\chi),\Lambda_k}(f, g)$$

be the walled ratio (Definition 15.8) of the configuration with the *two-level wall* $n\psi + K\chi$, which lies in $\mathcal{A}(Q_b)$; and let

$$\Psi_k(n, \infty) := e^{-a_k \mathsf{S}_{\Lambda_k}(g)} \big\langle \Theta(\mathrm{i}f, g) \big\rangle$$

in the walled torus state of $\mathfrak{c} = (J', n\psi)$ with the matter killed on $D_\chi \cap \Lambda_k$ (Definition 15.7(d)), the *dead-shell state*. Put $\mathrm{tail}_k(K) := \Psi_k(n, K) - \Psi_k(n, \infty)$. At $g = 0$ the multiplier is 1 and nothing depends on the scheme (Lemma 15.10(b)).

Lemma 19.4 (elementary properties)**.** (a) *$K \mapsto \Psi_k(n, K)$ is nonincreasing on $[0,\infty)$, $\Psi_k(n, K) \ge \Psi_k(n, \infty)$, and $\Psi_k(n, K) \to \Psi_k(n, \infty)$ as $K \to \infty$. Hence $\mathrm{tail}_k(K) \ge 0$ is nonincreasing in K and tends to 0 as $K \to \infty$, at every fixed k.*

(b) *(two-level sandwich) For $n, K \ge 0$: $\Psi_k(n + K, 0) \le \Psi_k(n, K) \le \Psi_k(n, 0)$.*

Proof. All states involved are BFS measures of King type (the paragraph after Definition 15.8), the observable $\Theta(\mathrm{i}f, g)$ is of class $\mathcal{I}$, and the multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is one positive constant for all of them (Lemma 2.12(a), Lemma 15.10(b)). (a) Increasing K to K' multiplies the single-site measures on D_χ by $e^{-(K'-K)\varepsilon^3 \chi(x) r_x^2}$, which does not increase the expectation by Lemma 2.8(c). Killing $D_\chi \cap \Lambda_k$ is the limit $K \to \infty$ (Lemma 2.10 with the rates $\chi > 0$ on D_χ; $|\Theta(\mathrm{i}f, g)| \le e^{C\sum_x r_x^2}$), and it does not increase the expectation (Lemma 2.8(c), killing clause, the state being of King type). (b) The wall of $\Psi_k(n + K, 0)$ exceeds that of $\Psi_k(n, K)$ by $K(\psi - \chi) \ge 0$, since $\chi \le \psi$; the wall of $\Psi_k(n, K)$ exceeds that of $\Psi_k(n, 0)$ by $K\chi \ge 0$. Apply Lemma 2.8(c). □

Part (b) is the two-level sandwich (Lemma 19.7 below in the form used); it is the only place where $\chi \le \psi$ is needed. In words: raising the matter mass on the shell alone, by any amount, gives a state above the one in which the whole plateau is raised by the same amount, and below the one with the shell untouched.

Definition 19.5 ((SH)$_a$ and (SH$_0$): shell-kill decoupling)**.** (SH)$_a$ is the property: for every $b > 0$, every $(f, g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there is a two-level configuration $(\mathfrak{s}, \chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ and

$$\liminf_{k\to\infty} \mathrm{tail}_k(0) = \liminf_{k\to\infty} \big[\Psi_k(n, 0) - \Psi_k(n, \infty)\big] \le \eta.$$

(SH$_0$) is the same statement for $g = 0$; it does not depend on the scheme.

Since $\tau(\mathfrak{s})$ depends only on H and the width $D(\mathfrak{s})$, the condition $\tau(\mathfrak{s}) \le \eta$ only asks the shell to be wide enough, logarithmically in b' and $1/\eta$. The property involves one lattice, a single walled torus and a single dead set, a fixed wall height, and the lim inf over the lattice spacings; it involves no limit $n \to \infty$ and no comparison between lattices.

Theorem 19.6 (shell-kill decoupling implies the closing estimate)**.** *Assume* (W)$_a$ *and* (K)$_a$*. Then* (SH)$_a$ *implies* (TIGHT)$_a$*. In the curvature sector,* (SH$_0$) *implies* (TIGHT$_0$)*. More precisely, for every two-level configuration* $(\mathfrak{s}, \chi)$ *for* Q_b*, every* $(f, g) \in \mathcal{K}$ *supported in* Q_b *and every* k *with* $\varepsilon_k \le \min(1/\mu_0, 1/4)$*,*

$$\Psi_k(n, \infty) \le Z^a_{k,L}(\mathrm{i}f, g) + Z^a_{k,L}(0, 2g)^{1/2}\big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}, \tag{45}$$

where $Z^a_{k,L}$ *is the functional of the comparison cube on* $X_L = C_L \cap \Lambda_k$*. Hence, under* (W)$_a$*,* $\limsup_k \Psi_k(n, \infty) \le Z^a_\infty(\mathrm{i}f, g) + Z^a_\infty(0, 2g)^{1/2}(e^{\tau(\mathfrak{s})} - 1)^{1/2}$*; and at* $g = 0$*, under* (E$_C$) *alone,* $\limsup_k \Psi_k(n, \infty) \le Z^K_\infty(\mathrm{i}f, 0) + (e^{\tau(\mathfrak{s})} - 1)^{1/2}$*.*

Proof. Fix k with $\varepsilon = \varepsilon_k \le \min(1/\mu_0, 1/4)$ and write $\Theta = \Theta(\mathrm{i}f, g)$. The multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is one positive constant for every state below (Lemma 2.12(a)); we compare the expectations of Θ and of $|\Theta|^2 = \Theta(0, 2g)$ and restore it at the end. Let E be the set of sites of the torus $D_{J'}$ outside C_L.

Step 1 (pinning). In the dead-shell state, add $s|A_\beta(y)|^2$ to the action for every photon variable on a bond with an endpoint in E, and let $s \to \infty$. By Lemma 2.8(a) the expectation of Θ does not decrease, and in the limit the photon is the conditioned law ($\natural$) of Lemma 19.1, the periodic photon of $D_{J'}$ (whose cell contains $N_1(C_L)$ by Definition 19.2) conditioned on the vanishing of all variables except the $A_\alpha(x)$ with $x, x + \varepsilon e_\alpha \in C_L$. In this state the hopping terms within E do not involve A, and there are no hopping terms between E and live sites: every site of C_L adjacent to E lies in D_χ and is dead. So the matter on E is a factor independent of A and cancels (Lemma 2.8(e)). What remains is the $\natural$-measure of Lemma 19.1, with the matter on H (dead on $D_\chi \cap \mathsf{H}$), the wall $n\psi$ in its single-site measures, and Dirichlet conditions at $\partial\mathsf{H}$, since $U \subset D_\chi$ is dead. Hence $\Psi_k(n, \infty) \le e^{-a_k \mathsf{S}_{\Lambda_k}(g)} \langle\Theta\rangle_\natural$.

Step 2 (photon switch). By Lemma 19.1, $\langle\Theta\rangle_\natural \le \langle\Theta\rangle_N + \langle|\Theta|^2\rangle_N^{1/2}(e^\tau - 1)^{1/2}$ with $|\Theta|^2 = \Theta(0, 2g)$; here f is supported on plaquettes of H and g in H because $b' \ge b + 3$, and $m_\varepsilon \ge 0.96\mu_0$ together with $\mathrm{vol}_\varepsilon(N_2(\mathsf{H})) \le \mathrm{vol}(N_3(\mathsf{H}))$ (for $\varepsilon \le 1$) give $\tau \le \tau(\mathfrak{s})$.

Step 3 (to the comparison cube). The N-measure is the limit $s \to \infty$ of the walled Neumann–Dirichlet state on X_L, with the wall $n\psi$ in its single-site measures, in which the single-site measures on $D_\chi \cap X_L$ are multiplied by $e^{-s\varepsilon^3\chi(x)r_x^2}$ (Lemma 2.10; the matter alive in the N-measure is that of $X_L \setminus D_\chi \subset \mathsf{H}$, its Dirichlet terms toward the dead sites being the diagonal parts of the bonds into them). By Lemma 2.8(c) the expectations of $\Theta(\mathrm{i}f, g)$ and $\Theta(0, 2g)$ do not decrease as s decreases to 0, nor when the wall is then removed. The result is the comparison cube $\langle\cdot\rangle^{(a)}_{k,J}$ on X_L (Definition 2.1), in which both observables are the same functions of the shared variables, every site with $g \ne 0$ and every plaquette of f lying in H, at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$. Restoring the multiplier (Lemma 2.12(a); for $\Theta(0, 2g)$ it is its square), this is (45).

Step 4 (limits). Under (W)$_a$ the cube functionals converge, $Z^a_{k,L} \to Z^a_L$, by Theorem 16.6(i), and $Z^a_L \le Z^a_\infty$ on $\mathcal{K}$ by Theorem 16.6(iii), with $(0, 2g) \in \mathcal{K}$. This gives the lim sup bound. Now let $(\mathfrak{s}, \chi)$ be as in (SH)$_a$ for (f, g) and η. Then $T^a_{\mathfrak{c},\Lambda_k}(f, g) = \Psi_k(n, 0) = \mathrm{tail}_k(0) + \Psi_k(n, \infty)$, the left side converges to $\mathcal{W}^a_\mathfrak{c}(f, g)$ by (K)$_a$ (Lemma 17.3(a), Definition 18.1), and taking $\liminf_k$, with $\liminf_k(x_k + y_k) \le \liminf_k x_k + \limsup_k y_k$,

$$\mathcal{W}^a_\mathfrak{c}(f, g) \le \eta + Z^a_\infty(\mathrm{i}f, g) + Z^a_\infty(0, 2g)^{1/2}\big(e^\eta - 1\big)^{1/2}.$$

Since $\mathfrak{c} \in \mathcal{A}(Q_b)$ and η is arbitrary, $I^a_b(f,g) \le Z^a_\infty(\mathrm{i}f,g)$ for every b, which is $(\mathrm{TIGHT})_a$.

At $g=0$ the multiplier is 1 and $\langle|\Theta|^2\rangle = 1$, so Step 3 gives $\Psi_k(n,\infty) \le Z_{k,L}(\mathrm{i}f) + (e^{\tau(\mathfrak{s})}-1)^{1/2}$ with $Z_{k,L}(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle^{(a)}_{k,J}$ at $J=L$. Under (E_C), since f is supported at ℓ^∞-distance >1 from $\mathbb{R}^3 \setminus C_L$, Corollary 4.4(ii)($\gamma$) gives $\limsup_k Z_{k,L}(\mathrm{i}f) \le Z^K_{L+1}(\mathrm{i}f)$, and $Z^K_{L+1}(\mathrm{i}f) \le Z^K_\infty(\mathrm{i}f,0)$ because $J \mapsto Z^K_J(\mathrm{i}f)$ is nondecreasing (Corollary 4.3, Theorem 7.6(b)) and converges to $Z^K_\infty(\mathrm{i}f,0)$ (Corollary 4.4(ii)(β)). This is the lim sup bound at $g=0$. With $(\mathfrak{s},\chi)$ as in (SH_0), $T_{\mathfrak{c},\Lambda_k}(f,0) = \Psi_k(n,0) \to \mathcal{W}_\mathfrak{c}(f,0)$ by (K_0) (Lemma 17.3(a)), and the same lim inf argument gives $\mathcal{W}_\mathfrak{c}(f,0) \le \eta + Z^K_\infty(\mathrm{i}f,0) + (e^\eta-1)^{1/2}$, hence $I_b(f,0) \le Z^K_\infty(\mathrm{i}f,0)$ for every b, which is (TIGHT_0). □

The theorem is stated for the cubic shell $U = C_L \setminus \mathsf{H}$ of Definition 19.2, in which C_L is a King cube and H a cube inside it. For the configurations of the next section, whose shell is the exterior of a box in the torus, the three steps of the proof are carried out in Proposition 20.1 ((DS)).

19.3. **Heavy-shell decoupling.**

Lemma 19.7 (two-level sandwich)**.** *For every two-level configuration, every $(f,g) \in \mathcal{K}$ supported in Q_b, every k with $\varepsilon_k \le \min(1/\mu_0, 1/4)$ and $n, K \ge 0$,*

$$\Psi_k(n+K,0) \;\le\; \Psi_k(n,K) \;=\; \Psi_k(n,\infty) + \mathrm{tail}_k(K) \;\le\; \Psi_k(n,0), \qquad \mathrm{tail}_k(K) \ge 0.$$

Proof. Lemma 19.4. □

Definition 19.8 ($(\mathrm{RES})_a$ and (RES_0): heavy-shell decoupling at one configuration)**.** $(\mathrm{RES})_a$ is the property: for every $b>0$, every $(f,g) \in \mathcal{K}$ supported in Q_b and every $\eta>0$ there are a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ and a shell mass $K \in [0,\infty)$ such that

$$\liminf_{k\to\infty} \mathrm{tail}_k(K) = \liminf_{k\to\infty} \big[\Psi_k(n,K) - \Psi_k(n,\infty)\big] \le \eta.$$

(RES_0) is the same statement for $g=0$; it does not depend on the scheme.

In words: in a walled torus, once the scalar mass squared on a thick shell of the heavy plateau has been raised by $2K$, removing the matter on the shell altogether changes the expectation of the observable by at most η in the continuum limit, for one geometry and one K, with no rate in K and no uniformity in anything. By Lemma 19.4(a) the bracket tends to 0 as $K\to\infty$ at every fixed k; the property says that the heavy-shell limit and the continuum limit may be interchanged at one configuration. The quantifiers are the weakest that Theorem 19.10 accepts: one configuration, one K (with $K=0$ allowed, which is shell-kill decoupling, Definition 19.5), the lim inf along the spacings, and no uniformity in n, L, $D(\mathfrak{s})$ or the torus.

Proposition 19.9 ($(\mathrm{SH})_a$ implies $(\mathrm{RES})_a$)**.** *For every two-level configuration and every $K \ge 0$, $\mathrm{tail}_k(K) \le \mathrm{tail}_k(0)$. Hence $(\mathrm{SH})_a$ implies $(\mathrm{RES})_a$ with the same configurations and every K, and likewise (SH_0) implies (RES_0).*

Proof. Lemma 19.4(a). □

Theorem 19.10 ($(\mathrm{RES})_a$ implies $(\mathrm{TIGHT})_a$)**.** *Assume $(\mathrm{W})_a$ and $(\mathrm{K})_a$. Then $(\mathrm{RES})_a$ implies $(\mathrm{TIGHT})_a$. In the curvature sector, (RES_0) implies (TIGHT_0). More precisely, if $(\mathfrak{s},\chi)$ and K are as in $(\mathrm{RES})_a$ for (f,g) and η, then the single-level configuration $\mathfrak{c}' := (J',\psi,n+K) \in \mathcal{A}(Q_b)$ satisfies*

$$\mathcal{W}^a_{\mathfrak{c}'}(f,g) \le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}\big(e^\eta-1\big)^{1/2} + \eta,$$

and at $g=0$, with $(\mathfrak{s},\chi)$ and K as in (RES_0), $\mathcal{W}_{\mathfrak{c}'}(f,0) \le Z^K_\infty(\mathrm{i}f,0) + (e^\eta-1)^{1/2} + \eta$.

Proof. By Lemma 19.7, $T^a_{\mathfrak{c}',\Lambda_k}(f,g) = \Psi_k(n+K,0) \le \Psi_k(n,\infty) + \mathrm{tail}_k(K)$ at every k. The left side converges to $\mathcal{W}^a_{\mathfrak{c}'}(f,g)$ by $(\mathrm{K})_a$, since $\mathfrak{c}'$ is a walled configuration in $\mathcal{A}(Q_b)$ (Lemma 17.3(a), Definition 18.1); hence, using $\liminf_k(x_k+y_k) \le \limsup_k x_k + \liminf_k y_k$,

$$\mathcal{W}^a_{\mathfrak{c}'}(f,g) = \liminf_k \Psi_k(n+K,0) \le \limsup_k \Psi_k(n,\infty) + \liminf_k \mathrm{tail}_k(K)$$

$$\le Z^a_\infty(\mathrm{i}f,g)+Z^a_\infty(0,2g)^{1/2}\big(e^{\tau(\mathfrak{s})}-1\big)^{1/2}+\eta$$

by Theorem 19.6 and $(\mathrm{RES})_a$. With $\tau(\mathfrak{s})\le\eta$ this is the displayed bound; as η is arbitrary and $\mathfrak{c}'\in\mathcal{A}(Q_b)$, $I^a_b(f,g)\le Z^a_\infty(\mathrm{i}f,g)$ for every b, which is $(\mathrm{TIGHT})_a$. At $g=0$ the same argument runs with the $g=0$ clause of Theorem 19.6 under (E_C), the convergence of the left side given by (K_0), and $Z(0,0)=1$; it gives $I_b(f,0)\le Z^K_\infty(\mathrm{i}f,0)$ for every b, which is (TIGHT_0). □

Remark 19.11 (on the proof). (i) The two-level state is only an intermediate inequality; the configuration that enters $(\mathrm{TIGHT})_a$, respectively (TIGHT_0), is the single-level wall $(n+K)\psi$, and the only limit input is the existence of its walled value, which is part of $(\mathrm{K})_a$, respectively (K_0) (the existence of the torus limits on the axis lattice, Remark 17.11). No existence statement is needed for the two-level state or for the dead-shell state. (ii) The lattice statements of this subsection hold verbatim for the sharp profile $\chi=\mathbf{1}_U$ (then $D_\chi=U$ and $\chi\le\psi$ needs only $\psi\ge1$ on U); smoothness of χ is used only in Theorem 20.18, where the two-level walled value must exist. (iii) Since $\mathrm{tail}_k(K)$ is nonincreasing in K, $(\mathrm{RES})_a$ for one K implies it for every larger K; so K may be taken as large as one likes, which is used to place $n+K$ in any prescribed window of wall heights (Remark 18.6).

20. The heavy shell II: the heavy-shell estimates and their continuum form

20.1. **The heavy-shell estimates.** In the curvature sector (RES_0) is proved at one configuration, from two estimates on King's walled torus model with the shell mass inside the matter covariance, the heavy-shell estimates (HS) of Theorem 20.2. The shell is now taken to be the whole exterior of the source box in the torus. Let $(\mathfrak{s},\chi)$ be a two-level configuration in the sense of Definitions 19.2 and 19.3, with the walled torus $T:=D_{J'}$ of $\mathfrak{c}=(J',\psi,n)$ and the source box $\mathsf{H}':=\mathsf{H}=Q_{b'}$, in which the shell $C_L\setminus\mathsf{H}$ is replaced by

$$U:=T\setminus\mathsf{H}',$$

so that $\partial U=\partial\mathsf{H}'$ is a single convex surface with no re-entrant edge (on a cubic shell $C_L\setminus\mathsf{H}$ the derivative clause of King's bound on the block propagators, [15, Thm. 3.3], fails at the re-entrant edges); the plateau condition $\psi\ge1$ on $N_1(U)$ is kept, the profile is the sharp one, $\chi=\mathbf{1}_U$, which Remark 19.11(ii) allows for every lattice statement of the preceding section, so that the dead set is $D_\chi=U$, and $\tau(\mathfrak{s})$ is computed, as in Definition 19.2, from H and a King cube C_L with $\mathsf{H}'\subset C_L\subset T$ at ℓ^∞-distance $D(\mathfrak{s})+3$ from H'. For $\varkappa\in[0,\infty)$ let $\Psi_k(n,\varkappa)$ be the two-level expectation of Definition 19.3 with the shell mass $\varkappa\mathbf{1}_U$ added to the wall: by Definition 15.7(c) the wall $n\psi+\varkappa\mathbf{1}_U$ raises the scalar mass squared by $2n\psi$ on the plateau and by a further $2\varkappa$ on U, and $\Psi_k(n,K)$, $\Psi_k(n,\infty)$ of Definition 19.8 are the cases $\varkappa=K$ and $\varkappa=\infty$ (the matter killed on U). Let $Z_\varkappa(zf,0)$ be the corresponding partition function with the curvature source zf, so that $Z_\varkappa(zf,0)/Z_\varkappa(0,0)=\langle e^{zF_{\Lambda_k}(f)}\rangle_\varkappa$ is the expectation of the source weight in the walled torus state with the shell mass $\varkappa\mathbf{1}_U$. Here $Z_\varkappa(\,\cdot\,,0)$ is the unnormalised functional of this walled state in the convention of Definition 15.8, with the wall $n\psi$ and the shell mass $\varkappa\mathbf{1}_U$ in its single-site measures; it is not the normalised functional Z^a of the first sections, and Theorem 20.14 uses it only through the ratios $Z_\varkappa(zf,0)/Z_\varkappa(0,0)$. The estimates of (HS) are read at plateau heights $n\ge1$ and for King's wall profile, $\psi\in C^\infty(D_{J'};[0,1])$ with $\psi=0$ on $N_1(Q_b)$ and $\psi=1$ off $N_2(Q_b)$ (so that $\psi=1$ on $N_1(U)$, since $b'\ge b+3$, and the plateau condition of Definition 19.2 holds); on the plateau the mass squared is then raised by exactly $2n\ge2$. The symbol n_* denotes in this part the number of large-field scales of Theorem 17.13, which enters the residue class $J'\in5^{n_*}\mathbb{N}$ of the torus below.

The first statement carries the three steps of the dead-shell chain, proved in the preceding section for the cubic shell $C_L\setminus\mathsf{H}$, to the exterior-shell geometry.

Proposition 20.1 ((DS): the dead-shell chain for the exterior shell)**.** *For the two-level configurations with the exterior shell $U=T\setminus\mathsf{H}'$ the three steps of the proof of Theorem 19.6 hold: pinning the photon outside a King cube $C_L\supset\mathsf{H}'$ all of whose sites outside H' are dead, the photon switch across the matter-free collar between H' and ∂C_L, and the passage to the comparison*

cube on $X_L = C_L \cap \Lambda_k$. Consequently the dead-shell bound (45) holds for them: at every k with ε_k small, $\Psi_k(n,\infty) \le Z_{k,L}(\mathrm{i}f) + (e^{\tau(\mathfrak{s})}-1)^{1/2}$ at $g=0$.

Idea of the proof. Adding $s|A_\beta(y)|^2$ to the action for every photon variable on a bond with an endpoint outside C_L does not decrease the expectation of $\Theta = e^{\mathrm{i}F(f)}$ (Lemma 2.8(a)), and as $s \to \infty$ the photon law becomes the law ($\natural$) of Lemma 19.1; the collar $C_L \setminus \mathsf{H}'$ carries no live matter, so Lemma 19.1 compares ($\natural$) with King's Neumann photon on C_L at the cost $(e^\tau-1)^{1/2}$, $\tau \le \tau(\mathfrak{s})$; and by Lemma 2.10 and Lemma 2.8(c), removing the killing weights on the collar and then the wall factors does not decrease the expectation, whence $\langle\Theta\rangle_N \le Z_{k,L}(\mathrm{i}f)$.

Theorem 20.2 ((HS): the heavy-shell estimates)**.** *Let $(\mathfrak{s},\chi)$ be a two-level configuration with the exterior shell $U = T \setminus \mathsf{H}'$, with H' and T unions of blocks of the final scale of King's block-spin flow, with H' of side at least two final blocks and at distance at least three final blocks from its periodic images in every direction, and with every corner coordinate of H', counted from the origin in units of the final block, congruent to $M'/2$ modulo M', where the large-block integer is $M = 5M'$ with M' even (the grid-offset condition; King's tori satisfy the clauses on T, and in each direction every interval of M' final blocks contains an admissible corner coordinate), with the torus $T = D_{J'}$ admitted by the class of Remark 18.6, that is $J' \in 5^{n_*}\mathbb{N}$ and $J' \ge \max(J_0(\delta_r), J_{\mathrm{pos}})$ (n_* and $J_0(\delta_r)$ as in Theorem 17.13, J_{pos} the threshold of Theorem 17.98 of the full version), with King's wall profile ($\psi \in C^\infty(D_{J'};[0,1])$, $\psi = 0$ on $N_1(Q_b)$, $\psi = 1$ off $N_2(Q_b)$) and plateau height $n \ge 1$, let $g = 0$ and let f be a real 2-form in C_c^∞ supported in Q_b. There are $\eta_1 > 0$ and, for every stopping scale $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \le \eta_1$: numbers $\mathfrak{a}(\eta_0), \mathfrak{b}(\eta_0) \in [0,\infty)$ with $\mathfrak{a}(\eta_0), \mathfrak{b}(\eta_0) \to 0$ as $\eta_0 \to 0$; a threshold $\varkappa_0(\eta_0) < \infty$; a spacing bound $\varepsilon_0(\eta_0) > 0$; a function $r(\,\cdot\,;\eta_0)$ on $[\varkappa_0,\infty)$ with $r(\varkappa;\eta_0) \to 0$ as $\varkappa \to \infty$; and, for every k with $\varepsilon_k \le \varepsilon_0(\eta_0)$, a measure space $(\tilde\chi, db)$ not depending on $\varkappa$ and measurable functions $\tilde S_\varkappa(\,\cdot\,;z)$ on $\tilde\chi$ for $\varkappa \ge \varkappa_0$ and $z \in [-1,1]$ (both may depend on k; the index is suppressed); such that, with*

$$I_\varkappa^{\mathrm{cap}}(z) := \int_{\tilde\chi} e^{-\tilde S_\varkappa(b;z)}\, db, \qquad \Delta_z(b) := \tilde S_{\varkappa'}(b;z) - \tilde S_\varkappa(b;z),$$

for every such k, all $\varkappa_0 \le \varkappa \le \varkappa' < \infty$ and all $z \in [-1,1]$:

(i) (the two-sided bound) $e^{-\mathfrak{a}}\, I_\varkappa^{\mathrm{cap}}(z) \le Z_\varkappa(zf,0) \le e^{\mathfrak{a}}\, I_\varkappa^{\mathrm{cap}}(z) + \mathfrak{b}\, Z_\varkappa(0,0)$;

(ii) (the $\varkappa$-Cauchy property) $\sup_{b,b'\in\tilde\chi} |\Delta_z(b) - \Delta_0(b')| \le 2\, r(\varkappa;\eta_0)$.

The shell model and the stopping-scale integrals. Fix the exterior-shell configuration of Theorem 20.2. The *shell model* at shell mass $\varkappa \in [0,\infty)$ is King's walled torus model with the wall $n\psi + \varkappa \mathbf{1}_U$, the matter operator $-\Delta_A + m^2 + \delta m_\varepsilon^2 + 2n\psi + 2\varkappa\mathbf{1}_U$ and $\varkappa\mathbf{1}_U$ kept inside the matter covariance of every block-spin step. King's flow is run on Λ_k through r steps down to the *stopping scale* $\eta_0 := 5^{r-k}$; the shell mass in block units there is $w_r := \varkappa\eta_0^2$, the *frozen regime* is $w_r \ge c_0$, and $\varkappa_0 = \varkappa_0(\eta_0) := \max(c_0,1)/\eta_0^2$. The output is the effective action $S^{(r),\varkappa}(A_r,\phi_r;zf,0)$ on the block fields, with the small-field indicator χ_r of [15, (3.2)], a *near-shell test* τ_r on the block fields of U, a *first-layer test* λ_r on the first light layer of H', and a weakened gradient family $\chi_r^\flat \ge \chi_r$ with no clause on the bonds inside U or crossing ∂U; $I_\varkappa^\sharp(z) := \int \chi_r\tau_r\lambda_r\, e^{-S^{(r),\varkappa}} \le I_\varkappa(z) := \int \chi_r^\flat\tau_r\lambda_r\, e^{-S^{(r),\varkappa}}$. For $\varkappa \ge \varkappa_0$ the thresholds of the tests are dominated by $\varkappa$-free caps $C_7 p_r$ on U and $C_8 p_r$ on the first layer, and the *capped* integral $I_\varkappa^{\mathrm{cap}}(z)$ is $I_\varkappa$ with the tests replaced by the indicators of these caps; with $b := (A_r, \phi_r|_{\mathsf{H}'})$ the *exterior block data*,

$$\tilde\chi^{\mathrm{cap}} := \{b :\ \chi_r^\flat \lambda_r^{\mathrm{cap}}(b) = 1\}, \qquad \tilde S_\varkappa^{\mathrm{cap}}(b;z) := -\ln \int \prod_{y\in U^{(r)}} d\phi_r(y)\, \mathbf{1}\{|\phi_r| \le C_7 p_r \text{ on } U^{(r)}\}\, e^{-S^{(r),\varkappa}(A_r,\phi_r;zf,0)},$$

so that $I_\varkappa^{\mathrm{cap}}(z) = \int_{\tilde\chi^{\mathrm{cap}}} e^{-\tilde S_\varkappa^{\mathrm{cap}}(b;z)}\, db$ exactly: this is the test set and the reduced action of Theorem 20.2, and $\tilde\chi^{\mathrm{cap}}$ is one and the same for every $\varkappa \ge \varkappa_0$.

Proposition 20.3 (the two-sided bound with the shell inside)**.** *Let $L = 5$, the couplings e, λ, m, μ_0 in King's range, (H', T) block-aligned with the grid offset as in Theorem 20.2, $n \ge 1$, and f as in Theorem 20.2. There are $\eta_1 > 0$, $\sigma > 0$, $c > 0$ and, for every $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \le \eta_1$,*

a spacing bound $\varepsilon_0(\eta_0) > 0$ and a constant $C < \infty$ depending on $(\lambda, e, m, \mu_0, |T|, n, f, \mathsf{H}', c_0)$ but not on η_0, $\varkappa$ or ε, such that for all $\varepsilon \le \varepsilon_0(\eta_0)$ and $z \in [-1,1]$, with

$$\mathfrak{a}(\eta_0) := C\eta_0^{\sigma}|T|, \qquad \mathfrak{b}(\eta_0) := e^{-c\,p(\eta_0)^2 + C|T|},$$

$$\begin{aligned}
Z_\varkappa(zf,0) &\ge e^{-\mathfrak{a}}\, I^{\sharp}_\varkappa(z) && (\varkappa \in [0,\infty)),\\
Z_\varkappa(zf,0) &\ge e^{-\mathfrak{a}}\, I^{\mathrm{cap}}_\varkappa(z) && (\varkappa \in [\varkappa_0,\infty)),\\
Z_\varkappa(zf,0) &\le e^{\mathfrak{a}}\, I_\varkappa(z) + \mathfrak{b}\, Z_\varkappa(0,0) \le e^{\mathfrak{a}}\, I^{\mathrm{cap}}_\varkappa(z) + \mathfrak{b}\, Z_\varkappa(0,0) && (\varkappa \in [0,\infty)).
\end{aligned}$$

In particular $\mathfrak{a}, \mathfrak{b} \to 0$ as $\eta_0 \to 0$ at fixed $(n, |T|)$, uniformly in $\varkappa$ and $\varepsilon \le \varepsilon_0(\eta_0)$, and clause (i) *of Theorem 20.2 holds for $\varkappa \ge \varkappa_0$ with the capped integral.*

Idea of the proof (four layers, each uniform in $\varkappa \in [0,\infty]$; the complete proof is in the full version). (1) The block-spin transformation is a Gaussian identity valid for every positive matter covariance, so with $\varkappa \mathbf{1}_U$ a non-negative diagonal operator the composition law, the slice decomposition and the Schur-complement monotonicity in $\varkappa$ hold verbatim. (2) King's Theorem 3.3 [15, (3.5)–(3.8)] for the shell model, uniformly in $\varkappa$ (coercivity, ℓ^2 decay by conjugation, Loewner monotonicity, Kato's inequality), Hölder bounds on the light side, barrier bounds on the heavy side, and the first- and mixed second-difference bounds of the line kernels on both sides of ∂U, where the convexity of ∂U enters. (3) Bałaban's renormalisation bound [3, Prop. 1, (1.33)] for the renormalised families of order at most 13 with the shell model's propagators and legs, the Ward–Takahashi identities for full step-j families (never for a fixed ordering of the lines), and the bound, uniform in the shell mass, on the first-order coefficient of the kept determinant polynomial on the propagator domains [2, (2.104)–(2.107)], which is Theorem 20.4 of the full version. (4) The lower bound is [1, §§3–4] (Jensen's restriction to the small-field region, King's Appendix class [15, (A.1)–(A.8)] for the shell model, the cumulant bound Theorem 17.116 of the full version), the restriction to the capped set being a positivity step inside the class; the upper bound is [2, §§2–3] (the large/small-field split at every scale, the propagator-domain comparisons in the form Lemma 5.1 with the regrouping Lemma 5.2, the continuation of the large-field terms past the stopping scale), with the remainder $\mathfrak{b}\, Z_\varkappa(0,0)$ in normalisation-free form by the ultraviolet-stability gap, as in (2).

The first-order determinant term on the sub-domains (Theorems 20.4 and 20.10, Lemmas 20.5 to 20.9, Corollary 20.11, and Remark 20.12 of the full version). In the upper bound [2, §2] the matter determinant of the composed steps on a propagator domain Ω, a union of unit blocks of any shape, is expanded in the photon fluctuation around the regular background; the first-order term vanishes at zero background and its size is governed by a derivative in the background that cannot be taken pointwise near the faces, holes and re-entrant edges. The theorem represents it exactly as a sum over the sites of the domain of second derivatives of the diagonal resolvent elements of the composed step operator, each gauge invariant at its site, so that every bound on the background is replaced by a bound on its first differences, and bounds it per unit block by $Ce_j^2 \sup_{Q_r}|dB| \sup_{Q_r}|da|$ plus exponentially small tails, on every block and uniformly in the shell mass, from the coercivity and the on-diagonal heat-kernel bound (Lemma F.152 and Lemma A.9 of the full version); both the first- and the second-order coefficient are $O(1)e_j^2 p^2$ per unit block, the form in which layer (3) consumes the kept vertices.

Proposition 20.13 (the $\varkappa$-Cauchy property of the reduced action)**.** *In the setting of Proposition 20.3 fix $\alpha_0 \in (0, \frac{1}{2}]$. There is $\eta_2 \le \eta_1$ such that for every $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \le \eta_2$ there is $C_{\mathfrak{c}}(\eta_0) < \infty$, depending on $(T, \mathsf{H}', n, \lambda, e, m, \mu_0, \alpha_0, c_0, \eta_0)$ and on f but not on ε, $\varkappa$, z, such that for all $\varepsilon \le \varepsilon_0(\eta_0)$, all $\varkappa_0 \le \varkappa \le \varkappa' < \infty$ and all $z \in [-1,1]$,*

$$\sup_{b,b' \in \tilde{\chi}^{\mathrm{cap}}} \big|\Delta_z(b) - \Delta_0(b')\big| \;\le\; 2R(\varkappa,\varkappa') \;\le\; 2\,r(\varkappa;\eta_0), \qquad R(\varkappa,\varkappa') := \int_{\varkappa\eta_0^2}^{\varkappa'\eta_0^2} \theta(w)\,dw + \tau(\varkappa),$$

$$\theta(w) := C_{\mathfrak{c}}\, w^{-1} w^{-(1-\alpha_0)/2} (1+\log w)^2, \qquad \tau(\varkappa) := C_{\mathfrak{c}}\,(\varkappa\eta_0^2)^{-\frac{1}{2}+\frac{\alpha_0}{4}}, \qquad r(K;\eta_0) := C'_{\mathfrak{c}}\,(K\eta_0^2)^{-(1-\alpha_0)/2}\big(1+\log(K$$

with $\Delta_z(b) = \tilde{S}^{\mathrm{cap}}_{\varkappa'}(b;z) - \tilde{S}^{\mathrm{cap}}_{\varkappa}(b;z)$ as in Theorem 20.2; $r(K;\eta_0) \to 0$ as $K \to \infty$ at fixed η_0. This is clause (ii) *of Theorem 20.2.*

Idea of the proof. The statement is derived from a bound on King's full explicit action in all block fields, $|S^{(r),\varkappa} - S^{(r),\varkappa'} - (c_\varkappa - c_{\varkappa'})| \le R(\varkappa,\varkappa')$ on the capped support, with a field-free function $c_\varkappa$ independent of z; on the capped set, whose U-section is the same cube for every exterior datum and every $\varkappa$, the reduced actions inherit the bound with the same $c_\varkappa$, which the mixed difference removes. The full-field bound is proved in derivative form: $\partial_\varkappa S^{(r),\varkappa}$ is King's sum of renormalised graphs with one insertion of $\mathbf{21}_U$ on a line, plus the $\varkappa$-derivative of the block-free part of the background determinant; for an insertion on a line with a scalar connection to the rest of the graph the heavy matter on U pays $w_j^{-1/2}$ per deep end at every frozen scale, and for an insertion whose matter component is closed and connects only through photon lines the members with the same full lines are grouped into one class, to which one gauge identity is applied before the expansion into King's members, the base layer being a first-moment bound, uniform in $\varkappa$, for the two-photon kernel. In the units of the stopping scale the sum over the members less its field-free part is at most $\theta(\hat\varkappa)$ per unit of $\hat\varkappa = \varkappa\eta_0^2$, integrable at infinity, and the block-free part of the determinant has the monotone tail $\tau(\varkappa)$; integrating in $\varkappa$ gives R, and the sum over dyadic intervals gives $r(K;\eta_0)$. The complete proof is in the full version.

Proof of Theorem 20.2. Take η_1 of the theorem to be the η_2 of Proposition 20.13, $\mathfrak{a}, \mathfrak{b}$ and $\varepsilon_0(\eta_0)$ as in Proposition 20.3, $\varkappa_0 = \varkappa_0$, $r(\cdot;\eta_0)$ as in Proposition 20.13, and $(\tilde\chi, db) = (\tilde\chi^{\mathrm{cap}}, db)$ with $\tilde S_\varkappa = \tilde S_\varkappa^{\mathrm{cap}}$. Clause (i) is the second and third lines of Proposition 20.3 for $\varkappa \ge \varkappa_0$; clause (ii) is Proposition 20.13; the test set does not depend on $\varkappa$ on $[\varkappa_0, \infty)$, as stated. □

In words. King's block-spin flow [15, §3] is run for the model in which $\varkappa\mathbf{1}_U$ is part of the matter covariance at every scale, down to a stopping scale η_0 at which the plateau is massive in block units; $\varkappa_0(\eta_0)$, of the order of η_0^{-2}, is the mass above which the shell is heavy in block units at that scale. $I_\varkappa^{\mathrm{cap}}(z)$ is the stopping-scale integral of the flow over a capped set $\tilde\chi$ of block fields, the *test set*, and $\tilde S_\varkappa(b;z)$ is the reduced small-field action at the block field b, shell mass $\varkappa$ and source coefficient z, both for the model at the spacing ε_k: the flow from ε_k to η_0 depends on ε_k, and so do the test set and the reduced action, whereas the constants $\mathfrak{a}, \mathfrak{b}, \varkappa_0, \varepsilon_0$ and the function r do not. Clause (i) is the two-sided bound of King's and Bałaban's type (small-field integral below, small-field integral plus a large-field remainder above) for the model with the shell mass inside the covariance, uniformly in $\varkappa \ge \varkappa_0$ and in the spacing; clause (ii) says that the dependence of the reduced action on the shell mass is, up to a quantity that tends to 0 as the shell gets heavier, the same at every block field and every source coefficient. That the test set is one and the same for all $\varkappa \ge \varkappa_0$ is where the threshold enters: no term from a change of representation of the stopping-scale integral arises between $\varkappa$ and $\varkappa'$, and (ii) compares two integrands on one set. The proof is Propositions 20.3 and 20.13; Remark 20.16 lists its inputs beyond the printed literature.

Theorem 20.14 (the heavy-shell estimate from the shell bounds)**.** *Let $g = 0$ and let $(\mathfrak{s}, \chi)$ be a two-level configuration with the exterior shell $U = T \setminus \mathsf{H}'$, the torus $T = D_{J'}$ with $J' \in 5^{n_*}\mathbb{N}$ and $J' \ge \max(J_0(\delta_r), J_{\mathrm{pos}})$, King's wall profile and plateau height $n \ge 1$, and H', T satisfying the block-alignment and grid-offset clauses of Theorem 20.2, and let f be a real 2-form in C_c^∞ supported in Q_b. Then for every $\eta > 0$ there are a stopping scale η_0 and a shell mass $K < \infty$ with*

$$\limsup_{k\to\infty} \mathrm{tail}_k(K) \le \eta.$$

The quantifiers of (RES$_0$) *are met in this order: given b, a source f supported in Q_b and $\eta > 0$, the configuration $(\mathfrak{s}, \chi)$ is chosen first, with $D(\mathfrak{s})$ so large that $\tau(\mathfrak{s}) \le \eta$, and then η_0 and K as above; so* (RES$_0$) *holds for $(f, 0)$ and η at $(\mathfrak{s}, \chi)$ with this K. Consequently the closing estimate* (TIGHT$_0$) *holds (Theorem 19.10, with Theorem 17.1 and Proposition 20.1), with the single-level configuration $(J', \psi, n + K)$ in the class of Remark 18.6, K being taken large enough for the wall condition of that class (Remark 19.11(iii)) and $J' \in 5^{n_*}\mathbb{N}$, $J' \ge \max(J_0(\delta_r), J_{\mathrm{pos}})$ by hypothesis; the vanishing of ψ near $R^{-1}C_L$, where C_L is the source cube of Theorem 18.4 and not the shell's cube of Definition 19.2, is the choice of b in Theorem 18.4.*

Proof. Fix $(\mathfrak{s},\chi)$, the source f and $g=0$. For $\varkappa\in[0,\infty]$ and $z\in\mathbb{C}$ put $\Psi_\varkappa(z):=Z_\varkappa(zf,0)/Z_\varkappa(0,0)=\langle e^{zF_{\Lambda_k}(f)}\rangle_\varkappa$, where for $\varkappa=\infty$ the state is the walled torus state with the matter killed on $D_\chi=U$. The shell mass $\varkappa\mathbf{1}_U$ is the second level $\varkappa\chi$ of the wall, so by Definition 19.3, at $g=0$, $\Psi_K(\mathrm{i})=\Psi_k(n,K)$, $\Psi_\infty(\mathrm{i})=\Psi_k(n,\infty)$ and $\mathrm{tail}_k(K)=\Psi_K(\mathrm{i})-\Psi_\infty(\mathrm{i})$. For $\varkappa\ge\varkappa_0$ and $z\in[-1,1]$ put $\mathcal{J}_\varkappa(z):=I^{\mathrm{cap}}_\varkappa(z)/I^{\mathrm{cap}}_\varkappa(0)$, with the test set and the reduced action of (HS) at the spacing ε_k under consideration. Every constant below is uniform in k with $\varepsilon_k\le\varepsilon_0(\eta_0)$.

Step 0: the disc bound. At fixed spacing $z\mapsto\Psi_\varkappa(z)$ is entire, a finite-dimensional integral of $e^{zF(f)}$ against a measure with Gaussian photon tails. For $|z|\le2$, $|e^{zF(f)}|\le e^{2F(f)}+e^{-2F(f)}$. Each state $\langle\cdot\rangle_\varkappa$, $\varkappa\in[0,\infty]$, is a BFS measure whose Gaussian is the periodic massive photon Gaussian of the torus, with covariance C; the wall, the shell mass and the killing act in the single-site measures only. So Lemma 2.9 with $s=\pm2$ and v the vector with $\langle v,A\rangle=F_{\Lambda_k}(f)$ gives $\langle e^{\pm2F(f)}\rangle_\varkappa\le e^{2\langle v,Cv\rangle}$, hence

$$|\Psi_\varkappa(z)|\le M_\circ:=2e^{2\langle v,Cv\rangle}\qquad(|z|\le2,\ \varkappa\in[0,\infty]).$$

Here $\langle v,Cv\rangle$, the variance of $F(f)$ under the free massive photon on the torus, is bounded uniformly in the spacing for smooth f: the periodic photon form is at least μ_0^2 times the ℓ^2 norm, so $\langle v,Cv\rangle\le\mu_0^{-2}\|d_k^*f\|^2_{2,k}$, a Riemann sum for $\mu_0^{-2}\|d^*f\|^2_{L^2}$ (as in the proof of Corollary 17.7). At $g=0$ no matter moment enters the disc bound. For real z, $\Psi_\varkappa(z)\to\Psi_\infty(z)$ as $\varkappa\to\infty$ at fixed spacing, by the substitution and dominated convergence of the proof of Lemma 2.10 applied with the set U, the rate $\chi\equiv1$ on U and $\Theta=e^{zF(f)}$, which is of the form admitted there ($|\Theta|=e^{\ell(A)}$ with $\ell=zF(f)$ real linear), the factor $e^{zF(f)}$ being integrable against the photon Gaussian by the same bound.

Step 1: the sandwich. Let $\eta_0\le\eta_1$ be so small that $\mathfrak{b}(\eta_0)\le\frac12$, fix k with $\varepsilon_k\le\varepsilon_0(\eta_0)$, and let $\varkappa\ge\varkappa_0$, $z\in[-1,1]$; write $\mathfrak{a}=\mathfrak{a}(\eta_0)$, $\mathfrak{b}=\mathfrak{b}(\eta_0)$. By (HS)(i) at $z=0$, $e^{-\mathfrak{a}}(1-\mathfrak{b})Z_\varkappa(0,0)\le I^{\mathrm{cap}}_\varkappa(0)\le e^{\mathfrak{a}}Z_\varkappa(0,0)$; by (HS)(i) at z and these two bounds,

$$e^{-2\mathfrak{a}}(1-\mathfrak{b})\,\mathcal{J}_\varkappa(z)\ \le\ \Psi_\varkappa(z)\ \le\ e^{2\mathfrak{a}}\mathcal{J}_\varkappa(z)+\mathfrak{b}.$$

Hence $\mathcal{J}_\varkappa(z)\le2e^{2\mathfrak{a}}M_\circ=:\bar{\mathcal{J}}$ and $|\Psi_\varkappa(z)-\mathcal{J}_\varkappa(z)|\le\bar{\mathcal{J}}(e^{2\mathfrak{a}}-e^{-2\mathfrak{a}}+\mathfrak{b})+\mathfrak{b}=:m_1(\eta_0)$, and $m_1(\eta_0)\to0$ as $\eta_0\to0$ because $\mathfrak{a},\mathfrak{b}\to0$.

Step 2: the $\varkappa$-Cauchy property. Let $\varkappa_0\le\varkappa\le\varkappa'<\infty$ and $z\in[-1,1]$. Since at this k the test set $\tilde\chi$ is the same for $\varkappa$ and $\varkappa'$,

$$\frac{\mathcal{J}_\varkappa(z)}{\mathcal{J}_{\varkappa'}(z)}=\frac{\iint_{\tilde\chi\times\tilde\chi}e^{-\tilde S_\varkappa(b;z)-\tilde S_{\varkappa'}(b';0)}\,db\,db'}{\iint_{\tilde\chi\times\tilde\chi}e^{-\tilde S_{\varkappa'}(b;z)-\tilde S_\varkappa(b';0)}\,db\,db'},$$

and the two integrands differ pointwise by the factor $e^{\Delta_z(b)-\Delta_0(b')}$, which lies in $[e^{-2r},e^{2r}]$ with $r:=r(\varkappa;\eta_0)$ by (HS)(ii). So $\mathcal{J}_\varkappa(z)/\mathcal{J}_{\varkappa'}(z)\in[e^{-2r},e^{2r}]$ and $|\mathcal{J}_\varkappa(z)-\mathcal{J}_{\varkappa'}(z)|\le\bar{\mathcal{J}}(e^{2r}-1)$. Put $\bar r(K;\eta_0):=\sup_{\varkappa\ge K}r(\varkappa;\eta_0)$, which tends to 0 as $K\to\infty$ at fixed η_0. With Step 1, for all $\varkappa'\ge\varkappa\ge K\ge\varkappa_0$,

$$|\Psi_\varkappa(z)-\Psi_{\varkappa'}(z)|\ \le\ m(K,\eta_0):=2m_1(\eta_0)+\bar{\mathcal{J}}\big(e^{2\bar r(K;\eta_0)}-1\big),\qquad z\in[-1,1],$$

uniformly in k with $\varepsilon_k\le\varepsilon_0(\eta_0)$.

Step 3. Letting $\varkappa'\to\infty$ at fixed spacing (Step 0): $|\Psi_K(z)-\Psi_\infty(z)|\le m(K,\eta_0)$ for $z\in[-1,1]$ and $K\ge\varkappa_0$.

Step 4: two constants. $\Xi:=\Psi_K-\Psi_\infty$ is entire, $|\Xi|\le2M_\circ$ on $\{|z|\le2\}$ and $|\Xi|\le m(K,\eta_0)$ on $[-1,1]$. If $m(K,\eta_0)=0$, then Ξ vanishes on $[-1,1]$ and so identically. Otherwise $\log|\Xi|$ is subharmonic on the domain $\Omega:=\{|z|<2\}\setminus[-1,1]$, bounded above by $\log(2M_\circ)$ there, with upper limit at most $\log m(K,\eta_0)$ at every point of the slit; by the two-constants theorem [21, §4.3] it is dominated on Ω by the harmonic function with boundary values $\log(2M_\circ)$ on $|z|=2$ and $\log m(K,\eta_0)$ on the slit. At the point $\mathrm{i}\in\Omega$,

$$|\Xi(\mathrm{i})|\le(2M_\circ)^{1-\omega}\,m(K,\eta_0)^{\omega},$$

where $\omega\in(0,1)$ is the harmonic measure of the slit $[-1,1]$ at i in Ω, an absolute constant.

Step 5: the real point. The charge conjugation $(A,\vartheta)\mapsto(-A,-\vartheta)$ preserves the action, the wall, the shell mass and the measure (Definition 15.7), and reverses the sign of $F(f)$; so

$\Psi_\varkappa(-z) = \Psi_\varkappa(z)$, and since $\Psi_\varkappa(\bar z) = \overline{\Psi_\varkappa(z)}$, the value $\Psi_\varkappa(\mathrm{i})$ is real. Thus $\Xi(\mathrm{i}) = \mathrm{tail}_k(K) \geq 0$ (Lemma 19.4(a)) and

$$0 \leq \mathrm{tail}_k(K) \leq (2M_\circ)^{1-\omega}\, m(K,\eta_0)^\omega \qquad \text{for every } \varepsilon_k \leq \varepsilon_0(\eta_0).$$

Choice of the parameters. Given $\eta > 0$: take $\delta > 0$ with $(2M_\circ)^{1-\omega}\delta^\omega \leq \eta$; then $\eta_0 \leq \eta_1$ with $\mathfrak{b}(\eta_0) \leq \frac{1}{2}$ and $2m_1(\eta_0) \leq \delta/2$, which is possible because $\mathfrak{a}, \mathfrak{b} \to 0$; then $K \geq \varkappa_0(\eta_0)$ with $\bar{\mathcal{J}}(e^{2\bar r(K;\eta_0)} - 1) \leq \delta/2$, which is possible because $r(K;\eta_0) \to 0$ as $K \to \infty$ at fixed η_0; and only then $\varepsilon_k \leq \varepsilon_0(\eta_0)$. Then $m(K,\eta_0) \leq \delta$ and $\mathrm{tail}_k(K) \leq \eta$ for all $\varepsilon_k \leq \varepsilon_0(\eta_0)$, so $\limsup_k \mathrm{tail}_k(K) \leq \eta$. This is the bracket of (RES_0) at $(\mathfrak{s}, \chi)$ with this K, in the quantifier order of Definition 19.8: one configuration, one K, no rate, the limit along the spacings last; since $\tau(\mathfrak{s})$ depends only on H and the width $D(\mathfrak{s})$, a configuration with $\tau(\mathfrak{s}) \leq \eta$, with b' in the residue class of the grid-offset condition and T a King torus, so that H' and T satisfy the alignment clauses of Theorem 20.2 (every interval of length M' final blocks contains an admissible b'), is available for every b, f and η, and (RES_0) holds. Since tail_k is nonincreasing in K (Lemma 19.4(a)), K may be enlarged at will (Remark 19.11(iii)), which, with $J' \in 5^{n_*}\mathbb{N}$ and $J' \geq \max(J_0(\delta_r), J_{\mathrm{pos}})$ (the hypothesis of the theorem), places $(J', \psi, n + K)$ in the class of Remark 18.6. The proof of Theorem 19.10 uses the dead-shell bound (45) at this configuration, which is (DS) (Proposition 20.1), and the existence of the walled value of $(J', \psi, n + K)$, which is part of (K_0) (Theorem 17.1), together with King's printed state $((\mathrm{E}_C))$; it then gives (TIGHT_0). □

The chain of the preceding theorems is run at the points of King's range off the set E_W of Remark 15.2. At a point of E_W the statement of Theorem 15.1 is obtained from the nearby scalar masses by the following theorem, whose proof is in the complete-proofs appendix.

Theorem 20.15 (right-continuity in the scalar mass at the exceptional set)**.** *Let (e, λ, m, μ_0) be a point of the set E_W of Remark 15.2. For $M > 0$ write $Z^K_\infty(\,\cdot\,, 0; M)$ for King's printed state at the couplings (e, λ, M, μ_0), the other three couplings fixed (Theorem 3.3, which holds at every point of King's range). Then there is $\delta > 0$ such that every scalar mass $M \in (m, m+\delta)$ is a point of King's range off E_W, and for every finite family $f_1, \dots, f_N$ of real 2-forms in $C^\infty_c(\mathbb{R}^3; \Lambda^2)$*

$$Z^K_\infty\Big(\sum_{j=1}^N z_j f_j, 0; M\Big) \;\longrightarrow\; Z^K_\infty\Big(\sum_{j=1}^N z_j f_j, 0; m\Big) \qquad (M \downarrow m),$$

locally uniformly in $z \in \mathbb{C}^N$. Consequently every identity between values of $Z^K_\infty(\,\cdot\,, 0)$ on finite families of 2-forms that holds at the masses $M \in (m, m+\delta)$ holds at m, and so does every identity between derivatives at $z = 0$ of such values: the conclusions of Theorem 18.12 and of Corollary 18.13, the Euclidean clause included, pass from the nearby scalar masses to m.

Proof. the full version. □

The proof rests on the monotonicity of the walled and the cube states in the bare mass coefficient and in the cube (Lemma 2.8, Lemma 4.2, Corollary 4.3); on the real-analyticity of α_∞ in the scalar mass (Lemma K.560 of the full version), which puts the nearby masses off E_W; on the comparison of King's cube with the dead-shell torus through the photon switch across a matter-free collar (Lemma 19.1) and the correlation inequalities; on the comparison, at a fixed stopping scale on the dead-shell torus, of the model at the mass M with the model at m, through the endpoint $\varkappa = \infty$ of the shell model of Proposition 20.3 with the counterterm offset of Proposition 17.126(b) of the full version graded in the couplings; and on (E_C) at the nearby masses. Neither the torus comparison (K_0) nor the closing estimate (TIGHT_0) is used at the point of E_W itself.

Proof of Theorem 15.1. Let first the couplings be a point of King's range off E_W (Remark 15.2). Theorem 20.14 gives, from the heavy-shell estimates (HS) (Theorem 20.2), the decoupling (RES_0) at one two-level configuration for each b, f and η, and its final clause, Theorem 19.10 with the torus comparison (K_0) (Theorem 17.1) and the dead-shell chain (DS) (Proposition 20.1),

gives the closing estimate (TIGHT$_0$). With (E$_C$) (Theorem 3.3), (K$_0$) and (TIGHT$_0$), Theorem 18.12 gives the invariance of $Z^K_\infty(\cdot,0)$ under $O(3)$ for all complex coefficients, and Corollary 8.3 adds the translations.

Let now the couplings be a point of E_W. The chain above is not run there: Theorem 17.1, Theorem 18.4 and Theorem 18.12 are stated off E_W. By Theorem 20.15 the scalar masses $M \in (m, m+\delta)$ lie off E_W, so by the first part the identity $Z^K_\infty(\sum_j z_j \tau_b R\cdot f_j, 0; M) = Z^K_\infty(\sum_j z_j f_j, 0; M)$ holds at them for every Euclidean motion $x \mapsto Rx + b$, every finite family $f_1, \dots, f_N$ of real 2-forms in C^∞_c and all $z \in \mathbb{C}^N$. As $M \downarrow m$ both sides converge, locally uniformly in z, to their values at m (Theorem 20.15, applied to the family (f_j) and to the family $(\tau_b R\cdot f_j)$), so the identity holds at m for all z; the covariance of the curvature Schwinger functions follows by differentiating at $z=0$, where the derivatives of locally uniformly convergent entire functions converge. □

Remark 20.16 (the inputs of the proof of (HS)). The two clauses of (HS) are proved by a renormalisation-group analysis of King's walled torus model [15, 16] with the shell mass kept inside the matter covariance at every scale, run down to a stopping scale at which the plateau is massive in block units. Its ingredients beyond the printed literature, by content: Bałaban's asserted block-factor-5 estimates re-run for the shell model; a frozen-scale sub-domain bound in scale-free form; the sub-domain form of the perturbative bound with a tangentially localised amplitude; pointwise first- and mixed second-difference bounds on the line kernels, on the torus and on the Neumann sub-domains used, uniformly in the shell mass and on both sides of ∂U; the light-side regularity estimates on sub-domains and a localisation lemma for the local inverses near the edges and corners of the source box; the tail of the block-free determinant factor with its bracket for the loops winding through U; heavy-kernel bounds for the matter operator with the shell mass; the $\varkappa$-derivative bound for clusters closed in matter, through a gauge identity for full lines; the restricted renormalisation theorem for the shell vertex set; the renormalised-degree clause, uniform in $\varkappa$, for the photon-end families without insertion, obtained by splitting their kernel into its part at zero shell mass and the shell-mass difference (Lemma K.375(c) of the full version, whose first-moment clause at zero shell mass is: at the wall $n\psi$ of the walled torus model of Definition 15.7 with the exterior shell U, the complement of the source box in the torus (the matter potential $m^2{+}2n\psi$ off U and $m^2{+}2n\psi{+}\varkappa\mathbf{1}_U$ on U, as in the full version), the first moment of Π^0_j, the two-photon-leg kernel of step j of King's block-spin flow for that model at zero shell mass $\varkappa = 0$, obeys $|\sum_y \vartheta^d\, \Pi^0_j(x,y)\,(y-x)_\sigma| \le C e_j^2 (1+a_j)^2 \big(1+\log_+(1/\max\{\mathrm{dist}(x,\partial U),\vartheta\})\big)$ in unit-j units, ϑ the lattice spacing in these units (the lattice regime $\vartheta<1$ and the continuum regime, in which the kernel is read on $\vartheta\mathbb{Z}^3$ uniformly as $\vartheta\to 0$), $e_j = e\ell_j^{1/2}$ the running charge and a_j the block-spin coefficient of step j, with C depending on d, a, L and on Bałaban's large-block integer M only, uniformly in ϑ in both regimes and in the small-field backgrounds $tA^{(r)}$, $t\in[0,1]$, of the data of [15, (3.2)]; and the two-photon-leg and mixed classes of that model at zero shell mass obey their renormalised-degree clause of [3, (3.26)–(3.30), (3.36)–(3.38)] with one difference placed on each photon leg, the first moment entering only through this bound); King's near-face analysis [16, §5.2] transplanted to the shell model; Bałaban's all-orders enumeration [3, §§2–3] with one inserted vertex; King's Appendix class [15, (A.1)–(A.8)] for the shell model. The counterterm offset between the torus and the free-boundary convention enters as a bounded constant. Of the printed literature it uses, besides the theorems as printed, the statements named in Remark 17.10 (among them the infrared stage the full version for the mass-counterterm units of order four, in the form stated there) together with [2, Lemma 2.7, (2.113)]; the shell analysis shares with the torus comparison and with the existence theorem (Section 5) Bałaban's same-scale estimates at block factor 5, their reading at a scalar wall with unwalled photon lines, and the orbit regrouping of the cut graph families. Several forms in which such shell estimates might be posed are false and are not used: a sub-domain clause with an absolute constant and a rate in the penetration depth, which fails when the heaviness threshold of the shell in block units is small (the scale-free form is used); the determinant tail without the winding bracket and with a right side proportional to the square of the background curvature, which fails on the

torus (a constant background has zero curvature while the winding loops contribute a positive tail); a Yukawa form of the heavy-pair clause, which fails for pairs in the first layer; and a supremum bound for the full covariance next to ∂U in the lattice regime. Theorem 20.14 uses the determinant tail only through a bound on it that tends to 0 as $\varkappa \to \infty$, so a change of its exponent or a logarithmic factor leaves the theorem unchanged. A photon Ward identity for each fixed ordering of King's graphs would not supply the $\varkappa$-derivative bound, since that identity is false; the identities that hold are those for full lines and for Bałaban's resummed organisation, which keeps a convergent remainder, and in its place the bound rests on that renormalised-degree clause for the photon-end families, in which the first moment of the zero-shell-mass part is the first-moment bound stated in Lemma K.375(c) of the full version and the shell-mass difference is of degree zero and bounded. The two caps on the stopping scale that the tail of the block-free determinant needs, $\varepsilon_0(\eta_0) \le w_U$ and $\varepsilon_0(\eta_0) \le \ell_H/2$ with w_U the width of the shell and ℓ_H the smallest side of H', hold automatically under the block alignment of Theorem 20.2: H' has side at least two final blocks and distance at least three final blocks from its periodic images, and $\varepsilon_0(\eta_0) \le \eta_0 \le \ell_f$, ℓ_f the side of a final block; they are recorded among the conditions of the full version. (DS) (Proposition 20.1) carries the three steps of Theorem 19.6, whose proof is written for a shell between two aligned cubes, to the geometry with one convex face.

20.2. **The continuum form.** This subsection is not on the chain of Theorem 15.1: it records an equivalent form of $(\mathrm{RES})_a$, which is not used, and the existence statement (E_∞) introduced here is used by nothing. Both sides of the bracket in $(\mathrm{RES})_a$ have continuum limits, the first by $(\mathrm{K})_a$ and the second under one further existence statement; in those terms the property says that two continuum objects coincide up to η. Throughout, $(\mathfrak{s}, \chi)$ is a two-level configuration for Q_b in the sense of Definition 19.2 (smooth profile χ, shell $U = C_L \setminus \mathsf{H}$), and $(f, g) \in \mathcal{K}$ is supported in Q_b.

Proposition 20.17 (the dead-shell state between two cube states)**.** *Let $B_k := e^{-a_k \mathsf{S}_{\Lambda_k}(g)} \langle \Theta(\mathrm{i}f, g) \rangle$ in the Neumann–Dirichlet state on $C_L \cap \Lambda_k$ (Neumann photon on C_L, Dirichlet matter outside C_L) whose single-site measures are those of the dead-shell walled torus: the wall $n\psi$, and the matter killed on $D_\chi \cap C_L$. Then at every k with ε_k small*

$$B_k \;\le\; \Psi_k(n, \infty) \;\le\; B_k + Z^a_{k,L}(0, 2g)^{1/2} \big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}, \qquad B_k \le Z^a_{k,L}(\mathrm{i}f, g).$$

Proof. The lower bound is Lemma 15.9 applied to the dead-shell torus state and $X = C_L \cap \Lambda_k$ (X and its neighbours lie in one fundamental cell, as $N_1(\bar{C}_L) \subset (-25MJ', 25MJ')^3$): the only couplings between X and its complement are photon gradient terms, since every site of C_L adjacent to the complement is dead, and deleting them lowers the expectation and factorises the measure, whose X-factor is the state defining B_k. The upper bound is Steps 1–2 of the proof of Theorem 19.6, since the N-measure there is exactly the state defining B_k, and $e^{-2a_k \mathsf{S}_{\Lambda_k}(g)} \langle \Theta(0, 2g) \rangle_N \le Z^a_{k,L}(0, 2g)$ by its Step 3. The last inequality is that Step 3. □

Theorem 20.18 (the continuum form)**.** *Assume $(\mathrm{K})_a$ if $g \neq 0$, and at $g = 0$ let the couplings be off E_W (Remark 15.2), where the torus comparison is Theorem 17.1; for the sandwich in* (d) *at $g \neq 0$ assume also $(\mathrm{W})_a$.*

(a) *For every $K \in [0, \infty)$ the limit $\Psi^0(n, K) := \lim_k \Psi_k(n, K)$ exists, equals the walled value $\mathcal{W}^a_{(J', n\psi + K\chi)}(f, g)$, and is nonincreasing in K; so $\Psi^0(n, \infty^-) := \lim_{K \to \infty} \Psi^0(n, K)$ exists.*

(b) $\liminf_k \mathrm{tail}_k(K) = \Psi^0(n, K) - \limsup_k \Psi_k(n, \infty)$ *for every K, and* $\limsup_k \Psi_k(n, \infty) \le \Psi^0(n, \infty^-)$.

(c) $(\mathrm{RES})_a$ *holds at $(\mathfrak{s}, \chi)$ with tolerance η for some K if and only if*

$$\Psi^0(n, \infty^-) = \lim_{K \to \infty} \lim_{k \to \infty} \Psi_k(n, K) \;\le\; \limsup_{k \to \infty} \Psi_k(n, \infty) + \eta.$$

(d) *If the* dead-shell torus value $\Psi^0(n, \infty) := \lim_k \Psi_k(n, \infty)$ *exists, then* (c) *reads* $\lim_{K \to \infty} \Psi^0(n, K) \le \Psi^0(n, \infty) + \eta$, *the reverse inequality* $\lim_K \Psi^0(n, K) \ge \Psi^0(n, \infty)$ *always holds, and*

$$B^0 \le \Psi^0(n, \infty) \le B^0 + Z^a_\infty(0, 2g)^{1/2} \big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}$$

for every limit point B^0 of (B_k); in particular $\Psi^0(n,\infty) \le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}(e^{\tau(\mathfrak{s})}-1)^{1/2}$. At $g=0$ the sandwich reads $B^0 \le \Psi^0(n,\infty) \le B^0 + (e^{\tau(\mathfrak{s})}-1)^{1/2}$ and $\Psi^0(n,\infty) \le Z^K_\infty(\mathrm{i}f,0) + (e^{\tau(\mathfrak{s})}-1)^{1/2}$.

Proof. (a) $(J', n\psi + K\chi)$ is a walled configuration in $\mathcal{A}(Q_b)$ (its wall is smooth and vanishes on $N_1(Q_b)$), so Lemma 17.3(a) gives the limit, under (K_0) at $g=0$; monotonicity in K passes to the limit (Lemma 19.4(a)). (b) $\liminf_k(x_k - y_k) = \lim_k x_k - \limsup_k y_k$ when x_k converges; and $\Psi_k(n,\infty) \le \Psi_k(n,K)$ for every k and K. (c) By (b), $(\mathrm{RES})_a$ for some K says $\Psi^0(n,K) \le \limsup_k \Psi_k(n,\infty) + \eta$ for some K, which by monotonicity in K is the same as $\inf_K \Psi^0(n,K) \le \limsup_k \Psi_k(n,\infty) + \eta$. (d) The first two statements are (b), (c) with the limit in place of the lim sup; the sandwich is Proposition 20.17 in the limit along a subsequence on which $B_k \to B^0$. At $g \neq 0$, $Z^a_{k,L} \to Z^a_L \le Z^a_\infty$ on $\mathcal{K}$ (Theorem 16.6(i),(iii), under $(\mathrm{W})_a$), with $(0,2g) \in \mathcal{K}$. At $g=0$, $Z^a_{k,L}(0,0) = 1$, and $B^0 \le \limsup_k Z_{k,L}(\mathrm{i}f) \le Z^K_{L+1}(\mathrm{i}f) \le Z^K_\infty(\mathrm{i}f,0)$ by Corollary 4.4(ii)(γ) and the monotonicity of $J \mapsto Z^K_J(\mathrm{i}f)$ (Theorem 7.6(b)); the support condition of that corollary holds since $\operatorname{supp} f \subset Q_b \subset \mathsf{H}$ and $\operatorname{dist}_\infty(\mathsf{H}, \mathbb{R}^3 \setminus C_L) \ge 4$. □

Definition 20.19 ((RES^0): the continuum form)**.** (RES^0) is the property: for every $b > 0$, $(f,g) \in \mathcal{K}$ supported in Q_b and $\eta > 0$ there is a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ for which the dead-shell torus value $\Psi^0(n,\infty)$ exists and

$$\lim_{K\to\infty} \Psi^0(n,K) \;\le\; \Psi^0(n,\infty) + \eta.$$

By Theorem 20.18 (under its hypotheses: $(\mathrm{K})_a$ at $g \neq 0$, the couplings off E_W at $g=0$), (RES^0) implies $(\mathrm{RES})_a$, and $(\mathrm{RES})_a$ implies (RES^0) whenever the dead-shell torus values of the configurations in question exist; the two are the same statement about the interchange of the limits $K \to \infty$ and $k \to \infty$, and in (RES^0) no lattice spacing appears. The existence of $\Psi^0(n,K)$ is covered by $(\mathrm{K})_a$ because the two-level wall $n\psi + K\chi$ is a smooth source; with the sharp profile $\mathbf{1}_U$ it would not be. The existence of $\Psi^0(n,\infty)$, the continuum limit of a torus state with an interior region of Dirichlet matter, is the statement (E_∞). It would be a theorem of King's type; King does not print it: the Dirichlet region is bounded by lattice hyperplanes only when $\partial\mathsf{H}$ is, and it would have to come from the methods of [16] for Dirichlet matter on a cube, applied to a shell between two aligned cubes; the statement is not in [15, 16]. Neither (E_∞) nor (RES^0) is consumed by any result of this paper, and the lattice form $(\mathrm{RES})_a$ needs neither of them nor any limit beyond $(\mathrm{K})_a$.

20.3. The fixed-spacing statement and the shape of a counterexample.

Proposition 20.20 (fixed spacing)**.** *For every two-level configuration, every $(f,g) \in \mathcal{K}$ supported in Q_b and every k, $\mathrm{tail}_k(K) \to 0$ as $K \to \infty$. Consequently, if $(\mathrm{RES})_a$ fails for some (f,g), η, then for every two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ and every K, $\mathrm{tail}_k(K) > \eta$ for all large k, while $\mathrm{tail}_k(K) \le \eta$ for $K \ge K_*(k) < \infty$; hence $K_*(k) \to \infty$ as $k \to \infty$.*

Proof. The first statement is Lemma 19.4(a). The second is its contrapositive form: the failure of $(\mathrm{RES})_a$ at $(\mathfrak{s},\chi)$ means $\liminf_k \mathrm{tail}_k(K) > \eta$ for every K, so $\mathrm{tail}_k(K) > \eta$ for all large k; and $K_*(k) := \inf\{K : \mathrm{tail}_k(K) \le \eta\}$ is finite by the first statement. If $K_*(k) \le K_0$ along a subsequence, then $\mathrm{tail}_k(K_0) \le \eta$ along it, contradicting the previous sentence. □

So any failure of $(\mathrm{RES})_a$ would be carried by shell matter whose mass squared tends to infinity with k: a lattice-scale effect that disappears at every fixed cutoff.

20.4. Why the wall, and why no soft argument.

Proposition 20.21 (no common lattice hyperplanes)**.** *Let $R = R_{\mu\nu}(\pm\theta_0) \in \mathcal{R}_0$ and let ρ be the third index.*

(a) *There is no lattice $P \subset \mathbb{R}^3$ of rank 3 with $RP = P$.*
(b) *An affine plane is a lattice hyperplane of both Λ_k and $R^{-1}\Lambda_k$ (that is, of the form $\{x_\alpha = c\}$ and of the form $\{(R^{-1}e_\beta)\cdot x = c'\}$) only if it is of the form $\{x_\rho = c\}$.*

(c) *Consequently no bounded region of* $\mathbb{R}^3$ *has all its faces on lattice hyperplanes of both lattices; the only unbounded regions with that property are the slabs* $\{c_1 \le x_\rho \le c_2\}$. *The bounded geometries to which both lattices descend are the tori* $\mathbb{R}^3/P$ *with* $P \subset \Lambda_k \cap R^{-1}\Lambda_k$, *for instance* $P = 50MJ\,\mathbb{Z}^3$; *by* (a), *no such* P *is* R*-invariant, so the rotated torus* $R^{-1}D_J$ *is never the torus* D_J.

Proof. (a) If $RP = P$, then in a $\mathbb{Z}$-basis of P the map R is an integer matrix, whose characteristic polynomial has integer coefficients; but the characteristic polynomial of R is $(x-1)(x^2-\frac{8}{5}x+1)$. (b) The normals are e_α and $R^{-1}e_\beta$; $e_\alpha = \pm R^{-1}e_\beta$ means $Re_\alpha = \pm e_\beta$, and for $\alpha \in \{\mu,\nu\}$ the vector Re_α has the components $\pm\frac{3}{5}, \pm\frac{4}{5}$ in the (μ,ν)-plane. (c) A bounded region with faces on hyperplanes needs face normals spanning $\mathbb{R}^3$; by (b) only e_ρ is available. Both lattices descend to $\mathbb{R}^3/P$ exactly when $P \subset \Lambda_k \cap R^{-1}\Lambda_k$, which contains $50MJ\,\mathbb{Z}^3$ by Lemma 15.4(b). □

The ultraviolet analysis of [15, 16] needs boundaries that are lattice hyperplanes of the lattice being analysed ([16, p. 326]: "These hold only when the boundary is a lattice hyperplane orthogonal to a lattice coordinate direction. Clearly the two lattices in Fig. 1 never share such a boundary. Our solution of this problem is to use periodic boundary conditions."). Proposition 20.21 makes this precise: the objects on which the two lattices can be compared by King's method are tori with smooth sources, and smooth walls are the only admissible way to localise a torus; a Dirichlet region cannot be aligned with both lattices. The correlation inequalities give exactly "cube $\le$ walled torus" on the cone and never the reverse (Lemma 15.9; Remark 18.8), so the comparison "walled torus $\le$ King's state $+\eta$" must be supplied by an estimate on King's measure; in the curvature sector that estimate is (HS), through Theorem 20.14.

Remark 20.22 (the content of $(\mathrm{RES})_a$). (i) At every fixed spacing $(\mathrm{RES})_a$ holds (Proposition 20.20); its content is uniformity in the spacing. (ii) It is a decoupling statement for a heavy, weakly coupled region: on the shell the dimensionless couplings are of order $\lambda/\sqrt{K}$ and $e^2/\sqrt{K}$, and the plateau mass n is free. The difficulty is not the shell but the comparison of two states that differ on the shell, seen through the interacting plateau and the light core, uniformly in the spacing. (iii) No correlation inequality bounds the tail from above: Lemma 2.8 gives signs and monotonicity, and the one comparison through a relative density that is uniform in ε is the Gaussian photon switch across a matter-free collar (Lemma 19.1); for the matter, the couplings across the boundary of the shell contribute to the logarithm of the relative density a term whose variance is of order $|\partial U|\varepsilon^{-2}$. (iv) In the curvature sector the estimate is obtained from (HS) in Theorem 20.14. The proof of (HS) is a renormalisation-group expansion of the walled model of [15, 16] with the shell mass kept inside the matter covariance at every scale, run down to a stopping scale at which the plateau is massive in block units, the dependence on the shell mass being controlled through the $\varkappa$-Cauchy property of the reduced small-field action on a test set that does not depend on $\varkappa$. No such expansion for the symmetric massive phase of the three-dimensional model is in the printed literature; Bałaban, Imbrie and Jaffe [6] develop a multiscale expansion of the high-frequency modes of the abelian Higgs model in the Higgs regime and postpone the large-field and decoupling estimates. (HS) carries a window $n \ge 1$ for the plateau height and King's wall profile, which cost nothing for the theorem since n and ψ are free. (v) At $e = 0$ the photon is independent of the matter, and (RES_0) holds trivially ($\mathrm{tail}_k \equiv 0$ at $g = 0$), while $(\mathrm{RES})_a$ for $g > 0$ is a decoupling statement for the two-component φ^4_3 lattice model with a heavy shell. Nothing established at $e = 0$ has been carried to $e > 0$ in this problem, since King's mass counterterm at order e^2 is negative and of order e^2/ε, so that every comparison of the $e > 0$ model with an $e = 0$ model at the bare level is a comparison with a model deep in its broken phase.

Remark 20.23 (the chain of the rotation part). Every implication in the chain

$$(\mathrm{HS}) \xrightarrow{\text{Theorem 20.14}} (\mathrm{RES}_0) \xrightarrow{(\mathrm{E}_C),\ (\mathrm{K}_0),\ (\mathrm{DS});\ \text{Theorem 19.10}} (\mathrm{TIGHT}_0),$$

$$(\mathrm{K}_0) + (\mathrm{TIGHT}_0) \xrightarrow{\text{Theorem 18.12}} \text{invariance under } \mathcal{R}_0 \xrightarrow{\text{Theorem 18.12}} O(3) \xrightarrow{\text{Corollary 8.3}} \mathbb{R}^3 \rtimes O(3)$$

is proved in this part, with (E_C) (Theorem 3.3) wherever King's printed state is named (Lemma 18.2, Theorem 18.12); Theorem 15.1 is the composition. Its two ends, the heavy-shell estimates (HS) (Theorem 20.2) and the torus comparison (K_0) (Theorem 17.1), are renormalisation-group statements about King's walled torus model, proved in Propositions 20.3 and 20.13 and in Section 17.3; (DS) (Proposition 20.1) is the dead-shell chain in the exterior-shell geometry. The chain is run at the points of King's range off E_W (Remark 15.2); at a point of E_W it is run at the nearby scalar masses $M \downarrow m$, which lie off E_W, and Theorem 20.15 carries its conclusion to m.

The main theorem

21. The main theorem for King's printed state

We state the main theorem in full and prove it by assembling Sections 4–12. The object is King's printed state $Z^K_\infty(f,0) = \lim_J \lim_k \langle e^{F(f)}\rangle^K_{k,J}$ on King's RG cube (Definition 2.2), block factor 5, at a fixed photon mass $\mu_0 > 0$ and all positive couplings e, λ, m (Lemma 22.1). Both sectors are proved on King's convention ND^K, on which the existence theorems Theorems 3.3 and 3.6 are posed. In the Wick sector (Leg W), proved in its tube form, Form (B), the tube statements Theorems 10.7 and 13.10 and Lemma 13.11 are written on the comparison convention $\mathrm{ND}^{(a)}$ (Definition 2.1) as templates; the two of them that the interlacing of Lemma 4.2 does not transfer, the bound (31) and the fixed-cube limits on the tube for the K-family functionals $Z^{a,K}_{k,J}$, are proved directly on King's convention in Theorem 21.3, after which the Wick sector follows on ND^K (Proposition 21.4); the comparison family enters through its LIM states, which coincide with King's tube object (Proposition 21.5). The torus inputs are Proposition 3.9 and Lemma 3.10, and the coupling condition is (T3) (Hypothesis 3.12), the coupling region of which contains the explicit regions of Proposition 22.14; $\mathcal{K} = \{(\mathrm{i}f, g) : g \geq 0\}$ is the cone. Theorem 1.1 is the theorem below.

Theorem 21.1 (the axioms for King's printed state at fixed photon mass)**.** *Let $e > 0$, $\lambda, m, \mu_0 > 0$, $\varepsilon_k = 5^{-k}$, $C_J = [-MJ, MJ)^3$ with King's large-block integer $M \in 50\mathbb{N}$, $M \geq M_{\min}$ (the objects below do not depend on such M, the tube limits of part C coinciding where both are defined, Remark 2.3), and let $\mathrm{ND}^K_{k,J}$ be King's cube state.*

A. The object. *For every finite family $f = (f_1, \dots, f_N)$ of real 2-forms in $C^\infty_c(\mathbb{R}^3; \Lambda^2)$ the limit*

$$Z^K_\infty\Big(\sum_j z_j f_j, 0\Big) = \lim_{J\to\infty} \lim_{k\to\infty} \langle e^{\sum_j z_j F_{\Lambda_k}(f_j)}\rangle^K_{k,J}$$

exists locally uniformly in $z \in \mathbb{C}^N$, is entire, depends on the family only through the total source, is real and ≥ 1 at real z, lies in $[0,1]$ at purely imaginary z, and satisfies $|Z^K_\infty(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|_2^2}$, $g(z) = \sum_j (\mathrm{Re}\, z_j) f_j$, with no dependence on μ_0, e, λ, m or M.

B. Leg C (the curvature field). *The Schwinger functions $S^K_n(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} Z^K_\infty(\sum_j z_j f_j, 0)|_{z=0}$ satisfy:*

(i) OS0. *$|Z^K_\infty(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|_2^2}$ for all $z \in \mathbb{C}^N$, and also $\leq \exp(\frac{1}{2}\mu_0^{-2}\|\sum_j (\mathrm{Re}\, z_j) d^* f_j\|_2^2)$; the same two bounds, with the lattice norms $\|\cdot\|_{2,k}$ and d^*_k, hold at every (k, J) for the lattice functionals $\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)}\rangle^K_{k,J}$. For real test functions*

$$|S^K_n(f_1, \dots, f_n)| \leq (\sqrt{2}\, e)^n (n!)^{1/2} \prod_{i=1}^n \|f_i\|_2 \leq (e\pi/2)^n (n!)^{1/2} \prod_{i=1}^n |f_i|_4;$$

each S^K_n extends to a continuous symmetric multilinear form on $\mathcal{S}(\mathbb{R}^3; \Lambda^2_{\mathbb{C}})^n$, a tempered distribution, the bound carrying the factor 2^n for complex test functions; so $\mathrm{E0}''$ holds with $s = 4$, $\sigma_n = (e\pi)^n (n!)^{1/2}$ and every $\beta > \frac{1}{2}$.

(ii) OS1. *The S^K_n are invariant under all translations of $\mathbb{R}^3$ and under the cubic group O_h about every point, F transforming as a 2-form. Moreover Z^K_∞ is invariant under the Euclidean group $\mathbb{R}^3 \rtimes O(3)$ and the S^K_n are Euclidean covariant.*

(iii) OS2. *Reflection positivity about every coordinate plane* $\{x_\nu = c\}$, $\nu \in \{1,2,3\}$, $c \in \mathbb{R}$: *for finite families* $\zeta_1, \dots, \zeta_N$ *of real 2-form sources supported in* $\{x_\nu > c\}$ *with arbitrary complex coefficients,* $\sum_{i,j} \bar\gamma_i \gamma_j Z^K_\infty(\theta\zeta_i^* \sqcup \zeta_j, 0) \geq 0$ *for all* $\gamma \in \mathbb{C}^N$ *(the matrix form for exponentials), and* $S(\theta(P)P) \geq 0$ *for every polynomial* P *in the smeared curvature with sources supported in* $\{x_\nu > c\}$ *(the polynomial form). The same holds about every plane* $\{x \cdot n = c\}$, n *a unit vector.*

(iv) OS3. S^K_n *is real for real test functions, symmetric, and zero for odd* n.

(v) The measure. *For every even* $r \geq 14$ *there is exactly one Borel probability measure* $\mu^{K,H}$ *on* $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ *whose moments on* C^∞_c *are the* S^K_n*; its characteristic functional is* $Z^K_\infty(\mathrm{i}\,\cdot, 0)$, *extended by continuity to* $\mathcal{S}$*; its image* μ^K_S *on* $H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2)$ *is the unique Borel probability measure there with this characteristic functional, and the unique one with these moments. For every real* $\chi \in C^\infty_c(\mathbb{R}^3)$ *and every box* Q *containing* $\operatorname{supp}\chi$ *in its interior, the laws of the localised lattice curvature fields* $\chi\Phi^K_{k,J}$ *under* $\mathrm{ND}^K_{k,J}$ *on* $H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$ *converge weakly, along the full sequence* $k \to \infty$ *at every fixed* $J \geq J_0(\chi)$ *and then as* $J \to \infty$, *to the law of* $\chi\Phi$ *under* μ^K_S. *Nothing is asserted about the laws of the unlocalised lattice fields at fixed* J.

(vi) Weak OS4. μ^K_S *is* $\mathbb{R}^3 \rtimes O_h$*-invariant and is an integral* $\int \mu_\xi \, d\nu(\xi)$ *of translation-ergodic, reflection-positive,* θ_j*-invariant components which cluster in the mean and under diagonal separation, have a unique joint vacuum in each coordinate direction, and satisfy the factorial growth bound* $|S^\xi_N(u_1,\dots,u_N)| \leq \sigma^\xi_N \prod_i \|u_i\|_{r+2}$, $\sigma^\xi_N \leq \alpha_\beta(\xi)(N!)^\beta$ *for every* $\beta > 1$, *with component constants (no tube bound); there is an* O_h*-invariant decomposition into mixtures of* $k \in \{1,2,3,6\}$ *ergodic measures; and* ν*-almost every component satisfies* OS4 *in the weak form (ergodicity under* $\mathbb{R}^3$*). Moreover* ν*-almost every component is* $\mathbb{R}^3 \rtimes O(3)$*-invariant, reflection positive about every plane, and satisfies* OS4 *in each of its forms for every time direction, with* $\ker H_n = \mathbb{C}\Omega_n$ *in every direction* n; *its moments are real and symmetric. The vanishing of a component's odd moments is not claimed.*

C. Leg W, Form (B), on King's convention. *Let the couplings lie in the coupling region of Hypothesis 3.12, the set on which* (T3) *holds, which contains the explicit regions of Proposition 22.14, and fix an admissible scheme* a *(Definition 3.7;* $z = a^\sharp$ *the reference scheme). Let* $\sigma_+ = s_1$, $H = s_1/16$, $\kappa = C_U/s_1$, K_- *be the constants of* (30) *(*K_- *built from* B_{cell}*; for* $a \neq z$ *with* C_U *replaced by* $C_U + s_1 \sup_k |z_k - a_k|$*),* $\sigma_0 = H/4$ *and* $K' = \kappa\pi^2 + K_-$. *Then for every finite family* $\mathcal{F} = (f_1,\dots,f_n;h_1,\dots,h_p)$ *of real 2-forms and real scalars the tube object*

$$G^K_\infty(\zeta) = \lim_{J\to\infty}\lim_{k\to\infty} Z^{a,K}_{k,J}\Big(\sum_i z_i f_i, \sum_j w_j h_j\Big), \qquad \zeta = (z,w) \in \mathcal{D}_{\sigma_+,H} = \{\|k_+(\zeta)\|_\infty < \sigma_+/2,\ \|k_-(\zeta)\|_\infty < H/2$$

exists locally uniformly on the tube and is holomorphic there; on the Wick tube of Definition 10.1 the limit $Z^{a,K}_{\infty,T} = \lim_J \lim_k Z^{a,K}_{k,J}$ *exists and equals* G^K_∞ *where both are defined. The joint Schwinger functions* $S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p) = \partial^{(n)}_z \partial^{(p)}_w G^K_\infty(0)$ *of the pair* (F, W^a) *satisfy:*

(i) OS0. $|G^K_\infty(\zeta)| \leq \exp(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w)$ *on* $\mathcal{D}_{\sigma_+,H}$, *and the same bound with lattice norms holds for* $Z^{a,K}_{k,J}$ *uniformly in* $k \geq 0$ *and* $J \geq J_1(R)$ *when the supports lie in* Q_R*; for real test functions*

$$|S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p)| \leq e^{K'\sigma_0}(\sqrt{2}\,e)^n (n!)^{1/2} (e/\sigma_0)^p\, p! \prod_{i=1}^n \|f_i\|_2 \prod_{j=1}^p |h_j|_w,$$

with the factor 2^{n+p} *for complex test functions; the* $S_{n,p}$ *are tempered distributions and* E0″ *holds with* $s = 4$ *and every* $\beta > 1$. *The constants* σ_+, H, κ, K_- *depend on* $(e, \lambda, m, \mu_0, M, a)$, *on the fixed auxiliary choices* M_1, χ_0 *and the amplitude* $s_1 \in (0, \sigma_1)$ *of* (I2), *and on nothing else; they are not uniform in* μ_0.

(ii) OS1. $G^K_\infty(\tau_b R \cdot \zeta) = G^K_\infty(\zeta)$ *on* $\mathcal{D}_{\sigma_+,H}$ *for every* $b \in \mathbb{R}^3$ *and* $R \in O_h$, F *transforming as a 2-form and* W^a *as a scalar (the transformed family has the same tube); hence the* $S_{n,p}$ *are invariant under translations and under* O_h *about every point. No Euclidean clause is asserted.*

(iii) OS2. *For finite families $\zeta_1, \dots, \zeta_N$ of real sources supported in $\{x_\nu > c\}$ with complex coefficients of amplitude $\mathsf{a}(\zeta_i) < \sigma_W := \sigma_+/8$ (in particular all purely imaginary Wick coefficients with arbitrary complex curvature coefficients), $\sum_{i,j} \bar\gamma_i \gamma_j Z^{a,K}_{\infty,T}(\theta\zeta_i^* \sqcup \zeta_j) \geq 0$ about every coordinate plane; and the polynomial form $S(\theta(P)P) \geq 0$ for polynomials in the smeared F and W^a with sources in $\{x_\nu > c\}$.*

(iv) OS3. *The $S_{n,p}$ are symmetric multilinear in the f's and in the h's, real for real test functions, and zero for odd n.*

(v),(vi) The measure and weak OS4. *Clauses (v) and (vi) of Leg C hold for the joint field on $\mathcal{H}^{-r}(\mathbb{R}^3; \Lambda^2 \oplus \mathbb{R})$ and $H^{-r}_{\mathrm{loc}}(\mathbb{R}^3; \Lambda^2 \oplus \mathbb{R})$, every even $r \geq 14$, with moments $S_{n,p}$, characteristic functional $(u, v) \mapsto G^K_\infty(\mathrm{i}u, \mathrm{i}v)$ on real test functions (purely imaginary coefficients have $k(\zeta) = 0$ and lie in the tube), the component growth bound with the constants of (i), and the local convergence of the laws of the scheme-shifted lattice fields $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ under $\mathrm{ND}^K_{k,J}$ along the full sequence $k \to \infty$ at every fixed $J \geq \max(J_0(\chi), J_1(R_\chi))$, then $J \to \infty$. No Euclidean clause is asserted. Two admissible schemes a, a' give objects differing by the shift $W \mapsto W - \kappa_{a'a} \int(\cdot)$ of the Wick component, $\kappa_{a'a} = \lim_k (a'_k - a_k)$; joint cumulants of order ≥ 2 are scheme-free.*

No bound at positive scalar sources beyond the cone $\sup q \leq s_1$ of Corollary 13.3 is used.

D. The comparison family. *Under the conditions of C, every increasing sequence of spacings has a continuum subsequence $\mathcal{K}$ for the Wick sector of the comparison family $\mathrm{ND}^{(a)}$ (Definition 14.1), and for every such $\mathcal{K}$ and every family the LIM state $Z^{a,\mathcal{K}}_\infty = \lim_J \lim_{k \in \mathcal{K}} Z^a_{k,J}$ exists on $\mathcal{D}_{\sigma_+,H}$ and equals G^K_∞ (Proposition 21.5). Hence clauses (i)–(vi) of C hold for the Wick-sector LIM states of the comparison family, which coincide with King's tube object and do not depend on $\mathcal{K}$; in (v) the laws of the localised lattice fields $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ under $\mathrm{ND}^{(a)}_{k,J}$ converge along $\mathcal{K}$ at every fixed J with $MJ > R(\chi) + \ell_c$, then as $J \to \infty$, to the law of the joint field under the measure of C(v).*

Proof. *A.* Theorem 7.6(b): by Theorem 3.3 the two limits exist locally uniformly on $\mathbb{C}^N$ and are entire, $|Z^K_J|, |Z^K_\infty| \leq e^{\|g(z)\|_2^2}$, $J \mapsto Z^K_J(\mathrm{i}t)$ is nondecreasing with values in $[0,1]$, and $Z^K_\infty(x) \geq 1$ at real x; realness at real z is the symmetry $(A, \vartheta) \mapsto (-A, -\vartheta)$ of every $\mathrm{ND}^K_{k,J}$, closed under limits. Each lattice functional depends on the family only through its total source, hence so do the limits. The bound is absolute (Remark 7.5).

B. (i) is Theorem 7.6: the finite-volume bound (a) with $\min\{\|g\|^2_{2,k}, \frac{1}{2}\mu_0^{-2}\|d^*_k g\|^2_{2,k}\}$ at every (k, J); the two limit bounds (b), the second being (b)'s second bound, carried from the lattice by $\|d^*_k g\|_{2,k} \to \|d^* g\|_2$, (1); the Schwinger bounds, the extension to $\mathcal{S}$, the factor 2^n and E0″ (c). (ii): translations are Corollary 8.3 (the lattice sandwich Lemma 8.2, the two limits, the identity theorem, density); the cubic group about every point is Theorem 8.5; the Euclidean clause is Theorem 8.7, that is Theorem 15.1 (the rotation theorem Theorem 3.14 at every point of King's range: Theorem 18.12 off the exceptional set E_W of Remark 15.2, Theorem 20.15 at the points of E_W) with Corollary 8.3. (iii): the matrix form about every coordinate plane and the polynomial form are Lemma 10.8 with Proposition 10.10; a plane $\{x \cdot n = c\}$ is the image of a coordinate plane under a Euclidean motion γ, and by (ii) $Z^K_\infty(\theta_\gamma \zeta_i^* \sqcup \zeta_j, 0) = Z^K_\infty(\gamma^{-1} \cdot (\theta_\gamma \zeta_i^* \sqcup \zeta_j), 0)$ is a Gram entry about the coordinate plane for the sources $\gamma^{-1} \cdot \zeta_i$, supported in the corresponding coordinate half-space. (iv) is Proposition 10.9 for $Z^K_\infty(\cdot, 0)$ (through Lemma 10.8; charge conjugation gives the vanishing for odd n). (v) is Theorem 11.1(a), including the local convergence of the lattice laws along the full sequence, with Remark 11.2 for the unlocalised fields; (vi) is Theorem 11.1(b) and (c), whose $O(3)$ input is (ii)'s Euclidean clause; the disclaimer on odd moments is that of Theorem 11.1(c).

C. Under (T3), Proposition 12.6 gives (I2) with amplitude $s_1 \in (0, \sigma_1)$ and constant C_U from the torus inputs Proposition 3.9 and Lemma 3.10; for $a \neq z$ the difference $z_k - a_k$ converges by admissibility, so $\sup_k |z_k - a_k| < \infty$ and C_U is replaced as stated (Lemma 14.5); (I3) is Proposition 12.2, the correlation inequalities are Lemma 2.8, and the cell constant $B^{(a)}_{\mathrm{cell}}$ is finite (Lemma 14.4). *(i)*: Theorem 21.3 gives the bound (56) for every $k \geq 0$ and

$J \geq J_1(R)$, the fixed-cube limits G_J^K and the cube limit G_∞^K locally uniformly on $\mathcal{D}_{\sigma_+,H}$, their holomorphy, the bound in the limit, the Schwinger bounds, the factor 2^{n+p} and E0″, with the stated dependence of the constants and no uniformity in μ_0. On the Wick tube $T_{\mathcal{F}'}$ of the enlarged family of Definition 10.1, whose scalar members are nonnegative and whose points have $2\|k_1\|_\infty < \sigma_+/2$, the fixed-cube limits of $Z_{k,J}^{a,K}$ exist locally uniformly by Theorem 6.17(i), whose domain $\{\sup_x 2k_1 < s_W\}$, $s_W = s_1$, contains $T_{\mathcal{F}'}$, for the reference scheme and hence for a (Definition 3.7); their cube limit $Z_{\infty,T}^{a,K}$ is Proposition 10.5(a) run on King's convention (Proposition 21.4(c)), and it equals G_∞^K on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ as the same J-limit of the same fixed-cube limits. *(ii)*: Proposition 21.4(b). *(iii)*: Proposition 21.4(c), that is Theorem 10.7 with Proposition 10.5 run on King's convention, and the polynomial form Proposition 10.10. *(iv)*: Proposition 10.9 for $Z_{\infty,T}^{a,K}$ and G_∞^K (Proposition 21.4(c)). *(v),(vi)*: Proposition 21.4(d). The Gram points of (v)'s characteristic functional have purely imaginary coefficients and lie in the tube, and the reflection positivity consumed there is (iii) at amplitude 0 (Remark 21.6). *The scheme clause*: for admissible a, a', $Z_{k,J}^{a',K}(\zeta) = e^{-(a'_k - a_k)\sum_j w_j \mathsf{S}_{\Lambda_k}(h_j)} Z_{k,J}^{a,K}(\zeta)$ (Definition 2.4), where $a'_k - a_k = (z_k - a_k) - (z_k - a'_k) \to \kappa_{a'a}$ and $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$; the factor converges locally uniformly in ζ, so the fixed-cube limits and then the cube limits satisfy $G_\infty^{K,a'}(\zeta) = e^{-\kappa_{a'a}\sum_j w_j \int h_j} G_\infty^{K,a}(\zeta)$ on the tube, the shift $W \mapsto W - \kappa_{a'a}\int(\cdot)$ of the Wick component; a factor $e^{\ell(w)}$ with ℓ linear changes only the first cumulants in w, so the joint cumulants of order ≥ 2 are scheme-free.

D. Proposition 21.5; the clauses of C then hold for $Z_\infty^{a,\mathcal{K}} = G_\infty^K$, and the convergence of the laws of the localised comparison-cube fields along $\mathcal{K}$ is clause (v) of the Wick sector of Theorem 14.9, whose hypotheses ($\mathrm{B}_{\mathrm{cell}}$) and (SCH) hold for every admissible scheme under (T3) (Lemmas 14.4 and 14.5). □

Remark 21.2 (the Euclidean clause). The Euclidean clauses of B(ii), (iii), (vi) rest on the rotation theorem Theorem 3.14, proved in the rotation part (Theorem 15.1): the walled tori on the two lattices have the same continuum limits (Theorem 17.1), King's printed state is the lower envelope of the walled tori (Theorem 18.4), and that closing estimate comes from the heavy-shell estimates on King's block-spin flow (Theorem 20.2), both the comparison and the closing estimate holding off the exceptional set E_W of Remark 15.2, the points of E_W being reached by right-continuity in the scalar mass (Theorem 20.15); the Wick sector carries no Euclidean clause (Theorem 23.7).

Theorem 21.3 (the Wick tube on King's convention)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme. Let $\sigma_+ = s_1$, H, κ be the constants of* (30) *and let K_- be the constant there, computed from the cell constant $B_{\mathrm{cell}}^{(a)}$ of the comparison convention (Lemma 14.4), which is finite. Then for every finite family of real sources with supports in Q_R, every $k \geq 0$, every $J \geq J_1(R)$ and every $\zeta \in \mathcal{D}_{\sigma_+,H}$,*

$$|Z_{k,J}^{a,K}(\zeta)| \;\leq\; Z_{k,J}^{a,K}(\operatorname{Re}\zeta) \;\leq\; \exp\big(\|g(\zeta)\|_{2,k}^2 + \kappa\|k_+(\zeta)\|_{1,k} + K_-|k_-(\zeta)|_w\big), \tag{56}$$

the bound (31) *for the K-family functionals with the constants of the comparison convention; and the fixed-cube limits $G_J^K(\zeta) = \lim_{k\to\infty} Z_{k,J}^{a,K}(\zeta)$ exist for every $\zeta \in \mathcal{D}_{\sigma_+,H}$, locally uniformly, so that Theorem 13.10(i),(ii) hold on ND^K: G_J^K and $G_\infty^K = \lim_J G_J^K$ exist locally uniformly on $\mathcal{D}_{\sigma_+,H}$ and are holomorphic, the bound* (56) *passes to the limits, and the Schwinger bounds and* E0″ *of Theorem 21.1C(i) hold for G_∞^K; moreover $G_\infty^K > 0$ at the real points of $\mathcal{D}_{\sigma_+,H}$ and $G_J^K \leq G_\infty^K$ at its cone points.*

Proof. Throughout, $\mathcal{F} = (f_1, \dots, f_n; h_1, \dots, h_p)$ is a family of real sources with supports in Q_R, $J \geq J_1(R)$, so that $MJ \geq R + \ell_c \geq R + 10$: every source lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, the support condition (S) holds for X_J, and $J \geq J_0(\mathcal{F})$. For $\zeta \in \mathcal{D}_{\sigma_+,H}$ write $g = g(\zeta)$ and $k = k(\zeta) = k_+ - k_-$ (Definition 2.6); by Lemma 2.7(E3) the sampled sources satisfy $0 \leq 2k_+ \leq s_1$ and $0 \leq 2k_- \leq H$ on Λ_k. The inputs are: (I2) with amplitude $s_1 \in (0, \sigma_1)$ and constant C_U, from (T3) and the torus inputs Proposition 3.9 and Lemma 3.10 through Proposition 12.6, for $a \neq z$ with the constant $C_U + s_1 \sup_k |z_k - a_k|$, finite by Definition 3.7 (Lemma 14.5); (I3),

which is Proposition 12.2; (I4), which is Lemma 2.8; and the cell constant $B^{(a)}_{\mathrm{cell}} < \infty$ of the comparison convention (Lemma 14.4). Under these, Corollary 13.3 and Proposition 13.9 hold on the comparison convention and Lemma 4.7(d) on King's.

Step 1 (modulus). Lemma 2.12(b) holds for every state of Definition 2.1, killed sites included: $|Z^{a,K}_{k,J}(\zeta)| \le Z^{a,K}_{k,J}(\mathrm{Re}\,\zeta)$. It therefore suffices to bound $Z^{a,K}_{k,J}$ at the real point $\mathrm{Re}\,\zeta$, where $Z^{a,K}_{k,J}(\mathrm{Re}\,\zeta) = N^{a,K}(g,k) := e^{-a_k \mathsf{S}(k)} \langle e^{F(g)+:\phi^2:(k)} \rangle^K_{k,J}$.

Step 2 (curvature off). Let $\langle\cdot\rangle_{K,k}$ be King's cube with each single-site measure at an alive site multiplied by $e^{\varepsilon^3 k(x) r_x^2}$; it is a BFS measure (the factor keeps $\int e^{\alpha r^2} d\rho_x < \infty$ because $\lambda > 0$, and the killed sites are unchanged), and $N^{a,K}(g,k) = N^{a,K}(0,k)\,\langle e^{F(g)}\rangle_{K,k}$. Its Gaussian is the Neumann photon Gaussian of X_J, the same for both conventions, with covariance C as in Lemma 7.1. By (S) every plaquette with $g \neq 0$ has its corners in X_J, so $F(g) = \langle v, A\rangle_k$ with $v = d_k^* g$, and Lemma 2.9 (with $s = 1$) and Lemma 7.1 give $\langle e^{F(g)}\rangle_{K,k} \le e^{\langle v, Cv\rangle_k/2} \le e^{\|g\|^2_{2,k}}$. This is the chain of Theorem 7.6(a) in the presence of the real scalar source k. Hence $N^{a,K}(g,k) \le e^{\|g\|^2_{2,k}} N^{a,K}(0,k)$.

Step 3 (the split). Lemma 2.12(d) at $f = 0$ with $k_1 = k_+$, $k_2 = k_-$: $N^{a,K}(0, k_+ - k_-) \le N^{a,K}(0, 2k_+)^{1/2} N^{a,K}(0, -2k_-)^{1/2}$.

Step 4 (the positive leg on King's convention). Put $q := 2k_+$, a lattice function with $0 \le q \le s_1$ supported in $Q_R \cap \Lambda_k \subset \mathrm{alive}^K_J$. The observable $e^{\sum_x \varepsilon^3 q(x) r_x^2}$ is $\Theta_{0,q'}$ with $q' = \varepsilon^3 q \ge 0$, of class $\mathcal{I}$, and (S) holds for X_J; Step 1 of the proof of Lemma 4.2 (the kill step on $\mathcal{I}$) gives $\langle\Theta_{0,q'}\rangle^K_{k,J} \le \langle\Theta_{0,q'}\rangle^{(a)}_{k,J}$, and the multiplier $e^{-(a_k + 2\mathsf{W}_\varepsilon)\mathsf{S}(q)}$ is one positive constant for both states (Lemma 2.12(a)). Hence

$$N^{a,K}_{k,C_J}(0,q) \;\le\; N^{a}_{k,C_J}(0,q) \;\le\; e^{\kappa \|q\|_{1,k}}$$

by Corollary 13.3, whose hypotheses (I2), (I3) hold (this is the cone instance of Remark 4.8, as in Remark 13.12). So $\frac{1}{2}\ln N^{a,K}(0,2k_+) \le \kappa \|k_+\|_{1,k}$.

Step 5 (the negative leg on King's convention). Put $q := 2k_-$, a lattice function with $0 \le q \le H$ supported in Q_R, and $J \ge J_1(R)$. Lemma 4.7(d) gives $\ln N^{a,K}_{k,C_J}(0,-q) \le K_- |q|_w$ with the printed constant K_- of (30), built from the cell constant $B^{(a)}_{\mathrm{cell}}$ of the comparison cell, which is finite by Lemma 14.4: by Lemma 4.7(c) every cell on which one of the cutoff sources $8\chi_s q$ does not vanish is a full cell inside C_J containing no killed site, so each factor of the wall insertion of Lemma 4.7(a) that carries a source is a comparison-cell functional, to which Lemmas 13.6 and 13.7 apply as printed, and the sum over cells is Lemma 13.8; the scheme multiplier factorises over the cells (Lemma 2.12(a)). The direction of the kill matters here and is the content of Lemma 4.7(e). Hence $\frac{1}{2}\ln N^{a,K}(0,-2k_-) \le \frac{1}{2} K_- |2k_-|_w = K_- |k_-|_w$.

Step 6 (assembly). Steps 1–5 give, for every $k \ge 0$, $J \ge J_1(R)$ and $\zeta \in \mathcal{D}_{\sigma_+,H}$,

$$|Z^{a,K}_{k,J}(\zeta)| \le Z^{a,K}_{k,J}(\mathrm{Re}\,\zeta) \le \exp\big(\|g(\zeta)\|^2_{2,k} + \kappa \|k_+(\zeta)\|_{1,k} + K_- |k_-(\zeta)|_w\big),$$

which is (56). The constants are those of (30), with $B^{(a)}_{\mathrm{cell}}$ in K_- and, for $a \neq z$, $C_U + s_1 \sup_k |z_k - a_k|$ in κ.

Step 7 (the fixed-cube limits). Each $Z^{a,K}_{k,J}$ is entire in $\zeta \in \mathbb{C}^{n+p}$: it is a finite-dimensional integral with Gaussian decay in A ($\mu_0 > 0$) and $e^{-\lambda\varepsilon^3 r^4}$ decay in r at every alive site, the killed sites carrying no variables, and differentiation under the integral sign is justified. On a compact $Q \subset \mathcal{D}_{\sigma_+,H}$ the family $\{Z^{a,K}_{k,J}\}_{k\ge 0}$ is uniformly bounded, by (56) and Lemma 2.7(E1)–(E3): $\|g\|^2_{2,k} \le C_0 |g|^2_4$, $\|k_+\|_{1,k} \le C_0' |k_+|_w$ and $|k_\pm|_w \le \sum_j |\mathrm{Re}\, w_j| |h_j|_w$, all continuous in ζ. At every real point of $\mathcal{D}_{\sigma_+,H}$ the sequence converges: there $2\|k_+\|_\infty < \sigma_+ = s_1 = s_W$ and the negative part is unrestricted, so Theorem 3.6, which is posed on King's convention (Corollary 6.18(a)), gives the limit in $(0,\infty)$ for the reference scheme; for the admissible scheme a, $Z^{a,K}_{k,J} = e^{-(a_k - z_k)\sum_j w_j \mathsf{S}_k(h_j)} Z^{z,K}_{k,J}$ with $a_k - z_k$ convergent and $\mathsf{S}_k(h_j) \to \int h_j$ (Definition 3.7, Remark 6.19). The real points of $\mathcal{D}_{\sigma_+,H}$ form a non-empty open subset of $\mathbb{R}^{n+p}$ (Definition 2.6), a set of uniqueness in the connected open $\mathcal{D}_{\sigma_+,H}$; by Montel's and Vitali's theorems (the fact

(T3) stated before Theorem 13.10), $Z^{a,K}_{k,J} \to G^K_J$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G^K_J is holomorphic there, and (56) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (k_+ is continuous with compact support) and the sampled $|k_-|_w$ at most the continuum one (E3):

$$|G^K_J(\zeta)| \le \exp\big(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w\big) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H},\ J \ge J_1(R)), \tag{57}$$

a bound free of J. At real points $G^K_J > 0$ (Theorem 3.6).

Step 8 (the cube limit on a sub-tube). Split $h_j = h_j^+ - h_j^-$ with $h_j^- := \|h_j\|_\infty \chi_j$, $\chi_j \in C_c^\infty$, $0 \le \chi_j \le 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset Q_{R+1/2}$, and $h_j^+ := h_j + h_j^- \ge 0$. For the split family $\mathcal{F}'' = (f; h^+, h^-)$ put $F^K_{k,J}(z, w^+, w^-) := Z^{a,K}_{k,J}(\sum_i z_i f_i, \sum_j (w_j^+ h_j^+ + w_j^- h_j^-))$ and $\mathcal{D}'' := \{(z, w^+, w^-) \in \mathbb{C}^{n+2p} : |\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j \delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) < H/2$. On $\mathcal{D}''$ the scalar source k'' of the split family has $\|k''_\pm\|_\infty \le \|k''\|_\infty < H/2 \le \sigma_+/2$, so $\mathcal{D}''$ lies in the tube of $\mathcal{F}''$, whose supports lie in $Q_{R+1/2}$; Steps 1–7 for $\mathcal{F}''$ and $J \ge J_1(R+\frac12)$ give $F^K_{k,J} \to F^K_J$ locally uniformly on $\mathcal{D}''$ with the J-free bound (57) for $\mathcal{F}''$. On $\Omega''_+ := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the curvature coefficients are imaginary and the total scalar source $\sum_j (w_j^+ h_j^+ + w_j^- h_j^-)$ is nonnegative, so the observable is of class $\mathcal{I}$ and (S) holds for X_J ($MJ \ge R + \frac12 + \ell_c$); Corollary 4.3 gives $F^K_{k,J} \le F^K_{k,J+1}$ at every k (the scheme multiplier is common, Lemma 2.12(a)), and $k \to \infty$ gives $F^K_J \le F^K_{J+1}$ on Ω''_+. Being nondecreasing and bounded there, $J \mapsto F^K_J$ converges on Ω''_+, a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$; $\mathcal{D}''$ is convex, hence connected, and the Montel–Vitali fact (T3) stated before Theorem 13.10 gives $F^K_J \to F^K_\infty$ locally uniformly on $\mathcal{D}''$. At $w_j^+ = w_j$, $w_j^- = -w_j$ the total scalar source is $\sum_j w_j(h_j^+ - h_j^-) = \sum_j w_j h_j$, so $F^K_{k,J}(z, w, -w) = Z^{a,K}_{k,J}(\zeta)$ exactly at every k; hence $G^K_J \to G^K_\infty$ locally uniformly on $\mathcal{D}' := \{\zeta : |\operatorname{Re} w_j| < \delta_j\}$, which lies in $\mathcal{D}_{\sigma_+,H}$ because there $\|k_\pm\|_\infty \le \|k\|_\infty < \sum_j \delta_j \|h_j\|_\infty < H/2$.

Step 9 (the whole tube). The family $\{G^K_J\}_{J \ge J_1(R+1/2)}$ is locally uniformly bounded on $\mathcal{D}_{\sigma_+,H}$ by (57); by Montel's theorem every subsequence has a further subsequence converging locally uniformly to a holomorphic function, any two such limits agree on the open set $\mathcal{D}'$ by Step 8, and $\mathcal{D}_{\sigma_+,H}$ is connected, so all subsequential limits coincide: $G^K_J \to G^K_\infty$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G^K_∞ is holomorphic, and (57) passes to G^K_∞. The finitely many indices $J_1(R) \le J < J_1(R+\frac12)$ do not affect the limit.

Step 10 (positivity and the cone). For a real $\zeta \in \mathcal{D}_{\sigma_+,H}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}_{\sigma_+,H}$. The function $\varphi_J(t) := \ln G^K_J(t\zeta)$ is finite and convex on $[-\epsilon, 1]$ (a pointwise limit of the convex functions $\ln Z^{a,K}_{k,J}(t\zeta)$, Lemma 2.12(b), with positive limits at real points) and $\varphi_J(0) = 0$, so $\varphi_J(1) \ge -\varphi_J(-\epsilon)/\epsilon \ge -C_\epsilon/\epsilon$, C_ϵ the exponent of (57) at $-\epsilon\zeta$; hence $G^K_\infty(\zeta) \ge e^{-C_\epsilon/\epsilon} > 0$. At the cone points of $\mathcal{D}_{\sigma_+,H}$ ($z \in (\mathrm{i}\mathbb{R})^n$, $w \in \mathbb{R}^p$, $k(\zeta) \ge 0$), Corollary 4.3 at every k and $k \to \infty$ give $G^K_J \le G^K_{J+1} \le G^K_\infty$.

Step 11 (the Schwinger bounds and E0″*).* Put $S^K_{n,p}(f; h) := \partial_z^{(n)} \partial_w^{(p)} G^K_\infty(0)$, $\sigma_0 = H/4$, $K' = \kappa\pi^2 + K_-$. Normalise $\|f_i\|_2 = 1$ and let $\rho_j := \sigma_0/(p|h_j|_w)$. On the polydisc $\{|w_j| \le \rho_j\} \times \mathbb{C}^n$ we have $\|k\|_\infty \le \sum_j |\operatorname{Re} w_j| \|h_j\|_\infty \le \sum_j \rho_j |h_j|_w = \sigma_0 < H/2 \le \sigma_+/2$ (Lemma 2.7(E2)), so the polydisc lies in $\mathcal{D}_{\sigma_+,H}$, where G^K_∞ exists with the bound (57), whose exponent is at most $\|g(z)\|_2^2 + \kappa\pi^2|k_+|_w + K_-|k_-|_w \le \|g(z)\|_2^2 + K'|k|_w \le \|g(z)\|_2^2 + K'\sigma_0$ (E2). Cauchy's estimate for the mixed first derivative in the w's at $w = 0$, z fixed, gives $|\partial_w^{(p)} G^K_\infty(z, 0)| \le \prod_j \rho_j^{-1} \sup|G^K_\infty| \le (p/\sigma_0)^p \prod_j |h_j|_w \, e^{K'\sigma_0} e^{\|g(z)\|_2^2}$, and $p^p \le e^p p!$ (E4). The function $z \mapsto \partial_w^{(p)} G^K_\infty(z, 0)$ is entire ($\mathbb{C}^n \times \{w\} \subset \mathcal{D}_{\sigma_+,H}$ for $|\operatorname{Re} w_j|$ small) and bounded by $A\, e^{\|g(z)\|_2^2}$ with $A := e^{K'\sigma_0}(e/\sigma_0)^p p! \prod_j |h_j|_w$; on the polydisc of radius $r = (2n)^{-1/2}$ about 0, $\|g(z)\|_2 \le \sum_i |\operatorname{Re} z_i| \le nr$, so Cauchy's estimate for the mixed first z-derivative gives $|S^K_{n,p}| \le A\, r^{-n} e^{n^2 r^2} = A\,(2n)^{n/2} e^{n/2} \le A\,(\sqrt{2}e)^n (n!)^{1/2}$ by (E4). Restoring the normalisation by multilinearity,

$$|S^K_{n,p}(f_1, \dots, f_n; h_1, \dots, h_p)| \le e^{K'\sigma_0} (\sqrt{2}e)^n (n!)^{1/2} (e/\sigma_0)^p \, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w.$$

By (E2), $\|f_i\|_2 \le \frac{\pi}{2\sqrt{2}}|f_i|_4$ and $|h_j|_w \le |h_j|_4$, and $(n!)^{1/2}p! \le (n+p)!$, so the right side is at most $\sigma_{n+p}\prod_i|f_i|_4\prod_j|h_j|_4$ with $\sigma_N = e^{K'\sigma_0}C_1^N N!$, $C_1 = 2\max(e\pi/2, e/\sigma_0)$; for complex test functions each argument is split into real and imaginary parts, which gives the factor $2^{n+p} \le 2^N$ absorbed in C_1. The $S^K_{n,p}$ are therefore continuous multilinear forms in the norms $|\cdot|_4$, extend to $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n \times \mathcal{S}(\mathbb{R}^3;\mathbb{C})^p$ and are tempered distributions, as in Theorem 7.4(c), and $C_1^N/(N!)^{\beta-1} \to 0$ for every $\beta > 1$: this is E0″ with $s = 4$ and every $\beta > 1$, with the constants of Theorem 21.1C(i). □

Proposition 21.4 (the Wick sector on King's convention)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme. Then, with the constants of Theorem 21.1C:*

(a) $G^K_J := \lim_k Z^{a,K}_{k,J}$ *exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ for every $J \ge J_1(R)$, $G^K_\infty := \lim_J G^K_J$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$, both are holomorphic, $|G^K_\infty(\zeta)| \le \exp(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w)$, and the Schwinger bounds and* E0″ *of Theorem 21.1C(i) follow.*

(b) $G^K_\infty(\tau_b R\cdot\zeta) = G^K_\infty(\zeta)$ *on $\mathcal{D}_{\sigma_+,H}$ for all $b \in \mathbb{R}^3$, $R \in O_h$, W^a a scalar.*

(c) *For every family $\mathcal{F}'$ with nonnegative scalar members the Wick-tube object $Z^{a,K}_{\infty,T} := \lim_J \lim_k Z^{a,K}_{k,J}$ exists locally uniformly on $T_{\mathcal{F}'}$ and equals G^K_∞ on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$; reflection positivity about every coordinate plane holds in the matrix form at amplitude $< \sigma_W$ and in the polynomial form, and* OS3*, for $Z^{a,K}_{\infty,T}$, G^K_∞ and their Schwinger functions.*

(d) *The measure and weak* OS4 *of Theorem 21.1C(v),(vi) for the joint field under* ND^K*, with the local convergence of the laws of $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ under $\mathrm{ND}^K_{k,J}$ along the full sequence $k \to \infty$ at every fixed $J \ge \max(J_0(\chi), J_1(R_\chi))$, then $J \to \infty$.*

Proof. Parts (a) and (c) are proved below; the complete proofs of parts (b) and (d) are given in the full version; the paragraphs (b) and (d) below summarise them.

(a) is Theorem 21.3, whose proof contains the fixed-cube limits (Step 7), the cube limit on the connected tube with the cube monotonicity of Corollary 4.3 in place of the first inequality of Lemma 2.11 (Steps 8–9), the positivity at real points and the cone monotonicity (Step 10), and the Schwinger bounds by Cauchy's estimates on a polydisc in the tube (Step 11).

(b) The steps of Theorem 8.5 with scalar sources. Coordinate permutations are exact on ND^K for (f,h) by Lemma 8.4. The reflection sandwich is Lemma 4.5(b) at the cone points of the tube: for $(f,g) \in \mathcal{K}$ with $\|g\|_\infty < \sigma_+/2$ the point $(\mathrm{i},1)$ lies in $\mathcal{D}_{\sigma_+,H}$, so (a) supplies the three k-limits and the two J-limits of the two-step sandwich $C^K_{J\mp2}$. Conjugation and group closure on the cone are unchanged. Off the cone, the identity theorem on the split family's tube of Lemma 13.11, where (a) gives the limits, extends the identity from the cone points to the connected tube; so Lemma 4.5 is needed on $\mathcal{K}$ only, and no class-$\mathcal{D}$ form of it enters (the proof splits $h = h^+ - h^-$ and never applies the sandwich to a negative scalar source). Translations are Lemma 8.2, stated on $\mathcal{K}$ and applied at the cone points of the tube with (a) for the limits, then the continuity route of Corollary 8.3 with the Wick-sector second-moment bound, which Cauchy's estimates give from (31) on ND^K; the translated family needs $J \ge \max(J_0+1, J_1(R+M+\frac12))$. Invariance about every point and differentiation concern the limit object.

(c) Theorem 10.7 and Proposition 10.5 on ND^K: in the sandwich of the J-limit the upper box is Lemma 4.6 (the printed boxes C_J, C_{J+2n}) and the lower end carries the kill step of Remark 4.8; the positive-leg bound Corollary 13.3 holds on ND^K by Remark 4.8 prepended to Proposition 13.2 (Remark 13.12); the cube monotonicity in both classes is Corollary 4.3; the fixed-cube limits on $T_{\mathcal{F}'}$ are Theorem 6.17(i), whose domain $\{\sup_x 2k_1 < s_W\}$, $s_W = s_1$, contains $T_{\mathcal{F}'}$ (for the reference scheme, hence for a by Definition 3.7), and on $\mathcal{D}_{\sigma_+,H}$ they are (a); the cube limit on $T_{\mathcal{F}'}$ is then Montel's theorem with the uniqueness set of Definition 10.1(iii), as in Proposition 10.5; the translations to the planes $c \notin D$ are (b). OS3 and the polynomial form use only the existence and holomorphy of the limit near 0 and charge conjugation (Propositions 10.9 and 10.10).

(d) The steps of Theorem 11.1 on $\mathcal{V} = \Lambda^2 \oplus \mathbb{R}$ with the scheme-shifted field: (31) on ND^K in the moment bounds, (a) for the limit functionals and the lattice laws, (b) and (c) for the invariance and the reflection positivity, which give (H1)–(H3) of the full version by Proposition P.9 there.

The second moments of the lattice field under $\mathrm{ND}^K_{k,J}$ come from (31) on ND^K by Cauchy's estimates, so tightness and the identification of the limit law run as in the curvature sector, with the fixed-cube limits of (a) along the full sequence, for $J \geq \max(J_0(\chi), J_1(R_\chi))$; the remaining steps concern the limit measure. □

Proposition 21.5 (the comparison family on the Wick tube)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme.*

(a) *Every increasing sequence of spacings has a continuum subsequence $\mathcal{K}$ for the Wick sector of the comparison family (Definition 14.1).*

(b) *For every such $\mathcal{K}$, every family $\mathcal{F}$ of real sources with supports in Q_R and every $\zeta \in \mathcal{D}_{\sigma_+,H}$, the LIM state of Definition 14.2 is King's tube object:*

$$Z^{a,\mathcal{K}}_\infty(\zeta) \;:=\; \lim_{J\to\infty}\lim_{m\to\infty} Z^a_{k_m,J}(\zeta) \;=\; G^K_\infty(\zeta) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H}).$$

In particular the Wick-sector LIM state does not depend on $\mathcal{K}$, and the cube limit of every subsequential fixed-cube limit of the comparison family on the tube is G^K_∞.

Proof. (a) By Lemmas 14.4 and 14.5 the scheme has the cell bound and the scheme bound, so Lemma 14.7(b) applies.

(b) Fix $\mathcal{K} = (k_m)$ and $\mathcal{F}$. By Lemma 14.8 and the Wick sector of Lemma 14.7(b), $Z^{a,\mathcal{K}}_\infty$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ and is holomorphic there; by Theorem 21.3 so is G^K_∞. Take the split family $\mathcal{F}'' = (f; h^+, h^-)$ of Step 8 of the proof of Theorem 21.3, with supports in $Q_{R+1/2}$, the domain $\mathcal{D}''$ and the set $\Omega''_+ = \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ defined there; write $F^K_{k,J}$ and $F^{(a)}_{k,J}$ for the functionals of $\mathcal{F}''$ on King's and on the comparison cube. Let J be so large that $MJ > R + \frac{1}{2} + \ell_c$. At a point of Ω''_+ the curvature coefficients are purely imaginary and the total scalar source is nonnegative, so the observable is of class $\mathcal{I}$, and the support condition (S) holds for X_J; Lemma 4.2 (with its clause for the scheme functionals) gives, at every k,

$$F^K_{k,J} \;\leq\; F^{(a)}_{k,J} \;\leq\; F^K_{k,J+1} \qquad \text{on } \Omega''_+.$$

Along $\mathcal{K}$ the two ends converge, by Theorem 21.3 for the family $\mathcal{F}''$ (whose fixed-cube limits exist along the full sequence, hence along $\mathcal{K}$), to the fixed-cube limits F^K_J, F^K_{J+1} of Step 8 of that proof, and the middle term converges to the fixed-cube limit $F^{(a),\mathcal{K}}_J$ of the comparison family along $\mathcal{K}$ (Definition 14.1 for $\mathcal{F}''$, whose tube contains $\mathcal{D}''$ and whose threshold is $MJ > R(\mathcal{F}'') + \ell_c$, $R(\mathcal{F}'') \leq R + \frac{1}{2}$). Letting $J \to \infty$, the two ends tend to F^K_∞ (Steps 8–9 of the proof of Theorem 21.3) and the middle term to the LIM state $F^{(a),\mathcal{K}}_\infty$ of $\mathcal{F}''$ (Lemma 14.8). Hence $F^{(a),\mathcal{K}}_\infty = F^K_\infty$ on Ω''_+, a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$, and both functions are holomorphic on the connected open set $\mathcal{D}''$; by the identity theorem (the fact (T2) stated before Theorem 13.10) they agree on $\mathcal{D}''$. At $w^+_j = w_j$, $w^-_j = -w_j$ the lattice functionals of $\mathcal{F}''$ coincide exactly with those of $\mathcal{F}$ at ζ, on both conventions and at every k (equal total sources); so $Z^{a,\mathcal{K}}_\infty = G^K_\infty$ on $\mathcal{D}' = \{\zeta : |\operatorname{Re} w_j| < \delta_j\} \subset \mathcal{D}_{\sigma_+,H}$. Both are holomorphic on the connected $\mathcal{D}_{\sigma_+,H}$ and agree on the open set $\mathcal{D}'$, hence on $\mathcal{D}_{\sigma_+,H}$. The last clause is Definition 14.2: every subsequential fixed-cube limit on the tube is the fixed-cube limit along some continuum subsequence. □

Remark 21.6 (the three amplitude constants). Three amplitudes occur in the Wick sector, all built from s_1: the matrix form of OS2 is proved at amplitude $\mathsf{a}(\zeta_i) < \sigma_W = \sigma_+/8$ on the Wick tube (Theorem 10.7); the tube $\mathcal{D}_{\sigma_+,H}$ has half-widths $\sigma_+/2$ for the positive and $H/2 = \sigma_+/32$ for the negative part of the real Wick source, and the form of OS2 proved for LIM states (Theorem 14.9) takes $\mathsf{a}(\zeta_i) < H/2 < \sigma_W$, so that every Gram point lies in $\mathcal{D}_{\sigma_+,H}$, where those limits exist; and $\sigma_0 = H/4$ is the radius of the polydisc in Cauchy's estimates for E0″. The consumer of OS2 in clauses (v),(vi), the reflection positivity of exponentials $e^{\mathrm{i}\Phi(u)}$ with real u that gives hypothesis (H3) of the ergodic decomposition and of the clustering theorems in the full version (Remark N.13 there), needs purely imaginary Wick coefficients only, that is amplitude 0 (Remark 10.11).

Remark 21.7 (the photon mass and the couplings). The photon mass $\mu_0 > 0$ is fixed throughout Theorem 21.1. No relation $\mu_0 \le m$ is used or needed: the lattice reflection positivity of King's tori holds for every $\mu_0 > 0$, its proof using no bound on μ_0 (Proposition 12.2, Lemma 22.1(d)). Reflection positivity of the cube limit needs the margin $R(\mu_0) = \pi\sqrt{3}/(2\mu_0)$ of (20) inside the upper half of the sandwich (Lemma 9.10) and nowhere in the conclusion. The constants of Leg C are absolute; those of Leg W depend on μ_0 and are not uniform in it. The couplings range over all of $e > 0$, $\lambda, m, \mu_0 > 0$: their quantifier is the sources' (Lemma 22.1(e), Remark 22.15), the stopping scale of the lattice constructions behind Theorems 3.3 and 3.6 and Proposition 3.9 being chosen after the point and the datum, with the mass cap of [2, Prop. 3.1] one more one-sided cap on that scale (Remark 22.2); so the theorem contains no sentence "e, λ sufficiently small", and the only coupling condition of the Wick sector is the one-sided sign clause (T3) (Lemma 22.1(b)), proved on the explicit regions of Proposition 22.14.

22. The coupling range

This section collects what the sources behind the inputs of Theorem 21.1 say about the couplings and shows that the only coupling condition anywhere in this paper is the one-sided sign clause (T3) of the Wick leg, which Proposition 22.14 proves on explicit regions.

King's range, units, two letters J. Let $e > 0$ and $\lambda, m, \mu_0 > 0$. Every constant of the cited constructions, including the number J_{K} of renormalisation steps after which the running couplings are small and the stopping scale $5^{-J_{\mathrm{K}}}$ (a member of the family $\{5^{-k}\}$), may depend on (e, λ, m, μ_0), on the large-block integer M and on the cube or torus; no numerical restriction is placed on the couplings themselves. That a stopping scale exists at every positive point is part of the cited constructions ([15, Thm. 3.1], [1, 2, 3, 4], and the existence and rotation parts of this paper), under the mass cap of [2, Prop. 3.1], a one-sided cap on the stopping scale (Remark 22.2). This is what "King's range" means here: all positive (e, λ, m, μ_0), with a stopping scale chosen after them. The cube index J of C_J is ours; King's stopping index, the "fixed integer J" of [15, Thm. 3.1], is written J_{K}. Couplings are measured in the units in which the coarsest spacing is 1; in $d = 3$ all four of e^2, λ, m, μ_0 have the dimension of a mass, rescaling to the unit lattice at spacing ε replacing m^2, λ, e by $m^2\varepsilon^2, \lambda\varepsilon, e\varepsilon^{1/2}$ [15, Sect. 0] and μ_0^2 like m^2.

The relative bare coefficient. Let $\alpha_k^\infty(M) = M^2 + \delta m_k^2(M)$, $s_k^\infty(M_1) = \frac{1}{2}(\alpha_k^\infty(m) - \alpha_k^\infty(M_1))$ and $\sigma_1(M_1) = \liminf_k s_k^\infty(M_1)$ be as in Lemma 3.10; the theorem is stated through σ_1 alone. Describing *where* (T3) fails uses one further statement about the continuum limit of this coefficient, which is of the same kind as Lemma 3.10 (Remark 3.11) and which we name rather than use tacitly:

(CT_∞) *For every $M > 0$ the limit*

$$\alpha_\infty(M) - \alpha_\infty(m) := \lim_{k\to\infty} \big[\alpha_k^\infty(M) - \alpha_k^\infty(m)\big]$$

exists and is a polynomial of degree ≤ 2 in (e^2, λ) with finite coefficients, differentiable in M^2 at $M = m$.

The divergent parts of the counterterm ($1/\varepsilon$ at one loop, $\ln(1/\varepsilon)$ at two loops) do not depend on the scalar mass and cancel in the difference. (CT_∞) is proved in Theorem F.168(b) of the full version; it is not a hypothesis of Theorem 21.1 and enters only the description of the failure set. Under (CT_∞) put

$$s_\infty(M_1) := \tfrac{1}{2}\big(\alpha_\infty(m) - \alpha_\infty(M_1)\big), \qquad s_* := \sup_{M>0} s_\infty(M) \in [0, \infty],$$

(58)

$$E_W := \{s_* = 0\} = \{m \text{ is a global minimiser of } M \mapsto \alpha_\infty(M) \text{ on } (0,\infty)\}, \qquad E_W^< := \{m \text{ minimises } \alpha_\infty \text{ on } (0,$$

subsets of the coupling space ($s_* \ge 0$ because $M = m$ gives 0; the value $+\infty$ is harmless below). The two sign clauses are

(NE) m is not a global minimiser of α_∞ on $(0,\infty)$, i.e. $s_* > 0$, i.e. $(e, \lambda, m, \mu_0) \notin E_W$;

($\mathrm{NE}^<$) some $M_1 \in (0, m)$ has $\alpha_\infty(M_1) < \alpha_\infty(m)$, i.e. $(e, \lambda, m, \mu_0) \notin E_W^<$ (the one-sided form; it implies (NE)).

Throughout, $'$ denotes d/dM^2, N is the number of real components of the scalar field, and "one-loop order" means the orders e^2 and λ.

Lemma 22.1 (the coupling range)**.** *Let $e > 0$ and $\lambda, m, \mu_0 > 0$.*

(a) The curvature sector. *The existence theorem of the curvature sector, Theorem 3.3 (the existence clause of* [16, Thm. 2.1] *at zero scalar source), carries no condition on e, λ, m, μ_0. The correlation inequalities it uses, Lemma 2.8 (Theorem 6.2(1),(2) of* [7]*), have the standing hypotheses of a finite single-site measure, positive hopping coefficients and a positive photon covariance, which for King's action are $\lambda > 0$ and $\mu_0 > 0$; the margin $R(\mu_0) = \pi\sqrt{3}/(2\mu_0)$ of Lemma 9.10 needs $\mu_0 > 0$ only; and no lemma proved in this paper for the curvature sector carries a condition on the couplings beyond those of its inputs. Hence the curvature sector of Theorem 21.1 holds on all of King's range, with no exceptional set. The one proof-internal proviso in the Bałaban antecedents of Theorem 3.3, the mass clause of* [2, Prop. 3.1]*, is a cap on the dimensionless scalar mass at the stopping scales and no condition on the bare couplings (Remark 22.2). Of the bounds of the paper only the absolute bounds of Theorem 7.6 are uniform in μ_0.*

(b) The Wick sector, Form (B). *The inputs of Form (B) beyond Theorem 3.6, Proposition 3.9, and Lemma 3.10 and* (T3) *— Proposition 12.2 and Lemmas 2.8, 12.4, 12.9 and 12.10 and the tube statements Theorems 10.7 and 13.10 and Lemma 13.11, whose conditions on the Wick amplitudes are not conditions on the couplings — print no condition on the bare couplings beyond positivity ($\lambda > 0$, $\mu_0 > 0$); Proposition 3.9 is used at the two masses m and M_1. The only coupling condition of Form (B) is therefore* (T3)*, which Proposition 22.14 proves on explicit regions. Under* (CT_∞)*:*

(i) *$\sigma_1(M_1) = \lim_k s_k^\infty(M_1) = s_\infty(M_1)$ for every $M_1 > 0$. Hence* (T3) *holds if and only if* $(\mathrm{NE}^<)$*: it fails exactly on $E_W^<$. If Proposition 3.9 and Lemma 3.10 are available also at auxiliary masses $M_1 > m$, then by Proposition 12.11 the two-sided clause* (NE) *replaces* (T3)*, with failure set exactly E_W; the branch $M_1 > m$ is needed exactly on $E_W^< \setminus E_W$, for instance when $\alpha'_\infty(m) < 0$ and m minimises α_∞ on $(0, m]$.*

(ii) *To one-loop order, for every $M_1 > 0$,*

$$s_\infty(M_1) = \tfrac{1}{2}(m^2 - M_1^2) + \frac{\lambda(N+2)(m-M_1)}{2\pi} - e^2\,\frac{(m-M_1)\,(m\mu_0 + mM_1 + M_1\mu_0)}{8\pi(\mu_0+M_1)(m+\mu_0)},$$

so that

$$\alpha'_\infty(m) = 1 + \frac{\lambda(N+2)}{2\pi m} - e^2\,\frac{m+2\mu_0}{8\pi(m+\mu_0)^2}, \qquad \frac{m+2\mu_0}{8\pi(m+\mu_0)^2} \le \frac{1}{4\pi(m+\mu_0)}, \tag{59}$$

and to one-loop order (T3) *holds if and only if*

$$e^2 < e_c^2(\lambda, m, \mu_0) := \max\Big\{\frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1 + \frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1 + \frac{\lambda(N+2)}{\pi m}\Big)\Big\}, \tag{60}$$

the first entry being the larger if and only if $\lambda(N+2)\mu_0 \le \pi m^2$. So the one-loop failure set of (T3) *is the half-line $e^2 \ge e_c^2$. The first entry is the zero $e_*^2 \ge 4\pi(m+\mu_0)$ of* (59)*; at $m = \mu_0 = 1$, $\lambda = 0$ it equals $32\pi/3 \approx 33.5$ (for orientation only). Nothing finer than one-loop order is claimed about $E_W^<$.*

(iii) *For the two-sided set, $E_W \subset \{\alpha'_\infty(m) = 0\}$; for each (λ, m, μ_0) at which the polynomial $e^2 \mapsto \alpha'_\infty(m)$, of degree ≤ 2, is not identically zero, this is at most two values of e^2. At $\mu_0 = m$ the box $0 < \lambda, e^2 \le 0.239\,m$ lies outside $E_W^<$, by (i) and (d).*

(c) The existence and rotation parts. *The existence theorems Theorems 3.3 and 3.6 and the rotation theorem Theorem 3.14 are proved for $e > 0$, $\lambda, m, \mu_0 > 0$ in King's range, and none carries a condition on the bare couplings beyond positivity: every smallness condition of* [1, 2, 3, 4] *that they use (listed in the full version) is a one-sided cap on a stopping scale $\beta \le 1$, imposed last in the schedule of constants, and an admissible β exists at every positive point.*

(d) The photon mass. *No relation $\mu_0 \le m$ is needed: Proposition 12.2 holds for every $\mu_0 > 0$. At $\mu_0 = m$,* (T3) *holds with $M_1 = 0.7m$ on the box $0 < \lambda, e^2 \le 0.239\,m$, and $\sigma_1(0.7m) \ge$*

$0.2209\,m^2$ *on* $0 < \lambda, e^2 \le 0.1m$ *(Proposition 22.14(iii)); at every* $\mu_0 \in (0, m]$ *the explicit regions are those of Proposition 22.14(iii),(iv).*

(e) The quantifier. *For all* $e > 0$, $\lambda, m, \mu_0 > 0$ *and each datum there are an integer* J_{K} *(*[15, Thm. 3.1]*) and a stopping scale* $\beta \le 1$ *such that the running couplings* $(e\beta^{1/2}, \lambda\beta, m\beta, \mu_0\beta)$ *satisfy finitely many one-sided caps on* β *and on three of the four of them (*$\mu_0\beta$ *is uncapped in the sources); no condition is placed on the bare couplings, and the theorems consumed from* [15, 16] *carry no coupling hypothesis. A sentence "*e, λ *sufficiently small,* $m, \mu_0 > 0$ *fixed", read as a hypothesis on the bare couplings, is strictly stronger and is printed in none of the sources (Remark 22.15).*

Proof. (a). Theorem 2.1 of [16] asserts, for g, h supported in C_J and all $z, w \in \mathbb{C}$, the existence of $\lim_K \langle \Theta(zg, wh)\rangle^{\varepsilon_K}_{N,D,C_J}$; no hypothesis on the couplings precedes it. Its proof rests on [15, Thm. 3.1] (the convergence of the effective action, referred there to [1, 2, 3, 4]), stated "for some fixed integer J", our J_{K}; neither it nor [15, Thms. 2.1, 3.4] carries a coupling hypothesis, the smallness in [15] being of the fields and of the stopping spacing. So the one proviso of [2, Prop. 3.1] is upstream of Theorem 3.3. For King's action the single-site measure $e^{-\varepsilon^3[\frac{1}{2}\alpha r^2 + \lambda r^4]} r\,dr$ is finite iff $\lambda > 0$ (or $\lambda = 0$, $\alpha > 0$), the hopping weights are positive, and the Feynman-form photon covariance is positive iff $\mu_0 > 0$; the "$g > 0$" of Theorem 6.2 of [7] is the sign of the scalar source, and Lemma 2.8 covers $g = 0$. The margin of Lemma 9.10 concerns the photon Gaussians of the two boxes alone, so μ_0 is its only parameter (no fixed margin works at $\mu_0 = 0$). The lemmas of this paper see only the form of the action.

The torus inputs. Proposition 3.9 is Bałaban's stability theorem [1, (1.14)], [2, (1.3)], in the form of Theorem F.1 of the full version, with constants independent of ε and of the torus, under $\lambda > 0$, $\mu_0^2 > 0$, $m^2 > 0$. Every printed smallness in [1, 2, 3, 4] is of one of three kinds. *Running charge:* the bounds hold "for $e(L^k\varepsilon)$ sufficiently small", $e(L^k\varepsilon) = e\,(L^k\varepsilon)^{(4-d)/2}$ [1, (2.23)]; [4] proves them for a unit-lattice model whose charge is this running charge (its theorem is stated there as Proposition 2.1 of [1]), so "e sufficiently small" there is a cap on $e\beta^{1/2}$ at the stopping scale β. *Scale:* the stopping scale ε_0 with $L^K\varepsilon \le \varepsilon_0 < L^{K+1}\varepsilon$ is "some fixed sufficiently small number" [3], and "$L^{k+1}\varepsilon \le 1$", "$L^k\varepsilon$ sufficiently small" [1, 2] are caps on it; Bałaban's units have $\lambda = 1$ (in $d = 3$: spacing $\lambda\varepsilon$, mass squared m^2/λ^2, charge $e\lambda^{-1/2}$, the running charge invariant), so his "ε small" is "$\lambda\varepsilon$ small" in ours, and his constants may depend on $e\lambda^{-1/2}$, m/λ, μ_0/λ: a dependence, not a condition. *Mass:* Proposition 3.1 of [2] carries the sentence "we assume $m^2 \le O(1)$ also"; this is the mass cap of that proposition (Remark 22.2). Lemma 3.10 carries no coupling condition: the bound on the relative bare coefficient holds for every mass in a compact interval containing m, on either side of m, with a constant depending on the couplings (Remark 3.11; the power-counting theorem of [22] that enters it has none). Lemma 12.10 needs a finite single-site integral ($\lambda > 0$) and a finite photon Gaussian on the torus ($\mu_0 > 0$; at $\mu_0 = 0$ the constant mode lies in the kernel); Lemma 12.9 needs massive propagators; Proposition 12.2 needs finite single-site moments and a positive Proca covariance and does not see the counterterm. The tube statements restrict Wick amplitudes only. So the only coupling condition of Form (B) is the sign clause of Proposition 12.6, i.e. (T3).

(b)(i). By Lemma 12.9, $s_k^\infty(M_1)$ is the fixed-k volume limit of the torus coefficient, the relative bare coefficient on Λ_k in the free convention; by (CT_∞) its limit $s_\infty(M_1)$ exists, hence $\sigma_1(M_1) = \liminf_k = \lim_k = s_\infty(M_1)$. Thus some $M_1 \in (0, m)$ has $\sigma_1(M_1) > 0$ iff $(\mathrm{NE}^<)$, and some $M_1 > 0$ has $\sigma_1(M_1) > 0$ iff (NE). Proposition 12.11 gives (I2) from Proposition 3.9 and Lemma 3.10 at an auxiliary mass on either side of m, so with those inputs (NE) suffices; on $E_W^< \setminus E_W$ every $M_1 < m$ fails and some $M_1 > m$ works.

(b)(ii). The closed form is the one-loop evaluation of King's counterterm [15, (3.32)–(3.33)] (tadpole and zero-momentum photon-exchange graphs; the $1/\varepsilon$ divergence cancels in the difference). It equals $\frac{1}{2}\int_{M_1^2}^{m^2} \alpha'_\infty(\mu)\,d\mu^2$ with $\alpha'_\infty(\mu) = 1 + \frac{\lambda(N+2)}{2\pi\mu} - e^2\frac{\mu+2\mu_0}{8\pi(\mu+\mu_0)^2}$, an orientation-free definite integral, valid on either side of m; at $\mu = m$ this is (59), whose inequality is $m \ge 0$. For

$M_1 \in (0,m)$ write $s_\infty(M_1) = \frac{m-M_1}{2}\beta(M_1)$ with

$$\beta(M_1) = (m+M_1) + c - e^2\,\frac{m\mu_0 + (m+\mu_0)M_1}{4\pi(m+\mu_0)(\mu_0+M_1)}, \qquad c := \frac{\lambda(N+2)}{\pi};$$

$s_\infty(M_1) > 0$ iff $\beta(M_1) > 0$ iff $e^2 < e_c^2(M_1) := \big[(m+M_1)+c\big]\frac{4\pi(m+\mu_0)(\mu_0+M_1)}{m\mu_0+(m+\mu_0)M_1}$. With $\mathcal{N}(M_1) := (m+M_1+c)(\mu_0+M_1)$ and $\mathcal{D}(M_1) := m\mu_0+(m+\mu_0)M_1$, the derivative of $e_c^2(M_1)$ has the sign of $\mathcal{N}'\mathcal{D}-\mathcal{N}\mathcal{D}' = 2m\mu_0M_1+(m+\mu_0)M_1^2 - c\mu_0^2$, positive on $(0,m)$ when $c=0$ and, when $c>0$, negative then positive with one zero. So $e_c^2(\cdot)$ is increasing, or decreasing then increasing, on $(0,m)$, and its supremum over the open interval is the larger of the two end values, not attained: $M_1 \to m^-$ gives $\frac{8\pi(m+\mu_0)^2}{m+2\mu_0}(1+\frac{c}{2m})$, the zero e_*^2 of (59), and $M_1 \to 0^+$ gives $4\pi(m+\mu_0)(1+\frac{c}{m})$. Hence to one-loop order (T3) holds iff $e^2 < e_c^2(\lambda,m,\mu_0)$ as in (60), and fails for every $e^2 \geq e_c^2$ (then $\beta \leq 0$ for every M_1); the first entry is the larger iff $(m+\mu_0)(2m+c) \geq (m+c)(m+2\mu_0)$, i.e. $c\mu_0 \leq m^2$.

(b)(iii). Under (CT_∞), $M^2 \mapsto \alpha_\infty(M)$ is differentiable at m, and a global minimum at the interior point m is a critical point: $E_W \subset \{\alpha_\infty'(m) = 0\}$. A polynomial of degree ≤ 2 in e^2 that is not identically zero has at most two zeros. For the clause at $\mu_0 = m$: on that box (T3) holds by (d), and under (CT_∞) (T3) is ($\mathrm{NE}^<$) by (i), so the box lies outside $E_W^<$.

(c). The list of the smallness conditions, the order in which the constants of the existence and rotation parts are fixed, and the proof that an admissible stopping scale exists at every positive point are given in the full version (Lemma K.15 of the full version, Proposition K.20 of the full version).

(d). The proof of Proposition 12.2 uses site-independent coefficients, $\lambda > 0$, $\mu_0 > 0$ and even sides, and no bound on μ_0. The statements at $\mu_0 = m$ and for $\mu_0 \in (0,m]$ are Proposition 22.14(iii),(iv), whose proof uses (b)(i) of this lemma and neither (b)(iii) nor (d).

(e). The quantifier is read off [15, Thm. 3.1] ("for some fixed integer J, for all $K \geq J$, and all $k \leq K - J$") and the stopping-scale caps listed above; the last sentence is Remark 22.15. □

Remark 22.2 (the mass clause of Bałaban's upper bound). Proposition 3.1 of [2] asserts an inequality (3.26), summed over the scales $k = 0,\dots,K$, with a constant $\gamma_0 = \gamma_0(d,a)$ and mass terms $(L^k\varepsilon)^d m^2|\phi_k|^2$; rescaled to the unit lattice, its k-th term is the per-scale inequality (3.29) with the mass term $(L^k\varepsilon)^2m^2|\phi_k'|^2$, so $(L^k\varepsilon)^2m^2$ is the dimensionless mass at scale k. The proposition carries the sentence "we assume $m^2 \leq O(1)$ also", m^2 being the bare scalar mass squared in Bałaban's units ($\lambda = 1$), i.e. m^2/λ^2 in ours. The clause under which this paper cites the proposition concerns the scales at which it is applied:

> *Inequality (3.29) of* [2] *holds, with $\gamma_0 = \gamma_0(d,a)$, at every scale $k \leq K$ at which $(L^k\varepsilon)^2m^2 \leq c_m(d,a)$; hence (3.26) holds whenever this is so for every $k \leq K$.*

Here $c_m(d,a)$ is a constant depending on d and a only. The clause belongs to the statement of the proposition: the mass that enters its printed proof, referred to [4, Sect. 4], is the unit-lattice mass at scale k, the text of [2] makes no further use of the sentence, and the proposition is cited here under the clause. It is then one more one-sided cap on the stopping scale $\beta = \varepsilon_0$ ($(L^k\varepsilon)^2m^2 \leq \beta^2m^2$ at every scale used), of the kind every other smallness condition of [1, 2, 3, 4] has; a further monotone cap on β leaves the existence of an admissible stopping scale unchanged, and nothing is imposed on the bare mass.

Read instead as an absolute bound on m^2/λ^2, the sentence would exclude small λ at fixed m, including every $\lambda < 0.1m$ ($m^2/\lambda^2 > 100$), which meets the explicit regions of Proposition 22.14, and would be an unprinted hypothesis of the theorem (1.3) of [2], whose only printed condition on m is $m^2 > 0$; that is not the reading under which the proposition is proved, and it is used nowhere. The clause sits upstream of Proposition 3.9 at both masses, of Theorems 3.3 and 3.6 (through [15, Thm. 3.1]) and of the steps of the rotation part that rest on [1, 2], as every cap on the stopping scale does; it is not a condition of this paper.

The explicit coupling region. The sign clause (T3) is proved below on an explicit region of the (λ,e^2)-plane at every photon mass $\mu_0 \in (0,m]$, with $N = 2$ and the auxiliary mass $M_1 = 0.7m$.

The argument has two parts: the three low orders of King's counterterm, whose contributions to $s_k^\infty(M_1)$ are lattice integrals with exact limits (Lemmas 22.3 and 22.4 and Lemmas 22.6 of the full version to 22.8 of the full version), and the two orders $e^2\lambda$ and e^4, whose contributions are bounded from below, uniformly in k, through the two-loop majorants of King's counterterm on periodic tori (the full version), read at the photon mass μ_0, with the torus floor removed (Lemmas 22.10 and 22.11 of the full version), made uniform on intervals of photon masses (Lemma 22.12 of the full version) and, in a weaker closed form, uniform in the photon mass (Lemma 22.13 of the full version). The resulting lower bound σ_{low} on $\sigma_1(0.7m)$, (66), is explicit through Table 1.

King's counterterm order by order. With $\varepsilon = \varepsilon_k$, $\Lambda = \Lambda_k$, the Brillouin zone $\mathsf{B} = [-\pi/\varepsilon, \pi/\varepsilon)^3$, $\int_{\mathsf{B}} := \int_{\mathsf{B}} \frac{d^3p}{(2\pi)^3}$, $\hat{p}_\mu := \frac{2}{\varepsilon}\sin\frac{\varepsilon p_\mu}{2}$, the free propagators $G_M(p) = (\hat{p}^2 + M^2)^{-1}$, $\hat{D}(p) = (\hat{p}^2+\mu_0^2)^{-1}$ of King's action on Λ and their position-space kernels C_M, D ($C_m(0) = \mathsf{W}_\varepsilon$), King's free-boundary counterterm [15, (3.32)–(3.33)], [16, p. 330] at scalar mass M is the polynomial

$$\delta m^2_\varepsilon(M) = \sum_{(\alpha,\beta)} e^\alpha \lambda^\beta \, \delta m^2_{(\alpha\beta)}(\varepsilon; M), \qquad (\alpha,\beta) \in \{(0,1),(2,0),(0,2),(2,1),(4,0)\}, \tag{61}$$

over the superficially divergent orders $2 \le \alpha + 2\beta \le 4$, whose coefficients are fixed recursively by the condition that the order-$e^\alpha\lambda^\beta$ part of the amputated one-particle-irreducible two-point function of ϕ at zero external momentum, with the counterterm vertices of lower order inserted, be cancelled by the counterterm vertex of that order: $\delta m^2_{(\alpha\beta)}(\varepsilon; M) = \Sigma^{(\alpha\beta)}(0; M)$ in the convention in which the inverse propagator per component is $\hat{p}^2 + M^2 - \Sigma(p)$. Each coefficient is a finite real number, smooth in M.

Lemma 22.3 (structure)**.** *Put $S_{\alpha\beta,k} := \frac{1}{2}\big[\delta m^2_{(\alpha\beta)}(\varepsilon_k; m) - \delta m^2_{(\alpha\beta)}(\varepsilon_k; M_1)\big]$, numbers free of the couplings. Then*

$$s_k^\infty(M_1) = \tfrac{1}{2}(m^2 - M_1^2) + \lambda S_{01,k} + e^2 S_{20,k} + \lambda^2 S_{02,k} + e^2\lambda S_{21,k} + e^4 S_{40,k}. \tag{62}$$

In $d = 3$, $S_{01,k}$ and $S_{20,k}$ have the dimension of a mass and $S_{02,k}$, $S_{21,k}$, $S_{40,k}$ are dimensionless.

Lemma 22.4 (the three low orders, exact)**.** *For every ε and $M > 0$:*

(i) $\delta m^2_{(01)}(\varepsilon; M) = -4(N+2)\, C_M(0)$;

(ii) $\delta m^2_{(20)}(\varepsilon; M) = \int_{\mathsf{B}} \hat{k}^2 \hat{D}(k) G_M(k)\, dk - 3D(0) = -2D(0) - M^2 \int_{\mathsf{B}} \hat{D} G_M$;

(iii) $\delta m^2_{(02)}(\varepsilon; M) = 32(N+2)\sum_{x\in\Lambda} \varepsilon^3 C_M(x)^3$.

Consequently, with $J_\varepsilon := \int_{\mathsf{B}} G_m G_{M_1}$, $I_\varepsilon := \int_{\mathsf{B}} \hat{k}^2 \hat{D}(k) G_m(k) G_{M_1}(k)\, dk$ and $F_\varepsilon := \sum_{x\in\Lambda} \varepsilon^3 [C_{M_1}(x)^3 - C_m(x)^3]$,

$$\begin{aligned} S_{01,k} &= 2(N+2)\big(C_{M_1}(0) - C_m(0)\big) = 2(N+2)(m^2 - M_1^2) J_{\varepsilon_k}, \\ S_{20,k} &= -\tfrac{1}{2}(m^2 - M_1^2) I_{\varepsilon_k}, \qquad S_{02,k} = -16(N+2) F_{\varepsilon_k}. \end{aligned} \tag{63}$$

Idea of the proof. Lemma 22.3 is (61) and the definition of s_k^∞. For Lemma 22.4, expand the hopping term of King's action into the free action and the vertices: (i) is the λ-tadpole with the $O(N)$ count $2(N+2)$; (ii) is the seagull $-3e^2D(0)$ plus the photon-exchange graph; (iii) is the sunset, the tadpole-insertion graph being cancelled exactly by the order-λ counterterm on the loop line; (63) follows with the resolvent identity. *Signs, k-uniform bounds and limits* (Lemmas 22.5 to 22.8 of the full version): from $\frac{4}{\pi^2}p^2 \le \hat{p}^2 \le p^2$ on B, the heat-kernel representation of C_M with a positive path expansion, the continuum integrals and dominated convergence, one gets for every k: $S_{01,k} > 0$ with $S_{01,k} \to \frac{(N+2)(m-M_1)}{2\pi}$; $S_{20,k} < 0$ with a μ_0- and k-free bound and $S_{20,k} \to -\frac{1}{2}(m^2 - M_1^2) I_\infty(\mu_0)$, $I_\infty(\mu_0) = \frac{m\mu_0 + mM_1 + M_1\mu_0}{4\pi(m+M_1)(\mu_0+M_1)(m+\mu_0)}$ strictly decreasing in μ_0 with $I_\infty \uparrow I^* = \frac{1}{4\pi(m+M_1)}$ as $\mu_0 \downarrow 0$; and $S_{02,k} < 0$ with a k-free bound and $S_{02,k} \to -\frac{(N+2)\ln(m/M_1)}{\pi^2}$ (a Plancherel argument and Frullani's integral), while each $\delta m^2_{(02)}(\varepsilon; M)$ alone diverges like $\ln(1/\varepsilon)$. Inserted into (62) these limits are the one-loop closed form of Lemma 22.1(b)(ii) and, at $e = 0$, the exact limit of Proposition 22.14(vi). At $N = 2$ the three coefficients are the

terms $-16\lambda C_T(M)$, $-2e^2D_T - e^2J_T(M)$, $128\lambda^2 S_T(M)$ of the exact two-loop form of the torus coefficient (Lemma J.4 of the full version), read on Λ; the orders $e^2\lambda$ and e^4 are the terms $16e^2\lambda F_T(M)$ and $e^4Q_T(M)$ there.

The two-loop orders at every photon mass. From here to the end of the section we use the units $m = 1$, $N = 2$ and $M_1 = 0.7$. The two-loop coefficients $S_{21,k}$, $S_{40,k}$ are bounded from below through the two-loop majorants of the full version, written on periodic tori of side $\ell_T \geq \ell_0$ with the photon mass $\mu_* \in (0,1]$ of the photon propagator as a parameter, read here at $\mu_* = \mu_0$ (Lemmas 22.10 to 22.13 of the full version): the torus coefficient $a_{K,T}(M)$, King's recursion (61) on the torus, which at $\mu_* = \mu_0$ is the bare coefficient of King's torus convention (Definition 12.1(ii)); the one-loop majorants $\mathcal{I}_{\alpha,\beta}(a,b;\ell)$, minima of finitely many closed forms in Γ and B functions, decreasing in ℓ and non-increasing in the masses; the slope bounds; the two-loop gap, order by order; and the cell rule, an upper sum over a partition of $[0.7,1]$ into $n = 400$ cells (Definition J.2, Lemmas J.5, J.7 and J.15, and Proposition J.6 of the full version). On King's tori T_J with $2MJ \geq \ell_0$ the gap gives $2S^T_{20,k} \geq -\bar{j}_n$, $2S^T_{02,k} \geq -\bar{s}_n$, $2S^T_{21,k} \geq -\bar{B}_{3,n}$, $2S^T_{40,k} \geq -\bar{B}^-_{4,n}$ uniformly in J, and the volume limit $J \to \infty$ at fixed k (Lemma 12.9, order by order) carries closed inequalities to $S_{\alpha\beta,k}$; the torus floor ℓ_0 is a free parameter and $\ell_0 \to \infty$ removes it; every majorant is non-increasing in the photon mass, so an *interval upper sum* evaluated at $\mu_{\rm lo}$ bounds the coefficients for all $\mu_0 \in [\mu_{\rm lo}, \mu_{\rm hi}]$ at once; and replacing every μ-dependent form by a μ-free one gives the closed form $2S_{21,k} \geq -B_3^*$, $2S_{40,k} \geq -(P^* + Q^*/\mu_0)$ with $B_3^* = 13.187549$, $P^* = 4.7629000$, $Q^* = 0.77120851$.

The numbers. Let $\hat{\mu} = \mu_0/m \in (0,1]$. The grid is $\{1, 0.99, 0.98, 0.95, 0.9, 0.85, \ldots, 0.15, 0.1, 0.08, 0.06, 0.05, 0.04$ (35 points), and the *rows* of Table 1 are the 34 intervals $[\mu_{\rm lo}, \mu_{\rm hi}]$ between consecutive grid points. For $\hat{\mu} \in [10^{-4}, 1]$ the *row of* $\hat{\mu}$ is the row containing $\hat{\mu}$ (at a grid point, where two rows contain $\hat{\mu}$, the row with $\mu_{\rm lo} = \hat{\mu}$), and $\bar{B}_3(\hat{\mu})$, $\bar{B}^-_4(\hat{\mu})$, $\bar{j}(\hat{\mu})$ denote the interval upper sums $\bar{B}^{\rm int}_{3,400}$, $\bar{B}^{-,\rm int}_{4,400}$, $\bar{j}^{\rm int}_{400}$ of that row (Lemma 22.12 of the full version), and $\bar{s} := \bar{s}_{400} = 6.5231964$; for $\hat{\mu} < 10^{-4}$, below the table, we read $\bar{B}_3(\hat{\mu}) := B_3^*$, $\bar{B}^-_4(\hat{\mu}) := P^* + Q^*/\hat{\mu}$ and $\bar{j}(\hat{\mu}) := 0.092610085$ (Lemma 22.13 of the full version for the first two; the last is the μ-free branch $\int 2M\bar{\tau}$ of $\bar{j}$, see the proof of Proposition 22.14(i)). With $A := \frac{1}{2}(1 - M_1^2) = 0.255$, $b := \frac{4(1-M_1)}{2\pi} = 0.1909859$, $L := \frac{4\ln(1/M_1)}{\pi^2} = 0.1445549$ (Lemmas 22.6 to 22.8 of the full version at $N = 2$, $M_1 = 0.7$), $I_\infty(\hat{\mu}) = \frac{0.7+1.7\hat{\mu}}{4\pi\cdot 1.7\,(\hat{\mu}+0.7)(1+\hat{\mu})}$ (so $I_\infty(1) = 0.0330425$) and $I^* = \frac{1}{4\pi\cdot 1.7} = 0.0468103$, put

$$\sigma_{\rm low}(\hat{\lambda}, \hat{e}^2; \hat{\mu}) := A\big(1 - \hat{e}^2 I_\infty(\mu_{\rm lo})\big) + b\hat{\lambda} - L\hat{\lambda}^2 - \tfrac{1}{2}\bar{B}_3(\hat{\mu})\,\hat{e}^2\hat{\lambda} - \tfrac{1}{2}\bar{B}^-_4(\hat{\mu})\,\hat{e}^4 \qquad (\hat{\mu} \in [10^{-4}, 1]), \tag{66}$$

$$\sigma^*(\hat{\lambda}, \hat{e}^2; \hat{\mu}) := A\big(1 - \hat{e}^2 I^*\big) + b\hat{\lambda} - L\hat{\lambda}^2 - \tfrac{1}{2}B_3^*\,\hat{e}^2\hat{\lambda} - \tfrac{1}{2}\big(P^* + Q^*/\hat{\mu}\big)\hat{e}^4 \qquad (\hat{\mu} \in (0,1]), \tag{67}$$

and $\sigma_{\rm low}(\cdot;\hat{\mu}) := \sigma^*(\cdot;\hat{\mu})$ for $\hat{\mu} < 10^{-4}$. The derived columns of the table are: $e^2_{\rm max}$, the positive root of $\sigma_{\rm low}(0,\cdot;\hat{\mu})$, $e^2_{\rm max} = \big(-AI_\infty + \sqrt{A^2I_\infty^2 + 2A\bar{B}^-_4}\big)/\bar{B}^-_4$; the side c of the largest box $[0,c]^2$ on which $\sigma_{\rm low} \geq 0$; and the value $\sigma_{\rm low}(0.1, 0.1; \hat{\mu})$, where a negative entry means only that the bound does not place $[0, 0.1]^2$ in the region.

How the entries were computed, and what is exact. The upper sums are the finite sums (65) of the full version over the 400 cells of the formulas (G2), (G3), with the interval rule of Lemma 22.12 of the full version; their validity as upper bounds is the inequality (65) of the full version, which is exact once the finitely many closed forms are evaluated correctly. At $\ell_0 = \infty$ only the $c_N C_3$ term of (64) of the full version and $\bar{\omega} = w \leq 0.25273101$ enter, so no series and no special function beyond Γ and B is involved. The evaluation was carried out in floating-point arithmetic with 50 significant digits; every final upper sum was multiplied by $1 + 10^{-30}$ and rounded upward to eight significant digits, and the derived region numbers were computed from the rounded upper sums with coefficients rounded up and sizes rounded down. The rounding of the elementary and Gamma functions at 50 digits is many orders below the safety factor and the final rounding; interval validity rests on Lemma 22.12 of the full version and finitely many evaluations, not on interval arithmetic. The same evaluation at $\hat{\mu} = 1$ and $\ell_0 = 96$, with $\bar{\omega} = w + \min\{\bar{E}, \tilde{E}\}$ as in (G2), gives 0.0886292849, 7.17906354, 9.3304261, 1.70378338, which

$[\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$	$\bar{j}$	$\bar{B}_3$	$\bar{B}_4^-$	$AI_\infty(\mu_{\mathrm{lo}})$	$e^2_{\max}(\lambda=0)$	c	$\sigma_{\mathrm{low}}(0.1,0.1)$
$[0.99, 1]$	0.085313936	8.5786210	1.5854190	0.008458	0.5618	0.2390	0.2209
$[0.98, 0.99]$	0.085757165	8.6036471	1.6023646	0.008491	0.5588	0.2385	0.2207
$[0.95, 0.98]$	0.087020288	8.6804035	1.6554211	0.008589	0.5498	0.2369	0.2201
$[0.9, 0.95]$	0.088887935	8.8139269	1.7519811	0.008757	0.5345	0.2341	0.2189
$[0.85, 0.9]$	0.090427040	8.9551127	1.8603008	0.008930	0.5188	0.2313	0.2176
$[0.8, 0.85]$	0.091595352	9.1048479	1.9826438	0.009108	0.5026	0.2283	0.2163
$[0.75, 0.8]$	0.092342924	9.2641791	2.1218953	0.009291	0.4858	0.2251	0.2147
$[0.7, 0.75]$	0.092610085	9.4343534	2.2812536	0.009480	0.4686	0.2217	0.2131
$[0.65, 0.7]$	0.092610085	9.6168755	2.4625429	0.009673	0.4511	0.2182	0.2112
$[0.6, 0.65]$	0.092610085	9.8135862	2.6698136	0.009871	0.4333	0.2145	0.2092
$[0.55, 0.6]$	0.092610085	10.026776	2.9084862	0.010073	0.4152	0.2105	0.2069
$[0.5, 0.55]$	0.092610085	10.259350	3.1856310	0.010279	0.3969	0.2063	0.2044
$[0.45, 0.5]$	0.092610085	10.515087	3.5155645	0.010488	0.3779	0.2017	0.2014
$[0.4, 0.45]$	0.092610085	10.799045	3.9183359	0.010697	0.3580	0.1967	0.1979
$[0.35, 0.4]$	0.092610085	11.118258	4.4224644	0.010906	0.3371	0.1912	0.1938
$[0.3, 0.35]$	0.092610085	11.477171	5.0733664	0.011111	0.3148	0.1850	0.1887
$[0.25, 0.3]$	0.092610085	11.851059	5.9437544	0.011309	0.2910	0.1781	0.1825
$[0.2, 0.25]$	0.092610085	12.193509	7.0632067	0.011495	0.2670	0.1709	0.1752
$[0.15, 0.2]$	0.092610085	12.433452	8.5440078	0.011662	0.2429	0.1635	0.1665
$[0.1, 0.15]$	0.092610085	12.659419	11.303445	0.011801	0.2113	0.1525	0.1516
$[0.08, 0.1]$	0.092610085	12.756284	13.316343	0.011846	0.1948	0.1460	0.1411
$[0.06, 0.08]$	0.092610085	12.857230	16.620716	0.011884	0.1744	0.1370	0.1240
$[0.05, 0.06]$	0.092610085	12.909321	19.239396	0.011899	0.1621	0.1310	0.1107
$[0.04, 0.05]$	0.092610085	12.962542	23.145186	0.011912	0.1479	0.1233	0.0909
$[0.03, 0.04]$	0.092610085	13.016932	29.623519	0.011923	0.1308	0.1132	0.0582
$[0.02, 0.03]$	0.092610085	13.072534	42.530531	0.011931	0.1092	0.0987	−0.0066
$[0.015, 0.02]$	0.092610085	13.100802	55.411533	0.011933	0.0957	0.0887	−0.0712
$[0.01, 0.015]$	0.092610085	13.129390	81.146532	0.011935	0.0791	0.0753	−0.2000
$[0.005, 0.01]$	0.092610085	13.158304	158.29597	0.011937	0.0566	0.0555	−0.5859
$[0.002, 0.005]$	0.092610085	13.175811	389.67594	0.011937	0.0361	0.0360	−1.7428
$[0.001, 0.002]$	0.092610085	13.181673	775.28604	0.011937	0.0256	0.0256	−3.6709
$[0.0005, 0.001]$	0.092610085	13.184609	1546.4975	0.011937	0.0181	0.0181	−7.5270
$[0.0002, 0.0005]$	0.092610085	13.186373	3860.1248	0.011937	0.0114	0.0114	−19.096
$[0.0001, 0.0002]$	0.092610085	13.186961	7716.1679	0.011937	0.0081	0.0081	−38.376

TABLE 1. The interval upper sums $\bar{j}$, $\bar{B}_3$, $\bar{B}_4^-$ of Lemma 22.12 of the full version on the 400-cell partition of $[0.7, 1]$ at $\ell_0 = \infty$ (Lemma 22.11 of the full version), valid for every $\hat{\mu}$ in the row; $\bar{s} = 6.5231964$ on every row. The three upper sums are rounded upward to eight significant digits; the column $AI_\infty(\mu_{\mathrm{lo}})$ is rounded up, and the derived columns $e^2_{\max}$, c and $\sigma_{\mathrm{low}}(0.1, 0.1)$ are rounded down (towards $-\infty$). Units $m = 1$, $N = 2$, $M_1 = 0.7$.

rounded upward are the four bounds 0.0886293, 7.179064, 9.330427, 1.703784 of Corollary J.8 of the full version; this evaluation at a finite torus floor exercises every admissible form that is active at $\mu = 1$ and the image constant $\hat{E}$ of (G2), which the table at $\ell_0 = \infty$ does not use. Internal consistency of the table: $\bar{s}_{400} = 6.5231964$ at every $\hat{\mu}$ ($\bar{s}$ is μ-free); $\bar{j} = 0.092610085$ for all $\hat{\mu} \le 0.7$, the μ-free branch $\int 2M\bar{\tau}$ of $\bar{j}$ (analytically $\frac{c_N}{4\pi}(1 - M_1) = 0.09253$ before the cell rounding); the interval values equal the point values at μ_{lo} except on $[0.99, 1]$ (Lemma 22.12 of the full version); $\bar{B}^-_{4,400}(\hat{\mu}, \infty) \le P^* + Q^*/\hat{\mu}$ at every grid point ($7716.17 \le 7716.85$ at $\hat{\mu} = 10^{-4}$, $1.5854 \le 5.534$ at $\hat{\mu} = 1$); and $\hat{\mu}\,\bar{B}_4^-(\hat{\mu}) \to 0.7716$ at $\hat{\mu} = 10^{-4}$ against $Q^* = 0.7712$. The closed form of Lemma 22.13 of the full version is weaker than the table near $\hat{\mu} = 1$ (it uses $\bar{\psi}^*$, $|\mu^2 - 2M^2| \le 2M^2$ and the first branch of the minimum) and is within one per cent of it for $\hat{\mu} \le 10^{-2}$.

Proposition 22.14 (the explicit coupling region of (T3) at every photon mass)**.** *Let $N = 2$, $M_1 = 0.7m$ and $\mu_0 \in (0, m]$; write $\hat{\lambda} = \lambda/m$, $\hat{e}^2 = e^2/m$, $\hat{\mu} = \mu_0/m$. Let $\delta m_k^2(M)$ be King's counterterm (61) on Λ_k at the photon mass μ_0, $S_{\alpha\beta,k}$ the coefficients of Lemma 22.3, $\sigma_1(M_1) = \liminf_k s_k^\infty(M_1)$ as in Lemma 3.10, and let σ_{low}, σ^*, $\bar{j}(\hat{\mu})$, $\bar{s}$, $\bar{B}_3(\hat{\mu})$, $\bar{B}_4^-(\hat{\mu})$, A, b,*

L, I_∞, I^* be as defined before Table 1, B_3^*, P^*, Q^* as in Lemma 22.13 of the full version, and $\lambda_c := \frac{b+\sqrt{b^2+4AL}}{2L} = 2.14398\ldots$.

(i) (The five orders.) *For every k with $\varepsilon_k \le 1/m$, in units $m = 1$: $S_{01,k} > 0$; $2S_{20,k} \ge -\bar{j}(\hat\mu) \ge -0.092611$; $2S_{02,k} \ge -\bar{s} = -6.5231964$; $2S_{21,k} \ge -\bar{B}_3(\hat\mu)$ and $2S_{21,k} \ge -B_3^*$; $2S_{40,k} \ge -\bar{B}_4^-(\hat\mu)$ and $2S_{40,k} \ge -(P^* + Q^*/\hat\mu)$.*

(ii) (The lower bound.) $\sigma_1(0.7m) \ge m^2 \max\{\sigma_{\text{low}}(\hat\lambda, \hat e^2; \hat\mu),\ \sigma^*(\hat\lambda, \hat e^2; \hat\mu)\}$.

(iii) (The region.) (T3) *holds on $R(\mu_0) := \{(\lambda, e^2) : \sigma_{\text{low}}(\hat\lambda, \hat e^2; \hat\mu) > 0\}$, with $\sigma_1(0.7m) \ge m^2\sigma_{\text{low}} > 0$ there. $R(\mu_0)$ contains the box $[0, c(\hat\mu)m]^2$ and the segment $\{\lambda = 0,\ e^2 < e^2_{\max}(\hat\mu)m\}$, with c and $e^2_{\max}$ from Table 1 for $\hat\mu \in [10^{-4}, 1]$, and the segment $\{e = 0,\ \lambda < \lambda_c m\}$ for every $\hat\mu$. The map $\hat\mu \mapsto \sigma_{\text{low}}(\hat\lambda, \hat e^2; \hat\mu)$ is non-decreasing (on $[10^{-4}, 1]$ a step function of $\hat\mu$, constant on each row of the table; below 10^{-4} the closed form σ^*), so $R(\mu_0) \supseteq R(\mu_0')$ for $\mu_0 \ge \mu_0'$. In particular $[0, 0.2m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.5m, m]$; $[0, 0.15m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.1m, m]$; $[0, 0.1m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.03m, m]$, with $\sigma_1(0.7m) \ge 0.0582\, m^2$ at $(\lambda, e^2) = (0.1m, 0.1m)$; $[0, 0.075m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.01m, m]$; and at $\mu_0 = m$, $R(m)$ contains the box $[0, 0.239m]^2$ and the segment $\{\lambda = 0,\ e^2 < 0.5618\, m\}$, and $\sigma_1(0.7m) \ge 0.2209\, m^2$ on $[0, 0.1m]^2$.*

(iv) (Uniform in the photon mass; closed form.) (T3) *holds whenever $\sigma^*(\hat\lambda, \hat e^2; \hat\mu) > 0$, with $\sigma_1(0.7m) \ge m^2\sigma^*$; in particular, for every $\mu_0 \in (0, m]$, on each of the boxes $\{\lambda \le \lambda_1 m,\ e^2 \le c_1\sqrt{\mu_0 m}\}$ with $(\lambda_1, c_1) \in \{(0, 0.3014), (0.05, 0.2533), (0.1, 0.2152), (0.2, 0.1616), (0.5, 0.0884), (1$*

(v) (Every k.) *For every k with $\varepsilon_k \le 1/m$: $s_k^\infty(0.7m) \ge m^2\big[A - \frac12\bar{j}(\hat\mu)\hat e^2 - \frac12\bar{s}\hat\lambda^2 - \frac12\bar{B}_3(\hat\mu)\hat e^2\hat\lambda - \frac12\bar{B}_4^-(\hat\mu)\hat e^4\big]$.*

(vi) ($e = 0$; exact.) *At $e = 0$, for every $M_1 \in (0, m)$, every $\mu_0 > 0$ and every $N \ge 1$: $\lim_k s_k^\infty(M_1) = \frac12(m^2 - M_1^2) + \frac{(N+2)\lambda(m-M_1)}{2\pi} - \frac{(N+2)\lambda^2}{\pi^2}\ln\frac{m}{M_1}$ exactly, and* (T3) *holds if and only if $\lambda < \frac{\pi}{4}\big(1 + \sqrt{1 + \frac{16}{N+2}}\big)m$, which is $\frac{\pi}{4}(1+\sqrt5)\, m = 2.5416\ldots\, m$ for $N = 2$; at $M_1 = 0.7m$, $N = 2$, the exact window is $\lambda < \lambda_c m$, $\lambda_c = 2.14398\ldots$, in particular $\lambda \le 2.1439\, m$.*

(vii) (Two-sided.) *On each region of (iii), (iv) and (vi) the two-sided clause* (NE) *holds as well, with no use of Proposition 12.11.*

Outside these regions the sign clause (T3) is not decided here. The region is explicit at every photon mass but not uniform in it: by (67), $e^2_{\max}(\hat\mu) \sim \sqrt{2A/Q^*}\sqrt{\hat\mu} = 0.81\sqrt{\hat\mu}$ as $\hat\mu \to 0$ (Table 1: $\hat\mu\,\bar{B}_4^-(\hat\mu) \to 0.77$), and $\bigcap_{\mu_0 \in (0,m]} R(\mu_0) = \{e = 0,\ \lambda < \lambda_c m\}$, since for $e > 0$ the term $-\frac12(Q^*/\hat\mu)\hat e^4$ of σ^* dominates as $\hat\mu \to 0$. The $1/\hat\mu$ sits in the order-e^4 majorant alone: the low orders and B_3^* are uniform in the photon mass.

Idea of the proof. (i) is Lemma 22.6(i) and the four bounds of Lemmas 22.11 to 22.13 of the full version, applied to the row of $\hat\mu$. (ii) By (62) and the limits of the first four coefficients, $\sigma_1(0.7) = A + b\hat\lambda - \hat e^2 A I_\infty(\hat\mu) - L\hat\lambda^2 + \liminf_k(\hat e^2\hat\lambda S_{21,k} + \hat e^4 S_{40,k})$, and (i) gives the two lower bounds, neither of which dominates the other on a whole row. (iii) σ_{low} is concave in each variable separately, so its minimum over a box is attained at a corner, which is how c and $e^2_{\max}$ are read off; the monotonicity in $\hat\mu$ holds because the three row values that enter with negative coefficients do not decrease down the table. (iv) At a corner $(\hat\lambda, c_1\sqrt{\hat\mu})$, $\sigma^* \ge \sigma^*(\hat\lambda, c_1; 1)$, and the listed c_1 were determined by bisection. (vi) At $e = 0$ the limit is $g_\lambda(M_1) = (m - M_1)[\frac{m+M_1}{2} + \frac{(N+2)\lambda}{2\pi}] - \frac{(N+2)\lambda^2}{\pi^2}\ln\frac{m}{M_1}$ exactly, strictly concave in M_1 with $g_\lambda(m) = 0$, so (T3) holds iff $g_\lambda'(m) < 0$. (vii) By Theorem F.168(b) of the full version, $\sigma_1(0.7m) > 0$ says $\alpha_\infty(0.7m) < \alpha_\infty(m)$, which is (NE$^<$). The complete proof, with the two-loop majorants, is in the full version.

Remark 22.15 (the quantifier over the couplings). The theorems of this paper print the sources' quantifier of Lemma 22.1(e), in the units of the coarsest spacing, with the cube family C_J not rescaled and $\beta = 5^{-J_K}$. A sentence "e, λ sufficiently small, $m, \mu_0 > 0$ fixed" describes *where* the smallness lives (the running couplings, through Bałaban's ε_0), but as a hypothesis on the bare couplings it is strictly stronger than what the sources need. (i) All four couplings have mass dimension one, so its dimensionless content is a bound on e^2/m and λ/m, while the

sources' hypothesis on the bare couplings is empty beyond positivity: the point $(e^2, \lambda, m, \mu_0) = (100, 100, 1, 1)$ is admitted by every source (with a large J_{K}) and excluded by the phrase in every system of units. (ii) In the units in which the stopping scale is 1, m and μ_0 are not fixed but rescaled to $m\beta$ and $\mu_0\beta$, of which $m\beta$ is capped (by the mass cap of [2, Prop. 3.1] and by a further cap of the schedule) and $\mu_0\beta$ is not: in those units the theorem reads "three couplings small, the fourth rescaled". (iii) Even at a fixed stopping scale β_0 the admissible set is not a product neighbourhood of $(0, 0)$ in (e, λ): one cap of the schedule of constants of the rotation part has the form $e \le \lambda^{1/4} c(\beta_0)$, a cusp that fails as $\lambda \to 0$ at fixed e. So the phrase names no set exactly, and the sources' form survives because β can be taken to 0.

The complete proofs: an outline of the appendices of the full version

The full version of this paper carries eighteen appendices, A to R, about a thousand pages, in which every statement of the main text whose proof is not printed above is proved in full; the main text and the numbering of this short form are those of the full version, so that each statement cited below can be found there under the same number. This section describes, appendix by appendix, what is proved there and by which idea, in the order of the appendices. Throughout this section the references are to the full version.

Appendix A: the small-field representation on King's cube. Appendix A proves Proposition 5.4, the two-sided small-field representation of the generating functional on King's cube at zero scalar source, including its clause (R_0). The proof is King's proof of [15, Thm. 3.1], that is, Bałaban's lower and upper bounds [1, 2] with the modifications of [15, Appendix], run on King's renormalisation-group cube C_J^K (Definition 2.2) with King's block-spin flow of Section 5.1. Two things change relative to the torus. The scalar field is killed on the Dirichlet planes and averaged with King's Dirichlet weight, so every place where the torus arguments use that the block average is uniform ($QQ^* = 1$, P a projector, constants reproduced by the average) is replaced by the corresponding fact for the weighted average, and every place where the torus has no boundary acquires a Dirichlet face at which the scalar background vanishes linearly (operator bounds on the cube and its sub-domains, the face-localised difference and the background of small-field data). The photon has Neumann conditions, and every photon object of the cube is the even restriction of the same object on a doubled torus, so every photon estimate of the torus analysis holds on the cube with its constant multiplied by at most 8. The large-field analysis near the Dirichlet faces uses King's face test [16, (5.41)] as a large-field family at every scale, paid by the Dirichlet-face positivity of [16, Prop. 5.4] in an elementary form valid for every gauge field. The lower bound carries King's Appendix class with a background-gradient clause and a block-mismatch clause in place of King's Laplacian clause; the upper bound runs Bałaban's large-field decomposition on the cube; the assembly gives the normalisation-free form of the remainder, in which the large-field term is measured against the partition function itself, which is the form consumed by the ratio argument of Section 5.

Appendix B: the one-step bound on the normalised action. Appendix B proves Proposition 5.5, the bound on the difference of the normalised effective actions of the two members of a diagonal pair, the k-model after j steps and the $(k+1)$-model after $j+1$ steps, at the rate $5^{-\gamma j}$ across the pair. After the decomposition of the effective action into the photon and the scalar quadratic forms, the background determinant and the graphs (units, operators, zones), each piece is compared across the pair: the quadratic forms through the comparison of the two cube flows at their common alive sites, the background determinant through its expansion in the regular background, and the graphs, King's renormalised families of order at most $\bar{n}$, through a degree count in which the Ward rows (the exact Ward identities applied family by family, never to a fixed ordering of the lines) raise every divergent or marginal subgraph to positive degree, and block vertices carry a split degree. The faces of the cube are treated by zones: away from the faces the comparison is that of the infinite lattice; in the face zone the scalar face remainders and the vacuum-polarisation face pieces are bounded by an ordered multiscale bound with layered

vertices. The assembly collects the rates into one exponent $\gamma < \min(\alpha, 1-\alpha)$, α the Hölder exponent of the line bounds [15, (3.62)].

Appendix C: the small-field lines at a non-negative scalar source. Appendix C proves Proposition 6.6: at a real source $s = (u, h)$ with $h \geq 0$ the lower line and the small-field half of the upper line hold on both members of every diagonal pair, uniformly on compact source families. The proof inserts the scalar source into the small-field chain of Appendix A, that is, into Bałaban's inductive lower bound [1, Sect. 3] and into the term of his upper bound [2, (2.43)] in which no large-field block occurs at any scale. The scalar source enters King's effective action through one vertex only, [15, (3.21)], whose coupling carries two powers of the running scale where the other vertices carry at most one and a half; this margin makes every step of the chain either blind to the source, or a bound on the interaction families which the margin leaves unchanged, or a piece of bookkeeping which the appendix writes out (where the sources sit, power counting with the source vertex, the two small-field lines, the insertion step by step).

Appendix D: the one-step bound with the scheme c-number. Appendix D proves Proposition 6.7. On the support of the small-field indicator the normalised action at the source (u, h) is split exactly into the normalised action at $(u, 0)$, the members of the family $\mathcal{N}_h$ of Lemma 6.1, and the h-linear vacuum constant of the three lowest orders. The one-step differences of the first two pieces are bounded by Proposition 5.5 and Lemma 6.1, uniformly on compact families by a polarisation argument. The h-linear constant plus the scheme term $c_k(h)$ is a local functional of h: it equals $\bar{\beta}_k^{(j)} \int h$ with $\bar{\beta}_k^{(j)}$ the block mean of an infinite-lattice one-point density, up to errors super-exponentially small in the inverse block scale, and this bulk sequence converges along the diagonal pairs at a power rate. The three pieces are assembled into the one-step bound with the scheme c-number.

Appendix E: the c-numbers and the growth of the bare coefficient with the mass. Appendix E proves Lemma 6.8 and Proposition 6.9, both statements about lattice Feynman integrals on the infinite lattice $\varepsilon\mathbb{Z}^3$ with $N = 2$ and the photon in Feynman form at mass μ_0. The first concerns the order-e^2 coefficient X_ε of Definition 2.4, which defines the reference scheme $z_k = e^2 X_{\varepsilon_k}$; its exact asymptotics as $\varepsilon \to 0$ are computed. The second concerns King's free-boundary mass counterterm $\delta m_\varepsilon^2(M)$ [15, (3.32)–(3.33)] at its five orders in (e^2, λ): the continuum limit of $\delta m_\varepsilon^2(M) - \delta m_\varepsilon^2(m)$ is computed in closed form (the divergent parts, $1/\varepsilon$ at one loop and $\ln(1/\varepsilon)$ at two loops, do not depend on the scalar mass and cancel), and the lower bound of Proposition 6.9 on the growth of the bare coefficient with the mass is derived from it.

Appendix F: Bałaban's torus estimates at block factor 5. Appendix F proves the three statements about King's torus model that the Wick leg takes from the torus analysis: the stability bounds Proposition 3.9 in King's normalisation at the two masses m and M_1, the c-number lemma Lemma 3.10, and the fixed-spacing volume limit Lemma 12.9 together with the boundedness clause used in Lemma 12.10. The first is Bałaban's theorem, [1, (1.14)] and [2, (1.3)], read at block factor 5 and in King's convention (Theorem F.1); the longest part of the appendix re-derives, at block factor 5 and with the constants scheduled once for all scales, the estimates which Bałaban's papers assert for a general block factor (the large-field chain, the renormalisation of the divergent classes, the orbit regrouping of the cut terms, the walled form of the perturbative bound, and the gate that removes the dangerous terms of the step-k remainder), each statement being proved in full in Appendix K. The other two are statements about finite lattice Feynman integrals, proved from Reisz's power counting in three dimensions, which gives the volume limit at fixed spacing order by order and the uniform bounds behind the c-number lemma.

Appendix G: two steps of the renormalisation-group analysis. Appendix G proves the two statements of Section 5.2. Both concern Bałaban's renormalised classes of graphs [3, Sects. 1–3] at the moment when some of their lines are cut, that is, replaced by a pair of external legs carrying a two-variable kernel, which happens when two propagator domains are compared and whenever a line object is telescoped. The orbit regrouping Lemma 5.2: the cut

terms of a renormalised family regroup into complete renormalised classes of the cut graphs, so that Bałaban's bound applies to them as it stands. The domain-change estimate Lemma 5.1, the interaction-term half of [2, Prop. 2.6] at a nonzero background: an exact regrouping of a second kind, into full classes of the cut components, to which Bałaban's renormalisation bound [3, Prop. 1, (1.33)] applies at every background. The appendix closes with the difference-kernel bound on a lattice torus fitted to the blocks, the form in which the torus analysis of Appendix F consumes the domain-change estimate.

Appendix H: corrections to King's papers. Appendix H lists the points of [15, 16] that affect the use of King's theorems in this paper, each with the corrected form that is used: the large-field term in [15, Thm. 3.1], whose printed form is replaced by the normalisation-free form of Proposition 5.4; King's torus theorem [16, Thm. 2.4] as printed, which is false already at $e = 0$ in the scalar reading of the source (Proposition H.2) and is replaced by the 2-form reading of Theorem 17.1; the bound (4.33) of [15], which holds although the argument printed for it treats the block average as a Fourier multiplier; [15, Thm. 3.5], which fails on the degree-zero members of the h-linear field-free family and holds, with King's power of the logarithm, on the families of positive ultraviolet degree whose infrared lines lie in mass-counterterm units (Corollary 17.42, Proposition 17.134); the inequalities (3.78) and (4.21) of [15]; and minor points. None of them affects the existence theorem (E_C) (Theorem 3.3), which is proved in Section 5 without them.

Appendix I: Bałaban's Proposition 1 at block factor 5. Appendix I writes out the proof of Bałaban's perturbative estimate for localised renormalised classes, [3, Prop. 1, (1.33)], at block factor 5, in the form in which Appendix F uses it. The printed proof [3, pp. 420–445] consists of twenty sketched steps and seven further sketched places; each is stated in the form in which the proposition uses it and proved, with constants independent of the lattice spacing, the step, the domains, the background and the torus. The four parts follow the order of the printed proof: the slices, the external-field kernels, the summation by parts and the induction; generalised graphs, power counting, the Ward–Takahashi identities and the estimate [3, (3.1)]; the component cube, the constant field and the expansion in the fluctuation; the cube at zero field, the norms and the identities of the local terms. The two steps that need more than a transcription of the printed text, the regrouping into renormalised classes after the separated lines are cut and the renormalisation of the divergent classes, are stated in Appendix F and proved in Appendix K.

Appendix J: two-loop majorants of the lattice mass counterterm. Appendix J proves the facts on King's counterterm on periodic tori that Proposition 22.14 uses through Lemmas 22.10 to 22.13: the exact two-loop form of the torus coefficient (King's recursion (61) on the torus, with $N = 2$ and the photon mass a free parameter, read at μ_0 in the proof of Proposition 22.14); slope bounds for each coupling order, from one-loop majorants which are minima of finitely many closed forms in Γ and B functions, decreasing in the torus side and non-increasing in the masses; the resulting two-loop gap between two scalar masses, order by order; and its evaluation by cell upper sums over a partition of $[M_1, m]$. The tools are a homogeneity under the change of units, lattice bounds on the free propagators, the graphs of the counterterm on a torus and, at order e^4, a Ward identity that removes the longitudinal part of the photon exchange. The numbers of Table 1 are the cell upper sums of this appendix.

Appendix K: complete proofs of the results used in the main text. Appendix K, the largest, contains the complete proofs of the results which the main text and the preceding appendices state with a proof that refers there; each of its sixty-two subsections restates the result in the form in which it is used and proves it, in the order of the results in the paper. For the sections on the axioms and the main theorem: the comparison of two states through the law of the field strength (the same box, and King's convention in the upper bound), the volume limit at fixed spacing, the axioms for the LIM states (Theorem 14.9), the Wick sector on King's convention (cubic symmetry on the tube, the measure and weak clustering of the joint field), the coupling range of the existence and rotation parts (Lemma 22.1(c)), and Form (A) of the Wick leg at arbitrary amplitudes. For the existence part: the complete proof of Lemma 6.1 with its

zone I and its tuples with several clusters (Proposition 6.4, Proposition 6.5); the perturbative bounds on King's cube and on its sub-domains, the lower bound on the cube, the changes of propagator domain, the upper line on the cube, the Dirichlet image effect on a sub-domain and the payment of the face family, the free weighted flow at zero field and the Dirichlet symbol, and, for Appendix B, the proof that the split degree governs the multiscale bound, zone I for the curvature classes and the source-free field-dependent classes, the Ward rows at every tuple, King's localisation argument with centres and the vacuum-polarisation difference pieces as layered vertices; for Appendices C to E, the complete proofs of the small-field lines at a source, of the averaging part of the bulk density with the Dirichlet weight, of the order e^4 of the bare coefficient and of the monotonicity of its order-e^4 function. For Appendices F and G: the asserted estimates of Bałaban's papers on the torus, the block-factor-5 reading of his theorems, the extensivity of the vacuum c-number, his Lemma 2.4 with scale-independent constants, the bad-localisation terms of the moved mass counterterm, the vacuum counterterm between scales, the inductive step of the large-field chain, the order-$(\bar{n}+1)$ remainder of the expansions, the cut kernels of the cluster cut rule, the bounds on the regrouped local coefficients, the renormalisation of the divergent classes, the walled slices and the walled form of Bałaban's Proposition 1, the scalar large-field bounds for the torus mass counterterm, Bałaban's Propositions 3.1–3.2 and the step-k remainders of the upper bound, the terms removed by the gate, Reisz's theorems in three dimensions with the uniform bounds, the printed inputs and the difference-kernel bound on a lattice torus fitted to the blocks, and the cut data of a per-label full class. For the rotation part: the two-sided bound with the shell inside (Proposition 20.3) and the $\varkappa$-Cauchy property of the reduced action (Proposition 20.13), the background determinant in the equal-spacing comparison (step (4) of the proof of Theorem 17.57), the hybrid lower bound (Proposition 17.117), the skew-torus instances (Proposition 17.121, Lemma 17.124), the torus-minus-free counterterm differences (Proposition 17.126(b)), the one-flow instance of the multiscale bound (Corollary 17.42), the restricted renormalisation theorem (Proposition 17.134), the floor shift (Theorem 17.120), the constant-background reduction (Lemma 17.41), the walk-sum bounds and the slice clauses of Proposition 17.73(a), the positivity condition on the constant of the convention bridge (Theorem 17.98), the right-continuity in the scalar mass at the exceptional set (Theorem 20.15), the fluctuation estimates for the centre-rooted and the hybrid contours (Proposition 17.18), and King's small-field bound at a fixed source with the inclusion of the sources.

Appendices M, N and O: the ergodic decomposition, reconstruction and clustering. These three appendices concern an abstract probability law μ of a field on a Polish space with three properties, (H1)–(H3): invariance under translations and coordinate reflections, a local structure of σ-algebras, and reflection positivity about the coordinate planes. Appendix M proves the ergodic decomposition of μ, with the coordinate reflections and reflection positivity inherited by almost every component; the correspondence of the decompositions of a law and of its image under a continuous injection; the regrouping of the components into O_h-invariant ones when μ is O_h-invariant, with the reconstruction and clustering of the regrouped components; and a growth bound for the moments of the components of a law on a Hermite–Sobolev space. Appendix N proves, for such a law, the Osterwalder–Schrader reconstruction in a coordinate direction, the description of its zero-energy and joint fixed vectors, clustering in the mean and under diagonal separation, and the relations between the forms of the cluster property OS4 used in the main text. Appendix O proves that a law with (H1)–(H3) which is moreover invariant under $O(3)$, through an admissible action, has almost all of its ergodic components invariant under the Euclidean group, reflection positive about every plane and clustering as the separation tends to infinity in any manner, so that each form of OS4 holds on almost every component for every choice of the time direction. No property of a particular model is used; where the hypotheses are met for the laws of King's state is recorded in Theorem 11.1, Proposition 21.4, Theorem 14.9 and Theorem 23.7.

Appendix P: probability measures on spaces of distributions. Appendix P contains the measure theory used in Theorem 11.1, Theorem 14.9 and Theorem 23.7: the setting (H1)–(H3) in which reflection positivity of the law of a field is formulated; the field spaces and the action

of the Euclidean group on them; the Hermite–Sobolev norms; the Borel structure of the field spaces and the density of the generating algebras; the passage from a measure on $\mathcal{S}'$ to a measure on a Hermite–Sobolev space; the uniqueness of a measure with a given characteristic functional or with given moments; and the verification of (H1)–(H3) from properties of the characteristic functional. Nothing in it uses the lattice.

Appendix Q: periodic tori with the counterterm at a fixed photon mass. Appendix Q defines the lattice states on periodic tori that enter the open problems of Section 23, and compares, at order e^2, the mass counterterm computed with the photon propagator at a fixed mass with King's counterterm, computed with the model's own photon mass μ_0. Nothing in the proof of Theorem 21.1 uses it.

Appendix R: proofs for the rotation part. Appendix R proves three groups of statements used in the rotation part and in Appendix K. The first group consists of steps of the proof of Proposition 20.13: the moment bound for the martingale part of the holonomy; the full-amplitude localisation of the leg of block data on the heavy set U; the deep clauses of legs and lines on U and the deep payment; the heavy differentiated point of a single-scale line in the lattice regime; the inserted line differenced at both ends; the U-cube remainder for lines used as legs; the mixed class with one shell-mass insertion; the block-free covariance next to the shell; the local vertex with lines as legs on the light side; the Hölder-leg remainder in the lattice regime; the one-loop z^2-term in the lattice-frozen regime; and Bałaban's enumeration with one inserted vertex. The second group is the torus input of Theorem F.1: the inputs of Appendices B and F restated in the forms in which they are used on King's torus and in the domain replacement, and the domain locality of the densities. The third group is the comparison of King's cubes with the periodic tori: the one translation of the fine lattice that carries King's corner-rooted hierarchy of blocks onto the centred blocks of Convention 17.17; the upper stability bound at a wall and the large-field envelope at a wall, with their consequences for Proposition 17.100 and Proposition 17.130; King's upper bound for the rotated lattice on the aligned and on the skew torus; the absorption of every polynomial loss by the large-field factor, used in the proof of Theorem 17.127 and in Proposition 17.139; the photon quadratic form across one step, the scalar quadratic form and the background determinant on King's torus; and the non-polynomial terms at the frozen wall, in the rate form.

23. Open problems

Throughout, K denotes King's printed state of Theorem 21.1: the Neumann–Dirichlet cube limit on King's RG cube at block factor 5, fixed photon mass $\mu_0 > 0$, with an admissible scheme a in the Wick-square sector; S denotes a periodic-torus limit in the sense of Definition Q.1(iii) of the full version (dyadic spacings, periodic tori, the mass counterterm computed with the photon propagator at the mass m), at couplings at which one exists. No identification between the two is proved, and the existence of S is not proved here. "Symmetric phase" is, for both, the expectation that the states lie above the surface predicted by one-loop perturbation theory; it is neither a hypothesis nor a conclusion of any theorem here, and Theorem 21.1 holds on all of King's coupling range (Lemma 22.1), the surface included. The problems below are extensions of the main theorem, not conditions on it.

23.1. **Relation to the periodic-torus states.** *Problem.* For which couplings do the periodic-torus states of Definition Q.1 of the full version have continuum limits S, and does such a limit coincide with K, possibly at a matched bare mass? Three statements bear on this, none of them proved: (FAM_L), King's cube construction exists with block factor 2 along the dyadic spacings ε'_K of Definition Q.1 of the full version, with the counterterm defined there; (KI_L), the fixed-torus continuum limit exists in that family; (CE), a convergent unit-scale cluster expansion with sources on the cubic tori, their tilted copies and their cuts along King's cubes, uniform in the lattice spacing. Whether (CE) holds anywhere at $e > 0$ is open. Two obstacles are structural. The counterterm of Definition Q.1 of the full version is computed with the photon propagator at the mass m, King's with the model's own μ_0; already at order e^2 the two relative bare coefficients,

$s^{\mathrm{per}}_{j,K}(M_1)$ and King's $s^{\infty}_{k}(M_1)$, have different limits unless $\mu_0 = m$ (the first as $j \to \infty$ and then $K \to \infty$), the difference being at most of order $e^2|m-\mu_0|$, of exactly that order for μ_0 in compact subsets of $(0,\infty)$, and not small when $|m-\mu_0|$ is comparable to m (Lemma Q.2(d) of the full version,(e), Remark Q.3 of the full version); and King's construction needs block factor 5 for the rotation by $\arctan(3/4)$, while no lattice family admits both a dyadic structure and a rotation of infinite order. On the one-loop surface $\lambda m^2 = e^6/(72\pi^2)(1+O(e^2/m))$ the two constructions are expected to select different phases (periodic tori the broken one, Dirichlet faces the symmetric one); this is a heuristic, not a theorem, and no statement of this paper uses the phase.

23.2. **Uniqueness and ergodicity of K.** Uniqueness of King's cube limit is part of the construction (monotonicity in the cube), but uniqueness of the infinite-volume state, in the sense of King's expected identity between cube limits and periodic-box limits, is not proved, and neither is ergodicity of K under translations: Theorem 21.1 gives clustering only on the ergodic components. The usual route, a cluster expansion with a mass gap, is (CE). This is distinct from the independence of a LIM state from its continuum subsequence, (UNIQ), which holds (Proposition 14.6, from Theorem 3.14 and from Theorem 3.3; Proposition 14.12, Corollary 4.4). Cubic symmetry of K is a theorem (Theorem 8.5): it needs only the uniqueness of the monotone cube limit and the reflection sandwich (Remark 8.6).

23.3. **Non-triviality.** For a state with Schwinger functions of the curvature and the Wick square let $M(f;h) := S^T_{2,1}(f,f;h)$ be the truncated function with two curvature arguments and one Wick-square argument. If the curvature and the Wick square are independent under the law of a state, all their mixed truncated functions vanish; if their joint law is Gaussian, all truncated functions of order three vanish. In either case $M \equiv 0$; so $M(f;h) \neq 0$ for one real 2-form f and one $h \geq 0$ would show that under K the curvature and the Wick square are neither independent ("decoupled") nor jointly Gaussian. On every BFS measure, in particular on every lattice approximant of K, $\langle \sum_x \varepsilon^3 h(x) r_x^2; F(f)^2 \rangle \leq 0$ for $h \geq 0$: this is Lemma 2.8(a) for $\Theta = e^{t\sum_x \varepsilon^3 h(x) r_x^2}$ and $\ell = F(f)$, divided by $t > 0$ and taken as $t \downarrow 0$ (dominated convergence, $\lambda > 0$). The sign is not strict. A route to $M(f;h) < 0$ at small couplings has two steps, both open: (C) at $e = \lambda = 0$ the coefficient of e^2 in that lattice covariance has a strictly negative continuum limit $c^{\mathrm{free}}(f,h)$ for $d^*f \not\equiv 0$ and $h \geq 0$, $h \not\equiv 0$ (a one-loop statement); (R) e^{-2} times the covariance differs from that coefficient by at most $\rho((e^2+\lambda)/m)\,\kappa(f,h)$, with $\rho(t) \to 0$ as $t \downarrow 0$, uniformly along the approximants of K. Given both, passing to the limit would give $M(f;h) \leq \frac{1}{2} e^2 c^{\mathrm{free}}(f,h) < 0$ for small $e^2 + \lambda$. The bounds of the existence theory are not uniform in the couplings, and the cited constructions state no asymptotics in (e,λ), so neither supplies (R). A non-vanishing curvature two-point function is also left open. For K nothing is proved here beyond the non-strict lattice sign above. Whether K is non-Gaussian, or even not decoupled, is open.

23.4. **The limit of vanishing photon mass.** Every constant of the Wick leg of Theorem 21.1 depends on μ_0 and is not uniform in it (Remark 21.7); the absolute bounds of the curvature leg are. The limit $\mu_0 \downarrow 0$, taken along a frozen family (counterterm and scheme frozen at a reference mass, so that the members are monotone in μ on the cone by the correlation inequalities), is left to later work: in the curvature sector it needs the existence theorem for every member and nothing else; in the Wick sector it needs a cell bound uniform in the photon mass (the analogue, along the family, of the cell bound ($\mathrm{B}_{\mathrm{cell}}$) of Lemma 14.4, which at a fixed mass is a lemma; the clause "several models at once" of Lemma 14.7 is formulated for this purpose), together with the torus inputs at the reference mass and the transfer of Theorem 21.3 for the members on King's convention; the identification of the members with King's models at shifted scalar masses, masslessness proper (the vanishing of the photon mass in the limit theory) and the two-point function are further questions there. None of these enters the present paper.

23.5. **The optional extension:** (W2) **at arbitrary amplitudes, Form (A), and the Euclidean clause of the Wick sector.** Form (B) of Theorem 21.1 lives on the tube $\mathcal{D}_{\sigma_+,H}$: the

only positive-source bound it uses is the one proved on the cone $\sup q \le s_1 < \sigma_1(M_1)$ (Corollary 13.3), and the only existence statement is Theorem 3.6, whose positive scalar part is below the same threshold. Above the threshold, the existence of the fixed-cube limits and the uniform positive-source bound are the hypothesis (W2) below, which is open; the Euclidean clause of Form (A) and its King-convention form are the further hypotheses (R_W) and (FA^K).

Hypothesis 23.1 ((W2): the Wick sector at arbitrary amplitudes)**.** Let a be an admissible scheme. (a) For every finite family of real 2-forms and real functions, every $J \ge J_0(f,h)$ and *every* real coefficient vector (z,w), the limit $\lim_k Z^a_{k,J}(z,w)$ exists and lies in $(0,\infty)$. (b) For every $q \in C^\infty_c(\mathbb{R}^3)$ with $q \ge 0$,

$$\sup_{J \ge J_q} \lim_{k\to\infty} Z^a_{k,J}(0,q) < \infty,$$

where J_q is the least J with $\operatorname{supp} q$ at ℓ^∞-distance greater than 1 from $\mathbb{R}^3 \setminus C_J$.

Remark 23.2 (on (W2)). (i) Clause (a) is Theorem 3.6 with the threshold on the positive scalar part removed. On the cone of amplitudes $\sup q \le s_1 < \sigma_1(M_1)$, clause (b) is proved in this paper in L^1 form with $\lim_k$ replaced by $\limsup_k$ (Corollary 13.3, on the comparison convention; on ND^K by Remark 4.8); with the limit itself it is proved on the comparison convention for $2\sup q < s_1$, under the convergence of Definition 3.2. Above the threshold both clauses are open; the exact obstruction is the free-energy bound $(\mathrm{B}^-)[u]$ at amplitude u above the threshold (Hypothesis 23.9 below): the ε-uniform stability of King's torus model with its bare coefficient shifted by $-2u$. (ii) No clause of Theorem 21.1 uses (W2). The Wick result of the main theorem is Form (B) on the tube $\mathcal{D}_{\sigma_+,H}$, where only the proved cone instance enters, through Theorem 13.10; (W2) is a hypothesis of Form (A) (Theorem 23.7) and of (R_W). In the rotation part, (W2) for a scheme a is the W-admissibility $(\mathrm{W})_a$ of Definition 16.3.

Hypothesis 23.3 ((R_W): rotation invariance of the Wick sector at arbitrary amplitudes)**.** For an admissible scheme a satisfying (W2), the full object $Z^a_\infty = \lim_J \lim_k Z^a_{k,J}$ of the comparison family (whose existence and entireness follow from (W2) as in Lemma 10.2) is invariant under every Euclidean motion for all complex coefficients, F transforming as a 2-form and W^a as a scalar.

Remark 23.4 (on (R_W)). (R_W) enters only in the Euclidean clause of Form (A) (Theorem 23.7). Form (B) has no Euclidean clause and does not assume it. The rotation part proves the curvature sector; its Wick-sector theorems (Theorems 18.9 and 18.11) give the $O(3)$ clause of (R_W) for a scheme a (its translations are Theorem 23.7(ii)) under the torus comparison with scalar sources and the closing hypothesis for that scheme (Definitions 17.2 and 18.5).

Hypothesis 23.5 ((FA^K): Form (A) on King's convention)**.** On King's convention ND^K, under (W2) read for ND^K: the limit functional $Z^{a,K}_\infty$ is entire in all coefficients; the lattice symmetries O_h about every point hold for the joint field at all amplitudes; reflection positivity holds at arbitrary complex coefficients; the measure exists off the tube; and the Euclidean clause of the Wick sector holds for the King-family object.

Remark 23.6 (on (FA^K)). (FA^K) is the King-convention counterpart of Form (A) as the comparison-convention results give it (Theorem 23.7, last paragraph). The main theorem makes no use of it: its Wick leg on King's convention is Form (B) on the tube (Theorem 21.3, Proposition 21.4).

Theorem 23.7 (Form (A): the Wick leg at arbitrary amplitudes)**.** *Let the couplings satisfy* (T3)*, and let a be an admissible scheme satisfying* (W2) *for the comparison convention* $\mathrm{ND}^{(a)}$*. Then for every finite family $(f_1,\dots,f_n;h_1,\dots,h_p)$ of real 2-forms and real scalars the full object $Z^a_\infty(\zeta) = \lim_J \lim_k Z^a_{k,J}(\sum_i z_i f_i, \sum_j w_j h_j)$ exists for every $\zeta = (z,w) \in \mathbb{C}^{n+p}$, locally uniformly, Z^a_∞ is entire, and $Z^a_\infty = G_\infty$ on $\mathcal{D}_{\sigma_+,H}$. Its Schwinger functions are those of Form (B), and:*

(i) OS0 *as in Form (B)(i): the tube bound on $\mathcal{D}_{\sigma_+,H}$ and the bounds on the $S_{n,p}$; no quantitative bound on Z^a_∞ off the tube is asserted.*

(ii) OS1*: $Z^a_\infty(\tau_b R\cdot\zeta) = Z^a_\infty(\zeta)$ for every $\zeta \in \mathbb{C}^{n+p}$, $b \in \mathbb{R}^3$, $R \in O_h$ (F a 2-form, W^a a scalar).*

(iii) OS2 *in matrix form about every coordinate plane for exponentials with arbitrary complex coefficients (Theorem 23.8), and in polynomial form (Proposition 10.10).*
(iv) OS3 *as in Form (B)(iv).*
(v),(vi) *As in Form (B), the characteristic functional* $(u,v)\mapsto Z^a_\infty(\mathrm{i}u,\mathrm{i}v)$ *being the restriction of the entire* Z^a_∞ *to the imaginary axes.*

Euclidean clause. *If in addition* (R_W) *holds for the comparison family,* Z^a_∞ *is invariant under every Euclidean motion, its mixed Schwinger functions are* $\mathbb{R}^3\rtimes O(3)$*-covariant, and* ν*-almost every ergodic component of the measure of (v) is* $\mathbb{R}^3\rtimes O(3)$*-invariant and reflection positive about every plane (no vanishing of odd moments is asserted for a component). This is the only Euclidean clause of the Wick sector.*

On King's convention. *The same statements for the K-family object* $Z^{a,K}_\infty$ *under* (W2) *read for* ND^K *are the content of* (FA^K) *(Hypothesis 23.5); nothing is derived for it from the comparison-family statements, since* (W2) *on* ND^K *does not give* (W2) *for the comparison family (Corollary 4.4(iv)).*

Indications. The complete proof is given in the full version; in outline: existence at all complex coefficients, locally uniform convergence and entireness follow from (W2) by the argument of Lemma 10.2 (version (A_∞) of Lemma 10.3), and $Z^a_\infty=G_\infty$ on $\mathcal{D}_{\sigma_+,H}$ as the same J-limit of the same Z^a_J (Theorem 13.10). (i) is Form (B)(i). (ii): Form (B)(ii) is the identity on $\mathcal{D}_{\sigma_+,H}$, an open neighbourhood of the origin of $\mathbb{C}^{n+p}$, and both sides are entire. (iii) is Theorem 23.8 with Proposition 10.10; (iv) is Proposition 10.9 for Z^a_∞; (v),(vi): the measure of Form (B) is built from G_∞ on the imaginary axes, which lie in $\mathcal{D}_{\sigma_+,H}$. The Euclidean clause is (R_W) itself; the component clause follows through the ergodic decomposition of Form (B)(vi) as in the curvature sector (Theorem 11.1(c)). □

For a LIM state of a continuum subsequence (Theorem 14.9) the corresponding Form (A) requires both clauses of (W2): the existence of the limits at every real coefficient vector and the uniform bound at every amplitude.

Theorem 23.8 (reflection positivity at arbitrary complex coefficients)**.** *Let a be an admissible scheme satisfying* (W2)*, let* $\nu\in\{1,2,3\}$*,* $c\in\mathbb{R}$*, and let* $\zeta_1,\dots,\zeta_N$ *be finite families of real sources* $f_{il}\in C^\infty_c(\mathbb{R}^3;\Lambda^2)$*,* $h_{il}\in C^\infty_c(\mathbb{R}^3)$ *supported in* $\{x_\nu>c\}$*, with arbitrary complex coefficients* z_{il},w_{il}*. Then*

$$\sum_{i,j=1}^{N}\bar\gamma_i\gamma_j\,Z^a_\infty\big(\theta\zeta_i^*\sqcup\zeta_j\big)\;\ge\;0\qquad\text{for all }\gamma\in\mathbb{C}^N.$$

Refinement. *Let* B_1*,* B''*, the vertex sources* q_v *(*$v\in\mathrm{vert}(B'')$*) and* q_ζ *be those of Definition 10.1 for the plane* $\{x_\nu=c\}$*. (a) If* $c\in D$*, the proof uses* (W2)*(b) only at the finitely many sources* q_v*, and it suffices to assume it at the single source* $(1+\eta)q_\zeta$ *for any one* $\eta>0$ *(Remark 10.4, one direction only); all other steps use* (W2)*(a) and the inequalities of Lemma 2.8, and the tube machinery is not needed. (b) If* $c\notin D$*, with* $c'\in D$ *and* $b:=(c'-c)e_\nu$*: the finitely many instances* q_v *and* $\tau_b q_v$*,* $v\in\mathrm{vert}(B'')$*, together with the tube inputs* (I2)*,* (I3) *of Theorem 10.7. Such a plane needs a non-lattice translation, which finitely many instances of* (W2)*(b) do not supply; it is carried by the translation invariance of the tube object (Proposition 10.5(b)).*

Proof. Let $c\in D$ and let $\mathcal{F}$, Φ_k, ζ^*_{ij}, $\mathcal{T}_1$, $T_{\mathcal{F}}$, U be as in Definition 10.1. By Lemma 10.2, under (W2)(a) and (W2)(b) at the q_v, $\{Z^a_J\}_{J\ge J_0}$ is uniformly bounded on $\mathcal{T}_1$, $Z^a_J\to Z^a_\infty$ locally uniformly on $\mathcal{T}_1$, Z^a_∞ is holomorphic there and $Z^a_J\le Z^a_\infty$ on $U\cap\mathcal{T}_1$. By Lemma 10.3(A), $\limsup_k\sup_{\mathcal{T}_1}|\Phi_k|\le C$ with C independent of $J\ge J_0$, and at fixed J the Φ_k are locally bounded on $\mathbb{C}^m$ ((W2)(a) only). Montel's theorem gives a subsequence along which $\Phi_k\to\Phi^J$ locally uniformly, Φ^J entire; the matrices $(\Phi_k(\zeta^*_{ij}))_{ij}$ are positive semidefinite for large k (Lemma 9.3(b)), hence so is their limit, and $\sup_{\mathcal{T}_1}|\Phi^J|\le C$ uniformly in J. On U every lattice value is real and the scheme multipliers of the two states agree, so the $\mathcal{I}$ halves of Lemmas 9.9 and 9.10 give $Z^a_{k,J}\le\Phi_k\le Z^a_{k,J+2n}$, hence $Z^a_J\le\Phi^J\le Z^a_{J+2n}$ on U. On $U\cap\mathcal{T}_1$, $Z^a_J\uparrow Z^a_\infty$ (Lemma 10.2), so $\Phi^J\to Z^a_\infty$ there; by the uniform bound on $\mathcal{T}_1$, Montel's theorem and the

uniqueness set $U \cap \mathcal{T}_1$ (Definition 10.1), $\Phi^J \to Z^a_\infty$ locally uniformly on the connected $\mathcal{T}_1$, in particular at the ζ^*_{ij}, and positivity passes to the limit. The only instances of (W2)(b) used are the q_v of Lemmas 10.2 and 10.3; Remark 10.4 replaces them by $(1+\eta)q_\zeta$.

Let $c \notin D$, $c' \in D$, $b := (c'-c)e_\nu$. The $\tau_b\zeta_i$ are supported in $\{x_\nu > c'\}$ and $\theta_{c'}(\tau_b\zeta)^* = \tau_b(\theta_c\zeta^*)$, so the display at c follows from the display at c' for the translated families once Z^a_∞ is translation invariant at the points in question. The enlarged family of the translated problem at c' is $\tau_b\mathcal{F}$, with the same coefficient space, matrix points, B_1, B'', and vertex sources $\tau_b q_v$; the argument above for $\tau_b\mathcal{F}$ gives the positivity of $(Z^a_\infty(\tau_b\cdot\zeta^*_{ij}))_{ij}$, and $\zeta \mapsto Z^a_\infty(\tau_b\cdot\zeta)$, $\zeta \mapsto Z^a_\infty(\zeta)$ are holomorphic on $\mathcal{T}_1$ by Lemma 10.2. On the non-empty open set $\mathcal{T}_1 \cap T_{\mathcal{F}}$ both equal the tube limits $Z^a_{\infty,T}(\tau_b\cdot\zeta)$, $Z^a_{\infty,T}(\zeta)$, which agree by Proposition 10.5(b); by the identity theorem they agree on $\mathcal{T}_1$, in particular at the ζ^*_{ij}. □

Hypothesis 23.9 ((B$^-$)[u]: stability of the shifted torus model)**.** Let $u > 0$. In the free convention and the scheme a,

$$\sup_{k\ge 0} \limsup_{J\to\infty} \alpha^a_{k,T_J}(u) < \infty, \qquad \alpha^a_{k,T}(u) = |T|^{-1} \ln N^a_{k,T}(0, u\mathbf{1}),$$

the free-energy density of the torus functional at the constant scalar source u; that is, (I2) of Definition 12.5 at the amplitude $s_1 = u$.

For u below the threshold s_* of Lemma 22.1, (B$^-$)[u] is Proposition 12.6 (or Proposition 12.11) at an auxiliary mass M_1 with $\sigma_1(M_1) > u$, so Form (B), whose constants are built from (I2) at one such amplitude, already contains every amplitude at which the positive-source bound is known. For $u \ge s_*$ the hypothesis is the exact obstruction. By Lemma 12.7, exact at fixed (k,T), $\alpha^z_{k,T}(u)$ is, up to explicit c-numbers, $|T|^{-1}$ times the difference of the free energies of King's torus model with bare scalar coefficient $\alpha^{\mathrm{tor}}_{k,T}(m) - 2u$ and with $\alpha^{\mathrm{tor}}_{k,T}(m)$, both with the counterterms of the mass m: the second is Proposition 3.9 at m, the first the ultraviolet stability of the shifted model. For $u < s_*$ the shifted coefficient is that of King's model at a positive mass M_1; for $u \ge s_*$ it lies, asymptotically in k, below $\inf_{M>0} \alpha^\infty_k(M)$, and no positive scalar mass realises it. The printed proofs of [1, 2] expand about the Gaussian at the positive scalar mass m and use its mass term in the small-field bounds; no printed theorem covers a bare coefficient below every positive-mass value. Conversely (B$^-$)[u] is sufficient for (W2)(b): Proposition 13.2 at $s_1 = u$, whose steps do not depend on the amplitude, gives $\ln N^a_{k,X}(0,q) \le (C_U/u)\|q\|_{1,k}$ for $0 \le q \le u$ on every Neumann–Dirichlet box, hence (W2)(b) in L^1 form on $\sup q \le u$. It is not asserted that (W2) implies (B$^-$)[u]: the inequalities available bound a constant-source density from below by local sources, never from above, so (B$^-$)[u] is sufficient and not shown necessary. Also open: a form of the tube bound (31) in which the negative part k_- of the Wick source enters through its L^1 norm in place of the weighted norm $|k_-|_w$.

23.6. **Rotation invariance of the Wick sector.** Theorem 21.1 gives invariance under $\mathbb{R}^3 \rtimes O(3)$ for the curvature sector; for the Wick sector it gives invariance under $\mathbb{R}^3 \rtimes O_h$ on the tube and no Euclidean clause. Euclidean invariance of the joint field is the hypothesis (R_W) of the optional extension (Hypothesis 23.3, Theorem 23.7), and on King's convention (FA^K) (Hypothesis 23.5); whether the heavy-shell comparison of the rotation part extends to the Wick square, on the tube or at arbitrary amplitudes, is open. Cubic symmetry is a theorem in every case. The two statements the rotation part would need for the Wick sector are open: the torus comparison with scalar sources $(\mathrm{K})_a$ (Definition 17.2) and the closing property $(\mathrm{TIGHT})_a$ (Definition 18.5) for a W-admissible scheme; given both, Theorems 18.9 and 18.11 give (R_W). In the curvature sector both are theorems off the set E_W (Theorems 17.1 and 18.4, Remark 15.2), and Theorem 15.1 holds at every point.

23.7. **The coupling condition.** The sign clause (T3) is proved on the explicit regions of Proposition 22.14 (at $\mu_0 = m$ on a region containing $0 \le \lambda, e^2 \le 0.1m$; at every $\mu_0 \in (0,m]$ on explicit boxes; exactly for $\lambda < \frac{\pi}{4}(1+\sqrt{5})\,m$ at $e = 0$), and the Wick leg is stated there. Outside those regions the exact position of the failure set $E^<_W$ is not determined; the two-sided clause that m is not a global minimiser of α_∞, with the auxiliary mass on either side of m (Proposition 12.11,

Lemma 22.1(b)), would enlarge the region. The cell bound (B_{cell}), the scheme bound (SCH) and the independence of the subsequence (UNIQ) of the statements on continuum subsequences are proved (Lemmas 14.4 and 14.5 and Proposition 14.6, Proposition 14.12); a LIM state of a scheme that is not admissible, or a LIM state along a family of photon masses, would require the first two to be re-established for it. The mass clause of [2, Prop. 3.1], a cap on the dimensionless scalar mass at the stopping scales where that proposition is used, belongs to the statement of the proposition, which is cited under it, and imposes nothing on the bare mass (Remark 22.2).

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